

johnston-linear-matrix-algebra: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

| Identifier                            | Page | Conf. | LaTeX source                          | Rendered                              | Image                                                                                                                                              |
|---------------------------------------|------|-------|---------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra_F00001 | 7    | 1.000 | $\cos (\pi / 6)=\sqrt{3} / 2$         | $\cos (\pi / 6)=\sqrt{3} / 2$         | For example, if we make use of the fact that $\cos (\pi / 6)=\sqrt{3} / 2$ in the middle of a long calculation, we                                 |
| johnston-linear-matrix-algebra_F00002 | 8    | 1.000 | $\mathbb{R}^n$                        | $\mathbb{R}^n$                        | this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$ , where $\mathbb{F}$ is a |
| johnston-linear-matrix-algebra_F00003 | 9    | 1.000 | $\mathbb{C}^n$                        | $\mathbb{C}^n$                        | this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$ , where $\mathbb{F}$ is a |
| johnston-linear-matrix-algebra_F00004 | 9    | 1.000 | $\mathbb{F}^n$                        | $\mathbb{F}^n$                        | this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$ , where $\mathbb{F}$ is a |
| johnston-linear-matrix-algebra_F00005 | 9    | 1.000 | $\mathbb{F}$                          | $\mathbb{F}$                          | this book considers the concrete vector spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ exclusively (and briefly $\mathbb{F}^n$ , where $\mathbb{F}$ is a |
| johnston-linear-matrix-algebra_F00006 | 16   | 1.000 | $a \in S$                             | $a \in S$                             | The notation $a \in S$                                                                                                                             |
| johnston-linear-matrix-algebra_F00007 | 16   | 1.000 | $a$                                   | $a$                                   | object $a$ is in the                                                                                                                               |
| johnston-linear-matrix-algebra_F00008 | 16   | 1.000 | $S$                                   | $S$                                   | set $S$ , so $\mathbf{v} \in \mathbb{R}^n$                                                                                                         |
| johnston-linear-matrix-algebra_F00009 | 16   | 1.000 | $\mathbf{v} \in \mathbb{R}^n$         | $\mathbf{v} \in \mathbb{R}^n$         | set $S$ , so $\mathbf{v} \in \mathbb{R}^n$                                                                                                         |
| johnston-linear-matrix-algebra_F00010 | 16   | 1.000 | $\mathbf{v}$                          | $\mathbf{v}$                          | vector $\mathbf{v}$ is in the set                                                                                                                  |
| johnston-linear-matrix-algebra_F00011 | 16   | 1.000 | $n$                                   | $n$                                   | $\mathbb{R}^n$ of $n$ -dimensional                                                                                                                 |
| johnston-linear-matrix-algebra_F00012 | 16   | 1.000 | $4 x-3=7$                             | $4 x-3=7$                             | single variable. For example, we learned how to solve equations like $4 x-3=7$                                                                     |
| johnston-linear-matrix-algebra_F00013 | 16   | 1.000 | $f(x)=3 x+8$                          | $f(x)=3 x+8$                          | and we learned about properties of functions like $f(x)=3 x+8$ , where in each                                                                     |
| johnston-linear-matrix-algebra_F00014 | 16   | 1.000 | $x$                                   | $x$                                   | case the one variable was called “ $x$ ”. One way of looking at linear algebra                                                                     |
| johnston-linear-matrix-algebra_F00015 | 16   | 1.000 | $3 x+2 y=1$                           | $3 x+2 y=1$                           | more variables. For example, we might try solving an equation like $3 x+2 y=1$ ,                                                                   |
| johnston-linear-matrix-algebra_F00016 | 16   | 0.679 | $\mathbf{w}$                          | $\mathbf{w}$                          | themselves by lowercase bold letters like $\mathbf{v}$ and $\mathbf{w}$ , and we write their entries                                               |
| johnston-linear-matrix-algebra_F00017 | 16   | 0.644 | $\mathbf{v}=(2,3) \in \mathbb{R}^2$   | $\mathbf{v}=(2,3) \in \mathbb{R}^2$   | within parentheses. For example, $\mathbf{v}=(2,3) \in \mathbb{R}^2$ is a 2-dimensional vector and                                                 |
| johnston-linear-matrix-algebra_F00018 | 16   | 0.611 | $\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ | $\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ | $\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ is a 3-dimensional vector (just like $4 \in \mathbb{R}$ is a real number).                                   |
| johnston-linear-matrix-algebra_F00019 | 16   | 0.611 | $4 \in \mathbb{R}$                    | $4 \in \mathbb{R}$                    | $\mathbf{w}=(1,3,2) \in \mathbb{R}^3$ is a 3-dimensional vector (just like $4 \in \mathbb{R}$ is a real number).                                   |
| johnston-linear-matrix-algebra_F00020 | 16   | 0.968 | $y$                                   | $y$                                   | ond entry represents displacement in the $y$ -direction, and in the 3-dimensional                                                                  |
| johnston-linear-matrix-algebra_F00021 | 16   | 1.000 | $z$                                   | $z$                                   | case its third entry represents displacement in the $z$ -direction, as in Figure 1.1.                                                              |
| johnston-linear-matrix-algebra_F00022 | 17   | 1.000 | $\vec{v}$                             | $\vec{v}$                             | with arrows like $\vec{v}$ , or                                                                                                                    |
| johnston-linear-matrix-algebra_F00023 | 17   | 1.000 | $\overrightarrow{A B}$                | $\overrightarrow{A B}$                | $\overrightarrow{A B}$ if they wish to                                                                                                             |
| johnston-linear-matrix-algebra_F00024 | 17   | 1.000 | $A$                                   | $A$                                   | $A$ and its head is                                                                                                                                |

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| johnston-linear-matrix-algebra-F00025 | 17   | ■ 1.000 | B                                                                                 | $B$                                                                               | located at point $B$ .                                                                                                             |
| johnston-linear-matrix-algebra-F00026 | 17   | ■ 0.968 | $(-1, 1)$                                                                         | $(-1, 1)$                                                                         | $(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$ . However, this is also the                                         |
| johnston-linear-matrix-algebra-F00027 | 17   | ■ 0.968 | $(2, 2)$                                                                          | $(2, 2)$                                                                          | $(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$ . However, this is also the                                         |
| johnston-linear-matrix-algebra-F00028 | 17   | ■ 0.968 | $(2, 2) - (-1, 1) = (3, 1)$                                                       | $(2, 2) - (-1, 1) = (3, 1)$                                                       | $(-1, 1)$ to the point $(2, 2)$ is $(2, 2) - (-1, 1) = (3, 1)$ . However, this is also the                                         |
| johnston-linear-matrix-algebra-F00029 | 17   | ■ 0.395 | $(4, 1) - (1, 0) = (3, 1)$                                                        | $(4, 1) - (1, 0) = (3, 1)$                                                        | same as the vector that points from $(1, 0)$ to $(4, 1)$ , since $(4, 1) - (1, 0) = (3, 1)$                                        |
| johnston-linear-matrix-algebra-F00030 | 17   | ■ 1.000 | $\mathbf{v} = (3, 1)$                                                             | $\mathbf{v} = (3, 1)$                                                             | <b>Figure 1.2:</b> Three copies of the vector $\mathbf{v} = (3, 1)$ located at different positions in                              |
| johnston-linear-matrix-algebra-F00031 | 17   | ■ 1.000 | $\mathbb{R}^7$                                                                    | $\mathbb{R}^7$                                                                    | and theorems themselves should work just as well in $\mathbb{R}^7$ (even though we can-                                            |
| johnston-linear-matrix-algebra-F00032 | 17   | ■ 1.000 | $\mathbb{R}^3$                                                                    | $\mathbb{R}^3$                                                                    | not really visualize this space) as they do in $\mathbb{R}^3$ . For this reason, we typically                                      |
| johnston-linear-matrix-algebra-F00033 | 18   | ■ 1.000 | $\mathbf{v} + \mathbf{w}$                                                         | $\mathbf{v} + \mathbf{w}$                                                         | vector $\mathbf{v} + \mathbf{w}$ is the                                                                                            |
| johnston-linear-matrix-algebra-F00034 | 18   | ■ 1.000 | $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$                            | $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$                            | Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ are vec- |
| johnston-linear-matrix-algebra-F00035 | 18   | ■ 1.000 | $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$                            | $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$                            | Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ are vec- |
| johnston-linear-matrix-algebra-F00036 | 18   | ■ 1.000 | $\mathbf{v} +$                                                                    | $\mathbf{v} +$                                                                    | If we instead work entirely with vectors in standard position, then $\mathbf{v} +$                                                 |
| johnston-linear-matrix-algebra-F00037 | 18   | ■ 1.000 | $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$                             | $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$                             | Suppose $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors. Then the following properties hold:                     |
| johnston-linear-matrix-algebra-F00038 | 18   | ■ 1.000 | $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$                               | $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$                               | a) $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$ , and                                                                       |
| johnston-linear-matrix-algebra-F00039 | 18   | ■ 1.000 | $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ | $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ | b) $(\mathbf{v} + \mathbf{w}) + \mathbf{x} = \mathbf{v} + (\mathbf{w} + \mathbf{x})$ .                                             |
| johnston-linear-matrix-algebra-F00040 | 18   | ■ 1.000 | $x + y = y + x$                                                                   | $x + y = y + x$                                                                   | $x + y = y + x$ for all $x, y \in \mathbb{R}$ :                                                                                    |
| johnston-linear-matrix-algebra-F00041 | 18   | ■ 1.000 | $x, y \in \mathbb{R}$                                                             | $x, y \in \mathbb{R}$                                                             | $x + y = y + x$ for all $x, y \in \mathbb{R}$ :                                                                                    |
| johnston-linear-matrix-algebra-F00042 | 19   | —       | $\square$                                                                         | $\square$                                                                         | —                                                                                                                                  |
| johnston-linear-matrix-algebra-F00043 | 19   | ■ 1.000 | $\mathbf{v} + \mathbf{w} + \mathbf{x}$                                            | $\mathbf{v} + \mathbf{w} + \mathbf{x}$                                            | this theorem shows that expressions like $\mathbf{v} + \mathbf{w} + \mathbf{x}$ make sense, since there is                         |
| johnston-linear-matrix-algebra-F00044 | 19   | ■ 1.000 | $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$                                          | $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$                                          | no need to question whether it means $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$ or $\mathbf{v} + (\mathbf{w} + \mathbf{x})$ .        |
| johnston-linear-matrix-algebra-F00045 | 19   | ■ 1.000 | $\mathbf{v} + (\mathbf{w} + \mathbf{x})$                                          | $\mathbf{v} + (\mathbf{w} + \mathbf{x})$                                          | no need to question whether it means $(\mathbf{v} + \mathbf{w}) + \mathbf{x}$ or $\mathbf{v} + (\mathbf{w} + \mathbf{x})$ .        |
| johnston-linear-matrix-algebra-F00046 | 19   | ■ 1.000 | $\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$                  | $\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$                  | commutative (i.e., the order of “multiplication” matters since $\mathbf{v} \times \mathbf{w} \neq \mathbf{w} \times \mathbf{v}$ ), |
| johnston-linear-matrix-algebra-F00047 | 19   | ■ 1.000 | $(2, 5, -1) + (1, -1, 2)$                                                         | $(2, 5, -1) + (1, -1, 2)$                                                         | a) $(2, 5, -1) + (1, -1, 2)$ ,                                                                                                     |
| johnston-linear-matrix-algebra-F00048 | 19   | ■ 1.000 | $(1, 2) + (3, 1) + (2, -1)$                                                       | $(1, 2) + (3, 1) + (2, -1)$                                                       | b) $(1, 2) + (3, 1) + (2, -1)$ , and                                                                                               |
| johnston-linear-matrix-algebra-F00049 | 19   | ■ 0.672 | $0, 0, 0$                                                                         | $0, 0, 0$                                                                         | a cube with opposite corners at $(0, 0, 0)$ and $(1, 1, 1)$ , as shown:                                                            |
| johnston-linear-matrix-algebra-F00050 | 19   | ■ 0.672 | $1, 1, 1$                                                                         | $1, 1, 1$                                                                         | a cube with opposite corners at $(0, 0, 0)$ and $(1, 1, 1)$ , as shown:                                                            |
| johnston-linear-matrix-algebra-F00051 | 19   | ■ 1.000 | $(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$                          | $(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$                          | a) $(2, 5, -1) + (1, -1, 2) = (2+1, 5-1, -1+2) = (3, 4, 1)$ .                                                                      |
| johnston-linear-matrix-algebra-F00052 | 19   | ■ 1.000 | $(1, 2) + (3, 1) + (2, -1) = (1+3+2, 2+1-1) = (6, 2)$                             | $(1, 2) + (3, 1) + (2, -1) = (1+3+2, 2+1-1) = (6, 2)$                             | b) $(1, 2) + (3, 1) + (2, -1) = (1+3+2, 2+1-1) = (6, 2)$ . Note that                                                               |

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| johnston-linear-matrix-algebra_F00053 | 20   | 1.000 | $c \in \mathbb{R}$                                                                         | $c \in \mathbb{R}$                                                                         | Suppose $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ is a vector and $c \in \mathbb{R}$ is a scalar. Then                                          |
| johnston-linear-matrix-algebra_F00054 | 20   | 1.000 | $c \mathbf{v}$                                                                             | $c\mathbf{v}$                                                                              | their <b>scalar multiplication</b> , denoted by $c\mathbf{v}$ , is the vector                                                                                |
| johnston-linear-matrix-algebra_F00055 | 20   | 0.531 | $c$                                                                                        | $c$                                                                                        | of $\mathbf{v}$ by $c$ seems like a rather natural operation, and it has the simple geometric                                                                |
| johnston-linear-matrix-algebra_F00056 | 20   | 0.967 | $ c  > 1$                                                                                  | $ c  > 1$                                                                                  | if $ c  > 1$ then scalar multiplication stretches $\mathbf{v}$ , but if $ c  < 1$ then it shrinks $\mathbf{v}$ .                                             |
| johnston-linear-matrix-algebra_F00057 | 20   | 0.967 | $ c  < 1$                                                                                  | $ c  < 1$                                                                                  | if $ c  > 1$ then scalar multiplication stretches $\mathbf{v}$ , but if $ c  < 1$ then it shrinks $\mathbf{v}$ .                                             |
| johnston-linear-matrix-algebra_F00058 | 20   | 0.967 | $c < 0$                                                                                    | $c < 0$                                                                                    | When $c < 0$ then this operation also reverses the direction of $\mathbf{v}$ , in addition to                                                                |
| johnston-linear-matrix-algebra_F00059 | 20   | 1.000 | $ c  \neq 1$                                                                               | $ c  \neq 1$                                                                               | any stretching or shrinking that it does if $ c  \neq 1$ .                                                                                                   |
| johnston-linear-matrix-algebra_F00060 | 20   | 0.772 | $c = 0$                                                                                    | $c = 0$                                                                                    | • If $c = 0$ then $c\mathbf{v}$ is the <b>zero vector</b> , all of whose entries are 0, which we                                                             |
| johnston-linear-matrix-algebra_F00061 | 20   | 1.000 | $\mathbf{0}$                                                                               | $\mathbf{0}$                                                                               | denote by $\mathbf{0}$ .                                                                                                                                     |
| johnston-linear-matrix-algebra_F00062 | 20   | 0.957 | $c = -1$                                                                                   | $c = -1$                                                                                   | • If $c = -1$ then $c\mathbf{v}$ is the vector whose entries are the negatives of $\mathbf{v}$ 's                                                            |
| johnston-linear-matrix-algebra_F00063 | 20   | 0.637 | $\mathrm{via} \mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$ | $\mathrm{via} \mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$ | We also define <b>vector subtraction</b> via $\mathbf{v} - \mathbf{w} \stackrel{\text{def}}{=} \mathbf{v} + (-\mathbf{w})$ , and we note that                |
| johnston-linear-matrix-algebra_F00064 | 20   | 1.000 | $\mathbf{v} - \mathbf{w}$                                                                  | $\mathbf{v} - \mathbf{w}$                                                                  | it has the geometric interpretation that $\mathbf{v} - \mathbf{w}$ is the vector pointing from the                                                           |
| johnston-linear-matrix-algebra_F00065 | 21   | 1.000 | $\mathbf{v} - \mathbf{v} = \mathbf{0}$                                                     | $\mathbf{v} - \mathbf{v} = \mathbf{0}$                                                     | such as the facts that $\mathbf{v} - \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every vector $\mathbf{v} \in \mathbb{R}^n$ , by |
| johnston-linear-matrix-algebra_F00066 | 21   | 1.000 | $\mathbf{v} + \mathbf{0} = \mathbf{v}$                                                     | $\mathbf{v} + \mathbf{0} = \mathbf{v}$                                                     | such as the facts that $\mathbf{v} - \mathbf{v} = \mathbf{0}$ and $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for every vector $\mathbf{v} \in \mathbb{R}^n$ , by |
| johnston-linear-matrix-algebra_F00067 | 21   | 1.000 | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$                                                  | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$                                                  | Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors and $c, d \in \mathbb{R}$ are scalars. Then the following                                      |
| johnston-linear-matrix-algebra_F00068 | 21   | 1.000 | $c, d \in \mathbb{R}$                                                                      | $c, d \in \mathbb{R}$                                                                      | Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors and $c, d \in \mathbb{R}$ are scalars. Then the following                                      |
| johnston-linear-matrix-algebra_F00069 | 21   | 1.000 | $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$                                   | $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$                                   | a) $c(\mathbf{v} + \mathbf{w}) = c\mathbf{v} + c\mathbf{w}$ ,                                                                                                |
| johnston-linear-matrix-algebra_F00070 | 21   | 1.000 | $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$                                            | $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$                                            | b) $(c + d)\mathbf{v} = c\mathbf{v} + d\mathbf{v}$ , and                                                                                                     |
| johnston-linear-matrix-algebra_F00071 | 21   | 1.000 | $c(d\mathbf{v}) = (cd)\mathbf{v}$                                                          | $c(d\mathbf{v}) = (cd)\mathbf{v}$                                                          | c) $c(d\mathbf{v}) = (cd)\mathbf{v}$ .                                                                                                                       |
| johnston-linear-matrix-algebra_F00072 | 21   | 1.000 | $3\mathbf{v} - 2\mathbf{w}$                                                                | $3\mathbf{v} - 2\mathbf{w}$                                                                | a) $3\mathbf{v} - 2\mathbf{w}$ , where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$ , and                                                         |
| johnston-linear-matrix-algebra_F00073 | 21   | 1.000 | $\mathbf{v} = (2, 1, -1)$                                                                  | $\mathbf{v} = (2, 1, -1)$                                                                  | a) $3\mathbf{v} - 2\mathbf{w}$ , where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$ , and                                                         |
| johnston-linear-matrix-algebra_F00074 | 21   | 1.000 | $\mathbf{w} = (-1, 0, 3)$                                                                  | $\mathbf{w} = (-1, 0, 3)$                                                                  | a) $3\mathbf{v} - 2\mathbf{w}$ , where $\mathbf{v} = (2, 1, -1)$ and $\mathbf{w} = (-1, 0, 3)$ , and                                                         |
| johnston-linear-matrix-algebra_F00075 | 21   | 1.000 | $(0, 0)$                                                                                   | $(0, 0)$                                                                                   | b) the sum of the 6 vectors that point from the center $(0, 0)$ of a regu-                                                                                   |
| johnston-linear-matrix-algebra_F00076 | 22   | 1.000 | $\mathbf{x} = (1, 1, 1)$                                                                   | $\mathbf{x} = (1, 1, 1)$                                                                   | $\mathbf{x} = (1, 1, 1)$ follows                                                                                                                             |
| johnston-linear-matrix-algebra_F00077 | 22   | 0.968 | $4\mathbf{x} = (4, 4, 4)$                                                                  | $4\mathbf{x} = (4, 4, 4)$                                                                  | $4\mathbf{x} = (4, 4, 4)$ .                                                                                                                                  |
| johnston-linear-matrix-algebra_F00078 | 22   | 1.000 | $3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$                         | $3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$                         | a) $3\mathbf{v} - 2\mathbf{w} = (6, 3, -3) - (-2, 0, 6) = (8, 3, -9)$ .                                                                                      |
| johnston-linear-matrix-algebra_F00079 | 22   | 0.980 | $\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$                | $\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$                | as $\mathbf{v}, \mathbf{w}, \mathbf{x}, -\mathbf{v}, -\mathbf{w}, -\mathbf{x}$ (since the final 3 vectors point in the op-                                   |
| johnston-linear-matrix-algebra_F00080 | 22   | 1.000 | $\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$ | $\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$ | $\mathbf{v} + \mathbf{w} + \mathbf{x} - \mathbf{v} - \mathbf{w} - \mathbf{x} = \mathbf{0}$ .                                                                 |

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| johnston-linear-matrix-algebra-F00081 | 22   | ■ 0.972 | $\mathbf{x}$                                                                                      | $\mathbf{x}$                                                                                      | Solve the following equations for the vector $\mathbf{x}$ :                                                                                        |
| johnston-linear-matrix-algebra-F00082 | 22   | ■ 1.000 | $\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3 \mathbf{x}$                                               | $\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3\mathbf{x}$                                                | a) $\mathbf{x} - (3, 2, 1) = (1, 2, 3) - 3\mathbf{x}$ , and                                                                                        |
| johnston-linear-matrix-algebra-F00083 | 22   | ■ 1.000 | $\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$              | $\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$              | b) $\mathbf{x} + 2(\mathbf{v} + \mathbf{w}) = -\mathbf{v} - 3(\mathbf{x} - \mathbf{w})$ .                                                          |
| johnston-linear-matrix-algebra-F00084 | 22   | ■ 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$                                                 | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$                                                 | vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ and then use vector addition and scalar multiplication to                                |
| johnston-linear-matrix-algebra-F00085 | 23   | ■ 1.000 | $\mathbf{e}_j$                                                                                    | $\mathbf{e}_j$                                                                                    | dimension of $\mathbf{e}_j$ will                                                                                                                   |
| johnston-linear-matrix-algebra-F00086 | 23   | ■ 0.574 | $\mathbf{e}_3 \in \mathbb{R}^7$                                                                   | $\mathbf{e}_3 \in \mathbb{R}^7$                                                                   | $\mathbf{e}_3 \in \mathbb{R}^7$ .                                                                                                                  |
| johnston-linear-matrix-algebra-F00087 | 23   | ■ 0.999 | $v_1, v_2, \dots, v_n$                                                                            | $v_1, v_2, \dots, v_n$                                                                            | object: $v_1, v_2, \dots, v_n$                                                                                                                     |
| johnston-linear-matrix-algebra-F00088 | 23   | ■ 1.000 | $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$                                                 | $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$                                                 | $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are                                                                                              |
| johnston-linear-matrix-algebra-F00089 | 23   | ■ 0.997 | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$                                | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$                                | A <b>linear combination</b> of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is any vector of                     |
| johnston-linear-matrix-algebra-F00090 | 23   | ■ 0.999 | $c_1, c_2, \dots, c_k \in \mathbb{R}$                                                             | $c_1, c_2, \dots, c_k \in \mathbb{R}$                                                             | where $c_1, c_2, \dots, c_k \in \mathbb{R}$ .                                                                                                      |
| johnston-linear-matrix-algebra-F00091 | 23   | ■ 0.647 | $(-1, 0, 1)$                                                                                      | $(-1, 0, 1)$                                                                                      | $(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$ . On the other hand, $(1, 2, 3)$ is                                                       |
| johnston-linear-matrix-algebra-F00092 | 23   | ■ 0.647 | $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$                                                             | $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$                                                             | $(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$ . On the other hand, $(1, 2, 3)$ is                                                       |
| johnston-linear-matrix-algebra-F00093 | 23   | ■ 0.647 | $(1, 2, 3)$                                                                                       | $(1, 2, 3)$                                                                                       | $(-1, 0, 1)$ since $(1, 2, 3) = 2(1, 1, 1) + (-1, 0, 1)$ . On the other hand, $(1, 2, 3)$ is                                                       |
| johnston-linear-matrix-algebra-F00094 | 23   | ■ 1.000 | $c_1(1, 1, 0) + c_2(2, 1, 0)$                                                                     | $c_1(1, 1, 0) + c_2(2, 1, 0)$                                                                     | of the form $c_1(1, 1, 0) + c_2(2, 1, 0)$ has a 0 in its third entry, and thus cannot                                                              |
| johnston-linear-matrix-algebra-F00095 | 23   | ■ 1.000 | $j=1, 2, \dots, n$                                                                                | $j = 1, 2, \dots, n$                                                                              | Specifically, for each $j = 1, 2, \dots, n$ , we define the vector $\mathbf{e}_j \in \mathbb{R}^n$ by                                              |
| johnston-linear-matrix-algebra-F00096 | 23   | ■ 1.000 | $\mathbf{e}_j \in \mathbb{R}^n$                                                                   | $\mathbf{e}_j \in \mathbb{R}^n$                                                                   | Specifically, for each $j = 1, 2, \dots, n$ , we define the vector $\mathbf{e}_j \in \mathbb{R}^n$ by                                              |
| johnston-linear-matrix-algebra-F00097 | 23   | ■ 1.000 | $\mathbb{R}^2$                                                                                    | $\mathbb{R}^2$                                                                                    | For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ .                                   |
| johnston-linear-matrix-algebra-F00098 | 23   | ■ 1.000 | $\mathbf{e}_1 = (1, 0)$                                                                           | $\mathbf{e}_1 = (1, 0)$                                                                           | For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ .                                   |
| johnston-linear-matrix-algebra-F00099 | 23   | ■ 1.000 | $\mathbf{e}_2 = (0, 1)$                                                                           | $\mathbf{e}_2 = (0, 1)$                                                                           | For example, in $\mathbb{R}^2$ there are two such vectors: $\mathbf{e}_1 = (1, 0)$ and $\mathbf{e}_2 = (0, 1)$ .                                   |
| johnston-linear-matrix-algebra-F00100 | 23   | ■ 1.000 | $\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0)$                                              | $\mathbf{e}_1 = (1, 0, 0), \mathbf{e}_2 = (0, 1, 0)$                                              | Similarly, in $\mathbb{R}^3$ there are three such vectors: $\mathbf{e}_1 = (1, 0, 0)$ , $\mathbf{e}_2 = (0, 1, 0)$ , and                           |
| johnston-linear-matrix-algebra-F00101 | 23   | ■ 1.000 | $\mathbf{e}_3 = (0, 0, 1)$                                                                        | $\mathbf{e}_3 = (0, 0, 1)$                                                                        | $\mathbf{e}_3 = (0, 0, 1)$ . In general, in $\mathbb{R}^n$ there are $n$ of these vectors, $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ , and |
| johnston-linear-matrix-algebra-F00102 | 23   | ■ 0.540 | $x - y -$                                                                                         | $x - y -$                                                                                         | <b>Figure 1.6:</b> The standard basis vectors point a distance of 1 along the $x$ -, $y$ -, and                                                    |
| johnston-linear-matrix-algebra-F00103 | 23   | ■ 1.000 | $\mathbf{v} = (v_1, v_2, \dots, v_n)$                                                             | $\mathbf{v} = (v_1, v_2, \dots, v_n)$                                                             | if $\mathbf{v} = (v_1, v_2, \dots, v_n)$ then                                                                                                      |
| johnston-linear-matrix-algebra-F00104 | 24   | ■ 1.000 | $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$                                   | $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$                                   | a) Compute $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 \in \mathbb{R}^3$ , and                                                                   |
| johnston-linear-matrix-algebra-F00105 | 24   | ■ 0.973 | $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$                         | $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$                         | b) Write $(3, 5, -2, -1)$ as a linear combination of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4 \in \mathbb{R}^4$ .                   |
| johnston-linear-matrix-algebra-F00106 | 24   | ■ 1.000 | $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$ | $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$ | a) $3\mathbf{e}_1 - 2\mathbf{e}_2 + \mathbf{e}_3 = 3(1, 0, 0) - 2(0, 1, 0) + (0, 0, 1) = (3, -2, 1)$ . In gen-                                     |
| johnston-linear-matrix-algebra-F00107 | 24   | ■ 1.000 | $\mathbf{e}_1$                                                                                    | $\mathbf{e}_1$                                                                                    | ing vector has the coefficient of $\mathbf{e}_1$ in its first entry, the coefficient                                                               |

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| johnston-linear-matrix-algebra-F00108 | 24   | ■ 1.000 | $\mathbf{e}_2$                                                                              | $\mathbf{e}_2$                                                                              | of $\mathbf{e}_2$ in its second entry, and so on.                                                      |
| johnston-linear-matrix-algebra-F00109 | 24   | ■ 1.000 | $(3, 5, -2, -1) = 3 \mathbf{e}_1 + 5 \mathbf{e}_2 - 2 \mathbf{e}_3 - \mathbf{e}_4$          | $(3, 5, -2, -1) = 3\mathbf{e}_1 + 5\mathbf{e}_2 - 2\mathbf{e}_3 - \mathbf{e}_4$             | linear combination: $(3, 5, -2, -1) = 3\mathbf{e}_1 + 5\mathbf{e}_2 - 2\mathbf{e}_3 - \mathbf{e}_4$ .  |
| johnston-linear-matrix-algebra-F00110 | 24   | ■ 1.000 | $\mathbf{v} = (3, 2)$                                                                       | $\mathbf{v} = (3, 2)$                                                                       | *(a) $\mathbf{v} = (3, 2)$                                                                             |
| johnston-linear-matrix-algebra-F00111 | 24   | ■ 1.000 | $\mathbf{w} = (-0.5, 3)$                                                                    | $\mathbf{w} = (-0.5, 3)$                                                                    | (b) $\mathbf{w} = (-0.5, 3)$                                                                           |
| johnston-linear-matrix-algebra-F00112 | 24   | ■ 0.998 | $\mathbf{x} = (1, -3)$                                                                      | $\mathbf{x} = (1, -3)$                                                                      | *(c) $\mathbf{x} = (1, -3)$                                                                            |
| johnston-linear-matrix-algebra-F00113 | 24   | ■ 1.000 | $\mathbf{y} = (-2, -1)$                                                                     | $\mathbf{y} = (-2, -1)$                                                                     | (d) $\mathbf{y} = (-2, -1)$                                                                            |
| johnston-linear-matrix-algebra-F00114 | 24   | ■ 1.000 | $\mathbf{v} = (0, 0, 2)$                                                                    | $\mathbf{v} = (0, 0, 2)$                                                                    | *(a) $\mathbf{v} = (0, 0, 2)$                                                                          |
| johnston-linear-matrix-algebra-F00115 | 24   | ■ 1.000 | $\mathbf{w} = (-1, 2, 1)$                                                                   | $\mathbf{w} = (-1, 2, 1)$                                                                   | (b) $\mathbf{w} = (-1, 2, 1)$                                                                          |
| johnston-linear-matrix-algebra-F00116 | 24   | ■ 0.999 | $\mathbf{x} = (1, 2, 0)$                                                                    | $\mathbf{x} = (1, 2, 0)$                                                                    | *(c) $\mathbf{x} = (1, 2, 0)$                                                                          |
| johnston-linear-matrix-algebra-F00117 | 24   | ■ 1.000 | $\mathbf{y} = (3, 2, -1)$                                                                   | $\mathbf{y} = (3, 2, -1)$                                                                   | (d) $\mathbf{y} = (3, 2, -1)$                                                                          |
| johnston-linear-matrix-algebra-F00118 | 24   | ■ 1.000 | $\mathbf{v}, \mathbf{w}, \mathbf{x}$                                                        | $\mathbf{v}, \mathbf{w}, \mathbf{x}$                                                        | 1.1.5 If the vectors $\mathbf{v}, \mathbf{w}, \mathbf{x}$ , and $\mathbf{y}$ are as in Exercise 1.1.1, |
| johnston-linear-matrix-algebra-F00119 | 24   | ■ 1.000 | $\mathbf{y}$                                                                                | $\mathbf{y}$                                                                                | 1.1.5 If the vectors $\mathbf{v}, \mathbf{w}, \mathbf{x}$ , and $\mathbf{y}$ are as in Exercise 1.1.1, |
| johnston-linear-matrix-algebra-F00120 | 24   | ■ 1.000 | $\mathbf{v} + \mathbf{w} + \mathbf{y}$                                                      | $\mathbf{v} + \mathbf{w} + \mathbf{y}$                                                      | (b) $\mathbf{v} + \mathbf{w} + \mathbf{y}$                                                             |
| johnston-linear-matrix-algebra-F00121 | 24   | ■ 0.967 | $\mathbf{y} - 2\mathbf{x}$                                                                  | $\mathbf{y} - 2\mathbf{x}$                                                                  | *(c) $\mathbf{y} - 2\mathbf{x}$                                                                        |
| johnston-linear-matrix-algebra-F00122 | 24   | ■ 1.000 | $\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$                                      | $\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$                                      | (d) $\mathbf{v} + 2\mathbf{w} + 2\mathbf{x} + 2\mathbf{y}$                                             |
| johnston-linear-matrix-algebra-F00123 | 24   | ■ 1.000 | $\mathbf{v} + \mathbf{y}$                                                                   | $\mathbf{v} + \mathbf{y}$                                                                   | *(a) $\mathbf{v} + \mathbf{y}$                                                                         |
| johnston-linear-matrix-algebra-F00124 | 24   | ■ 1.000 | $4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$                                   | $4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$                                   | (b) $4\mathbf{w} + 3\mathbf{w} - (2\mathbf{w} + 6\mathbf{w})$                                          |
| johnston-linear-matrix-algebra-F00125 | 24   | ■ 0.979 | $4\mathbf{x} - 2\mathbf{w}$                                                                 | $4\mathbf{x} - 2\mathbf{w}$                                                                 | *(c) $4\mathbf{x} - 2\mathbf{w}$                                                                       |
| johnston-linear-matrix-algebra-F00126 | 24   | ■ 0.992 | $2\mathbf{x} - \mathbf{w} - \mathbf{y}$                                                     | $2\mathbf{x} - \mathbf{w} - \mathbf{y}$                                                     | (d) $2\mathbf{x} - \mathbf{w} - \mathbf{y}$                                                            |
| johnston-linear-matrix-algebra-F00127 | 24   | ■ 1.000 | $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$                                 | $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$                                 | vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 \in \mathbb{R}^3$ .                                  |
| johnston-linear-matrix-algebra-F00128 | 24   | ■ 1.000 | $\mathbf{v} = (1, 4)$                                                                       | $\mathbf{v} = (1, 4)$                                                                       | $\mathbf{v} = (1, 4)$ and $\mathbf{w} = (-2, 1)$ . Find vectors describing both of                     |
| johnston-linear-matrix-algebra-F00129 | 24   | ■ 1.000 | $\mathbf{w} = (-2, 1)$                                                                      | $\mathbf{w} = (-2, 1)$                                                                      | $\mathbf{v} = (1, 4)$ and $\mathbf{w} = (-2, 1)$ . Find vectors describing both of                     |
| johnston-linear-matrix-algebra-F00130 | 24   | ■ 1.000 | $\mathbf{x} = (3, -2)$                                                                      | $\mathbf{x} = (3, -2)$                                                                      | gram are $\mathbf{x} = (3, -2)$ and $\mathbf{y} = (1, 4)$ . Find vectors describing                    |
| johnston-linear-matrix-algebra-F00131 | 24   | ■ 1.000 | $\mathbf{y} = (1, 4)$                                                                       | $\mathbf{y} = (1, 4)$                                                                       | gram are $\mathbf{x} = (3, -2)$ and $\mathbf{y} = (1, 4)$ . Find vectors describing                    |
| johnston-linear-matrix-algebra-F00132 | 24   | ■ 0.630 | $(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$                                                | $(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$                                                | *(a) $(1, 2) - \mathbf{x} = (3, 4) - 2\mathbf{x}$                                                      |
| johnston-linear-matrix-algebra-F00133 | 24   | ■ 1.000 | $3((1, -1) + \mathbf{x}) = 2\mathbf{x}$                                                     | $3((1, -1) + \mathbf{x}) = 2\mathbf{x}$                                                     | (b) $3((1, -1) + \mathbf{x}) = 2\mathbf{x}$                                                            |
| johnston-linear-matrix-algebra-F00134 | 24   | ■ 1.000 | $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$ | $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$ | *(c) $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 3(\mathbf{x} + 3(\mathbf{x} + 3\mathbf{x}))$       |
| johnston-linear-matrix-algebra-F00135 | 24   | ■ 1.000 | $-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$                            | $-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$                            | (d) $-2(\mathbf{x} - (1, -2)) = \mathbf{x} + 2(\mathbf{x} + (1, 1))$                                   |
| johnston-linear-matrix-algebra-F00136 | 25   | ■ 1.000 | $\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$                                         | $\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$                                         | *(a) $\mathbf{v} - \mathbf{x} = \mathbf{w} + \mathbf{x}$                                               |

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| johnston-linear-matrix-algebra-F00137 | 25   | ■ 1.000 | $2 \mathbf{v} - 3 \mathbf{x} = 4 \mathbf{x} - 5 \mathbf{w}$                         | $2\mathbf{v} - 3\mathbf{x} = 4\mathbf{x} - 5\mathbf{w}$                             | (b) $2\mathbf{v} - 3\mathbf{x} = 4\mathbf{x} - 5\mathbf{w}$                                                                                           |
| johnston-linear-matrix-algebra-F00138 | 25   | ■ 1.000 | $4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$              | $4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$              | *(c) $4(\mathbf{x} + \mathbf{v}) - \mathbf{x} = 2(\mathbf{w} + \mathbf{x})$                                                                           |
| johnston-linear-matrix-algebra-F00139 | 25   | ■ 1.000 | $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$         | $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$         | (d) $2(\mathbf{x} + 2(\mathbf{x} + 2\mathbf{x})) = 2(\mathbf{v} + 2\mathbf{v})$                                                                       |
| johnston-linear-matrix-algebra-F00140 | 25   | ■ 0.856 | $c(1, 2) =$                                                                         | $c(1, 2) =$                                                                         | *1.1.12 Does there exist a scalar $c \in \mathbb{R}$ such that $c(1, 2) =$                                                                            |
| johnston-linear-matrix-algebra-F00141 | 25   | ■ 1.000 | $n \geq 3$                                                                          | $n \geq 3$                                                                          | 1.1.13 Let $n \geq 3$ be an integer and consider the set of $n$                                                                                       |
| johnston-linear-matrix-algebra-F00142 | 25   | ■ 0.844 | $n=6$                                                                               | $n = 6$                                                                             | tors is <b>0</b> . [Hint: We solved the $n = 6$ case in Exam-                                                                                         |
| johnston-linear-matrix-algebra-F00143 | 25   | ■ 0.985 | $\mathbf{v} \cdot \mathbf{w}$                                                       | $\mathbf{v} \cdot \mathbf{w}$                                                       | tors. Then their <b>dot product</b> , denoted by $\mathbf{v} \cdot \mathbf{w}$ , is the quantity                                                      |
| johnston-linear-matrix-algebra-F00144 | 25   | ■ 1.000 | $\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$                                    | $\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$                                    | not a vector. So, for example, the expression $\mathbf{v} \cdot (\mathbf{w} \cdot \mathbf{x})$ does not make sense,                                   |
| johnston-linear-matrix-algebra-F00145 | 25   | ■ 0.995 | $\mathbf{w} \cdot \mathbf{x}$                                                       | $\mathbf{w} \cdot \mathbf{x}$                                                       | since $\mathbf{w} \cdot \mathbf{x}$ is a number, and so we cannot take its dot product with $\mathbf{v}$ . On the                                     |
| johnston-linear-matrix-algebra-F00146 | 25   | ■ 1.000 | $\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$                                        | $\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$                                        | other hand, the expression $\mathbf{v} / (\mathbf{w} \cdot \mathbf{x})$ <i>does</i> make sense, since dividing a vector by                            |
| johnston-linear-matrix-algebra-F00147 | 25   | ■ 1.000 | $(1, 2, 3) \cdot (4, -3, 2)$                                                        | $(1, 2, 3) \cdot (4, -3, 2)$                                                        | a) $(1, 2, 3) \cdot (4, -3, 2)$ ,                                                                                                                     |
| johnston-linear-matrix-algebra-F00148 | 25   | ■ 0.945 | $(3, 6, 2) \cdot (-1, 5, 2, 1)$                                                     | $(3, 6, 2) \cdot (-1, 5, 2, 1)$                                                     | b) $(3, 6, 2) \cdot (-1, 5, 2, 1)$ , and                                                                                                              |
| johnston-linear-matrix-algebra-F00149 | 26   | ■ 1.000 | $j$                                                                                 | $j$                                                                                 | $j$ -th entry and 0s                                                                                                                                  |
| johnston-linear-matrix-algebra-F00150 | 26   | ■ 1.000 | $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$                                         | $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$                                         | c) $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$ , where $1 \leq j \leq n$ .                                                                            |
| johnston-linear-matrix-algebra-F00151 | 26   | ■ 1.000 | $1 \leq j \leq n$                                                                   | $1 \leq j \leq n$                                                                   | c) $(v_1, v_2, \dots, v_n) \cdot \mathbf{e}_j$ , where $1 \leq j \leq n$ .                                                                            |
| johnston-linear-matrix-algebra-F00152 | 26   | ■ 1.000 | $(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$ | $(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$ | a) $(1, 2, 3) \cdot (4, -3, 2) = 1 \cdot 4 + 2 \cdot (-3) + 3 \cdot 2 = 4 - 6 + 6 = 4$ .                                                              |
| johnston-linear-matrix-algebra-F00153 | 26   | ■ 0.997 | $(v_1, v_2, \dots, v_n)$                                                            | $(v_1, v_2, \dots, v_n)$                                                            | $\mathbf{e}_j$ has $n$ entries (the same number of entries as $(v_1, v_2, \dots, v_n)$ ). Then                                                        |
| johnston-linear-matrix-algebra-F00154 | 26   | ■ 0.996 | $\mathbf{v} = \mathbf{w} = (1, 0)$                                                  | $\mathbf{v} = \mathbf{w} = (1, 0)$                                                  | the amount of overlap between $\mathbf{v}$ and $\mathbf{w}$ . For example, if $\mathbf{v} = \mathbf{w} = (1, 0)$ then                                 |
| johnston-linear-matrix-algebra-F00155 | 26   | ■ 0.755 | $\mathbf{v} \cdot \mathbf{w} = 1$                                                   | $\mathbf{v} \cdot \mathbf{w} = 1$                                                   | $\mathbf{v} \cdot \mathbf{w} = 1$ , but as we rotate $\mathbf{w}$ away from $\mathbf{v}$ , their dot product decreases down                           |
| johnston-linear-matrix-algebra-F00156 | 26   | ■ 0.982 | $\mathbf{w} = (0, 1)$                                                               | $\mathbf{w} = (0, 1)$                                                               | to 0 when $\mathbf{v}$ and $\mathbf{w}$ are perpendicular (i.e., when $\mathbf{w} = (0, 1)$ or $\mathbf{w} = (0, -1)$ ),                              |
| johnston-linear-matrix-algebra-F00157 | 26   | ■ 0.982 | $\mathbf{w} = (0, -1)$                                                              | $\mathbf{w} = (0, -1)$                                                              | to 0 when $\mathbf{v}$ and $\mathbf{w}$ are perpendicular (i.e., when $\mathbf{w} = (0, 1)$ or $\mathbf{w} = (0, -1)$ ),                              |
| johnston-linear-matrix-algebra-F00158 | 26   | ■ 0.963 | $\mathbf{w} = (-1, 0)$                                                              | $\mathbf{w} = (-1, 0)$                                                              | points in the opposite direction of $\mathbf{v}$ (i.e., when $\mathbf{w} = (-1, 0)$ ).                                                                |
| johnston-linear-matrix-algebra-F00159 | 26   | ■ 1.000 | $\theta$                                                                            | $\theta$                                                                            | $\theta$ then its coordinates become $\mathbf{w} = (\cos(\theta), \sin(\theta))$ . The dot product between                                            |
| johnston-linear-matrix-algebra-F00160 | 26   | ■ 1.000 | $\mathbf{w} = (\cos(\theta), \sin(\theta))$                                         | $\mathbf{w} = (\cos(\theta), \sin(\theta))$                                         | $\theta$ then its coordinates become $\mathbf{w} = (\cos(\theta), \sin(\theta))$ . The dot product between                                            |
| johnston-linear-matrix-algebra-F00161 | 26   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$      | $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$      | $\mathbf{v}$ and $\mathbf{w}$ is then $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) = \cos(\theta)$ , which is largest when $\theta$ |
| johnston-linear-matrix-algebra-F00162 | 26   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$                   | $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$                   | each other. Here, the dot product between $\mathbf{v}$ and $\mathbf{w}$ is $\mathbf{v} \cdot \mathbf{w} = 1 \cos(\theta) + 0 \sin(\theta) =$          |
| johnston-linear-matrix-algebra-F00163 | 26   | ■ 1.000 | $\cos(\theta)$                                                                      | $\cos(\theta)$                                                                      | $\cos(\theta)$ , which is largest when $\theta$ is small.                                                                                             |
| johnston-linear-matrix-algebra-F00164 | 26   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$                         | $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$                         | a) $\mathbf{v} \cdot \mathbf{w} = \mathbf{w} \cdot \mathbf{v}$ ,                                                                                      |

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| johnston-linear-matrix-algebra_F00165 | 26   | ■ 1.000 | $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ | $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ | b) $\mathbf{v} \cdot (\mathbf{w} + \mathbf{x}) = \mathbf{v} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{x}$ , and             |
| johnston-linear-matrix-algebra_F00166 | 26   | ■ 1.000 | $\mathbf{v} \cdot (c \mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$                                       | $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$                                        | c) $\mathbf{v} \cdot (c\mathbf{w}) = c(\mathbf{v} \cdot \mathbf{w})$ .                                                        |
| johnston-linear-matrix-algebra_F00167 | 27   | ■ 1.000 | $(x+2)(x^2+3x)$                                                                                          | $(x+2)(x^2+3x)$                                                                                          | $(x+2)(x^2+3x)$ , the                                                                                                         |
| johnston-linear-matrix-algebra_F00168 | 27   | ■ 1.000 | $b$                                                                                                      | $b$                                                                                                      | and $b$ , then                                                                                                                |
| johnston-linear-matrix-algebra_F00169 | 27   | ■ 1.000 | $c^2 = a^2 + b^2$                                                                                        | $c^2 = a^2 + b^2$                                                                                        | $c^2 = a^2 + b^2$ (so                                                                                                         |
| johnston-linear-matrix-algebra_F00170 | 27   | ■ 1.000 | $c = \sqrt{a^2 + b^2}$                                                                                   | $c = \sqrt{a^2 + b^2}$                                                                                   | $c = \sqrt{a^2 + b^2}$ .                                                                                                      |
| johnston-linear-matrix-algebra_F00171 | 27   | ■ 0.992 | $\ \mathbf{v}\ $                                                                                         | $\ \mathbf{v}\ $                                                                                         | of a vector $\mathbf{v}$ , which we represent by $\ \mathbf{v}\ $ . The length of a vector $\mathbf{v} = (v_1, v_2) \in$      |
| johnston-linear-matrix-algebra_F00172 | 27   | ■ 0.992 | $\mathbf{v} = (v_1, v_2) \in$                                                                            | $\mathbf{v} = (v_1, v_2) \in$                                                                            | of a vector $\mathbf{v}$ , which we represent by $\ \mathbf{v}\ $ . The length of a vector $\mathbf{v} = (v_1, v_2) \in$      |
| johnston-linear-matrix-algebra_F00173 | 27   | ■ 1.000 | $\mathbf{v} = (v_1, 0) + (0, v_2)$                                                                       | $\mathbf{v} = (v_1, 0) + (0, v_2)$                                                                       | $\mathbb{R}^2$ can be computed by noticing that $\mathbf{v} = (v_1, 0) + (0, v_2)$ , so $\mathbf{v}$ forms the                |
| johnston-linear-matrix-algebra_F00174 | 27   | ■ 1.000 | $(v_1, 0)$                                                                                               | $(v_1, 0)$                                                                                               | $(v_1, 0)$ and $(0, v_2)$ , as illustrated in Figure 1.8(a). Since the length of $(v_1, 0)$ is                                |
| johnston-linear-matrix-algebra_F00175 | 27   | ■ 1.000 | $(0, v_2)$                                                                                               | $(0, v_2)$                                                                                               | $(v_1, 0)$ and $(0, v_2)$ , as illustrated in Figure 1.8(a). Since the length of $(v_1, 0)$ is                                |
| johnston-linear-matrix-algebra_F00176 | 27   | ■ 1.000 | $ v_1 $                                                                                                  | $ v_1 $                                                                                                  | $ v_1 $ and the length of $(0, v_2)$ is $ v_2 $ , the Pythagorean theorem tells us that                                       |
| johnston-linear-matrix-algebra_F00177 | 27   | ■ 1.000 | $ v_2 $                                                                                                  | $ v_2 $                                                                                                  | $ v_1 $ and the length of $(0, v_2)$ is $ v_2 $ , the Pythagorean theorem tells us that                                       |
| johnston-linear-matrix-algebra_F00178 | 27   | ■ 1.000 | $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$                                                          | $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$                                                          | $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$ . In this case, we instead write $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$ , |
| johnston-linear-matrix-algebra_F00179 | 27   | ■ 1.000 | $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$                                                               | $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$                                                               | $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$ . In this case, we instead write $\mathbf{v} = (v_1, v_2, 0) + (0, 0, v_3)$ , |
| johnston-linear-matrix-algebra_F00180 | 27   | ■ 0.995 | $(v_1, v_2, 0)$                                                                                          | $(v_1, v_2, 0)$                                                                                          | given by the vectors $(v_1, v_2, 0)$ and $(0, 0, v_3)$ , as in Figure 1.8(b). Since the                                       |
| johnston-linear-matrix-algebra_F00181 | 27   | ■ 0.995 | $(0, 0, v_3)$                                                                                            | $(0, 0, v_3)$                                                                                            | given by the vectors $(v_1, v_2, 0)$ and $(0, 0, v_3)$ , as in Figure 1.8(b). Since the                                       |
| johnston-linear-matrix-algebra_F00182 | 27   | ■ 1.000 | $\sqrt{v_1^2 + v_2^2}$                                                                                   | $\sqrt{v_1^2 + v_2^2}$                                                                                   | length of $(v_1, v_2, 0)$ is $\sqrt{v_1^2 + v_2^2}$ (it is just a vector in $\mathbb{R}^2$ with an extra “0”                  |
| johnston-linear-matrix-algebra_F00183 | 28   | ■ 1.000 | $ v_3 $                                                                                                  | $ v_3 $                                                                                                  | entry tacked on) and the length of $(0, 0, v_3)$ is $ v_3 $ , the Pythagorean theorem                                         |
| johnston-linear-matrix-algebra_F00184 | 28   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \dots + v_n^2$                                            | $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \dots + v_n^2$                                            | quantity $\mathbf{v} \cdot \mathbf{v} = v_1^2 + v_2^2 + \dots + v_n^2$ is non-negative, so we can take its square root.       |
| johnston-linear-matrix-algebra_F00185 | 28   | ■ 0.963 | $(\cos(\theta), \sin(\theta))$                                                                           | $(\cos(\theta), \sin(\theta))$                                                                           | b) $(\cos(\theta), \sin(\theta))$ , and                                                                                       |
| johnston-linear-matrix-algebra_F00186 | 28   | ■ 1.000 | $\ (2, -5, 4, 6)\  = \sqrt{2^2 + (-5)^2 + 4^2 + 6^2} = \sqrt{81} = 9$                                    | $\ (2, -5, 4, 6)\  = \sqrt{2^2 + (-5)^2 + 4^2 + 6^2} = \sqrt{81} = 9$                                    | a) $\ (2, -5, 4, 6)\  = \sqrt{2^2 + (-5)^2 + 4^2 + 6^2} = \sqrt{81} = 9$ .                                                    |
| johnston-linear-matrix-algebra_F00187 | 28   | ■ 1.000 | $\ (\cos(\theta), \sin(\theta))\  = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \sqrt{1} = 1$               | $\ (\cos(\theta), \sin(\theta))\  = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \sqrt{1} = 1$               | b) $\ (\cos(\theta), \sin(\theta))\  = \sqrt{\cos^2(\theta) + \sin^2(\theta)} = \sqrt{1} = 1$ .                               |
| johnston-linear-matrix-algebra_F00188 | 29   | ■ 1.000 | $\sqrt{c^2} =  c $                                                                                       | $\sqrt{c^2} =  c $                                                                                       | fact that $\sqrt{c^2} =  c $ .                                                                                                |
| johnston-linear-matrix-algebra_F00189 | 29   | ■ 1.000 | $v_1^2 + v_2^2 + \dots + v_n^2$                                                                          | $v_1^2 + v_2^2 + \dots + v_n^2$                                                                          | $v_1^2 + v_2^2 + \dots + v_n^2$                                                                                               |
| johnston-linear-matrix-algebra_F00190 | 29   | ■ 1.000 | $\mathbf{v} = \mathbf{0}$                                                                                | $\mathbf{v} = \mathbf{0}$                                                                                | If $\mathbf{v} = \mathbf{0}$ then we                                                                                          |
| johnston-linear-matrix-algebra_F00191 | 29   | ■ 1.000 | $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$                                                                  | $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$                                                                  | $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$ where $\mathbf{u}$ is                                                                 |

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|---------------------------------------|------|---------|--------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra_F00192 | 29   | ■ 1.000 | $\mathbf{u}$                                                       | $\mathbf{u}$                                                       | $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$ where $\mathbf{u}$ is                                                                                 |
| johnston-linear-matrix-algebra_F00193 | 29   | ■ 1.000 | $\mathbf{v} = (1, 1, 1)$                                           | $\mathbf{v} = (1, 1, 1)$                                           | The main diagonal of this cube is the vector $\mathbf{v} = (1, 1, 1)$ , which has                                                             |
| johnston-linear-matrix-algebra_F00194 | 29   | ■ 1.000 | $\ \mathbf{v}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$               | $\ \mathbf{v}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$               | length $\ \mathbf{v}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ .                                                                                 |
| johnston-linear-matrix-algebra_F00195 | 29   | ■ 0.998 | $\ c \mathbf{v}\  =  c  \ \mathbf{v}\ $                            | $\ c\mathbf{v}\  =  c  \ \mathbf{v}\ $                             | a) $\ c\mathbf{v}\  =  c  \ \mathbf{v}\ $ , and                                                                                               |
| johnston-linear-matrix-algebra_F00196 | 29   | ■ 1.000 | $\ \mathbf{v}\  \geq 0$                                            | $\ \mathbf{v}\  \geq 0$                                            | b) $\ \mathbf{v}\  \geq 0$ , with equality if and only if $\mathbf{v} = \mathbf{0}$ .                                                         |
| johnston-linear-matrix-algebra_F00197 | 29   | ■ 1.000 | $\ \mathbf{0}\  = 0$                                               | $\ \mathbf{0}\  = 0$                                               | straightforward to show that $\ \mathbf{0}\  = 0$ , so to complete the proof we just need to                                                  |
| johnston-linear-matrix-algebra_F00198 | 29   | ■ 1.000 | $\ \mathbf{v}\  = 0$                                               | $\ \mathbf{v}\  = 0$                                               | show that if $\ \mathbf{v}\  = 0$ then $\mathbf{v} = \mathbf{0}$ . Well, if $\ \mathbf{v}\  = 0$ then $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$ ,  |
| johnston-linear-matrix-algebra_F00199 | 29   | ■ 1.000 | $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$                               | $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$                               | show that if $\ \mathbf{v}\  = 0$ then $\mathbf{v} = \mathbf{0}$ . Well, if $\ \mathbf{v}\  = 0$ then $v_1^2 + v_2^2 + \cdots + v_n^2 = 0$ ,  |
| johnston-linear-matrix-algebra_F00200 | 29   | ■ 1.000 | $v_j^2 \geq 0$                                                     | $v_j^2 \geq 0$                                                     | and since $v_j^2 \geq 0$ for each $1 \leq j \leq n$ , with equality if and only if $v_j = 0$ , we see                                         |
| johnston-linear-matrix-algebra_F00201 | 29   | ■ 1.000 | $v_j = 0$                                                          | $v_j = 0$                                                          | and since $v_j^2 \geq 0$ for each $1 \leq j \leq n$ , with equality if and only if $v_j = 0$ , we see                                         |
| johnston-linear-matrix-algebra_F00202 | 29   | ■ 1.000 | $v_1 = v_2 = \cdots = v_n = 0$                                     | $v_1 = v_2 = \cdots = v_n = 0$                                     | that it must be the case that $v_1 = v_2 = \cdots = v_n = 0$ (i.e., $\mathbf{v} = \mathbf{0}$ ).                                              |
| johnston-linear-matrix-algebra_F00203 | 30   | ■ 1.000 | $\mathbf{u} = \mathbf{v} / \ \mathbf{v}\ $                         | $\mathbf{u} = \mathbf{v} / \ \mathbf{v}\ $                         | in the form $\mathbf{v} = \ \mathbf{v}\ \mathbf{u}$ , where $\mathbf{u} = \mathbf{v} / \ \mathbf{v}\ $ is the unique unit vector pointing in  |
| johnston-linear-matrix-algebra_F00204 | 30   | ■ 1.000 | $\mathbf{v} = (1, 0)$                                              | $\mathbf{v} = (1, 0)$                                              | Figure 1.7, where we noticed that the dot product of the vector $\mathbf{v} = (1, 0)$ with                                                    |
| johnston-linear-matrix-algebra_F00205 | 30   | ■ 1.000 | $ \mathbf{v} \cdot \mathbf{w}  \leq \ \mathbf{v}\  \ \mathbf{w}\ $ | $ \mathbf{v} \cdot \mathbf{w}  \leq \ \mathbf{v}\  \ \mathbf{w}\ $ | Suppose that $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors. Then $ \mathbf{v} \cdot \mathbf{w}  \leq \ \mathbf{v}\  \ \mathbf{w}\ $ . |
| johnston-linear-matrix-algebra_F00206 | 30   | ■ 0.994 | $\ c\mathbf{v} + d\mathbf{w}\ ^2$                                  | $\ c\mathbf{v} + d\mathbf{w}\ ^2$                                  | tion of $\mathbf{v}$ and $\mathbf{w}$ . Specifically, if $c, d \in \mathbb{R}$ are any real numbers then $\ c\mathbf{v} + d\mathbf{w}\ ^2$    |
| johnston-linear-matrix-algebra_F00207 | 30   | ■ 1.000 | $d$                                                                | $d$                                                                | for all real numbers $c$ and $d$ . Well, if $\mathbf{w} = \mathbf{0}$ then the Cauchy–Schwarz inequality                                      |
| johnston-linear-matrix-algebra_F00208 | 30   | ■ 1.000 | $\mathbf{w} = \mathbf{0}$                                          | $\mathbf{w} = \mathbf{0}$                                          | for all real numbers $c$ and $d$ . Well, if $\mathbf{w} = \mathbf{0}$ then the Cauchy–Schwarz inequality                                      |
| johnston-linear-matrix-algebra_F00209 | 30   | ■ 1.000 | $0 \leq 0$                                                         | $0 \leq 0$                                                         | follows trivially since it just says that $0 \leq 0$ , and otherwise we can choose                                                            |
| johnston-linear-matrix-algebra_F00210 | 30   | ■ 1.000 | $c = \ \mathbf{w}\ $                                               | $c = \ \mathbf{w}\ $                                               | $c = \ \mathbf{w}\ $ and $d = -(\mathbf{v} \cdot \mathbf{w}) / \ \mathbf{w}\ $ in the above inequality to see that                            |
| johnston-linear-matrix-algebra_F00211 | 30   | ■ 1.000 | $d = -(\mathbf{v} \cdot \mathbf{w}) / \ \mathbf{w}\ $              | $d = -(\mathbf{v} \cdot \mathbf{w}) / \ \mathbf{w}\ $              | $c = \ \mathbf{w}\ $ and $d = -(\mathbf{v} \cdot \mathbf{w}) / \ \mathbf{w}\ $ in the above inequality to see that                            |
| johnston-linear-matrix-algebra_F00212 | 30   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\ $                                      | $\ \mathbf{v} + \mathbf{w}\ $                                      | lengths—the fact that $\ \mathbf{v} + \mathbf{w}\ $ is never larger than $\ \mathbf{v}\  + \ \mathbf{w}\ $ . To get some                      |
| johnston-linear-matrix-algebra_F00213 | 30   | ■ 1.000 | $\ \mathbf{v}\  + \ \mathbf{w}\ $                                  | $\ \mathbf{v}\  + \ \mathbf{w}\ $                                  | lengths—the fact that $\ \mathbf{v} + \mathbf{w}\ $ is never larger than $\ \mathbf{v}\  + \ \mathbf{w}\ $ . To get some                      |
| johnston-linear-matrix-algebra_F00214 | 30   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\  \leq \ \mathbf{v}\  + \ \mathbf{w}\ $ | $\ \mathbf{v} + \mathbf{w}\  \leq \ \mathbf{v}\  + \ \mathbf{w}\ $ | The inequality $\ \mathbf{v} + \mathbf{w}\  \leq \ \mathbf{v}\  + \ \mathbf{w}\ $ thus simply says that the length of one                     |
| johnston-linear-matrix-algebra_F00215 | 31   | ■ 1.000 | $a, b$                                                             | $a, b$                                                             | triangle are $a, b$                                                                                                                           |
| johnston-linear-matrix-algebra_F00216 | 31   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\ ^2$                                    | $\ \mathbf{v} + \mathbf{w}\ ^2$                                    | <i>Proof.</i> We start by expanding $\ \mathbf{v} + \mathbf{w}\ ^2$ in terms of the dot product:                                              |
| johnston-linear-matrix-algebra_F00217 | 31   | ■ 0.971 | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$                          | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$                          | the angle between two vectors $\mathbf{v}$ and $\mathbf{w}$ . If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ then we can place $\mathbf{v}$ and |
| johnston-linear-matrix-algebra_F00218 | 31   | ■ 1.000 | $\mathbf{v}, \mathbf{w}$                                           | $\mathbf{v}, \mathbf{w}$                                           | $\mathbf{w}$ in standard position, so that the vectors $\mathbf{v}$ , $\mathbf{w}$ , and $\mathbf{v} - \mathbf{w}$ form the sides of a        |

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| johnston-linear-matrix-algebra_F00219 | 31   | ■ 1.000 | $\ \mathbf{v}\ , \ \mathbf{w}\ , \ \mathbf{v} - \mathbf{w}\ $         | $\ \mathbf{v}\ , \ \mathbf{w}\ , \ \mathbf{v} - \mathbf{w}\ $         | We can then use the law of cosines to relate $\ \mathbf{v}\ $ , $\ \mathbf{w}\ $ , $\ \mathbf{v} - \mathbf{w}\ $ , and the        |
| johnston-linear-matrix-algebra_F00220 | 32   | ■ 1.000 | $\theta \leq \theta \leq \pi$                                         | $0 \leq \theta \leq \pi$                                              | $0 \leq \theta \leq \pi$ , then                                                                                                   |
| johnston-linear-matrix-algebra_F00221 | 32   | ■ 1.000 | $\arccos(x) = \theta$                                                 | $\arccos(x) = \theta$                                                 | $\arccos(x) = \theta$ is                                                                                                          |
| johnston-linear-matrix-algebra_F00222 | 32   | ■ 1.000 | $\cos(\theta) = x$                                                    | $\cos(\theta) = x$                                                    | $\cos(\theta) = x$ . It is                                                                                                        |
| johnston-linear-matrix-algebra_F00223 | 32   | ■ 1.000 | $\cos^{-1}$                                                           | $\cos^{-1}$                                                           | as $\cos^{-1}$ or $\arccos$ .                                                                                                     |
| johnston-linear-matrix-algebra_F00224 | 32   | ■ 1.000 | $\ \mathbf{v}\ , \ \mathbf{w}\ $                                      | $\ \mathbf{v}\ , \ \mathbf{w}\ $                                      | $\ \mathbf{v}\ $ , $\ \mathbf{w}\ $ , and $\ \mathbf{v} - \mathbf{w}\ $ via the law of cosines.                                   |
| johnston-linear-matrix-algebra_F00225 | 32   | ■ 1.000 | $\ \mathbf{v} - \mathbf{w}\ $                                         | $\ \mathbf{v} - \mathbf{w}\ $                                         | $\ \mathbf{v}\ $ , $\ \mathbf{w}\ $ , and $\ \mathbf{v} - \mathbf{w}\ $ via the law of cosines.                                   |
| johnston-linear-matrix-algebra_F00226 | 32   | ■ 1.000 | $\ \mathbf{v} - \mathbf{w}\ ^2$                                       | $\ \mathbf{v} - \mathbf{w}\ ^2$                                       | By setting these two expressions for $\ \mathbf{v} - \mathbf{w}\ ^2$ equal to each other, we see that                             |
| johnston-linear-matrix-algebra_F00227 | 32   | ■ 0.651 | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$                             | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$                             | with vector $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ that are 3-dimensional. In this case, we can still arrange                  |
| johnston-linear-matrix-algebra_F00228 | 32   | ■ 1.000 | $(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\  \ \mathbf{w}\ )$     | $(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\  \ \mathbf{w}\ )$     | between vectors in this way, since it ensures that the fraction $(\mathbf{v} \cdot \mathbf{w}) / (\ \mathbf{v}\  \ \mathbf{w}\ )$ |
| johnston-linear-matrix-algebra_F00229 | 32   | ■ 1.000 | $\mathbf{v} = (1, 2)$                                                 | $\mathbf{v} = (1, 2)$                                                 | a) $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, 4)$ ,                                                                              |
| johnston-linear-matrix-algebra_F00230 | 32   | ■ 1.000 | $\mathbf{w} = (3, 4)$                                                 | $\mathbf{w} = (3, 4)$                                                 | a) $\mathbf{v} = (1, 2)$ and $\mathbf{w} = (3, 4)$ ,                                                                              |
| johnston-linear-matrix-algebra_F00231 | 32   | ■ 1.000 | $\mathbf{v} = (1, 2, -1, -2)$                                         | $\mathbf{v} = (1, 2, -1, -2)$                                         | b) $\mathbf{v} = (1, 2, -1, -2)$ and $\mathbf{w} = (1, -1, 1, -1)$ , and                                                          |
| johnston-linear-matrix-algebra_F00232 | 32   | ■ 1.000 | $\mathbf{w} = (1, -1, 1, -1)$                                         | $\mathbf{w} = (1, -1, 1, -1)$                                         | b) $\mathbf{v} = (1, 2, -1, -2)$ and $\mathbf{w} = (1, -1, 1, -1)$ , and                                                          |
| johnston-linear-matrix-algebra_F00233 | 33   | ■ 1.000 | $\arccos(11 / (5 \sqrt{5}))$                                          | $\arccos(11 / (5 \sqrt{5}))$                                          | $\arccos(11 / (5 \sqrt{5}))$ .                                                                                                    |
| johnston-linear-matrix-algebra_F00234 | 33   | ■ 1.000 | $\arccos(x)$                                                          | $\arccos(x)$                                                          | values of $\arccos(x)$                                                                                                            |
| johnston-linear-matrix-algebra_F00235 | 33   | ■ 1.000 | $\arccos(\sqrt{0}/2) = \pi/2$                                         | $\arccos(\sqrt{0}/2) = \pi/2$                                         | $\arccos(\sqrt{0}/2) = \pi/2$                                                                                                     |
| johnston-linear-matrix-algebra_F00236 | 33   | ■ 1.000 | $\arccos(\sqrt{1}/2) = \pi/3$                                         | $\arccos(\sqrt{1}/2) = \pi/3$                                         | $\arccos(\sqrt{1}/2) = \pi/3$                                                                                                     |
| johnston-linear-matrix-algebra_F00237 | 33   | ■ 1.000 | $\arccos(\sqrt{2}/2) = \pi/4$                                         | $\arccos(\sqrt{2}/2) = \pi/4$                                         | $\arccos(\sqrt{2}/2) = \pi/4$                                                                                                     |
| johnston-linear-matrix-algebra_F00238 | 33   | ■ 1.000 | $\arccos(\sqrt{3}/2) = \pi/6$                                         | $\arccos(\sqrt{3}/2) = \pi/6$                                         | $\arccos(\sqrt{3}/2) = \pi/6$                                                                                                     |
| johnston-linear-matrix-algebra_F00239 | 33   | ■ 1.000 | $\arccos(\sqrt{4}/2) = 0$                                             | $\arccos(\sqrt{4}/2) = 0$                                             | $\arccos(\sqrt{4}/2) = 0$                                                                                                         |
| johnston-linear-matrix-algebra_F00240 | 33   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11, \ \mathbf{v}\  = \sqrt{5}$ | $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11, \ \mathbf{v}\  = \sqrt{5}$ | a) $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11$ , $\ \mathbf{v}\  = \sqrt{5}$ , and $\ \mathbf{w}\  = 5$ , so the angle between     |
| johnston-linear-matrix-algebra_F00241 | 33   | ■ 1.000 | $\ \mathbf{w}\  = 5$                                                  | $\ \mathbf{w}\  = 5$                                                  | a) $\mathbf{v} \cdot \mathbf{w} = 3 + 8 = 11$ , $\ \mathbf{v}\  = \sqrt{5}$ , and $\ \mathbf{w}\  = 5$ , so the angle between     |
| johnston-linear-matrix-algebra_F00242 | 33   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$                     | $\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$                     | b) $\mathbf{v} \cdot \mathbf{w} = 1 - 2 - 1 + 2 = 0$ , so the angle between $\mathbf{v}$ and $\mathbf{w}$ is                      |
| johnston-linear-matrix-algebra_F00243 | 33   | ■ 1.000 | $\ \mathbf{w}\ $                                                      | $\ \mathbf{w}\ $                                                      | $\mathbf{v}$ and $\mathbf{w}$ without even computing $\ \mathbf{v}\ $ or $\ \mathbf{w}\ $ . For this reason, it is a              |
| johnston-linear-matrix-algebra_F00244 | 33   | ■ 0.999 | $\mathbf{v} \cdot \mathbf{w} = 0$                                     | $\mathbf{v} \cdot \mathbf{w} = 0$                                     | good idea to compute $\mathbf{v} \cdot \mathbf{w}$ first (as we did here)—if $\mathbf{v} \cdot \mathbf{w} = 0$ then               |
| johnston-linear-matrix-algebra_F00245 | 33   | ■ 1.000 | $\theta = \pi/2$                                                      | $\theta = \pi/2$                                                      | we know right away that the angle is $\theta = \pi/2$ .                                                                           |
| johnston-linear-matrix-algebra_F00246 | 33   | ■ 0.503 | $(0, 0, 0)$                                                           | $(0, 0, 0)$                                                           | vertex at $(0, 0, 0)$ and its opposite vertex at $(1, 1, 1)$ . There are lots                                                     |

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| johnston-linear-matrix-algebra_F00247 | 33   | 0.503 | $(1,1,1)$                                                                                                                                   | $(1,1,1)$                                                                               | vertex at $(0,0,0)$ and its opposite vertex at $(1,1,1)$ . There are lots                                                                           |
| johnston-linear-matrix-algebra_F00248 | 33   | 1.000 | $\mathbf{v}=(1,0,1)-(1,1,0)=(0,-1,1)$                                                                                                       | $\mathbf{v}=(1,0,1)-(1,1,0)=(0,-1,1)$                                                   | choose the face diagonals $\mathbf{v}=(1,0,1)-(1,1,0)=(0,-1,1)$ and                                                                                 |
| johnston-linear-matrix-algebra_F00249 | 33   | 1.000 | $\mathbf{w}=(0,1,1)-(1,1,0)=(-1,0,1)$                                                                                                       | $\mathbf{w}=(0,1,1)-(1,1,0)=(-1,0,1)$                                                   | $\mathbf{w}=(0,1,1)-(1,1,0)=(-1,0,1)$ , as shown below.                                                                                             |
| johnston-linear-matrix-algebra_F00250 | 33   | 1.000 | $\mathbf{v} \cdot \mathbf{w} = 0+0+1=1, \ \mathbf{v}\  = \sqrt{2}$                                                                          | $\mathbf{v} \cdot \mathbf{w} = 0+0+1=1, \ \mathbf{v}\  = \sqrt{2}$                      | Then $\mathbf{v} \cdot \mathbf{w} = 0+0+1=1, \ \mathbf{v}\  = \sqrt{2}$ , and $\ \mathbf{w}\  = \sqrt{2}$ , so the angle                            |
| johnston-linear-matrix-algebra_F00251 | 33   | 1.000 | $\ \mathbf{w}\  = \sqrt{2}$                                                                                                                 | $\ \mathbf{w}\  = \sqrt{2}$                                                             | Then $\mathbf{v} \cdot \mathbf{w} = 0+0+1=1, \ \mathbf{v}\  = \sqrt{2}$ , and $\ \mathbf{w}\  = \sqrt{2}$ , so the angle                            |
| johnston-linear-matrix-algebra_F00252 | 33   | 0.988 | $\pi / 2$                                                                                                                                   | $\pi/2$                                                                                 | gle between $\mathbf{v}$ and $\mathbf{w}$ was $\pi/2$ based only on the fact that $\mathbf{v} \cdot \mathbf{w} = 0$ (since                          |
| johnston-linear-matrix-algebra_F00253 | 33   | 0.953 | $\arccos (0)=\pi / 2$                                                                                                                       | $\arccos(0) = \pi/2$                                                                    | $\arccos(0) = \pi/2$ ). This implication goes both ways (i.e., if the angle between                                                                 |
| johnston-linear-matrix-algebra_F00254 | 33   | 1.000 | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$                                                                                                   | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$                                               | also applies to higher-dimensional situations (e.g., the two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^4$                                      |
| johnston-linear-matrix-algebra_F00255 | 34   | 0.998 | $\mathbf{v}=(-2,4), \mathbf{w}=(2,1)$                                                                                                       | $\mathbf{v}=(-2,4), \mathbf{w}=(2,1)$                                                   | *(a) $\mathbf{v}=(-2,4), \mathbf{w}=(2,1)$                                                                                                          |
| johnston-linear-matrix-algebra_F00256 | 34   | 1.000 | $\mathbf{v}=(1,2,3), \mathbf{w}=(-3,2,-1)$                                                                                                  | $\mathbf{v}=(1,2,3), \mathbf{w}=(-3,2,-1)$                                              | (b) $\mathbf{v}=(1,2,3), \mathbf{w}=(-3,2,-1)$                                                                                                      |
| johnston-linear-matrix-algebra_F00257 | 34   | 0.998 | $\mathbf{v}=(3,-1,0,1), \mathbf{w}=(0,2,1,3)$                                                                                               | $\mathbf{v}=(3,-1,0,1), \mathbf{w}=(0,2,1,3)$                                           | *(c) $\mathbf{v}=(3,-1,0,1), \mathbf{w}=(0,2,1,3)$                                                                                                  |
| johnston-linear-matrix-algebra_F00258 | 34   | 1.000 | $\mathbf{v}=(\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w}=(\sqrt{2}, \sqrt{3}, \sqrt{5})$                                                      | $\mathbf{v}=(\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w}=(\sqrt{2}, \sqrt{3}, \sqrt{5})$  | (d) $\mathbf{v}=(\sqrt{2}, \sqrt{3}, \sqrt{5}), \mathbf{w}=(\sqrt{2}, \sqrt{3}, \sqrt{5})$                                                          |
| johnston-linear-matrix-algebra_F00259 | 34   | 0.999 | $\mathbf{v}=\mathbf{0} \in \mathbb{R}^9, \mathbf{w}=(8,1,5,-7,3,9,1,-3,2)$                                                                  | $\mathbf{v}=\mathbf{0} \in \mathbb{R}^9, \mathbf{w}=(8,1,5,-7,3,9,1,-3,2)$              | *(e) $\mathbf{v}=\mathbf{0} \in \mathbb{R}^9, \mathbf{w}=(8,1,5,-7,3,9,1,-3,2)$                                                                     |
| johnston-linear-matrix-algebra_F00260 | 34   | 0.999 | $\mathbf{v}=(3,4)$                                                                                                                          | $\mathbf{v}=(3,4)$                                                                      | *(a) $\mathbf{v}=(3,4)$                                                                                                                             |
| johnston-linear-matrix-algebra_F00261 | 34   | 1.000 | $\mathbf{v}=(2,1,-2)$                                                                                                                       | $\mathbf{v}=(2,1,-2)$                                                                   | (b) $\mathbf{v}=(2,1,-2)$                                                                                                                           |
| johnston-linear-matrix-algebra_F00262 | 34   | 1.000 | $\mathbf{v}=(-2 \sqrt{2},-3, \sqrt{10}, 3)$                                                                                                 | $\mathbf{v}=(-2 \sqrt{2},-3, \sqrt{10}, 3)$                                             | *(c) $\mathbf{v}=(-2 \sqrt{2},-3, \sqrt{10}, 3)$                                                                                                    |
| johnston-linear-matrix-algebra_F00263 | 34   | 1.000 | $\mathbf{v}=(\cos (\theta), \sin (\theta))$                                                                                                 | $\mathbf{v}=(\cos (\theta), \sin (\theta))$                                             | (d) $\mathbf{v}=(\cos (\theta), \sin (\theta))$                                                                                                     |
| johnston-linear-matrix-algebra_F00264 | 34   | 1.000 | $\mathbf{v}=(1, \sqrt{3}), \mathbf{w}=(\sqrt{3}, 1)$                                                                                        | $\mathbf{v}=(1, \sqrt{3}), \mathbf{w}=(\sqrt{3}, 1)$                                    | *(a) $\mathbf{v}=(1, \sqrt{3}), \mathbf{w}=(\sqrt{3}, 1)$                                                                                           |
| johnston-linear-matrix-algebra_F00265 | 34   | 1.000 | $\mathbf{v}=(0,-2,2), \mathbf{w}=(1,0,1)$                                                                                                   | $\mathbf{v}=(0,-2,2), \mathbf{w}=(1,0,1)$                                               | (b) $\mathbf{v}=(0,-2,2), \mathbf{w}=(1,0,1)$                                                                                                       |
| johnston-linear-matrix-algebra_F00266 | 34   | 1.000 | $\mathbf{v}=(1,1,1,1), \mathbf{w}=(-2,0,-2,0)$                                                                                              | $\mathbf{v}=(1,1,1,1), \mathbf{w}=(-2,0,-2,0)$                                          | *(c) $\mathbf{v}=(1,1,1,1), \mathbf{w}=(-2,0,-2,0)$                                                                                                 |
| johnston-linear-matrix-algebra_F00267 | 34   | 1.000 | $\mathbf{v}=(2,1,-3), \mathbf{w}=(-1,2,3)$                                                                                                  | $\mathbf{v}=(2,1,-3), \mathbf{w}=(-1,2,3)$                                              | (d) $\mathbf{v}=(2,1,-3), \mathbf{w}=(-1,2,3)$                                                                                                      |
| johnston-linear-matrix-algebra_F00268 | 34   | 0.842 | $\mathbf{v}=(\cos (\boldsymbol{\theta}), \sin (\boldsymbol{\theta})), \mathbf{w}=(-\sin (\boldsymbol{\theta}), \cos (\boldsymbol{\theta}))$ | $\mathbf{v}=(\cos (\theta), \sin (\theta)), \mathbf{w}=(-\sin (\theta), \cos (\theta))$ | *(e) $\mathbf{v}=(\cos (\theta), \sin (\theta)), \mathbf{w}=(-\sin (\theta), \cos (\theta))$                                                        |
| johnston-linear-matrix-algebra_F00269 | 34   | 1.000 | $\mathbf{v} \cdot \mathbf{w}=\mathbf{v} \cdot \mathbf{x}$                                                                                   | $\mathbf{v} \cdot \mathbf{w}=\mathbf{v} \cdot \mathbf{x}$                               | *(a) If $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors with $\mathbf{v} \cdot \mathbf{w}=\mathbf{v} \cdot \mathbf{x}$ , then     |
| johnston-linear-matrix-algebra_F00270 | 34   | 0.785 | $\mathbf{w}=\mathbf{x}$                                                                                                                     | $\mathbf{w}=\mathbf{x}$                                                                 | $\mathbf{w}=\mathbf{x}$ .                                                                                                                           |
| johnston-linear-matrix-algebra_F00271 | 34   | 1.000 | $\mathbf{w} \cdot \mathbf{x}=0$                                                                                                             | $\mathbf{w} \cdot \mathbf{x}=0$                                                         | (b) If $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^n$ are vectors with $\mathbf{v} \cdot \mathbf{w}=0$ and $\mathbf{w} \cdot \mathbf{x}=0$ , |
| johnston-linear-matrix-algebra_F00272 | 34   | 1.000 | $\mathbf{v} \cdot \mathbf{x}=0$                                                                                                             | $\mathbf{v} \cdot \mathbf{x}=0$                                                         | then $\mathbf{v} \cdot \mathbf{x}=0$ too.                                                                                                           |
| johnston-linear-matrix-algebra_F00273 | 34   | 0.760 | $\ \mathbf{v}\ +\ \mathbf{w}\  \leq 2$                                                                                                      | $\ \mathbf{v}\ +\ \mathbf{w}\  \leq 2$                                                  | *(c) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors with $\ \mathbf{v}\ +\ \mathbf{w}\  \leq 2$ , then                                    |
| johnston-linear-matrix-algebra_F00274 | 34   | 1.000 | $\ \mathbf{v}+\mathbf{w}\  \leq 2$                                                                                                          | $\ \mathbf{v}+\mathbf{w}\  \leq 2$                                                      | $\ \mathbf{v}+\mathbf{w}\  \leq 2$ too.                                                                                                             |

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| johnston-linear-matrix-algebra_F00275 | 34   | 1.000 | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$                              | $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$                              | (d) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$ such that $\ \mathbf{v}\  = 2$ ,                                       |
| johnston-linear-matrix-algebra_F00276 | 34   | 1.000 | $\ \mathbf{v}\  = 2$                                                   | $\ \mathbf{v}\  = 2$                                                   | (d) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^5$ such that $\ \mathbf{v}\  = 2$ ,                                       |
| johnston-linear-matrix-algebra_F00277 | 34   | 1.000 | $\ \mathbf{w}\  = 4$                                                   | $\ \mathbf{w}\  = 4$                                                   | $\ \mathbf{w}\  = 4$ , and $\ \mathbf{v} - \mathbf{w}\  = 1$ .                                                                           |
| johnston-linear-matrix-algebra_F00278 | 34   | 1.000 | $\ \mathbf{v} - \mathbf{w}\  = 1$                                      | $\ \mathbf{v} - \mathbf{w}\  = 1$                                      | $\ \mathbf{w}\  = 4$ , and $\ \mathbf{v} - \mathbf{w}\  = 1$ .                                                                           |
| johnston-linear-matrix-algebra_F00279 | 34   | 0.995 | $\ \mathbf{v}\  = 1$                                                   | $\ \mathbf{v}\  = 1$                                                   | *(e) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ such that $\ \mathbf{v}\  = 1$ ,                                      |
| johnston-linear-matrix-algebra_F00280 | 34   | 1.000 | $\ \mathbf{w}\  = 2$                                                   | $\ \mathbf{w}\  = 2$                                                   | $\ \mathbf{w}\  = 2$ , and $\mathbf{v} \cdot \mathbf{w} = -1$ .                                                                          |
| johnston-linear-matrix-algebra_F00281 | 34   | 1.000 | $\mathbf{v} \cdot \mathbf{w} = -1$                                     | $\mathbf{v} \cdot \mathbf{w} = -1$                                     | $\ \mathbf{w}\  = 2$ , and $\mathbf{v} \cdot \mathbf{w} = -1$ .                                                                          |
| johnston-linear-matrix-algebra_F00282 | 34   | 1.000 | $ \mathbf{v} \cdot \mathbf{w}  \leq 1$                                 | $ \mathbf{v} \cdot \mathbf{w}  \leq 1$                                 | (f) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are unit vectors, then $ \mathbf{v} \cdot \mathbf{w}  \leq 1$ .                         |
| johnston-linear-matrix-algebra_F00283 | 34   | 0.598 | $\ \mathbf{v}\  \leq 1$                                                | $\ \mathbf{v}\  \leq 1$                                                | *(g) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ are vectors with $ \mathbf{v} \cdot \mathbf{w}  \leq 1$ , then $\ \mathbf{v}\  \leq 1$ |
| johnston-linear-matrix-algebra_F00284 | 34   | 1.000 | $\ \mathbf{w}\  \leq 1$                                                | $\ \mathbf{w}\  \leq 1$                                                | or $\ \mathbf{w}\  \leq 1$ (or both).                                                                                                    |
| johnston-linear-matrix-algebra_F00285 | 34   | 0.965 | $\mathbf{v} = (3, \sqrt{3})$                                           | $\mathbf{v} = (3, \sqrt{3})$                                           | *1.2.6 Suppose $\mathbf{v} = (3, \sqrt{3})$ . Find all vectors $\mathbf{w} \in \mathbb{R}^2$ such                                        |
| johnston-linear-matrix-algebra_F00286 | 34   | 0.965 | $\mathbf{w} \in \mathbb{R}^2$                                          | $\mathbf{w} \in \mathbb{R}^2$                                          | *1.2.6 Suppose $\mathbf{v} = (3, \sqrt{3})$ . Find all vectors $\mathbf{w} \in \mathbb{R}^2$ such                                        |
| johnston-linear-matrix-algebra_F00287 | 34   | 1.000 | $\theta = \pi/3$                                                       | $\theta = \pi/3$                                                       | that $\ \mathbf{w}\  = 2$ and the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\theta = \pi/3$ .                                      |
| johnston-linear-matrix-algebra_F00288 | 34   | 0.997 | $\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$                           | $\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$                           | *(a) $\mathbf{v} \cdot (\mathbf{w} - \mathbf{x})$                                                                                        |
| johnston-linear-matrix-algebra_F00289 | 34   | 1.000 | $(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$                              | $(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$                              | (b) $(\mathbf{v} \cdot \mathbf{w})\mathbf{x}$                                                                                            |
| johnston-linear-matrix-algebra_F00290 | 34   | 1.000 | $\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$                           | $\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$                           | *(c) $\mathbf{v} + (\mathbf{w} \cdot \mathbf{x})$                                                                                        |
| johnston-linear-matrix-algebra_F00291 | 34   | 1.000 | $\mathbf{v} / \ \mathbf{v}\ $                                          | $\mathbf{v} / \ \mathbf{v}\ $                                          | (d) $\mathbf{v} / \ \mathbf{v}\ $                                                                                                        |
| johnston-linear-matrix-algebra_F00292 | 34   | 0.647 | $\mathbf{v}^2$                                                         | $\mathbf{v}^2$                                                         | *(e) $\mathbf{v}^2$                                                                                                                      |
| johnston-linear-matrix-algebra_F00293 | 34   | 1.000 | $(\mathbf{v} + \mathbf{w})/\mathbf{x}$                                 | $(\mathbf{v} + \mathbf{w})/\mathbf{x}$                                 | (f) $(\mathbf{v} + \mathbf{w})/\mathbf{x}$                                                                                               |
| johnston-linear-matrix-algebra_F00294 | 34   | 1.000 | $\ \mathbf{e}_j\ $                                                     | $\ \mathbf{e}_j\ $                                                     | (a) Compute $\ \mathbf{e}_j\ $ .                                                                                                         |
| johnston-linear-matrix-algebra_F00295 | 34   | 1.000 | $1 \leq i, j \leq n$                                                   | $1 \leq i, j \leq n$                                                   | (b) Suppose $1 \leq i, j \leq n$ . Compute $\mathbf{e}_i \cdot \mathbf{e}_j$ . [Side note: You                                           |
| johnston-linear-matrix-algebra_F00296 | 34   | 1.000 | $\mathbf{e}_i \cdot \mathbf{e}_j$                                      | $\mathbf{e}_i \cdot \mathbf{e}_j$                                      | (b) Suppose $1 \leq i, j \leq n$ . Compute $\mathbf{e}_i \cdot \mathbf{e}_j$ . [Side note: You                                           |
| johnston-linear-matrix-algebra_F00297 | 34   | 1.000 | $i = j$                                                                | $i = j$                                                                | $i = j$ or $i \neq j$ .]                                                                                                                 |
| johnston-linear-matrix-algebra_F00298 | 34   | 1.000 | $i \neq j$                                                             | $i \neq j$                                                             | $i = j$ or $i \neq j$ .]                                                                                                                 |
| johnston-linear-matrix-algebra_F00299 | 34   | 1.000 | $\mathbf{v} = (1, 2) \in \mathbb{R}^2$                                 | $\mathbf{v} = (1, 2) \in \mathbb{R}^2$                                 | *1.2.9 Let $\mathbf{v} = (1, 2) \in \mathbb{R}^2$ .                                                                                      |
| johnston-linear-matrix-algebra_F00300 | 34   | 1.000 | $\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$                              | $\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$                              | 1.2.10 Let $\mathbf{v} = (1, 2, 3) \in \mathbb{R}^3$ .                                                                                   |
| johnston-linear-matrix-algebra_F00301 | 35   | 1.000 | $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$                              | $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$                              | (instead of real) entries, we write $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$ and we define                                              |
| johnston-linear-matrix-algebra_F00302 | 35   | 1.000 | $\overline{a + ib} = a - ib$                                           | $\overline{a + ib} = a - ib$                                           | where $\overline{a + ib} = a - ib$ is the complex conjugate (see Ap-                                                                     |
| johnston-linear-matrix-algebra_F00303 | 35   | 1.000 | $\mathbf{v} \cdot \mathbf{w} = \overline{\mathbf{w}} \cdot \mathbf{v}$ | $\mathbf{v} \cdot \mathbf{w} = \overline{\mathbf{w}} \cdot \mathbf{v}$ | (a) Show that $\mathbf{v} \cdot \mathbf{w} = \overline{\mathbf{w}} \cdot \mathbf{v}$ .                                                   |

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| johnston-linear-matrix-algebra_F00304 | 35   | ■ 0.988 | $c \in \mathbb{C}$                                                                                      | $c \in \mathbb{C}$                                                                                      | $c \in \mathbb{C}$ , but $(c\mathbf{v}) \cdot \mathbf{w} = \overline{c}(\mathbf{v} \cdot \mathbf{w})$ .                               |
| johnston-linear-matrix-algebra_F00305 | 35   | ■ 0.988 | $(c \mathbf{v}) \cdot \mathbf{w} = \overline{c}(\mathbf{v} \cdot \mathbf{w})$                           | $(c\mathbf{v}) \cdot \mathbf{w} = \overline{c}(\mathbf{v} \cdot \mathbf{w})$                            | $c \in \mathbb{C}$ , but $(c\mathbf{v}) \cdot \mathbf{w} = \overline{c}(\mathbf{v} \cdot \mathbf{w})$ .                               |
| johnston-linear-matrix-algebra_F00306 | 35   | ■ 1.000 | $\mathbf{x} \in \mathbb{R}^n$                                                                           | $\mathbf{x} \in \mathbb{R}^n$                                                                           | <b>1.2.16</b> Suppose that $\mathbf{x} \in \mathbb{R}^n$ . Show that $\mathbf{x} \cdot \mathbf{y} = 0$ for all                        |
| johnston-linear-matrix-algebra_F00307 | 35   | ■ 1.000 | $\mathbf{x} \cdot \mathbf{y} = 0$                                                                       | $\mathbf{x} \cdot \mathbf{y} = 0$                                                                       | <b>1.2.16</b> Suppose that $\mathbf{x} \in \mathbb{R}^n$ . Show that $\mathbf{x} \cdot \mathbf{y} = 0$ for all                        |
| johnston-linear-matrix-algebra_F00308 | 35   | ■ 1.000 | $\mathbf{y} \in \mathbb{R}^n$                                                                           | $\mathbf{y} \in \mathbb{R}^n$                                                                           | $\mathbf{y} \in \mathbb{R}^n$ if and only if $\mathbf{x} = \mathbf{0}$ .                                                              |
| johnston-linear-matrix-algebra_F00309 | 35   | ■ 1.000 | $\mathbf{x} = \mathbf{0}$                                                                               | $\mathbf{x} = \mathbf{0}$                                                                               | $\mathbf{y} \in \mathbb{R}^n$ if and only if $\mathbf{x} = \mathbf{0}$ .                                                              |
| johnston-linear-matrix-algebra_F00310 | 35   | ■ 1.000 | $ \mathbf{v} \cdot \mathbf{w}  = \ \mathbf{v}\  \ \mathbf{w}\ $                                         | $ \mathbf{v} \cdot \mathbf{w}  = \ \mathbf{v}\  \ \mathbf{w}\ $                                         | <b>(a)</b> Prove that $ \mathbf{v} \cdot \mathbf{w}  = \ \mathbf{v}\  \ \mathbf{w}\ $ if and only if either $\mathbf{w} = \mathbf{0}$ |
| johnston-linear-matrix-algebra_F00311 | 35   | ■ 1.000 | $\mathbf{v} = c\mathbf{w}$                                                                              | $\mathbf{v} = c\mathbf{w}$                                                                              | or there exists a scalar $c \in \mathbb{R}$ such that $\mathbf{v} = c\mathbf{w}$ .                                                    |
| johnston-linear-matrix-algebra_F00312 | 35   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $                                         | $\ \mathbf{v} + \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $                                         | <b>(b)</b> Prove that $\ \mathbf{v} + \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $ if and only if either                           |
| johnston-linear-matrix-algebra_F00313 | 35   | ■ 1.000 | $0 \leq c \in \mathbb{R}$                                                                               | $0 \leq c \in \mathbb{R}$                                                                               | $\mathbf{w} = \mathbf{0}$ or there exists a scalar $0 \leq c \in \mathbb{R}$ such that                                                |
| johnston-linear-matrix-algebra_F00314 | 35   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2$                                   | $\ \mathbf{v} + \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2$                                   | other. Prove that $\ \mathbf{v} + \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 + \ \mathbf{w}\ ^2$ .                                             |
| johnston-linear-matrix-algebra_F00315 | 35   | ■ 1.000 | $\ \mathbf{v} + \mathbf{w}\ ^2 + \ \mathbf{v} - \mathbf{w}\ ^2 = 2\ \mathbf{v}\ ^2 + 2\ \mathbf{w}\ ^2$ | $\ \mathbf{v} + \mathbf{w}\ ^2 + \ \mathbf{v} - \mathbf{w}\ ^2 = 2\ \mathbf{v}\ ^2 + 2\ \mathbf{w}\ ^2$ | <b>(a)</b> Show that $\ \mathbf{v} + \mathbf{w}\ ^2 + \ \mathbf{v} - \mathbf{w}\ ^2 = 2\ \mathbf{v}\ ^2 + 2\ \mathbf{w}\ ^2$ .        |
| johnston-linear-matrix-algebra_F00316 | 35   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} =$                                                                         | $\mathbf{v} \cdot \mathbf{w} =$                                                                         | <b>1.2.20</b> Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be vectors. Prove that $\mathbf{v} \cdot \mathbf{w} =$                    |
| johnston-linear-matrix-algebra_F00317 | 35   | ■ 1.000 | $\frac{1}{4} (\ \mathbf{v} + \mathbf{w}\ ^2 - \ \mathbf{v} - \mathbf{w}\ ^2)$                           | $\frac{1}{4} (\ \mathbf{v} + \mathbf{w}\ ^2 - \ \mathbf{v} - \mathbf{w}\ ^2)$                           | $\frac{1}{4} (\ \mathbf{v} + \mathbf{w}\ ^2 - \ \mathbf{v} - \mathbf{w}\ ^2)$ .                                                       |
| johnston-linear-matrix-algebra_F00318 | 35   | ■ 0.816 | $\ \mathbf{v} - \mathbf{w}\  \geq$                                                                      | $\ \mathbf{v} - \mathbf{w}\  \geq$                                                                      | <b>*1.2.21</b> Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ be vectors. Prove that $\ \mathbf{v} - \mathbf{w}\  \geq$                |
| johnston-linear-matrix-algebra_F00319 | 35   | ■ 1.000 | $\ \mathbf{v}\  - \ \mathbf{w}\ $                                                                       | $\ \mathbf{v}\  - \ \mathbf{w}\ $                                                                       | $\ \mathbf{v}\  - \ \mathbf{w}\ $ .                                                                                                   |
| johnston-linear-matrix-algebra_F00320 | 35   | ■ 1.000 | $d = \ \mathbf{v}\ $                                                                                    | $d = \ \mathbf{v}\ $                                                                                    | $d = \ \mathbf{v}\ $ in the proof?                                                                                                    |
| johnston-linear-matrix-algebra_F00321 | 35   | ■ 1.000 | $d = -\ \mathbf{v}\ $                                                                                   | $d = -\ \mathbf{v}\ $                                                                                   | $d = -\ \mathbf{v}\ $ in the proof?                                                                                                   |
| johnston-linear-matrix-algebra_F00322 | 35   | ■ 1.000 | $f(x) = \ \mathbf{v} - x\mathbf{w}\ ^2$                                                                 | $f(x) = \ \mathbf{v} - x\mathbf{w}\ ^2$                                                                 | and consider the function $f(x) = \ \mathbf{v} - x\mathbf{w}\ ^2$ .                                                                   |
| johnston-linear-matrix-algebra_F00323 | 35   | ■ 1.000 | $ax^2 + bx + c$                                                                                         | $ax^2 + bx + c$                                                                                         | that the discriminant of the quadratic $ax^2 + bx + c$ is                                                                             |
| johnston-linear-matrix-algebra_F00324 | 35   | ■ 1.000 | $b^2 - 4ac$                                                                                             | $b^2 - 4ac$                                                                                             | $b^2 - 4ac$ .                                                                                                                         |
| johnston-linear-matrix-algebra_F00325 | 35   | ■ 0.777 | $\leq 0$                                                                                                | $\leq 0$                                                                                                | <b>(c)</b> Why must the discriminant from part (b) be $\leq 0$ ?                                                                      |
| johnston-linear-matrix-algebra_F00326 | 35   | ■ 1.000 | $f$                                                                                                     | $f$                                                                                                     | [Hint: How many roots does $f$ have?]                                                                                                 |
| johnston-linear-matrix-algebra_F00327 | 35   | ■ 1.000 | $ x + y  \leq  x  +  y $                                                                                | $ x + y  \leq  x  +  y $                                                                                | <b>(a)</b> Show that if $x, y \in \mathbb{R}$ then $ x + y  \leq  x  +  y $ . Do <i>not</i>                                           |
| johnston-linear-matrix-algebra_F00328 | 35   | ■ 1.000 | $x_1, \dots, x_n \in \mathbb{R}$                                                                        | $x_1, \dots, x_n \in \mathbb{R}$                                                                        | <b>(b)</b> Show that if $x_1, \dots, x_n \in \mathbb{R}$ then $ x_1 + \dots + x_n  \leq$                                              |
| johnston-linear-matrix-algebra_F00329 | 35   | ■ 1.000 | $ x_1 + \dots + x_n  \leq$                                                                              | $ x_1 + \dots + x_n  \leq$                                                                              | <b>(b)</b> Show that if $x_1, \dots, x_n \in \mathbb{R}$ then $ x_1 + \dots + x_n  \leq$                                              |
| johnston-linear-matrix-algebra_F00330 | 35   | ■ 1.000 | $ x_1  + \dots +  x_n $                                                                                 | $ x_1  + \dots +  x_n $                                                                                 | $ x_1  + \dots +  x_n $ .                                                                                                             |
| johnston-linear-matrix-algebra_F00331 | 35   | ■ 1.000 | $xy \leq \frac{1}{2}(x^2 + y^2)$                                                                        | $xy \leq \frac{1}{2}(x^2 + y^2)$                                                                        | <b>(c)</b> Show that if $x, y \in \mathbb{R}$ then $xy \leq \frac{1}{2}(x^2 + y^2)$ .                                                 |
| johnston-linear-matrix-algebra_F00332 | 35   | ■ 1.000 | $ \mathbf{v} \cdot \mathbf{w}  / (\ \mathbf{v}\  \ \mathbf{w}\ ) \leq 1$                                | $ \mathbf{v} \cdot \mathbf{w}  / (\ \mathbf{v}\  \ \mathbf{w}\ ) \leq 1$                                | $ \mathbf{v} \cdot \mathbf{w}  / (\ \mathbf{v}\  \ \mathbf{w}\ ) \leq 1$ . [Hint: Start by writing out                                |

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|---------------------------------------|------|-------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F00333 | 35   | 1.000 | $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$ | $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$ | $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + \cdots + v_n w_n$ in the fraction on the left.                    |
| johnston-linear-matrix-algebra-F00334 | 36   | 1.000 | $a_{i,j}$                                                  | $a_{i,j}$                                                  | columns: $a_{i,j}$                                                                                         |
| johnston-linear-matrix-algebra-F00335 | 36   | 1.000 | $i$                                                        | $i$                                                        | the $i$ -th row and                                                                                        |
| johnston-linear-matrix-algebra-F00336 | 36   | 0.623 | $3 \times 4$                                               | $3 \times 4$                                               | $3 \times 4$ matrix is one                                                                                 |
| johnston-linear-matrix-algebra-F00337 | 36   | 1.000 | $A, B, C, \ldots$                                          | $A, B, C, \ldots$                                          | We typically denote matrices by uppercase letters like $A, B, C, \ldots$ , and                             |
| johnston-linear-matrix-algebra-F00338 | 36   | 1.000 | $a_{1,2}=3$                                                | $a_{1,2} = 3$                                              | from. For example, if $A$ and $B$ are as above, then $a_{1,2} = 3$ , since the entry of                    |
| johnston-linear-matrix-algebra-F00339 | 36   | 0.998 | $a_{2,1}=2$                                                | $a_{2,1} = 2$                                              | $A$ in the 1st row and 2nd column is 3. Similarly, $a_{2,1} = 2$ and $b_{2,3} = 1$ . The                   |
| johnston-linear-matrix-algebra-F00340 | 36   | 0.998 | $b_{2,3}=1$                                                | $b_{2,3} = 1$                                              | $A$ in the 1st row and 2nd column is 3. Similarly, $a_{2,1} = 2$ and $b_{2,3} = 1$ . The                   |
| johnston-linear-matrix-algebra-F00341 | 36   | 0.999 | $(i, j)$                                                   | $(i, j)$                                                   | entry in the $i$ -th row and $j$ -th column is also called its “ $(i, j)$ -entry”, and we                  |
| johnston-linear-matrix-algebra-F00342 | 36   | 1.000 | $[A]_{i,j}$                                                | $[A]_{i,j}$                                                | sometimes alternatively denote it using square brackets like $[A]_{i,j}$ . For example,                    |
| johnston-linear-matrix-algebra-F00343 | 36   | 0.633 | $b_{2,1}=[B]_{2,1}=0$                                      | $b_{2,1} = [B]_{2,1} = 0$                                  | the $(2, 1)$ -entry of $B$ is $b_{2,1} = [B]_{2,1} = 0$ .                                                  |
| johnston-linear-matrix-algebra-F00344 | 36   | 1.000 | $m$                                                        | $m$                                                        | matrices with $m$ rows and $n$ columns is denoted by $\mathcal{M}_{m,n}$ , or simply by $\mathcal{M}_n$ if |
| johnston-linear-matrix-algebra-F00345 | 36   | 1.000 | $\mathcal{M}_{m,n}$                                        | $\mathcal{M}_{m,n}$                                        | matrices with $m$ rows and $n$ columns is denoted by $\mathcal{M}_{m,n}$ , or simply by $\mathcal{M}_n$ if |
| johnston-linear-matrix-algebra-F00346 | 36   | 1.000 | $\mathcal{M}_n$                                            | $\mathcal{M}_n$                                            | matrices with $m$ rows and $n$ columns is denoted by $\mathcal{M}_{m,n}$ , or simply by $\mathcal{M}_n$ if |
| johnston-linear-matrix-algebra-F00347 | 36   | 1.000 | $m=n$                                                      | $m = n$                                                    | $m = n$ (in which case the matrix is called <b>square</b> ). So if $A$ and $B$ again refer to              |
| johnston-linear-matrix-algebra-F00348 | 36   | 1.000 | $A \in \mathcal{M}_2$                                      | $A \in \mathcal{M}_2$                                      | the matrices displayed above, then $A \in \mathcal{M}_2$ and $B \in \mathcal{M}_{2,3}$ .                   |
| johnston-linear-matrix-algebra-F00349 | 36   | 1.000 | $B \in \mathcal{M}_{2,3}$                                  | $B \in \mathcal{M}_{2,3}$                                  | the matrices displayed above, then $A \in \mathcal{M}_2$ and $B \in \mathcal{M}_{2,3}$ .                   |
| johnston-linear-matrix-algebra-F00350 | 36   | 1.000 | $A, B \in \mathcal{M}_{m,n}$                               | $A, B \in \mathcal{M}_{m,n}$                               | Suppose $A, B \in \mathcal{M}_{m,n}$ are matrices and $c \in \mathbb{R}$ is a scalar. Then the <b>sum</b>  |
| johnston-linear-matrix-algebra-F00351 | 36   | 0.985 | $A+B$                                                      | $A + B$                                                    | $A + B$ and <b>scalar multiplication</b> $cA$ are the $m \times n$ matrices whose $(i, j)$ -               |
| johnston-linear-matrix-algebra-F00352 | 36   | 0.985 | $c A$                                                      | $cA$                                                       | $A + B$ and <b>scalar multiplication</b> $cA$ are the $m \times n$ matrices whose $(i, j)$ -               |
| johnston-linear-matrix-algebra-F00353 | 36   | 0.985 | $m \times n$                                               | $m \times n$                                               | $A + B$ and <b>scalar multiplication</b> $cA$ are the $m \times n$ matrices whose $(i, j)$ -               |
| johnston-linear-matrix-algebra-F00354 | 36   | 1.000 | $1 \leq i \leq m$                                          | $1 \leq i \leq m$                                          | entries, for each $1 \leq i \leq m$ and $1 \leq j \leq n$ , are                                            |
| johnston-linear-matrix-algebra-F00355 | 36   | 1.000 | $0$                                                        | $O$                                                        | We also use $O$ to denote the <b>zero matrix</b> whose entries all equal 0 (or                             |
| johnston-linear-matrix-algebra-F00356 | 36   | 1.000 | $O_{m,n}$                                                  | $O_{m,n}$                                                  | $O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$ , or $O_n$ if it is $n \times n$ ).   |
| johnston-linear-matrix-algebra-F00357 | 36   | 1.000 | $O_n$                                                      | $O_n$                                                      | $O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$ , or $O_n$ if it is $n \times n$ ).   |
| johnston-linear-matrix-algebra-F00358 | 36   | 1.000 | $n \times n$                                               | $n \times n$                                               | $O_{m,n}$ if we wish to emphasize or clarify that it is $m \times n$ , or $O_n$ if it is $n \times n$ ).   |
| johnston-linear-matrix-algebra-F00359 | 36   | 0.971 | $(A-B=A+(-1) B)$                                           | $(A - B = A + (-1)B)$                                      | Similarly, we define matrix subtraction $(A - B = A + (-1)B)$ and the negative of                          |
| johnston-linear-matrix-algebra-F00360 | 36   | 1.000 | $(-A=(-1) A)$                                              | $(-A = (-1)A)$                                             | a matrix $(-A = (-1)A)$ in the obvious entrywise ways. There is nothing fancy                              |

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|---------------------------------------|------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra_F00361 | 36   | 0.755 | $A=\left[\begin{array}{cc}1 & 3 \\ 2 & -1\end{array}\right]$ , $B=\left[\begin{array}{cc}2 & 1 \\ 0 & 1\end{array}\right]$                                                         | $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}, B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$                                                   | Suppose $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ , $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ , and $C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$ . Compute |
| johnston-linear-matrix-algebra_F00362 | 36   | 0.755 | $C=\left[\begin{array}{ccc}1 & 0 & 1 \\ 0 & -1 & 1\end{array}\right]$                                                                                                              | $C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$                                                                                             | Suppose $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ , $B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$ , and $C = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$ . Compute |
| johnston-linear-matrix-algebra_F00363 | 36   | 1.000 | $2A-3B$                                                                                                                                                                            | $2A-3B$                                                                                                                                                 | b) $2A-3B$ , and                                                                                                                                                                             |
| johnston-linear-matrix-algebra_F00364 | 36   | 1.000 | $A+2C$                                                                                                                                                                             | $A+2C$                                                                                                                                                  | c) $A+2C$ .                                                                                                                                                                                  |
| johnston-linear-matrix-algebra_F00365 | 37   | 1.000 | $A+B=\left[\begin{array}{cc}1 & 3 \\ 2 & -1\end{array}\right]+\left[\begin{array}{cc}2 & 1 \\ 0 & 1\end{array}\right]=\left[\begin{array}{cc}3 & 4 \\ 2 & 0\end{array}\right]$     | $A+B = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 0 \end{bmatrix}$     | a) $A+B = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 2 & 0 \end{bmatrix}$ .                                     |
| johnston-linear-matrix-algebra_F00366 | 37   | 0.994 | $2A-3B=\left[\begin{array}{cc}2 & 6 \\ 4 & -2\end{array}\right]-\left[\begin{array}{cc}6 & 3 \\ 0 & 3\end{array}\right]=\left[\begin{array}{cc}-4 & 3 \\ 4 & -5\end{array}\right]$ | $2A-3B = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 6 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 4 & -5 \end{bmatrix}$ | b) $2A-3B = \begin{bmatrix} 2 & 6 \\ 4 & -2 \end{bmatrix} - \begin{bmatrix} 6 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 4 & -5 \end{bmatrix}$ .                                 |
| johnston-linear-matrix-algebra_F00367 | 37   | 0.999 | $2 \times 2$                                                                                                                                                                       | $2 \times 2$                                                                                                                                            | matrices that have different sizes ( $A$ is a $2 \times 2$ matrix, but $2C$ is a                                                                                                             |
| johnston-linear-matrix-algebra_F00368 | 37   | 0.999 | $2C$                                                                                                                                                                               | $2C$                                                                                                                                                    | matrices that have different sizes ( $A$ is a $2 \times 2$ matrix, but $2C$ is a                                                                                                             |
| johnston-linear-matrix-algebra_F00369 | 37   | 0.991 | $2 \times 3$                                                                                                                                                                       | $2 \times 3$                                                                                                                                            | $2 \times 3$ matrix).                                                                                                                                                                        |
| johnston-linear-matrix-algebra_F00370 | 37   | 1.000 | $A, B, C \in \mathcal{M}_{m,n}$                                                                                                                                                    | $A, B, C \in \mathcal{M}_{m,n}$                                                                                                                         | Suppose $A, B, C \in \mathcal{M}_{m,n}$ are $m \times n$ matrices and $c, d \in \mathbb{R}$ are scalars. Then                                                                                |
| johnston-linear-matrix-algebra_F00371 | 37   | 1.000 | $A+B=B+A$                                                                                                                                                                          | $A+B=B+A$                                                                                                                                               | a) $A+B=B+A$ ,                                                                                                                                                                               |
| johnston-linear-matrix-algebra_F00372 | 37   | 1.000 | $(A+B)+C=A+(B+C)$                                                                                                                                                                  | $(A+B)+C=A+(B+C)$                                                                                                                                       | b) $(A+B)+C=A+(B+C)$ ,                                                                                                                                                                       |
| johnston-linear-matrix-algebra_F00373 | 37   | 1.000 | $c(A+B)=cA+cB$                                                                                                                                                                     | $c(A+B)=cA+cB$                                                                                                                                          | c) $c(A+B)=cA+cB$ ,                                                                                                                                                                          |
| johnston-linear-matrix-algebra_F00374 | 37   | 1.000 | $(c+d)A=cA+dA$                                                                                                                                                                     | $(c+d)A=cA+dA$                                                                                                                                          | d) $(c+d)A=cA+dA$ , and                                                                                                                                                                      |
| johnston-linear-matrix-algebra_F00375 | 37   | 1.000 | $c(dA)=(cd)A$                                                                                                                                                                      | $c(dA)=(cd)A$                                                                                                                                           | e) $c(dA)=(cd)A$ .                                                                                                                                                                           |
| johnston-linear-matrix-algebra_F00376 | 37   | 1.000 | $a+b=b+a$                                                                                                                                                                          | $a+b=b+a$                                                                                                                                               | that $a+b=b+a$                                                                                                                                                                               |
| johnston-linear-matrix-algebra_F00377 | 37   | 1.000 | $a, b \in \mathbb{R}$                                                                                                                                                              | $a, b \in \mathbb{R}$                                                                                                                                   | when $a, b \in \mathbb{R}$ . We                                                                                                                                                              |
| johnston-linear-matrix-algebra_F00378 | 37   | 1.000 | $mn$                                                                                                                                                                               | $mn$                                                                                                                                                    | each of the $mn$                                                                                                                                                                             |
| johnston-linear-matrix-algebra_F00379 | 37   | 1.000 | $A+O=A$                                                                                                                                                                            | $A+O=A$                                                                                                                                                 | are the facts that $A+O=A$ and $A-A=O$ for all matrices $A \in \mathcal{M}_{m,n}$ .                                                                                                          |
| johnston-linear-matrix-algebra_F00380 | 37   | 1.000 | $A-A=O$                                                                                                                                                                            | $A-A=O$                                                                                                                                                 | are the facts that $A+O=A$ and $A-A=O$ for all matrices $A \in \mathcal{M}_{m,n}$ .                                                                                                          |
| johnston-linear-matrix-algebra_F00381 | 37   | 1.000 | $A \in \mathcal{M}_{m,n}$                                                                                                                                                          | $A \in \mathcal{M}_{m,n}$                                                                                                                               | are the facts that $A+O=A$ and $A-A=O$ for all matrices $A \in \mathcal{M}_{m,n}$ .                                                                                                          |
| johnston-linear-matrix-algebra_F00382 | 38   | 1.000 | $r \times p$                                                                                                                                                                       | $r \times p$                                                                                                                                            | matrix by an $r \times p$                                                                                                                                                                    |
| johnston-linear-matrix-algebra_F00383 | 38   | 0.910 | $n=r$                                                                                                                                                                              | $n=r$                                                                                                                                                   | $n=r$ .                                                                                                                                                                                      |

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|---------------------------------------|------|---------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra_F00384 | 38   | ■ 1.000 | $B \in \mathcal{M}_{n,p}$                                                | $B \in \mathcal{M}_{n,p}$                                                | If $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,p}$ are matrices, then their <b>product</b> $AB$ is the   |
| johnston-linear-matrix-algebra_F00385 | 38   | ■ 1.000 | $A B$                                                                    | $AB$                                                                     | If $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{n,p}$ are matrices, then their <b>product</b> $AB$ is the   |
| johnston-linear-matrix-algebra_F00386 | 38   | ■ 0.918 | $m \times p$                                                             | $m \times p$                                                             | $m \times p$ matrix whose $(i, j)$ -entry, for each $1 \leq i \leq m$ and $1 \leq j \leq p$ , is                 |
| johnston-linear-matrix-algebra_F00387 | 38   | ■ 0.918 | $1 \leq j \leq p$                                                        | $1 \leq j \leq p$                                                        | $m \times p$ matrix whose $(i, j)$ -entry, for each $1 \leq i \leq m$ and $1 \leq j \leq p$ , is                 |
| johnston-linear-matrix-algebra_F00388 | 38   | ■ 0.981 | $3 \times 7$                                                             | $3 \times 7$                                                             | $3 \times 7$ matrix.                                                                                             |
| johnston-linear-matrix-algebra_F00389 | 38   | ■ 0.554 | $3 \times 2$                                                             | $3 \times 2$                                                             | <b>Figure 1.13:</b> A visualization of the multiplication of a $3 \times 2$ matrix $A$ and a $2 \times 4$ matrix |
| johnston-linear-matrix-algebra_F00390 | 38   | ■ 0.554 | $2 \times 4$                                                             | $2 \times 4$                                                             | <b>Figure 1.13:</b> A visualization of the multiplication of a $3 \times 2$ matrix $A$ and a $2 \times 4$ matrix |
| johnston-linear-matrix-algebra_F00391 | 38   | ■ 1.000 | $A C$                                                                    | $AC$                                                                     | b) $AC$ ,                                                                                                        |
| johnston-linear-matrix-algebra_F00392 | 38   | ■ 1.000 | $B A$                                                                    | $BA$                                                                     | c) $BA$ , and                                                                                                    |
| johnston-linear-matrix-algebra_F00393 | 38   | ■ 1.000 | $B C$                                                                    | $BC$                                                                     | d) $BC$ .                                                                                                        |
| johnston-linear-matrix-algebra_F00394 | 38   | ■ 1.000 | $B : (1, 2) \cdot (5, 8) = 1 \cdot 5 + 2 \cdot 8 = 21$                   | $B : (1, 2) \cdot (5, 8) = 1 \cdot 5 + 2 \cdot 8 = 21$                   | the first column of $B$ : $(1, 2) \cdot (5, 8) = 1 \cdot 5 + 2 \cdot 8 = 21$ . Similarly                         |
| johnston-linear-matrix-algebra_F00395 | 38   | ■ 0.998 | $A B^{\prime}$                                                           | $AB'$                                                                    | $AB$ 's size consists of                                                                                         |
| johnston-linear-matrix-algebra_F00396 | 39   | ■ 1.000 | $n \times p$                                                             | $n \times p$                                                             | and $B$ is $n \times p$ then                                                                                     |
| johnston-linear-matrix-algebra_F00397 | 39   | ■ 1.000 | $C$                                                                      | $C$                                                                      | b) $AC$ does not exist since $A$ has 2 columns but $C$ has 3 rows, and those                                     |
| johnston-linear-matrix-algebra_F00398 | 39   | ■ 1.000 | $C : (5, 6, 7) \cdot (1, 0, 2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$ | $C : (5, 6, 7) \cdot (1, 0, 2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$ | the first column of $C$ : $(5, 6, 7) \cdot (1, 0, 2) = 5 \cdot 1 + 6 \cdot 0 + 7 \cdot 2 = 19$ .                 |
| johnston-linear-matrix-algebra_F00399 | 40   | ■ 1.000 | $A, B$                                                                   | $A, B$                                                                   | Let $A, B$ , and $C$ be matrices (with sizes such that the multiplications and                                   |
| johnston-linear-matrix-algebra_F00400 | 40   | ■ 1.000 | $(A B) C = A(B C)$                                                       | $(AB)C = A(BC)$                                                          | a) $(AB)C = A(BC)$ ,                                                                                             |
| johnston-linear-matrix-algebra_F00401 | 40   | ■ 1.000 | $A(B+C) = A B + A C$                                                     | $A(B + C) = AB + AC$                                                     | b) $A(B + C) = AB + AC$ ,                                                                                        |
| johnston-linear-matrix-algebra_F00402 | 40   | ■ 1.000 | $(A+B) C = A C + B C$                                                    | $(A + B)C = AC + BC$                                                     | c) $(A + B)C = AC + BC$ , and                                                                                    |
| johnston-linear-matrix-algebra_F00403 | 40   | ■ 1.000 | $c(A B) = (c A) B$                                                       | $c(AB) = (cA)B$                                                          | d) $c(AB) = (cA)B$ .                                                                                             |
| johnston-linear-matrix-algebra_F00404 | 40   | ■ 0.619 | $A(B+C)$                                                                 | $A(B + C)$                                                               | To this end we note that, for each $i$ and $j$ , the $(i, j)$ -entry of $A(B + C)$ is                            |
| johnston-linear-matrix-algebra_F00405 | 40   | ■ 0.998 | $A B + A C$                                                              | $AB + AC$                                                                | whereas the $(i, j)$ -entry of $AB + AC$ is                                                                      |
| johnston-linear-matrix-algebra_F00406 | 40   | ■ 1.000 | $A B =$                                                                  | $AB =$                                                                   | It is worth stressing the fact that we did <i>not</i> list commutativity (i.e., $AB =$                           |
| johnston-linear-matrix-algebra_F00407 | 40   | ■ 1.000 | $I_{\{n\}}$                                                              | $I_n$                                                                    | size $n \times n$ then it is denoted by $I_n$ (or if its size is clear from context or irrelevant,               |
| johnston-linear-matrix-algebra_F00408 | 40   | ■ 1.000 | $I$                                                                      | $I$                                                                      | we denote it just by $I$ ). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are        |
| johnston-linear-matrix-algebra_F00409 | 40   | ■ 1.000 | $\mathcal{M}_{\{2\}}$                                                    | $\mathcal{M}_2$                                                          | we denote it just by $I$ ). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are        |
| johnston-linear-matrix-algebra_F00410 | 40   | ■ 1.000 | $\mathcal{M}_{\{3\}}$                                                    | $\mathcal{M}_3$                                                          | we denote it just by $I$ ). For example, the identity matrices in $\mathcal{M}_2$ and $\mathcal{M}_3$ are        |
| johnston-linear-matrix-algebra_F00411 | 41   | ■ 1.000 | $x \in \mathbb{R}$                                                       | $x \in \mathbb{R}$                                                       | Recall that if $x \in \mathbb{R}$                                                                                |

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| johnston-linear-matrix-algebra-F00412 | 41   | ■ 1.000 | $x^{\{0\}}=1$                                                  | $x^0 = 1$                                           | then $x^0 = 1$ . The                                                                                               |
| johnston-linear-matrix-algebra-F00413 | 41   | ■ 1.000 | $A^{\{0\}}=I$                                                  | $A^0 = I$                                           | definition $A^0 = I$ is                                                                                            |
| johnston-linear-matrix-algebra-F00414 | 41   | ■ 1.000 | $A I_{\{n\}}=A=I_{\{m\}} A$                                    | $AI_n = A = I_m A$                                  | If $A \in \mathcal{M}_{m,n}$ then $AI_n = A = I_m A$ .                                                             |
| johnston-linear-matrix-algebra-F00415 | 41   | ■ 0.996 | $A I_{\{n\}}$                                                  | $AI_n$                                              | <i>Proof.</i> For the first equality, we recall that the $(i, j)$ -entry of $AI_n$ is the dot                      |
| johnston-linear-matrix-algebra-F00416 | 41   | ■ 1.000 | $A I_{\{n\}}=A$                                                | $AI_n = A$                                          | and $A$ coincide, so $AI_n = A$ .                                                                                  |
| johnston-linear-matrix-algebra-F00417 | 41   | ■ 1.000 | $A^{\{2\}}=A A, A^{\{3\}}=A A A$                               | $A^2 = AA, A^3 = AAA$                               | obtain <b>powers</b> of $A$ . That is, we define $A^2 = AA, A^3 = AAA$ , and in general                            |
| johnston-linear-matrix-algebra-F00418 | 41   | ■ 1.000 | $A^{\{k+\ell\}}=A^{\{k\}} A^{\{\ell\}}$                        | $A^{k+\ell} = A^k A^\ell$                           | to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We |
| johnston-linear-matrix-algebra-F00419 | 41   | ■ 1.000 | $\left(A^{\{k\}}\right)^{\{\ell\}}=A^{\{k \ell\}}$             | $(A^k)^\ell = A^{k\ell}$                            | to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We |
| johnston-linear-matrix-algebra-F00420 | 41   | ■ 1.000 | $k, \ell \geq 1$                                               | $k, \ell \geq 1$                                    | to, like $A^{k+\ell} = A^k A^\ell$ and $(A^k)^\ell = A^{k\ell}$ for all $k, \ell \geq 1$ (see Exercise 1.3.22). We |
| johnston-linear-matrix-algebra-F00421 | 41   | ■ 1.000 | $k=0$                                                          | $k = 0$                                             | also define $A^0 = I$ so that these same properties even hold if $k = 0$ or $\ell = 0$ .                           |
| johnston-linear-matrix-algebra-F00422 | 41   | ■ 1.000 | $\ell=0$                                                       | $\ell = 0$                                          | also define $A^0 = I$ so that these same properties even hold if $k = 0$ or $\ell = 0$ .                           |
| johnston-linear-matrix-algebra-F00423 | 41   | ■ 1.000 | $A=\left[\begin{array}{cc} 2 & 1 \\ -1 & 3 \end{array}\right]$ | $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ | Compute each of the indicated matrix powers if $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ .               |
| johnston-linear-matrix-algebra-F00424 | 41   | ■ 1.000 | $A^{\{2\}}$                                                    | $A^2$                                               | a) $A^2$ ,                                                                                                         |
| johnston-linear-matrix-algebra-F00425 | 41   | ■ 1.000 | $A^{\{4\}}$                                                    | $A^4$                                               | b) $A^4$ , and                                                                                                     |
| johnston-linear-matrix-algebra-F00426 | 41   | ■ 1.000 | $I^{\{7\}}$                                                    | $I^7$                                               | c) $I^7$ .                                                                                                         |
| johnston-linear-matrix-algebra-F00427 | 41   | ■ 1.000 | $A^{\{4\}}=\left(A^{\{2\}}\right)^{\{2\}}$                     | $A^4 = (A^2)^2$                                     | that $A^4 = (A^2)^2$ , so we just need to square our answer from part (a):                                         |
| johnston-linear-matrix-algebra-F00428 | 41   | ■ 1.000 | $I^{\{k\}}=I$                                                  | $I^k = I$                                           | c) We can repeatedly use Theorem 1.3.3 to see that $I^k = I$ for all non-                                          |
| johnston-linear-matrix-algebra-F00429 | 41   | ■ 1.000 | $k$                                                            | $k$                                                 | negative integers $k$ , so $I^7 = I$ .                                                                             |
| johnston-linear-matrix-algebra-F00430 | 41   | ■ 1.000 | $I^{\{7\}}=I$                                                  | $I^7 = I$                                           | negative integers $k$ , so $I^7 = I$ .                                                                             |
| johnston-linear-matrix-algebra-F00431 | 41   | ■ 1.000 | $(A+B)^{\{2\}}$                                                | $(A+B)^2$                                           | when expanding the expression $(A+B)^2$ , the reader might be tempted to write                                     |
| johnston-linear-matrix-algebra-F00432 | 41   | ■ 1.000 | $(A+B)^{\{2\}}=A^{\{2\}}+2 A B+B^{\{2\}}$                      | $(A+B)^2 = A^2 + 2AB + B^2$                         | $(A+B)^2 = A^2 + 2AB + B^2$ . However, if we go through the calculation a bit                                      |
| johnston-linear-matrix-algebra-F00433 | 42   | ■ 1.000 | $(A+B)^{\{2\}}=$                                               | $(A+B)^2 =$                                         | $(A+B)^2 =$                                                                                                        |
| johnston-linear-matrix-algebra-F00434 | 42   | ■ 1.000 | $A^{\{2\}}+2 A B+B^{\{2\}}$                                    | $A^2 + 2AB + B^2$                                   | $A^2 + 2AB + B^2$ in                                                                                               |
| johnston-linear-matrix-algebra-F00435 | 42   | ■ 1.000 | $(A+I)^{\{2\}}=A^{\{2\}}+2 A I$                                | $(A+I)^2 = A^2 + 2A + I$                            | $(A+I)^2 = A^2 + 2A + I$ ,                                                                                         |
| johnston-linear-matrix-algebra-F00436 | 42   | ■ 0.994 | $1 \times n$                                                   | $1 \times n$                                        | vectors as $1 \times n$ or                                                                                         |
| johnston-linear-matrix-algebra-F00437 | 42   | ■ 1.000 | $m \times 1$                                                   | $m \times 1$                                        | $m \times 1$ matrices.                                                                                             |
| johnston-linear-matrix-algebra-F00438 | 42   | ■ 1.000 | $A, B \in \mathcal{M}_{\{n\}}$                                 | $A, B \in \mathcal{M}_n$                            | Suppose $A, B \in \mathcal{M}_n$ . Expand and simplify the following expressions:                                  |

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| johnston-linear-matrix-algebra_F00439 | 42   | ■ 1.000 | $(A+I)^2(A-I)$                                                | $(A+I)^2(A-I)$                                                | a) $(A+I)^2(A-I)$ , and                                                                                                                            |
| johnston-linear-matrix-algebra_F00440 | 42   | ■ 1.000 | $(A-A)(A+B-I)^7$                                              | $(A-A)(A+B-I)^7$                                              | b) $(A-A)(A+B-I)^7$ .                                                                                                                              |
| johnston-linear-matrix-algebra_F00441 | 42   | ■ 1.000 | $A+I$                                                         | $A+I$                                                         | a) We first square the $A+I$ and then multiply it by $A-I$ :                                                                                       |
| johnston-linear-matrix-algebra_F00442 | 42   | ■ 1.000 | $A-I$                                                         | $A-I$                                                         | a) We first square the $A+I$ and then multiply it by $A-I$ :                                                                                       |
| johnston-linear-matrix-algebra_F00443 | 42   | ■ 1.000 | $(A+B-I)^7$                                                   | $(A+B-I)^7$                                                   | b) We could expand $(A+B-I)^7$ (which would be ugly!) and then                                                                                     |
| johnston-linear-matrix-algebra_F00444 | 42   | ■ 1.000 | $(A-A)(A+$                                                    | $(A-A)(A+$                                                    | that $O$ times <i>any</i> matrix is equal to $O$ . It follows that $(A-A)(A+$                                                                      |
| johnston-linear-matrix-algebra_F00445 | 42   | ■ 0.950 | $B-I)^7=O(A+B-I)^7=O$                                         | $B-I)^7=O(A+B-I)^7=O$                                         | $B-I)^7=O(A+B-I)^7=O$ .                                                                                                                            |
| johnston-linear-matrix-algebra_F00446 | 42   | ■ 1.000 | $\mathbf{v} \in \mathcal{M}_{n,1}$                            | $\mathbf{v} \in \mathcal{M}_{n,1}$                            | move vectors around. In particular, if $A \in \mathcal{M}_{m,n}$ and $\mathbf{v} \in \mathcal{M}_{n,1}$ then $A\mathbf{v} \in \mathcal{M}_{m,1}$ , |
| johnston-linear-matrix-algebra_F00447 | 42   | ■ 1.000 | $A \mathbf{v} \in \mathcal{M}_{m,1}$                          | $A\mathbf{v} \in \mathcal{M}_{m,1}$                           | move vectors around. In particular, if $A \in \mathcal{M}_{m,n}$ and $\mathbf{v} \in \mathcal{M}_{n,1}$ then $A\mathbf{v} \in \mathcal{M}_{m,1}$ , |
| johnston-linear-matrix-algebra_F00448 | 42   | ■ 0.942 | $A \mathbf{v}$                                                | $A\mathbf{v}$                                                 | and since each of $\mathbf{v}$ and $A\mathbf{v}$ have just one column, we can think of them as vectors                                             |
| johnston-linear-matrix-algebra_F00449 | 42   | ■ 1.000 | $\mathbf{v}=(2,-1)$                                           | $\mathbf{v}=(2,-1)$                                           | so we think of the matrix $A$ as transforming $\mathbf{v}=(2,-1)$ into $A\mathbf{v}=(0,2)$ , but                                                   |
| johnston-linear-matrix-algebra_F00450 | 42   | ■ 1.000 | $A \mathbf{v}=(0,2)$                                          | $A\mathbf{v}=(0,2)$                                           | so we think of the matrix $A$ as transforming $\mathbf{v}=(2,-1)$ into $A\mathbf{v}=(0,2)$ , but                                                   |
| johnston-linear-matrix-algebra_F00451 | 43   | ■ 1.000 | $[1,2,3] \in \mathbb{R}^3$                                    | $[1,2,3] \in \mathbb{R}^3$                                    | $[1,2,3] \in \mathbb{R}^3$ (even                                                                                                                   |
| johnston-linear-matrix-algebra_F00452 | 43   | ■ 1.000 | $\mathcal{M}_{1,3}$                                           | $\mathcal{M}_{1,3}$                                           | lives in $\mathcal{M}_{1,3}$ ).                                                                                                                    |
| johnston-linear-matrix-algebra_F00453 | 43   | ■ 1.000 | $\mathbf{v}=[1,2,3] \in \mathcal{M}_{1,3}$                    | $\mathbf{v}=[1,2,3] \in \mathcal{M}_{1,3}$                    | (as we did in the previous two sections). For example, $\mathbf{v}=[1,2,3] \in \mathcal{M}_{1,3}$ is a                                             |
| johnston-linear-matrix-algebra_F00454 | 43   | ■ 1.000 | $\mathbf{x}=(1,2,3) \in \mathbb{R}^3$                         | $\mathbf{x}=(1,2,3) \in \mathbb{R}^3$                         | is a column vector, and $\mathbf{x}=(1,2,3) \in \mathbb{R}^3$ is a vector for which we do not care                                                 |
| johnston-linear-matrix-algebra_F00455 | 43   | ■ 1.000 | $\mathbf{v} \mapsto A \mathbf{v}$                             | $\mathbf{v} \mapsto A\mathbf{v}$                              | used as a function $\mathbf{v} \mapsto A\mathbf{v}$ that moves vectors around $\mathbb{R}^n$ . We introduce this                                   |
| johnston-linear-matrix-algebra_F00456 | 43   | ■ 0.774 | $A^T$                                                         | $A^T$                                                         | denote by $A^T$ , is the $n \times m$ matrix whose $(i,j)$ -entry is $a_{j,i}$ .                                                                   |
| johnston-linear-matrix-algebra_F00457 | 43   | ■ 0.774 | $n \times m$                                                  | $n \times m$                                                  | denote by $A^T$ , is the $n \times m$ matrix whose $(i,j)$ -entry is $a_{j,i}$ .                                                                   |
| johnston-linear-matrix-algebra_F00458 | 43   | ■ 0.774 | $a_{j,i}$                                                     | $a_{j,i}$                                                     | denote by $A^T$ , is the $n \times m$ matrix whose $(i,j)$ -entry is $a_{j,i}$ .                                                                   |
| johnston-linear-matrix-algebra_F00459 | 43   | ■ 1.000 | $B^T$                                                         | $B^T$                                                         | b) $B^T$ ,                                                                                                                                         |
| johnston-linear-matrix-algebra_F00460 | 43   | ■ 1.000 | $(AB)^T$                                                      | $(AB)^T$                                                      | c) $(AB)^T$ , and                                                                                                                                  |
| johnston-linear-matrix-algebra_F00461 | 43   | ■ 1.000 | $B^T A^T$                                                     | $B^T A^T$                                                     | d) $B^T A^T$ .                                                                                                                                     |
| johnston-linear-matrix-algebra_F00462 | 43   | ■ 1.000 | $A^T=\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$ | $A^T=\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$ | a) $A^T=\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$ .                                                                                 |
| johnston-linear-matrix-algebra_F00463 | 44   | ■ 1.000 | $\left[(AB)^T\right]_{i,j}$                                   | $\left[(AB)^T\right]_{i,j}$                                   | $\left[(AB)^T\right]_{i,j}$ means the                                                                                                              |
| johnston-linear-matrix-algebra_F00464 | 44   | ■ 0.994 | $\mathbf{v}^T \mathbf{w}$                                     | $\mathbf{v}^T \mathbf{w}$                                     | $\mathbf{v}^T \mathbf{w}$ is a $1 \times 1$ matrix,                                                                                                |
| johnston-linear-matrix-algebra_F00465 | 44   | ■ 0.994 | $1 \times 1$                                                  | $1 \times 1$                                                  | $\mathbf{v}^T \mathbf{w}$ is a $1 \times 1$ matrix,                                                                                                |

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| johnston-linear-matrix-algebra_F00466 | 44   | 1.000 | $B^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$                                                                                                            | $B^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$                                                                                                                                                                                      | b) $B^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$ .                                                                                                                                                                                      |
| johnston-linear-matrix-algebra_F00467 | 44   | 0.971 | $(A B)^T = \begin{bmatrix} 1 & 2 & 3 & 4 \\ -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 1 & -3 & 7 & 3 \end{bmatrix}$                                                                  | $(AB)^T = \left( \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right)^T = \begin{bmatrix} -1 & 3 & 1 \\ -3 & 7 & 3 \end{bmatrix}^T = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$ | c) $(AB)^T = \left( \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \right)^T = \begin{bmatrix} -1 & 3 & 1 \\ -3 & 7 & 3 \end{bmatrix}^T = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra_F00468 | 44   | 1.000 | $B^T A^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$ | $B^T A^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$                                                                           | d) $B^T A^T = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ 3 & 7 \\ 1 & 3 \end{bmatrix}$ .                                                                           |
| johnston-linear-matrix-algebra_F00469 | 44   | 1.000 | $(A B)^T = B^T A^T$                                                                                                                                                       | $(AB)^T = B^T A^T$                                                                                                                                                                                                                                  | Parts (c) and (d) of the previous example suggest that $(AB)^T = B^T A^T$ might                                                                                                                                                                          |
| johnston-linear-matrix-algebra_F00470 | 44   | 1.000 | $\left( A^T \right)^T = A$                                                                                                                                                | $(A^T)^T = A$                                                                                                                                                                                                                                       | a) $(A^T)^T = A$ ,                                                                                                                                                                                                                                       |
| johnston-linear-matrix-algebra_F00471 | 44   | 1.000 | $(A+B)^T = A^T + B^T$                                                                                                                                                     | $(A+B)^T = A^T + B^T$                                                                                                                                                                                                                               | b) $(A+B)^T = A^T + B^T$ ,                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra_F00472 | 44   | 1.000 | $(c A)^T = c A^T$                                                                                                                                                         | $(cA)^T = cA^T$                                                                                                                                                                                                                                     | d) $(cA)^T = cA^T$ .                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra_F00473 | 44   | 1.000 | $\left[ (A B)^T \right]_{i,j} = \left[ B^T A^T \right]_{i,j}$                                                                                                             | $[(AB)^T]_{i,j} = [B^T A^T]_{i,j}$                                                                                                                                                                                                                  | Since these two quantities are the same, we conclude that $[(AB)^T]_{i,j} = [B^T A^T]_{i,j}$                                                                                                                                                             |
| johnston-linear-matrix-algebra_F00474 | 45   | 1.000 | $C D$                                                                                                                                                                     | $CD$                                                                                                                                                                                                                                                | is $CD$ , not $DC$ .                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra_F00475 | 45   | 1.000 | $D C$                                                                                                                                                                     | $DC$                                                                                                                                                                                                                                                | is $CD$ , not $DC$ .                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra_F00476 | 45   | 0.722 | $I_3$                                                                                                                                                                     | $I_3$                                                                                                                                                                                                                                               | We then recognize two of the blocks of $A$ as $I_3$ (the $3 \times 3$ identity matrix)                                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00477 | 45   | 0.722 | $3 \times 3$                                                                                                                                                              | $3 \times 3$                                                                                                                                                                                                                                        | We then recognize two of the blocks of $A$ as $I_3$ (the $3 \times 3$ identity matrix)                                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00478 | 45   | 1.000 | $D$                                                                                                                                                                       | $D$                                                                                                                                                                                                                                                 | then we can write $A$ and $B$ in terms of $C$ and $D$ as follows:                                                                                                                                                                                        |
| johnston-linear-matrix-algebra_F00479 | 45   | 1.000 | $4 \cdot 5 = 20$                                                                                                                                                          | $4 \cdot 5 = 20$                                                                                                                                                                                                                                    | the long way (i.e., computing $4 \cdot 5 = 20$ dot products of 6-dimensional vectors),                                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00480 | 46   | 1.000 | $I_2$                                                                                                                                                                     | $I_2$                                                                                                                                                                                                                                               | Recall that $I_2$ and                                                                                                                                                                                                                                    |
| johnston-linear-matrix-algebra_F00481 | 46   | 0.926 | $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                                                                                                                      | $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                                                                                                                                                                                                | $I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ .                                                                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00482 | 46   | 1.000 | $C D = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$                                                                                                                     | $CD = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$                                                                                                                                                                                                | It is not difficult to compute $CD = \begin{bmatrix} 5 & 4 \\ -7 & 1 \end{bmatrix}$ , so it follows that                                                                                                                                                 |
| johnston-linear-matrix-algebra_F00483 | 46   | 0.996 | $\left( \begin{bmatrix} A & B \\ B & A \end{bmatrix} \right)^2$                                                                                                           | $\begin{bmatrix} A & B \\ B & A \end{bmatrix}^2$                                                                                                                                                                                                    | a) $\begin{bmatrix} A & B \\ B & A \end{bmatrix}^2$ ,                                                                                                                                                                                                    |

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| johnston-linear-matrix-algebra_F00484 | 46   | 0.863 | $\left[\begin{array}{ll}A & B \\ 0 & I_3\end{array}\right]$ | $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix}$            | b) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \\ I_2 & O \end{bmatrix}$ ,                                     |
| johnston-linear-matrix-algebra_F00485 | 46   | 0.434 | $\left[\begin{array}{ll}A & B \\ 0 & I_3\end{array}\right]$ | $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix}$            | c) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} A & A \\ O & A \end{bmatrix}$ , and                                            |
| johnston-linear-matrix-algebra_F00486 | 46   | 0.995 | $\left[\begin{array}{ll}A & B \\ 0 & I_3\end{array}\right]$ | $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix}$            | d) $\begin{bmatrix} A & B \\ O & I_3 \end{bmatrix} \begin{bmatrix} B & O \\ I_3 & I_3 \end{bmatrix}$ .                                            |
| johnston-linear-matrix-algebra_F00487 | 46   | 0.969 | $\left[\begin{array}{ll}A & B \\ B & A\end{array}\right]$   | $\begin{bmatrix} A & B \\ B & A \end{bmatrix}$              | a) The block matrix $\begin{bmatrix} A & B \\ B & A \end{bmatrix}$ itself makes no sense, since $A$ has two                                       |
| johnston-linear-matrix-algebra_F00488 | 47   | 1.000 | $A + B$                                                     | $AB + B$                                                    | The computation of the top-left block $AB + B$ did take some effort,                                                                              |
| johnston-linear-matrix-algebra_F00489 | 47   | 1.000 | $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$           | $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$           | Suppose $A \in \mathcal{M}_{m,n}$ has columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ (in that order) and $\mathbf{v} \in \mathbb{R}^n$ |
| johnston-linear-matrix-algebra_F00490 | 47   | 0.813 | $n \times 1$                                                | $n \times 1$                                                | $\mathbf{v}$ as an $n \times 1$ matrix.                                                                                                           |
| johnston-linear-matrix-algebra_F00491 | 47   | 1.000 | $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$           | $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$           | $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ (in that order), then                                                                           |
| johnston-linear-matrix-algebra_F00492 | 47   | 1.000 | $1 \times p$                                                | $1 \times p$                                                | $B$ as a $1 \times p$ block matrix.                                                                                                               |
| johnston-linear-matrix-algebra_F00493 | 48   | 1.000 | $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$          | $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$          | Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ . Compute $AB$ via     |
| johnston-linear-matrix-algebra_F00494 | 48   | 1.000 | $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ | $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ | Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ . Compute $AB$ via     |
| johnston-linear-matrix-algebra_F00495 | 48   | 1.000 | $2B - 3A$                                                   | $2B - 3A$                                                   | (b) $2B - 3A$                                                                                                                                     |
| johnston-linear-matrix-algebra_F00496 | 48   | 0.881 | $A - B$                                                     | $A - B$                                                     | *(c) $A - B$                                                                                                                                      |
| johnston-linear-matrix-algebra_F00497 | 48   | 1.000 | $2(A + B) - A$                                              | $2(A + B) - A$                                              | (d) $2(A + B) - A$                                                                                                                                |
| johnston-linear-matrix-algebra_F00498 | 48   | 1.000 | $CD^T$                                                      | $CD^T$                                                      | (b) $CD^T$                                                                                                                                        |
| johnston-linear-matrix-algebra_F00499 | 48   | 1.000 | $C^T D$                                                     | $C^T D$                                                     | (d) $C^T D$                                                                                                                                       |
| johnston-linear-matrix-algebra_F00500 | 48   | 1.000 | $(C^T D)^2$                                                 | $(C^T D)^2$                                                 | (f) $(C^T D)^2$                                                                                                                                   |
| johnston-linear-matrix-algebra_F00501 | 48   | 1.000 | $(C - D)^T (A - B^T)$                                       | $(C - D)^T (A - B^T)$                                       | (h) $(C - D)^T (A - B^T)$                                                                                                                         |
| johnston-linear-matrix-algebra_F00502 | 48   | 0.997 | $(A + B)(C + D)$                                            | $(A + B)(C + D)$                                            | *(i) $(A + B)(C + D)$                                                                                                                             |
| johnston-linear-matrix-algebra_F00503 | 48   | 1.000 | $D^T (A^T + B)^T C$                                         | $D^T (A^T + B)^T C$                                         | (j) $D^T (A^T + B)^T C$                                                                                                                           |
| johnston-linear-matrix-algebra_F00504 | 49   | 1.000 | $A^T A$                                                     | $A^T A$                                                     | (b) The matrices $A^T A$ and $AA^T$ are always defined, re-                                                                                       |
| johnston-linear-matrix-algebra_F00505 | 49   | 1.000 | $A A^T$                                                     | $AA^T$                                                      | (b) The matrices $A^T A$ and $AA^T$ are always defined, re-                                                                                       |

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| johnston-linear-matrix-algebra_F00506 | 49   | 0.749 | $(A \ B)^2 = A^2 \ B^2$                             | $(AB)^2 = A^2 B^2$                                  | $(AB)^2 = A^2 B^2$ .                                                                                 |
| johnston-linear-matrix-algebra_F00507 | 49   | 1.000 | $A \ B = 0$                                         | $AB = O$                                            | (f) If $A$ and $B$ are matrices such that $AB = O$ and $A \neq O$ ,                                  |
| johnston-linear-matrix-algebra_F00508 | 49   | 1.000 | $A \neq 0$                                          | $A \neq O$                                          | (f) If $A$ and $B$ are matrices such that $AB = O$ and $A \neq O$ ,                                  |
| johnston-linear-matrix-algebra_F00509 | 49   | 1.000 | $B = 0$                                             | $B = O$                                             | then $B = O$ .                                                                                       |
| johnston-linear-matrix-algebra_F00510 | 49   | 0.998 | $A^2 = I$                                           | $A^2 = I$                                           | *(g) If $A$ is a square matrix for which $A^2 = I$ , then $A = I$                                    |
| johnston-linear-matrix-algebra_F00511 | 49   | 0.998 | $A = I$                                             | $A = I$                                             | *(g) If $A$ is a square matrix for which $A^2 = I$ , then $A = I$                                    |
| johnston-linear-matrix-algebra_F00512 | 49   | 1.000 | $A = -I$                                            | $A = -I$                                            | or $A = -I$ .                                                                                        |
| johnston-linear-matrix-algebra_F00513 | 49   | 1.000 | $A^2 = A$                                           | $A^2 = A$                                           | (h) If $A$ is a square matrix for which $A^2 = A$ , then either                                      |
| johnston-linear-matrix-algebra_F00514 | 49   | 1.000 | $A = 0$                                             | $A = O$                                             | $A = I$ or $A = O$ .                                                                                 |
| johnston-linear-matrix-algebra_F00515 | 49   | 1.000 | $A \in \mathcal{M}_{n,m}$                           | $A \in \mathcal{M}_{n,m}$                           | *(i) If $A \in \mathcal{M}_{n,m}$ and $\mathbf{b} \in \mathbb{R}^m$ is a column vector, then         |
| johnston-linear-matrix-algebra_F00516 | 49   | 1.000 | $\mathbf{b} \in \mathbb{R}^m$                       | $\mathbf{b} \in \mathbb{R}^m$                       | *(i) If $A \in \mathcal{M}_{n,m}$ and $\mathbf{b} \in \mathbb{R}^m$ is a column vector, then         |
| johnston-linear-matrix-algebra_F00517 | 49   | 1.000 | $A \mathbf{b}$                                      | $A\mathbf{b}$                                       | $A\mathbf{b}$ is a linear combination of the rows of $A$ .                                           |
| johnston-linear-matrix-algebra_F00518 | 49   | 1.000 | $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$  | $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$  | 1.3.4 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$ , and $C \in \mathcal{M}_{5,2}$ are |
| johnston-linear-matrix-algebra_F00519 | 49   | 1.000 | $C \in \mathcal{M}_{5,2}$                           | $C \in \mathcal{M}_{5,2}$                           | 1.3.4 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,5}$ , and $C \in \mathcal{M}_{5,2}$ are |
| johnston-linear-matrix-algebra_F00520 | 49   | 1.000 | $A^7$                                               | $A^7$                                               | (b) $A^7$                                                                                            |
| johnston-linear-matrix-algebra_F00521 | 49   | 1.000 | $B^2$                                               | $B^2$                                               | (d) $B^2$                                                                                            |
| johnston-linear-matrix-algebra_F00522 | 49   | 1.000 | $B \ B^T$                                           | $BB^T$                                              | (f) $BB^T$                                                                                           |
| johnston-linear-matrix-algebra_F00523 | 49   | 0.999 | $A \ B \ C$                                         | $ABC$                                               | *(g) $ABC$                                                                                           |
| johnston-linear-matrix-algebra_F00524 | 49   | 1.000 | $B^T \ B + C^T \ C$                                 | $B^T B + C^T C$                                     | (h) $B^T B + C^T C$                                                                                  |
| johnston-linear-matrix-algebra_F00525 | 49   | 0.963 | $A + B \ C$                                         | $A + BC$                                            | *(i) $A + BC$                                                                                        |
| johnston-linear-matrix-algebra_F00526 | 49   | 1.000 | $\left( (A + C^T C) \ (B + C^T) \right)$            | $(A + C^T C) (B + C^T)$                             | (j) $(A + C^T C) (B + C^T)$                                                                          |
| johnston-linear-matrix-algebra_F00527 | 49   | 0.830 | $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ | $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ | **1.3.5 Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ . Compute $A^{1000}$ .               |
| johnston-linear-matrix-algebra_F00528 | 49   | 0.830 | $A^{1000}$                                          | $A^{1000}$                                          | **1.3.5 Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ . Compute $A^{1000}$ .               |
| johnston-linear-matrix-algebra_F00529 | 49   | 1.000 | $A^2, A^3, \dots$                                   | $A^2, A^3, \dots$                                   | [Hint: Start by computing $A^2, A^3, \dots$ , and see if you notice                                  |
| johnston-linear-matrix-algebra_F00530 | 49   | 1.000 | $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ | $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ | 1.3.6 Let $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ . Compute $B^{1000}$ .                 |
| johnston-linear-matrix-algebra_F00531 | 49   | 1.000 | $B^{1000}$                                          | $B^{1000}$                                          | 1.3.6 Let $B = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ . Compute $B^{1000}$ .                 |

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| johnston-linear-matrix-algebra_F00532 | 49   | ■ 1.000 | $B^{\{2\}}, B^{\{3\}}, \ldots$                                                                                                               | $B^2, B^3, \dots$                                                                                     | [Hint: Start by computing $B^2, B^3, \dots$ , and see if you notice                                        |
| johnston-linear-matrix-algebra_F00533 | 49   | ■ 0.972 | $A=\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]$                                                                                  | $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$                                                    | *1.3.7 Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ .                                            |
| johnston-linear-matrix-algebra_F00534 | 49   | ■ 1.000 | $A^{\{3\}}$                                                                                                                                  | $A^3$                                                                                                 | (b) Compute $A^3$ .                                                                                        |
| johnston-linear-matrix-algebra_F00535 | 49   | ■ 1.000 | $A^{\{k\}}$                                                                                                                                  | $A^k$                                                                                                 | (c) Find a general formula for $A^k$ (where $k \geq 0$ is an                                               |
| johnston-linear-matrix-algebra_F00536 | 49   | ■ 1.000 | $k \geq 0$                                                                                                                                   | $k \geq 0$                                                                                            | (c) Find a general formula for $A^k$ (where $k \geq 0$ is an                                               |
| johnston-linear-matrix-algebra_F00537 | 49   | ■ 1.000 | $B=\left[\begin{array}{lll}1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1\end{array}\right]$                                                            | $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$                               | 1.3.8 Let $B = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ .                        |
| johnston-linear-matrix-algebra_F00538 | 49   | ■ 1.000 | $B^{\{3\}}$                                                                                                                                  | $B^3$                                                                                                 | (b) Compute $B^3$ .                                                                                        |
| johnston-linear-matrix-algebra_F00539 | 49   | ■ 1.000 | $B^{\{k\}}$                                                                                                                                  | $B^k$                                                                                                 | (c) Find a general formula for $B^k$ (where $k \geq 0$ is an                                               |
| johnston-linear-matrix-algebra_F00540 | 49   | ■ 1.000 | $C=\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$                                                                                | $C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$                                                  | 1.3.9 Let $C = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ .                                           |
| johnston-linear-matrix-algebra_F00541 | 49   | ■ 1.000 | $C^{\{2\}}$                                                                                                                                  | $C^2$                                                                                                 | (a) Compute $C^2$ .                                                                                        |
| johnston-linear-matrix-algebra_F00542 | 49   | ■ 1.000 | $C^{\{3\}}$                                                                                                                                  | $C^3$                                                                                                 | (b) Compute $C^3$ .                                                                                        |
| johnston-linear-matrix-algebra_F00543 | 49   | ■ 1.000 | $C^{\{k\}}$                                                                                                                                  | $C^k$                                                                                                 | (c) Find a general formula for $C^k$ (where $k \geq 0$ is an                                               |
| johnston-linear-matrix-algebra_F00544 | 49   | ■ 0.342 | $h \in \mathbb{R}$                                                                                                                           | $h \in \mathbb{R}$                                                                                    | 1.3.10 Let $h \in \mathbb{R}$ and                                                                          |
| johnston-linear-matrix-algebra_F00545 | 49   | ■ 1.000 | $A^{\{2500\}}$                                                                                                                               | $A^{2500}$                                                                                            | Use computer software to compute $A^{2500}$ when $h = 0.39$ ,                                              |
| johnston-linear-matrix-algebra_F00546 | 49   | ■ 1.000 | $h=0.39$                                                                                                                                     | $h = 0.39$                                                                                            | Use computer software to compute $A^{2500}$ when $h = 0.39$ ,                                              |
| johnston-linear-matrix-algebra_F00547 | 49   | ■ 1.000 | $h=0.40$                                                                                                                                     | $h = 0.40$                                                                                            | $h = 0.40$ , and $h = 0.41$ . What do you think happens as                                                 |
| johnston-linear-matrix-algebra_F00548 | 49   | ■ 1.000 | $h=0.41$                                                                                                                                     | $h = 0.41$                                                                                            | $h = 0.40$ , and $h = 0.41$ . What do you think happens as                                                 |
| johnston-linear-matrix-algebra_F00549 | 49   | ■ 0.948 | $J_{\{n\}}$                                                                                                                                  | $J_n$                                                                                                 | *1.3.11 Suppose $J_n$ is the $n \times n$ matrix all of whose entries                                      |
| johnston-linear-matrix-algebra_F00550 | 49   | ■ 1.000 | $J_{\{n\}}^{\{2\}}$                                                                                                                          | $J_n^2$                                                                                               | Compute $J_n^2$ .                                                                                          |
| johnston-linear-matrix-algebra_F00551 | 49   | ■ 1.000 | $A \in \mathcal{M}_{\{2,2\}}, B \in \mathcal{M}_{\{2,3\}}$                                                                                   | $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$                                                    | 1.3.12 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$ , and $C \in \mathcal{M}_{3,4}$ are      |
| johnston-linear-matrix-algebra_F00552 | 49   | ■ 1.000 | $C \in \mathcal{M}_{\{3,4\}}$                                                                                                                | $C \in \mathcal{M}_{3,4}$                                                                             | 1.3.12 Suppose $A \in \mathcal{M}_{2,2}, B \in \mathcal{M}_{2,3}$ , and $C \in \mathcal{M}_{3,4}$ are      |
| johnston-linear-matrix-algebra_F00553 | 49   | ■ 0.885 | $\left[\begin{array}{cc}A & A \\ A & A\end{array}\right]^{\{2\}}$                                                                            | $\begin{bmatrix} A & A \\ A & A \end{bmatrix}^2$                                                      | *(a) $\begin{bmatrix} A & A \\ A & A \end{bmatrix}^2$                                                      |
| johnston-linear-matrix-algebra_F00554 | 49   | ■ 0.999 | $\left[\begin{array}{cc}C & I_{\{3\}} \\ 0 & I_{\{3\}}\end{array}\right]\left[\begin{array}{cc}0 & I_{\{3\}} \\ I_4 & C^T\end{array}\right]$ | $\begin{bmatrix} C & I_3 \\ I_4 & O \end{bmatrix} \begin{bmatrix} O & I_3 \\ I_4 & C^T \end{bmatrix}$ | (b) $\begin{bmatrix} C & I_3 \\ I_4 & O \end{bmatrix} \begin{bmatrix} O & I_3 \\ I_4 & C^T \end{bmatrix}$  |
| johnston-linear-matrix-algebra_F00555 | 49   | ■ 1.000 | $\left[\begin{array}{cc}A & B \\ B^T & I_{\{3\}}\end{array}\right]\left[\begin{array}{cc}0 & C^T \\ B^T & I_3\end{array}\right]$             | $\begin{bmatrix} A & B \\ B^T & I_3 \end{bmatrix} \begin{bmatrix} O & C^T \\ B^T & I_3 \end{bmatrix}$ | *(c) $\begin{bmatrix} A & B \\ B^T & I_3 \end{bmatrix} \begin{bmatrix} O & C^T \\ B^T & I_3 \end{bmatrix}$ |

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| johnston-linear-matrix-algebra_F00556 | 49   | ■ 1.000 | $\left[\begin{array}{cc} A & B \\ 0 & C^T \end{array}\right] \left[\begin{array}{cc} I_2 & B \\ B^T & O \end{array}\right]$                                                                                                        | $\begin{bmatrix} A & B \\ O & C^T \end{bmatrix} \begin{bmatrix} I_2 & B \\ B^T & O \end{bmatrix}$                                                                                                                                                          | (d) $\begin{bmatrix} A & B \\ O & C^T \end{bmatrix} \begin{bmatrix} I_2 & B \\ B^T & O \end{bmatrix}$                                                                                                                                          |
| johnston-linear-matrix-algebra_F00557 | 49   | ■ 1.000 | $A=\left[\begin{array}{ll ll} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 2 & 1 \end{array}\right], B=\left[\begin{array}{ll ll} 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 2 \\ \hline 2 & 1 & 2 & 1 \end{array}\right]$                   | $A = \left[ \begin{array}{cc cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 2 & 1 \end{array} \right], B = \left[ \begin{array}{cc cc} 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ \hline 1 & 2 & 2 & 1 \end{array} \right]$                                   | *(a) $A = \left[ \begin{array}{cc cc} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 2 & 1 & 1 & 2 \end{array} \right], B = \left[ \begin{array}{cc cc} 1 & 2 & 2 & 1 \\ 2 & 1 & 1 & 2 \\ \hline 1 & 2 & 2 & 1 \end{array} \right]$                  |
| johnston-linear-matrix-algebra_F00558 | 49   | ■ 0.999 | $A=\left[\begin{array}{ll l} 1 & 2 & 0 \\ \hline 0 & 0 & 2 \end{array}\right], B=\left[\begin{array}{cc cc} -1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 \end{array}\right]$                                            | $A = \left[ \begin{array}{cc c} 1 & 2 & 0 \\ 3 & 1 & 0 \\ \hline 0 & 0 & 2 \end{array} \right], B = \left[ \begin{array}{cc cc} -1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 \end{array} \right]$                                               | (b) $A = \left[ \begin{array}{cc c} 1 & 2 & 0 \\ 3 & 1 & 0 \\ \hline 0 & 0 & 2 \end{array} \right], B = \left[ \begin{array}{cc cc} -1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 \end{array} \right]$                               |
| johnston-linear-matrix-algebra_F00559 | 49   | ■ 1.000 | $A=\left[\begin{array}{llll l} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 2 \end{array}\right], B=\left[\begin{array}{ll ll} 1 & 2 & 3 & 4 \\ 2 & 3 & 2 & 3 \\ \hline 3 & 4 & 5 & 3 \end{array}\right]$ | $A = \left[ \begin{array}{cccc c} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ \hline 0 & 0 & 0 & 2 & 2 \end{array} \right], B = \left[ \begin{array}{cc cc} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ \hline 1 & 2 & 3 & 3 \end{array} \right]$ | *(c) $A = \left[ \begin{array}{ccc c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ \hline 0 & 0 & 0 & 2 \end{array} \right], B = \left[ \begin{array}{ccc c} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 5 \\ \hline 1 & 2 & 3 & 3 \end{array} \right]$ |
| johnston-linear-matrix-algebra_F00560 | 50   | ■ 0.991 | $7 \times 7$                                                                                                                                                                                                                       | $7 \times 7$                                                                                                                                                                                                                                               | 1.3.14 Consider the $7 \times 7$ matrix                                                                                                                                                                                                        |
| johnston-linear-matrix-algebra_F00561 | 50   | ■ 0.999 | $B \in \mathcal{M}_{r,p}$                                                                                                                                                                                                          | $B \in \mathcal{M}_{r,p}$                                                                                                                                                                                                                                  | **1.3.15 Suppose $A \in \mathcal{M}_{m,n}$ and $B \in \mathcal{M}_{r,p}$ are matrices.                                                                                                                                                         |
| johnston-linear-matrix-algebra_F00562 | 50   | ■ 1.000 | $m, n, r$                                                                                                                                                                                                                          | $m, n, r$                                                                                                                                                                                                                                                  | (a) What restrictions must be placed on $m, n, r$ , and $p$ to                                                                                                                                                                                 |
| johnston-linear-matrix-algebra_F00563 | 50   | ■ 1.000 | $p$                                                                                                                                                                                                                                | $p$                                                                                                                                                                                                                                                        | (a) What restrictions must be placed on $m, n, r$ , and $p$ to                                                                                                                                                                                 |
| johnston-linear-matrix-algebra_F00564 | 50   | ■ 1.000 | $\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$                                                                                                                                                                            | $\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$                                                                                                                                                                                                    | 1.3.16 Let $A, B \in \mathcal{M}_{m,n}$ . Show that $\mathbf{v}^T A \mathbf{w} = \mathbf{v}^T B \mathbf{w}$ for                                                                                                                                |
| johnston-linear-matrix-algebra_F00565 | 50   | ■ 1.000 | $\mathbf{v} \in \mathbb{R}^m$                                                                                                                                                                                                      | $\mathbf{v} \in \mathbb{R}^m$                                                                                                                                                                                                                              | all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$ .                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00566 | 50   | ■ 1.000 | $\mathbf{w} \in \mathbb{R}^n$                                                                                                                                                                                                      | $\mathbf{w} \in \mathbb{R}^n$                                                                                                                                                                                                                              | all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$ .                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00567 | 50   | ■ 1.000 | $A=B$                                                                                                                                                                                                                              | $A = B$                                                                                                                                                                                                                                                    | all $\mathbf{v} \in \mathbb{R}^m$ and $\mathbf{w} \in \mathbb{R}^n$ if and only if $A = B$ .                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00568 | 50   | ■ 1.000 | $\mathbf{e}_i^T A \mathbf{e}_j$                                                                                                                                                                                                    | $\mathbf{e}_i^T A \mathbf{e}_j$                                                                                                                                                                                                                            | [Hint: What is $\mathbf{e}_i^T A \mathbf{e}_j$ ?]                                                                                                                                                                                              |
| johnston-linear-matrix-algebra_F00569 | 50   | ■ 1.000 | $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$                                                                                                                                                              | $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$                                                                                                                                                                                      | (a) Show that $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and                                                                                                                  |
| johnston-linear-matrix-algebra_F00570 | 50   | ■ 1.000 | $\mathbf{x} \in \mathbb{R}^m$                                                                                                                                                                                                      | $\mathbf{x} \in \mathbb{R}^m$                                                                                                                                                                                                                              | (a) Show that $\mathbf{x} \cdot (A \mathbf{y}) = (A^T \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and                                                                                                                  |
| johnston-linear-matrix-algebra_F00571 | 50   | ■ 1.000 | $\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$                                                                                                                                                                | $\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$                                                                                                                                                                                        | (b) Show that if $\mathbf{x} \cdot (A \mathbf{y}) = (B \mathbf{x}) \cdot \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and                                                                                                                 |
| johnston-linear-matrix-algebra_F00572 | 50   | ■ 1.000 | $B=A^T$                                                                                                                                                                                                                            | $B = A^T$                                                                                                                                                                                                                                                  | $\mathbf{y} \in \mathbb{R}^n$ then $B = A^T$ .                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra_F00573 | 50   | ■ 0.919 | $\mathbf{e}_i$                                                                                                                                                                                                                     | $\mathbf{e}_i$                                                                                                                                                                                                                                             | **1.3.18 Let $\mathbf{e}_i$ be the standard basis vector with 1 in its                                                                                                                                                                         |
| johnston-linear-matrix-algebra_F00574 | 50   | ■ 1.000 | $A \mathbf{e}_i$                                                                                                                                                                                                                   | $A \mathbf{e}_i$                                                                                                                                                                                                                                           | (a) Show that $A \mathbf{e}_i$ is the $i$ -th column of $A$ .                                                                                                                                                                                  |
| johnston-linear-matrix-algebra_F00575 | 50   | ■ 1.000 | $\mathbf{e}_i^T A$                                                                                                                                                                                                                 | $\mathbf{e}_i^T A$                                                                                                                                                                                                                                         | (b) Show that $\mathbf{e}_i^T A$ is the $i$ -th row of $A$ .                                                                                                                                                                                   |
| johnston-linear-matrix-algebra_F00576 | 50   | ■ 1.000 | $A \mathbf{v} = B \mathbf{v}$                                                                                                                                                                                                      | $A \mathbf{v} = B \mathbf{v}$                                                                                                                                                                                                                              | (d) Use part (a) to show that $A \mathbf{v} = B \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ if                                                                                                                                           |
| johnston-linear-matrix-algebra_F00577 | 50   | ■ 1.000 | $I_m A = A$                                                                                                                                                                                                                        | $I_m A = A$                                                                                                                                                                                                                                                | That is, show that if $A \in \mathcal{M}_{m,n}$ then $I_m A = A$ .                                                                                                                                                                             |

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| johnston-linear-matrix-algebra-F00578 | 50   | ■ 1.000 | $A \in \mathcal{M}_n$                                                                              | $A \in \mathcal{M}_n$                                        | <b>**1.3.22</b> Suppose $A \in \mathcal{M}_n$ and $k$ and $\ell$ are non-negative                                                                                 |
| johnston-linear-matrix-algebra-F00579 | 50   | ■ 1.000 | $\ell$                                                                                             | $\ell$                                                       | <b>**1.3.22</b> Suppose $A \in \mathcal{M}_n$ and $k$ and $\ell$ are non-negative                                                                                 |
| johnston-linear-matrix-algebra-F00580 | 50   | ■ 1.000 | $A_{\{1\}}, A_{\{2\}}, \ldots, A_{\{k\}}$                                                          | $A_1, A_2, \dots, A_k$                                       | <b>1.3.24</b> Suppose that $A_1, A_2, \dots, A_k$ are matrices whose                                                                                              |
| johnston-linear-matrix-algebra-F00581 | 50   | ■ 1.000 | $A_{\{1\}} A_{\{2\}} \ldots A_{\{k\}}$                                                             | $A_1 A_2 \dots A_k$                                          | sizes are such that the product $A_1 A_2 \dots A_k$ makes sense.                                                                                                  |
| johnston-linear-matrix-algebra-F00582 | 50   | ■ 1.000 | $\left(A_{\{1\}} A_{\{2\}} \cdots A_{\{k\}}\right)^T = A_{\{k\}}^T \cdots A_{\{2\}}^T A_{\{1\}}^T$ | $(A_1 A_2 \dots A_k)^T = A_k^T \dots A_2^T A_1^T$            | Prove that $(A_1 A_2 \dots A_k)^T = A_k^T \dots A_2^T A_1^T$ .                                                                                                    |
| johnston-linear-matrix-algebra-F00583 | 50   | ■ 1.000 | $\left(A^{\{n\}}\right)^T = \left(A^T\right)^{\{n\}}$                                              | $(A^n)^T = (A^T)^n$                                          | <b>1.3.25</b> Suppose that $A \in \mathcal{M}_n$ . Show that $(A^n)^T = (A^T)^n$                                                                                  |
| johnston-linear-matrix-algebra-F00584 | 50   | ■ 1.000 | $n \geq 0$                                                                                         | $n \geq 0$                                                   | for all integers $n \geq 0$ .                                                                                                                                     |
| johnston-linear-matrix-algebra-F00585 | 50   | ■ 1.000 | $\mathbf{a}_{\{j\}}^T$                                                                             | $\mathbf{a}_j^T$                                             | and let $\mathbf{a}_j^T$ be the $j$ -th row of $A$ . Show that                                                                                                    |
| johnston-linear-matrix-algebra-F00586 | 51   | ■ 1.000 | $T: \mathbb{R}^{\{n\}} \rightarrow \mathbb{R}^{\{m\}}$                                             | $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                   | $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ means                                                                                                                  |
| johnston-linear-matrix-algebra-F00587 | 51   | ■ 1.000 | $T$                                                                                                | $T$                                                          | that $T$ is a function                                                                                                                                            |
| johnston-linear-matrix-algebra-F00588 | 51   | ■ 1.000 | $\mathbb{R}^{\{m\}}$                                                                               | $\mathbb{R}^m$                                               | $\mathbb{R}^m$ . More generally,                                                                                                                                  |
| johnston-linear-matrix-algebra-F00589 | 51   | ■ 1.000 | $f: X \rightarrow Y$                                                                               | $f: X \rightarrow Y$                                         | $f: X \rightarrow Y$ means                                                                                                                                        |
| johnston-linear-matrix-algebra-F00590 | 51   | ■ 1.000 | $X$                                                                                                | $X$                                                          | the set $X$ to the set $Y$ .                                                                                                                                      |
| johnston-linear-matrix-algebra-F00591 | 51   | ■ 1.000 | $Y$                                                                                                | $Y$                                                          | the set $X$ to the set $Y$ .                                                                                                                                      |
| johnston-linear-matrix-algebra-F00592 | 51   | ■ 1.000 | $T(\mathbf{0}) = \mathbf{0}$                                                                       | $T(\mathbf{0}) = \mathbf{0}$                                 | case that $T(\mathbf{0}) = \mathbf{0}$                                                                                                                            |
| johnston-linear-matrix-algebra-F00593 | 51   | ■ 1.000 | $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$                                       | $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ | a) $T(\mathbf{v} + \mathbf{w}) = T(\mathbf{v}) + T(\mathbf{w})$ for all vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ , and                                   |
| johnston-linear-matrix-algebra-F00594 | 51   | ■ 1.000 | $T(c \mathbf{v}) = c T(\mathbf{v})$                                                                | $T(c\mathbf{v}) = cT(\mathbf{v})$                            | b) $T(c\mathbf{v}) = cT(\mathbf{v})$ for all vectors $\mathbf{v} \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$ .                                           |
| johnston-linear-matrix-algebra-F00595 | 51   | ■ 1.000 | $A \mathbf{v} \in \mathbb{R}^{\{m\}}$                                                              | $A\mathbf{v} \in \mathbb{R}^m$                               | sends $\mathbf{v} \in \mathbb{R}^n$ to $A\mathbf{v} \in \mathbb{R}^m$ preserves vector addition (i.e., $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$ ) |
| johnston-linear-matrix-algebra-F00596 | 51   | ■ 1.000 | $A(\mathbf{v} + \mathbf{w}) = A \mathbf{v} + A \mathbf{w}$                                         | $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$     | sends $\mathbf{v} \in \mathbb{R}^n$ to $A\mathbf{v} \in \mathbb{R}^m$ preserves vector addition (i.e., $A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w}$ ) |
| johnston-linear-matrix-algebra-F00597 | 51   | ■ 0.999 | $A(c \mathbf{v}) = c(A \mathbf{v})$                                                                | $A(c\mathbf{v}) = c(A\mathbf{v})$                            | and scalar multiplication (i.e., $A(c\mathbf{v}) = c(A\mathbf{v})$ ).                                                                                             |
| johnston-linear-matrix-algebra-F00598 | 51   | ■ 1.000 | $T: \mathbb{R}^{\{2\}} \rightarrow \mathbb{R}^{\{2\}}$                                             | $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$                   | transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ can rotate, stretch, shrink, and/or reflect that grid,                                                  |
| johnston-linear-matrix-algebra-F00599 | 51   | ■ 1.000 | $T(\mathbf{v})$                                                                                    | $T(\mathbf{v})$                                              | then $T(\mathbf{v})$ is located in the $x$ -th cell in the direction of $T(\mathbf{e}_1)$ and the $y$ -th cell                                                    |
| johnston-linear-matrix-algebra-F00600 | 51   | ■ 1.000 | $T(\mathbf{e}_{\{1\}})$                                                                            | $T(\mathbf{e}_1)$                                            | then $T(\mathbf{v})$ is located in the $x$ -th cell in the direction of $T(\mathbf{e}_1)$ and the $y$ -th cell                                                    |
| johnston-linear-matrix-algebra-F00601 | 51   | ■ 1.000 | $T(\mathbf{e}_{\{2\}})$                                                                            | $T(\mathbf{e}_2)$                                            | in the direction of $T(\mathbf{e}_2)$ (again, see Figure 1.14).                                                                                                   |
| johnston-linear-matrix-algebra-F00602 | 51   | ■ 1.000 | $T(v_{\{1\}}, v_{\{2\}}) = (1 + v_{\{1\}}, 2 + v_{\{2\}})$                                         | $T(v_1, v_2) = (1 + v_1, 2 + v_2)$                           | a) $T(v_1, v_2) = (1 + v_1, 2 + v_2)$ ,                                                                                                                           |
| johnston-linear-matrix-algebra-F00603 | 51   | ■ 1.000 | $T(v_{\{1\}}, v_{\{2\}}) = (v_{\{1\}} - v_{\{2\}}, v_{\{1\}} v_{\{2\}})$                           | $T(v_1, v_2) = (v_1 - v_2, v_1 v_2)$                         | b) $T(v_1, v_2) = (v_1 - v_2, v_1 v_2)$ , and                                                                                                                     |
| johnston-linear-matrix-algebra-F00604 | 51   | ■ 1.000 | $T(v_{\{1\}}, v_{\{2\}}) = (v_{\{1\}} - v_{\{2\}}, v_{\{1\}} + v_{\{2\}})$                         | $T(v_1, v_2) = (v_1 - v_2, v_1 + v_2)$                       | c) $T(v_1, v_2) = (v_1 - v_2, v_1 + v_2)$ .                                                                                                                       |
| johnston-linear-matrix-algebra-F00605 | 51   | ■ 1.000 | $2 T(0, 0) = 2(1, 2) = (2, 4)$                                                                     | $2T(0, 0) = 2(1, 2) = (2, 4)$                                | that $2T(0, 0) = 2(1, 2) = (2, 4)$ , but $T(2(0, 0)) = T(0, 0) = (1, 2)$ .                                                                                        |

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| johnston-linear-matrix-algebra-F00606 | 51   | ■ 1.000 | $T(2(0,0))=T(0,0)=(1,2)$                                   | $T(2(0,0)) = T(0,0) = (1,2)$                                      | that $2T(0,0) = 2(1,2) = (2,4)$ , but $T(2(0,0)) = T(0,0) = (1,2)$ .                                                                                             |
| johnston-linear-matrix-algebra-F00607 | 52   | ■ 1.000 | $2\ T(1,1)=2(0,1)=(0,2)$                                   | $2T(1,1) = 2(0,1) = (0,2)$                                        | that $2T(1,1) = 2(0,1) = (0,2)$ , but $T(2(1,1)) = T(2,2) = (0,4)$ .                                                                                             |
| johnston-linear-matrix-algebra-F00608 | 52   | ■ 1.000 | $T(2(1,1))=T(2,2)=(0,4)$                                   | $T(2(1,1)) = T(2,2) = (0,4)$                                      | that $2T(1,1) = 2(0,1) = (0,2)$ , but $T(2(1,1)) = T(2,2) = (0,4)$ .                                                                                             |
| johnston-linear-matrix-algebra-F00609 | 52   | ■ 1.000 | $\sqrt{2}$                                                 | $\sqrt{2}$                                                        | factor of $\sqrt{2}$ :                                                                                                                                           |
| johnston-linear-matrix-algebra-F00610 | 53   | ■ 1.000 | $\mathbf{v}=v_1\ \mathbf{e}_1+v_2\ \mathbf{e}_2+\cdots+$   | $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots +$       | for every vector $\mathbf{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ , we can write $\mathbf{v} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + \cdots +$               |
| johnston-linear-matrix-algebra-F00611 | 53   | ■ 1.000 | $v_n\ \mathbf{e}_n$                                        | $v_n\mathbf{e}_n$                                                 | $v_n\mathbf{e}_n$ , where $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are the standard basis vectors. By using the fact that                               |
| johnston-linear-matrix-algebra-F00612 | 53   | ■ 1.000 | $\mathbf{v} \in \mathbb{R}^2$                              | $\mathbf{v} \in \mathbb{R}^2$                                     | which is exactly what we said before: if $\mathbf{v} \in \mathbb{R}^2$ extends a distance of $v_1$ in                                                            |
| johnston-linear-matrix-algebra-F00613 | 53   | ■ 1.000 | $v_1$                                                      | $v_1$                                                             | which is exactly what we said before: if $\mathbf{v} \in \mathbb{R}^2$ extends a distance of $v_1$ in                                                            |
| johnston-linear-matrix-algebra-F00614 | 53   | ■ 1.000 | $v_2$                                                      | $v_2$                                                             | the direction of $\mathbf{e}_1$ and a distance of $v_2$ in the direction of $\mathbf{e}_2$ , then $T(\mathbf{v})$ ex-                                            |
| johnston-linear-matrix-algebra-F00615 | 53   | ■ 1.000 | $\mathbf{e}_3, \dots, \mathbf{e}_n$                        | $\mathbf{e}_3, \dots, \mathbf{e}_n$                               | similarly for $\mathbf{e}_3, \dots, \mathbf{e}_n$ in higher dimensions). We thus learn the very impor-                                                           |
| johnston-linear-matrix-algebra-F00616 | 53   | ■ 1.000 | $T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$ | $T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$        | $T(\mathbf{e}_1), T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$ : if we know what $T$ does to the standard basis vectors,                                             |
| johnston-linear-matrix-algebra-F00617 | 53   | ■ 1.000 | $T(\mathbf{e}_1)=(1,1)$                                    | $T(\mathbf{e}_1) = (1,1)$                                         | Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation for which $T(\mathbf{e}_1) = (1,1)$                                               |
| johnston-linear-matrix-algebra-F00618 | 53   | ■ 1.000 | $T(\mathbf{e}_2)=(-1,1)$                                   | $T(\mathbf{e}_2) = (-1,1)$                                        | and $T(\mathbf{e}_2) = (-1,1)$ .                                                                                                                                 |
| johnston-linear-matrix-algebra-F00619 | 53   | ■ 1.000 | $T(2,3)$                                                   | $T(2,3)$                                                          | a) Compute $T(2,3)$ , and                                                                                                                                        |
| johnston-linear-matrix-algebra-F00620 | 53   | ■ 1.000 | $T(v_1, v_2)$                                              | $T(v_1, v_2)$                                                     | b) Find a general formula for $T(v_1, v_2)$ .                                                                                                                    |
| johnston-linear-matrix-algebra-F00621 | 53   | ■ 1.000 | $(2,3)=2\ \mathbf{e}_1+3\ \mathbf{e}_2$                    | $(2,3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$                           | a) Since $(2,3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$ , we know that $T(2,3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$                                                    |
| johnston-linear-matrix-algebra-F00622 | 53   | ■ 1.000 | $T(2,3)=T(2\ \mathbf{e}_1+3\ \mathbf{e}_2)=$               | $T(2,3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$                     | a) Since $(2,3) = 2\mathbf{e}_1 + 3\mathbf{e}_2$ , we know that $T(2,3) = T(2\mathbf{e}_1 + 3\mathbf{e}_2) =$                                                    |
| johnston-linear-matrix-algebra-F00623 | 53   | ■ 1.000 | $2\ T(1,1)+3\ T(-1,1)=(-1,5)$                              | $2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = 2(1,1) + 3(-1,1) = (-1,5)$ | $2T(\mathbf{e}_1) + 3T(\mathbf{e}_2) = 2(1,1) + 3(-1,1) = (-1,5)$ .                                                                                              |
| johnston-linear-matrix-algebra-F00624 | 53   | ■ 1.000 | $(v_1, v_2)=v_1\ \mathbf{e}_1+v_2\ \mathbf{e}_2$           | $(v_1, v_2) = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$                  | b) We can mimic the computation from part (a): $(v_1, v_2) = v_1\mathbf{e}_1 + v_2\mathbf{e}_2$ ,                                                                |
| johnston-linear-matrix-algebra-F00625 | 54   | ■ 1.000 | $[T]$                                                      | $[T]$                                                             | In other words, $[T]$                                                                                                                                            |
| johnston-linear-matrix-algebra-F00626 | 54   | ■ 1.000 | $T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$                  | $T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$                         | $T(\mathbf{e}_2), \dots, T(\mathbf{e}_n)$ as its                                                                                                                 |
| johnston-linear-matrix-algebra-F00627 | 54   | ■ 1.000 | $T(v_1, v_2)=v_1-$                                         | $T(v_1, v_2) = (v_1 -$                                            | effect as the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(v_1, v_2) = (v_1 -$                                                |
| johnston-linear-matrix-algebra-F00628 | 54   | ■ 0.951 | $v_2, v_1+v_2)$                                            | $v_2, v_1 + v_2)$                                                 | $v_2, v_1 + v_2)$ (i.e., the linear transformation from Examples 1.4.1(c) and 1.4.2(b)).                                                                         |
| johnston-linear-matrix-algebra-F00629 | 54   | ■ 1.000 | $[T] \in \mathcal{M}_{m,n}$                                | $[T] \in \mathcal{M}_{m,n}$                                       | exists a matrix $[T] \in \mathcal{M}_{m,n}$ such that                                                                                                            |
| johnston-linear-matrix-algebra-F00630 | 54   | ■ 0.623 | $[T]\mathbf{v}$                                            | $[T]\mathbf{v}$                                                   | function that sends $\mathbf{v}$ to $[T]\mathbf{v}$ is a linear transformation. We thus only have to                                                             |
| johnston-linear-matrix-algebra-F00631 | 54   | ■ 1.000 | $[T]\mathbf{v}=T(\mathbf{v})$                              | $[T]\mathbf{v} = T(\mathbf{v})$                                   | satisfies $[T]\mathbf{v} = T(\mathbf{v})$ , and no other matrix has this property.                                                                               |
| johnston-linear-matrix-algebra-F00632 | 54   | ■ 1.000 | $T(\mathbf{v})=A\ \mathbf{v}$                              | $T(\mathbf{v}) = A\mathbf{v}$                                     | the property that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ . Then $T(\mathbf{v}) = [T]\mathbf{v}$ and $T(\mathbf{v}) = A\mathbf{v}$ , |

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| johnston-linear-matrix-algebra-F00633 | 54   | 1.000 | $T(\mathbf{v})=[T] \mathbf{v}$                                              | $T(\mathbf{v}) = [T]\mathbf{v}$                                             | the property that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ . Then $T(\mathbf{v}) = [T]\mathbf{v}$ and $T(\mathbf{v}) = A\mathbf{v}$ ,           |
| johnston-linear-matrix-algebra-F00634 | 54   | 1.000 | $[T] \mathbf{v}=A \mathbf{v}$                                               | $[T]\mathbf{v} = A\mathbf{v}$                                               | so $[T]\mathbf{v} = A\mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ as well. It follows from Exercise 1.3.18(d) that                                                    |
| johnston-linear-matrix-algebra-F00635 | 54   | 1.000 | $A=[T]$                                                                     | $A = [T]$                                                                   | $A = [T]$ , which completes the proof.                                                                                                                                     |
| johnston-linear-matrix-algebra-F00636 | 55   | 1.000 | $T(v_1, v_2)=\left(v_1+2 v_2, 3 v_1+4 v_2\right)$                           | $T\left(v_1, v_2\right)=\left(v_1+2 v_2, 3 v_1+4 v_2\right)$                | a) $T\left(v_1, v_2\right)=\left(v_1+2 v_2, 3 v_1+4 v_2\right)$ , and                                                                                                      |
| johnston-linear-matrix-algebra-F00637 | 55   | 1.000 | $T\left(v_1, v_2, v_3\right)=\left(3 v_1-v_2+v_3, 2 v_1+4 v_2-2 v_3\right)$ | $T\left(v_1, v_2, v_3\right)=\left(3 v_1-v_2+v_3, 2 v_1+4 v_2-2 v_3\right)$ | b) $T\left(v_1, v_2, v_3\right)=\left(3 v_1-v_2+v_3, 2 v_1+4 v_2-2 v_3\right)$ .                                                                                           |
| johnston-linear-matrix-algebra-F00638 | 55   | 1.000 | $T\left(\mathbf{e}_1\right)=(1,3)$                                          | $T\left(\mathbf{e}_1\right)=(1,3)$                                          | a) We use Theorem 1.4.1, which tells us to compute $T\left(\mathbf{e}_1\right)=(1,3)$                                                                                      |
| johnston-linear-matrix-algebra-F00639 | 55   | 1.000 | $T\left(\mathbf{e}_2\right)=(2,4)$                                          | $T\left(\mathbf{e}_2\right)=(2,4)$                                          | and $T\left(\mathbf{e}_2\right)=(2,4)$ and place these as columns into a matrix, in that                                                                                   |
| johnston-linear-matrix-algebra-F00640 | 55   | 1.000 | $T\left(\mathbf{e}_1\right), T\left(\mathbf{e}_2\right)$                    | $T\left(\mathbf{e}_1\right), T\left(\mathbf{e}_2\right)$                    | b) We could explicitly compute $T\left(\mathbf{e}_1\right), T\left(\mathbf{e}_2\right)$ , and $T\left(\mathbf{e}_3\right)$ and place                                       |
| johnston-linear-matrix-algebra-F00641 | 55   | 1.000 | $T\left(\mathbf{e}_3\right)$                                                | $T\left(\mathbf{e}_3\right)$                                                | b) We could explicitly compute $T\left(\mathbf{e}_1\right), T\left(\mathbf{e}_2\right)$ , and $T\left(\mathbf{e}_3\right)$ and place                                       |
| johnston-linear-matrix-algebra-F00642 | 55   | 1.000 | $v_1, v_2$                                                                  | $v_1, v_2$                                                                  | simply place the coefficients of $v_1, v_2$ , and $v_3$ in the output of $T$ , in                                                                                          |
| johnston-linear-matrix-algebra-F00643 | 55   | 1.000 | $v_3$                                                                       | $v_3$                                                                       | simply place the coefficients of $v_1, v_2$ , and $v_3$ in the output of $T$ , in                                                                                          |
| johnston-linear-matrix-algebra-F00644 | 55   | 1.000 | $A=\left[\begin{array}{ll} 2.1 & 0.3 \\ 0.2 & 1.2 \end{array}\right]$       | $A=\left[\begin{array}{ll} 2.1 & 0.3 \\ 0.2 & 1.2 \end{array}\right]$       | a) $A=\left[\begin{array}{ll} 2.1 & 0.3 \\ 0.2 & 1.2 \end{array}\right]$                                                                                                   |
| johnston-linear-matrix-algebra-F00645 | 55   | 1.000 | $B=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{array}\right]$      | $B=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{array}\right]$      | b) $B=\left[\begin{array}{ll} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{array}\right]$                                                                                                  |
| johnston-linear-matrix-algebra-F00646 | 55   | 0.508 | $A \mathbf{e}_1$                                                            | $A\mathbf{e}_1$                                                             | a) Since $A\mathbf{e}_1$ (i.e., the first column of $A$ ) equals $(2.1, 0.2)$ and $A\mathbf{e}_2$ (i.e.,                                                                   |
| johnston-linear-matrix-algebra-F00647 | 55   | 0.508 | $A \mathbf{e}_2$                                                            | $A\mathbf{e}_2$                                                             | a) Since $A\mathbf{e}_1$ (i.e., the first column of $A$ ) equals $(2.1, 0.2)$ and $A\mathbf{e}_2$ (i.e.,                                                                   |
| johnston-linear-matrix-algebra-F00648 | 55   | 0.530 | $B \mathbf{e}_1$                                                            | $B\mathbf{e}_1$                                                             | b) Since $B\mathbf{e}_1$ (i.e., the first column of $B$ ) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,                                                                    |
| johnston-linear-matrix-algebra-F00649 | 55   | 0.530 | $2, 1, 1$                                                                   | $2, 1, 1$                                                                   | b) Since $B\mathbf{e}_1$ (i.e., the first column of $B$ ) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,                                                                    |
| johnston-linear-matrix-algebra-F00650 | 55   | 0.530 | $B \mathbf{e}_2$                                                            | $B\mathbf{e}_2$                                                             | b) Since $B\mathbf{e}_1$ (i.e., the first column of $B$ ) equals $(2, 1, 1)$ and $B\mathbf{e}_2$ (i.e.,                                                                    |
| johnston-linear-matrix-algebra-F00651 | 55   | 0.637 | $(1, 2, 1)$                                                                 | $(1, 2, 1)$                                                                 | its second column) equals $(1, 2, 1)$ , we can think of this matrix as                                                                                                     |
| johnston-linear-matrix-algebra-F00652 | 56   | 1.000 | $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                               | $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                               | $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S+T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are |
| johnston-linear-matrix-algebra-F00653 | 56   | 1.000 | $S+T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                | $S+T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                | $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S+T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are |
| johnston-linear-matrix-algebra-F00654 | 56   | 1.000 | $c T: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                | $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                 | $S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $c \in \mathbb{R}$ then $S+T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are |
| johnston-linear-matrix-algebra-F00655 | 56   | 1.000 | $S+T$                                                                       | $S+T$                                                                       | ticular, the standard matrix of $S+T$ is $[S+T]=[S]+[T]$ and similarly                                                                                                     |
| johnston-linear-matrix-algebra-F00656 | 56   | 1.000 | $[S+T]=[S]+[T]$                                                             | $[S+T]=[S]+[T]$                                                             | ticular, the standard matrix of $S+T$ is $[S+T]=[S]+[T]$ and similarly                                                                                                     |
| johnston-linear-matrix-algebra-F00657 | 56   | 1.000 | $c T$                                                                       | $cT$                                                                        | the standard matrix of $cT$ is $[cT]=c[T]$ (see Exercise 1.4.11), so we can                                                                                                |
| johnston-linear-matrix-algebra-F00658 | 56   | 1.000 | $[c T]=c[T]$                                                                | $[cT]=c[T]$                                                                 | the standard matrix of $cT$ is $[cT]=c[T]$ (see Exercise 1.4.11), so we can                                                                                                |

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| johnston-linear-matrix-algebra-F00659 | 56   | ■ 1.000 | $0: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                     | $O: \mathbb{R}^n \rightarrow \mathbb{R}^m$                                     | <b>mation</b> $O: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , defined by $O(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$ , and the <b>identity</b> |
| johnston-linear-matrix-algebra-F00660 | 56   | ■ 1.000 | $0(\mathbf{v})=\mathbf{0}$                                                     | $O(\mathbf{v}) = \mathbf{0}$                                                   | <b>mation</b> $O: \mathbb{R}^n \rightarrow \mathbb{R}^m$ , defined by $O(\mathbf{v}) = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$ , and the <b>identity</b> |
| johnston-linear-matrix-algebra-F00661 | 56   | ■ 1.000 | $I: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                     | $I: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                     | <b>transformation</b> $I: \mathbb{R}^n \rightarrow \mathbb{R}^n$ , defined by $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ . It is perhaps   |
| johnston-linear-matrix-algebra-F00662 | 56   | ■ 1.000 | $I(\mathbf{v})=\mathbf{v}$                                                     | $I(\mathbf{v}) = \mathbf{v}$                                                   | <b>transformation</b> $I: \mathbb{R}^n \rightarrow \mathbb{R}^n$ , defined by $I(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ . It is perhaps   |
| johnston-linear-matrix-algebra-F00663 | 56   | ■ 1.000 | $0 \in \mathcal{M}_{m,n}$                                                      | $O \in \mathcal{M}_{m,n}$                                                      | that if $O \in \mathcal{M}_{m,n}$ and $I \in \mathcal{M}_n$ are the zero matrix and the identity matrix, then                                                      |
| johnston-linear-matrix-algebra-F00664 | 56   | ■ 1.000 | $I \in \mathcal{M}_n$                                                          | $I \in \mathcal{M}_n$                                                          | that if $O \in \mathcal{M}_{m,n}$ and $I \in \mathcal{M}_n$ are the zero matrix and the identity matrix, then                                                      |
| johnston-linear-matrix-algebra-F00665 | 56   | ■ 1.000 | $0 \mathbf{v}=\mathbf{0}$                                                      | $O\mathbf{v} = \mathbf{0}$                                                     | $O\mathbf{v} = \mathbf{0}$ and $I\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ too.                                                               |
| johnston-linear-matrix-algebra-F00666 | 56   | ■ 1.000 | $I \mathbf{v}=\mathbf{v}$                                                      | $I\mathbf{v} = \mathbf{v}$                                                     | $O\mathbf{v} = \mathbf{0}$ and $I\mathbf{v} = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ too.                                                               |
| johnston-linear-matrix-algebra-F00667 | 56   | ■ 1.000 | $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                     | $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                     | The next simplest type of linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is one that does                                                        |
| johnston-linear-matrix-algebra-F00668 | 56   | ■ 1.000 | $c_1, c_2, \dots, c_n \in \mathbb{R}^n$                                        | $c_1, c_2, \dots, c_n \in \mathbb{R}^n$                                        | formations are the ones for which there exist scalars $c_1, c_2, \dots, c_n \in \mathbb{R}^n$ such                                                                 |
| johnston-linear-matrix-algebra-F00669 | 56   | ■ 0.995 | $T(v_1, v_2, \dots, v_n) = (c_1 v_1, c_2 v_2, \dots, c_n v_n)$                 | $T(v_1, v_2, \dots, v_n) = (c_1 v_1, c_2 v_2, \dots, c_n v_n)$                 | that $T(v_1, v_2, \dots, v_n) = (c_1 v_1, c_2 v_2, \dots, c_n v_n)$ .                                                                                              |
| johnston-linear-matrix-algebra-F00670 | 57   | ■ 1.000 | $DA$                                                                           | $DA$                                                                           | matrix, then the matrix multiplication $DA$ simply multiplies each row of $A$ by                                                                                   |
| johnston-linear-matrix-algebra-F00671 | 57   | ■ 1.000 | $AD$                                                                           | $AD$                                                                           | the corresponding diagonal entry of $D$ . Similarly, the product $AD$ multiplies                                                                                   |
| johnston-linear-matrix-algebra-F00672 | 57   | ■ 0.999 | $\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$                                     | $\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$                                     | the <b>projection onto the x-axis</b> sends every vector $\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2$ to $(v_1, 0)$                                                  |
| johnston-linear-matrix-algebra-F00673 | 58   | ■ 1.000 | $\mathbf{u}^T \mathbf{u}$                                                      | $\mathbf{u}^T \mathbf{u}$                                                      | Be careful: $\mathbf{u}^T \mathbf{u}$ is a                                                                                                                         |
| johnston-linear-matrix-algebra-F00674 | 58   | ■ 1.000 | $\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$                           | $\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$                           | $\mathbf{u} \cdot \mathbf{u} = \ \mathbf{u}\ ^2 = 1$ ), but                                                                                                        |
| johnston-linear-matrix-algebra-F00675 | 58   | ■ 1.000 | $\mathbf{u} \mathbf{u}^T$                                                      | $\mathbf{u} \mathbf{u}^T$                                                      | $\mathbf{u} \mathbf{u}^T$ is a matrix!                                                                                                                             |
| johnston-linear-matrix-algebra-F00676 | 58   | ■ 1.000 | $\mathbf{u}^T$                                                                 | $\mathbf{u}^T$                                                                 | $\mathbf{u}^T$ is $1 \times n$ , $\mathbf{u} \mathbf{u}^T$ is                                                                                                      |
| johnston-linear-matrix-algebra-F00677 | 58   | ■ 1.000 | $1 \times n, \mathbf{u} \mathbf{u}^T$                                          | $1 \times n, \mathbf{u} \mathbf{u}^T$                                          | $\mathbf{u}^T$ is $1 \times n$ , $\mathbf{u} \mathbf{u}^T$ is                                                                                                      |
| johnston-linear-matrix-algebra-F00678 | 58   | ■ 1.000 | $P_{\mathbf{u}}$                                                               | $P_{\mathbf{u}}$                                                               | think of unit vectors as specifying directions) and let $P_{\mathbf{u}}$ denote the projection                                                                     |
| johnston-linear-matrix-algebra-F00679 | 58   | ■ 0.999 | $P_{\mathbf{u}}(\mathbf{v})$                                                   | $P_{\mathbf{u}}(\mathbf{v})$                                                   | denoted by $P_{\mathbf{u}}(\mathbf{v})$ , and it points a distance of $\ \mathbf{v}\  \cos(\theta)$ in the direction of $\mathbf{u}$ , where                       |
| johnston-linear-matrix-algebra-F00680 | 58   | ■ 0.999 | $\ \mathbf{v}\  \cos(\theta)$                                                  | $\ \mathbf{v}\  \cos(\theta)$                                                  | denoted by $P_{\mathbf{u}}(\mathbf{v})$ , and it points a distance of $\ \mathbf{v}\  \cos(\theta)$ in the direction of $\mathbf{u}$ , where                       |
| johnston-linear-matrix-algebra-F00681 | 58   | ■ 1.000 | $P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\  \cos(\theta))$         | $P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\  \cos(\theta))$         | words, $P_{\mathbf{u}}(\mathbf{v}) = \mathbf{u}(\ \mathbf{v}\  \cos(\theta))$ .                                                                                    |
| johnston-linear-matrix-algebra-F00682 | 58   | ■ 1.000 | $\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\  \ \mathbf{v}\ )$ | $\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\  \ \mathbf{v}\ )$ | We then recall from Definition 1.2.3 that $\cos(\theta) = \mathbf{u} \cdot \mathbf{v} / (\ \mathbf{u}\  \ \mathbf{v}\ )$ . By                                      |
| johnston-linear-matrix-algebra-F00683 | 58   | ■ 1.000 | $\ \mathbf{u}\  = 1$                                                           | $\ \mathbf{u}\  = 1$                                                           | plugging this into our formula for $P_{\mathbf{u}}(\mathbf{v})$ and using the fact that $\ \mathbf{u}\  = 1$ , we                                                  |
| johnston-linear-matrix-algebra-F00684 | 58   | ■ 1.000 | $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$                        | $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$                        | Finally, if we recall from Section 1.3.3 that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$ , it follows that $P_{\mathbf{u}}(\mathbf{v}) =$             |
| johnston-linear-matrix-algebra-F00685 | 58   | ■ 1.000 | $P_{\mathbf{u}}(\mathbf{v}) =$                                                 | $P_{\mathbf{u}}(\mathbf{v}) =$                                                 | Finally, if we recall from Section 1.3.3 that $\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$ , it follows that $P_{\mathbf{u}}(\mathbf{v}) =$             |
| johnston-linear-matrix-algebra-F00686 | 58   | ■ 1.000 | $\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$   | $\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$   | $\mathbf{u}(\mathbf{u}^T \mathbf{v}) = (\mathbf{u} \mathbf{u}^T) \mathbf{v}$ . In other words, $P_{\mathbf{u}}$ is indeed a linear transformation, and it          |

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| johnston-linear-matrix-algebra_F00687 | 58   | 1.000 | $\mathbf{u}=(1,0) \in \mathbb{R}^2$                                                                 | $\mathbf{u} = (1, 0) \in \mathbb{R}^2$                                                 | a) $\mathbf{u} = (1, 0) \in \mathbb{R}^2$ , and                                                                                             |
| johnston-linear-matrix-algebra_F00688 | 58   | 1.000 | $\mathbf{w}=(1,2,3) \in \mathbb{R}^3$                                                               | $\mathbf{w} = (1, 2, 3) \in \mathbb{R}^3$                                              | b) $\mathbf{w} = (1, 2, 3) \in \mathbb{R}^3$ .                                                                                              |
| johnston-linear-matrix-algebra_F00689 | 59   | 1.000 | $\mathbf{u} \in \mathbb{R}^n$                                                                       | $\mathbf{u} \in \mathbb{R}^n$                                                          | In general, if $\mathbf{u} \in \mathbb{R}^n$                                                                                                |
| johnston-linear-matrix-algebra_F00690 | 59   | 1.000 | $P_{\{\mathbf{u}\}}(\mathbf{u})=\mathbf{u}$ : $P_{\{\mathbf{u}\}}$                                  | $P_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : P_{\mathbf{u}}$                             | then $P_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : P_{\mathbf{u}}$                                                                             |
| johnston-linear-matrix-algebra_F00691 | 59   | 1.000 | $F$                                                                                                 | $F$                                                                                    | The “ $F$ ” in $F_{\mathbf{u}}$ stands                                                                                                      |
| johnston-linear-matrix-algebra_F00692 | 59   | 1.000 | $F_{\{\mathbf{u}\}}$                                                                                | $F_{\mathbf{u}}$                                                                       | The “ $F$ ” in $F_{\mathbf{u}}$ stands                                                                                                      |
| johnston-linear-matrix-algebra_F00693 | 59   | 1.000 | $\left[P_{\{\mathbf{u}\}}\right] \mathbf{e}_1=\mathbf{e}_1$                                         | $[P_{\mathbf{u}}]\mathbf{e}_1 = \mathbf{e}_1$                                          | onto the $x$ -axis (in particular, $[P_{\mathbf{u}}]\mathbf{e}_1 = \mathbf{e}_1$ and $[P_{\mathbf{u}}]\mathbf{e}_2 = \mathbf{0}$ ):         |
| johnston-linear-matrix-algebra_F00694 | 59   | 1.000 | $\left[P_{\{\mathbf{u}\}}\right] \mathbf{e}_2=\mathbf{0}$                                           | $[P_{\mathbf{u}}]\mathbf{e}_2 = \mathbf{0}$                                            | onto the $x$ -axis (in particular, $[P_{\mathbf{u}}]\mathbf{e}_1 = \mathbf{e}_1$ and $[P_{\mathbf{u}}]\mathbf{e}_2 = \mathbf{0}$ ):         |
| johnston-linear-matrix-algebra_F00695 | 59   | 1.000 | $ \mathbf{w} =\sqrt{1^2+2^2+3^2}=\sqrt{14}$                                                         | $\ \mathbf{w}\  = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$                                  | same direction. Well, $\ \mathbf{w}\  = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$ , so we let $\mathbf{u} =$                                      |
| johnston-linear-matrix-algebra_F00696 | 59   | 1.000 | $\mathbf{u} =$                                                                                      | $\mathbf{u} =$                                                                         | same direction. Well, $\ \mathbf{w}\  = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$ , so we let $\mathbf{u} =$                                      |
| johnston-linear-matrix-algebra_F00697 | 59   | 0.987 | $\mathbf{w} /  \mathbf{w} =(1,2,3) / \sqrt{14}$                                                     | $\mathbf{w} / \ \mathbf{w}\  = (1, 2, 3) / \sqrt{14}$                                  | $\mathbf{w} / \ \mathbf{w}\  = (1, 2, 3) / \sqrt{14}$ . The standard matrix of $P_{\mathbf{u}}$ is then                                     |
| johnston-linear-matrix-algebra_F00698 | 59   | 1.000 | $\mathbf{e}_1, \mathbf{e}_2$                                                                        | $\mathbf{e}_1, \mathbf{e}_2$                                                           | This projection sends each of $\mathbf{e}_1, \mathbf{e}_2$ , and $\mathbf{e}_3$ to different multiples of                                   |
| johnston-linear-matrix-algebra_F00699 | 59   | 1.000 | $\mathbf{e}_3$                                                                                      | $\mathbf{e}_3$                                                                         | This projection sends each of $\mathbf{e}_1, \mathbf{e}_2$ , and $\mathbf{e}_3$ to different multiples of                                   |
| johnston-linear-matrix-algebra_F00700 | 59   | 0.999 | $F_{\{\mathbf{u}\}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                         | $F_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                | The function $F_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ that reflects space through the line in the direc-                     |
| johnston-linear-matrix-algebra_F00701 | 59   | 1.000 | $F_{\{\mathbf{u}\}}(\mathbf{v})=\mathbf{v}+2\left(P_{\{\mathbf{u}\}}(\mathbf{v})-\mathbf{v}\right)$ | $F_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$ | and determine its standard matrix, we note that $F_{\mathbf{u}}(\mathbf{v}) = \mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$ , as |
| johnston-linear-matrix-algebra_F00702 | 60   | 1.000 | $F_{\{\mathbf{u}\}}(\mathbf{v})$                                                                    | $F_{\mathbf{u}}(\mathbf{v})$                                                           | denoted by $F_{\mathbf{u}}(\mathbf{v})$ and is equal to $\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$ .                         |
| johnston-linear-matrix-algebra_F00703 | 60   | 1.000 | $\mathbf{v}+2\left(P_{\{\mathbf{u}\}}(\mathbf{v})-\mathbf{v}\right)$                                | $\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$                              | denoted by $F_{\mathbf{u}}(\mathbf{v})$ and is equal to $\mathbf{v} + 2(P_{\mathbf{u}}(\mathbf{v}) - \mathbf{v})$ .                         |
| johnston-linear-matrix-algebra_F00704 | 60   | 1.000 | $\mathbf{u}=(0,1) \in \mathbb{R}^2$                                                                 | $\mathbf{u} = (0, 1) \in \mathbb{R}^2$                                                 | a) $\mathbf{u} = (0, 1) \in \mathbb{R}^2$ , and                                                                                             |
| johnston-linear-matrix-algebra_F00705 | 60   | 1.000 | $\mathbf{w}=(1,1,1) \in \mathbb{R}^3$                                                               | $\mathbf{w} = (1, 1, 1) \in \mathbb{R}^3$                                              | b) $\mathbf{w} = (1, 1, 1) \in \mathbb{R}^3$ .                                                                                              |
| johnston-linear-matrix-algebra_F00706 | 60   | 1.000 | $\left[F_{\{\mathbf{u}\}}\right]$                                                                   | $[F_{\mathbf{u}}]$                                                                     | We can verify our computation of $[F_{\mathbf{u}}]$ by noting that multiplication                                                           |
| johnston-linear-matrix-algebra_F00707 | 61   | 1.000 | $F_{\{\mathbf{u}\}}(\mathbf{u})=\mathbf{u}$ : $F_{\{\mathbf{u}\}}$                                  | $F_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : F_{\mathbf{u}}$                             | then $F_{\mathbf{u}}(\mathbf{u}) = \mathbf{u} : F_{\mathbf{u}}$                                                                             |
| johnston-linear-matrix-algebra_F00708 | 61   | 1.000 | $ \mathbf{w} =\sqrt{1^2+1^2+1^2}=\sqrt{3}$                                                          | $\ \mathbf{w}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$                                   | rection. Well, $\ \mathbf{w}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ , so we let $\mathbf{u} = \mathbf{w} / \ \mathbf{w}\  =$                |
| johnston-linear-matrix-algebra_F00709 | 61   | 1.000 | $\mathbf{u}=\mathbf{w} /  \mathbf{w}  =$                                                            | $\mathbf{u} = \mathbf{w} / \ \mathbf{w}\  =$                                           | rection. Well, $\ \mathbf{w}\  = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$ , so we let $\mathbf{u} = \mathbf{w} / \ \mathbf{w}\  =$                |
| johnston-linear-matrix-algebra_F00710 | 61   | 0.999 | $(1,1,1) / \sqrt{3}$                                                                                | $(1, 1, 1) / \sqrt{3}$                                                                 | $(1, 1, 1) / \sqrt{3}$ . The standard matrix of $F_{\mathbf{u}}$ is then                                                                    |
| johnston-linear-matrix-algebra_F00711 | 61   | 1.000 | $\mathbf{v}=(-1,3)$                                                                                 | $\mathbf{v} = (-1, 3)$                                                                 | Find the entries of the vector that is obtained by reflecting $\mathbf{v} = (-1, 3)$                                                        |
| johnston-linear-matrix-algebra_F00712 | 61   | 1.000 | $\pi / 3$                                                                                           | $\pi / 3$                                                                              | through the line going through the origin at an angle of $\pi / 3$ counter-                                                                 |
| johnston-linear-matrix-algebra_F00713 | 62   | 1.000 | $R^{\theta}$                                                                                        | $R^{\theta}$                                                                           | angle $\theta$ in $\mathbb{R}^{\theta}$ might                                                                                               |

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| johnston-linear-matrix-algebra-F00714 | 62   | ■ 1.000 | $\mathbf{u}=(\cos (\pi / 3), \sin (\pi / 3))=(1, \sqrt{3}) / 2$                   | $\mathbf{u}=(\cos (\pi / 3), \sin (\pi / 3))=(1, \sqrt{3}) / 2$                   | unit vector is $\mathbf{u}=(\cos (\pi / 3), \sin (\pi / 3))=(1, \sqrt{3}) / 2$ , so                                                                                        |
| johnston-linear-matrix-algebra-F00715 | 62   | ■ 1.000 | $R^{\theta}: \mathbb{R}^2 \rightarrow$                                            | $R^{\theta}: \mathbb{R}^2 \rightarrow$                                            | (counter-clockwise) <b>rotations by an angle <math>\theta</math></b> , which we denote by $R^{\theta}: \mathbb{R}^2 \rightarrow$                                           |
| johnston-linear-matrix-algebra-F00716 | 62   | ■ 1.000 | $R^{\theta}(\mathbf{0})=\mathbf{0}$                                               | $R^{\theta}(\mathbf{0})=\mathbf{0}$                                               | origin so that $R^{\theta}(\mathbf{0})=\mathbf{0}$ . To see that $R^{\theta}$ then is indeed linear, we simply note                                                        |
| johnston-linear-matrix-algebra-F00717 | 62   | ■ 1.000 | $R^{\theta}(\mathbf{v}+\mathbf{w})=R^{\theta}(\mathbf{v})+R^{\theta}(\mathbf{w})$ | $R^{\theta}(\mathbf{v}+\mathbf{w})=R^{\theta}(\mathbf{v})+R^{\theta}(\mathbf{w})$ | that it is geometrically clear that $R^{\theta}(\mathbf{v}+\mathbf{w})=R^{\theta}(\mathbf{v})+R^{\theta}(\mathbf{w})$ for all $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$ ,  |
| johnston-linear-matrix-algebra-F00718 | 62   | ■ 1.000 | $R^{\theta}(c \mathbf{v})=c R^{\theta}(\mathbf{v})$                               | $R^{\theta}(c \mathbf{v})=c R^{\theta}(\mathbf{v})$                               | the requirement that $R^{\theta}(c \mathbf{v})=c R^{\theta}(\mathbf{v})$ for all $\mathbf{v} \in \mathbb{R}^2$ and all $c \in \mathbb{R}$ follows from                     |
| johnston-linear-matrix-algebra-F00719 | 62   | ■ 1.000 | $R^{\theta}\left(\mathbf{e}_1\right)$                                             | $R^{\theta}\left(\mathbf{e}_1\right)$                                             | tion of how it acts on arbitrary vectors), it is enough to compute $R^{\theta}\left(\mathbf{e}_1\right)$ and                                                               |
| johnston-linear-matrix-algebra-F00720 | 63   | ■ 1.000 | $\sin (-x)=-\sin (x)$                                                             | $\sin (-x)=-\sin (x)$                                                             | $\sin (-x)=-\sin (x)$                                                                                                                                                      |
| johnston-linear-matrix-algebra-F00721 | 63   | ■ 1.000 | $\cos (-x)=\cos (x)$                                                              | $\cos (-x)=\cos (x)$                                                              | and $\cos (-x)=\cos (x)$                                                                                                                                                   |
| johnston-linear-matrix-algebra-F00722 | 63   | ■ 1.000 | $R^{\theta}\left(\mathbf{e}_2\right)$                                             | $R^{\theta}\left(\mathbf{e}_2\right)$                                             | $R^{\theta}\left(\mathbf{e}_2\right)$ . By regarding $R^{\theta}\left(\mathbf{e}_1\right)$ and $R^{\theta}\left(\mathbf{e}_2\right)$ as the hypotenuses (with length 1) of |
| johnston-linear-matrix-algebra-F00723 | 63   | ■ 1.000 | $R^{\theta}\left(\mathbf{e}_1\right)=(\cos (\theta), \sin (\theta))$              | $R^{\theta}\left(\mathbf{e}_1\right)=(\cos (\theta), \sin (\theta))$              | right-angled triangles as in Figure 1.19, we see that $R^{\theta}\left(\mathbf{e}_1\right)=(\cos (\theta), \sin (\theta))$                                                 |
| johnston-linear-matrix-algebra-F00724 | 63   | ■ 1.000 | $R^{\theta}\left(\mathbf{e}_2\right)=(-\sin (\theta), \cos (\theta))$             | $R^{\theta}\left(\mathbf{e}_2\right)=(-\sin (\theta), \cos (\theta))$             | and $R^{\theta}\left(\mathbf{e}_2\right)=(-\sin (\theta), \cos (\theta))$ .                                                                                                |
| johnston-linear-matrix-algebra-F00725 | 63   | ■ 1.000 | $\pi / 4$                                                                         | $\pi / 4$                                                                         | a) $\pi / 4$ radians counter-clockwise, and                                                                                                                                |
| johnston-linear-matrix-algebra-F00726 | 63   | ■ 1.000 | $\pi / 6$                                                                         | $\pi / 6$                                                                         | b) $\pi / 6$ radians clockwise.                                                                                                                                            |
| johnston-linear-matrix-algebra-F00727 | 63   | ■ 1.000 | $-\pi / 6$                                                                        | $-\pi / 6$                                                                        | clockwise rotation by $-\pi / 6$ radians, so we compute                                                                                                                    |
| johnston-linear-matrix-algebra-F00728 | 63   | ■ 1.000 | $\mathbf{v}=(1,3)$                                                                | $\mathbf{v}=(1,3)$                                                                | a) Rotate $\mathbf{v}=(1,3)$ by $\pi / 4$ radians counter-clockwise, and                                                                                                   |
| johnston-linear-matrix-algebra-F00729 | 63   | ■ 1.000 | $\mathbf{w}=(\sqrt{3}, 3)$                                                        | $\mathbf{w}=(\sqrt{3}, 3)$                                                        | b) rotate $\mathbf{w}=(\sqrt{3}, 3)$ by $\pi / 6$ radians clockwise.                                                                                                       |
| johnston-linear-matrix-algebra-F00730 | 64   | ■ 1.000 | $R^{\pi / 6}(\mathbf{w})$                                                         | $R^{\pi / 6}(\mathbf{w})$                                                         | ample 1.4.8(b), so we can simply compute $R^{\pi / 6}(\mathbf{w})$ via matrix mul-                                                                                         |
| johnston-linear-matrix-algebra-F00731 | 64   | ■ 1.000 | $\mathbf{v}=(1,2,3)$                                                              | $\mathbf{v}=(1,2,3)$                                                              | “rotate $\mathbf{v}=(1,2,3)$ counter-clockwise by an angle of $\pi / 4$ ”—this instruction is                                                                              |
| johnston-linear-matrix-algebra-F00732 | 64   | ■ 1.000 | $R_{yz}^{\theta}$                                                                 | $R_{yz}^{\theta}$                                                                 | To illustrate how to do this, consider the linear transformation $R_{yz}^{\theta}$ that                                                                                    |
| johnston-linear-matrix-algebra-F00733 | 65   | ■ 1.000 | $\left[R_{zx}^{\theta}\right]$                                                    | $\left[R_{zx}^{\theta}\right]$                                                    | We list $\left[R_{zx}^{\theta}\right]$ here                                                                                                                                |
| johnston-linear-matrix-algebra-F00734 | 65   | ■ 1.000 | $\left[R_{xz}^{\theta}\right]$                                                    | $\left[R_{xz}^{\theta}\right]$                                                    | instead of $\left[R_{xz}^{\theta}\right]$ to                                                                                                                               |
| johnston-linear-matrix-algebra-F00735 | 65   | ■ 1.000 | $-\sin (\theta)$                                                                  | $-\sin (\theta)$                                                                  | keep $-\sin (\theta)$ in the                                                                                                                                               |
| johnston-linear-matrix-algebra-F00736 | 65   | ■ 1.000 | $R_{yz}^{\theta}\left(\mathbf{e}_1\right)=\mathbf{e}_1$                           | $R_{yz}^{\theta}\left(\mathbf{e}_1\right)=\mathbf{e}_1$                           | has no effect at all (i.e., $R_{yz}^{\theta}\left(\mathbf{e}_1\right)=\mathbf{e}_1$ ), as shown in Figure 1.20. Similarly,                                                 |
| johnston-linear-matrix-algebra-F00737 | 65   | ■ 1.000 | $yz$                                                                              | $yz$                                                                              | to rotate $\mathbf{e}_2$ or $\mathbf{e}_3$ we can just treat the $yz$ -plane as if it were $\mathbb{R}^2$ and repeat                                                       |
| johnston-linear-matrix-algebra-F00738 | 65   | ■ 1.000 | $R_{yz}^{\theta}\left(\mathbf{e}_2\right)=(0, \cos (\theta), \sin (\theta))$      | $R_{yz}^{\theta}\left(\mathbf{e}_2\right)=(0, \cos (\theta), \sin (\theta))$      | the derivation from Figure 1.18 to see that $R_{yz}^{\theta}\left(\mathbf{e}_2\right)=(0, \cos (\theta), \sin (\theta))$ and                                               |
| johnston-linear-matrix-algebra-F00739 | 65   | ■ 0.551 | $R_{yz}^{\theta}\left(\mathbf{e}_3\right)=(0,-\sin (\theta), \cos (\theta))$      | $R_{yz}^{\theta}\left(\mathbf{e}_3\right)=(0,-\sin (\theta), \cos (\theta))$      | $R_{yz}^{\theta}\left(\mathbf{e}_3\right)=(0,-\sin (\theta), \cos (\theta))$ .                                                                                             |
| johnston-linear-matrix-algebra-F00740 | 65   | ■ 0.999 | $\mathbf{v}=(3,-1,2)$                                                             | $\mathbf{v}=(3,-1,2)$                                                             | Rotate $\mathbf{v}=(3,-1,2)$ around the $z$ -axis by an angle of $\theta=2 \pi / 3$ in the                                                                                 |

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| johnston-linear-matrix-algebra-F00741 | 65   | ■ 0.999 | $\theta=2\pi/3$                                    | $\theta = 2\pi/3$                                  | Rotate $\mathbf{v} = (3, -1, 2)$ around the $z$ -axis by an angle of $\theta = 2\pi/3$ in the                                          |
| johnston-linear-matrix-algebra-F00742 | 65   | ■ 1.000 | $R_{xy}^{2\pi/3}(\mathbf{v})$                      | $R_{xy}^{2\pi/3}(\mathbf{v})$                      | Since our goal is to compute $R_{xy}^{2\pi/3}(\mathbf{v})$ , we start by constructing the                                              |
| johnston-linear-matrix-algebra-F00743 | 65   | ■ 1.000 | $R_{xy}^{2\pi/3}$                                  | $R_{xy}^{2\pi/3}$                                  | standard matrix of $R_{xy}^{2\pi/3}$ :                                                                                                 |
| johnston-linear-matrix-algebra-F00744 | 66   | ■ 1.000 | $S \circ T$                                        | $S \circ T$                                        | Be careful: $S \circ T$ is                                                                                                             |
| johnston-linear-matrix-algebra-F00745 | 66   | ■ 0.907 | $(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$       | $(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$       | $(S \circ T)(\mathbf{v}) = S(T(\mathbf{v}))$ .                                                                                         |
| johnston-linear-matrix-algebra-F00746 | 66   | ■ 0.999 | $S, T: \mathbb{R}^n \rightarrow$                   | $S, T: \mathbb{R}^n \rightarrow$                   | transformations to create new ones. In particular, we noted that if $S, T: \mathbb{R}^n \rightarrow$                                   |
| johnston-linear-matrix-algebra-F00747 | 66   | ■ 1.000 | $S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$    | $S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$    | $\mathbb{R}^m$ are linear transformations and $c \in \mathbb{R}$ then $S + cT: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is also a linear |
| johnston-linear-matrix-algebra-F00748 | 66   | ■ 1.000 | $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$         | $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$         | Suppose $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S: \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations. Then     |
| johnston-linear-matrix-algebra-F00749 | 66   | ■ 1.000 | $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ | $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ | their <b>composition</b> is the function $S \circ T: \mathbb{R}^n \rightarrow \mathbb{R}^p$ defined by                                 |
| johnston-linear-matrix-algebra-F00750 | 66   | ■ 1.000 | $\mathbb{R}^p$                                     | $\mathbb{R}^p$                                     | plying $S$ . In other words, while $T$ sends $\mathbb{R}^n$ to $\mathbb{R}^m$ and $S$ sends $\mathbb{R}^m$ to $\mathbb{R}^p$ , the     |
| johnston-linear-matrix-algebra-F00751 | 67   | ■ 1.000 | $[S \circ T] = [S][T]$                             | $[S \circ T] = [S][T]$                             | $[S \circ T] = [S][T]$ is a                                                                                                            |
| johnston-linear-matrix-algebra-F00752 | 67   | ■ 1.000 | $p \times n$                                       | $p \times n$                                       | $p \times n$ matrix.                                                                                                                   |
| johnston-linear-matrix-algebra-F00753 | 67   | ■ 1.000 | $[D][F_{\mathbf{u}}]$                              | $[D][F_{\mathbf{u}}]$                              | are after is $[D][F_{\mathbf{u}}]$ ,                                                                                                   |
| johnston-linear-matrix-algebra-F00754 | 67   | ■ 0.980 | $\text{not}[F_{\mathbf{u}}][D]$                    | $\text{not}[F_{\mathbf{u}}][D]$                    | not $[F_{\mathbf{u}}][D]$ .                                                                                                            |
| johnston-linear-matrix-algebra-F00755 | 67   | ■ 0.998 | $S(T(\mathbf{v}))$                                 | $S(T(\mathbf{v}))$                                 | mation that sends $\mathbf{v}$ to $S(T(\mathbf{v}))$ .                                                                                 |
| johnston-linear-matrix-algebra-F00756 | 67   | ■ 1.000 | $[S] \in \mathcal{M}_{p,m}$                        | $[S] \in \mathcal{M}_{p,m}$                        | standard matrices $[T] \in \mathcal{M}_{m,n}$ and $[S] \in \mathcal{M}_{p,m}$ , respectively. Then $S \circ T$ :                       |
| johnston-linear-matrix-algebra-F00757 | 67   | ■ 1.000 | $\mathbb{R}^n \rightarrow \mathbb{R}^p$            | $\mathbb{R}^n \rightarrow \mathbb{R}^p$            | $\mathbb{R}^n \rightarrow \mathbb{R}^p$ is a linear transformation, and its standard matrix is                                         |
| johnston-linear-matrix-algebra-F00758 | 67   | ■ 1.000 | $(S \circ T)(\mathbf{v})$                          | $(S \circ T)(\mathbf{v})$                          | <i>Proof.</i> Let $\mathbf{v} \in \mathbb{R}^n$ and compute $(S \circ T)(\mathbf{v})$ :                                                |
| johnston-linear-matrix-algebra-F00759 | 67   | ■ 1.000 | $[S][T]$                                           | $[S][T]$                                           | multiplication by the matrix $[S][T]$ . It thus follows from Theorem 1.4.1 that                                                        |
| johnston-linear-matrix-algebra-F00760 | 67   | ■ 1.000 | $y = \frac{4}{3}x$                                 | $y = \frac{4}{3}x$                                 | through the line $y = \frac{4}{3}x$ and then stretches it in the $x$ -direction by a factor                                            |
| johnston-linear-matrix-algebra-F00761 | 67   | ■ 0.960 | $\mathbf{u} = (3/5, 4/5)$                          | $\mathbf{u} = (3/5, 4/5)$                          | them together. A unit vector on the line $y = \frac{4}{3}x$ is $\mathbf{u} = (3/5, 4/5)$ , and the                                     |
| johnston-linear-matrix-algebra-F00762 | 67   | ■ 1.000 | $T = D \circ F_{\mathbf{u}}$                       | $T = D \circ F_{\mathbf{u}}$                       | The standard matrix of the composite linear transformation $T = D \circ F_{\mathbf{u}}$ is                                             |
| johnston-linear-matrix-algebra-F00763 | 68   | ■ 0.999 | $\phi$                                             | $\phi$                                             | Explain why, if $\theta$ and $\phi$ are any two angles, then $R^\theta \circ R^\phi = R^{\theta+\phi}$ . Use                           |
| johnston-linear-matrix-algebra-F00764 | 68   | ■ 0.999 | $R^\theta \circ R^\phi = R^{\theta+\phi}$          | $R^\theta \circ R^\phi = R^{\theta+\phi}$          | Explain why, if $\theta$ and $\phi$ are any two angles, then $R^\theta \circ R^\phi = R^{\theta+\phi}$ . Use                           |
| johnston-linear-matrix-algebra-F00765 | 69   | ■ 1.000 | $\phi = \theta$                                    | $\phi = \theta$                                    | If we choose $\phi = \theta$                                                                                                           |
| johnston-linear-matrix-algebra-F00766 | 69   | ■ 1.000 | $\sin(2\theta) =$                                  | $\sin(2\theta) =$                                  | that $\sin(2\theta) =$                                                                                                                 |
| johnston-linear-matrix-algebra-F00767 | 69   | ■ 1.000 | $2 \sin(\theta) \cos(\theta)$                      | $2 \sin(\theta) \cos(\theta)$                      | $2 \sin(\theta) \cos(\theta)$ and                                                                                                      |

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| johnston-linear-matrix-algebra_F00768 | 69   | ■ 1.000 | $\cos (2 \theta)=$                                                           | $\cos(2\theta) =$                                                   | $\cos(2\theta) =$                                                                                                               |
| johnston-linear-matrix-algebra_F00769 | 69   | ■ 0.980 | $\cos ^{2}(\theta)-\sin ^{2}(\theta)$                                        | $\cos ^2(\theta)-\sin ^2(\theta)$                                   | $\cos ^2(\theta)-\sin ^2(\theta).$                                                                                              |
| johnston-linear-matrix-algebra_F00770 | 69   | ■ 0.995 | $\theta+\phi$                                                                | $\theta+\phi$                                                       | by $\theta+\phi$ radians:                                                                                                       |
| johnston-linear-matrix-algebra_F00771 | 69   | ■ 1.000 | $R^{\theta} \circ R^{\phi}$                                                  | $R^{\theta} \circ R^{\phi}$                                         | of $R^{\theta} \circ R^{\phi}$ and $R^{\theta+\phi}$ :                                                                          |
| johnston-linear-matrix-algebra_F00772 | 69   | ■ 1.000 | $R^{\theta+\phi}$                                                            | $R^{\theta+\phi}$                                                   | of $R^{\theta} \circ R^{\phi}$ and $R^{\theta+\phi}$ :                                                                          |
| johnston-linear-matrix-algebra_F00773 | 69   | ■ 1.000 | $\left[R^{\theta} \circ R^{\phi}\right]$                                     | $\left[R^{\theta} \circ R^{\phi}\right]$                            | The matrices $\left[R^{\theta} \circ R^{\phi}\right]$ and $\left[R^{\theta+\phi}\right]$ must be equal (since they are standard |
| johnston-linear-matrix-algebra_F00774 | 69   | ■ 1.000 | $\left[R^{\theta+\phi}\right]$                                               | $\left[R^{\theta+\phi}\right]$                                      | The matrices $\left[R^{\theta} \circ R^{\phi}\right]$ and $\left[R^{\theta+\phi}\right]$ must be equal (since they are standard |
| johnston-linear-matrix-algebra_F00775 | 69   | ■ 0.983 | $(2,1)$                                                                      | $(2,1)$                                                             | entries must be equal. By looking at their $(2,1)$ -entries, we see that                                                        |
| johnston-linear-matrix-algebra_F00776 | 69   | ■ 0.831 | $\left[R^{\theta}\right]$                                                    | $\left[R^{\theta}\right]$                                           | at all— $\left[R^{\theta}\right]$ really is the $\theta$ -th power of the matrix                                                |
| johnston-linear-matrix-algebra_F00777 | 69   | ■ 1.000 | $R$                                                                          | $R$                                                                 | For now, this interpretation of $R^{\theta}$ as the $\theta$ -th power of $R$ only makes                                        |
| johnston-linear-matrix-algebra_F00778 | 70   | ■ 1.000 | $R, S$                                                                       | $R, S$                                                              | two of them: if $R, S$ , and $T$ are linear transformations then their composition                                              |
| johnston-linear-matrix-algebra_F00779 | 70   | ■ 1.000 | $R \circ S \circ T$                                                          | $R \circ S \circ T$                                                 | $R \circ S \circ T$ is the linear transformation that applies $T$ , then $S$ , then $R$ , and its                               |
| johnston-linear-matrix-algebra_F00780 | 70   | ■ 1.000 | $[R \circ S \circ T]=[R][S][T]$                                              | $[R \circ S \circ T]=[R][S][T]$                                     | standard matrix is $[R \circ S \circ T]=[R][S][T]$ .                                                                            |
| johnston-linear-matrix-algebra_F00781 | 70   | ■ 0.999 | $T\left(v_1, v_2\right)=\left(v_1+2 v_2, 3 v_1-v_2\right)$                   | $T\left(v_1, v_2\right)=\left(v_1+2 v_2, 3 v_1-v_2\right)$          | * (a) $T\left(v_1, v_2\right)=\left(v_1+2 v_2, 3 v_1-v_2\right)$ .                                                              |
| johnston-linear-matrix-algebra_F00782 | 70   | ■ 1.000 | $T\left(v_1, v_2\right)=\left(v_1+v_2, 2 v_1-v_2,-v_1+3 v_2\right)$          | $T\left(v_1, v_2\right)=\left(v_1+v_2, 2 v_1-v_2,-v_1+3 v_2\right)$ | (b) $T\left(v_1, v_2\right)=\left(v_1+v_2, 2 v_1-v_2,-v_1+3 v_2\right)$ .                                                       |
| johnston-linear-matrix-algebra_F00783 | 70   | ■ 0.924 | $T\left(v_1, v_2, v_3\right)=\left(v_1+v_2, v_1+v_2-v_3, v_1+v_2-v_3\right)$ | $T\left(v_1, v_2, v_3\right)=\left(v_1+v_2, v_1+v_2-v_3\right)$     | * (c) $T\left(v_1, v_2, v_3\right)=\left(v_1+v_2, v_1+v_2-v_3\right)$ .                                                         |
| johnston-linear-matrix-algebra_F00784 | 70   | ■ 1.000 | $T\left(v_1, v_2, v_3\right)=\left(v_2, 2 v_1+v_3, v_2-v_3\right)$           | $T\left(v_1, v_2, v_3\right)=\left(v_2, 2 v_1+v_3, v_2-v_3\right)$  | (d) $T\left(v_1, v_2, v_3\right)=\left(v_2, 2 v_1+v_3, v_2-v_3\right)$ .                                                        |
| johnston-linear-matrix-algebra_F00785 | 70   | ■ 0.669 | $T(1,0)=(3,-1)$                                                              | $T(1,0)=(3,-1)$                                                     | * (a) $T(1,0)=(3,-1)$ and $T(0,1)=(1,2)$ .                                                                                      |
| johnston-linear-matrix-algebra_F00786 | 70   | ■ 0.669 | $T(0,1)=(1,2)$                                                               | $T(0,1)=(1,2)$                                                      | * (a) $T(1,0)=(3,-1)$ and $T(0,1)=(1,2)$ .                                                                                      |
| johnston-linear-matrix-algebra_F00787 | 70   | ■ 1.000 | $T(1,0)=(1,3)$                                                               | $T(1,0)=(1,3)$                                                      | (b) $T(1,0)=(1,3)$ and $T(1,1)=(3,7)$ .                                                                                         |
| johnston-linear-matrix-algebra_F00788 | 70   | ■ 1.000 | $T(1,1)=(3,7)$                                                               | $T(1,1)=(3,7)$                                                      | (b) $T(1,0)=(1,3)$ and $T(1,1)=(3,7)$ .                                                                                         |
| johnston-linear-matrix-algebra_F00789 | 70   | ■ 0.971 | $T(1,0)=(-1,0,1)$                                                            | $T(1,0)=(-1,0,1)$                                                   | * (c) $T(1,0)=(-1,0,1)$ and $T(0,1)=(2,3,0)$ .                                                                                  |
| johnston-linear-matrix-algebra_F00790 | 70   | ■ 0.971 | $T(0,1)=(2,3,0)$                                                             | $T(0,1)=(2,3,0)$                                                    | * (c) $T(1,0)=(-1,0,1)$ and $T(0,1)=(2,3,0)$ .                                                                                  |
| johnston-linear-matrix-algebra_F00791 | 70   | ■ 0.975 | $T(1,0,0)=(2,1), \quad T(0,1,0)=(-1,1)$                                      | $T(1,0,0)=(2,1), \quad T(0,1,0)=(-1,1)$                             | (d) $T(1,0,0)=(2,1), \quad T(0,1,0)=(-1,1),$ and                                                                                |
| johnston-linear-matrix-algebra_F00792 | 70   | ■ 1.000 | $T(0,0,1)=(0,3)$                                                             | $T(0,0,1)=(0,3)$                                                    | $T(0,0,1)=(0,3)$ .                                                                                                              |
| johnston-linear-matrix-algebra_F00793 | 70   | ■ 0.539 | $T(1,0,0)=(1,2,3), \quad T(1,1,0)=(0,1,2)$                                   | $T(1,0,0)=(1,2,3), T(1,1,0)=(0,1,2)$                                | *(e) $T(1,0,0)=(1,2,3), \quad T(1,1,0)=(0,1,2),$ and                                                                            |
| johnston-linear-matrix-algebra_F00794 | 70   | ■ 0.976 | $T(1,1,1)=(0,0,1)$                                                           | $T(1,1,1)=(0,0,1)$                                                  | $T(1,1,1)=(0,0,1)$ .                                                                                                            |
| johnston-linear-matrix-algebra_F00795 | 70   | ■ 1.000 | $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$                                   | $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$                          | *(c) Every linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^4$ is com-                                             |

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| johnston-linear-matrix-algebra_F00796 | 70   | 1.000 | $\mathbb{R}^4 \rightarrow \mathbb{R}^3$                     | $\mathbb{R}^4 \rightarrow \mathbb{R}^3$                     | $\mathbb{R}^4 \rightarrow \mathbb{R}^3$ has size $4 \times 3$ .                                         |
| johnston-linear-matrix-algebra_F00797 | 70   | 1.000 | $4 \times 3$                                                | $4 \times 3$                                                | $\mathbb{R}^4 \rightarrow \mathbb{R}^3$ has size $4 \times 3$ .                                         |
| johnston-linear-matrix-algebra_F00798 | 70   | 0.999 | $T: \mathbb{R}^2 \rightarrow$                               | $T: \mathbb{R}^2 \rightarrow$                               | *(e) There exists a linear transformation $T: \mathbb{R}^2 \rightarrow$                                 |
| johnston-linear-matrix-algebra_F00799 | 70   | 1.000 | $T(\mathbf{e}_1)=(2,1), T(\mathbf{e}_2)=(1,3)$              | $T(\mathbf{e}_1)=(2,1), T(\mathbf{e}_2)=(1,3)$              | $\mathbb{R}^2$ such that $T(\mathbf{e}_1)=(2,1), T(\mathbf{e}_2)=(1,3)$ , and                           |
| johnston-linear-matrix-algebra_F00800 | 70   | 1.000 | $T(1,1)=(3,3)$                                              | $T(1,1)=(3,3)$                                              | $T(1,1)=(3,3)$ .                                                                                        |
| johnston-linear-matrix-algebra_F00801 | 70   | 0.966 | $*(\mathbf{g})$                                             | *(g)                                                        | *(g) If $R_{xy}^\theta, R_{yz}^\theta$ , and $R_{xz}^\theta$ are rotations around the co-               |
| johnston-linear-matrix-algebra_F00802 | 70   | 0.966 | $R_{xy}^\theta, R_{yz}^\theta$                              | $R_{xy}^\theta, R_{yz}^\theta$                              | *(g) If $R_{xy}^\theta, R_{yz}^\theta$ , and $R_{xz}^\theta$ are rotations around the co-               |
| johnston-linear-matrix-algebra_F00803 | 70   | 0.966 | $R_{xz}^\theta$                                             | $R_{xz}^\theta$                                             | *(g) If $R_{xy}^\theta, R_{yz}^\theta$ , and $R_{xz}^\theta$ are rotations around the co-               |
| johnston-linear-matrix-algebra_F00804 | 70   | 0.860 | $R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta$         | $R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta$         | $R_{xy}^\theta \circ R_{yz}^\theta = R_{xz}^\theta$ .                                                   |
| johnston-linear-matrix-algebra_F00805 | 70   | 1.000 | $\mathbb{R}^2 \rightarrow \mathbb{R}^2$                     | $\mathbb{R}^2 \rightarrow \mathbb{R}^2$                     | $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ are and are not linear transformations. If $T$ is a             |
| johnston-linear-matrix-algebra_F00806 | 70   | 0.972 | $T(v_1, v_2) = (v_1^2, v_2)$                                | $T(v_1, v_2) = (v_1^2, v_2)$                                | *(a) $T(v_1, v_2) = (v_1^2, v_2)$ .                                                                     |
| johnston-linear-matrix-algebra_F00807 | 70   | 1.000 | $T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1)$                     | $T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1)$                     | (b) $T(v_1, v_2) = (v_1 + 2v_2, v_2 - v_1)$ .                                                           |
| johnston-linear-matrix-algebra_F00808 | 70   | 1.000 | $T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2))$          | $T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2))$          | *(c) $T(v_1, v_2) = (\sin(v_1) + v_2, v_1 - \cos(v_2))$ .                                               |
| johnston-linear-matrix-algebra_F00809 | 70   | 1.000 | $T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2)$        | $T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2)$        | (d) $T(v_1, v_2) = (\sin(3)v_1 + v_2, v_1 - \cos(2)v_2)$ .                                              |
| johnston-linear-matrix-algebra_F00810 | 70   | 1.000 | $T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2})$ | $T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2})$ | *(c) $T(v_1, v_2) = (\sqrt{v_1} + \sqrt{v_2}, \sqrt{v_1 + v_2})$ .                                      |
| johnston-linear-matrix-algebra_F00811 | 70   | 1.000 | $T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1)$    | $T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1)$    | (f) $T(v_1, v_2) = (\sqrt{3}v_1, \sqrt{2}v_2 - \sqrt{5}v_1)$ .                                          |
| johnston-linear-matrix-algebra_F00812 | 70   | 0.993 | $T(v_1, v_2) = ( v_1 ,  v_2 )$                              | $T(v_1, v_2) = ( v_1 ,  v_2 )$                              | *(g) $T(v_1, v_2) = ( v_1 ,  v_2 )$ .                                                                   |
| johnston-linear-matrix-algebra_F00813 | 70   | 1.000 | $y = 3x$                                                    | $y = 3x$                                                    | *(a) The projection onto the line $y = 3x$ .                                                            |
| johnston-linear-matrix-algebra_F00814 | 70   | 1.000 | $y = 3x - 2$                                                | $y = 3x - 2$                                                | (b) The projection onto the line $y = 3x - 2$ .                                                         |
| johnston-linear-matrix-algebra_F00815 | 70   | 1.000 | $y = 2x$                                                    | $y = 2x$                                                    | *(c) The reflection across the line $y = 2x$ .                                                          |
| johnston-linear-matrix-algebra_F00816 | 70   | 1.000 | $y = 2x + 1$                                                | $y = 2x + 1$                                                | (d) The reflection across the line $y = 2x + 1$ .                                                       |
| johnston-linear-matrix-algebra_F00817 | 70   | 0.999 | $\pi/5$                                                     | $\pi/5$                                                     | $(0,0)$ by $\pi/5$ radians.                                                                             |
| johnston-linear-matrix-algebra_F00818 | 70   | 1.000 | $(0,1)$                                                     | $(0,1)$                                                     | $(0,1)$ by $\pi/5$ radians.                                                                             |
| johnston-linear-matrix-algebra_F00819 | 70   | 0.981 | $S(v_1, v_2) = (2v_2, v_1 + v_2)$                           | $S(v_1, v_2) = (2v_2, v_1 + v_2)$                           | *(a) $S(v_1, v_2) = (2v_2, v_1 + v_2)$ ,                                                                |
| johnston-linear-matrix-algebra_F00820 | 70   | 1.000 | $S(v_1, v_2) = (v_1 - 2v_2, 3v_1 + v_2)$                    | $S(v_1, v_2) = (v_1 - 2v_2, 3v_1 + v_2)$                    | (b) $S(v_1, v_2) = (v_1 - 2v_2, 3v_1 + v_2)$ ,                                                          |
| johnston-linear-matrix-algebra_F00821 | 70   | 0.999 | $S(v_1, v_2, v_3) = (v_1, v_1 + v_2, v_1 + v_2 + v_3)$      | $S(v_1, v_2, v_3) = (v_1, v_1 + v_2, v_1 + v_2 + v_3)$      | *(c) $S(v_1, v_2, v_3) = (v_1, v_1 + v_2, v_1 + v_2 + v_3)$ ,                                           |
| johnston-linear-matrix-algebra_F00822 | 70   | 0.999 | $y = x$                                                     | $y = x$                                                     | *(b) Projects $\mathbb{R}^2$ onto the line $y = x$ and then rotates                                     |
| johnston-linear-matrix-algebra_F00823 | 70   | 1.000 | $y = x/2$                                                   | $y = x/2$                                                   | $\mathbb{R}^2$ through the line $y = x/2$ .                                                             |
| johnston-linear-matrix-algebra_F00824 | 70   | 1.000 | $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$        | $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$        | for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and all $c_1, \dots, c_k \in \mathbb{R}$ . |

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| johnston-linear-matrix-algebra_F00825 | 70   | 1.000 | $c_{\{1\}}, \ldots, c_{\{k\}} \in \mathbb{R}$                           | $c_1, \dots, c_k \in \mathbb{R}$                        | for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and all $c_1, \dots, c_k \in \mathbb{R}$ .    |
| johnston-linear-matrix-algebra_F00826 | 70   | 1.000 | $D \in \mathcal{M}_{\{n\}}$                                             | $D \in \mathcal{M}_n$                                   | <b>** 1.4.10</b> Let $D \in \mathcal{M}_n$ be a diagonal matrix and let                                    |
| johnston-linear-matrix-algebra_F00827 | 70   | 1.000 | $1 \leq i \leq n, \ldots$                                               | $1 \leq i \leq n, \dots$                                | tiplication to show that, for all $1 \leq i \leq n, \dots$                                                 |
| johnston-linear-matrix-algebra_F00828 | 70   | 1.000 | $d_{\{i, i\}}$                                                          | $d_{i,i}$                                               | (a) the product $DA$ multiplies the $i$ -th row of $A$ by $d_{i,i}$ ,                                      |
| johnston-linear-matrix-algebra_F00829 | 71   | 1.000 | $Q, S, T: \mathbb{R}^{\{n\}} \rightarrow \mathbb{R}^{\{m\}}$            | $Q, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$        | <b>1.4.12</b> Let $Q, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear transformations.              |
| johnston-linear-matrix-algebra_F00830 | 71   | 1.000 | $S+T=T+S$                                                               | $S + T = T + S$                                         | (a) $S + T = T + S$ , and                                                                                  |
| johnston-linear-matrix-algebra_F00831 | 71   | 1.000 | $Q+(S+T)=(Q+S)+T$                                                       | $Q + (S + T) = (Q + S) + T$                             | (b) $Q + (S + T) = (Q + S) + T$ .                                                                          |
| johnston-linear-matrix-algebra_F00832 | 71   | 1.000 | $c(S+T)=c S+c T$                                                        | $c(S + T) = cS + cT$                                    | (a) $c(S + T) = cS + cT$ ,                                                                                 |
| johnston-linear-matrix-algebra_F00833 | 71   | 1.000 | $(c+d) T=c T+d T$                                                       | $(c + d)T = cT + dT$                                    | (b) $(c + d)T = cT + dT$ , and                                                                             |
| johnston-linear-matrix-algebra_F00834 | 71   | 1.000 | $c(d T)=(c d) T$                                                        | $c(dT) = (cd)T$                                         | (c) $c(dT) = (cd)T$ .                                                                                      |
| johnston-linear-matrix-algebra_F00835 | 71   | 1.000 | $A^{\{160\}}$                                                           | $A^{160}$                                               | Compute the matrix $A^{160}$ .                                                                             |
| johnston-linear-matrix-algebra_F00836 | 71   | 1.000 | $\theta \in \mathbb{R}$                                                 | $\theta \in \mathbb{R}$                                 | for some $\theta \in \mathbb{R}$ . Show that $A^T A = I$ .                                                 |
| johnston-linear-matrix-algebra_F00837 | 71   | 1.000 | $A^{\{T\}} A=I$                                                         | $A^T A = I$                                             | for some $\theta \in \mathbb{R}$ . Show that $A^T A = I$ .                                                 |
| johnston-linear-matrix-algebra_F00838 | 71   | 0.999 | $A=\mathbf{u} \mathbf{u}^{\{T\}} \in$                                   | $A = \mathbf{u} \mathbf{u}^T \in$                       | <b>1.4.16</b> Suppose $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A = \mathbf{u} \mathbf{u}^T \in$ |
| johnston-linear-matrix-algebra_F00839 | 71   | 1.000 | $A=$                                                                    | $A =$                                                   | <b>** 1.4.17</b> Suppose $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A =$                          |
| johnston-linear-matrix-algebra_F00840 | 71   | 0.481 | $2 \mathbf{u} \mathbf{u}^{\{T\}}-I \in \mathcal{M}_{\{n\}}$             | $2\mathbf{u} \mathbf{u}^T - I \in \mathcal{M}_n$        | $2\mathbf{u} \mathbf{u}^T - I \in \mathcal{M}_n$ . Show that $A^2 = I$ . Provide a geometric               |
| johnston-linear-matrix-algebra_F00841 | 71   | 1.000 | $P_{\{\mathbf{u}\}}: \mathbb{R}^{\{n\}} \rightarrow \mathbb{R}^{\{n\}}$ | $P_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ | <b>1.4.18</b> Show that if $P_{\mathbf{u}}: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a projection onto a  |
| johnston-linear-matrix-algebra_F00842 | 71   | 1.000 | $\left\ P_{\{\mathbf{u}\}}(\mathbf{v})\right\  \leq\ \mathbf{v}\ $      | $\ P_{\mathbf{u}}(\mathbf{v})\  \leq \ \mathbf{v}\ $    | line and $\mathbf{v} \in \mathbb{R}^n$ then $\ P_{\mathbf{u}}(\mathbf{v})\  \leq \ \mathbf{v}\ $ .         |
| johnston-linear-matrix-algebra_F00843 | 71   | 1.000 | $y=m x$                                                                 | $y = mx$                                                | $\mathbb{R}^2$ through the line $y = mx$ has standard matrix                                               |
| johnston-linear-matrix-algebra_F00844 | 71   | 1.000 | $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{\{2\}}$                         | $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$               | <b>1.4.21</b> Suppose $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ are vectors with an angle of $\theta$      |
| johnston-linear-matrix-algebra_F00845 | 71   | 1.000 | $F_{\{\mathbf{u}\}} \circ F_{\{\mathbf{v}\}}=R^{\{2 \theta\}}$          | $F_{\mathbf{u}} \circ F_{\mathbf{v}} = R^{2\theta}$     | between them. Show that $F_{\mathbf{u}} \circ F_{\mathbf{v}} = R^{2\theta}$ .                              |
| johnston-linear-matrix-algebra_F00846 | 71   | 1.000 | $S_{\{i, j\}}^{\{c\}}$                                                  | $S_{i,j}^c$                                             | for some non-zero scalar $c \in \mathbb{R}$ . Let $S_{i,j}^c$ denote the shear                             |
| johnston-linear-matrix-algebra_F00847 | 71   | 1.000 | $S_{\{1,2\}}^{\{c\}}$                                                   | $S_{1,2}^c$                                             | matrices $S_{1,2}^c$ and $S_{2,1}^c$ shown above as linear trans-                                          |
| johnston-linear-matrix-algebra_F00848 | 71   | 1.000 | $S_{\{2,1\}}^{\{c\}}$                                                   | $S_{2,1}^c$                                             | matrices $S_{1,2}^c$ and $S_{2,1}^c$ shown above as linear trans-                                          |
| johnston-linear-matrix-algebra_F00849 | 71   | 1.000 | $\left(S_{\{i, j\}}^{\{c\}}\right)^n=S_{\{i, j\}}^{\{n c\}}$            | $\left(S_{i,j}^c\right)^n = S_{i,j}^{nc}$               | (b) Show that $\left(S_{i,j}^c\right)^n = S_{i,j}^{nc}$ for all integers $n \geq 1$ (and                   |
| johnston-linear-matrix-algebra_F00850 | 71   | 1.000 | $n \geq 1$                                                              | $n \geq 1$                                              | (b) Show that $\left(S_{i,j}^c\right)^n = S_{i,j}^{nc}$ for all integers $n \geq 1$ (and                   |
| johnston-linear-matrix-algebra_F00851 | 71   | 1.000 | $S_{\{i, j\}}^{\{c\}}=I+c E_{\{i, j\}}$                                 | $S_{i,j}^c = I + cE_{i,j}$                              | $S_{i,j}^c = I + cE_{i,j}$ , where $E_{i,j}$ is the matrix with a 1 in                                     |
| johnston-linear-matrix-algebra_F00852 | 71   | 1.000 | $E_{\{i, j\}}$                                                          | $E_{i,j}$                                               | $S_{i,j}^c = I + cE_{i,j}$ , where $E_{i,j}$ is the matrix with a 1 in                                     |
| johnston-linear-matrix-algebra_F00853 | 71   | 1.000 | $E_{\{i, j\}}^{\{2\}}$                                                  | $E_{i,j}^2$                                             | its $(i, j)$ -entry and zeros elsewhere. What is $E_{i,j}^2$ ?                                             |

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| johnston-linear-matrix-algebra_F00854 | 71   | ■ 0.999 | $R_{\{\mathbf{u}\}^{\{\theta\}}}$                                                                       | $R_{\mathbf{u}}^{\theta}$                                                          | <b>**1.4.24</b> Let $R_{\mathbf{u}}^{\theta}$ be the linear transformation that rotates                              |
| johnston-linear-matrix-algebra_F00855 | 71   | ■ 1.000 | $\left[R_{\{\mathbf{u}\}^{\{\theta\}}}\right]$                                                          | $[R_{\mathbf{u}}^{\theta}]$                                                        | <b>(a)</b> Compute $[R_{\mathbf{u}}^{\theta}]$ if $\mathbf{u} = (1, 0, 0)$ . Which matrix that we                    |
| johnston-linear-matrix-algebra_F00856 | 71   | ■ 1.000 | $\mathbf{u} = (1, 0, 0)$                                                                                | $\mathbf{u} = (1, 0, 0)$                                                           | <b>(a)</b> Compute $[R_{\mathbf{u}}^{\theta}]$ if $\mathbf{u} = (1, 0, 0)$ . Which matrix that we                    |
| johnston-linear-matrix-algebra_F00857 | 71   | ■ 1.000 | $R_{\{\mathbf{u}\}^{\{\pi/4\}}}(\mathbf{v})$                                                            | $R_{\mathbf{u}}^{\pi/4}(\mathbf{v})$                                               | <b>(b)</b> Compute $R_{\mathbf{u}}^{\pi/4}(\mathbf{v})$ if $\mathbf{u} = (1, 2, 2)/3$ and $\mathbf{v} = (3, 2, 1)$ . |
| johnston-linear-matrix-algebra_F00858 | 71   | ■ 1.000 | $\mathbf{u} = (1, 2, 2) / 3$                                                                            | $\mathbf{u} = (1, 2, 2)/3$                                                         | <b>(b)</b> Compute $R_{\mathbf{u}}^{\pi/4}(\mathbf{v})$ if $\mathbf{u} = (1, 2, 2)/3$ and $\mathbf{v} = (3, 2, 1)$ . |
| johnston-linear-matrix-algebra_F00859 | 71   | ■ 1.000 | $\mathbf{v} = (3, 2, 1)$                                                                                | $\mathbf{v} = (3, 2, 1)$                                                           | <b>(b)</b> Compute $R_{\mathbf{u}}^{\pi/4}(\mathbf{v})$ if $\mathbf{u} = (1, 2, 2)/3$ and $\mathbf{v} = (3, 2, 1)$ . |
| johnston-linear-matrix-algebra_F00860 | 71   | ■ 1.000 | $R_{\{y\ z\}^{\{\pi/3\}}} \circ R_{\{x\ y\}^{\{\pi/3\}}}$                                               | $R_{yz}^{\pi/3} \circ R_{xy}^{\pi/3}$                                              | <b>(c)</b> Show that $R_{yz}^{\pi/3} \circ R_{xy}^{\pi/3}$ is a rotation matrix. Further-                            |
| johnston-linear-matrix-algebra_F00861 | 71   | ■ 1.000 | $R_{\{\mathbf{u}\}^{\{\theta\}}} = R_{\{y\ z\}^{\{\pi/3\}}} \circ R_{\{x\ y\}^{\{\pi/3\}}}$             | $R_{\mathbf{u}}^{\theta} = R_{yz}^{\pi/3} \circ R_{xy}^{\pi/3}$                    | $R_{\mathbf{u}}^{\theta} = R_{yz}^{\pi/3} \circ R_{xy}^{\pi/3}$ .                                                    |
| johnston-linear-matrix-algebra_F00862 | 73   | ■ 1.000 | $(A\ B)^{\{T\}} = A^{\{T\}}\ B^{\{T\}}$                                                                 | $(AB)^T = A^T B^T$                                                                 | then $(AB)^T = A^T B^T$ .                                                                                            |
| johnston-linear-matrix-algebra_F00863 | 73   | ■ 1.000 | $(A\ B)^{\{3\}} = A^{\{3\}}\ B^{\{3\}}$                                                                 | $(AB)^3 = A^3 B^3$                                                                 | then $(AB)^3 = A^3 B^3$ .                                                                                            |
| johnston-linear-matrix-algebra_F00864 | 73   | ■ 1.000 | $(A+B)(A-B) = A^{\{2\}} - B^{\{2\}}$                                                                    | $(A+B)(A-B) = A^2 - B^2$                                                           | then $(A+B)(A-B) = A^2 - B^2$ .                                                                                      |
| johnston-linear-matrix-algebra_F00865 | 73   | ■ 0.993 | $A^{\{2\}} = 0$                                                                                         | $A^2 = O$                                                                          | <b>*(e)</b> If $A$ is a square matrix for which $A^2 = O$ , then                                                     |
| johnston-linear-matrix-algebra_F00866 | 73   | ■ 0.998 | $R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                                        | $R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$                                   | <b>*(g)</b> If $R, S, T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations, then                     |
| johnston-linear-matrix-algebra_F00867 | 73   | ■ 1.000 | $R \circ (S \circ T) = (R \circ S) \circ T$                                                             | $R \circ (S \circ T) = (R \circ S) \circ T$                                        | $R \circ (S \circ T) = (R \circ S) \circ T$ .                                                                        |
| johnston-linear-matrix-algebra_F00868 | 73   | ■ 1.000 | $\ A\ \mathbf{v}\  = \ \mathbf{v}\ $                                                                    | $\ A\mathbf{v}\  = \ \mathbf{v}\ $                                                 | <b>(a)</b> Show (algebraically) that $\ A\mathbf{v}\  = \ \mathbf{v}\ $ for all $\mathbf{v} \in \mathbb{R}^2$ .      |
| johnston-linear-matrix-algebra_F00869 | 73   | ■ 0.717 | $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$                                     | $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$                                     | <b>*(a)</b> $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$                                                           |
| johnston-linear-matrix-algebra_F00870 | 73   | ■ 1.000 | $\left[\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$                                    | $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$                                    | <b>(b)</b> $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$                                                           |
| johnston-linear-matrix-algebra_F00871 | 73   | ■ 0.822 | $\frac{1}{\sqrt{2}} \left[\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right]$                 | $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$                 | <b>*(c)</b> $\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$                                       |
| johnston-linear-matrix-algebra_F00872 | 73   | ■ 1.000 | $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right]$                                     | $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                                     | <b>(d)</b> $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$                                                            |
| johnston-linear-matrix-algebra_F00873 | 73   | ■ 0.902 | $\left[\begin{smallmatrix} 1 & 2 \\ 3 & 4 \end{smallmatrix}\right]$                                     | $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                                     | <b>*(e)</b> $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                                                           |
| johnston-linear-matrix-algebra_F00874 | 73   | ■ 1.000 | $\frac{1}{2} \left[\begin{smallmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{smallmatrix}\right]$          | $\frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$          | <b>(f)</b> $\frac{1}{2} \begin{bmatrix} 1 & \sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$                                 |
| johnston-linear-matrix-algebra_F00875 | 73   | ■ 0.931 | $\frac{1}{9} \left[\begin{smallmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{smallmatrix}\right]$    | $\frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$    | <b>*(g)</b> $\frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}$                          |
| johnston-linear-matrix-algebra_F00876 | 73   | ■ 1.000 | $\frac{1}{9} \left[\begin{smallmatrix} -1 & 4 & 8 \\ 4 & -7 & 4 \\ 8 & 4 & -1 \end{smallmatrix}\right]$ | $\frac{1}{9} \begin{bmatrix} -1 & 4 & 8 \\ 4 & -7 & 4 \\ 8 & 4 & -1 \end{bmatrix}$ | <b>(h)</b> $\frac{1}{9} \begin{bmatrix} -1 & 4 & 8 \\ 4 & -7 & 4 \\ 8 & 4 & -1 \end{bmatrix}$                        |

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| johnston-linear-matrix-algebra_F00877 | 73   | 1.000 | $A^{500} \mathbf{v}$                                                                                        | $A^{500} \mathbf{v}$                                                                                        | Compute $A^{500} \mathbf{v}$ .                                                                                                          |
| johnston-linear-matrix-algebra_F00878 | 73   | 1.000 | $\mathbf{w} = \left( v_2, -v_1 \right)$                                                                     | $\mathbf{w} = (v_2, -v_1)$                                                                                  | vector that works is $\mathbf{w} = (v_2, -v_1)$ , since then $\mathbf{v} \cdot \mathbf{w} = v_1 v_2 - v_2 v_1 = 0$ (see                 |
| johnston-linear-matrix-algebra_F00879 | 73   | 1.000 | $\mathbf{v} \cdot \mathbf{w} = v_1 v_2 - v_2 v_1 = 0$                                                       | $\mathbf{v} \cdot \mathbf{w} = v_1 v_2 - v_2 v_1 = 0$                                                       | vector that works is $\mathbf{w} = (v_2, -v_1)$ , since then $\mathbf{v} \cdot \mathbf{w} = v_1 v_2 - v_2 v_1 = 0$ (see                 |
| johnston-linear-matrix-algebra_F00880 | 73   | 1.000 | $\mathbf{x} \in \mathbb{R}^3$                                                                               | $\mathbf{x} \in \mathbb{R}^3$                                                                               | ask for a vector $\mathbf{x} \in \mathbb{R}^3$ that is orthogonal to <i>two</i> vectors $\mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$ |
| johnston-linear-matrix-algebra_F00881 | 73   | 1.000 | $\mathbf{w} = \left( w_1, w_2, w_3 \right) \in \mathbb{R}^3$                                                | $\mathbf{w} = (w_1, w_2, w_3) \in \mathbb{R}^3$                                                             | and $\mathbf{w} = (w_1, w_2, w_3) \in \mathbb{R}^3$ . It is much more difficult to eyeball a solution in                                |
| johnston-linear-matrix-algebra_F00882 | 73   | 1.000 | $\mathbf{v} \times \mathbf{w}$                                                                              | $\mathbf{v} \times \mathbf{w}$                                                                              | <b>cross product</b> , denoted by $\mathbf{v} \times \mathbf{w}$ , is defined by                                                        |
| johnston-linear-matrix-algebra_F00883 | 74   | 0.999 | $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w})$                                                           | $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w})$                                                           | The dot product $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w})$ can similarly be shown to equal 0 (see Exer-                          |
| johnston-linear-matrix-algebra_F00884 | 74   | 0.717 | $v_1, v_2, v_3, v_1, v_2, v_3$                                                                              | $v_1, v_2, v_3, v_1, v_2, v_3$                                                                              | write the entries of $\mathbf{v}$ in order twice, like $v_1, v_2, v_3, v_1, v_2, v_3$ , and then similarly                              |
| johnston-linear-matrix-algebra_F00885 | 74   | 0.539 | $2 \times 6$                                                                                                | $2 \times 6$                                                                                                | write the entries of $\mathbf{w}$ twice underneath, creating a $2 \times 6$ array. We completely                                        |
| johnston-linear-matrix-algebra_F00886 | 75   | 0.986 | $(2, -1)$                                                                                                   | $(2, -1)$                                                                                                   | of one of them: $(2, -1)$ works, since $(1, 2) \cdot (2, -1) = 2 - 2 = 0$ .                                                             |
| johnston-linear-matrix-algebra_F00887 | 75   | 0.986 | $(1, 2) \cdot (2, -1) = 2 - 2 = 0$                                                                          | $(1, 2) \cdot (2, -1) = 2 - 2 = 0$                                                                          | of one of them: $(2, -1)$ works, since $(1, 2) \cdot (2, -1) = 2 - 2 = 0$ .                                                             |
| johnston-linear-matrix-algebra_F00888 | 75   | 1.000 | $(1, 2)$                                                                                                    | $(1, 2)$                                                                                                    | Alternatively, we could think of $(1, 2)$ as living in $\mathbb{R}^3$ (with $z$ -entry                                                  |
| johnston-linear-matrix-algebra_F00889 | 75   | 0.999 | $(1, 2, 0) \times (0, 0, 1) = (2, -1, 0)$                                                                   | $(1, 2, 0) \times (0, 0, 1) = (2, -1, 0)$                                                                   | the direction of the $z$ -axis: $(1, 2, 0) \times (0, 0, 1) = (2, -1, 0)$ , which is                                                    |
| johnston-linear-matrix-algebra_F00890 | 75   | 1.000 | $(1, 2, 3) \times (-1, 0, 1) = (2, -4, 2)$                                                                  | $(1, 2, 3) \times (-1, 0, 1) = (2, -4, 2)$                                                                  | b) $(1, 2, 3) \times (-1, 0, 1) = (2, -4, 2)$ is orthogonal to both of these vec-                                                       |
| johnston-linear-matrix-algebra_F00891 | 75   | 1.000 | $(2, -4, 2) = 0$                                                                                            | $(2, -4, 2) = 0$                                                                                            | $(2, -4, 2) = 0$ and $(-1, 0, 1) \cdot (2, -4, 2) = 0$ .                                                                                |
| johnston-linear-matrix-algebra_F00892 | 75   | 1.000 | $(-1, 0, 1) \cdot (2, -4, 2) = 0$                                                                           | $(-1, 0, 1) \cdot (2, -4, 2) = 0$                                                                           | $(2, -4, 2) = 0$ and $(-1, 0, 1) \cdot (2, -4, 2) = 0$ .                                                                                |
| johnston-linear-matrix-algebra_F00893 | 75   | 1.000 | $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$                                                       | $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$                                                       | Let $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ be vectors and let $c \in \mathbb{R}$ be a scalar. Then                       |
| johnston-linear-matrix-algebra_F00894 | 75   | 1.000 | $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$                                            | $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$                                            | a) $\mathbf{v} \times \mathbf{w} = -(\mathbf{w} \times \mathbf{v})$ ,                                                                   |
| johnston-linear-matrix-algebra_F00895 | 75   | 1.000 | $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ | $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ | b) $\mathbf{v} \times (\mathbf{w} + \mathbf{x}) = \mathbf{v} \times \mathbf{w} + \mathbf{v} \times \mathbf{x}$ ,                        |
| johnston-linear-matrix-algebra_F00896 | 75   | 1.000 | $\mathbf{v} \times \mathbf{v} = \mathbf{0}$                                                                 | $\mathbf{v} \times \mathbf{v} = \mathbf{0}$                                                                 | c) $\mathbf{v} \times \mathbf{v} = \mathbf{0}$ , and                                                                                    |
| johnston-linear-matrix-algebra_F00897 | 75   | 1.000 | $(c \mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$                                        | $(c \mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$                                        | d) $(c \mathbf{v}) \times \mathbf{w} = c(\mathbf{v} \times \mathbf{w})$ .                                                               |
| johnston-linear-matrix-algebra_F00898 | 75   | 1.000 | $\mathbf{w} \times \mathbf{v}$                                                                              | $\mathbf{w} \times \mathbf{v}$                                                                              | For property (a), we just compute $\mathbf{v} \times \mathbf{w}$ and $\mathbf{w} \times \mathbf{v}$ directly from the defini-           |
| johnston-linear-matrix-algebra_F00899 | 76   | 1.000 | $v_1^2$                                                                                                     | $v_1^2$                                                                                                     | The terms $v_1^2$                                                                                                                       |
| johnston-linear-matrix-algebra_F00900 | 76   | 1.000 | $w_1^2, v_2^2, w_2^2$                                                                                       | $w_1^2, v_2^2, w_2^2$                                                                                       | $w_1^2, v_2^2, w_2^2$ , and $v_3^2, w_3^2$                                                                                              |
| johnston-linear-matrix-algebra_F00901 | 76   | 1.000 | $v_3^2, w_3^2$                                                                                              | $v_3^2, w_3^2$                                                                                              | $w_1^2, v_2^2, w_2^2$ , and $v_3^2, w_3^2$                                                                                              |
| johnston-linear-matrix-algebra_F00902 | 76   | 1.000 | $\ \mathbf{v} \times \mathbf{w}\ $                                                                          | $\ \mathbf{v} \times \mathbf{w}\ $                                                                          | a) $\ \mathbf{v} \times \mathbf{w}\ $ ,                                                                                                 |
| johnston-linear-matrix-algebra_F00903 | 76   | 1.000 | $\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$                                | $\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$                                | b) $\sqrt{\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2}$ ,                                                       |

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| johnston-linear-matrix-algebra_F00904 | 76   | ■ 0.934 | $\ \mathbf{v}\ \ \mathbf{w}\  \sin(\theta)$                                                                        | $\ \mathbf{v}\ \ \mathbf{w}\  \sin(\theta)$                                                                        | c) $\ \mathbf{v}\ \ \mathbf{w}\  \sin(\theta)$ , and                                                                                                            |
| johnston-linear-matrix-algebra_F00905 | 76   | ■ 0.990 | $\ \mathbf{w}\  \sin(\theta)$                                                                                      | $\ \mathbf{w}\  \sin(\theta)$                                                                                      | $\ \mathbf{w}\  \sin(\theta)$ , and thus area equal to $\ \mathbf{v}\ \ \mathbf{w}\  \sin(\theta)$ .                                                            |
| johnston-linear-matrix-algebra_F00906 | 76   | ■ 1.000 | $\ \mathbf{w}\  \cos(\theta)$                                                                                      | $\ \mathbf{w}\  \cos(\theta)$                                                                                      | This process creates a right-angled triangle whose legs have lengths $\ \mathbf{w}\  \cos(\theta)$                                                              |
| johnston-linear-matrix-algebra_F00907 | 76   | ■ 1.000 | $\ \mathbf{v}\  \times \ \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$     | $\ \mathbf{v} \times \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$         | that $\ \mathbf{v} \times \mathbf{w}\ ^2 = \ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$ and then taking the square root of both         |
| johnston-linear-matrix-algebra_F00908 | 76   | ■ 1.000 | $\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$                                              | $\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$                                              | On the other hand, we can also expand $\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 - (\mathbf{v} \cdot \mathbf{w})^2$ to get                                              |
| johnston-linear-matrix-algebra_F00909 | 76   | ■ 1.000 | $\ \mathbf{v} \times \mathbf{w}\ ^2$                                                                               | $\ \mathbf{v} \times \mathbf{w}\ ^2$                                                                               | which is the exact same as the expanded form of $\ \mathbf{v} \times \mathbf{w}\ ^2$ that we computed                                                           |
| johnston-linear-matrix-algebra_F00910 | 77   | ■ 1.000 | $\sin^2(\theta) + \cos^2(\theta) = 1$                                                                              | $\sin^2(\theta) + \cos^2(\theta) = 1$                                                                              | $\sin^2(\theta) + \cos^2(\theta) = 1$ ,                                                                                                                         |
| johnston-linear-matrix-algebra_F00911 | 77   | ■ 0.718 | $1 - \cos^2(\theta) = \sin^2(\theta)$                                                                              | $1 - \cos^2(\theta) = \sin^2(\theta)$                                                                              | $1 - \cos^2(\theta) = \sin^2(\theta)$ .                                                                                                                         |
| johnston-linear-matrix-algebra_F00912 | 77   | ■ 0.754 | $\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\  \ \mathbf{w}\ ))$                                  | $\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\  \ \mathbf{w}\ ))$                                  | fact that the angle between $\mathbf{v}$ and $\mathbf{w}$ is $\theta = \arccos(\mathbf{v} \cdot \mathbf{w} / (\ \mathbf{v}\  \ \mathbf{w}\ ))$ (recall          |
| johnston-linear-matrix-algebra_F00913 | 77   | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\  \cos(\theta)$                                         | $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\  \cos(\theta)$                                         | Definition 1.2.3), which can be rearranged into the form $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\  \cos(\theta)$ .                           |
| johnston-linear-matrix-algebra_F00914 | 77   | ■ 1.000 | $\sqrt{\sin^2(\theta)} =  \sin(\theta)  = \sin(\theta)$                                                            | $\sqrt{\sin^2(\theta)} =  \sin(\theta)  = \sin(\theta)$                                                            | where the final equality makes use of the fact that $\sqrt{\sin^2(\theta)} =  \sin(\theta)  = \sin(\theta)$                                                     |
| johnston-linear-matrix-algebra_F00915 | 77   | ■ 1.000 | $\sin(\theta) \geq 0$                                                                                              | $\sin(\theta) \geq 0$                                                                                              | since $0 \leq \theta \leq \pi$ , so $\sin(\theta) \geq 0$ . This completes the proof.                                                                           |
| johnston-linear-matrix-algebra_F00916 | 77   | ■ 0.883 | $(3, 1)$                                                                                                           | $(3, 1)$                                                                                                           | b) the parallelogram in $\mathbb{R}^2$ with sides $(1, 2)$ and $(3, 1)$ ,                                                                                       |
| johnston-linear-matrix-algebra_F00917 | 77   | ■ 0.882 | $\mathbb{R}^4$                                                                                                     | $\mathbb{R}^4$                                                                                                     | c) the parallelogram in $\mathbb{R}^4$ with sides $(2, 2, -1, 1)$ and $(-1, 1, 2, 1)$ , and                                                                     |
| johnston-linear-matrix-algebra_F00918 | 77   | ■ 1.000 | $(1, 2, -1) \times (2, 1, 2) =$                                                                                    | $(1, 2, -1) \times (2, 1, 2) =$                                                                                    | a) We compute the length of the cross product: $(1, 2, -1) \times (2, 1, 2) =$                                                                                  |
| johnston-linear-matrix-algebra_F00919 | 77   | ■ 0.450 | $\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$                                                                         | $\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$                                                                         | $(5, -4, -3)$ , which has length $\sqrt{5^2 + (-4)^2 + (-3)^2} = 5\sqrt{2}$ .                                                                                   |
| johnston-linear-matrix-algebra_F00920 | 77   | ■ 1.000 | $(1, 2, 0) \times$                                                                                                 | $(1, 2, 0) \times$                                                                                                 | a $z$ -entry equal to 0 and then using the cross product: $(1, 2, 0) \times$                                                                                    |
| johnston-linear-matrix-algebra_F00921 | 77   | ■ 0.991 | $(3, 1, 0) = (0, 0, -5)$                                                                                           | $(3, 1, 0) = (0, 0, -5)$                                                                                           | $(3, 1, 0) = (0, 0, -5)$ , which has length 5.                                                                                                                  |
| johnston-linear-matrix-algebra_F00922 | 78   | ■ 1.000 | $ \cos(\theta) $                                                                                                   | $ \cos(\theta) $                                                                                                   | $ \cos(\theta) $ rather than                                                                                                                                    |
| johnston-linear-matrix-algebra_F00923 | 78   | ■ 1.000 | $\mathbf{w} \times \mathbf{x}$                                                                                     | $\mathbf{w} \times \mathbf{x}$                                                                                     | $\mathbf{w} \times \mathbf{x}$ might point                                                                                                                      |
| johnston-linear-matrix-algebra_F00924 | 78   | ■ 1.000 | $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$                                                                  | $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$                                                                  | $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})$ is called                                                                                                     |
| johnston-linear-matrix-algebra_F00925 | 78   | ■ 0.960 | $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $                                                                | $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $                                                                | sides $\mathbf{v}$ , $\mathbf{w}$ , and $\mathbf{x}$ is $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) $ .                                                   |
| johnston-linear-matrix-algebra_F00926 | 78   | ■ 0.935 | $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})  = \ \mathbf{v}\  \ \mathbf{w} \times \mathbf{x}\  \cos(\theta)$ | $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})  = \ \mathbf{v}\  \ \mathbf{w} \times \mathbf{x}\  \cos(\theta)$ | that $ \mathbf{v} \cdot (\mathbf{w} \times \mathbf{x})  = \ \mathbf{v}\  \ \mathbf{w} \times \mathbf{x}\  \cos(\theta)$ .                                       |
| johnston-linear-matrix-algebra_F00927 | 78   | ■ 0.999 | $\ \mathbf{w} \times \mathbf{x}\ $                                                                                 | $\ \mathbf{w} \times \mathbf{x}\ $                                                                                 | Figure 1.26) is a parallelogram with sides $\mathbf{w}$ and $\mathbf{x}$ , and thus has area $\ \mathbf{w} \times \mathbf{x}\ $                                 |
| johnston-linear-matrix-algebra_F00928 | 78   | ■ 0.878 | $\ \mathbf{v}\  \cos(\theta)$                                                                                      | $\ \mathbf{v}\  \cos(\theta)$                                                                                      | its base $\ \mathbf{w} \times \mathbf{x}\ $ times its height $\ \mathbf{v}\  \cos(\theta)$ , where $\theta$ is the angle between $\mathbf{w} \times \mathbf{x}$ |
| johnston-linear-matrix-algebra_F00929 | 79   | ■ 0.998 | $\mathbf{w} = (3, 2, 1)$                                                                                           | $\mathbf{w} = (3, 2, 1)$                                                                                           | *(a) $\mathbf{v} = (1, 2, 3)$ and $\mathbf{w} = (3, 2, 1)$                                                                                                      |
| johnston-linear-matrix-algebra_F00930 | 79   | ■ 1.000 | $\mathbf{v} = (87, 17, -43)$                                                                                       | $\mathbf{v} = (87, 17, -43)$                                                                                       | (b) $\mathbf{v} = (87, 17, -43)$ and $\mathbf{w} = (87, 17, -43)$                                                                                               |

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| johnston-linear-matrix-algebra_F00931 | 79   | ■ 1.000 | $\mathbf{w}=(87,17,-43)$                             | $\mathbf{w}=(87,17,-43)$                             | <b>(b)</b> $\mathbf{v}=(87,17,-43)$ and $\mathbf{w}=(87,17,-43)$                    |
| johnston-linear-matrix-algebra_F00932 | 79   | ■ 1.000 | $\mathbf{v}=(-1,2,0)$                                | $\mathbf{v}=(-1,2,0)$                                | *( <b>(c)</b> $\mathbf{v}=(-1,2,0)$ and $\mathbf{w}=(-4,1,-2)$                      |
| johnston-linear-matrix-algebra_F00933 | 79   | ■ 1.000 | $\mathbf{w}=(-4,1,-2)$                               | $\mathbf{w}=(-4,1,-2)$                               | *( <b>(c)</b> $\mathbf{v}=(-1,2,0)$ and $\mathbf{w}=(-4,1,-2)$                      |
| johnston-linear-matrix-algebra_F00934 | 79   | ■ 1.000 | $\mathbf{v}=(1,0,0)$                                 | $\mathbf{v}=(1,0,0)$                                 | <b>(d)</b> $\mathbf{v}=(1,0,0)$ and $\mathbf{w}=(0,1,0)$                            |
| johnston-linear-matrix-algebra_F00935 | 79   | ■ 1.000 | $\mathbf{w}=(0,1,0)$                                 | $\mathbf{w}=(0,1,0)$                                 | <b>(d)</b> $\mathbf{v}=(1,0,0)$ and $\mathbf{w}=(0,1,0)$                            |
| johnston-linear-matrix-algebra_F00936 | 79   | ■ 0.994 | $\mathbf{v}=(2,1)$                                   | $\mathbf{v}=(2,1)$                                   | *( <b>(a)</b> $\mathbf{v}=(2,1)$ and $\mathbf{w}=(-2,3)$                            |
| johnston-linear-matrix-algebra_F00937 | 79   | ■ 0.994 | $\mathbf{w}=(-2,3)$                                  | $\mathbf{w}=(-2,3)$                                  | *( <b>(a)</b> $\mathbf{v}=(2,1)$ and $\mathbf{w}=(-2,3)$                            |
| johnston-linear-matrix-algebra_F00938 | 78   | ■ 1.000 | $\mathbf{v}=(3,0,0)$                                 | $\mathbf{v}=(3,0,0)$                                 | <b>(b)</b> $\mathbf{v}=(3,0,0)$ and $\mathbf{w}=(0,4,0)$                            |
| johnston-linear-matrix-algebra_F00939 | 79   | ■ 1.000 | $\mathbf{w}=(0,4,0)$                                 | $\mathbf{w}=(0,4,0)$                                 | <b>(b)</b> $\mathbf{v}=(3,0,0)$ and $\mathbf{w}=(0,4,0)$                            |
| johnston-linear-matrix-algebra_F00940 | 79   | ■ 1.000 | $\mathbf{w}=(3,-1,2)$                                | $\mathbf{w}=(3,-1,2)$                                | *( <b>(c)</b> $\mathbf{v}=(1,2,3)$ and $\mathbf{w}=(3,-1,2)$                        |
| johnston-linear-matrix-algebra_F00941 | 79   | ■ 1.000 | $\mathbf{v}=(1,0,1,-2)$                              | $\mathbf{v}=(1,0,1,-2)$                              | <b>(d)</b> $\mathbf{v}=(1,0,1,-2)$ and $\mathbf{w}=(-2,1,3,1)$                      |
| johnston-linear-matrix-algebra_F00942 | 79   | ■ 1.000 | $\mathbf{w}=(-2,1,3,1)$                              | $\mathbf{w}=(-2,1,3,1)$                              | <b>(d)</b> $\mathbf{v}=(1,0,1,-2)$ and $\mathbf{w}=(-2,1,3,1)$                      |
| johnston-linear-matrix-algebra_F00943 | 78   | ■ 0.954 | $\mathbf{v}=(0,4)$                                   | $\mathbf{v}=(0,4)$                                   | * <b>(a)</b> $\mathbf{v}=(0,4)$ and $\mathbf{w}=(1,1)$                              |
| johnston-linear-matrix-algebra_F00944 | 79   | ■ 0.954 | $\mathbf{w}=(1,1)$                                   | $\mathbf{w}=(1,1)$                                   | * <b>(a)</b> $\mathbf{v}=(0,4)$ and $\mathbf{w}=(1,1)$                              |
| johnston-linear-matrix-algebra_F00945 | 79   | ■ 1.000 | $\mathbf{v}=(0,2,2)$                                 | $\mathbf{v}=(0,2,2)$                                 | <b>(b)</b> $\mathbf{v}=(0,2,2)$ and $\mathbf{w}=(1,-2,1)$                           |
| johnston-linear-matrix-algebra_F00946 | 79   | ■ 1.000 | $\mathbf{w}=(1,-2,1)$                                | $\mathbf{w}=(1,-2,1)$                                | <b>(b)</b> $\mathbf{v}=(0,2,2)$ and $\mathbf{w}=(1,-2,1)$                           |
| johnston-linear-matrix-algebra_F00947 | 79   | ■ 0.650 | $\mathbf{v}=(-1,1,-1)$                               | $\mathbf{v}=(-1,1,-1)$                               | * <b>(c)</b> $\mathbf{v}=(-1,1,-1)$ and $\mathbf{w}=(3,2,1)$                        |
| johnston-linear-matrix-algebra_F00948 | 79   | ■ 1.000 | $\mathbf{v}=(1,2,1,2)$                               | $\mathbf{v}=(1,2,1,2)$                               | <b>(d)</b> $\mathbf{v}=(1,2,1,2)$ and $\mathbf{w}=(0,-1,2,1)$                       |
| johnston-linear-matrix-algebra_F00949 | 79   | ■ 1.000 | $\mathbf{w}=(0,-1,2,1)$                              | $\mathbf{w}=(0,-1,2,1)$                              | <b>(d)</b> $\mathbf{v}=(1,2,1,2)$ and $\mathbf{w}=(0,-1,2,1)$                       |
| johnston-linear-matrix-algebra_F00950 | 79   | ■ 0.988 | $\mathbf{v}=(1,0,0), \mathbf{w}=(0,2,0)$             | $\mathbf{v}=(1,0,0), \mathbf{w}=(0,2,0)$             | * <b>(a)</b> $\mathbf{v}=(1,0,0), \mathbf{w}=(0,2,0)$ , and $\mathbf{x}=(0,0,3)$    |
| johnston-linear-matrix-algebra_F00951 | 79   | ■ 0.988 | $\mathbf{x}=(0,0,3)$                                 | $\mathbf{x}=(0,0,3)$                                 | * <b>(a)</b> $\mathbf{v}=(1,0,0), \mathbf{w}=(0,2,0)$ , and $\mathbf{x}=(0,0,3)$    |
| johnston-linear-matrix-algebra_F00952 | 79   | ■ 1.000 | $\mathbf{v}=(0,4,1), \mathbf{w}=(1,1,0)$             | $\mathbf{v}=(0,4,1), \mathbf{w}=(1,1,0)$             | <b>(b)</b> $\mathbf{v}=(0,4,1), \mathbf{w}=(1,1,0)$ , and $\mathbf{x}=(2,0,-1)$     |
| johnston-linear-matrix-algebra_F00953 | 79   | ■ 1.000 | $\mathbf{x}=(2,0,-1)$                                | $\mathbf{x}=(2,0,-1)$                                | <b>(b)</b> $\mathbf{v}=(0,4,1), \mathbf{w}=(1,1,0)$ , and $\mathbf{x}=(2,0,-1)$     |
| johnston-linear-matrix-algebra_F00954 | 78   | ■ 0.913 | $\mathbf{v}=(1,1,1), \mathbf{w}=(2,-1,1)$            | $\mathbf{v}=(1,1,1), \mathbf{w}=(2,-1,1)$            | * <b>(c)</b> $\mathbf{v}=(1,1,1), \mathbf{w}=(2,-1,1)$ , and $\mathbf{x}=(2,-2,-1)$ |
| johnston-linear-matrix-algebra_F00955 | 79   | ■ 0.913 | $\mathbf{x}=(2,-2,-1)$                               | $\mathbf{x}=(2,-2,-1)$                               | * <b>(c)</b> $\mathbf{v}=(1,1,1), \mathbf{w}=(2,-1,1)$ , and $\mathbf{x}=(2,-2,-1)$ |
| johnston-linear-matrix-algebra_F00956 | 79   | ■ 1.000 | $\mathbf{v}=(-2,1,3), \mathbf{w}=(1,-4,2)$           | $\mathbf{v}=(-2,1,3), \mathbf{w}=(1,-4,2)$           | <b>(d)</b> $\mathbf{v}=(-2,1,3), \mathbf{w}=(1,-4,2)$ , and $\mathbf{x}=(3,3,-2)$   |
| johnston-linear-matrix-algebra_F00957 | 79   | ■ 1.000 | $\mathbf{x}=(3,3,-2)$                                | $\mathbf{x}=(3,3,-2)$                                | <b>(d)</b> $\mathbf{v}=(-2,1,3), \mathbf{w}=(1,-4,2)$ , and $\mathbf{x}=(3,3,-2)$   |
| johnston-linear-matrix-algebra_F00958 | 79   | ■ 1.000 | $\left.\begin{array}{c} -2 \\ 3 \end{array}\right\}$ | $\left.\begin{array}{c} -2 \\ 3 \end{array}\right\}$ | $\left.\begin{array}{c} -2 \\ 3 \end{array}\right\}$                                |

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| johnston-linear-matrix-algebra_F00959 | 79   | 0.455 | $\begin{aligned} & \left[ \begin{array}{ll} 1 & 7 \\ 0 & 1 \end{array} \right] \text{ (b) } \left[ \begin{array}{ll} 1 & 2 \\ 3 & 4 \end{array} \right] \\ & \end{aligned}$ | $\begin{bmatrix} 1 & 7 \\ 0 & 1 \end{bmatrix}$<br>(b) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                                                  | (b) $\begin{bmatrix} 1 & 7 \\ 0 & 1 \\ 1 & 2 \\ 3 & 4 \end{bmatrix}$                                                                                                                  |
| johnston-linear-matrix-algebra_F00960 | 79   | 1.000 | $\left[ \begin{array}{ccc} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{array} \right]$                                                                                       | $\begin{bmatrix} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{bmatrix}$                                                                                 | $\begin{bmatrix} 1 & -3 & 2 \\ 0 & 2 & -7 \\ 0 & 0 & 3 \end{bmatrix}$                                                                                                                 |
| johnston-linear-matrix-algebra_F00961 | 79   | 1.000 | $\left[ \begin{array}{lll} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{array} \right]$                                                                                         | $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$                                                                                   | $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$                                                                                                                   |
| johnston-linear-matrix-algebra_F00962 | 79   | 1.000 | $ \mathbf{v} \times \mathbf{w}  =  \mathbf{w} \times \mathbf{v} $                                                                                                           | $\ \mathbf{v} \times \mathbf{w}\  = \ \mathbf{w} \times \mathbf{v}\ $                                                                                 | (b) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\ \mathbf{v} \times \mathbf{w}\  = \ \mathbf{w} \times \mathbf{v}\ $ .                                                         |
| johnston-linear-matrix-algebra_F00963 | 79   | 0.991 | $\mathbf{v} \in \mathbb{R}^3$                                                                                                                                               | $\mathbf{v} \in \mathbb{R}^3$                                                                                                                         | * (c) If $\mathbf{v} \in \mathbb{R}^3$ then $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$ .                                                                                           |
| johnston-linear-matrix-algebra_F00964 | 79   | 0.991 | $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$                                                                                                                               | $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$                                                                                                         | * (c) If $\mathbf{v} \in \mathbb{R}^3$ then $\mathbf{v} \times \mathbf{v} = \mathbf{v}^2$ .                                                                                           |
| johnston-linear-matrix-algebra_F00965 | 79   | 1.000 | $(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$                                                                                             | $(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$                                                                       | (d) If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $(\mathbf{v} + \mathbf{v}) \times \mathbf{w} = 2(\mathbf{v} \times \mathbf{w})$ .                                               |
| johnston-linear-matrix-algebra_F00966 | 79   | 1.000 | $ \mathbf{v}  = 1,  \mathbf{w}  = 1$                                                                                                                                        | $\ \mathbf{v}\  = 1, \ \mathbf{w}\  = 1$                                                                                                              | (f) There exist vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ with $\ \mathbf{v}\  = 1, \ \mathbf{w}\  = 1$ ,                                                                     |
| johnston-linear-matrix-algebra_F00967 | 79   | 1.000 | $\mathbf{v} \times \mathbf{w} = (1, 1, 1)$                                                                                                                                  | $\mathbf{v} \times \mathbf{w} = (1, 1, 1)$                                                                                                            | and $\mathbf{v} \times \mathbf{w} = (1, 1, 1)$ .                                                                                                                                      |
| johnston-linear-matrix-algebra_F00968 | 79   | 1.000 | $\mathbf{e}_1 \times \mathbf{e}_2, \mathbf{e}_2 \times \mathbf{e}_3, \mathbf{e}_3 \times \mathbf{e}_1$                                                                      | $\mathbf{e}_1 \times \mathbf{e}_2, \mathbf{e}_2 \times \mathbf{e}_3$                                                                                  | products: $\mathbf{e}_1 \times \mathbf{e}_2$ , $\mathbf{e}_2 \times \mathbf{e}_3$ , and $\mathbf{e}_3 \times \mathbf{e}_1$ .                                                          |
| johnston-linear-matrix-algebra_F00969 | 79   | 1.000 | $\mathbf{e}_3 \times \mathbf{e}_1$                                                                                                                                          | $\mathbf{e}_3 \times \mathbf{e}_1$                                                                                                                    | products: $\mathbf{e}_1 \times \mathbf{e}_2$ , $\mathbf{e}_2 \times \mathbf{e}_3$ , and $\mathbf{e}_3 \times \mathbf{e}_1$ .                                                          |
| johnston-linear-matrix-algebra_F00970 | 79   | 1.000 | $\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$                                                                                                                            | $\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$                                                                                                      | $\mathbf{v} \times \mathbf{w} = (1/3, 2/3, 2/3)$ . What is the angle between $\mathbf{v}$ and $\mathbf{w}$ ?                                                                          |
| johnston-linear-matrix-algebra_F00971 | 80   | 0.999 | $ \mathbf{v} \times \mathbf{w}  = \ \mathbf{v}\  \ \mathbf{w}\ $                                                                                                            | $\ \mathbf{v} \times \mathbf{w}\  = \ \mathbf{v}\  \ \mathbf{w}\ $                                                                                    | $\ \mathbf{v} \times \mathbf{w}\  = \ \mathbf{v}\  \ \mathbf{w}\ $ if and only if $\mathbf{v}$ and $\mathbf{w}$ are orthogonal.                                                       |
| johnston-linear-matrix-algebra_F00972 | 80   | 0.560 | $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                                       | $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                 | **1.A.10 We showed that if $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ .                                                     |
| johnston-linear-matrix-algebra_F00973 | 80   | 0.560 | $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                                       | $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                 | **1.A.10 We showed that if $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ .                                                     |
| johnston-linear-matrix-algebra_F00974 | 80   | 1.000 | $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                                       | $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$                                                                                                 | Show that $\mathbf{w} \cdot (\mathbf{v} \times \mathbf{w}) = 0$ too.                                                                                                                  |
| johnston-linear-matrix-algebra_F00975 | 80   | 1.000 | $\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$                                                                                                         | $\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$                                                                                   | a scalar. Show that $\mathbf{v} \times (c\mathbf{w}) = c(\mathbf{v} \times \mathbf{w})$ .                                                                                             |
| johnston-linear-matrix-algebra_F00976 | 80   | 1.000 | $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$                                                                       | $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$                                                 | is it true that $(\mathbf{v} \times \mathbf{w}) \times \mathbf{x} = \mathbf{v} \times (\mathbf{w} \times \mathbf{x})$ for all $\mathbf{v}, \mathbf{w}, \mathbf{x} \in \mathbb{R}^3$ ? |
| johnston-linear-matrix-algebra_F00977 | 80   | 1.000 | $\mathbf{v} \times \mathbf{w} = \mathbf{0}$                                                                                                                                 | $\mathbf{v} \times \mathbf{w} = \mathbf{0}$                                                                                                           | the stronger statement that $\mathbf{v} \times \mathbf{w} = \mathbf{0}$ if and only if $\mathbf{w} = \mathbf{0}$                                                                      |
| johnston-linear-matrix-algebra_F00978 | 80   | 1.000 | $\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$                                                                                                               | $\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$                                                                                         | $\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$ even if $\mathbf{w} \neq \mathbf{x}$ .                                                                                  |
| johnston-linear-matrix-algebra_F00979 | 80   | 1.000 | $\mathbf{w} \neq \mathbf{x}$                                                                                                                                                | $\mathbf{w} \neq \mathbf{x}$                                                                                                                          | $\mathbf{v} \times \mathbf{w} = \mathbf{v} \times \mathbf{x}$ even if $\mathbf{w} \neq \mathbf{x}$ .                                                                                  |
| johnston-linear-matrix-algebra_F00980 | 80   | 1.000 | $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$                       | $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$ | (a) Show that $\mathbf{v} \cdot (\mathbf{w} \times \mathbf{x}) = \mathbf{w} \cdot (\mathbf{x} \times \mathbf{v}) = \mathbf{x} \cdot (\mathbf{v} \times \mathbf{w})$ .                 |
| johnston-linear-matrix-algebra_F00981 | 80   | 1.000 | $\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$                                      | $\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$                | $\mathbf{v} \times (\mathbf{w} \times \mathbf{x}) = (\mathbf{v} \cdot \mathbf{x})\mathbf{w} - (\mathbf{v} \cdot \mathbf{w})\mathbf{x}$ . [Side note: This is called                   |
| johnston-linear-matrix-algebra_F00982 | 81   | 1.000 | A-D-B                                                                                                                                                                       | $A - D - B$                                                                                                                                           | path: $A - D - B$ .                                                                                                                                                                   |

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|---------------------------------------|------|---------|---------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F00983 | 82   | ■ 1.000 | $a_{\{i, j\}}=a_{\{j, i\}}$                                                     | $a_{i,j} = a_{j,i}$                   | $a_{i,j} = a_{j,i}$ for all $i, j$                                                         |
| johnston-linear-matrix-algebra-F00984 | 82   | ■ 1.000 | $i, j$                                                                          | $i, j$                                | $a_{i,j} = a_{j,i}$ for all $i, j$                                                         |
| johnston-linear-matrix-algebra-F00985 | 82   | ■ 0.999 | $\left.A=A^{\{T\}}\right)$                                                      | $A = A^T)$                            | $A = A^T)$ .                                                                               |
| johnston-linear-matrix-algebra-F00986 | 82   | ■ 1.000 | $a_{\{i, j\}}=1$                                                                | $a_{i,j} = 1$                         | the number of paths of length 1 between vertices $i$ and $j$ (since $a_{i,j} = 1$ if there |
| johnston-linear-matrix-algebra-F00987 | 82   | ■ 1.000 | $a_{\{i, j\}}=0$                                                                | $a_{i,j} = 0$                         | is such a path and $a_{i,j} = 0$ otherwise). Remarkably, the entries of higher powers      |
| johnston-linear-matrix-algebra-F00988 | 83   | ■ 1.000 | $\left[A^{\{k\}}\right]_{\{i, j\}}$                                             | $[A^k]_{i,j}$                         | notation $[A^k]_{i,j}$                                                                     |
| johnston-linear-matrix-algebra-F00989 | 83   | ■ 1.000 | $k \geq 1$                                                                      | $k \geq 1$                            | Suppose $A \in \mathcal{M}_n$ is the adjacency matrix of a graph, and let $k \geq 1$ be    |
| johnston-linear-matrix-algebra-F00990 | 83   | ■ 1.000 | $k=1$                                                                           | $k = 1$                               | <i>Proof.</i> We already noted why this theorem is true when $k = 1$ . To show that it     |
| johnston-linear-matrix-algebra-F00991 | 83   | ■ 1.000 | $k=2$                                                                           | $k = 2$                               | is true when $k = 2$ , recall from the definition of matrix multiplication that            |
| johnston-linear-matrix-algebra-F00992 | 83   | ■ 0.981 | $a_{\{i, 1\}} a_{\{1, j\}}$                                                     | $a_{i,1}a_{1,j}$                      | The first term in the sum above ( $a_{i,1}a_{1,j}$ ) equals 1 exactly if there is an edge  |
| johnston-linear-matrix-algebra-F00993 | 83   | ■ 1.000 | $a_{\{i, 2\}} a_{\{2, j\}}$                                                     | $a_{i,2}a_{2,j}$                      | Similarly, $a_{i,2}a_{2,j}$ counts the number of paths of length 2 between vertices        |
| johnston-linear-matrix-algebra-F00994 | 83   | ■ 1.000 | $\left[A^{\{2\}}\right]_{\{i, j\}}$                                             | $[A^2]_{i,j}$                         | terms up, we see that $[A^2]_{i,j}$ counts the number of paths of length 2 between         |
| johnston-linear-matrix-algebra-F00995 | 84   | ■ 1.000 | $A^{\{6\}}$                                                                     | $A^6$                                 | entire matrix $A^6$                                                                        |
| johnston-linear-matrix-algebra-F00996 | 84   | ■ 0.968 | (A)                                                                             | (A)                                   | from the first vertex (A) to the fourth vertex (D). Notice that this                       |
| johnston-linear-matrix-algebra-F00997 | 84   | ■ 0.968 | (D)                                                                             | (D)                                   | from the first vertex (A) to the fourth vertex (D). Notice that this                       |
| johnston-linear-matrix-algebra-F00998 | 84   | ■ 1.000 | $A^{\{6\}}=A A A A A A$                                                         | $A^6 = AAAAAA$                        | laboriously multiply five times to get $A^6 = AAAAAA$ , it is quicker                      |
| johnston-linear-matrix-algebra-F00999 | 84   | ■ 1.000 | $A^{\{6\}}=\left(A^{\{3\}}\right)^{\{2\}}$                                      | $A^6 = \left(A^3\right)^2$            | to recall that $A^6 = (A^3)^2$ , so we can just square the matrix that we                  |
| johnston-linear-matrix-algebra-F01000 | 84   | ■ 1.000 | (B)                                                                             | (B)                                   | length 6 from the second vertex (B) to the third vertex (C).                               |
| johnston-linear-matrix-algebra-F01001 | 84   | ■ 1.000 | (C)                                                                             | (C)                                   | length 6 from the second vertex (B) to the third vertex (C).                               |
| johnston-linear-matrix-algebra-F01002 | 85   | ■ 1.000 | $\left[A^{\{2\}}\right]_{\{5,1\}}$                                              | $[A^2]_{5,1}$                         | stopover in between is then $[A^2]_{5,1}$ , so we compute $A^2$ :                          |
| johnston-linear-matrix-algebra-F01003 | 85   | ■ 1.000 | $\left[A^{\{3\}}\right]_{\{4,6\}}+\left[A^{\{2\}}\right]_{\{4,6\}}+a_{\{4,6\}}$ | $[A^3]_{4,6} + [A^2]_{4,6} + a_{4,6}$ | $[A^3]_{4,6} + [A^2]_{4,6} + a_{4,6}$ , so we start by computing $[A^3]_{4,6}$ , which is  |
| johnston-linear-matrix-algebra-F01004 | 85   | ■ 1.000 | $\left[A^{\{3\}}\right]_{\{4,6\}}$                                              | $[A^3]_{4,6}$                         | $[A^3]_{4,6} + [A^2]_{4,6} + a_{4,6}$ , so we start by computing $[A^3]_{4,6}$ , which is  |
| johnston-linear-matrix-algebra-F01005 | 86   | ■ 1.000 | E                                                                               | $E$                                   | b) of length 4 from C to E, and                                                            |
| johnston-linear-matrix-algebra-F01006 | 86   | ■ 1.000 | $\left[A^{\{4\}}\right]_{\{3,5\}}$                                              | $[A^4]_{3,5}$                         | b) We need to compute $[A^4]_{3,5}$ , which is the dot product of the 3rd row              |
| johnston-linear-matrix-algebra-F01007 | 86   | ■ 0.995 | $A^{\{87\}}$                                                                    | $A^{87}$                              | c) We could compute $A^{87}$ and look at its (1,3)-entry (ugh!), but an                    |
| johnston-linear-matrix-algebra-F01008 | 87   | ■ 0.995 | $0,0,1,0,0$                                                                     | $0,0,1,0,0$                           | path of length $k$ from B to C) and $(0,0,1,0,0)$ when $k$ is even (and                    |
| johnston-linear-matrix-algebra-F01009 | 87   | ■ 1.000 | $\left[A^{\{2\}}\right]_{\{1,4\}}$                                              | $[A^2]_{1,4}$                         | D, we compute $[A^2]_{1,4}$ , which is the dot product of the 1st row of                   |

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| johnston-linear-matrix-algebra-F01010 | 88   | ■ 1.000 | $\left[A^{\{3\}}\right]_{\{2,3\}}$                                                                                                          | $[A^3]_{2,3}$                                                                                                                         | $[A^3]_{2,3}$ is also the                                                                                                             |
| johnston-linear-matrix-algebra-F01011 | 88   | ■ 1.000 | $(0,1,1,1) \cdot (1,1,2,1)=4$                                                                                                               | $(0,1,1,1) \cdot (1,1,2,1) = 4$                                                                                                       | $(0,1,1,1) \cdot (1,1,2,1) = 4$                                                                                                       |
| johnston-linear-matrix-algebra-F01012 | 88   | ■ 1.000 | $\sum_{j=1}^n w_{\{j\}}$                                                                                                                    | $\sum_{j=1}^n w_j$                                                                                                                    | $\sum_{j=1}^n w_j$ means                                                                                                              |
| johnston-linear-matrix-algebra-F01013 | 88   | ■ 1.000 | $w_{\{1\}}+w_{\{2\}}+\cdots+w_{\{n\}}$                                                                                                      | $w_1 + w_2 + \cdots + w_n$                                                                                                            | $w_1 + w_2 + \cdots + w_n$ ,                                                                                                          |
| johnston-linear-matrix-algebra-F01014 | 88   | ■ 0.964 | $n \approx 7.5$                                                                                                                             | $n \approx 7.5$                                                                                                                       | in the world ( $n \approx 7.5$ billion) and let $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$ be such                        |
| johnston-linear-matrix-algebra-F01015 | 88   | ■ 1.000 | $1 \leq j \leq n, w_{\{j\}}$                                                                                                                | $1 \leq j \leq n, w_j$                                                                                                                | that, for each $1 \leq j \leq n$ , $w_j$ is the number of friends that the $j$ -th person                                             |
| johnston-linear-matrix-algebra-F01016 | 89   | ■ 1.000 | $(3+2+1+2) / 4=2$                                                                                                                           | $(3 + 2 + 1 + 2)/4 = 2$                                                                                                               | $(3 + 2 + 1 + 2)/4 = 2$ .                                                                                                             |
| johnston-linear-matrix-algebra-F01017 | 89   | ■ 1.000 | $2 \leq 2.25$                                                                                                                               | $2 \leq 2.25$                                                                                                                         | Since $2 \leq 2.25$ , we                                                                                                              |
| johnston-linear-matrix-algebra-F01018 | 89   | ■ 0.999 | $2 \times 2=4$                                                                                                                              | $2 \times 2 = 4$                                                                                                                      | $2 \times 2 = 4$ names,                                                                                                               |
| johnston-linear-matrix-algebra-F01019 | 89   | ■ 0.835 | $1 \times 1=1$                                                                                                                              | $1 \times 1 = 1$                                                                                                                      | $1 \times 1 = 1$ .                                                                                                                    |
| johnston-linear-matrix-algebra-F01020 | 89   | ■ 1.000 | $(2+1+2+3+2+3+3+2) / 8=$                                                                                                                    | $(2 + 1 + 2 + 3 + 2 + 3 + 3 + 2)/8 =$                                                                                                 | (excluding the bold headers), which is $(2 + 1 + 2 + 3 + 2 + 3 + 3 + 2)/8 =$                                                          |
| johnston-linear-matrix-algebra-F01021 | 89   | ■ 1.000 | $w_{\{j\}}$                                                                                                                                 | $w_j$                                                                                                                                 | In general, the $j$ -th person creates $w_j$ lists—one for each of their                                                              |
| johnston-linear-matrix-algebra-F01022 | 89   | ■ 1.000 | $w_{\{j\}}^{\{2\}}$                                                                                                                         | $w_j^2$                                                                                                                               | of those lists contain $w_j$ entries. It follows that person $j$ contributes $w_j^2$                                                  |
| johnston-linear-matrix-algebra-F01023 | 89   | ■ 1.000 | $\sum_{j=1}^n w_{\{j\}}^{\{2\}}$                                                                                                            | $\sum_{j=1}^n w_j^2$                                                                                                                  | entries to these lists, so there are $\sum_{j=1}^n w_j^2$ total friends of friends listed.                                            |
| johnston-linear-matrix-algebra-F01024 | 89   | ■ 1.000 | $\mathbf{v}=(1,1,\ldots,1) \in \mathbb{R}^{\{n\}}$                                                                                          | $\mathbf{v} = (1,1,\dots,1) \in \mathbb{R}^n$                                                                                         | If we let $\mathbf{v} = (1,1,\dots,1) \in \mathbb{R}^n$ and then apply the Cauchy–Schwarz                                             |
| johnston-linear-matrix-algebra-F01025 | 89   | ■ 1.000 | $n \sum_{j=1}^n w_{\{j\}}$                                                                                                                  | $n \sum_{j=1}^n w_j$                                                                                                                  | by $n \sum_{j=1}^n w_j$ , results in the inequality                                                                                   |
| johnston-linear-matrix-algebra-F01026 | 90   | ■ 1.000 | $A=A^{\{T\}}$                                                                                                                               | $A = A^T$                                                                                                                             | is symmetric (i.e., has the property $A = A^T$ ).                                                                                     |
| johnston-linear-matrix-algebra-F01027 | 90   | ■ 1.000 | $\left[\begin{array}{llll}0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0\end{array}\right]$                               | $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$                                      | $\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$                                      |
| johnston-linear-matrix-algebra-F01028 | 90   | ■ 1.000 | $\left[\begin{array}{lllll}0 & 1 & 1 & 1 & 1 \\ & 1 & 0 & 1 & 1 \\ & 1 & 1 & 0 & 1 \\ & 1 & 1 & 1 & 0 \\ & 1 & 1 & 1 & 0\end{array}\right]$ | $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$ | $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F01029 | 90   | ■ 1.000 | $\left[\begin{array}{llll}2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1\end{array}\right]$                               | $\begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \end{bmatrix}$                                      | $\begin{bmatrix} 2 & 0 & 0 & 1 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \end{bmatrix}$                                      |
| johnston-linear-matrix-algebra-F01030 | 91   | ■ 1.000 | $A^{\{n\}}$                                                                                                                                 | $A^n$                                                                                                                                 | (c) Use part (b) to find a formula for $A^n$ . [Hint: Your                                                                            |
| johnston-linear-matrix-algebra-F01031 | 92   | ■ 1.000 | $x, y$                                                                                                                                      | $x, y$                                                                                                                                | use $x, y$ , and $z$ if                                                                                                               |

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| johnston-linear-matrix-algebra-F01032 | 92   | ■ 1.000 | $x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}$                                                          | $x_1, x_2, \dots, x_n$                                          | use $x_1, x_2, \dots, x_n$ if                                                                                                                              |
| johnston-linear-matrix-algebra-F01033 | 92   | ■ 1.000 | $a_{\{1\}}, a_{\{2\}}, \ldots, a_{\{n\}}$                                                          | $a_1, a_2, \dots, a_n$                                          | where $a_1, a_2, \dots, a_n$ and $b$ are constants called the <b>coefficients</b> of the linear                                                            |
| johnston-linear-matrix-algebra-F01034 | 92   | ■ 1.000 | 4 x-6 y=-3                                                                                         | $4x - 6y = -3$                                                  | is linear. In particular, it is equivalent to the equation $4x - 6y = -3$ , which is in                                                                    |
| johnston-linear-matrix-algebra-F01035 | 93   | ■ 1.000 | y=m x+b                                                                                            | $y = mx + b$                                                    | the form $y = mx + b$ ,                                                                                                                                    |
| johnston-linear-matrix-algebra-F01036 | 93   | ■ 1.000 | x=2                                                                                                | $x = 2$                                                         | Try plugging $x = 2$ ,                                                                                                                                     |
| johnston-linear-matrix-algebra-F01037 | 93   | ■ 1.000 | y=1                                                                                                | $y = 1$                                                         | $y = 1$ back into the                                                                                                                                      |
| johnston-linear-matrix-algebra-F01038 | 93   | ■ 1.000 | a x+b y=c                                                                                          | $ax + by = c$                                                   | example, the general equation of a line is $ax + by = c$ , and the general equation                                                                        |
| johnston-linear-matrix-algebra-F01039 | 93   | ■ 1.000 | a x+b y+c z=d                                                                                      | $ax + by + cz = d$                                              | of a plane is $ax + by + cz = d$ , both of which are linear equations (see Figure 2.1).                                                                    |
| johnston-linear-matrix-algebra-F01040 | 93   | ■ 1.000 | $\mathbf{x} =$                                                                                     | $\mathbf{x} =$                                                  | • a <b>solution</b> of a system of linear equations is a vector $\mathbf{x} =$                                                                             |
| johnston-linear-matrix-algebra-F01041 | 93   | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right)$                                             | $(x_1, x_2, \dots, x_n)$                                        | $(x_1, x_2, \dots, x_n)$ whose entries satisfy all of the linear equations in the                                                                          |
| johnston-linear-matrix-algebra-F01042 | 94   | ■ 0.834 | $\mathbf{x} = (2, 1)$                                                                              | $\mathbf{x} = (2, 1)$                                           | it is located at the point $(2, 1)$ . The vector $\mathbf{x} = (2, 1)$ thus seems to be the                                                                |
| johnston-linear-matrix-algebra-F01043 | 94   | ■ 1.000 | 3 y=3                                                                                              | $3y = 3$                                                        | new equation $3y = 3$ , which tells us that $y = 1$ . Plugging $y = 1$ back into the                                                                       |
| johnston-linear-matrix-algebra-F01044 | 94   | ■ 1.000 | x+2 y=4                                                                                            | $x + 2y = 4$                                                    | original equation $x + 2y = 4$ then tells us that $x = 2$ .                                                                                                |
| johnston-linear-matrix-algebra-F01045 | 94   | ■ 1.000 | x+2 y=4, -x+y=-1                                                                                   | $x + 2y = 4, -x + y = -1$                                       | <b>Figure 2.2:</b> The system of linear equations $x + 2y = 4, -x + y = -1$ is represented graph-                                                          |
| johnston-linear-matrix-algebra-F01046 | 94   | ■ 0.943 | x+2 y                                                                                              | $x + 2y$                                                        | $x + 2y$ cannot simultaneously equal 4 and 3. Geometrically, these two equations                                                                           |
| johnston-linear-matrix-algebra-F01047 | 95   | ■ 1.000 | $A \mathbf{x} = \mathbf{b}$                                                                        | $A\mathbf{x} = \mathbf{b}$                                      | can be rewritten as the single matrix equation $A\mathbf{x} = \mathbf{b}$ , where $A \in \mathcal{M}_{m,n}$ is the <b>co-</b>                              |
| johnston-linear-matrix-algebra-F01048 | 95   | ■ 0.950 | $a_{\{i, j\}}, \mathbf{b} = \left(b_{\{1\}}, b_{\{2\}}, \ldots, b_{\{m\}}\right) \in \mathbb{R}^m$ | $a_{i,j}, \mathbf{b} = (b_1, b_2, \dots, b_m) \in \mathbb{R}^m$ | <b>efficient matrix</b> whose $(i, j)$ -entry is $a_{i,j}$ , $\mathbf{b} = (b_1, b_2, \dots, b_m) \in \mathbb{R}^m$ is a vector                            |
| johnston-linear-matrix-algebra-F01049 | 95   | ■ 0.997 | $\mathbf{x} = \left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right) \in \mathbb{R}^n$               | $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$          | containing the constants from the right-hand side, and $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$                                              |
| johnston-linear-matrix-algebra-F01050 | 96   | ■ 0.743 | $\mathbf{x}_{\{1\}}$                                                                               | $\mathbf{x}_1$                                                  | heads of $\mathbf{x}_1$ and $\mathbf{x}_2$                                                                                                                 |
| johnston-linear-matrix-algebra-F01051 | 96   | ■ 0.743 | $\mathbf{x}_{\{2\}}$                                                                               | $\mathbf{x}_2$                                                  | heads of $\mathbf{x}_1$ and $\mathbf{x}_2$                                                                                                                 |
| johnston-linear-matrix-algebra-F01052 | 96   | ■ 1.000 | $A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$                                             | $A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$          | If $A\mathbf{x} = \mathbf{b}$ is the matrix form of the linear system (where $A \in \mathcal{M}_{m,n}, \mathbf{x} \in \mathbb{R}^n$ ,                      |
| johnston-linear-matrix-algebra-F01053 | 96   | ■ 1.000 | $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$                                                  | $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$               | means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$ . |
| johnston-linear-matrix-algebra-F01054 | 96   | ■ 1.000 | $A \mathbf{x}_1 = \mathbf{b}$                                                                      | $A\mathbf{x}_1 = \mathbf{b}$                                    | means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$ . |
| johnston-linear-matrix-algebra-F01055 | 96   | ■ 1.000 | $A \mathbf{x}_2 = \mathbf{b}$                                                                      | $A\mathbf{x}_2 = \mathbf{b}$                                    | means that there exist vectors $\mathbf{x}_1 \neq \mathbf{x}_2 \in \mathbb{R}^n$ such that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$ . |
| johnston-linear-matrix-algebra-F01056 | 96   | ■ 1.000 | $(1 - c) \mathbf{x}_{\{1\}} + c \mathbf{x}_{\{2\}}$                                                | $(1 - c)\mathbf{x}_1 + c\mathbf{x}_2$                           | so every vector of the form $(1 - c)\mathbf{x}_1 + c\mathbf{x}_2$ is a solution of the linear system.                                                      |
| johnston-linear-matrix-algebra-F01057 | 96   | ■ 1.000 | 3 z=6                                                                                              | $3z = 6$                                                        | The final equation in this system tells us that $3z = 6$ , so $z = 2$ . Plugging $z = 2$                                                                   |
| johnston-linear-matrix-algebra-F01058 | 96   | ■ 1.000 | z=2                                                                                                | $z = 2$                                                         | The final equation in this system tells us that $3z = 6$ , so $z = 2$ . Plugging $z = 2$                                                                   |
| johnston-linear-matrix-algebra-F01059 | 97   | ■ 1.000 | 2 y=16                                                                                             | $2y = 16$                                                       | equation: $2y = 16$ , so $y = 8$ . Finally, plugging $y = 8$ into the first equation then                                                                  |

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| johnston-linear-matrix-algebra-F01060 | 97   | 1.000 | y=8                                                                                                                                            | $y = 8$                                                                                       | equation: $2y = 16$ , so $y = 8$ . Finally, plugging $y = 8$ into the first equation then                        |
| johnston-linear-matrix-algebra-F01061 | 97   | 1.000 | x+24=9                                                                                                                                         | $x + 24 = 9$                                                                                  | tells us that $x + 24 = 9$ , so $x = -15$ . This system of linear equations thus has a                           |
| johnston-linear-matrix-algebra-F01062 | 97   | 1.000 | x=- 15                                                                                                                                         | $x = -15$                                                                                     | tells us that $x + 24 = 9$ , so $x = -15$ . This system of linear equations thus has a                           |
| johnston-linear-matrix-algebra-F01063 | 97   | 1.000 | (x, y, z)=(-15,8,2)                                                                                                                            | $(x, y, z) = (-15, 8, 2)$                                                                     | unique solution, and it is $(x, y, z) = (-15, 8, 2)$ .                                                           |
| johnston-linear-matrix-algebra-F01064 | 98   | 1.000 | y-3 z=2                                                                                                                                        | $y - 3z = 2$                                                                                  | $y - 3z = 2$ if we                                                                                               |
| johnston-linear-matrix-algebra-F01065 | 98   | 1.000 | [A \mid \mathbf{b}]                                                                                                                            | $[A \mid \mathbf{b}]$                                                                         | $[A \mid \mathbf{b}]$ , so that we only have to write down the coefficients in the linear system                 |
| johnston-linear-matrix-algebra-F01066 | 99   | 1.000 | c R_{j}                                                                                                                                        | $cR_j$                                                                                        | <b>Multiplication.</b> Multiplying row $j$ by a non-zero scalar $c \in \mathbb{R}$ is denoted by $cR_j$ .        |
| johnston-linear-matrix-algebra-F01067 | 99   | 1.000 | R_{i} \leftrightarrow R_{j}                                                                                                                    | $R_i \leftrightarrow R_j$                                                                     | <b>Swap.</b> Swapping rows $i$ and $j$ is denoted by $R_i \leftrightarrow R_j$ .                                 |
| johnston-linear-matrix-algebra-F01068 | 99   | 0.974 | )+c(                                                                                                                                           | $) + c($                                                                                      | <b>Addition.</b> For any scalar $c \in \mathbb{R}$ , replacing row $i$ by (row i) + c(row j) is denoted by       |
| johnston-linear-matrix-algebra-F01069 | 99   | 0.974 | )                                                                                                                                              | $)$                                                                                           | <b>Addition.</b> For any scalar $c \in \mathbb{R}$ , replacing row $i$ by (row i) + c(row j) is denoted by       |
| johnston-linear-matrix-algebra-F01070 | 99   | 0.996 | R_{i}+c R_{j}                                                                                                                                  | $R_i + cR_j$                                                                                  | $R_i + cR_j$ .                                                                                                   |
| johnston-linear-matrix-algebra-F01071 | 100  | 0.997 | (x, y, z)=                                                                                                                                     | $(x, y, z) =$                                                                                 | off from the right-hand side of the bottom augmented matrix: $(x, y, z) =$                                       |
| johnston-linear-matrix-algebra-F01072 | 100  | 1.000 | \frac{1}{c} R_{j}                                                                                                                              | $\frac{1}{c}R_j$                                                                              | particular, the multiplication operation $cR_j$ is reversed by $\frac{1}{c}R_j$ , the swap                       |
| johnston-linear-matrix-algebra-F01073 | 100  | 1.000 | R_{i}-c R_{j}                                                                                                                                  | $R_i - cR_j$                                                                                  | is undone by $R_i - cR_j$ .                                                                                      |
| johnston-linear-matrix-algebra-F01074 | 100  | 1.000 | c \neq 0                                                                                                                                       | $c \neq 0$                                                                                    | undoing the row operations). In fact, this is exactly why we require $c \neq 0$                                  |
| johnston-linear-matrix-algebra-F01075 | 100  | 1.000 | \mathbf{x}=(1,1)                                                                                                                               | $\mathbf{x} = (1, 1)$                                                                         | the system on the left has a unique solution $\mathbf{x} = (1, 1)$ , but the system on                           |
| johnston-linear-matrix-algebra-F01076 | 100  | 1.000 | \mathbf{x}=(3,0)                                                                                                                               | $\mathbf{x} = (3, 0)$                                                                         | the right has infinitely many others, such as $\mathbf{x} = (3, 0)$ .                                            |
| johnston-linear-matrix-algebra-F01077 | 101  | 1.000 | \left[\begin{array}{lll}1 & 2 & 3 \\ 0 & 1 & 0\end{array}\right]                                                                               | $\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 1 & 0\end{array}\right]$                            | a) $\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 1 & 0\end{array}\right]$                                            |
| johnston-linear-matrix-algebra-F01078 | 101  | 1.000 | \left[\begin{array}{lll}1 & 0 & 2 \\ 0 & 1 & 4\end{array}\right]                                                                               | $\left[\begin{array}{ccc}1 & 0 & 2 \\ 0 & 1 & 4\end{array}\right]$                            | b) $\left[\begin{array}{ccc}1 & 0 & 2 \\ 0 & 1 & 4\end{array}\right]$                                            |
| johnston-linear-matrix-algebra-F01079 | 101  | 1.000 | \left[\begin{array}{cccc}2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0\end{array}\right]                                                    | $\left[\begin{array}{cccc}2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0\end{array}\right]$ | c) $\left[\begin{array}{cccc}2 & 0 & -1 & 5 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0\end{array}\right]$                 |
| johnston-linear-matrix-algebra-F01080 | 101  | 1.000 | \left[\begin{array}{cccc}1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0\end{array}\right]                                                    | $\left[\begin{array}{cccc}1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0\end{array}\right]$ | d) $\left[\begin{array}{cccc}1 & 2 & 3 & 4 \\ 0 & 0 & 2 & -1 \\ 0 & 0 & 1 & 0\end{array}\right]$                 |
| johnston-linear-matrix-algebra-F01081 | 104  | 1.000 | 0=1                                                                                                                                            | $0 = 1$                                                                                       | Of particular note is the third equation, which says $0 = 1$ . This tells us that the                            |
| johnston-linear-matrix-algebra-F01082 | 104  | 0.513 | \left[\begin{array}{cccc}0 & 0 & \cdots & 0\end{array}\right] \rightarrow \left[\begin{array}{cccc}0 & 0 & \cdots & 0\end{array}\right] \mid b | $\left[\begin{array}{cccc}0 & 0 & \cdots & 0\end{array}\right] \mid b$                        | forms have a row that looks like $\left[\begin{array}{cccc}0 & 0 & \cdots & 0\end{array}\right] \mid b$ for some |
| johnston-linear-matrix-algebra-F01083 | 104  | 1.000 | b \neq 0                                                                                                                                       | $b \neq 0$                                                                                    | scalar $b \neq 0$ .                                                                                              |

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| johnston-linear-matrix-algebra-F01084 | 105  | 1.000 | $x=3-4y$                                                                                                            | $x = 3 - 4y$                                                                                                   | that $x = 3 - 4y$ and                                                                                                 |
| johnston-linear-matrix-algebra-F01085 | 105  | 1.000 | $y=y$                                                                                                               | $y = y$                                                                                                        | $y = y$ .                                                                                                             |
| johnston-linear-matrix-algebra-F01086 | 106  | 0.996 | $x+4y=3$                                                                                                            | $x + 4y = 3$                                                                                                   | <b>Figure 2.5:</b> Geometrically, the solution set to the system of linear equations $x + 4y = 3$ ,                   |
| johnston-linear-matrix-algebra-F01087 | 107  | 1.000 | $\left[\begin{array}{cccc c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$      | $\left[\begin{array}{cccc c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ | a) $\left[\begin{array}{cccc c} 1 & -1 & 0 & 1 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$     |
| johnston-linear-matrix-algebra-F01088 | 107  | 0.997 | $\left[\begin{array}{ccc c} 1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8 \end{array}\right]$                  | $\left[\begin{array}{ccc c} 1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8 \end{array}\right]$             | b) $\left[\begin{array}{ccc c} 1 & 2 & -4 & -4 \\ 0 & 3 & -1 & 2 \\ 0 & 0 & 8 & 8 \end{array}\right]$                 |
| johnston-linear-matrix-algebra-F01089 | 107  | 1.000 | $w, x, y, z$                                                                                                        | $w, x, y, z$                                                                                                   | corresponding to the columns from left-to-right as $w, x, y, z$ , then $w$                                            |
| johnston-linear-matrix-algebra-F01090 | 107  | 1.000 | $w$                                                                                                                 | $w$                                                                                                            | corresponding to the columns from left-to-right as $w, x, y, z$ , then $w$                                            |
| johnston-linear-matrix-algebra-F01091 | 107  | 1.000 | $w=2+x-z$                                                                                                           | $w = 2 + x - z$                                                                                                | hand-side) $w = 2 + x - z$ , and the second equation says $y = 1 + z$                                                 |
| johnston-linear-matrix-algebra-F01092 | 107  | 1.000 | $y=1+z$                                                                                                             | $y = 1 + z$                                                                                                    | hand-side) $w = 2 + x - z$ , and the second equation says $y = 1 + z$                                                 |
| johnston-linear-matrix-algebra-F01093 | 107  | 0.818 | $\left[\begin{array}{l} 0 & 0 & 0 \end{array}\right] \begin{array}{l} \text{ } \\ \text{ } \\ \text{ } \end{array}$ | $\left[\begin{array}{ccc c} 0 & 0 & 0 & b \end{array}\right]$                                                  | the form $\left[\begin{array}{ccc c} 0 & 0 & 0 & b \end{array}\right]$ , we know the system has a unique solution. To |
| johnston-linear-matrix-algebra-F01094 | 107  | 1.000 | $x, y, z$                                                                                                           | $x, y, z$                                                                                                      | as $x, y, z$ , then the third equation says $8z = 8$ , so $z = 1$ . Plugging that                                     |
| johnston-linear-matrix-algebra-F01095 | 107  | 1.000 | $8z=8$                                                                                                              | $8z = 8$                                                                                                       | as $x, y, z$ , then the third equation says $8z = 8$ , so $z = 1$ . Plugging that                                     |
| johnston-linear-matrix-algebra-F01096 | 107  | 1.000 | $z=1$                                                                                                               | $z = 1$                                                                                                        | as $x, y, z$ , then the third equation says $8z = 8$ , so $z = 1$ . Plugging that                                     |
| johnston-linear-matrix-algebra-F01097 | 107  | 1.000 | $3y-1=2$                                                                                                            | $3y - 1 = 2$                                                                                                   | into the second equation tells us $3y - 1 = 2$ , so $y = 1$ . Plugging those                                          |
| johnston-linear-matrix-algebra-F01098 | 107  | 1.000 | $x+2-4=-4$                                                                                                          | $x + 2 - 4 = -4$                                                                                               | values into the first equation tells us $x + 2 - 4 = -4$ , so $x = -2$ . The                                          |
| johnston-linear-matrix-algebra-F01099 | 107  | 1.000 | $x=-2$                                                                                                              | $x = -2$                                                                                                       | values into the first equation tells us $x + 2 - 4 = -4$ , so $x = -2$ . The                                          |
| johnston-linear-matrix-algebra-F01100 | 107  | 1.000 | $(x, y, z)=(-2, 1, 1)$                                                                                              | $(x, y, z) = (-2, 1, 1)$                                                                                       | unique solution is thus $(x, y, z) = (-2, 1, 1)$ .                                                                    |
| johnston-linear-matrix-algebra-F01101 | 107  | 1.000 | $0x+0y+0z=-42/5$                                                                                                    | $0x + 0y + 0z = -42/5$                                                                                         | equation $0x + 0y + 0z = -42/5$ , we see that this system of equations has                                            |
| johnston-linear-matrix-algebra-F01102 | 107  | 1.000 | $A \mathbf{x}=\mathbf{0}$                                                                                           | $A\mathbf{x} = \mathbf{0}$                                                                                     | if the linear system has a zero right-hand side (i.e., it is of the form $A\mathbf{x} = \mathbf{0}$ )                 |
| johnston-linear-matrix-algebra-F01103 | 108  | 1.000 | $m<n$                                                                                                               | $m < n$                                                                                                        | Suppose $A \in \mathcal{M}_{m,n}$ and $m < n$ . Then the linear system $A\mathbf{x} = \mathbf{b}$ has either          |
| johnston-linear-matrix-algebra-F01104 | 108  | 1.000 | $v_1=-2v_2-3v_3$                                                                                                    | $v_1 = -2v_2 - 3v_3$                                                                                           | form $v_1 = -2v_2 - 3v_3$ .                                                                                           |
| johnston-linear-matrix-algebra-F01105 | 108  | 1.000 | $v_2=v_3$                                                                                                           | $v_2 = v_3$                                                                                                    | $v_2 = v_3$ to replace                                                                                                |
| johnston-linear-matrix-algebra-F01106 | 108  | 1.000 | $\mathbf{v}=\left(v_1, v_2, v_3\right)$                                                                             | $\mathbf{v} = (v_1, v_2, v_3)$                                                                                 | vector $\mathbf{v} = (v_1, v_2, v_3)$ such that                                                                       |
| johnston-linear-matrix-algebra-F01107 | 109  | 1.000 | $v_4$                                                                                                               | $v_4$                                                                                                          | can choose $v_4$ to                                                                                                   |
| johnston-linear-matrix-algebra-F01108 | 109  | 1.000 | $\left(v_1, v_2, v_3\right)$                                                                                        | $(v_1, v_2, v_3)$                                                                                              | the vectors $(v_1, v_2, v_3)$ with $v_2 = v_3$ and $v_1 = -2v_2 - 3v_3 = -5v_3$ ,                                     |
| johnston-linear-matrix-algebra-F01109 | 109  | 1.000 | $v_1=-2v_2-3v_3=-5v_3$                                                                                              | $v_1 = -2v_2 - 3v_3 = -5v_3$                                                                                   | the vectors $(v_1, v_2, v_3)$ with $v_2 = v_3$ and $v_1 = -2v_2 - 3v_3 = -5v_3$ ,                                     |

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|---------------------------------------|------|-------|-----------------------------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F01110 | 109  | 1.000 | $v_{\{3\}}=1$                                             | $v_3 = 1$                          | choose $v_3 = 1$ to get $\mathbf{v} = (-5, 1, 1)$ .                                                           |
| johnston-linear-matrix-algebra-F01111 | 109  | 1.000 | $\mathbf{v}=(-5,1,1)$                                     | $\mathbf{v} = (-5, 1, 1)$          | choose $v_3 = 1$ to get $\mathbf{v} = (-5, 1, 1)$ .                                                           |
| johnston-linear-matrix-algebra-F01112 | 109  | 0.999 | $v_{\{1\}}, v_{\{2\}}, v_{\{3\}}$                         | $v_1, v_2, v_3$                    | We thus see that $v_4$ is a free variable, and $v_1, v_2, v_3$ are all leading                                |
| johnston-linear-matrix-algebra-F01113 | 109  | 1.000 | $\left(v_{\{1\}}, v_{\{2\}}, v_{\{3\}}, v_{\{4\}}\right)$ | $(v_1, v_2, v_3, v_4)$             | variables. Solutions $(v_1, v_2, v_3, v_4)$ to this system have the form $v_1 =$                              |
| johnston-linear-matrix-algebra-F01114 | 109  | 1.000 | $v_{\{1\}}=$                                              | $v_1 =$                            | variables. Solutions $(v_1, v_2, v_3, v_4)$ to this system have the form $v_1 =$                              |
| johnston-linear-matrix-algebra-F01115 | 109  | 0.991 | $0, v_{\{2\}}=-v_{\{4\}} / 2$                             | $0, v_2 = -v_4/2$                  | $0, v_2 = -v_4/2$ , and $v_3 = -v_4/2$ , with $v_4$ arbitrary. If we choose                                   |
| johnston-linear-matrix-algebra-F01116 | 109  | 0.991 | $v_{\{3\}}=-v_{\{4\}} / 2$                                | $v_3 = -v_4/2$                     | $0, v_2 = -v_4/2$ , and $v_3 = -v_4/2$ , with $v_4$ arbitrary. If we choose                                   |
| johnston-linear-matrix-algebra-F01117 | 109  | 1.000 | $v_{\{4\}}=2$                                             | $v_4 = 2$                          | $v_4 = 2$ then we find the specific vector $\mathbf{v} = (0, -1, -1, 2)$ , which is                           |
| johnston-linear-matrix-algebra-F01118 | 109  | 1.000 | $\mathbf{v}=(0,-1,-1,2)$                                  | $\mathbf{v} = (0, -1, -1, 2)$      | $v_4 = 2$ then we find the specific vector $\mathbf{v} = (0, -1, -1, 2)$ , which is                           |
| johnston-linear-matrix-algebra-F01119 | 109  | 1.000 | $c_{\{1\}}, c_{\{2\}} \in \mathbb{R}$                     | $c_1, c_2 \in \mathbb{R}$          | $c_1, c_2 \in \mathbb{R}$ such that                                                                           |
| johnston-linear-matrix-algebra-F01120 | 110  | 1.000 | $c_{\{1\}}, c_{\{2\}}$                                    | $c_1, c_2$                         | coefficients $c_1, c_2$                                                                                       |
| johnston-linear-matrix-algebra-F01121 | 110  | 1.000 | $(1,3,1)=\frac{4}{5}(-1,3,2)+$                            | $(1,3,1) = \frac{4}{5}(-1, 3, 2)+$ | $(1,3,1) = \frac{4}{5}(-1, 3, 2) +$                                                                           |
| johnston-linear-matrix-algebra-F01122 | 110  | 1.000 | $\frac{3}{5}(3,1,-1)$                                     | $\frac{3}{5}(3, 1, -1)$            | $\frac{3}{5}(3, 1, -1)$ . The hard                                                                            |
| johnston-linear-matrix-algebra-F01123 | 110  | 1.000 | $c_{\{1\}}$                                               | $c_1$                              | part is finding $c_1$                                                                                         |
| johnston-linear-matrix-algebra-F01124 | 110  | 1.000 | $c_{\{2\}}$                                               | $c_2$                              | and $c_2$ in the first                                                                                        |
| johnston-linear-matrix-algebra-F01125 | 110  | 1.000 | $0=3$                                                     | $0 = 3$                            | the final row in this row echelon form says that $0 = 3$ . It follows                                         |
| johnston-linear-matrix-algebra-F01126 | 110  | 1.000 | $c_{\{2\}}=3 / 5$                                         | $c_2 = 3/5$                        | This linear system can now be solved via back substitution: $c_2 = 3/5$ ,                                     |
| johnston-linear-matrix-algebra-F01127 | 110  | 1.000 | $-c_{\{1\}}+3 c_{\{2\}}=1$                                | $-c_1 + 3c_2 = 1$                  | so $-c_1 + 3c_2 = 1$ implies $c_1 = 4/5$ . We thus conclude that $(1, 3, 1)$                                  |
| johnston-linear-matrix-algebra-F01128 | 110  | 1.000 | $c_{\{1\}}=4 / 5$                                         | $c_1 = 4/5$                        | so $-c_1 + 3c_2 = 1$ implies $c_1 = 4/5$ . We thus conclude that $(1, 3, 1)$                                  |
| johnston-linear-matrix-algebra-F01129 | 110  | 1.000 | $0.02 \times 500=10$                                      | $0.02 \times 500 = 10$             | We similarly want there to be a total of $0.02 \times 500 = 10$ liters of butterfat                           |
| johnston-linear-matrix-algebra-F01130 | 111  | 1.000 | $x=200$                                                   | $x = 200$                          | We thus see that the farmer should mix together $x = 200$ liters of the                                       |
| johnston-linear-matrix-algebra-F01131 | 111  | 0.996 | $3.5 \%$                                                  | $3.5\%$                            | $3.5\%$ butterfat milk and $y = 300$ liters of the $1\%$ butterfat milk.                                      |
| johnston-linear-matrix-algebra-F01132 | 111  | 0.996 | $y=300$                                                   | $y = 300$                          | $3.5\%$ butterfat milk and $y = 300$ liters of the $1\%$ butterfat milk.                                      |
| johnston-linear-matrix-algebra-F01133 | 111  | 0.996 | $1 \%$                                                    | $1\%$                              | $3.5\%$ butterfat milk and $y = 300$ liters of the $1\%$ butterfat milk.                                      |
| johnston-linear-matrix-algebra-F01134 | 111  | 1.000 | $(x, y)=(200,300)$                                        | $(x, y) = (200, 300)$              | changes considerably, from $(x, y) = (200, 300)$ to $(x, y) = (125, 375)$ . For                               |
| johnston-linear-matrix-algebra-F01135 | 111  | 1.000 | $(x, y)=(125,375)$                                        | $(x, y) = (125, 375)$              | changes considerably, from $(x, y) = (200, 300)$ to $(x, y) = (125, 375)$ . For                               |
| johnston-linear-matrix-algebra-F01136 | 112  | 1.000 | $\mathrm{H}_{\{2\}} \mathrm{H}_{\{0\}}$                   | $\mathrm{H}_2\mathrm{O}$           | formula $\mathrm{H}_2\mathrm{O}$ for                                                                          |
| johnston-linear-matrix-algebra-F01137 | 112  | 0.996 | $\left(\mathrm{C}_{\{4\}} \mathrm{H}_{\{10\}}\right)$     | $(\mathrm{C}_4\mathrm{H}_{10})$    | When butane ( $\mathrm{C}_4\mathrm{H}_{10}$ ) burns in the presence of oxygen gas ( $\mathrm{O}_2$ ), it pro- |

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| johnston-linear-matrix-algebra-F01138 | 112  | 0.996 | $\left(\mathrm{O}_2\right)$                                                                       | (O <sub>2</sub> )                                                                                 | When butane (C <sub>4</sub> H <sub>10</sub> ) burns in the presence of oxygen gas (O <sub>2</sub> ), it pro- |
| johnston-linear-matrix-algebra-F01139 | 112  | 0.890 | $\left(\mathrm{CO}_2\right)$                                                                      | (CO <sub>2</sub> )                                                                                | duces carbon dioxide (CO <sub>2</sub> ) and water vapor (H <sub>2</sub> O). We thus say that the             |
| johnston-linear-matrix-algebra-F01140 | 112  | 0.890 | $\left(\mathrm{H}_2\right) \mathrm{O}$                                                            | (H <sub>2</sub> O)                                                                                | duces carbon dioxide (CO <sub>2</sub> ) and water vapor (H <sub>2</sub> O). We thus say that the             |
| johnston-linear-matrix-algebra-F01141 | 112  | 1.000 | $w, x, y$                                                                                         | $w, x, y$                                                                                         | Find the corresponding “balanced” equation. That is, find integers $w, x, y$ ,                               |
| johnston-linear-matrix-algebra-F01142 | 112  | 1.000 | $2 x$                                                                                             | $2x$                                                                                              | On the left-hand side of this chemical equation, we have $2x$ atoms of                                       |
| johnston-linear-matrix-algebra-F01143 | 112  | 1.000 | $2 y+z$                                                                                           | $2y+z$                                                                                            | oxygen (O) and on the right-hand side we have $2y+z$ atoms of oxygen, so                                     |
| johnston-linear-matrix-algebra-F01144 | 112  | 1.000 | $z=10$                                                                                            | $z=10$                                                                                            | In particular, if we choose $z=10$ then we get $w=2, x=13$ , and $y=8$ ,                                     |
| johnston-linear-matrix-algebra-F01145 | 112  | 1.000 | $w=2, x=13$                                                                                       | $w=2, x=13$                                                                                       | In particular, if we choose $z=10$ then we get $w=2, x=13$ , and $y=8$ ,                                     |
| johnston-linear-matrix-algebra-F01146 | 113  | 1.000 | $\left[\begin{array}{ccc} 4 & 0 & 8 \\ 4 & 1 & 11 \\ 1 & 2 & 8 \\ -1 & 0 & -2 \end{array}\right]$ | $\left[\begin{array}{ccc} 4 & 0 & 8 \\ 4 & 1 & 11 \\ 1 & 2 & 8 \\ -1 & 0 & -2 \end{array}\right]$ |                                                                                                              |
| johnston-linear-matrix-algebra-F01147 | 113  | 1.000 | $4 x-\sin (1) y+2 z=\sqrt[3]{5}$                                                                  | $4x-\sin(1)y+2z=\sqrt[3]{5}$                                                                      | *(a) The equation $4x-\sin(1)y+2z=\sqrt[3]{5}$ is linear.                                                    |
| johnston-linear-matrix-algebra-F01148 | 113  | 1.000 | $x y+4 z=4$                                                                                       | $xy+4z=4$                                                                                         | (b) The equation $xy+4z=4$ is linear.                                                                        |
| johnston-linear-matrix-algebra-F01149 | 114  | 1.000 | $t \in \mathbb{R}$                                                                                | $t \in \mathbb{R}$                                                                                | *2.1.8 Let $t \in \mathbb{R}$ and consider the linear system described                                       |
| johnston-linear-matrix-algebra-F01150 | 114  | 1.000 | $t$                                                                                               | $t$                                                                                               | For which values of $t$ does this system have (i) no solutions,                                              |
| johnston-linear-matrix-algebra-F01151 | 114  | 1.000 | $h, k \in \mathbb{R}$                                                                             | $h, k \in \mathbb{R}$                                                                             | 2.1.9 Let $h, k \in \mathbb{R}$ and consider the following system of                                         |
| johnston-linear-matrix-algebra-F01152 | 114  | 1.000 | $h$                                                                                               | $h$                                                                                               | For which values of $h$ and $k$ does this system have (i) no                                                 |
| johnston-linear-matrix-algebra-F01153 | 114  | 1.000 | $h=0.95, h=1.00$                                                                                  | $h=0.95, h=1.00$                                                                                  | $h=0.95, h=1.00$ , and $h=1.05$ . Do you notice anything                                                     |
| johnston-linear-matrix-algebra-F01154 | 114  | 1.000 | $h=1.05$                                                                                          | $h=1.05$                                                                                          | $h=0.95, h=1.00$ , and $h=1.05$ . Do you notice anything                                                     |
| johnston-linear-matrix-algebra-F01155 | 114  | 1.000 | $a, b, c, d \in \mathbb{R}$                                                                       | $a, b, c, d \in \mathbb{R}$                                                                       | *2.1.11 Let $a, b, c, d \in \mathbb{R}$ be scalars and consider the system                                   |
| johnston-linear-matrix-algebra-F01156 | 114  | 1.000 | $a d-b c \neq 0$                                                                                  | $ad-bc \neq 0$                                                                                    | tion whenever $ad-bc \neq 0$ .                                                                               |
| johnston-linear-matrix-algebra-F01157 | 114  | 1.000 | $A, B, R \in \mathcal{M}_{m, n}$                                                                  | $A, B, R \in \mathcal{M}_{m, n}$                                                                  | **2.1.13 Suppose $A, B, R \in \mathcal{M}_{m, n}$ .                                                          |
| johnston-linear-matrix-algebra-F01158 | 114  | 1.000 | $R_2-R_1$                                                                                         | $R_2-R_1$                                                                                         | (a) Perform the row operation $R_2-R_1$ , followed by the                                                    |
| johnston-linear-matrix-algebra-F01159 | 114  | 1.000 | $R_1-R_2$                                                                                         | $R_1-R_2$                                                                                         | row operation $R_1-R_2$ , on this augmented matrix.                                                          |
| johnston-linear-matrix-algebra-F01160 | 114  | 0.998 | $\mathbf{b}$                                                                                      | <b>b</b>                                                                                          | 2.1.17 Determine whether or not <b>b</b> is a linear combination                                             |
| johnston-linear-matrix-algebra-F01161 | 114  | 0.999 | $\mathbf{b}=(3,4), \mathbf{v}_1=(6,8)$                                                            | $\mathbf{b}=(3,4), \mathbf{v}_1=(6,8)$                                                            | *(a) $\mathbf{b}=(3,4), \mathbf{v}_1=(6,8)$                                                                  |
| johnston-linear-matrix-algebra-F01162 | 114  | 1.000 | $\mathbf{b}=(2,-3), \mathbf{v}_1=(1,2), \mathbf{v}_2=(4,-1)$                                      | $\mathbf{b}=(2,-3), \mathbf{v}_1=(1,2), \mathbf{v}_2=(4,-1)$                                      | (b) $\mathbf{b}=(2,-3), \mathbf{v}_1=(1,2), \mathbf{v}_2=(4,-1)$                                             |
| johnston-linear-matrix-algebra-F01163 | 114  | 1.000 | $\mathbf{b}=(1,5,-6), \mathbf{v}_1=(1,0,0), \mathbf{v}_2=(0,1,0), \mathbf{v}_3=(0,1,0)$           | $\mathbf{b}=(1,5,-6), \mathbf{v}_1=(1,0,0), \mathbf{v}_2=(0,1,0), \mathbf{v}_3=(0,1,0)$           | *(c) $\mathbf{b}=(1,5,-6), \mathbf{v}_1=(1,0,0), \mathbf{v}_2=(0,1,0), \mathbf{v}_3=(0,1,0)$                 |
| johnston-linear-matrix-algebra-F01164 | 114  | 1.000 | $\mathbf{b}=(2,1,2), \mathbf{v}_1=(-2,2,1), \mathbf{v}_2=(1,2,3)$                                 | $\mathbf{b}=(2,1,2), \mathbf{v}_1=(-2,2,1), \mathbf{v}_2=(1,2,3)$                                 | (d) $\mathbf{b}=(2,1,2), \mathbf{v}_1=(-2,2,1), \mathbf{v}_2=(1,2,3)$                                        |

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| johnston-linear-matrix-algebra-F01165 | 114  | 1.000 | $\mathbf{b}=(2,1,2), \mathbf{v}_1=(-4,4,-1), \mathbf{v}_2=(2,-1,1)$      | $\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-4, 4, -1), \mathbf{v}_2 = (2, -1, 1)$         | <b>(e)</b> $\mathbf{b} = (2, 1, 2), \mathbf{v}_1 = (-4, 4, -1), \mathbf{v}_2 = (2, -1, 1)$          |
| johnston-linear-matrix-algebra-F01166 | 114  | 1.000 | $\mathbf{b}=(2,1,2,3), \mathbf{v}_1=(1,2,3,4), \mathbf{v}_2=(4,3,2,1)$   | $\mathbf{b} = (2, 1, 2, 3), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$   | <b>(f)</b> $\mathbf{b} = (2, 1, 2, 3), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1),$   |
| johnston-linear-matrix-algebra-F01167 | 114  | 1.000 | $\mathbf{v}_3=(1,-1,1,-1)$                                               | $\mathbf{v}_3 = (1, -1, 1, -1)$                                                         | $\mathbf{v}_3 = (1, -1, 1, -1)$                                                                     |
| johnston-linear-matrix-algebra-F01168 | 114  | 0.831 | $\mathbf{b}=(1,3,-3,-1), \mathbf{v}_1=(1,2,3,4), \mathbf{v}_2=(4,3,2,1)$ | $\mathbf{b} = (1, 3, -3, -1), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1)$ | <b>(g)</b> $\mathbf{b} = (1, 3, -3, -1), \mathbf{v}_1 = (1, 2, 3, 4), \mathbf{v}_2 = (4, 3, 2, 1),$ |
| johnston-linear-matrix-algebra-F01169 | 114  | 1.000 | $\mathbf{w}=\left(w_1, w_2, w_3\right)$                                  | $\mathbf{w} = (w_1, w_2, w_3)$                                                          | <b>2.1.18</b> Let $\mathbf{v} = (v_1, v_2, v_3)$ and $\mathbf{w} = (w_1, w_2, w_3)$ be non-         |
| johnston-linear-matrix-algebra-F01170 | 115  | 1.000 | $\left(\mathrm{SO}_2\right)$                                             | $(\mathrm{SO}_2)$                                                                       | fur dioxide ( $\mathrm{SO}_2$ ). Determine how many of each of these                                |
| johnston-linear-matrix-algebra-F01171 | 115  | 0.996 | $\left(\mathrm{C}_2 \mathrm{H}_6\right)$                                 | $(\mathrm{C}_2\mathrm{H}_6)$                                                            | <b>2.1.22</b> When ethane ( $\mathrm{C}_2\mathrm{H}_6$ ) burns in the presence of oxy-              |
| johnston-linear-matrix-algebra-F01172 | 115  | 1.000 | $P_{\mathbf{u}}(\mathbf{v})=(2,4,4)$                                     | $P_{\mathbf{u}}(\mathbf{v}) = (2, 4, 4)$                                                | property that $P_{\mathbf{u}}(\mathbf{v}) = (2, 4, 4)$ .                                            |
| johnston-linear-matrix-algebra-F01173 | 115  | 1.000 | $(3,-2,1)=c_1(1,4,2)+$                                                   | $(3, -2, 1) = c_1(1, 4, 2) +$                                                           | [Hint: Rewrite the equation $(3, -2, 1) = c_1(1, 4, 2) +$                                           |
| johnston-linear-matrix-algebra-F01174 | 115  | 1.000 | $c_2(2,-1,1)$                                                            | $c_2(2, -1, 1)$                                                                         | $c_2(2, -1, 1)$ as a system of linear equations.]                                                   |
| johnston-linear-matrix-algebra-F01175 | 115  | 0.566 | $T(1,1)=(3,7), T(1,-1)=(-1,-1)$                                          | $T(1, 1) = (3, 7), T(1, -1) = (-1, -1)$                                                 | <b>(a)</b> $T(1, 1) = (3, 7), T(1, -1) = (-1, -1)$                                                  |
| johnston-linear-matrix-algebra-F01176 | 115  | 1.000 | $T(1,2)=(5,3,4), T(2,-1)=(0,1,3)$                                        | $T(1, 2) = (5, 3, 4), T(2, -1) = (0, 1, 3)$                                             | <b>(b)</b> $T(1, 2) = (5, 3, 4), T(2, -1) = (0, 1, 3)$                                              |
| johnston-linear-matrix-algebra-F01177 | 115  | 0.992 | $T(1,1,1)=(4,6,1), \quad T(2,-1,1)=(1,1,-4)$                             | $T(1, 1, 1) = (4, 6, 1), \quad T(2, -1, 1) = (1, 1, -4)$                                | <b>(c)</b> $T(1, 1, 1) = (4, 6, 1), \quad T(2, -1, 1) = (1, 1, -4),$                                |
| johnston-linear-matrix-algebra-F01178 | 115  | 1.000 | $T(0,0,1)=(1,2,0)$                                                       | $T(0, 0, 1) = (1, 2, 0)$                                                                | $T(0, 0, 1) = (1, 2, 0)$                                                                            |
| johnston-linear-matrix-algebra-F01179 | 115  | 1.000 | $A B=B A$                                                                | $AB = BA$                                                                               | $AB = BA$ , and we can find matrices with this property by                                          |
| johnston-linear-matrix-algebra-F01180 | 115  | 1.000 | $B \in \mathcal{M}_2$                                                    | $B \in \mathcal{M}_2$                                                                   | Find all $B \in \mathcal{M}_2$ with the property that $AB = BA$ .                                   |
| johnston-linear-matrix-algebra-F01181 | 115  | 1.000 | $D \in \mathcal{M}_2$                                                    | $D \in \mathcal{M}_2$                                                                   | Find all $D \in \mathcal{M}_2$ with the property that $CD = DC$ .                                   |
| johnston-linear-matrix-algebra-F01182 | 115  | 1.000 | $C D=D C$                                                                | $CD = DC$                                                                               | Find all $D \in \mathcal{M}_2$ with the property that $CD = DC$ .                                   |
| johnston-linear-matrix-algebra-F01183 | 115  | 1.000 | $c I$                                                                    | $cI$                                                                                    | ces of the form $cI$ , where $c \in \mathbb{R}$ . [Side note: These                                 |
| johnston-linear-matrix-algebra-F01184 | 115  | 1.000 | $2=m+b$                                                                  | $2 = m + b$                                                                             | us that $2 = m + b$ and the fact that the line goes                                                 |
| johnston-linear-matrix-algebra-F01185 | 115  | 1.000 | $(3,8)$                                                                  | $(3, 8)$                                                                                | through $(3, 8)$ tells us that $8 = 3m + b$ .]                                                      |
| johnston-linear-matrix-algebra-F01186 | 115  | 1.000 | $8=3 m+b$                                                                | $8 = 3m + b$                                                                            | through $(3, 8)$ tells us that $8 = 3m + b$ .]                                                      |
| johnston-linear-matrix-algebra-F01187 | 115  | 0.977 | $y=a x^2+b x+c$                                                          | $y = ax^2 + bx + c$                                                                     | $y = ax^2 + bx + c$ goes through the points $(1, 3), (2, 6),$                                       |
| johnston-linear-matrix-algebra-F01188 | 115  | 1.000 | $(2,6)$                                                                  | $(2, 6)$                                                                                | [Hint: The fact that the parabola goes through $(2, 6)$                                             |
| johnston-linear-matrix-algebra-F01189 | 115  | 1.000 | $6=4 a+2 b+c$                                                            | $6 = 4a + 2b + c$                                                                       | tells us that $6 = 4a + 2b + c$ , and so on.]                                                       |
| johnston-linear-matrix-algebra-F01190 | 115  | 1.000 | $y=a x^3+b x^2+c x+d$                                                    | $y = ax^3 + bx^2 + cx + d$                                                              | $y = ax^3 + bx^2 + cx + d$ that goes through the 3 points?                                          |
| johnston-linear-matrix-algebra-F01191 | 116  | 1.000 | $a, b, c, d$                                                             | $a, b, c, d$                                                                            | nomial for this data. That is, find scalars $a, b, c, d,$                                           |
| johnston-linear-matrix-algebra-F01192 | 116  | 1.000 | $e$                                                                      | $e$                                                                                     | $e$ , and $f$ such that if $x$ is the number of decades that                                        |
| johnston-linear-matrix-algebra-F01193 | 116  | 1.000 | $p(0)=3.70, p(1)=4.46$                                                   | $p(0) = 3.70, p(1) = 4.46$                                                              | the decade (i.e., $p(0) = 3.70, p(1) = 4.46$ , and so                                               |

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| johnston-linear-matrix-algebra-F01194 | 117  | 1.000 | $R_{\{1\}} \leftrightarrow R_{\{2\}}$                                       | $R_1 \leftrightarrow R_2$                          | For example, we carried out the “swap” row operation $R_1 \leftrightarrow R_2$ in Equa-                                            |
| johnston-linear-matrix-algebra-F01195 | 117  | 1.000 | $R_{\{3\}} - 3 R_{\{1\}}$                                                   | $R_3 - 3R_1$                                       | row operations $R_3 - 3R_1$ and $\frac{1}{2}R_2$ to the identity matrix. We will make extensive                                    |
| johnston-linear-matrix-algebra-F01196 | 117  | 1.000 | $\frac{1}{2} R_{\{2\}}$                                                     | $\frac{1}{2}R_2$                                   | row operations $R_3 - 3R_1$ and $\frac{1}{2}R_2$ to the identity matrix. We will make extensive                                    |
| johnston-linear-matrix-algebra-F01197 | 117  | 1.000 | $R_{\{i\}} \leftrightarrow R_{\{j\}}$                                       | $R_i \leftrightarrow R_j$                          | The elementary matrices corresponding to the “swap” row operation $R_i \leftrightarrow R_j$ look like                              |
| johnston-linear-matrix-algebra-F01198 | 117  | 1.000 | $R_{\{j\}}$                                                                 | $R_j$                                              |                                                                                                                                    |
| johnston-linear-matrix-algebra-F01199 | 118  | 0.844 | $A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$                          | $A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$ | <b>Figure 2.10:</b> The “addition” matrix $A = \begin{bmatrix} 1 & c \\ 0 & 1 \end{bmatrix}$ shears space in the $x$ -direction. A |
| johnston-linear-matrix-algebra-F01200 | 119  | 0.512 | $EA = R$                                                                    | $EA = R$                                           | Find a $3 \times 3$ matrix $E$ such that $EA = R$ , where $R$ is the reduced row                                                   |
| johnston-linear-matrix-algebra-F01201 | 119  | 1.000 | $R_{\{3\}} - R_{\{2\}}$                                                     | $R_3 - R_2$                                        | elementary row operations we perform are $R_1 - R_2$ and $R_3 - R_2$ , which                                                       |
| johnston-linear-matrix-algebra-F01202 | 119  | 1.000 | $\frac{1}{4} R_{\{3\}}$                                                     | $\frac{1}{4}R_3$                                   | Finally, we apply the elementary row operations $\frac{1}{4}R_3$ and then $R_1 + R_3$ ,                                            |
| johnston-linear-matrix-algebra-F01203 | 119  | 1.000 | $R_{\{1\}} + R_{\{3\}}$                                                     | $R_1 + R_3$                                        | Finally, we apply the elementary row operations $\frac{1}{4}R_3$ and then $R_1 + R_3$ ,                                            |
| johnston-linear-matrix-algebra-F01204 | 119  | 1.000 | $E = E_{\{7\}} E_{\{6\}} E_{\{5\}} E_{\{4\}} E_{\{3\}} E_{\{2\}} E_{\{1\}}$ | $E = E_7 E_6 E_5 E_4 E_3 E_2 E_1$                  | $E = E_7 E_6 E_5 E_4 E_3 E_2 E_1$ , which can be computed by straightforward (albeit                                               |
| johnston-linear-matrix-algebra-F01205 | 120  | 0.999 | $E_{\{1\}}, E_{\{2\}}, \dots, E_{\{7\}}$                                    | $E_1, E_2, \dots, E_7$                             | The matrices $E_1, E_2, \dots, E_7$ in the previous example can be thought of like                                                 |
| johnston-linear-matrix-algebra-F01206 | 120  | 1.000 | $E_{\{1\}}$                                                                 | $E_1$                                              | its reduced row echelon form $R$ —first the row operation corresponding to $E_1$                                                   |
| johnston-linear-matrix-algebra-F01207 | 120  | 1.000 | $E_{\{2\}}$                                                                 | $E_2$                                              | was performed, then the one corresponding to $E_2$ , and so on. Similarly, the                                                     |
| johnston-linear-matrix-algebra-F01208 | 120  | 0.516 | $-5 / 8$                                                                    | $-5/8$                                             | $-5/8$ of $A$ ’s first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.                                             |
| johnston-linear-matrix-algebra-F01209 | 120  | 0.516 | $2 / 8 = 1 / 4$                                                             | $2/8 = 1/4$                                        | $-5/8$ of $A$ ’s first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.                                             |
| johnston-linear-matrix-algebra-F01210 | 120  | 0.516 | $1 / 4$                                                                     | $1/4$                                              | $-5/8$ of $A$ ’s first row, $2/8 = 1/4$ of its second row, and $1/4$ of its third row.                                             |
| johnston-linear-matrix-algebra-F01211 | 120  | 1.000 | $E =$                                                                       | $E =$                                              | To actually compute the product of all of those elementary matrices $E =$                                                          |
| johnston-linear-matrix-algebra-F01212 | 120  | 0.984 | $E_{\{7\}} E_{\{6\}} E_{\{5\}} E_{\{4\}} E_{\{3\}} E_{\{2\}} E_{\{1\}}$     | $E_7 E_6 E_5 E_4 E_3 E_2 E_1$                      | $E_7 E_6 E_5 E_4 E_3 E_2 E_1$ , we could perform six matrix multiplications (ugh!), but a                                          |
| johnston-linear-matrix-algebra-F01213 | 120  | 0.913 | $[A \mid I]$                                                                | $[A \mid I]$                                       | Compute the reduced row echelon form of the $1 \times 2$ block matrix $[A \mid I]$ .                                               |
| johnston-linear-matrix-algebra-F01214 | 121  | 1.000 | $[R \mid E]$                                                                | $[R \mid E]$                                       | to some other form $[R \mid E]$ then these blocks must satisfy the property $EA = R$ .                                             |
| johnston-linear-matrix-algebra-F01215 | 121  | 1.000 | $A, R \in \mathcal{M}_{m,n}$                                                | $A, R \in \mathcal{M}_{m,n}$                       | If $A, R \in \mathcal{M}_{m,n}$ and $E \in \mathcal{M}_m$ are matrices such that the block matrix $[A \mid I]$                     |
| johnston-linear-matrix-algebra-F01216 | 121  | 1.000 | $E \in \mathcal{M}_m$                                                       | $E \in \mathcal{M}_m$                              | If $A, R \in \mathcal{M}_{m,n}$ and $E \in \mathcal{M}_m$ are matrices such that the block matrix $[A \mid I]$                     |
| johnston-linear-matrix-algebra-F01217 | 121  | 1.000 | $R = EA$                                                                    | $R = EA$                                           | can be row-reduced to $[R \mid E]$ , then $R = EA$ .                                                                               |
| johnston-linear-matrix-algebra-F01218 | 121  | 0.706 | $E_{\{1\}}, E_{\{2\}}, \dots, E_{\{k\}}$                                    | $E_1, E_2, \dots, E_k$                             | corresponding to elementary matrices $E_1, E_2, \dots, E_k$ , in that order, then                                                  |
| johnston-linear-matrix-algebra-F01219 | 121  | 1.000 | $E = E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}}$                                  | $E = E_k \cdots E_2 E_1$                           | $E = E_k \cdots E_2 E_1$ , which then implies (by looking at the left half of the block                                            |
| johnston-linear-matrix-algebra-F01220 | 121  | 1.000 | $E_{\{1\}} E_{\{2\}} = I$                                                   | $E_1 E_2 = I$                                      | then it is straightforward to check that $E_1 E_2 = I$ and $E_2 E_1 = I$ . We thus say that                                        |

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| johnston-linear-matrix-algebra-F01221 | 121  | 1.000 | $E_{\{2\}} E_{\{1\}}=I$                                                                               | $E_2 E_1 = I$                                                                                 | then it is straightforward to check that $E_1 E_2 = I$ and $E_2 E_1 = I$ . We thus say that                                                                                            |
| johnston-linear-matrix-algebra-F01222 | 122  | 1.000 | $R_{\{3\}}+3 R_{\{1\}}$                                                                               | $R_3 + 3R_1$                                                                                  | operation $R_3 + 3R_1$ , and performing these row operations one after another has                                                                                                     |
| johnston-linear-matrix-algebra-F01223 | 122  | 1.000 | $A^{\{-1\}}$                                                                                          | $A^{-1}$                                                                                      | we denote by $A^{-1}$ and call the <b>inverse</b> of $A$ , such that                                                                                                                   |
| johnston-linear-matrix-algebra-F01224 | 122  | 1.000 | $B, C \in \mathcal{M}_n$                                                                              | $B, C \in \mathcal{M}_n$                                                                      | had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$ ). It would                                                                                    |
| johnston-linear-matrix-algebra-F01225 | 122  | 1.000 | $A B=B A=I$                                                                                           | $AB = BA = I$                                                                                 | had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$ ). It would                                                                                    |
| johnston-linear-matrix-algebra-F01226 | 122  | 1.000 | $A C=C A=I$                                                                                           | $AC = CA = I$                                                                                 | had <i>two</i> inverses $B, C \in \mathcal{M}_n$ (i.e., $AB = BA = I$ and $AC = CA = I$ ). It would                                                                                    |
| johnston-linear-matrix-algebra-F01227 | 122  | 1.000 | $\frac{1}{2}\left[\begin{array}{cc}-4 & 2 \\ 3 & -1\end{array}\right]$                                | $\frac{1}{2}\left[\begin{array}{cc}-4 & 2 \\ 3 & -1\end{array}\right]$                        | a) $\left[\begin{array}{cc}1 & 2 \\ 3 & 4\end{array}\right]$ and $\frac{1}{2}\left[\begin{array}{cc}-4 & 2 \\ 3 & -1\end{array}\right]$ ,                                              |
| johnston-linear-matrix-algebra-F01228 | 122  | 1.000 | $\left[\begin{array}{ll}1 & 2 \\ 2 & 4\end{array}\right]$                                             | $\left[\begin{array}{cc}1 & 2 \\ 2 & 4\end{array}\right]$                                     | b) $\left[\begin{array}{cc}1 & 2 \\ 2 & 4\end{array}\right]$ and $\left[\begin{array}{cc}-4 & 2 \\ 2 & -1\end{array}\right]$ , and                                                     |
| johnston-linear-matrix-algebra-F01229 | 122  | 1.000 | $\left[\begin{array}{cc}-4 & 2 \\ 2 & -1\end{array}\right]$                                           | $\left[\begin{array}{cc}-4 & 2 \\ 2 & -1\end{array}\right]$                                   | b) $\left[\begin{array}{cc}1 & 2 \\ 2 & 4\end{array}\right]$ and $\left[\begin{array}{cc}-4 & 2 \\ 2 & -1\end{array}\right]$ , and                                                     |
| johnston-linear-matrix-algebra-F01230 | 122  | 0.991 | $\left[\begin{array}{lll}0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0\end{array}\right]$                       | $\left[\begin{array}{ccc}0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0\end{array}\right]$               | c) $\left[\begin{array}{ccc}0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0\end{array}\right]$ and $\frac{1}{2}\left[\begin{array}{ccc}-1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1\end{array}\right]$ . |
| johnston-linear-matrix-algebra-F01231 | 122  | 0.991 | $\frac{1}{2}\left[\begin{array}{ccc}-1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1\end{array}\right]$         | $\frac{1}{2}\left[\begin{array}{ccc}-1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1\end{array}\right]$ | c) $\left[\begin{array}{ccc}0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0\end{array}\right]$ and $\frac{1}{2}\left[\begin{array}{ccc}-1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1\end{array}\right]$ . |
| johnston-linear-matrix-algebra-F01232 | 123  | 1.000 | $T^{\{-1\}}$                                                                                          | $T^{-1}$                                                                                      | denote by $T^{-1}$ , such that $T \circ T^{-1} = T^{-1} \circ T = I$ . In other words, $T^{-1}$ is the                                                                                 |
| johnston-linear-matrix-algebra-F01233 | 123  | 1.000 | $T \circ T^{\{-1\}}=T^{\{-1\}} \circ T=I$                                                             | $T \circ T^{-1} = T^{-1} \circ T = I$                                                         | denote by $T^{-1}$ , such that $T \circ T^{-1} = T^{-1} \circ T = I$ . In other words, $T^{-1}$ is the                                                                                 |
| johnston-linear-matrix-algebra-F01234 | 123  | 1.000 | $T^{\{-1\}}(T(\mathbf{v}))=\mathbf{v}$                                                                | $T^{-1}(T(\mathbf{v})) = \mathbf{v}$                                                          | $T^{-1}(T(\mathbf{v})) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ (and $T(T^{-1}(\mathbf{v})) = \mathbf{v}$ too).                                                             |
| johnston-linear-matrix-algebra-F01235 | 123  | 1.000 | $T\left(T^{\{-1\}}(\mathbf{v})\right)=\mathbf{v}$                                                     | $T\left(T^{-1}(\mathbf{v})\right) = \mathbf{v}$                                               | $T^{-1}(T(\mathbf{v})) = \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ (and $T(T^{-1}(\mathbf{v})) = \mathbf{v}$ too).                                                             |
| johnston-linear-matrix-algebra-F01236 | 124  | 1.000 | $P_{\{\mathbf{u}\}}\left(\mathbf{v}_{\{1\}}\right)=P_{\{\mathbf{u}\}}\left(\mathbf{v}_{\{2\}}\right)$ | $P_{\mathbf{u}}(\mathbf{v}_1) = P_{\mathbf{u}}(\mathbf{v}_2)$                                 | single vector on the dashed line on the right is sent to $P_{\mathbf{u}}(\mathbf{v}_1) = P_{\mathbf{u}}(\mathbf{v}_2)$ .                                                               |
| johnston-linear-matrix-algebra-F01237 | 124  | 0.966 | $0 x$                                                                                                 | $0x$                                                                                          | $0x$ , we cannot determine $x$ : it could be absolutely anything. Multiplication by 0                                                                                                  |
| johnston-linear-matrix-algebra-F01238 | 124  | 1.000 | $\left(A^{\{-1\}}\right)^{\{-1\}}=A$                                                                  | $\left(A^{-1}\right)^{-1} = A$                                                                | a) $A^{-1}$ is invertible and $(A^{-1})^{-1} = A$ .                                                                                                                                    |
| johnston-linear-matrix-algebra-F01239 | 124  | 1.000 | $(c A)^{\{-1\}}=\frac{1}{c} A^{\{-1\}}$                                                               | $(cA)^{-1} = \frac{1}{c}A^{-1}$                                                               | b) $cA$ is invertible and $(cA)^{-1} = \frac{1}{c}A^{-1}$ .                                                                                                                            |
| johnston-linear-matrix-algebra-F01240 | 124  | 1.000 | $\left(A^{\{T\}}\right)^{\{-1\}}=\left(A^{\{-1\}}\right)^{\{T\}}$                                     | $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$                                               | c) $A^T$ is invertible and $(A^T)^{-1} = (A^{-1})^T$ .                                                                                                                                 |
| johnston-linear-matrix-algebra-F01241 | 124  | 1.000 | $\left(A^{\{k\}}\right)^{\{-1\}}=\left(A^{\{-1\}}\right)^{\{k\}}$                                     | $\left(A^k\right)^{-1} = \left(A^{-1}\right)^k$                                               | d) $A^k$ is invertible and $(A^k)^{-1} = (A^{-1})^k$ .                                                                                                                                 |
| johnston-linear-matrix-algebra-F01242 | 124  | 1.000 | $\frac{1}{c} A^{\{-1\}}$                                                                              | $\frac{1}{c}A^{-1}$                                                                           | which shows that $cA$ has $\frac{1}{c}A^{-1}$ as its inverse, as desired.                                                                                                              |
| johnston-linear-matrix-algebra-F01243 | 124  | 1.000 | $A^{\{k \ell\}}=$                                                                                     | $A^{k\ell} =$                                                                                 | then matrix powers still have the “nice” properties like $A^{k+\ell} = A^k A^\ell$ and $A^{k\ell} =$                                                                                   |

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| johnston-linear-matrix-algebra-F01244 | 124  | 1.000 | $\left(A^{\{k\}}\right)^{\{ell\}}$                                                                                                                                             | $(A^k)^\ell$                                                                                                                              | $(A^k)^\ell$ even when $k$ and $\ell$ are allowed to be negative (see Exercise 2.2.18).                                                                                             |
| johnston-linear-matrix-algebra-F01245 | 124  | 0.965 | $A^{\{-2\}}$                                                                                                                                                                   | $A^{-2}$                                                                                                                                  | Compute $A^{-2}$ if $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ .                                                                                                            |
| johnston-linear-matrix-algebra-F01246 | 125  | 1.000 | $c\ R_{\{i\}}$                                                                                                                                                                 | $cR_i$                                                                                                                                    | <ul style="list-style-type: none"> <li>The row operation <math>cR_i</math> is undone by the row operation <math>\frac{1}{c}R_j</math>.</li> </ul>                                   |
| johnston-linear-matrix-algebra-F01247 | 125  | 1.000 | $\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]$                                                                                                                    | $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$                                                                                            | a) $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$                                                                                                                                   |
| johnston-linear-matrix-algebra-F01248 | 125  | 1.000 | $\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]$                                                                                                                    | $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                                                            | b) $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                                                                                                   |
| johnston-linear-matrix-algebra-F01249 | 125  | 1.000 | $\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$                                                                                              | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$                                                                       | c) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$                                                                                                              |
| johnston-linear-matrix-algebra-F01250 | 125  | 1.000 | $\left[\begin{array}{ccc} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$                                                                                             | $\begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                                                      | d) $\begin{bmatrix} 1 & 0 & 0 \\ -5 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                                                                                             |
| johnston-linear-matrix-algebra-F01251 | 125  | 1.000 | $3\ R_{\{2\}}$                                                                                                                                                                 | $3R_2$                                                                                                                                    | $3R_2$ , so its inverse is the elementary matrix for the row operation                                                                                                              |
| johnston-linear-matrix-algebra-F01252 | 125  | 1.000 | $\frac{1}{3}\ R_{\{2\}}$                                                                                                                                                       | $\frac{1}{3}R_2$                                                                                                                          | $\frac{1}{3}R_2$ :                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F01253 | 125  | 1.000 | $R_{\{1\}}+2\ R_{\{2\}}$                                                                                                                                                       | $R_1 + 2R_2$                                                                                                                              | b) This is the elementary matrix for the row operation $R_1 + 2R_2$ , so its                                                                                                        |
| johnston-linear-matrix-algebra-F01254 | 125  | 1.000 | $R_{\{1\}}-2\ R_{\{2\}}$                                                                                                                                                       | $R_1 - 2R_2$                                                                                                                              | inverse is the elementary matrix for the row operation $R_1 - 2R_2$ :                                                                                                               |
| johnston-linear-matrix-algebra-F01255 | 125  | 1.000 | $R_{\{2\}}\ \leftrightarrow\ R_{\{3\}}$                                                                                                                                        | $R_2 \leftrightarrow R_3$                                                                                                                 | c) This is the elementary matrix for the row operation $R_2 \leftrightarrow R_3$ , so its                                                                                           |
| johnston-linear-matrix-algebra-F01256 | 125  | 1.000 | $R_{\{2\}}-5\ R_{\{1\}}$                                                                                                                                                       | $R_2 - 5R_1$                                                                                                                              | d) This is the elementary matrix for to the row operation $R_2 - 5R_1$ , so                                                                                                         |
| johnston-linear-matrix-algebra-F01257 | 126  | 1.000 | $R_{\{2\}}+5\ R_{\{1\}}$                                                                                                                                                       | $R_2 + 5R_1$                                                                                                                              | its inverse is the elementary matrix for the row operation $R_2 + 5R_1$ :                                                                                                           |
| johnston-linear-matrix-algebra-F01258 | 126  | 1.000 | $B^{\{-1\}}\ A^{\{-1\}}$                                                                                                                                                       | $B^{-1}A^{-1}$                                                                                                                            | invertible, and its inverse is $B^{-1}A^{-1}$ , as claimed.                                                                                                                         |
| johnston-linear-matrix-algebra-F01259 | 126  | 1.000 | $\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]\left[\begin{array}{ll} 1 & 0 \\ 0 & 3 \end{array}\right]=\left[\begin{array}{ll} 1 & 6 \\ 0 & 3 \end{array}\right]$ | $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}$ | Find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 6 \\ 0 & 3 \end{bmatrix}$ .          |
| johnston-linear-matrix-algebra-F01260 | 127  | 1.000 | $\mathbf{b}\ \in\ \mathbb{R}^{\{n\}}$                                                                                                                                          | $\mathbf{b} \in \mathbb{R}^n$                                                                                                             | b) The linear system $A\mathbf{x} = \mathbf{b}$ has a solution for all $\mathbf{b} \in \mathbb{R}^n$ .                                                                              |
| johnston-linear-matrix-algebra-F01261 | 127  | 1.000 | $\mathbf{x}=A^{\{-1\}}\ \mathbf{b}$                                                                                                                                            | $\mathbf{x} = A^{-1}\mathbf{b}$                                                                                                           | To see that (a) $\implies$ (c), we note that if $A$ is invertible then $\mathbf{x} = A^{-1}\mathbf{b}$ is a                                                                         |
| johnston-linear-matrix-algebra-F01262 | 127  | 1.000 | $\mathbf{x},\mathbf{y}\ \in\ \mathbb{R}^{\{n\}}$                                                                                                                               | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$                                                                                                 | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$ , so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$ . It |
| johnston-linear-matrix-algebra-F01263 | 127  | 1.000 | $A\ \mathbf{y}=\mathbf{b}$                                                                                                                                                     | $A\mathbf{y} = \mathbf{b}$                                                                                                                | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$ , so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$ . It |
| johnston-linear-matrix-algebra-F01264 | 127  | 1.000 | $A(\mathbf{x}-\mathbf{y})=\mathbf{0}$                                                                                                                                          | $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$                                                                                                 | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Then $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{y} = \mathbf{b}$ , so subtracting gives us $A(\mathbf{x} - \mathbf{y}) = \mathbf{0}$ . It |
| johnston-linear-matrix-algebra-F01265 | 127  | 1.000 | $\mathbf{x}=\mathbf{y}$                                                                                                                                                        | $\mathbf{x} = \mathbf{y}$                                                                                                                 | so $\mathbf{x} = \mathbf{y}$ (i.e., the solution is unique).                                                                                                                        |

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| johnston-linear-matrix-algebra-F01266 | 127  | 0.842 | $\mathbf{b}=\mathbf{0}$                                                             | $\mathbf{b} = \mathbf{0}$                                                           | The implication (c) $\implies$ (d) follows simply by choosing $\mathbf{b} = \mathbf{0}$ , so we jump |
| johnston-linear-matrix-algebra-F01267 | 127  | 1.000 | $[A \mid \mathbf{0}]$                                                               | $[A \mid \mathbf{0}]$                                                               | in augmented matrix form $[A \mid \mathbf{0}]$ and then apply Gauss–Jordan elimination,              |
| johnston-linear-matrix-algebra-F01268 | 127  | 1.000 | $[R \mid \mathbf{0}]$                                                               | $[R \mid \mathbf{0}]$                                                               | we get the augmented matrix $[R \mid \mathbf{0}]$ , where $R$ is the reduced row echelon form        |
| johnston-linear-matrix-algebra-F01269 | 127  | 1.000 | $R \neq I$                                                                          | $R \neq I$                                                                          | of $A$ . Since $R$ is square, if $R \neq I$ then it must have a row of zeros at the bottom,          |
| johnston-linear-matrix-algebra-F01270 | 128  | 0.999 | $\left[R \mid \mathbf{e}_n\right]$                                                  | $[R \mid \mathbf{e}_n]$                                                             | produces $[R \mid \mathbf{e}_n]$ ,                                                                   |
| johnston-linear-matrix-algebra-F01271 | 128  | 1.000 | $\mathbf{e}_n$                                                                      | $\mathbf{e}_n$                                                                      | row operations to $\mathbf{e}_n$ .                                                                   |
| johnston-linear-matrix-algebra-F01272 | 128  | 1.000 | $R=I$                                                                               | $R = I$                                                                             | a unique solution (i.e., (d) holds) then $R = I$ (i.e., (e) holds).                                  |
| johnston-linear-matrix-algebra-F01273 | 128  | 0.996 | $E_{\{k\}} \cdots E_{\{2\}} E_{\{1\}} A=I$                                          | $E_k \cdots E_2 E_1 A = I$                                                          | sequence of elementary matrices $E_1, E_2, \dots, E_k$ such that $E_k \cdots E_2 E_1 A = I$ . By     |
| johnston-linear-matrix-algebra-F01274 | 128  | 1.000 | $E_{\{k\}}^{-1}, E_{\{k-1\}}^{-1}$                                                  | $E_k^{-1}, E_{k-1}^{-1}$                                                            | multiplying on the left by $E_k^{-1}, E_{k-1}^{-1}$ , and so on to $E_1^{-1}$ , we see that          |
| johnston-linear-matrix-algebra-F01275 | 128  | 1.000 | $E_{\{1\}}^{-1}$                                                                    | $E_1^{-1}$                                                                          | multiplying on the left by $E_k^{-1}, E_{k-1}^{-1}$ , and so on to $E_1^{-1}$ , we see that          |
| johnston-linear-matrix-algebra-F01276 | 128  | 1.000 | $\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$ | $\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$ | b) $\left[\begin{array}{ccc} 1 & 2 & -2 \\ 0 & 1 & -2 \\ 1 & 1 & 1 \end{array}\right]$               |
| johnston-linear-matrix-algebra-F01277 | 129  | 1.000 | $A^{-1}[A \mid I]=\left[I \mid A^{-1}\right]$                                       | $A^{-1}[A \mid I] = [I \mid A^{-1}]$                                                | tiplication shows that $A^{-1}[A \mid I] = [I \mid A^{-1}]$ . However, multiplication on             |
| johnston-linear-matrix-algebra-F01278 | 129  | 1.000 | $\left[I \mid A^{-1}\right]$                                                        | $[I \mid A^{-1}]$                                                                   | $[A \mid I]$ , so we conclude that $[A \mid I]$ can be row-reduced to $[I \mid A^{-1}]$ . This leads |
| johnston-linear-matrix-algebra-F01279 | 129  | 1.000 | $E \in \mathcal{M}_n$                                                               | $E \in \mathcal{M}_n$                                                               | $E \in \mathcal{M}_n$ such that the RREF of the block matrix $[A \mid I]$ is $[I \mid E]$ .          |
| johnston-linear-matrix-algebra-F01280 | 129  | 1.000 | $[I \mid E]$                                                                        | $[I \mid E]$                                                                        | $E \in \mathcal{M}_n$ such that the RREF of the block matrix $[A \mid I]$ is $[I \mid E]$ .          |
| johnston-linear-matrix-algebra-F01281 | 129  | 1.000 | $A^{-1}=E$                                                                          | $A^{-1} = E$                                                                        | Furthermore, if $A$ is invertible then it is necessarily the case that $A^{-1} = E$ .                |
| johnston-linear-matrix-algebra-F01282 | 129  | 1.000 | $I=E A$                                                                             | $I = EA$                                                                            | $I = EA$ (and thus $E = A^{-1}$ ).                                                                   |
| johnston-linear-matrix-algebra-F01283 | 129  | 1.000 | $E=A^{-1}$                                                                          | $E = A^{-1}$                                                                        | $I = EA$ (and thus $E = A^{-1}$ ).                                                                   |
| johnston-linear-matrix-algebra-F01284 | 129  | 0.995 | $\left[\begin{array}{ll} 2 & 2 \\ 4 & 5 \end{array}\right]$                         | $\left[\begin{array}{cc} 2 & 2 \\ 4 & 5 \end{array}\right]$                         | a) $\left[\begin{array}{cc} 2 & 2 \\ 4 & 5 \end{array}\right]$                                       |
| johnston-linear-matrix-algebra-F01285 | 129  | 1.000 | $\left[\begin{array}{cc} 1 & -2 \\ -3 & 6 \end{array}\right]$                       | $\left[\begin{array}{cc} 1 & -2 \\ -3 & 6 \end{array}\right]$                       | b) $\left[\begin{array}{cc} 1 & -2 \\ -3 & 6 \end{array}\right]$                                     |
| johnston-linear-matrix-algebra-F01286 | 129  | 1.000 | $\left[\begin{array}{lll} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$   | $\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$   | d) $\left[\begin{array}{ccc} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{array}\right]$                 |
| johnston-linear-matrix-algebra-F01287 | 131  | 0.500 | $\mathbf{2} \times \mathbf{2}$                                                      | $2 \times 2$                                                                        | $2 \times 2$ Matrix                                                                                  |
| johnston-linear-matrix-algebra-F01288 | 131  | 1.000 | $a \text{ d-} b \text{ c}$                                                          | $ad - bc$                                                                           | The quantity $ad - bc$                                                                               |

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| johnston-linear-matrix-algebra-F01289 | 131  | 0.924 | $A=\left[\begin{array}{ll}a & b \\ c & d\end{array}\right]$ | $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$      | The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$ , and                                            |
| johnston-linear-matrix-algebra-F01290 | 131  | 1.000 | $a - b - c = 0$                                             | $ad - bc = 0$                                           | On the other hand, if $ad - bc = 0$ then $ad = bc$ . From here we split into                                                                                 |
| johnston-linear-matrix-algebra-F01291 | 131  | 1.000 | $a = b - c$                                                 | $ad = bc$                                               | On the other hand, if $ad - bc = 0$ then $ad = bc$ . From here we split into                                                                                 |
| johnston-linear-matrix-algebra-F01292 | 131  | 1.000 | $a = 0$                                                     | $a = 0$                                                 | <b>Case 1:</b> If either $a = 0$ or $b = 0$ then $ad = bc$ implies that two of the entries                                                                   |
| johnston-linear-matrix-algebra-F01293 | 131  | 1.000 | $b = 0$                                                     | $b = 0$                                                 | <b>Case 1:</b> If either $a = 0$ or $b = 0$ then $ad = bc$ implies that two of the entries                                                                   |
| johnston-linear-matrix-algebra-F01294 | 131  | 1.000 | $a, b \neq 0$                                               | $a, b \neq 0$                                           | <b>Case 2:</b> If $a, b \neq 0$ then $ad = bc$ implies $d/b = c/a$ , so the second row of $A$ is                                                             |
| johnston-linear-matrix-algebra-F01295 | 131  | 1.000 | $d/b = c/a$                                                 | $d/b = c/a$                                             | <b>Case 2:</b> If $a, b \neq 0$ then $ad = bc$ implies $d/b = c/a$ , so the second row of $A$ is                                                             |
| johnston-linear-matrix-algebra-F01296 | 131  | 1.000 | $a - b - c = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$   | $ad - bc = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$ | a) We first compute $ad - bc = 1 \times 4 - 2 \times 3 = 4 - 6 = -2 \neq 0$ , so this                                                                        |
| johnston-linear-matrix-algebra-F01297 | 131  | 1.000 | $a - b - c = 1 \times 4 - 2 \times 2 = 4 - 4 = 0$           | $ad - bc = 1 \times 4 - 2 \times 2 = 4 - 4 = 0$         | b) Again, we start by computing $ad - bc = 1 \times 4 - 2 \times 2 = 4 - 4 = 0$ .                                                                            |
| johnston-linear-matrix-algebra-F01298 | 131  | 1.000 | $a - b - c = 2 \times 5 - 2 \times 4 = 10 - 8 = 2 \neq 0$   | $ad - bc = 2 \times 5 - 2 \times 4 = 10 - 8 = 2 \neq 0$ | c) Yet again, we first compute $ad - bc = 2 \times 5 - 2 \times 4 = 10 - 8 = 2 \neq 0$ ,                                                                     |
| johnston-linear-matrix-algebra-F01299 | 132  | 1.000 | $a - b - c = 1 \times 6 - (-2) \times (-3) = 6 - 6 = 0$     | $ad - bc = 1 \times 6 - (-2) \times (-3) = 6 - 6 = 0$   | d) Once again, computing $ad - bc = 1 \times 6 - (-2) \times (-3) = 6 - 6 = 0$                                                                               |
| johnston-linear-matrix-algebra-F01300 | 133  | 1.000 | $AA^{-1} = A^{-1}A = I$                                     | $AA^{-1} = A^{-1}A = I$                                 | $A^{-1}$ as being inverses of each other if $AA^{-1} = A^{-1}A = I$ , it is actually enough                                                                  |
| johnston-linear-matrix-algebra-F01301 | 133  | 1.000 | $B \in \mathcal{M}_n$                                       | $B \in \mathcal{M}_n$                                   | Suppose $A \in \mathcal{M}_n$ . If there exists a matrix $B \in \mathcal{M}_n$ such that $AB = I$ or                                                         |
| johnston-linear-matrix-algebra-F01302 | 133  | 1.000 | $AB = I$                                                    | $AB = I$                                                | Suppose $A \in \mathcal{M}_n$ . If there exists a matrix $B \in \mathcal{M}_n$ such that $AB = I$ or                                                         |
| johnston-linear-matrix-algebra-F01303 | 133  | 1.000 | $BA = I$                                                    | $BA = I$                                                | $BA = I$ then $A$ is invertible and $A^{-1} = B$ .                                                                                                           |
| johnston-linear-matrix-algebra-F01304 | 133  | 1.000 | $A^{-1} = B$                                                | $A^{-1} = B$                                            | $BA = I$ then $A$ is invertible and $A^{-1} = B$ .                                                                                                           |
| johnston-linear-matrix-algebra-F01305 | 133  | 0.974 | $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$                   | $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$               | shows that $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$ , but also $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$ , so $\mathbf{x} = \mathbf{0}$ . The linear system |
| johnston-linear-matrix-algebra-F01306 | 133  | 0.974 | $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$                   | $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$               | shows that $BA\mathbf{x} = B\mathbf{0} = \mathbf{0}$ , but also $BA\mathbf{x} = I\mathbf{x} = \mathbf{x}$ , so $\mathbf{x} = \mathbf{0}$ . The linear system |
| johnston-linear-matrix-algebra-F01307 | 133  | 1.000 | $B = A^{-1}$                                                | $B = A^{-1}$                                            | that $B = A^{-1}$ , as desired.                                                                                                                              |
| johnston-linear-matrix-algebra-F01308 | 133  | 0.966 | $AC = I$                                                    | $AC = I$                                                | $AC = I$ , since the only way that $AC$ could be square is if $C$ is $2 \times 3$ , in                                                                       |
| johnston-linear-matrix-algebra-F01309 | 133  | 1.000 | $xy$                                                        | $xy$                                                    | that sends $\mathbb{R}^2$ to the $xy$ -plane in $\mathbb{R}^3$ , and there is another linear transforma-                                                     |
| johnston-linear-matrix-algebra-F01310 | 134  | 1.000 | $E \in \mathcal{M}_3$                                       | $E \in \mathcal{M}_3$                                   | <b>2.2.1</b> Write down the elementary matrix $E \in \mathcal{M}_3$ corre-                                                                                   |
| johnston-linear-matrix-algebra-F01311 | 134  | 0.975 | $R_1 \leftrightarrow R_3$                                   | $R_1 \leftrightarrow R_3$                               | *(a) $R_1 \leftrightarrow R_3$                                                                                                                               |
| johnston-linear-matrix-algebra-F01312 | 134  | 1.000 | $R_1 + 3R_2$                                                | $R_1 + 3R_2$                                            | *(c) $R_1 + 3R_2$                                                                                                                                            |
| johnston-linear-matrix-algebra-F01313 | 134  | 1.000 | $R_3 - 4R_1$                                                | $R_3 - 4R_1$                                            | (d) $R_3 - 4R_1$                                                                                                                                             |
| johnston-linear-matrix-algebra-F01314 | 134  | 1.000 | $3R_1$                                                      | $3R_1$                                                  | *(e) $3R_1$                                                                                                                                                  |
| johnston-linear-matrix-algebra-F01315 | 134  | 1.000 | $6R_3$                                                      | $6R_3$                                                  | (f) $6R_3$                                                                                                                                                   |

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| johnston-linear-matrix-aligned-<br>F01316 | 134  | ■ 0.995 | $R_{\{2\}}-2 R_{\{3\}}$                                                                                                                                                                                             | $R_2-2R_3$                                                                                                                                                                                                          | *(g) $R_2-2R_3$                                                                                                                                                                                                     |
| johnston-linear-matrix-aligned-<br>F01317 | 134  | ■ 1.000 | $R_{\{2\}}+3 R_{\{1\}}$                                                                                                                                                                                             | $R_2+3R_1$                                                                                                                                                                                                          | (h) $R_2+3R_1$                                                                                                                                                                                                      |
| johnston-linear-matrix-aligned-<br>F01318 | 134  | ■ 1.000 | $\left[\begin{array}{ll} 6 & 3 \\ 2 & 1 \end{array}\right]$                                                                                                                                                         | $\left[\begin{array}{cc} 6 & 3 \\ 2 & 1 \end{array}\right]$                                                                                                                                                         | (b) $\left[\begin{array}{cc} 6 & 3 \\ 2 & 1 \end{array}\right]$                                                                                                                                                     |
| johnston-linear-matrix-aligned-<br>F01319 | 134  | ■ 0.892 | $\left[\begin{array}{ll} 2 & 3 \\ 3 & 2 \end{array}\right]$                                                                                                                                                         | $\left[\begin{array}{cc} 2 & 3 \\ 3 & 2 \end{array}\right]$                                                                                                                                                         | *(c) $\left[\begin{array}{cc} 2 & 3 \\ 3 & 2 \end{array}\right]$                                                                                                                                                    |
| johnston-linear-matrix-aligned-<br>F01320 | 134  | ■ 0.995 | $\left[\begin{array}{lll} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array}\right]$                                                                                                                                                | $\left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array}\right]$                                                                                                                                                | (d) $\left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 1 & 2 \end{array}\right]$                                                                                                                                            |
| johnston-linear-matrix-aligned-<br>F01321 | 134  | ■ 0.887 | $\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array}\right]$                                                                                                                                 | $\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array}\right]$                                                                                                                                 | *(e) $\left[\begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array}\right]$                                                                                                                            |
| johnston-linear-matrix-aligned-<br>F01322 | 134  | ■ 0.997 | $\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array}\right]$                                                                                                                                  | $\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array}\right]$                                                                                                                                  | (f) $\left[\begin{array}{ccc} 2 & 6 & 1 \\ 0 & 0 & 0 \\ 3 & -2 & 7 \end{array}\right]$                                                                                                                              |
| johnston-linear-matrix-aligned-<br>F01323 | 134  | ■ 0.980 | $\left[\begin{array}{ll} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array}\right]$                                                                                                                                                | $\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array}\right]$                                                                                                                                                | *(g) $\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{array}\right]$                                                                                                                                           |
| johnston-linear-matrix-aligned-<br>F01324 | 134  | ■ 1.000 | $\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array}\right]$                                                                                                                             | $\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array}\right]$                                                                                                                             | (h) $\left[\begin{array}{ccc} 1 & -2 & -2 \\ -2 & 1 & -2 \\ -2 & -2 & 1 \end{array}\right]$                                                                                                                         |
| johnston-linear-matrix-aligned-<br>F01325 | 134  | ■ 0.997 | $\left[\begin{array}{llll} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array}\right]$                                                                                                     | $\left[\begin{array}{cccc} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array}\right]$                                                                                                     | $\left[\begin{array}{cccc} 0 & 1 & 1 & 2 \\ 4 & 2 & 3 & 0 \\ 3 & 1 & 5 & 5 \\ 5 & 0 & 5 & 2 \end{array}\right]$                                                                                                     |
| johnston-linear-matrix-aligned-<br>F01326 | 134  | ■ 1.000 | $\left[\begin{array}{cccc} -1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0 \end{array}\right]$                                                                                                  | $\left[\begin{array}{cccc} -1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0 \end{array}\right]$                                                                                                  | (b) $\left[\begin{array}{cccc} -1 & -2 & 2 & 2 \\ -1 & 1 & 1 & 0 \\ 3 & 3 & 5 & 1 \\ 5 & 2 & 2 & 0 \end{array}\right]$                                                                                              |
| johnston-linear-matrix-aligned-<br>F01327 | 134  | ■ 1.000 | $\left[\begin{array}{ccccc} -1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5 \end{array}\right]$                                                          | $\left[\begin{array}{ccccc} -1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5 \end{array}\right]$                                                          | $\left[\begin{array}{ccccc} -1 & -3 & 3 & 5 & 4 \\ 1 & 5 & -2 & 4 & 5 \\ 5 & -3 & 0 & 1 & 0 \\ 4 & 1 & -2 & 1 & 1 \\ 4 & 4 & 5 & 3 & 5 \end{array}\right]$                                                          |
| johnston-linear-matrix-aligned-<br>F01328 | 134  | ■ 1.000 | $\left[\begin{array}{cccccc} -1 & -2 & -1 & 1 & 2 & 4 \\ -1 & 0 & 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -2 & -5 & -3 \\ 1 & 0 & -2 & -5 & -3 & -6 \\ 1 & -5 & 3 & 1 & -4 & 7 \\ 1 & -5 & 3 & 1 & -4 & 7 \end{array}\right]$ | $\left[\begin{array}{cccccc} -1 & -2 & -1 & 1 & 2 & 4 \\ -1 & 0 & 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -2 & -5 & -3 \\ 1 & 0 & -2 & -5 & -3 & -6 \\ 1 & -5 & 3 & 1 & -4 & 7 \\ 1 & -5 & 3 & 1 & -4 & 7 \end{array}\right]$ | $\left[\begin{array}{cccccc} -1 & -2 & -1 & 1 & 2 & 4 \\ -1 & 0 & 0 & -1 & 0 & 2 \\ 1 & 0 & 0 & -2 & -5 & -3 \\ 1 & 0 & -2 & -5 & -3 & -6 \\ 1 & -5 & 3 & 1 & -4 & 7 \\ 1 & -5 & 3 & 1 & -4 & 7 \end{array}\right]$ |
| johnston-linear-matrix-aligned-<br>F01329 | 134  | ■ 1.000 | E A                                                                                                                                                                                                                 | EA                                                                                                                                                                                                                  | a matrix $E \in \mathcal{M}_m$ such that EA equals the reduced row                                                                                                                                                  |

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| johnston-linear-matrix-align=left<br>F01330 | 134  | 1.000 | $\left[\begin{array}{cc}2 & -1 \\ -4 & 2\end{array}\right]$                                    | $\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$                                  | (b) $\begin{bmatrix} 2 & -1 \\ -4 & 2 \end{bmatrix}$                                  |
| johnston-linear-matrix-align=left<br>F01331 | 134  | 0.897 | $\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 6\end{array}\right]$                             | $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$                            | *(c) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$                           |
| johnston-linear-matrix-align=left<br>F01332 | 134  | 0.992 | $\left[\begin{array}{ccc}0 & -1 & 3 \\ 0 & 2 & 1\end{array}\right]$                            | $\begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}$                           | (d) $\begin{bmatrix} 0 & -1 & 3 \\ 0 & 2 & 1 \end{bmatrix}$                           |
| johnston-linear-matrix-align=left<br>F01333 | 134  | 0.876 | $\left[\begin{array}{cc}2 & 1 \\ -1 & 2 \\ 1 & 1\end{array}\right]$                            | $\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$                          | *(e) $\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}$                         |
| johnston-linear-matrix-align=left<br>F01334 | 134  | 1.000 | $\left[\begin{array}{ccc}1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2\end{array}\right]$              | $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$             | (f) $\begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & -1 \\ 0 & 1 & 2 \end{bmatrix}$             |
| johnston-linear-matrix-align=left<br>F01335 | 134  | 0.999 | $\left[\begin{array}{ccc}0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3\end{array}\right]$               | $\begin{bmatrix} 0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3 \end{bmatrix}$              | *(g) $\begin{bmatrix} 0 & 1 & 3 \\ 3 & -2 & 3 \\ 1 & 0 & 3 \end{bmatrix}$             |
| johnston-linear-matrix-align=left<br>F01336 | 134  | 1.000 | $\left[\begin{array}{cccc}4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2\end{array}\right]$ | $\begin{bmatrix} 4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2 \end{bmatrix}$ | (h) $\begin{bmatrix} 4 & 0 & 2 & 4 \\ -1 & -1 & 1 & 2 \\ 5 & 0 & 0 & 2 \end{bmatrix}$ |
| johnston-linear-matrix-align=left<br>F01337 | 134  | 1.000 | $A^6=I$                                                                                        | $A^6 = I$                                                                         | (h) If $A^6 = I$ then $A$ is invertible.                                              |
| johnston-linear-matrix-align=left<br>F01338 | 134  | 1.000 | $A^7=0$                                                                                        | $A^7 = O$                                                                         | *(i) If $A^7 = O$ then $A$ is not invertible.                                         |
| johnston-linear-matrix-align=left<br>F01339 | 134  | 1.000 | $X A=B$                                                                                        | $XA = B$                                                                          | $XA = B$ , then $X = A^{-1}B$ .                                                       |
| johnston-linear-matrix-align=left<br>F01340 | 134  | 1.000 | $X=A^{-1} B$                                                                                   | $X = A^{-1}B$                                                                     | $XA = B$ , then $X = A^{-1}B$ .                                                       |
| johnston-linear-matrix-align=left<br>F01341 | 134  | 1.000 | $a \neq 0$                                                                                     | $a \neq 0$                                                                        | 2.2.6 Let $a \neq 0$ be a real number. Show that the following                        |
| johnston-linear-matrix-align=left<br>F01342 | 134  | 1.000 | $a=b$                                                                                          | $a = b$                                                                           | (a) Show that $A$ is not invertible if $a = b$ .                                      |
| johnston-linear-matrix-align=left<br>F01343 | 134  | 1.000 | $a \neq b$                                                                                     | $a \neq b$                                                                        | vertible) if $a \neq b$ . [Hint: Try some small examples                              |
| johnston-linear-matrix-align=left<br>F01344 | 134  | 1.000 | $A^{-4}$                                                                                       | $A^{-4}$                                                                          | (d) $A^{-4}$                                                                          |
| johnston-linear-matrix-align=left<br>F01345 | 134  | 1.000 | $A^k=0$                                                                                        | $A^k = O$                                                                         | some positive integer $k$ such that $A^k = O$ . In this exercise,                     |
| johnston-linear-matrix-align=left<br>F01346 | 135  | 1.000 | $I-A$                                                                                          | $I - A$                                                                           | (a) Suppose $A^2 = O$ . Prove that $I - A$ is invertible, and                         |
| johnston-linear-matrix-align=left<br>F01347 | 135  | 1.000 | $I+A$                                                                                          | $I + A$                                                                           | its inverse is $I + A$ .                                                              |
| johnston-linear-matrix-align=left<br>F01348 | 135  | 1.000 | $A^3=0$                                                                                        | $A^3 = O$                                                                         | (b) Suppose $A^3 = O$ . Prove that $I - A$ is invertible, and                         |
| johnston-linear-matrix-align=left<br>F01349 | 135  | 1.000 | $I+A+A^2$                                                                                      | $I + A + A^2$                                                                     | its inverse is $I + A + A^2$ .                                                        |
| johnston-linear-matrix-align=left<br>F01350 | 135  | 1.000 | $P=A\left(A^T A\right)^{-1} A^T$                                                               | $P = A \left(A^T A\right)^{-1} A^T$                                               | and let $P = \tilde{A} \left(\tilde{A}^T \tilde{A}\right)^{-1} \tilde{A}^T$ .         |
| johnston-linear-matrix-align=left<br>F01351 | 135  | 1.000 | $P$                                                                                            | $P$                                                                               | (a) What is the size of $P$ ?                                                         |
| johnston-linear-matrix-align=left<br>F01352 | 135  | 1.000 | $P=P^T$                                                                                        | $P = P^T$                                                                         | (b) Show that $P = P^T$ . [Side note: Matrices with this                              |

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| johnston-linear-matrix-augmented<br>F01353 | 135  | 0.999 | $P^2=P$                                                                                                                                                  | $P^2 = P$                                                                                                                                                | (c) Show that $P^2 = P$ . [Side note: Matrices such that                                                                                                     |
| johnston-linear-matrix-augmented<br>F01354 | 135  | 1.000 | $P^T=P^2=P$                                                                                                                                              | $P^T = P^2 = P$                                                                                                                                          | $P^T = P^2 = P$ are called <b>orthogonal projections</b> .]                                                                                                  |
| johnston-linear-matrix-augmented<br>F01355 | 135  | 0.997 | $P, Q \in \mathcal{M}_n$                                                                                                                                 | $P, Q \in \mathcal{M}_n$                                                                                                                                 | *2.2.12 Suppose that $P, Q \in \mathcal{M}_n$ and $P$ is invertible. Show                                                                                    |
| johnston-linear-matrix-augmented<br>F01356 | 135  | 0.996 | $[P \mid Q]$                                                                                                                                             | $[P \mid Q]$                                                                                                                                             | that row-reducing the augmented matrix $[P \mid Q]$ produces                                                                                                 |
| johnston-linear-matrix-augmented<br>F01357 | 135  | 1.000 | $\left[ I \mid P^{-1}Q \right]$                                                                                                                          | $[I \mid P^{-1}Q]$                                                                                                                                       | the matrix $[I \mid P^{-1}Q]$ .                                                                                                                              |
| johnston-linear-matrix-augmented<br>F01358 | 135  | 1.000 | $AX=B$                                                                                                                                                   | $AX = B$                                                                                                                                                 | *(a) $AX = B$                                                                                                                                                |
| johnston-linear-matrix-augmented<br>F01359 | 135  | 1.000 | $AX=0$                                                                                                                                                   | $AXB = O$                                                                                                                                                | (b) $AXB = O$                                                                                                                                                |
| johnston-linear-matrix-augmented<br>F01360 | 135  | 1.000 | $AX=B$                                                                                                                                                   | $AXB = I$                                                                                                                                                | *(c) $AXB = I$                                                                                                                                               |
| johnston-linear-matrix-augmented<br>F01361 | 135  | 1.000 | $AX+B=A-B$                                                                                                                                               | $AX + BX = A - B$                                                                                                                                        | (d) $AX + BX = A - B$                                                                                                                                        |
| johnston-linear-matrix-augmented<br>F01362 | 135  | 1.000 | $B^{-1}$                                                                                                                                                 | $B^{-1}$                                                                                                                                                 | tained by swapping two of the rows of $A$ . Explain why $B^{-1}$                                                                                             |
| johnston-linear-matrix-augmented<br>F01363 | 135  | 0.996 | $A_1, A_2, \ldots, A_n$                                                                                                                                  | $A_1, A_2, \ldots, A_n$                                                                                                                                  | is invertible if and only if each of $A_1, A_2, \ldots, A_n$ are                                                                                             |
| johnston-linear-matrix-augmented<br>F01364 | 135  | 0.959 | $\left[ \begin{array}{cc} A & I \\ I & O \end{array} \right]^{-1} = \left[ \begin{array}{cc} O & I \\ I & -A \end{array} \right]$                        | $\left[ \begin{array}{cc} A & I \\ I & O \end{array} \right]^{-1} = \left[ \begin{array}{cc} O & I \\ I & -A \end{array} \right]$                        | *(a) $\left[ \begin{array}{cc} A & I \\ I & O \end{array} \right]^{-1} = \left[ \begin{array}{cc} O & I \\ I & -A \end{array} \right]$                       |
| johnston-linear-matrix-augmented<br>F01365 | 135  | 0.998 | $\left[ \begin{array}{cc} I & B \\ O & I \end{array} \right]^{-1} = \left[ \begin{array}{cc} I & -B \\ O & I \end{array} \right]$                        | $\left[ \begin{array}{cc} I & B \\ O & I \end{array} \right]^{-1} = \left[ \begin{array}{cc} I & -B \\ O & I \end{array} \right]$                        | (b) $\left[ \begin{array}{cc} I & B \\ O & I \end{array} \right]^{-1} = \left[ \begin{array}{cc} I & -B \\ O & I \end{array} \right]$                        |
| johnston-linear-matrix-augmented<br>F01366 | 135  | 0.946 | $\left[ \begin{array}{ll} A & B \\ O & D \end{array} \right]^{-1} = \left[ \begin{array}{ll} A^{-1} & -A^{-1}BD^{-1} \\ O & D^{-1} \end{array} \right]$  | $\left[ \begin{array}{cc} A & B \\ O & D \end{array} \right]^{-1} = \left[ \begin{array}{cc} A^{-1} & -A^{-1}BD^{-1} \\ O & D^{-1} \end{array} \right]$  | *(c) $\left[ \begin{array}{cc} A & B \\ O & D \end{array} \right]^{-1} = \left[ \begin{array}{cc} A^{-1} & -A^{-1}BD^{-1} \\ O & D^{-1} \end{array} \right]$ |
| johnston-linear-matrix-augmented<br>F01367 | 135  | 1.000 | $S=D-CA^{-1}B$                                                                                                                                           | $S = D - CA^{-1}B$                                                                                                                                       | (d) If we define $S = D - CA^{-1}B$ , then                                                                                                                   |
| johnston-linear-matrix-augmented<br>F01368 | 135  | 1.000 | $\left[ \begin{array}{cccc} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right]$                                        | $\left[ \begin{array}{cccc} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right]$                                        | $\left[ \begin{array}{cccc} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right]$                                            |
| johnston-linear-matrix-augmented<br>F01369 | 135  | 1.000 | $\left[ \begin{array}{cccc} 1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$                                        | $\left[ \begin{array}{cccc} 1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$                                        | $\left[ \begin{array}{cccc} 1 & 0 & 4 & 6 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$                                            |
| johnston-linear-matrix-augmented<br>F01370 | 135  | 0.999 | $\left[ \begin{array}{ccccc} 1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array} \right]$ | $\left[ \begin{array}{ccccc} 1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array} \right]$ | $\left[ \begin{array}{ccccc} 1 & 1 & 2 & -1 & 0 \\ 1 & 2 & 2 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 3 \end{array} \right]$     |

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| johnston-linear-matrix-algebra-F01371 | 135  | 1.000 | $\left[\begin{array}{llll}2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2\end{array}\right]$ | $\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$ | $\begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 1 & 1 & 2 \end{bmatrix}$                                               |
| johnston-linear-matrix-algebra-F01372 | 136  | 1.000 | $A + \mathbf{v} \mathbf{w}^T$                                                                                 | $A + \mathbf{v} \mathbf{w}^T$                                                                    | that this formula holds by multiplying $A + \mathbf{v} \mathbf{w}^T$ by its pro-                                                               |
| johnston-linear-matrix-algebra-F01373 | 136  | 0.520 | $A_{\{n\}}$                                                                                                   | $A_n$                                                                                            | <a href="#">2.2.24</a> Let $A_n$ be the $n \times n$ matrix                                                                                    |
| johnston-linear-matrix-algebra-F01374 | 136  | 1.000 | $A_{\{n\}}^{-1}$                                                                                              | $A_n^{-1}$                                                                                       | (a) Use computer software to compute $A_n^{-1}$ when $n =$                                                                                     |
| johnston-linear-matrix-algebra-F01375 | 136  | 1.000 | $n =$                                                                                                         | $n =$                                                                                            | (a) Use computer software to compute $A_n^{-1}$ when $n =$                                                                                     |
| johnston-linear-matrix-algebra-F01376 | 136  | 0.998 | $\mathcal{S}$                                                                                                 | $\mathcal{S}$                                                                                    | A <b>subspace</b> of $\mathbb{R}^n$ is a non-empty set $\mathcal{S}$ of vectors in $\mathbb{R}^n$ with the properties                          |
| johnston-linear-matrix-algebra-F01377 | 136  | 1.000 | $\mathbf{v}, \mathbf{w} \in \mathcal{S}$                                                                      | $\mathbf{v}, \mathbf{w} \in \mathcal{S}$                                                         | a) if $\mathbf{v}, \mathbf{w} \in \mathcal{S}$ then $\mathbf{v} + \mathbf{w} \in \mathcal{S}$ , and                                            |
| johnston-linear-matrix-algebra-F01378 | 136  | 1.000 | $\mathbf{v} + \mathbf{w} \in \mathcal{S}$                                                                     | $\mathbf{v} + \mathbf{w} \in \mathcal{S}$                                                        | a) if $\mathbf{v}, \mathbf{w} \in \mathcal{S}$ then $\mathbf{v} + \mathbf{w} \in \mathcal{S}$ , and                                            |
| johnston-linear-matrix-algebra-F01379 | 136  | 1.000 | $\mathbf{v} \in \mathcal{S}$                                                                                  | $\mathbf{v} \in \mathcal{S}$                                                                     | b) if $\mathbf{v} \in \mathcal{S}$ and $c \in \mathbb{R}$ then $c\mathbf{v} \in \mathcal{S}$ .                                                 |
| johnston-linear-matrix-algebra-F01380 | 136  | 1.000 | $c \mathbf{v} \in \mathcal{S}$                                                                                | $c\mathbf{v} \in \mathcal{S}$                                                                    | b) if $\mathbf{v} \in \mathcal{S}$ and $c \in \mathbb{R}$ then $c\mathbf{v} \in \mathcal{S}$ .                                                 |
| johnston-linear-matrix-algebra-F01381 | 137  | 1.000 | $y = x / 2 + 1$                                                                                               | $y = x/2 + 1$                                                                                    | origin, so the line (a) $y = x/2$ is a subspace but the line (b) $y = x/2 + 1$ is not.                                                         |
| johnston-linear-matrix-algebra-F01382 | 137  | 0.993 | $1 / 2$                                                                                                       | $1/2$                                                                                            | b) The line in $\mathbb{R}^2$ with slope $1/2$ and y-intercept $-1$ .                                                                          |
| johnston-linear-matrix-algebra-F01383 | 137  | 1.000 | $x + y - 3z = 0$                                                                                              | $x + y - 3z = 0$                                                                                 | c) The plane in $\mathbb{R}^3$ with equation $x + y - 3z = 0$ .                                                                                |
| johnston-linear-matrix-algebra-F01384 | 137  | 0.995 | $(2, 0)$                                                                                                      | $(2, 0)$                                                                                         | by noticing that $(2, 0)$ is on the line, but $2(2, 0) = (4, 0)$ is not, so it                                                                 |
| johnston-linear-matrix-algebra-F01385 | 137  | 0.995 | $2(2, 0) = (4, 0)$                                                                                            | $2(2, 0) = (4, 0)$                                                                               | by noticing that $(2, 0)$ is on the line, but $2(2, 0) = (4, 0)$ is not, so it                                                                 |
| johnston-linear-matrix-algebra-F01386 | 138  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots$                                                                          | $\mathbf{v}_1, \mathbf{v}_2, \dots$                                                              | Be careful: $\mathbf{v}_1, \mathbf{v}_2, \dots$                                                                                                |
| johnston-linear-matrix-algebra-F01387 | 138  | 1.000 | $\mathbf{v}_k$                                                                                                | $\mathbf{v}_k$                                                                                   | $\mathbf{v}_k$ are vectors, not                                                                                                                |
| johnston-linear-matrix-algebra-F01388 | 138  | 0.600 | $\ldots, v_{\{n\}}$                                                                                           | $\dots, v_n$                                                                                     | $\dots, v_n$ .                                                                                                                                 |
| johnston-linear-matrix-algebra-F01389 | 138  | 0.999 | $\{(x, y) \in$                                                                                                | $\{(x, y) \in$                                                                                   | The notation $\{(x, y) \in$                                                                                                                    |
| johnston-linear-matrix-algebra-F01390 | 138  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathcal{S}$                                            | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathcal{S}$                                | for all $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathcal{S}$ and all $c_1, c_2, \dots, c_k \in \mathbb{R}$ (see Exercise 2.3.15). |
| johnston-linear-matrix-algebra-F01391 | 138  | 1.000 | $m \leq n$                                                                                                    | $m \leq n$                                                                                       | of $\mathbb{R}^m$ (for some $m \leq n$ ) going through the origin. For example, lines look like                                                |
| johnston-linear-matrix-algebra-F01392 | 138  | 0.993 | $\mathbb{R}^{\{1\}}$                                                                                          | $\mathbb{R}^1$                                                                                   | copies or $\mathbb{R}^1$ , planes look like copies of $\mathbb{R}^2$ , and so on. The following example                                        |
| johnston-linear-matrix-algebra-F01393 | 138  | 1.000 | $y = x^{\{2\}}$                                                                                               | $y = x^2$                                                                                        | a) The graph of the function $y = x^2$ in $\mathbb{R}^2$ .                                                                                     |
| johnston-linear-matrix-algebra-F01394 | 138  | 1.000 | $\left\{ (x, y) \in \mathbb{R}^{\{2\}} : x \geq 0, y \geq 0 \right\}$                                         | $\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$                                               | b) The set $\{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\}$ .                                                                                |
| johnston-linear-matrix-algebra-F01395 | 138  | 1.000 | $\mathbf{v} =$                                                                                                | $\mathbf{v} =$                                                                                   | a) This set is <i>not</i> a subspace, since (for example) the vectors $\mathbf{v} =$                                                           |
| johnston-linear-matrix-algebra-F01396 | 138  | 0.867 | $-1, 1$                                                                                                       | $-1, 1$                                                                                          | $(-1, 1)$ and $\mathbf{w} = (1, 1)$ are in the set, but $\mathbf{v} + \mathbf{w} = (0, 2)$ is not, since                                       |

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| johnston-linear-matrix-algebra-F01397 | 138  | 0.867 | $\mathbf{v}+\mathbf{w}=(0,2)$                                                | $\mathbf{v}+\mathbf{w}=(0,2)$                                                | $(-1,1)$ and $\mathbf{w}=(1,1)$ are in the set, but $\mathbf{v}+\mathbf{w}=(0,2)$ is not, since                                                           |
| johnston-linear-matrix-algebra-F01398 | 138  | 1.000 | $2=0^2$                                                                      | $2=0^2$                                                                      | it is not the case that $2=0^2$ :                                                                                                                         |
| johnston-linear-matrix-algebra-F01399 | 139  | 1.000 | $\{\mathbf{0}\}$                                                             | $\{\mathbf{0}\}$                                                             | origin, $\{\mathbf{0}\}$ , and all                                                                                                                        |
| johnston-linear-matrix-algebra-F01400 | 139  | 1.000 | $\mathbf{b}\neq\mathbf{0}$                                                   | $\mathbf{b}\neq\mathbf{0}$                                                   | when $\mathbf{b}\neq\mathbf{0}$ (see                                                                                                                      |
| johnston-linear-matrix-algebra-F01401 | 139  | 1.000 | $\operatorname{ker}(A)$                                                      | $\ker(A)$                                                                    | denoted by $\ker(A)$ .                                                                                                                                    |
| johnston-linear-matrix-algebra-F01402 | 139  | 1.000 | $\mathbf{v}=(1,1)$                                                           | $\mathbf{v}=(1,1)$                                                           | b) This set is <i>not</i> a subspace, since (for example) the vector $\mathbf{v}=(1,1)$                                                                   |
| johnston-linear-matrix-algebra-F01403 | 139  | 1.000 | $-\mathbf{v}=(-1,-1)$                                                        | $-\mathbf{v}=(-1,-1)$                                                        | is in the set, but $-\mathbf{v}=(-1,-1)$ is not, since the set only contains                                                                              |
| johnston-linear-matrix-algebra-F01404 | 139  | 1.000 | $v_1+v_2-3v_3=0$                                                             | $v_1+v_2-3v_3=0$                                                             | are both on the plane, so $v_1+v_2-3v_3=0$ and $w_1+w_2-3w_3=0$ .                                                                                         |
| johnston-linear-matrix-algebra-F01405 | 139  | 1.000 | $w_1+w_2-3w_3=0$                                                             | $w_1+w_2-3w_3=0$                                                             | are both on the plane, so $v_1+v_2-3v_3=0$ and $w_1+w_2-3w_3=0$ .                                                                                         |
| johnston-linear-matrix-algebra-F01406 | 139  | 1.000 | $A\mathbf{x}\in\mathbb{R}^m$                                                 | $A\mathbf{x}\in\mathbb{R}^m$                                                 | transformation that sends $\mathbf{x}\in\mathbb{R}^n$ to $A\mathbf{x}\in\mathbb{R}^m$ . The set of possible outputs of                                    |
| johnston-linear-matrix-algebra-F01407 | 139  | 1.000 | $A\mathbf{x}$                                                                | $A\mathbf{x}$                                                                | this linear transformation—that is, the set of vectors of the form $A\mathbf{x}$ —is also a                                                               |
| johnston-linear-matrix-algebra-F01408 | 139  | 0.751 | $\operatorname{range}(A)$                                                    | $\operatorname{range}(A)$                                                    | a) The <b>range</b> of $A$ is the subspace of $\mathbb{R}^m$ , denoted by $\operatorname{range}(A)$ , that                                                |
| johnston-linear-matrix-algebra-F01409 | 139  | 1.000 | $\operatorname{null}(A)$                                                     | $\operatorname{null}(A)$                                                     | b) The <b>null space</b> of $A$ is the subspace of $\mathbb{R}^n$ , denoted by $\operatorname{null}(A)$ , that                                            |
| johnston-linear-matrix-algebra-F01410 | 139  | 1.000 | $A\mathbf{0}=\mathbf{0}$                                                     | $A\mathbf{0}=\mathbf{0}$                                                     | Definition 2.3.1. It is non-empty because $A\mathbf{0}=\mathbf{0}$ , so $\mathbf{0}\in\operatorname{null}(A)$ . To see that it                            |
| johnston-linear-matrix-algebra-F01411 | 139  | 1.000 | $\mathbf{0}\in\operatorname{null}(A)$                                        | $\mathbf{0}\in\operatorname{null}(A)$                                        | Definition 2.3.1. It is non-empty because $A\mathbf{0}=\mathbf{0}$ , so $\mathbf{0}\in\operatorname{null}(A)$ . To see that it                            |
| johnston-linear-matrix-algebra-F01412 | 140  | 1.000 | $\mathbf{v},\mathbf{w}\in\operatorname{null}(A)$                             | $\mathbf{v},\mathbf{w}\in\operatorname{null}(A)$                             | satisfies property (a), let $\mathbf{v},\mathbf{w}\in\operatorname{null}(A)$ , so that $A\mathbf{v}=\mathbf{0}$ and $A\mathbf{w}=\mathbf{0}$ . Then       |
| johnston-linear-matrix-algebra-F01413 | 140  | 1.000 | $A\mathbf{v}=\mathbf{0}$                                                     | $A\mathbf{v}=\mathbf{0}$                                                     | satisfies property (a), let $\mathbf{v},\mathbf{w}\in\operatorname{null}(A)$ , so that $A\mathbf{v}=\mathbf{0}$ and $A\mathbf{w}=\mathbf{0}$ . Then       |
| johnston-linear-matrix-algebra-F01414 | 140  | 1.000 | $A\mathbf{w}=\mathbf{0}$                                                     | $A\mathbf{w}=\mathbf{0}$                                                     | satisfies property (a), let $\mathbf{v},\mathbf{w}\in\operatorname{null}(A)$ , so that $A\mathbf{v}=\mathbf{0}$ and $A\mathbf{w}=\mathbf{0}$ . Then       |
| johnston-linear-matrix-algebra-F01415 | 140  | 1.000 | $\mathbf{v}+\mathbf{w}\in\operatorname{null}(A)$                             | $\mathbf{v}+\mathbf{w}\in\operatorname{null}(A)$                             | so $\mathbf{v}+\mathbf{w}\in\operatorname{null}(A)$ too. Similarly, for property (b) we note that if $\mathbf{v}\in\operatorname{null}(A)$                |
| johnston-linear-matrix-algebra-F01416 | 140  | 1.000 | $\mathbf{v}\in\operatorname{null}(A)$                                        | $\mathbf{v}\in\operatorname{null}(A)$                                        | so $\mathbf{v}+\mathbf{w}\in\operatorname{null}(A)$ too. Similarly, for property (b) we note that if $\mathbf{v}\in\operatorname{null}(A)$                |
| johnston-linear-matrix-algebra-F01417 | 140  | 0.984 | $c\mathbf{v}\in\operatorname{null}(A)$                                       | $c\mathbf{v}\in\operatorname{null}(A)$                                       | so $c\mathbf{v}\in\operatorname{null}(A)$ too. It follows that $\operatorname{null}(A)$ is indeed a subspace.                                             |
| johnston-linear-matrix-algebra-F01418 | 140  | 1.000 | $A\mathbf{0}=\mathbf{0}\in\operatorname{range}(A)$                           | $A\mathbf{0}=\mathbf{0}\in\operatorname{range}(A)$                           | empty since $A\mathbf{0}=\mathbf{0}\in\operatorname{range}(A)$ . Furthermore, if $A\mathbf{x},A\mathbf{y}\in\operatorname{range}(A)$ and $c\in\mathbb{R}$ |
| johnston-linear-matrix-algebra-F01419 | 140  | 1.000 | $A\mathbf{x},A\mathbf{y}\in\operatorname{range}(A)$                          | $A\mathbf{x},A\mathbf{y}\in\operatorname{range}(A)$                          | empty since $A\mathbf{0}=\mathbf{0}\in\operatorname{range}(A)$ . Furthermore, if $A\mathbf{x},A\mathbf{y}\in\operatorname{range}(A)$ and $c\in\mathbb{R}$ |
| johnston-linear-matrix-algebra-F01420 | 140  | 1.000 | $A(\mathbf{x}+\mathbf{y})=A\mathbf{x}+A\mathbf{y}\in\operatorname{range}(A)$ | $A\mathbf{x}+A\mathbf{y}=A(\mathbf{x}+\mathbf{y})\in\operatorname{range}(A)$ | then $A\mathbf{x}+A\mathbf{y}=A(\mathbf{x}+\mathbf{y})\in\operatorname{range}(A)$ as well, as is $A(c\mathbf{x})=c(A\mathbf{x})$ .                        |
| johnston-linear-matrix-algebra-F01421 | 140  | 1.000 | $A(c\mathbf{x})=c(A\mathbf{x})$                                              | $A(c\mathbf{x})=c(A\mathbf{x})$                                              | then $A\mathbf{x}+A\mathbf{y}=A(\mathbf{x}+\mathbf{y})\in\operatorname{range}(A)$ as well, as is $A(c\mathbf{x})=c(A\mathbf{x})$ .                        |
| johnston-linear-matrix-algebra-F01422 | 140  | 0.899 | $A=\begin{bmatrix}2&-2\\1&-1\end{bmatrix}$                                   | $A=\begin{bmatrix}2&-2\\1&-1\end{bmatrix}$                                   | Describe and plot the range and null space of the matrix $A=\begin{bmatrix}2&-2\\1&-1\end{bmatrix}$ .                                                     |
| johnston-linear-matrix-algebra-F01423 | 140  | 1.000 | $\mathbf{x}=(x,y)$                                                           | $\mathbf{x}=(x,y)$                                                           | To get our hands on the range of $A$ , we let $\mathbf{x}=(x,y)$ and compute                                                                              |

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| johnston-linear-matrix-algebra-F01424 | 140  | 1.000 | $x - y$                                                                       | $x - y$                                                                       | Notice that $A\mathbf{x}$ only depends on $x - y$ , not $x$ and $y$ individually. We can                                               |
| johnston-linear-matrix-algebra-F01425 | 140  | 1.000 | $z = x - y$                                                                   | $z = x - y$                                                                   | thus define a new variable $z = x - y$ and see that $A\mathbf{x} = (2z, z) = z(2, 1)$ , so                                             |
| johnston-linear-matrix-algebra-F01426 | 140  | 1.000 | $A \mathbf{x} = (2z, z) = z(2, 1)$                                            | $A\mathbf{x} = (2z, z) = z(2, 1)$                                             | thus define a new variable $z = x - y$ and see that $A\mathbf{x} = (2z, z) = z(2, 1)$ , so                                             |
| johnston-linear-matrix-algebra-F01427 | 140  | 1.000 | $x = y$                                                                       | $x = y$                                                                       | $x = y$ . It follows that $\mathbf{x} = (x, x) = x(1, 1)$ , so the null space is the line going                                        |
| johnston-linear-matrix-algebra-F01428 | 140  | 1.000 | $\mathbf{x} = (x, x) = x(1, 1)$                                               | $\mathbf{x} = (x, x) = x(1, 1)$                                               | $x = y$ . It follows that $\mathbf{x} = (x, x) = x(1, 1)$ , so the null space is the line going                                        |
| johnston-linear-matrix-algebra-F01429 | 141  | 1.000 | $\mathbf{v}_1, \dots, \mathbf{v}_k$                                           | $\mathbf{v}_1, \dots, \mathbf{v}_k$                                           | vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is                                                                                         |
| johnston-linear-matrix-algebra-F01430 | 141  | 1.000 | $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$                                 | $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$                                 | form $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k$ ,                                                                                   |
| johnston-linear-matrix-algebra-F01431 | 141  | 0.724 | $2(2, 1) = (4, 2)$                                                            | $2(2, 1) = (4, 2)$                                                            | vector $(2, 1)$ is not a subspace of $\mathbb{R}^2$ because, for example, $2(2, 1) = (4, 2)$ is                                        |
| johnston-linear-matrix-algebra-F01432 | 141  | 1.000 | $(4, 2)$                                                                      | $(4, 2)$                                                                      | not in the set. To fix this problem, we could simply add $(4, 2)$ to the set and ask                                                   |
| johnston-linear-matrix-algebra-F01433 | 141  | 1.000 | $\{(2, 1), (4, 2)\}$                                                          | $\{(2, 1), (4, 2)\}$                                                          | whether or not $\{(2, 1), (4, 2)\}$ is a subspace. This larger set also fails to be a                                                  |
| johnston-linear-matrix-algebra-F01434 | 141  | 1.000 | $\{(2, 1)\}$                                                                  | $\{(2, 1)\}$                                                                  | <b>Figure 2.14:</b> To turn the set $\{(2, 1)\}$ into a subspace, we have to add all of the scalar                                     |
| johnston-linear-matrix-algebra-F01435 | 141  | 1.000 | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                     | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                     | If $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in $\mathbb{R}^n$ , then the set of all linear        |
| johnston-linear-matrix-algebra-F01436 | 141  | 0.511 | $((2, 1))$                                                                    | $((2, 1))$                                                                    | For example, $\text{span}((2, 1))$ is the line through the origin and the point $(2, 1)$ ,                                             |
| johnston-linear-matrix-algebra-F01437 | 141  | 1.000 | $\text{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$                      | $\text{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$                      | $\text{span}(\mathbf{e}_1, \mathbf{e}_2) = \mathbb{R}^2$ . To verify this claim, we just need to show that every vector                |
| johnston-linear-matrix-algebra-F01438 | 141  | 1.000 | $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$ | $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$ | dimensions also holds: $\text{span}(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) = \mathbb{R}^n$ for all $n \geq 1$ , since we can |
| johnston-linear-matrix-algebra-F01439 | 142  | 1.000 | $\text{span}((1, 2), (2, 1))$                                                 | $\text{span}((1, 2), (2, 1))$                                                 | a) $\text{span}((1, 2), (2, 1))$                                                                                                       |
| johnston-linear-matrix-algebra-F01440 | 142  | 1.000 | $\text{span}((1, 2, 1), (2, 1, 1))$                                           | $\text{span}((1, 2, 1), (2, 1, 1))$                                           | b) $\text{span}((1, 2, 1), (2, 1, 1))$                                                                                                 |
| johnston-linear-matrix-algebra-F01441 | 142  | 1.000 | $\text{span}((1, 2), (2, 1)) = \mathbb{R}^2$                                  | $\text{span}((1, 2), (2, 1)) = \mathbb{R}^2$                                  | To verify that $\text{span}((1, 2), (2, 1)) = \mathbb{R}^2$ algebraically, we show that                                                |
| johnston-linear-matrix-algebra-F01442 | 142  | 1.000 | $(x, y) \in \mathbb{R}^2$                                                     | $(x, y) \in \mathbb{R}^2$                                                     | for every $(x, y) \in \mathbb{R}^2$ we can find coefficients $c_1, c_2 \in \mathbb{R}$ such that                                       |
| johnston-linear-matrix-algebra-F01443 | 142  | 0.726 | $\left[ \begin{array}{cc c} 0 & 0 & b \end{array} \right]$                    | $\left[ \begin{array}{cc c} 0 & 0 & b \end{array} \right]$                    | no row of the form $\left[ \begin{array}{cc c} 0 & 0 & b \end{array} \right]$ , so it has a solution regardless of                     |
| johnston-linear-matrix-algebra-F01444 | 142  | 1.000 | $(x, y)$                                                                      | $(x, y)$                                                                      | the values of $x$ and $y$ . That is, we can always write $(x, y)$ as a linear                                                          |
| johnston-linear-matrix-algebra-F01445 | 143  | 1.000 | $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$                | $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$                | Of $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k)$                                                                      |
| johnston-linear-matrix-algebra-F01446 | 143  | 1.000 | $(x, y, z) \in \mathbb{R}^3$                                                  | $(x, y, z) \in \mathbb{R}^3$                                                  | $(x, y, z) \in \mathbb{R}^3$ and we try to find $c_1, c_2 \in \mathbb{R}$ such that                                                    |
| johnston-linear-matrix-algebra-F01447 | 143  | 1.000 | $z = \frac{1}{3}x + \frac{1}{3}y$                                             | $z = \frac{1}{3}x + \frac{1}{3}y$                                             | a solution if and only if the bottom-right entry, $z = \frac{1}{3}x + \frac{1}{3}y$ , equals                                           |
| johnston-linear-matrix-algebra-F01448 | 143  | 0.955 | $(x, y, z)$                                                                   | $(x, y, z)$                                                                   | $x + y - 3z = 0$ , so the vector $(x, y, z)$ is a linear combination of                                                                |
| johnston-linear-matrix-algebra-F01449 | 143  | 0.703 | $(2, 1, 1)$                                                                   | $(2, 1, 1)$                                                                   | $(1, 2, 1)$ and $(2, 1, 1)$ if and only if $x + y - 3z = 0$ .                                                                          |

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| johnston-linear-matrix-algebra-F01450 | 143  | ■ 1.000 | $\mathbf{v}, \mathbf{w} \in \operatorname{span}\left(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right)$  | $\mathbf{v}, \mathbf{w} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$  | of subspaces (Definition 2.3.1) holds, suppose that $\mathbf{v}, \mathbf{w} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ .                                           |
| johnston-linear-matrix-algebra-F01451 | 143  | ■ 0.991 | $c_1, c_2, \ldots, c_k, d_1, d_2, \ldots, d_k \in \mathbb{R}$                                                  | $c_1, c_2, \ldots, c_k, d_1, d_2, \ldots, d_k \in \mathbb{R}$                                       | Then there exist $c_1, c_2, \ldots, c_k, d_1, d_2, \ldots, d_k \in \mathbb{R}$ such that                                                                                                           |
| johnston-linear-matrix-algebra-F01452 | 143  | ■ 1.000 | $\mathbf{v} + \mathbf{w} \in \operatorname{span}\left(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right)$ | $\mathbf{v} + \mathbf{w} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ | $\mathbf{v} + \mathbf{w} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ too.                                                                                           |
| johnston-linear-matrix-algebra-F01453 | 144  | ■ 1.000 | $\mathcal{S} = \operatorname{span}\left(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right)$               | $\mathcal{S} = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$               | $\mathcal{S} = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$                                                                                                              |
| johnston-linear-matrix-algebra-F01454 | 144  | ■ 1.000 | $\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right\}$                                              | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$                                              | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ spans                                                                                                                                       |
| johnston-linear-matrix-algebra-F01455 | 144  | ■ 1.000 | $\operatorname{col}(A)$                                                                                        | $\operatorname{col}(A)$                                                                             | denoted by $\operatorname{col}(A)$ .                                                                                                                                                               |
| johnston-linear-matrix-algebra-F01456 | 144  | ■ 0.997 | $c \mathbf{v} \in \operatorname{span}\left(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right)$            | $c \mathbf{v} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$            | so $c \mathbf{v} \in \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ , which shows that $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k)$ is indeed a |
| johnston-linear-matrix-algebra-F01457 | 144  | ■ 0.982 | $(2, 3)$                                                                                                       | $(2, 3)$                                                                                            | things like “the line going through the origin and $(2, 3)$ ” or “the plane going                                                                                                                  |
| johnston-linear-matrix-algebra-F01458 | 144  | ■ 1.000 | $\operatorname{span}\left((1, 2, 3), (3, -1, 2)\right)$                                                        | $\operatorname{span}((1, 2, 3), (3, -1, 2))$                                                        | $\operatorname{span}((1, 2, 3), (3, -1, 2))$ , respectively. Being able to describe subspaces like                                                                                                 |
| johnston-linear-matrix-algebra-F01459 | 144  | ■ 1.000 | $\operatorname{span}\left(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\right)$                             | $\operatorname{span}(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$                             | of all linear combinations of $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ , which is $\operatorname{span}(\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$ .                              |
| johnston-linear-matrix-algebra-F01460 | 145  | ■ 0.958 | $A \operatorname{span} \mathbb{R}^n$                                                                           | $A \operatorname{span} \mathbb{R}^n$                                                                | c) The rows of $A \operatorname{span} \mathbb{R}^n$ .                                                                                                                                              |
| johnston-linear-matrix-algebra-F01461 | 145  | ■ 0.755 | $\operatorname{span} \mathbb{R}^n$                                                                             | $\operatorname{span} \mathbb{R}^n$                                                                  | $\operatorname{span} \mathbb{R}^n$ .                                                                                                                                                               |
| johnston-linear-matrix-algebra-F01462 | 146  | ■ 1.000 | $\left\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\right\} \subseteq \mathbb{R}^n$                       | $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \subseteq \mathbb{R}^n$                       | $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \subseteq \mathbb{R}^n$ is                                                                                                                   |
| johnston-linear-matrix-algebra-F01463 | 146  | ■ 1.000 | $-x + y = -2$                                                                                                  | $-x + y = -2$                                                                                       | above, the second equation of the associated linear system says $-x + y = -2$ ,                                                                                                                    |
| johnston-linear-matrix-algebra-F01464 | 146  | ■ 1.000 | $x - y = 2$                                                                                                    | $x - y = 2$                                                                                         | which is redundant since the first equation already said that $x - y = 2$ , and these                                                                                                              |
| johnston-linear-matrix-algebra-F01465 | 146  | ■ 1.000 | $\{(2, 3), (1, 0), (0, 1)\}$                                                                                   | $\{(2, 3), (1, 0), (0, 1)\}$                                                                        | For example, the set of vectors $\{(2, 3), (1, 0), (0, 1)\}$ is linearly dependent                                                                                                                 |
| johnston-linear-matrix-algebra-F01466 | 146  | ■ 1.000 | $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$                                                                          | $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$                                                               | On the other hand, the set of vectors $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is linearly                                                                                                            |
| johnston-linear-matrix-algebra-F01467 | 146  | ■ 1.000 | $c_1, c_2, \ldots, c_k$                                                                                        | $c_1, c_2, \ldots, c_k$                                                                             | and then try to solve for the scalars $c_1, c_2, \ldots, c_k$ . If they must all equal 0                                                                                                           |
| johnston-linear-matrix-algebra-F01468 | 146  | ■ 0.998 | $\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$                                                                       | $\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$                                                            | a) $\{(1, -1, 0), (-2, 1, 2), (1, 1, -4)\}$                                                                                                                                                        |
| johnston-linear-matrix-algebra-F01469 | 146  | ■ 1.000 | $\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$                                                                         | $\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$                                                              | b) $\{(1, 2, 3), (1, 0, 1), (0, -1, 2)\}$                                                                                                                                                          |
| johnston-linear-matrix-algebra-F01470 | 147  | ■ 1.000 | $c_3$                                                                                                          | $c_3$                                                                                               | chosen $c_3$ to be any                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F01471 | 147  | ■ 1.000 | $c_1, c_2, c_3 \in \mathbb{R}$                                                                                 | $c_1, c_2, c_3 \in \mathbb{R}$                                                                      | all zero) scalars $c_1, c_2, c_3 \in \mathbb{R}$ such that                                                                                                                                         |
| johnston-linear-matrix-algebra-F01472 | 147  | ■ 1.000 | $c_3 = 1$                                                                                                      | $c_3 = 1$                                                                                           | $c_3 = 1$ and then solve by back substitution, we find that $c_2 = 2$ and                                                                                                                          |
| johnston-linear-matrix-algebra-F01473 | 147  | ■ 1.000 | $c_2 = 2$                                                                                                      | $c_2 = 2$                                                                                           | $c_3 = 1$ and then solve by back substitution, we find that $c_2 = 2$ and                                                                                                                          |
| johnston-linear-matrix-algebra-F01474 | 147  | ■ 1.000 | $c_1 = 3$                                                                                                      | $c_1 = 3$                                                                                           | $c_1 = 3$ , which tells us that                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F01475 | 147  | ■ 1.000 | $c_1 = c_2 = c_3 = 0$                                                                                          | $c_1 = c_2 = c_3 = 0$                                                                               | $c_1 = c_2 = c_3 = 0$ is a solution, it must be the only solution, so                                                                                                                              |
| johnston-linear-matrix-algebra-F01476 | 148  | ■ 0.997 | $\left\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\right\}$                                              | $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$                                              | a) The set $\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$ is linearly dependent.                                                                                                           |

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| johnston-linear-matrix-algebra-F01477 | 148  | 0.998 | $A=\left[\mathbf{a}_{\{1\}}\left \mathbf{a}_{\{2\}}\right \cdots\right.\left.\mid\mathbf{a}_{\{n\}}\right]$       | $A = [\mathbf{a}_1   \mathbf{a}_2   \cdots   \mathbf{a}_n]$      | <i>Proof.</i> We use block matrix multiplication. If $A = \left[ \mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_n \right]$ then |
| johnston-linear-matrix-algebra-F01478 | 148  | 0.778 | $x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}($                                                                        | $x_1, x_2, \ldots, x_n($                                         | there exist constants $x_1, x_2, \ldots, x_n$ (not all 0) such that $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n =$        |
| johnston-linear-matrix-algebra-F01479 | 148  | 0.778 | $x_{\{1\}}\mathbf{a}_{\{1\}}+x_{\{2\}}\mathbf{a}_{\{2\}}+\cdots+x_{\{n\}}\mathbf{a}_{\{n\}}=$                     | $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n =$ | there exist constants $x_1, x_2, \ldots, x_n$ (not all 0) such that $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n =$        |
| johnston-linear-matrix-algebra-F01480 | 148  | 0.999 | $\mathbf{v}_{\{1\}}$                                                                                              | $\mathbf{v}_1$                                                   | so as to solve for one of the vectors (say $\mathbf{v}_1$ ) in terms of the others. In other                                                |
| johnston-linear-matrix-algebra-F01481 | 148  | 1.000 | $\{(1,2,1),(2,4,2),(2,1,1)\}$                                                                                     | $\{(1,2,1), (2,4,2), (2,1,1)\}$                                  | set. To illustrate why, consider the set $\{(1,2,1), (2,4,2), (2,1,1)\}$ , which is                                                         |
| johnston-linear-matrix-algebra-F01482 | 149  | 0.863 | $\{(1,2),(3,1)\}$                                                                                                 | $\{(1,2), (3,1)\}$                                               | a) $\{(1,2), (3,1)\}$                                                                                                                       |
| johnston-linear-matrix-algebra-F01483 | 149  | 0.985 | $\{(1,2,3),(2,4,6)\}$                                                                                             | $\{(1,2,3), (2,4,6)\}$                                           | b) $\{(1,2,3), (2,4,6)\}$                                                                                                                   |
| johnston-linear-matrix-algebra-F01484 | 149  | 0.861 | $(3,1)=c(1,2)$                                                                                                    | $(3,1) = c(1,2)$                                                 | is no $c \in \mathbb{R}$ such that $(3,1) = c(1,2)$ ), we conclude that this set                                                            |
| johnston-linear-matrix-algebra-F01485 | 150  | 1.000 | $(2,4,6)=2(1,2,3)$                                                                                                | $(2,4,6) = 2(1,2,3)$                                             | b) Since $(2,4,6) = 2(1,2,3)$ , these two vectors are multiples of each                                                                     |
| johnston-linear-matrix-algebra-F01486 | 150  | 1.000 | $(n-1)$                                                                                                           | $(n-1)$                                                          | only if they lie on a common $(n-1)$ -dimensional hyperplane. More generally,                                                               |
| johnston-linear-matrix-algebra-F01487 | 151  | 1.000 | $a_{\{0\}}=1$                                                                                                     | $a_0 = 1$                                                        | matrix with $a_0 = 1$ ,                                                                                                                     |
| johnston-linear-matrix-algebra-F01488 | 151  | 1.000 | $a_{\{1\}}=2$                                                                                                     | $a_1 = 2$                                                        | $a_1 = 2$ , and $a_2 = 3$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F01489 | 151  | 1.000 | $a_{\{2\}}=3$                                                                                                     | $a_2 = 3$                                                        | $a_1 = 2$ , and $a_2 = 3$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F01490 | 151  | 1.000 | $V$                                                                                                               | $V$                                                              | Be careful: $V$ is an                                                                                                                       |
| johnston-linear-matrix-algebra-F01491 | 151  | 1.000 | $(n+1)\times(n+1)$                                                                                                | $(n+1) \times (n+1)$                                             | $(n+1) \times (n+1)$                                                                                                                        |
| johnston-linear-matrix-algebra-F01492 | 151  | 1.000 | $x_{\{0\}}, x_{\{1\}}, \ldots, x_{\{n\}}$                                                                         | $x_0, x_1, \ldots, x_n$                                          | $x_0, x_1, \ldots, x_n$ are                                                                                                                 |
| johnston-linear-matrix-algebra-F01493 | 151  | 1.000 | $a_{\{0\}}, a_{\{1\}}, \ldots, a_{\{n\}}$                                                                         | $a_0, a_1, \ldots, a_n$                                          | where $a_0, a_1, \ldots, a_n$ are real numbers. Show that $V$ is invertible if and only                                                     |
| johnston-linear-matrix-algebra-F01494 | 151  | 0.998 | $a_{\{i\}} \neq a_{\{j\}}$                                                                                        | $a_i \neq a_j$                                                   | if $a_0, a_1, \ldots, a_n$ are distinct (i.e., $a_i \neq a_j$ whenever $i \neq j$ ).                                                        |
| johnston-linear-matrix-algebra-F01495 | 151  | 0.989 | $c_{\{0\}}=c_{\{1\}}=\cdots=c_{\{n\}}=0$                                                                          | $c_0 = c_1 = \cdots = c_n = 0$                                   | implies $c_0 = c_1 = \cdots = c_n = 0$ .                                                                                                    |
| johnston-linear-matrix-algebra-F01496 | 151  | 1.000 | $p(x)=c_{\{0\}}+c_{\{1\}}x+c_{\{2\}}x^2+\cdots+c_{\{n\}}x^n$                                                      | $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$                   | If we define the polynomial $p(x) = c_0 + c_1x + c_2x^2 + \cdots + c_nx^n$ , then                                                           |
| johnston-linear-matrix-algebra-F01497 | 151  | 1.000 | $p\left(a_{\{0\}}\right)=0$                                                                                       | $p(a_0) = 0$                                                     | the first entry of Equation (2.3.1) says that $p(a_0) = 0$ , the next entry says                                                            |
| johnston-linear-matrix-algebra-F01498 | 151  | 1.000 | $p\left(a_{\{1\}}\right)=0$                                                                                       | $p(a_1) = 0$                                                     | that $p(a_1) = 0$ , and so on to $p(a_n) = 0$ . Since $p$ is a degree- $n$ polynomial,                                                      |
| johnston-linear-matrix-algebra-F01499 | 151  | 1.000 | $p\left(a_{\{n\}}\right)=0$                                                                                       | $p(a_n) = 0$                                                     | that $p(a_1) = 0$ , and so on to $p(a_n) = 0$ . Since $p$ is a degree- $n$ polynomial,                                                      |
| johnston-linear-matrix-algebra-F01500 | 151  | 1.000 | $n+1$                                                                                                             | $n+1$                                                            | and we just showed that it has $n+1$ distinct roots (since $a_0, a_1, \ldots, a_n$ are                                                      |
| johnston-linear-matrix-algebra-F01501 | 151  | 0.982 | $c_{\{0\}}=c_{\{1\}}=$                                                                                            | $c_0 = c_1 =$                                                    | zero polynomial. That is, its coefficients must all equal 0, so $c_0 = c_1 =$                                                               |
| johnston-linear-matrix-algebra-F01502 | 151  | 1.000 | $\cdots=c_{\{n\}}=0$                                                                                              | $\cdots = c_n = 0$                                               | $\cdots = c_n = 0$ , so the columns of $A$ form a linearly independent set, which                                                           |
| johnston-linear-matrix-algebra-F01503 | 151  | 1.000 | $\left(x_{\{0\}}, y_{\{0\}}\right), \left(x_{\{1\}}, y_{\{1\}}\right), \ldots, \left(x_{\{n\}}, y_{\{n\}}\right)$ | $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$                     | Suppose we are given a set of points $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$ in $\mathbb{R}^2$                                         |
| johnston-linear-matrix-algebra-F01504 | 151  | 0.960 | $p\left(x_{\{0\}}\right)=y_{\{0\}}, p\left(x_{\{1\}}\right)=y_{\{1\}}, \ldots, p\left(x_{\{n\}}\right)=y_{\{n\}}$ | $p(x_0) = y_0, p(x_1) = y_1, \ldots, p(x_n) = y_n$               | $p(x_0) = y_0, p(x_1) = y_1, \ldots, p(x_n) = y_n$ .                                                                                        |

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| johnston-linear-matrix-align=left F01505              | 151  | 1.000 | n=1                                                                              | $n = 1$                                                                    | Since we can find a line going through any two points, the $n = 1$ case                                              |
| johnston-linear-matrix-align=left F01506              | 152  | 1.000 | $p\left(x_{\{0\}}\right)=y_{\{0\}}$                                              | $p(x_0) = y_0$                                                             | $p(x_0) = y_0$ can be                                                                                                |
| johnston-linear-matrix-align=left F01507              | 152  | 0.623 | $c_{\{n\}}\ x_{\{0\}}^{\{n\}}+\cdots+c_{\{0\}}=y_{\{0\}}$                        | $c_nx_0^n + \cdots + c_0 = y_0$                                            | $c_nx_0^n + \cdots + c_0 = y_0,$                                                                                     |
| johnston-linear-matrix-align=left F01508              | 152  | 1.000 | $c_{\{0\}},\ c_{\{1\}},\ \ldots,\ c_{\{n\}}$                                     | $c_0, c_1, \ldots, c_n$                                                    | for some scalars $c_0, c_1, \ldots, c_n$ . The fact that we are requiring $p(x_0) = y_0,$                            |
| johnston-linear-matrix-align=left F01509              | 152  | 1.000 | $p\left(x_{\{1\}}\right)=y_{\{1\}},\ \ldots,\ p\left(x_{\{n\}}\right)=y_{\{n\}}$ | $p(x_1) = y_1, \ldots, p(x_n) = y_n$                                       | $p(x_1) = y_1, \ldots, p(x_n) = y_n,$ can be written explicitly as the linear system                                 |
| johnston-linear-matrix-align=left F01510              | 153  | 1.000 | $\left(c_{\{0\}},\ c_{\{1\}},\ c_{\{2\}}\right)=(-1,-3,2)$                       | $(c_0, c_1, c_2) = (-1, -3, 2)$                                            | which has unique solution $(c_0, c_1, c_2) = (-1, -3, 2)$ . The desired quadratic                                    |
| johnston-linear-matrix-align=left F01511              | 153  | 1.000 | $p(-2)=13$                                                                       | $p(-2) = 13$                                                               | we could plug values into $p$ to see that it is indeed the case that $p(-2) = 13,$                                   |
| johnston-linear-matrix-align=left F01512              | 153  | 1.000 | $p(1)=-2$                                                                        | $p(1) = -2$                                                                | $p(1) = -2,$ and $p(3) = 8.$                                                                                         |
| johnston-linear-matrix-align=left F01513              | 153  | 1.000 | $p(3)=8$                                                                         | $p(3) = 8$                                                                 | $p(1) = -2,$ and $p(3) = 8.$                                                                                         |
| johnston-linear-matrix-align=left F01514              | 153  | 1.000 | $y=x+1$                                                                          | $y = x + 1$                                                                | *(a) The graph of the function $y = x + 1$ in $\mathbb{R}^2.$                                                        |
| johnston-linear-matrix-align=left F01515              | 153  | 1.000 | $y=\sin (x)$                                                                     | $y = \sin(x)$                                                              | *(c) The graph of the function $y = \sin(x)$ in $\mathbb{R}^2.$                                                      |
| johnston-linear-matrix-align=left F01516              | 153  | 0.995 | $x+2\ y+3\ z=4$                                                                  | $x + 2y + 3z = 4$                                                          | *(e) The set of solutions $(x,y,z)$ to $x + 2y + 3z = 4.$                                                            |
| johnston-linear-matrix-align=left F01517              | 153  | 1.000 | $x-y+8\ z=0$                                                                     | $x - y + 8z = 0$                                                           | (f) The set of solutions $(x,y,z)$ to $x - y + 8z = 0.$                                                              |
| johnston-linear-matrix-align=left F01518              | 153  | 1.000 | $\left\{(x,\ y)\in\mathbb{R}^2:\ x+2\ y=0\right\}$                               | $\{(x,y) \in \mathbb{R}^2 : x + 2y = 0\}$                                  | *(g) $\{(x,y) \in \mathbb{R}^2 : x + 2y = 0\}$                                                                       |
| johnston-linear-matrix-align=left F01519              | 153  | 1.000 | $\left\{(x,\ y)\in\mathbb{R}^2:\ x+y\geq 0\right\}$                              | $\{(x,y) \in \mathbb{R}^2 : x + y \geq 0\}$                                | (h) $\{(x,y) \in \mathbb{R}^2 : x + y \geq 0\}$                                                                      |
| johnston-linear-matrix-align=left F01520              | 153  | 1.000 | $\left\{(x,\ y)\in\mathbb{R}^2:\ x\ y\geq 0\right\}$                             | $\{(x,y) \in \mathbb{R}^2 : xy \geq 0\}$                                   | *(i) $\{(x,y) \in \mathbb{R}^2 : xy \geq 0\}$                                                                        |
| johnston-linear-matrix-align=left F01521              | 153  | 1.000 | $\left\{(x,\ y,\ z)\in\mathbb{R}^3:\ x\ y+y\ z=0\right\}$                        | $\{(x,y,z) \in \mathbb{R}^3 : xy + yz = 0\}$                               | (j) $\{(x,y,z) \in \mathbb{R}^3 : xy + yz = 0\}$                                                                     |
| johnston-linear-matrix-align=left F01522              | 153  | 0.995 | $\{(2,3),(0,0)\}$                                                                | $\{(2,3), (0,0)\}$                                                         | (b) $\{(2,3), (0,0)\}$                                                                                               |
| johnston-linear-matrix-align=left F01523              | 153  | 1.000 | $\{(0,1,-1),(1,2,1),(3,-1,4)\}$                                                  | $\{(0,1,-1), (1,2,1), (3,-1,4)\}$                                          | (b) $\{(0,1,-1), (1,2,1), (3,-1,4)\}$                                                                                |
| johnston-linear-matrix-align=left F01524              | 153  | 1.000 | $\{(1,2,1),(0,1,-1),(2,5,1)\}$                                                   | $\{(1,2,1), (0,1,-1), (2,5,1)\}$                                           | *(c) $\{(1,2,1), (0,1,-1), (2,5,1)\}$                                                                                |
| johnston-linear-matrix-align=left F01525              | 153  | 1.000 | $\{(1,1,0),(1,0,-1),(0,1,1),(1,2,1)\}$                                           | $\{(1,1,0), (1,0,-1), (0,1,1), (1,2,1)\}$                                  | (d) $\{(1,1,0), (1,0,-1), (0,1,1), (1,2,1)\}$                                                                        |
| johnston-linear-matrix-align=left F01526              | 153  | 0.359 | $\left(\mathbf{a}\right)\left[\begin{array}{rr}1&1\\1&1\end{array}\right]$       | $\left(\mathbf{a}\right)\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$ | *(a) $\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$<br>*(c) $\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$ |
| johnston-linear-matrix-align=left F01527              | 153  | 0.359 | $\left(\mathbf{c}\right)\left[\begin{array}{ll}0&1\\2&3\end{array}\right]$       | $\left(\mathbf{c}\right)\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$ | *(a) $\left[\begin{array}{cc}1&1\\1&1\end{array}\right]$<br>*(c) $\left[\begin{array}{cc}0&1\\2&3\end{array}\right]$ |
| johnston-linear-matrix-align=left F01528 [conf 0.083] | 153  | 0.083 | $\left[\begin{array}{r}{11}\\{22}\end{array}\right]\text{ (d)}$                  | $\left[\begin{array}{c}11\\22\end{array}\right]$<br>(d)                    | (b) $\left[\begin{array}{cc}1&2\\2&4\end{array}\right]$<br>(d) $\left[\begin{array}{cc}0&0\\0&0\end{array}\right]$   |

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|------------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra<br>F01529 | 153  | 0.420 | $*(\mathbf{a}) \quad \left[ \begin{array}{lll} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 1 & 2 & 2 \end{array} \right]$ | $*(\mathbf{a}) \quad \left[ \begin{array}{lll} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{array} \right]$ | $*(\mathbf{a}) \quad \left[ \begin{array}{lll} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{array} \right]$<br>$*(\mathbf{c}) \quad \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$ |
| johnston-linear-matrix-algebra<br>F01530 | 153  | 0.420 | $*(\mathbf{c}) \quad \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$ | $*(\mathbf{c}) \quad \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$ | $*(\mathbf{a}) \quad \left[ \begin{array}{lll} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{array} \right]$<br>$*(\mathbf{c}) \quad \left[ \begin{array}{lll} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{array} \right]$ |
| johnston-linear-matrix-algebra<br>F01531 | 153  | 1.000 | $\left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array} \right]$                     | $\left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array} \right]$                     | $(\mathbf{b}) \quad \left[ \begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{array} \right]$                                                                                                             |
| johnston-linear-matrix-algebra<br>F01532 | 153  | 0.999 | $\left[ \begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{array} \right]$                     | $\left[ \begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{array} \right]$                     | $(\mathbf{d}) \quad \left[ \begin{array}{lll} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{array} \right]$                                                                                                             |
| johnston-linear-matrix-algebra<br>F01533 | 153  | 1.000 | $\{(1,0,1), (1,1,1)\}$                                                                                  | $\{(1,0,1), (1,1,1)\}$                                                                                  | $(\mathbf{b}) \quad \{(1,0,1), (1,1,1)\}$                                                                                                                                                                          |
| johnston-linear-matrix-algebra<br>F01534 | 153  | 0.640 | $\{(1,0,-1), (1,1,1), (1,2,-1)\}$                                                                       | $\{(1,0,-1), (1,1,1), (1,2,-1)\}$                                                                       | $*(\mathbf{c}) \quad \{(1,0,-1), (1,1,1), (1,2,-1)\}$                                                                                                                                                              |
| johnston-linear-matrix-algebra<br>F01535 | 153  | 1.000 | $\{(1,2,3), (4,5,6), (7,8,9)\}$                                                                         | $\{(1,2,3), (4,5,6), (7,8,9)\}$                                                                         | $(\mathbf{d}) \quad \{(1,2,3), (4,5,6), (7,8,9)\}$                                                                                                                                                                 |
| johnston-linear-matrix-algebra<br>F01536 | 153  | 0.969 | $\{(1,1), (2,1), (3,-2)\}$                                                                              | $\{(1,1), (2,1), (3,-2)\}$                                                                              | $*(\mathbf{e}) \quad \{(1,1), (2,1), (3,-2)\}$                                                                                                                                                                     |
| johnston-linear-matrix-algebra<br>F01537 | 153  | 0.990 | $\{(2,1,0), (0,0,0), (1,1,2)\}$                                                                         | $\{(2,1,0), (0,0,0), (1,1,2)\}$                                                                         | $(\mathbf{f}) \quad \{(2,1,0), (0,0,0), (1,1,2)\}$                                                                                                                                                                 |
| johnston-linear-matrix-algebra<br>F01538 | 153  | 0.516 | $\{(1,2,4,1), (2,4,-1,3), (-1,1,1,-1)\}$                                                                | $\{(1,2,4,1), (2,4,-1,3), (-1,1,1,-1)\}$                                                                | $*(\mathbf{g}) \quad \{(1,2,4,1), (2,4,-1,3), (-1,1,1,-1)\}$                                                                                                                                                       |
| johnston-linear-matrix-algebra<br>F01539 | 153  | 1.000 | $\{(0,1,1,1), (1,0,1,1), (1,1,0,1), (1,1,1,0)\}$                                                        | $\{(0,1,1,1), (1,0,1,1), (1,1,0,1), (1,1,1,0)\}$                                                        | $(\mathbf{h}) \quad \{(0,1,1,1), (1,0,1,1), (1,1,0,1), (1,1,1,0)\}$                                                                                                                                                |
| johnston-linear-matrix-algebra<br>F01540 | 153  | 0.963 | $\{(4,2,5,2), (3,1,2,4), (1,4,2,3), (3,1,4,2)\}$                                                        | $\{(4,2,5,2), (3,1,2,4), (1,4,2,3), (3,1,4,2)\}$                                                        | $(\mathbf{b}) \quad \{(4,2,5,2), (3,1,2,4), (1,4,2,3), (3,1,4,2)\}$                                                                                                                                                |
| johnston-linear-matrix-algebra<br>F01541 | 153  | 0.998 | $\{(4,4,4,3), (3,3,-1,1), (-1,2,1,2)\}$                                                                 | $\{(4,4,4,3), (3,3,-1,1), (-1,2,1,2)\}$                                                                 | $*(\mathbf{c}) \quad \{(4,4,4,3), (3,3,-1,1), (-1,2,1,2),$                                                                                                                                                         |
| johnston-linear-matrix-algebra<br>F01542 | 153  | 0.999 | $\{(2,-1,4,-1), (3,1,2,1), (0,1,3,4)\}$                                                                 | $\{(2,-1,4,-1), (3,1,2,1), (0,1,3,4)\}$                                                                 | $(\mathbf{d}) \quad \{(2,-1,4,-1), (3,1,2,1), (0,1,3,4),$                                                                                                                                                          |
| johnston-linear-matrix-algebra<br>F01543 | 153  | 1.000 | $\{(3,5,1,4), (4,4,5,5), (5,0,4,3), (1,1,5,-1)\}$                                                       | $\{(3,5,1,4), (4,4,5,5), (5,0,4,3), (1,1,5,-1)\}$                                                       | $(\mathbf{b}) \quad \{(3,5,1,4), (4,4,5,5), (5,0,4,3), (1,1,5,-1)\}$                                                                                                                                               |
| johnston-linear-matrix-algebra<br>F01544 | 153  | 1.000 | $\{(5,4,5,1,5), (4,3,3,0,4), (-1,0,3,-1,4)\}$                                                           | $\{(5,4,5,1,5), (4,3,3,0,4), (-1,0,3,-1,4)\}$                                                           | $*(\mathbf{c}) \quad \{(5,4,5,1,5), (4,3,3,0,4), (-1,0,3,-1,4)\}$                                                                                                                                                  |
| johnston-linear-matrix-algebra<br>F01545 | 153  | 0.999 | $\{(5,-1,2,4,3), (4,-8,1,-4,-9), (2,2,1,4,5)\}$                                                         | $\{(5,-1,2,4,3), (4,-8,1,-4,-9), (2,2,1,4,5)\}$                                                         | $(\mathbf{d}) \quad \{(5,-1,2,4,3), (4,-8,1,-4,-9), (2,2,1,4,5)\}$                                                                                                                                                 |
| johnston-linear-matrix-algebra<br>F01546 | 154  | 1.000 | $\mathbf{v}_2$                                                                                          | $\mathbf{v}_2$                                                                                          | head-to-tail (i.e., the head of $\mathbf{v}_1$ at the tail of $\mathbf{v}_2$ , the                                                                                                                                 |
| johnston-linear-matrix-algebra<br>F01547 | 154  | 1.000 | $\mathbf{v}_3$                                                                                          | $\mathbf{v}_3$                                                                                          | head of $\mathbf{v}_2$ at the tail of $\mathbf{v}_3$ , and so on). If the vectors                                                                                                                                  |
| johnston-linear-matrix-algebra<br>F01548 | 154  | 0.999 | $\operatorname{span}(\mathbf{v}, \mathbf{w})$                                                           | $\operatorname{span}(\mathbf{v}, \mathbf{w})$                                                           | $*(\mathbf{e})$ If $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ then $\operatorname{span}(\mathbf{v}, \mathbf{w})$ is a plane through the                                                                             |
| johnston-linear-matrix-algebra<br>F01549 | 154  | 1.000 | $k \geq n$                                                                                              | $k \geq n$                                                                                              | $(\mathbf{f})$ Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ be vectors in $\mathbb{R}^n$ . If $k \geq n$ then it                                                                                          |
| johnston-linear-matrix-algebra<br>F01550 | 154  | 1.000 | $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$                   | $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$                   | must be the case that $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$ .                                                                                                      |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-aligned-F01551 | 154  | 0.534 | $(k, -1, 4)$                                                                    | $(k, -1, 4)$                                                                    | *(a) $(0, 1, -1), (1, 2, 1), (k, -1, 4)$                                                                            |
| johnston-linear-matrix-aligned-F01552 | 154  | 0.976 | $(1, 2, 3), (3, k, k+3), (2, 4, k)$                                             | $(1, 2, 3), (3, k, k + 3), (2, 4, k)$                                           | (b) $(1, 2, 3), (3, k, k + 3), (2, 4, k)$                                                                           |
| johnston-linear-matrix-aligned-F01553 | 154  | 0.870 | $(5, 7)$                                                                        | $(5, 7)$                                                                        | *(a) $(2, 1), (5, 7)$                                                                                               |
| johnston-linear-matrix-aligned-F01554 | 154  | 0.721 | $(4, 4)$                                                                        | $(4, 4)$                                                                        | (b) $(1, 1), (2, 2), (4, 4)$                                                                                        |
| johnston-linear-matrix-aligned-F01555 | 154  | 0.772 | $(4, 10)$                                                                       | $(4, 10)$                                                                       | *(c) $(1, 1), (2, 2), (4, 10)$                                                                                      |
| johnston-linear-matrix-aligned-F01556 | 154  | 0.999 | $1, 2, \ldots, n^2$                                                             | $1, 2, \dots, n^2$                                                              | **2.3.14 Let $A_n$ be the $n \times n$ matrix with entries $1, 2, \dots, n^2$                                       |
| johnston-linear-matrix-aligned-F01557 | 154  | 1.000 | $n \leq 2$                                                                      | $n \leq 2$                                                                      | Show that $A_n$ is invertible if and only if $n \leq 2$ .                                                           |
| johnston-linear-matrix-aligned-F01558 | 154  | 1.000 | $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathcal{S}$                            | $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathcal{S}$                             | for all $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathcal{S}$ and all $c_1, \dots, c_k \in \mathbb{R}$ .              |
| johnston-linear-matrix-aligned-F01559 | 154  | 1.000 | $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                              | $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$                              | 2.3.16 Suppose $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ .                                                 |
| johnston-linear-matrix-aligned-F01560 | 154  | 1.000 | $\{\mathbf{v}, A\mathbf{v}\}$                                                   | $\{\mathbf{v}, A\mathbf{v}\}$                                                   | (a) Find all vectors $\mathbf{v} \in \mathbb{R}^2$ with the property that $\{\mathbf{v}, A\mathbf{v}\}$             |
| johnston-linear-matrix-aligned-F01561 | 154  | 1.000 | $\mathbf{v} = (1, 2, 3, 4, 5)$                                                  | $\mathbf{v} = (1, 2, 3, 4, 5)$                                                  | not $\mathbf{v} = (1, 2, 3, 4, 5)$ is in the range of the given matrix.                                             |
| johnston-linear-matrix-aligned-F01562 | 154  | 1.000 | $A\mathbf{v} = \mathbf{v}$                                                      | $A\mathbf{v} = \mathbf{v}$                                                      | satisfying $A\mathbf{v} = \mathbf{v}$ is a subspace of $\mathbb{R}^n$ .                                             |
| johnston-linear-matrix-aligned-F01563 | 154  | 1.000 | $\{\mathbf{v}, \mathbf{w}\} \subset \mathbb{R}^n$                               | $\{\mathbf{v}, \mathbf{w}\} \subset \mathbb{R}^n$                               | **2.3.20 Prove that a set of two vectors $\{\mathbf{v}, \mathbf{w}\} \subset \mathbb{R}^n$ is                       |
| johnston-linear-matrix-aligned-F01564 | 154  | 1.000 | $\{\mathbf{v}, \mathbf{w}, \mathbf{x}\} \subseteq \mathbb{R}^n$                 | $\{\mathbf{v}, \mathbf{w}, \mathbf{x}\} \subseteq \mathbb{R}^n$                 | 2.3.22 Suppose that the set of vectors $\{\mathbf{v}, \mathbf{w}, \mathbf{x}\} \subseteq \mathbb{R}^n$ is           |
| johnston-linear-matrix-aligned-F01565 | 154  | 1.000 | $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} - \mathbf{x}\}$ | $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} - \mathbf{x}\}$ | (a) Show that the set $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} - \mathbf{x}\}$ is linearly   |
| johnston-linear-matrix-aligned-F01566 | 154  | 1.000 | $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} + \mathbf{x}\}$ | $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} + \mathbf{x}\}$ | (b) Show that the set $\{\mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{x}, \mathbf{w} + \mathbf{x}\}$ is linearly   |
| johnston-linear-matrix-aligned-F01567 | 154  | 1.000 | $\{\mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} + \mathbf{x}\}$ | $\{\mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} + \mathbf{x}\}$ | (c) Determine whether or not $\{\mathbf{v}, \mathbf{v} + \mathbf{w}, \mathbf{v} + \mathbf{w} + \mathbf{x}\}$ is     |
| johnston-linear-matrix-aligned-F01568 | 154  | 1.000 | $B \subseteq \mathbb{R}^n$                                                      | $B \subseteq \mathbb{R}^n$                                                      | **2.3.23 Suppose that $B \subseteq \mathbb{R}^n$ is a finite set of vectors.                                        |
| johnston-linear-matrix-aligned-F01569 | 154  | 0.822 | $\mathbf{0} \in B$                                                              | $\mathbf{0} \in B$                                                              | Show that if $\mathbf{0} \in B$ then $B$ is linearly dependent.                                                     |
| johnston-linear-matrix-aligned-F01570 | 154  | 1.000 | $\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ | $\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$  | 2.3.24 Suppose that $\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are vectors such |
| johnston-linear-matrix-aligned-F01571 | 154  | 1.000 | $B \subseteq C \subseteq \mathbb{R}^n$                                          | $B \subseteq C \subseteq \mathbb{R}^n$                                          | *2.3.25 Suppose that $B \subseteq C \subseteq \mathbb{R}^n$ are finite sets of vectors.                             |
| johnston-linear-matrix-aligned-F01572 | 154  | 1.000 | $n > m$                                                                         | $n > m$                                                                         | pendent whenever $n > m$ .                                                                                          |
| johnston-linear-matrix-aligned-F01573 | 155  | 0.943 | $(AB) \subseteq \text{range}(A)$                                                | $(AB) \subseteq \text{range}(A)$                                                | (a) Show that $\text{range}(AB) \subseteq \text{range}(A)$ .                                                        |
| johnston-linear-matrix-aligned-F01574 | 155  | 1.000 | $\text{null}(B) \subseteq \text{null}(AB)$                                      | $\text{null}(B) \subseteq \text{null}(AB)$                                      | (b) Show that $\text{null}(B) \subseteq \text{null}(AB)$ .                                                          |
| johnston-linear-matrix-aligned-F01575 | 155  | 1.000 | $(AB)$                                                                          | $(AB)$                                                                          | (c) Provide an example to show that $\text{range}(AB)$ might                                                        |
| johnston-linear-matrix-aligned-F01576 | 155  | 1.000 | $\text{range}(B)$                                                               | $\text{range}(B)$                                                               | not be contained in $\text{range}(B)$ , and $\text{null}(A)$ might not                                              |
| johnston-linear-matrix-aligned-F01577 | 155  | 0.988 | $\text{null}(AB)$                                                               | $\text{null}(AB)$                                                               | be contained in $\text{null}(AB)$ .                                                                                 |
| johnston-linear-matrix-aligned-F01578 | 155  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n \in \mathbb{R}^n$             | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$              | ** 2.3.28 Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^n$ are vectors                    |
| johnston-linear-matrix-aligned-F01579 | 155  | 1.000 | $c_1, c_2, \ldots, c_n \in \mathbb{R}$                                          | $c_1, c_2, \dots, c_n \in \mathbb{R}$                                           | and $c_1, c_2, \dots, c_n \in \mathbb{R}$ are non-zero scalars. Show that                                           |

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| johnston-linear-matrix-algebra-F01580 | 155  | 1.000 | $\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\right\}$                        | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$                      | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is linearly independent if and only if                                                                 |
| johnston-linear-matrix-algebra-F01581 | 155  | 1.000 | $\left\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \ldots, c_n \mathbf{v}_n\right\}$            | $\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \ldots, c_n \mathbf{v}_n\}$          | $\{c_1 \mathbf{v}_1, c_2 \mathbf{v}_2, \ldots, c_n \mathbf{v}_n\}$ is linearly independent.                                                                   |
| johnston-linear-matrix-algebra-F01582 | 155  | 0.402 | $V_n$                                                                                    | $V_n$                                                                       |  <b>2.3.29</b> Let $V_n$ be the $n \times n$ Vandermonde matrix (see         |
| johnston-linear-matrix-algebra-F01583 | 155  | 1.000 | $V_n^{-1}$                                                                               | $V_n^{-1}$                                                                  | <b>(a)</b> Use computer software to compute $V_n^{-1}$ when $n =$                                                                                             |
| johnston-linear-matrix-algebra-F01584 | 155  | 0.812 | $(3, 3, 2) =$                                                                            | $(3, 3, 2) =$                                                               | $(3, 3, 2) =$                                                                                                                                                 |
| johnston-linear-matrix-algebra-F01585 | 155  | 0.898 | $(1, 2, 1) + (2, 1, 1)$                                                                  | $(1, 2, 1) + (2, 1, 1)$                                                     | $(1, 2, 1) + (2, 1, 1)$ .                                                                                                                                     |
| johnston-linear-matrix-algebra-F01586 | 155  | 0.993 | $\operatorname{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$                                   | $\operatorname{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$                      | $x + y - 3z = 0$ in $\mathbb{R}^3$ (see Figure 2.17) as $\operatorname{span}((1, 2, 1), (2, 1, 1), (3, 3, 2))$ , since                                        |
| johnston-linear-matrix-algebra-F01587 | 155  | 0.972 | $(1, 2, 1), (2, 1, 1)$                                                                   | $(1, 2, 1), (2, 1, 1)$                                                      | <b>Figure 2.17:</b> The plane $x + y - 3z = 0$ is the span of the three vectors $(1, 2, 1), (2, 1, 1),$                                                       |
| johnston-linear-matrix-algebra-F01588 | 156  | 0.988 | $\{\mathbf{v}, \mathbf{w}\}$                                                             | $\{\mathbf{v}, \mathbf{w}\}$                                                | $\{\mathbf{v}, \mathbf{w}\}$ with two                                                                                                                         |
| johnston-linear-matrix-algebra-F01589 | 156  | 1.000 | $\mathcal{S} \subseteq \mathbb{R}^n$                                                     | $\mathcal{S} \subseteq \mathbb{R}^n$                                        | A <b>basis</b> of a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ is a set of vectors in $\mathcal{S}$ that                                                   |
| johnston-linear-matrix-algebra-F01590 | 156  | 1.000 | $\left\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\right\} \subseteq \mathbb{R}^n$ | $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \subset \mathbb{R}^n$ | example, the set $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\} \subset \mathbb{R}^n$ of standard basis vectors is (as their name                      |
| johnston-linear-matrix-algebra-F01591 | 156  | 1.000 | $\left\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\right\}$                        | $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$                      | $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is linearly independent. To see linear independence, suppose                                           |
| johnston-linear-matrix-algebra-F01592 | 156  | 0.742 | $\left(c_1, c_2, \ldots, c_n\right)$                                                     | $(c_1, c_2, \ldots, c_n)$                                                   | The vector on the left is just $(c_1, c_2, \ldots, c_n)$ , and if that vector equals $\mathbf{0}$ then                                                        |
| johnston-linear-matrix-algebra-F01593 | 156  | 0.999 | $c_1 = c_2 = \cdots = c_n = 0$                                                           | $c_1 = c_2 = \cdots = c_n = 0$                                              | $c_1 = c_2 = \cdots = c_n = 0$ , which means exactly that $\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_n\}$ is linearly                                  |
| johnston-linear-matrix-algebra-F01594 | 156  | 0.999 | $B = \{(1, 2, 1), (2, 1, 1)\}, \mathcal{S}$                                              | $B = \{(1, 2, 1), (2, 1, 1)\}, \mathcal{S}$                                 | a) $B = \{(1, 2, 1), (2, 1, 1)\}, \mathcal{S}$ is the plane $x + y - 3z = 0$ in $\mathbb{R}^3$ .                                                              |
| johnston-linear-matrix-algebra-F01595 | 156  | 1.000 | $B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, \mathcal{S} = \mathbb{R}^3$                    | $B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, \mathcal{S} = \mathbb{R}^3$       | b) $B = \{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}, \mathcal{S} = \mathbb{R}^3$ .                                                                                    |
| johnston-linear-matrix-algebra-F01596 | 156  | 1.000 | $\operatorname{span}(B) = \mathbb{R}^3$                                                  | $\operatorname{span}(B) = \mathbb{R}^3$                                     | b) We have to show that $\operatorname{span}(B) = \mathbb{R}^3$ and that $B$ is linearly indepen-                                                             |
| johnston-linear-matrix-algebra-F01597 | 157  | 0.973 | $\{\mathbf{0}\}$                                                                         | $\{\mathbf{0}\}$                                                            | The only subsets of $\{\mathbf{0}\}$ are $\{\mathbf{0}\}$ and $\{\mathbf{0}\}$ itself, so these are the only sets                                             |
| johnston-linear-matrix-algebra-F01598 | 157  | 0.999 | $B = \{\mathbf{0}\}$                                                                     | $B = \{\mathbf{0}\}$                                                        | independent, notice that if we choose $B = \{\mathbf{0}\}$ (i.e., $k = 0$ ) in Definition 2.3.4                                                               |
| johnston-linear-matrix-algebra-F01599 | 157  | 1.000 | $\left\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\right\}$                                | $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$                              | $\mathbb{R}^3$ has at least two bases: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$ (from                         |
| johnston-linear-matrix-algebra-F01600 | 157  | 1.000 | $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$                                                    | $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$                                       | $\mathbb{R}^3$ has at least two bases: $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ and $\{(1, 1, 2), (1, 2, 1), (2, 1, 1)\}$ (from                         |
| johnston-linear-matrix-algebra-F01601 | 158  | 1.000 | $ B $                                                                                    | $ B $                                                                       | The notation $ B $                                                                                                                                            |
| johnston-linear-matrix-algebra-F01602 | 158  | 0.520 | $(1, 1, 2)$                                                                              | $(1, 1, 2)$                                                                 | of $\mathbb{R}^3$ even if we change $(1, 1, 2)$ to any vector that is not a linear combination of                                                             |
| johnston-linear-matrix-algebra-F01603 | 158  | 1.000 | $B, C \subseteq \mathcal{S}$                                                             | $B, C \subseteq \mathcal{S}$                                                | Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ , and suppose that $B, C \subseteq \mathcal{S}$ are finite sets with                                        |
| johnston-linear-matrix-algebra-F01604 | 158  | 1.000 | $\operatorname{span}(C) = \mathcal{S}$                                                   | $\operatorname{span}(C) = \mathcal{S}$                                      | the properties that $B$ is linearly independent and $\operatorname{span}(C) = \mathcal{S}$ . Then                                                             |
| johnston-linear-matrix-algebra-F01605 | 158  | 1.000 | $B =$                                                                                    | $B =$                                                                       | <i>Proof.</i> We begin by giving names to the vectors in $B$ and $C$ : we write $B =$                                                                         |
| johnston-linear-matrix-algebra-F01606 | 158  | 1.000 | $C = \left\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell\right\}$                 | $C = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell\}$               | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell\}$ , where $k =  B $ and $\ell =  C $ . |
| johnston-linear-matrix-algebra-F01607 | 158  | 1.000 | $k =  B $                                                                                | $k =  B $                                                                   | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_\ell\}$ , where $k =  B $ and $\ell =  C $ . |

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| johnston-linear-matrix-algebra-F01608 | 158  | 1.000 | $\ell= C $                                                           | $\ell =  C $                                               | $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell\}$ , where $k =  B $ and $\ell =  C $ .       |
| johnston-linear-matrix-algebra-F01609 | 158  | 1.000 | $\ell < k$                                                           | $\ell < k$                                                 | We prove the result by showing that if $\ell < k$ and $C$ spans $S$ , then $B$ must in                                                                            |
| johnston-linear-matrix-algebra-F01610 | 158  | 1.000 | $\mathbf{v}_i \in B \subseteq \mathcal{S}$                           | $\mathbf{v}_i \in B \subseteq S$                           | $S$ , and $\mathbf{v}_i \in B \subseteq S$ for all $1 \leq i \leq k$ , we can write each $\mathbf{v}_i$ as a linear combination                                   |
| johnston-linear-matrix-algebra-F01611 | 158  | 1.000 | $1 \leq i \leq k$                                                    | $1 \leq i \leq k$                                          | $S$ , and $\mathbf{v}_i \in B \subseteq S$ for all $1 \leq i \leq k$ , we can write each $\mathbf{v}_i$ as a linear combination                                   |
| johnston-linear-matrix-algebra-F01612 | 158  | 1.000 | $\mathbf{v}_i$                                                       | $\mathbf{v}_i$                                             | $S$ , and $\mathbf{v}_i \in B \subseteq S$ for all $1 \leq i \leq k$ , we can write each $\mathbf{v}_i$ as a linear combination                                   |
| johnston-linear-matrix-algebra-F01613 | 158  | 1.000 | $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$                 | $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$       | of $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$ :                                                                                                         |
| johnston-linear-matrix-algebra-F01614 | 158  | 0.999 | $(1 \leq i \leq k, 1 \leq j \leq \ell)$                              | $(1 \leq i \leq k, 1 \leq j \leq \ell)$                    | where each of the $a_{i,j}$ ’s ( $1 \leq i \leq k, 1 \leq j \leq \ell$ ) is a scalar.                                                                             |
| johnston-linear-matrix-algebra-F01615 | 158  | 1.000 | $A \in \mathcal{M}_{k, \ell}$                                        | $A \in \mathcal{M}_{k, \ell}$                              | and matrices in the “usual” way—we define $A \in \mathcal{M}_{k, \ell}$ to be the matrix whose                                                                    |
| johnston-linear-matrix-algebra-F01616 | 158  | 0.997 | $a_{i,j}, V \in \mathcal{M}_{n,k}$                                   | $a_{i,j}, V \in \mathcal{M}_{n,k}$                         | $(i, j)$ -entry is $a_{i,j}$ , $V \in \mathcal{M}_{n,k}$ to be the matrix whose columns are $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ ,                   |
| johnston-linear-matrix-algebra-F01617 | 158  | 1.000 | $W \in \mathcal{M}_{n, \ell}$                                        | $W \in \mathcal{M}_{n, \ell}$                              | and $W \in \mathcal{M}_{n, \ell}$ to be the matrix whose columns are $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_\ell$ . We can                                |
| johnston-linear-matrix-algebra-F01618 | 158  | 1.000 | $V = WA^T$                                                           | $V = WA^T$                                                 | $V = WA^T$ , since block matrix multiplication shows that the $i$ -th column of $WA^T$                                                                            |
| johnston-linear-matrix-algebra-F01619 | 158  | 1.000 | $WA^T$                                                               | $WA^T$                                                     | $V = WA^T$ , since block matrix multiplication shows that the $i$ -th column of $WA^T$                                                                            |
| johnston-linear-matrix-algebra-F01620 | 158  | 1.000 | $\mathbf{x} \in \mathbb{R}^k$                                        | $\mathbf{x} \in \mathbb{R}^k$                              | Next, Corollary 2.1.4 tells us that there is a non-zero solution $\mathbf{x} \in \mathbb{R}^k$ to the                                                             |
| johnston-linear-matrix-algebra-F01621 | 158  | 1.000 | $A^T \mathbf{x} = \mathbf{0}$                                        | $A^T \mathbf{x} = \mathbf{0}$                              | linear system $A^T \mathbf{x} = \mathbf{0}$ , since $A^T$ has more columns than rows. It follows that                                                             |
| johnston-linear-matrix-algebra-F01622 | 159  | 1.000 | $V \mathbf{x} = \mathbf{0}$                                          | $V \mathbf{x} = \mathbf{0}$                                | Finally, since the linear system $V \mathbf{x} = \mathbf{0}$ has a non-zero solution, Theorem 2.3.4                                                               |
| johnston-linear-matrix-algebra-F01623 | 159  | 1.000 | $B \subseteq \mathcal{S}$                                            | $B \subseteq S$                                            | <i>Proof.</i> This result follows immediately from Theorem 2.4.1: if $B \subseteq S$ and                                                                          |
| johnston-linear-matrix-algebra-F01624 | 159  | 1.000 | $C \subseteq \mathcal{S}$                                            | $C \subseteq S$                                            | $C \subseteq S$ are bases of $S$ then $ B  \leq  C $ since $B$ is linearly independent and $C$ spans                                                              |
| johnston-linear-matrix-algebra-F01625 | 159  | 1.000 | $ B  \leq  C $                                                       | $ B  \leq  C $                                             | $C \subseteq S$ are bases of $S$ then $ B  \leq  C $ since $B$ is linearly independent and $C$ spans                                                              |
| johnston-linear-matrix-algebra-F01626 | 159  | 1.000 | $ C  \leq  B $                                                       | $ C  \leq  B $                                             | $S$ . On the other hand, $ C  \leq  B $ since $C$ is linearly independent and $B$ spans $S$ .                                                                     |
| johnston-linear-matrix-algebra-F01627 | 159  | 1.000 | $ B  =  C $                                                          | $ B  =  C $                                                | It follows that $ B  =  C $ .                                                                                                                                     |
| johnston-linear-matrix-algebra-F01628 | 159  | 1.000 | $\dim(\mathcal{S})$                                                  | $\dim(S)$                                                  | called the <b>dimension</b> of $S$ and is denoted by $\dim(S)$ .                                                                                                  |
| johnston-linear-matrix-algebra-F01629 | 159  | 1.000 | $\dim(\mathbb{R}^n) = n$                                             | $\dim(\mathbb{R}^n) = n$                                   | $\dim(\mathbb{R}^n) = n$ since the standard basis of $\mathbb{R}^n$ (i.e., the set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ ) con-                   |
| johnston-linear-matrix-algebra-F01630 | 160  | 0.994 | $\leq \dim(\mathcal{S}) \leq$                                        | $\leq \dim(S) \leq$                                        |  size of a linearly independent set $\leq \dim(S) \leq$ size of a spanning set |
| johnston-linear-matrix-algebra-F01631 | 160  | 1.000 | $\mathcal{S} = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$ | $S = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$ | a) $S = \operatorname{span}((1, 1, 1), (1, 2, 3), (3, 2, 1))$ .                                                                                                   |
| johnston-linear-matrix-algebra-F01632 | 160  | 0.998 | $2x - y + 3z = 0$                                                    | $2x - y + 3z = 0$                                          | b) The set $S$ of vectors $(x, y, z)$ satisfying the equation $2x - y + 3z = 0$ .                                                                                 |
| johnston-linear-matrix-algebra-F01633 | 160  | 1.000 | $B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$                            | $B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$                  | a) It seems natural to first guess that $B = \{(1, 1, 1), (1, 2, 3), (3, 2, 1)\}$                                                                                 |
| johnston-linear-matrix-algebra-F01634 | 160  | 1.000 | $\mathcal{S} = \operatorname{span}(B)$                               | $S = \operatorname{span}(B)$                               | know that $S = \operatorname{span}(B)$ , so we just need to check linear indepen-                                                                                 |

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| johnston-linear-matrix-algebra-F01635 | 160  | 0.671 | $\operatorname{span}(B)$                                                     | $\operatorname{span}(B)$                                          | $\operatorname{span}(B)$ . If we (arbitrarily) decide to discard $(1, 1, 1)$ then we arrive                                                    |
| johnston-linear-matrix-algebra-F01636 | 160  | 1.000 | $C=\{(1,2,3),(3,2,1)\}$                                                      | $C = \{(1, 2, 3), (3, 2, 1)\}$                                    | at the set $C = \{(1, 2, 3), (3, 2, 1)\}$ , which has the same span as $B$ .                                                                   |
| johnston-linear-matrix-algebra-F01637 | 161  | 0.856 | $(3,3,-1) \in \mathcal{S}$                                                   | $(3, 3, -1) \in \mathcal{S}$                                      | $(3, 3, -1) \in \mathcal{S}$ we                                                                                                                |
| johnston-linear-matrix-algebra-F01638 | 161  | 0.992 | $2(3)-(3)+3(-1)=0$                                                           | $2(3) - (3) + 3(-1) = 0$                                          | $2(3) - (3) + 3(-1) = 0$ .                                                                                                                     |
| johnston-linear-matrix-algebra-F01639 | 161  | 0.889 | $(1,-1,-1)$                                                                  | $(1, -1, -1)$                                                     | the vector $(1, -1, -1)$ , which we can see is in $\mathcal{S}$ since $2(1) - (-1) +$                                                          |
| johnston-linear-matrix-algebra-F01640 | 161  | 0.889 | $2(1)-(-1)+$                                                                 | $2(1) - (-1) +$                                                   | the vector $(1, -1, -1)$ , which we can see is in $\mathcal{S}$ since $2(1) - (-1) +$                                                          |
| johnston-linear-matrix-algebra-F01641 | 161  | 1.000 | $3(-1)=0$                                                                    | $3(-1) = 0$                                                       | $3(-1) = 0$ . However, this vector by itself does not span $\mathcal{S}$ since we                                                              |
| johnston-linear-matrix-algebra-F01642 | 161  | 0.809 | $\operatorname{span}((1,-1,-1))$                                             | $\operatorname{span}((1, -1, -1))$                                | can find a vector in $\mathcal{S}$ but not in $\operatorname{span}((1, -1, -1))$ —in particular,                                               |
| johnston-linear-matrix-algebra-F01643 | 161  | 0.979 | $(3,3,-1)$                                                                   | $(3, 3, -1)$                                                      | $(3, 3, -1)$ is one such vector. If we set $B = \{(1, -1, -1), (3, 3, -1)\}$ ,                                                                 |
| johnston-linear-matrix-algebra-F01644 | 161  | 0.979 | $B=\{(1,-1,-1),(3,3,-1)\}$                                                   | $B = \{(1, -1, -1), (3, 3, -1)\}$                                 | $(3, 3, -1)$ is one such vector. If we set $B = \{(1, -1, -1), (3, 3, -1)\}$ ,                                                                 |
| johnston-linear-matrix-algebra-F01645 | 161  | 0.655 | $\operatorname{span}(B)=\mathcal{S}$                                         | $\operatorname{span}(B) = \mathcal{S}$                            | $\operatorname{span}(B) = \mathcal{S}$ , we check which vectors $(x,y,z)$ are linear combina-                                                  |
| johnston-linear-matrix-algebra-F01646 | 161  | 1.000 | $\frac{2}{3}x-\frac{1}{3}y+z=0$                                              | $\frac{2}{3}x - \frac{1}{3}y + z = 0$                             | is in $\operatorname{span}(B)$ ) if and only if $\frac{2}{3}x - \frac{1}{3}y + z = 0$ . Multiplying this equa-                                 |
| johnston-linear-matrix-algebra-F01647 | 161  | 1.000 | $\operatorname{dim}(\mathcal{S})=2$                                          | $\dim(\mathcal{S}) = 2$                                           | subspace $\mathcal{S}$ and $B$ is a basis of $\mathcal{S}$ . Furthermore, $\dim(\mathcal{S}) = 2$ since $B$                                    |
| johnston-linear-matrix-algebra-F01648 | 161  | 1.000 | $B \subseteq C$                                                              | $B \subseteq C$                                                   | a) If $B$ is linearly independent then there is a basis $C$ of $\mathcal{S}$ with $B \subseteq C$ .                                            |
| johnston-linear-matrix-algebra-F01649 | 161  | 1.000 | $C \subseteq B$                                                              | $C \subseteq B$                                                   | b) If $B$ spans $\mathcal{S}$ then there is a basis $C$ of $\mathcal{S}$ with $C \subseteq B$ .                                                |
| johnston-linear-matrix-algebra-F01650 | 161  | 1.000 | $\operatorname{span}(B)=$                                                    | $\operatorname{span}(B) =$                                        | <i>Proof.</i> Suppose $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ . To prove part (a), notice that if $\operatorname{span}(B) =$ |
| johnston-linear-matrix-algebra-F01651 | 162  | 1.000 | $\mathbf{w} \in \mathcal{S}$                                                 | $\mathbf{w} \in \mathcal{S}$                                      | of it. On the other hand, if it does not span $\mathcal{S}$ then there exists a vector $\mathbf{w} \in \mathcal{S}$                            |
| johnston-linear-matrix-algebra-F01652 | 162  | 1.000 | $\left\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\right\}$ | $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ | Our next step is to show that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is linearly indepen-                   |
| johnston-linear-matrix-algebra-F01653 | 162  | 1.000 | $d=c_1=c_2=\cdots=c_k=0$                                                     | $d = c_1 = c_2 = \cdots = c_k = 0$                                | with the goal of showing that $d = c_1 = c_2 = \cdots = c_k = 0$ . Well, $d = 0$ since oth-                                                    |
| johnston-linear-matrix-algebra-F01654 | 162  | 1.000 | $d=0$                                                                        | $d = 0$                                                           | with the goal of showing that $d = c_1 = c_2 = \cdots = c_k = 0$ . Well, $d = 0$ since oth-                                                    |
| johnston-linear-matrix-algebra-F01655 | 162  | 0.999 | $c_1=c_2=\cdots=c_k=0$                                                       | $c_1 = c_2 = \cdots = c_k = 0$                                    | $c_1 = c_2 = \cdots = c_k = 0$ too, so $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is indeed linearly independent.       |
| johnston-linear-matrix-algebra-F01656 | 162  | 1.000 | $k \neq \operatorname{dim}(\mathcal{S})$                                     | $k \neq \dim(\mathcal{S})$                                        | a) If $k \neq \dim(\mathcal{S})$ then $B$ is not a basis of $\mathcal{S}$ .                                                                    |
| johnston-linear-matrix-algebra-F01657 | 162  | 1.000 | $k=\operatorname{dim}(\mathcal{S})$                                          | $k = \dim(\mathcal{S})$                                           | b) If $k = \dim(\mathcal{S})$ then the following are equivalent:                                                                               |
| johnston-linear-matrix-algebra-F01658 | 162  | 1.000 | $\operatorname{dim}(\mathcal{S})=k$                                          | $\dim(\mathcal{S}) = k$                                           | create a basis of $\mathcal{S}$ . However, $B$ already has $k$ vectors and $\dim(\mathcal{S}) = k$ , so all                                    |
| johnston-linear-matrix-algebra-F01659 | 163  | 1.000 | $B=\{(1,2,3),(4,1,2)\}$                                                      | $B = \{(1, 2, 3), (4, 1, 2)\}$                                    | a) $B = \{(1, 2, 3), (4, 1, 2)\}$                                                                                                              |
| johnston-linear-matrix-algebra-F01660 | 163  | 1.000 | $B=\{(1,-1,-1),(2,2,-1),(1,1,0)\}$                                           | $B = \{(1, -1, -1), (2, 2, -1), (1, 1, 0)\}$                      | b) $B = \{(1, -1, -1), (2, 2, -1), (1, 1, 0)\}$                                                                                                |
| johnston-linear-matrix-algebra-F01661 | 163  | 0.913 | $B=\{(1,-1,-1),(2,2,-1)\}$                                                   | $B = \{(1, -1, -1), (2, 2, -1)\}$                                 | Determine whether or not $B = \{(1, -1, -1), (2, 2, -1)\}$ is a basis of the                                                                   |

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| johnston-linear-matrix-algebra-F01662 | 163  | 1.000 | $\mathcal{S} \subset \mathbb{R}^3$                                       | $\mathcal{S} \subset \mathbb{R}^3$                                       | plane $\mathcal{S} \subset \mathbb{R}^3$ defined by the equation $3x - y + 4z = 0$ .                                                                   |
| johnston-linear-matrix-algebra-F01663 | 163  | 1.000 | $3x - y + 4z = 0$                                                        | $3x - y + 4z = 0$                                                        | plane $\mathcal{S} \subset \mathbb{R}^3$ defined by the equation $3x - y + 4z = 0$ .                                                                   |
| johnston-linear-matrix-algebra-F01664 | 163  | 0.624 | $(2, 2, -1)$                                                             | $(2, 2, -1)$                                                             | $(1, -1, -1)$ and $(2, 2, -1)$ are not multiples of each other, so $B$ is indeed a                                                                     |
| johnston-linear-matrix-algebra-F01665 | 164  | 0.993 | $\mathcal{S} = \{\mathbf{0}\}$                                           | $\mathcal{S} = \{\mathbf{0}\}$                                           | If $\mathcal{S} = \{\mathbf{0}\}$ is the zero                                                                                                          |
| johnston-linear-matrix-algebra-F01666 | 164  | 1.000 | $(2, 1, 1) = c_1(1, 2, 1) +$                                             | $(2, 1, 1) = c_1(1, 2, 1) +$                                             | $(2, 1, 1) = c_1(1, 2, 1) +$                                                                                                                           |
| johnston-linear-matrix-algebra-F01667 | 164  | 0.975 | $c_2(1, -1, 0)$                                                          | $c_2(1, -1, 0)$                                                          | $c_2(1, -1, 0)$ .                                                                                                                                      |
| johnston-linear-matrix-algebra-F01668 | 164  | 1.000 | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$                | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$                | vectors in the subspace $\mathcal{S}$ , and calling it $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ (a largest such set                   |
| johnston-linear-matrix-algebra-F01669 | 164  | 1.000 | $\mathbf{w} \in \operatorname{span}(B)$                                  | $\mathbf{w} \in \operatorname{span}(B)$                                  | $\mathbf{w} \in \operatorname{span}(B)$ . We know that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent |
| johnston-linear-matrix-algebra-F01670 | 164  | 1.000 | $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$        | $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$        | $\mathbf{w} \in \operatorname{span}(B)$ . We know that the set $\{\mathbf{w}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m\}$ is linearly dependent |
| johnston-linear-matrix-algebra-F01671 | 164  | 1.000 | $m + 1$                                                                  | $m + 1$                                                                  | since it contains $m + 1$ vectors, and the largest linearly independent set in $\mathcal{S}$                                                           |
| johnston-linear-matrix-algebra-F01672 | 164  | 1.000 | $d, c_1, c_2, \dots, c_m$                                                | $d, c_1, c_2, \dots, c_m$                                                | contains $m$ vectors. We can thus find scalars $d, c_1, c_2, \dots, c_m$ (not all equal to                                                             |
| johnston-linear-matrix-algebra-F01673 | 164  | 1.000 | $c_1 = c_2 = \dots =$                                                    | $c_1 = c_2 = \dots =$                                                    | Since $B$ is linearly independent, we know that if $d = 0$ then $c_1 = c_2 = \dots =$                                                                  |
| johnston-linear-matrix-algebra-F01674 | 164  | 1.000 | $c_m = 0$                                                                | $c_m = 0$                                                                | $c_m = 0$ too, which contradicts our assumption that not all of these scalars equal                                                                    |
| johnston-linear-matrix-algebra-F01675 | 164  | 0.991 | $d \neq 0$                                                               | $d \neq 0$                                                               | 0. We thus conclude that $d \neq 0$ , so we can rearrange this equation into the form                                                                  |
| johnston-linear-matrix-algebra-F01676 | 164  | 1.000 | $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ | $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ | Find bases of the range and null space of $A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ .                                   |
| johnston-linear-matrix-algebra-F01677 | 164  | 1.000 | $(2, 1, 1) = (1, 2, 1) + (1, -1, 0)$                                     | $(2, 1, 1) = (1, 2, 1) + (1, -1, 0)$                                     | To this end, we note that $(2, 1, 1) = (1, 2, 1) + (1, -1, 0)$ , so we can                                                                             |
| johnston-linear-matrix-algebra-F01678 | 164  | 1.000 | $\{(1, 2, 1), (1, -1, 0)\}$                                              | $\{(1, 2, 1), (1, -1, 0)\}$                                              | indeed, it is straightforward to see that $\{(1, 2, 1), (1, -1, 0)\}$ is linearly                                                                      |
| johnston-linear-matrix-algebra-F01679 | 165  | 1.000 | $2 + 1 = 3$                                                              | $2 + 1 = 3$                                                              | $2 + 1 = 3$ for $\alpha 3 \times 3$                                                                                                                    |
| johnston-linear-matrix-algebra-F01680 | 165  | 1.000 | $x_3$                                                                    | $x_3$                                                                    | From here we see that $x_3$ is a free variable, and we can solve for $x_1$ and $x_2$                                                                   |
| johnston-linear-matrix-algebra-F01681 | 165  | 1.000 | $x_1$                                                                    | $x_1$                                                                    | From here we see that $x_3$ is a free variable, and we can solve for $x_1$ and $x_2$                                                                   |
| johnston-linear-matrix-algebra-F01682 | 165  | 1.000 | $x_2$                                                                    | $x_2$                                                                    | From here we see that $x_3$ is a free variable, and we can solve for $x_1$ and $x_2$                                                                   |
| johnston-linear-matrix-algebra-F01683 | 165  | 1.000 | $x_1 + x_3 = 0$                                                          | $x_1 + x_3 = 0$                                                          | (the leading variables) in terms of $x_3$ : the first row tells us that $x_1 + x_3 = 0$                                                                |
| johnston-linear-matrix-algebra-F01684 | 165  | 1.000 | $x_1 = -x_3$                                                             | $x_1 = -x_3$                                                             | (i.e., $x_1 = -x_3$ ) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$ ).                                                          |
| johnston-linear-matrix-algebra-F01685 | 165  | 1.000 | $x_2 + x_3 = 0$                                                          | $x_2 + x_3 = 0$                                                          | (i.e., $x_1 = -x_3$ ) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$ ).                                                          |
| johnston-linear-matrix-algebra-F01686 | 165  | 1.000 | $x_2 = -x_3$                                                             | $x_2 = -x_3$                                                             | (i.e., $x_1 = -x_3$ ) and the second row tells us that $x_2 + x_3 = 0$ (i.e., $x_2 = -x_3$ ).                                                          |
| johnston-linear-matrix-algebra-F01687 | 165  | 1.000 | $(x_1, x_2, x_3) =$                                                      | $(x_1, x_2, x_3) =$                                                      | It follows that $\operatorname{null}(A)$ consists of the vectors of the form $(x_1, x_2, x_3) =$                                                       |

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| johnston-linear-matrix-algebra-F01688 | 165  | ■ 1.000 | $\backslash\left(-x_{\{3\}},-x_{\{3\}}, x_{\{3\}}\backslash\right)=x_{\{3\}}(-1,-1,1)$                                          | $(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$                                                      | $(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$ , so $\{(-1, -1, 1)\}$ is a basis of $\text{null}(A)$ .                                                  |
| johnston-linear-matrix-algebra-F01689 | 165  | ■ 1.000 | $\backslash\{(-1,-1,1)\}$                                                                                                       | $\{(-1, -1, 1)\}$                                                                         | $(-x_3, -x_3, x_3) = x_3(-1, -1, 1)$ , so $\{(-1, -1, 1)\}$ is a basis of $\text{null}(A)$ .                                                  |
| johnston-linear-matrix-algebra-F01690 | 165  | ■ 1.000 | $R \ \mathbf{x}=\mathbf{0}$                                                                                                     | $R\mathbf{x} = \mathbf{0}$                                                                | the linear system $R\mathbf{x} = \mathbf{0}$ (this is why we started looking at row echelon forms                                             |
| johnston-linear-matrix-algebra-F01691 | 165  | ■ 0.837 | $\operatorname{range}(R)$                                                                                                       | $\text{range}(R)$                                                                         | if and only if that same subset of columns of $R$ is a basis of $\text{range}(R)$ . Since                                                     |
| johnston-linear-matrix-algebra-F01692 | 165  | ■ 0.572 | $(R)$                                                                                                                           | $(R)$                                                                                     | the leading columns of $R$ form a basis of $\text{range}(R)$ (after all, the non-leading                                                      |
| johnston-linear-matrix-algebra-F01693 | 166  | ■ 0.984 | $\backslash\{(1,1,-1,2),(0,1,0,1),(0,0,1,-1)\}$                                                                                 | $\{(1, 1, -1, 2), (0, 1, 0, 1), (0, 0, 1, -1)\}$                                          | $\{(1, 1, -1, 2), (0, 1, 0, 1), (0, 0, 1, -1)\}$ is a basis of $\text{range}(A)$ .                                                            |
| johnston-linear-matrix-algebra-F01694 | 166  | ■ 1.000 | $x_{\{5\}}$                                                                                                                     | $x_5$                                                                                     | that the RREF tells us that $x_3$ and $x_5$ are free variables, so we solve for                                                               |
| johnston-linear-matrix-algebra-F01695 | 166  | ■ 1.000 | $x_{\{1\}}, x_{\{2\}}$                                                                                                          | $x_1, x_2$                                                                                | $x_1, x_2$ , and $x_4$ (the leading variables) in terms of them. Explicitly, we get                                                           |
| johnston-linear-matrix-algebra-F01696 | 166  | ■ 1.000 | $x_{\{4\}}$                                                                                                                     | $x_4$                                                                                     | $x_1, x_2$ , and $x_4$ (the leading variables) in terms of them. Explicitly, we get                                                           |
| johnston-linear-matrix-algebra-F01697 | 167  | ■ 1.000 | $\backslash\{(-1,1,1,0,0),(1,-2,0,-3,1)\}$                                                                                      | $\{(-1, 1, 1, 0, 0), (1, -2, 0, -3, 1)\}$                                                 | We thus conclude that $\{(-1, 1, 1, 0, 0), (1, -2, 0, -3, 1)\}$ is a basis of                                                                 |
| johnston-linear-matrix-algebra-F01698 | 167  | ■ 0.977 | $\operatorname{range}\backslash\left(A^{\{T\}}\backslash\right)$                                                                | $\text{range}\left(A^T\right)$                                                            | It turns out that the subspaces $\text{range}(A^T)$ and $\text{null}(A^T)$ play just as important                                             |
| johnston-linear-matrix-algebra-F01699 | 167  | ■ 0.977 | $\operatorname{null}\backslash\left(A^{\{T\}}\backslash\right)$                                                                 | $\text{null}\left(A^T\right)$                                                             | It turns out that the subspaces $\text{range}(A^T)$ and $\text{null}(A^T)$ play just as important                                             |
| johnston-linear-matrix-algebra-F01700 | 168  | ■ 0.719 | $\backslash\left(A^{\{T\}}\backslash\right)$                                                                                    | $(A^T)$                                                                                   | why $\text{range}(A^T)$ and                                                                                                                   |
| johnston-linear-matrix-algebra-F01701 | 168  | ■ 1.000 | $\backslash\{(0,-1,1,1)\}$                                                                                                      | $\{(0, -1, 1, 1)\}$                                                                       | so $\{(0, -1, 1, 1)\}$ is a basis of $\text{null}(A^T)$ .                                                                                     |
| johnston-linear-matrix-algebra-F01702 | 169  | ■ 1.000 | $\operatorname{row}(A)$                                                                                                         | $\text{row}(A)$                                                                           | $\text{row}(A)$ .                                                                                                                             |
| johnston-linear-matrix-algebra-F01703 | 169  | ■ 1.000 | $\mathbf{a}_{\{1\}}, \mathbf{a}_{\{2\}}, \ldots, \mathbf{a}_{\{m\}}$                                                            | $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$                                         | Suppose $A \in \mathcal{M}_{m,n}$ has rows $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m$ . Then                                           |
| johnston-linear-matrix-algebra-F01704 | 169  | ■ 1.000 | $\operatorname{range}\backslash\left(A^{\{T\}}\backslash\right)=\operatorname{range}\backslash\left(R^{\{T\}}\backslash\right)$ | $\text{range}\left(A^T\right)=\text{range}\left(R^T\right)$                               | It follows that $\text{range}(A^T) = \text{range}(R^T)$ , which is the span of the rows of $R$ . Since                                        |
| johnston-linear-matrix-algebra-F01705 | 169  | ■ 0.999 | $A^{\{T\}} E^{\{T\}}=R^{\{T\}}$                                                                                                 | $A^T E^T = R^T$                                                                           | Taking the transpose of both sides of that equation gives $A^T E^T = R^T$ . If we                                                             |
| johnston-linear-matrix-algebra-F01706 | 169  | ■ 1.000 | $E^{\{T\}}$                                                                                                                     | $E^T$                                                                                     | write $E^T$ in terms of its columns $\mathbf{v}_1, \dots, \mathbf{v}_m$ (i.e., the rows of $E$ ), then we see                                 |
| johnston-linear-matrix-algebra-F01707 | 169  | ■ 1.000 | $\mathbf{v}_{\{1\}}, \ldots, \mathbf{v}_{\{m\}}$                                                                                | $\mathbf{v}_1, \dots, \mathbf{v}_m$                                                       | write $E^T$ in terms of its columns $\mathbf{v}_1, \dots, \mathbf{v}_m$ (i.e., the rows of $E$ ), then we see                                 |
| johnston-linear-matrix-algebra-F01708 | 169  | ■ 1.000 | $R^{\{T\}}$                                                                                                                     | $R^T$                                                                                     | $k$ columns of $R^T$ are $\mathbf{0}$ as well, so $A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$ . |
| johnston-linear-matrix-algebra-F01709 | 169  | ■ 1.000 | $A^{\{T\}} \ \mathbf{v}_{\{m-k+1\}}=A^{\{T\}} \ \mathbf{v}_{\{m-k+2\}}=\cdots=A^{\{T\}} \ \mathbf{v}_{\{m\}}=\mathbf{0}$        | $A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$ | $k$ columns of $R^T$ are $\mathbf{0}$ as well, so $A^T \mathbf{v}_{m-k+1} = A^T \mathbf{v}_{m-k+2} = \dots = A^T \mathbf{v}_m = \mathbf{0}$ . |
| johnston-linear-matrix-algebra-F01710 | 170  | ■ 1.000 | $A=\backslash\left[\backslash\begin{array}{cccc}1&1&1&-1\\0&1&1&0\\-1&1&1&1\end{array}\backslash\right]$                        | $A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$     | Find bases of the four fundamental subspaces of $A = \begin{bmatrix} 1 & 1 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{bmatrix}$ .       |
| johnston-linear-matrix-algebra-F01711 | 170  | ■ 1.000 | $\operatorname{range}(A):\backslash\{(1,0,-1),(1,1,1)\}$                                                                        | $\text{range}(A) : \{(1, 0, -1), (1, 1, 1)\}$                                             | • $\text{range}(A)$ : $\{(1, 0, -1), (1, 1, 1)\}$ ,                                                                                           |
| johnston-linear-matrix-algebra-F01712 | 170  | ■ 1.000 | $\operatorname{null}\backslash\left(A^{\{T\}}\backslash\right):\backslash\{(1,-2,1)\}$                                          | $\text{null}\left(A^T\right) : \{(1, -2, 1)\}$                                            | • $\text{null}(A^T)$ : $\{(1, -2, 1)\}$ , and                                                                                                 |

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| johnston-linear-matrix-algebra-F01713 | 170  | 1.000 | $\operatorname{range}(A^T): \{(1,0,0,-1), (0,1,1,0)\}$                                                                | $\operatorname{range}(A^T) : \{(1,0,0,-1), (0,1,1,0)\}$                                                | <ul style="list-style-type: none"> <li><math>\operatorname{range}(A^T): \{(1,0,0,-1), (0,1,1,0)\}</math>.</li> </ul>                           |
| johnston-linear-matrix-algebra-F01714 | 170  | 1.000 | $x_1 - x_4 = 0$                                                                                                       | $x_1 - x_4 = 0$                                                                                        | Furthermore, $x_1 - x_4 = 0$ and $x_2 + x_3 = 0$ , so writing the solution in vector                                                           |
| johnston-linear-matrix-algebra-F01715 | 170  | 0.995 | $\{(0,-1,1,0), (1,0,0,1)\}$                                                                                           | $\{(0,-1,1,0), (1,0,0,1)\}$                                                                            | It follows that one basis of $\operatorname{null}(A)$ is $\{(0,-1,1,0), (1,0,0,1)\}$ .                                                         |
| johnston-linear-matrix-algebra-F01716 | 171  | 1.000 | $\operatorname{range}(A) = \mathbb{R}^n$                                                                              | $\operatorname{range}(A) = \mathbb{R}^n$                                                               | b) $\operatorname{range}(A) = \mathbb{R}^n$ .                                                                                                  |
| johnston-linear-matrix-algebra-F01717 | 171  | 1.000 | $\operatorname{null}(A) = \{\mathbf{0}\}$                                                                             | $\operatorname{null}(A) = \{\mathbf{0}\}$                                                              | c) $\operatorname{null}(A) = \{\mathbf{0}\}$ .                                                                                                 |
| johnston-linear-matrix-algebra-F01718 | 171  | 0.990 | $\left(A^T\right)=\mathbb{R}^n$                                                                                       | $(A^T) = \mathbb{R}^n$                                                                                 | We could also add “ $\operatorname{range}(A^T) = \mathbb{R}^n$ ” and “ $\operatorname{null}(A^T) = \{\mathbf{0}\}$ ” as additional             |
| johnston-linear-matrix-algebra-F01719 | 171  | 0.990 | $\left(A^T\right)=\{\mathbf{0}\}$                                                                                     | $(A^T) = \{\mathbf{0}\}$                                                                               | We could also add “ $\operatorname{range}(A^T) = \mathbb{R}^n$ ” and “ $\operatorname{null}(A^T) = \{\mathbf{0}\}$ ” as additional             |
| johnston-linear-matrix-algebra-F01720 | 171  | 0.531 | $\operatorname{rank}(A)$                                                                                              | $\operatorname{rank}(A)$                                                                               | Suppose $A$ is a matrix. Its <b>rank</b> , denoted by $\operatorname{rank}(A)$ , is the dimension of                                           |
| johnston-linear-matrix-algebra-F01721 | 171  | 1.000 | $A=\mathbf{u} \mathbf{u}^T$                                                                                           | $A = \mathbf{u} \mathbf{u}^T$                                                                          | For example, if $\mathbf{u} \in \mathbb{R}^n$ is a unit vector and $A = \mathbf{u} \mathbf{u}^T$ is the standard matrix                        |
| johnston-linear-matrix-algebra-F01722 | 171  | 0.823 | $\operatorname{rank}(A)=1$                                                                                            | $\operatorname{rank}(A) = 1$                                                                           | 1-dimensional, so $\operatorname{rank}(A) = 1$ . Similarly, we showed that the matrix                                                          |
| johnston-linear-matrix-algebra-F01723 | 171  | 0.406 | $((1,2,1),(1,-1,0))$                                                                                                  | $((1,2,1), (1,-1,0))$                                                                                  | from Example 2.4.5 has range equal to $\operatorname{span}((1,2,1), (1,-1,0))$ . Since this                                                    |
| johnston-linear-matrix-algebra-F01724 | 172  | 1.000 | $\operatorname{rank}(A) \leq \min \{m, n\}$                                                                           | $\operatorname{rank}(A) \leq \min \{m, n\}$                                                            | $\operatorname{rank}(A) \leq \min \{m, n\}$                                                                                                    |
| johnston-linear-matrix-algebra-F01725 | 172  | 1.000 | $\operatorname{rank}(A)=\min \{m, n\}$                                                                                | $\operatorname{rank}(A) = \min \{m, n\}$                                                               | $\operatorname{rank}(A) = \min \{m, n\}$ .                                                                                                     |
| johnston-linear-matrix-algebra-F01726 | 172  | 0.994 | $\operatorname{rank}(A) \leq n$                                                                                       | $\operatorname{rank}(A) \leq n$                                                                        | we conclude that $A$ is invertible if and only if its rank is $n$ . Since $\operatorname{rank}(A) \leq n$ for                                  |
| johnston-linear-matrix-algebra-F01727 | 172  | 1.000 | $\operatorname{rank}\left(A^T\right)$                                                                                 | $\operatorname{rank}\left(A^T\right)$                                                                  | b) $\operatorname{rank}\left(A^T\right)$                                                                                                       |
| johnston-linear-matrix-algebra-F01728 | 172  | 1.000 | $C=\left[\begin{array}{ccccc} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{array}\right]$  | $C = \begin{bmatrix} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{bmatrix}$ | c) $C = \begin{bmatrix} 0 & 0 & -2 & 2 & -2 \\ 2 & -2 & -1 & 3 & 3 \\ -1 & 1 & -1 & 0 & -3 \end{bmatrix}$                                      |
| johnston-linear-matrix-algebra-F01729 | 173  | 1.000 | $\operatorname{rank}(A)=n$                                                                                            | $\operatorname{rank}(A) = n$                                                                           | that $\operatorname{rank}(A) = n$ .                                                                                                            |
| johnston-linear-matrix-algebra-F01730 | 173  | 1.000 | $\operatorname{rank}(B)=2$                                                                                            | $\operatorname{rank}(B) = 2$                                                                           | dimensional. It follows that $\operatorname{rank}(B) = 2$ .                                                                                    |
| johnston-linear-matrix-algebra-F01731 | 173  | 1.000 | $\operatorname{rank}(C)=2$                                                                                            | $\operatorname{rank}(C) = 2$                                                                           | $\operatorname{rank}(C) = 2$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F01732 | 173  | 1.000 | $\operatorname{nullity}(A)=0$                                                                                         | $\operatorname{nullity}(A) = 0$                                                                        | c) $\operatorname{nullity}(A) = 0$ .                                                                                                           |
| johnston-linear-matrix-algebra-F01733 | 173  | 1.000 | $\operatorname{rank}(A)+\operatorname{nullity}(A)=n$                                                                  | $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$                                               | Suppose $A \in \mathcal{M}_{m,n}$ is a matrix. Then $\operatorname{rank}(A) + \operatorname{nullity}(A) = n$ .                                 |
| johnston-linear-matrix-algebra-F01734 | 173  | 1.000 | $r=\operatorname{rank}(A)$                                                                                            | $r = \operatorname{rank}(A)$                                                                           | Let $r = \operatorname{rank}(A)$ and consider the linear system $A\mathbf{x} = \mathbf{0}$ , which has $r$ leading                             |
| johnston-linear-matrix-algebra-F01735 | 173  | 1.000 | $r$                                                                                                                   | $r$                                                                                                    | Let $r = \operatorname{rank}(A)$ and consider the linear system $A\mathbf{x} = \mathbf{0}$ , which has $r$ leading                             |
| johnston-linear-matrix-algebra-F01736 | 173  | 1.000 | $n-r$                                                                                                                 | $n - r$                                                                                                | variables. Since this linear system has $n$ variables total, it must have $n - r$ free                                                         |
| johnston-linear-matrix-algebra-F01737 | 173  | 0.996 | $\operatorname{nullity}(A)=\operatorname{dim}(\operatorname{null}(A))=\operatorname{dim}(\operatorname{null}(A))=n-r$ | $\operatorname{nullity}(A) = \operatorname{dim}(\operatorname{null}(A)) = n - r$                       | of a basis of $\operatorname{null}(A)$ , so $\operatorname{nullity}(A) = \operatorname{dim}(\operatorname{null}(A)) = n - r$ . It follows that |
| johnston-linear-matrix-algebra-F01738 | 174  | 1.000 | $\operatorname{rank}(A)=3$                                                                                            | $\operatorname{rank}(A) = 3$                                                                           | $\operatorname{rank}(A) = 3$ and $\operatorname{nullity}(A) = 5 - 3 = 2$ .                                                                     |

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| johnston-linear-matrix-algebra-F01739 | 174  | 1.000 | $\operatorname{nullity}(A)=5-3=2$                                                               | $\operatorname{nullity}(A) = 5 - 3 = 2$                                                         | $\operatorname{rank}(A) = 3$ and $\operatorname{nullity}(A) = 5 - 3 = 2$ .                                                                         |
| johnston-linear-matrix-algebra-F01740 | 174  | 1.000 | $\mathbb{R}^5$                                                                                  | $\mathbb{R}^5$                                                                                  | transformation from $\mathbb{R}^5$ to $\mathbb{R}^4$ . Since its rank is 3, it sends a 3-dimensional                                               |
| johnston-linear-matrix-algebra-F01741 | 175  | 1.000 | $\operatorname{rank}(A\ B)=\operatorname{rank}(A)$                                              | $\operatorname{rank}(AB) = \operatorname{rank}(A)$                                              | $\operatorname{rank}(AB) = \operatorname{rank}(A)$                                                                                                 |
| johnston-linear-matrix-algebra-F01742 | 175  | 1.000 | $\sum_{j=1}^n$                                                                                  | $\sum_{j=1}^n$                                                                                  | The notation $\sum_{j=1}^n$                                                                                                                        |
| johnston-linear-matrix-algebra-F01743 | 175  | 1.000 | $\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$                                                           | $\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$                                                           | $\sum_{j=1}^n b_{j,1} \mathbf{a}_j =$                                                                                                              |
| johnston-linear-matrix-algebra-F01744 | 175  | 1.000 | $b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$                                          | $b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$                                          | $b_{1,1} \mathbf{a}_1 + \cdots + b_{n,1} \mathbf{a}_n$ .                                                                                           |
| johnston-linear-matrix-algebra-F01745 | 175  | 1.000 | $A \in \mathcal{M}_{3,2}$                                                                       | $A \in \mathcal{M}_{3,2}$                                                                       | <b>Figure 2.25:</b> A non-zero matrix $A \in \mathcal{M}_{3,2}$ either has rank 1 and nullity 1 (as in (a)) or                                     |
| johnston-linear-matrix-algebra-F01746 | 175  | 1.000 | $\operatorname{rank}(A+B)=\operatorname{rank}(A)+\operatorname{rank}(B)$                        | $\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B)$                    | it is not the case that $\operatorname{rank}(A+B) = \operatorname{rank}(A) + \operatorname{rank}(B)$ , nor is it the case that                     |
| johnston-linear-matrix-algebra-F01747 | 175  | 0.813 | $\operatorname{rank}(A\ B)=\operatorname{rank}(A) \times \operatorname{rank}(B)$                | $\operatorname{rank}(AB) = \operatorname{rank}(A) \times \operatorname{rank}(B)$                | $\operatorname{rank}(AB) = \operatorname{rank}(A) \times \operatorname{rank}(B)$ ). However, there are some <i>inequalities</i> that at            |
| johnston-linear-matrix-algebra-F01748 | 175  | 0.994 | $\operatorname{rank}(A+B)$                                                                      | $\operatorname{rank}(A+B)$                                                                      | least loosely relate $\operatorname{rank}(A+B)$ and $\operatorname{rank}(AB)$ to the ranks of $A$ and $B$ :                                        |
| johnston-linear-matrix-algebra-F01749 | 175  | 0.994 | $\operatorname{rank}(A\ B)$                                                                     | $\operatorname{rank}(AB)$                                                                       | least loosely relate $\operatorname{rank}(A+B)$ and $\operatorname{rank}(AB)$ to the ranks of $A$ and $B$ :                                        |
| johnston-linear-matrix-algebra-F01750 | 175  | 1.000 | $\operatorname{rank}(A+B) \leq \operatorname{rank}(A)+\operatorname{rank}(B)$                   | $\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$                 | a) $\operatorname{rank}(A+B) \leq \operatorname{rank}(A) + \operatorname{rank}(B)$                                                                 |
| johnston-linear-matrix-algebra-F01751 | 175  | 1.000 | $\operatorname{rank}(A\ B) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$      | $\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$        | b) $\operatorname{rank}(AB) \leq \min \{ \operatorname{rank}(A), \operatorname{rank}(B) \}$                                                        |
| johnston-linear-matrix-algebra-F01752 | 175  | 0.996 | $\mathbf{a}_1, \mathbf{a}_2, \ldots$                                                            | $\mathbf{a}_1, \mathbf{a}_2, \ldots$                                                            | <i>Proof.</i> Denote the columns of $A$ and $B$ by $\mathbf{a}_1, \mathbf{a}_2, \ldots$ and $\mathbf{b}_1, \mathbf{b}_2, \ldots$ , respectively.   |
| johnston-linear-matrix-algebra-F01753 | 175  | 0.996 | $\mathbf{b}_1, \mathbf{b}_2, \ldots$                                                            | $\mathbf{b}_1, \mathbf{b}_2, \ldots$                                                            | <i>Proof.</i> Denote the columns of $A$ and $B$ by $\mathbf{a}_1, \mathbf{a}_2, \ldots$ and $\mathbf{b}_1, \mathbf{b}_2, \ldots$ , respectively.   |
| johnston-linear-matrix-algebra-F01754 | 175  | 1.000 | $\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$ | $\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$ | of $A+B$ are $\mathbf{a}_1 + \mathbf{b}_1, \mathbf{a}_2 + \mathbf{b}_2, \ldots, \mathbf{a}_n + \mathbf{b}_n$ , which are each contained within the |
| johnston-linear-matrix-algebra-F01755 | 175  | 0.999 | $\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$    | $\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$    | span of $\mathbf{a}_1, \mathbf{b}_1, \mathbf{a}_2, \mathbf{b}_2, \ldots, \mathbf{a}_n, \mathbf{b}_n$ . It follows that                             |
| johnston-linear-matrix-algebra-F01756 | 175  | 0.955 | $\operatorname{range}(A+B)$                                                                     | $\operatorname{range}(A+B)$                                                                     | as well. The span on the left equals $\operatorname{range}(A+B)$ , so its dimension is $\operatorname{rank}(A+$                                    |
| johnston-linear-matrix-algebra-F01757 | 175  | 0.955 | $\operatorname{rank}(A+$                                                                        | $\operatorname{rank}(A+$                                                                        | as well. The span on the left equals $\operatorname{range}(A+B)$ , so its dimension is $\operatorname{rank}(A+$                                    |
| johnston-linear-matrix-algebra-F01758 | 175  | 0.863 | $\operatorname{rank}(A)+\operatorname{rank}(B)$                                                 | $\operatorname{rank}(A) + \operatorname{rank}(B)$                                               | $B)$ . The span on the right cannot have dimension larger than $\operatorname{rank}(A) + \operatorname{rank}(B)$ ,                                 |
| johnston-linear-matrix-algebra-F01759 | 176  | 1.000 | $\operatorname{range}(A\ B) \subseteq \operatorname{range}(A)$                                  | $\operatorname{range}(AB) \subseteq \operatorname{range}(A)$                                    | $\operatorname{range}(AB) \subseteq \operatorname{range}(A)$                                                                                       |
| johnston-linear-matrix-algebra-F01760 | 176  | 0.998 | $\operatorname{rank}(A\ B) \leq \operatorname{rank}(A)$                                         | $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$                                           | the dimension of its range, it follows that $\operatorname{rank}(AB) \leq \operatorname{rank}(A)$ . To see that                                    |
| johnston-linear-matrix-algebra-F01761 | 176  | 1.000 | $\operatorname{rank}(A\ B) \leq \operatorname{rank}(B)$                                         | $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$                                           | $\operatorname{rank}(AB) \leq \operatorname{rank}(B)$ as well (thus completing the proof), we could apply this                                     |
| johnston-linear-matrix-algebra-F01762 | 176  | 0.999 | $B=\{(1,1),(1,2),(3,4)\}, \mathcal{S}=\mathbb{R}^2$                                             | $B = \{(1,1), (1,2), (3,4)\}, \mathcal{S} = \mathbb{R}^2$                                       | <b>(a)</b> $B = \{(1,1), (1,2), (3,4)\}, \mathcal{S} = \mathbb{R}^2$                                                                               |
| johnston-linear-matrix-algebra-F01763 | 176  | 1.000 | $B=\{(1,1),(1,-1)\}, \mathcal{S}=\mathbb{R}^2$                                                  | $B = \{(1,1), (1,-1)\}, \mathcal{S} = \mathbb{R}^2$                                             | <b>(b)</b> $B = \{(1,1), (1,-1)\}, \mathcal{S} = \mathbb{R}^2$                                                                                     |
| johnston-linear-matrix-algebra-F01764 | 176  | 0.963 | $B=\{(1,1,1),(2,1,-1)\}, \mathcal{S}=\mathbb{R}^3$                                              | $B = \{(1,1,1), (2,1,-1)\}, \mathcal{S} = \mathbb{R}^3$                                         | <b>(c)</b> $B = \{(1,1,1), (2,1,-1)\}, \mathcal{S} = \mathbb{R}^3$                                                                                 |
| johnston-linear-matrix-algebra-F01765 | 176  | 0.993 | $B=\{(2,2,1),(0,1,1)\}, \mathcal{S}$                                                            | $B = \{(2,2,1), (0,1,1)\}, \mathcal{S}$                                                         | <b>(d)</b> $B = \{(2,2,1), (0,1,1)\}, \mathcal{S}$ is the plane in $\mathbb{R}^3$ with                                                             |

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|--------------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-augmented<br>F01766 | 176  | 1.000 | $x+y-z=0$                                                                                                                             | $x+y-z=0$                                                                                                                             | equation $x+y-z=0$ .                                                                                                                  |
| johnston-linear-matrix-augmented<br>F01767 | 176  | 0.997 | $B=\{(1,0,1),(2,1,3)\}, \mathcal{S}$                                                                                                  | $B=\{(1,0,1),(2,1,3)\}, \mathcal{S}$                                                                                                  | *(e) $B=\{(1,0,1),(2,1,3)\}$ , $\mathcal{S}$ is the plane in $\mathbb{R}^3$ with                                                      |
| johnston-linear-matrix-augmented<br>F01768 | 176  | 1.000 | $B=\{(1,3,2),(-1,-2,1),(1,4,5)\}, \mathcal{S}=\operatorname{span}(B)$                                                                 | $B=\{(1,3,2),(-1,-2,1),(1,4,5)\}, \mathcal{S}=\operatorname{span}(B)$                                                                 | (f) $B=\{(1,3,2),(-1,-2,1),(1,4,5)\}, \mathcal{S}=\operatorname{span}(B)$ .                                                           |
| johnston-linear-matrix-augmented<br>F01769 | 176  | 0.958 | $2x+y+z=0$                                                                                                                            | $2x+y+z=0$                                                                                                                            | *(c) The plane in $\mathbb{R}^3$ with equation $2x+y+z=0$ .                                                                           |
| johnston-linear-matrix-augmented<br>F01770 | 176  | 1.000 | $z=4x-3y$                                                                                                                             | $z=4x-3y$                                                                                                                             | (d) The plane in $\mathbb{R}^3$ with equation $z=4x-3y$ .                                                                             |
| johnston-linear-matrix-augmented<br>F01771 | 176  | 1.000 | $2x-y+2z=0$                                                                                                                           | $2x-y+2z=0$                                                                                                                           | $x+y-z=0$ and $2x-y+2z=0$ .                                                                                                           |
| johnston-linear-matrix-augmented<br>F01772 | 176  | 1.000 | $\operatorname{span}(\{(1,3,2,1),(-1,-2,1,0),(1,4,-1,1)\})$                                                                           | $\operatorname{span}(\{(1,3,2,1),(-1,-2,1,0),(1,4,-1,1)\})$                                                                           | (f) $\operatorname{span}(\{(1,3,2,1),(-1,-2,1,0),(1,4,-1,1)\})$                                                                       |
| johnston-linear-matrix-augmented<br>F01773 | 176  | 1.000 | $\{(5,5,5,1),(3,6,3,2),(4,1,4,4),(1,4,1,4)\}$                                                                                         | $\{(5,5,5,1),(3,6,3,2),(4,1,4,4),(1,4,1,4)\}$                                                                                         | (b) $\{(5,5,5,1),(3,6,3,2),(4,1,4,4),(1,4,1,4)\}$                                                                                     |
| johnston-linear-matrix-augmented<br>F01774 | 176  | 0.942 | $\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3)\}$                                                                                            | $\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3)\}$                                                                                            | *(c) $\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3),$                                                                                        |
| johnston-linear-matrix-augmented<br>F01775 | 176  | 1.000 | $\{(3,4,-1,1,3),(-1,1,3,0,1),(4,6,-5,3,6)\}$                                                                                          | $\{(3,4,-1,1,3),(-1,1,3,0,1),(4,6,-5,3,6)\}$                                                                                          | (d) $\{(3,4,-1,1,3),(-1,1,3,0,1),(4,6,-5,3,6),$                                                                                       |
| johnston-linear-matrix-augmented<br>F01776 | 176  | 0.811 | $\{(2,3,1,1),(3,3,2,3),(2,1,2,3)\}$                                                                                                   | $\{(2,3,1,1),(3,3,2,3),(2,1,2,3)\}$                                                                                                   | *(a) $\{(2,3,1,1),(3,3,2,3),(2,1,2,3)\}$                                                                                              |
| johnston-linear-matrix-augmented<br>F01777 | 176  | 0.993 | $\{(3,2,0,2),(0,1,3,3),(2,1,1,1),(0,2,3,2)\}$                                                                                         | $\{(3,2,0,2),(0,1,3,3),(2,1,1,1),(0,2,3,2)\}$                                                                                         | (b) $\{(3,2,0,2),(0,1,3,3),(2,1,1,1),(0,2,3,2)\}$                                                                                     |
| johnston-linear-matrix-augmented<br>F01778 | 176  | 1.000 | $\{(2,4,3,4),(3,2,-1,-1)\}$                                                                                                           | $\{(2,4,3,4),(3,2,-1,-1)\}$                                                                                                           | *(c) $\{(2,4,3,4),(3,2,-1,-1)\}$                                                                                                      |
| johnston-linear-matrix-augmented<br>F01779 | 176  | 1.000 | $\{(-1,3,6,3),(1,3,5,0),(3,0,-1,-3)\}$                                                                                                | $\{(-1,3,6,3),(1,3,5,0),(3,0,-1,-3)\}$                                                                                                | (d) $\{(-1,3,6,3),(1,3,5,0),(3,0,-1,-3)\}$                                                                                            |
| johnston-linear-matrix-augmented<br>F01780 | 176  | 1.000 | $\operatorname{range}(A), \operatorname{null}(A), \operatorname{range}(A^T)$                                                          | $\operatorname{range}(A), \operatorname{null}(A), \operatorname{range}(A^T)$                                                          | each of $\operatorname{range}(A)$ , $\operatorname{null}(A)$ , $\operatorname{range}(A^T)$ , and $\operatorname{null}(A^T)$ .         |
| johnston-linear-matrix-augmented<br>F01781 | 176  | 1.000 | $\left[\begin{array}{lll}0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0\end{array}\right]$                                                       | $\left[\begin{array}{ccc}0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0\end{array}\right]$                                                       | $\left[\begin{array}{ccc}0 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0\end{array}\right]$                                                       |
| johnston-linear-matrix-augmented<br>F01782 | 176  | 1.000 | $\left[\begin{array}{ccccc}1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1\end{array}\right]$                            | $\left[\begin{array}{ccccc}1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1\end{array}\right]$                            | $\left[\begin{array}{ccccc}1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1\end{array}\right]$                            |
| johnston-linear-matrix-augmented<br>F01783 | 176  | 1.000 | $\left[\begin{array}{ccccc}0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ -2 & 0 & 2 & 6 & 3\end{array}\right]$                          | $\left[\begin{array}{ccccc}0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ -2 & 0 & 2 & 6 & 3\end{array}\right]$                          | $\left[\begin{array}{ccccc}0 & -4 & 0 & 2 & 1 \\ -1 & 2 & 1 & 2 & 1 \\ -2 & 0 & 2 & 6 & 3\end{array}\right]$                          |
| johnston-linear-matrix-augmented<br>F01784 | 176  | 1.000 | $\left[\begin{array}{ccccc}2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1\end{array}\right]$ | $\left[\begin{array}{ccccc}2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1\end{array}\right]$ | $\left[\begin{array}{ccccc}2 & -1 & 2 & 3 & 5 \\ -1 & 1 & -2 & -2 & -2 \\ 1 & -1 & 2 & 1 & 2 \\ 1 & -2 & 4 & 2 & 1\end{array}\right]$ |
| johnston-linear-matrix-augmented<br>F01785 | 176  | 0.651 | $\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$                                                                           | $\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$                                                                           | *(a) $\left[\begin{array}{cc}1 & -1 \\ -1 & 1\end{array}\right]$                                                                      |
| johnston-linear-matrix-augmented<br>F01786 | 176  | 1.000 | $\left[\begin{array}{cccc}5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2\end{array}\right]$                                                          | $\left[\begin{array}{cccc}5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2\end{array}\right]$                                                          | $\left[\begin{array}{cccc}5 & 4 & 8 & 1 \\ 3 & -2 & 7 & 2\end{array}\right]$                                                          |
| johnston-linear-matrix-augmented<br>F01787 | 176  | 1.000 | $\left[\begin{array}{ll}1 & 0 \\ 0 & 2 \\ 3 & 1\end{array}\right]$                                                                    | $\left[\begin{array}{ll}1 & 0 \\ 0 & 2 \\ 3 & 1\end{array}\right]$                                                                    | $\left[\begin{array}{ll}1 & 0 \\ 0 & 2 \\ 3 & 1\end{array}\right]$                                                                    |

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| johnston-linear-matrix-algebra-F01788 | 176  | 1.000 | $\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$ | $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$ |                                                                                                           |
| johnston-linear-matrix-algebra-F01789 | 177  | 0.902 | $A \in \mathcal{M}_{3,7}$                                                                                       | $A \in \mathcal{M}_{3,7}$                                                                          | (h) If $A \in \mathcal{M}_{3,7}$ then $\text{rank}(A) \leq 3$ .                                           |
| johnston-linear-matrix-algebra-F01790 | 177  | 0.902 | $(A) \leq 3$                                                                                                    | $(A) \leq 3$                                                                                       | (h) If $A \in \mathcal{M}_{3,7}$ then $\text{rank}(A) \leq 3$ .                                           |
| johnston-linear-matrix-algebra-F01791 | 177  | 1.000 | $\text{rank}(A \cdot B) = \text{rank}(A) \cdot \text{rank}(B)$                                                  | $\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$                                            | $\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$ .                                                 |
| johnston-linear-matrix-algebra-F01792 | 177  | 1.000 | $\text{rank}(A \cdot B) = \min\{\text{rank}(A), \text{rank}(B)\}$                                               | $\text{rank}(AB) = \min\{\text{rank}(A), \text{rank}(B)\}$                                         | $\text{rank}(AB) = \min\{\text{rank}(A), \text{rank}(B)\}$ .                                              |
| johnston-linear-matrix-algebra-F01793 | 177  | 1.000 | $\text{null}(A) =$                                                                                              | $\text{null}(A) =$                                                                                 | (n) If $A, B \in \mathcal{M}_{m,n}$ are row equivalent then $\text{null}(A) =$                            |
| johnston-linear-matrix-algebra-F01794 | 177  | 1.000 | $\text{null}(B)$                                                                                                | $\text{null}(B)$                                                                                   | $\text{null}(B)$ .                                                                                        |
| johnston-linear-matrix-algebra-F01795 | 177  | 0.603 | $\text{nullity}(A) = \text{nullity}(A^T)$                                                                       | $\text{nullity}(A) = \text{nullity}(A^T)$                                                          | (o) If $A \in \mathcal{M}_{m,n}$ then $\text{nullity}(A) = \text{nullity}(A^T)$ .                         |
| johnston-linear-matrix-algebra-F01796 | 177  | 1.000 | $A \in \mathcal{M}_4$                                                                                           | $A \in \mathcal{M}_4$                                                                              | (a) Find a matrix $A \in \mathcal{M}_4$ whose range is the same as                                        |
| johnston-linear-matrix-algebra-F01797 | 177  | 1.000 | $A \in \mathcal{M}_3$                                                                                           | $A \in \mathcal{M}_3$                                                                              | (b) Show that there does not exist a matrix $A \in \mathcal{M}_3$                                         |
| johnston-linear-matrix-algebra-F01798 | 177  | 0.715 | $\text{rank}(J_n)$                                                                                              | $\text{rank}(J_n)$                                                                                 | 1. Compute $\text{rank}(J_n)$ .                                                                           |
| johnston-linear-matrix-algebra-F01799 | 177  | 1.000 | $A \mathbf{v} \neq B \mathbf{v}$                                                                                | $A \mathbf{v} \neq B \mathbf{v}$                                                                   | erty that $A \mathbf{v} \neq B \mathbf{v}$ for all $\mathbf{v} \neq \mathbf{0}$ . What is the rank of the |
| johnston-linear-matrix-algebra-F01800 | 177  | 1.000 | $\mathbf{v} \neq \mathbf{0}$                                                                                    | $\mathbf{v} \neq \mathbf{0}$                                                                       | erty that $A \mathbf{v} \neq B \mathbf{v}$ for all $\mathbf{v} \neq \mathbf{0}$ . What is the rank of the |
| johnston-linear-matrix-algebra-F01801 | 177  | 0.884 | $\text{rank}(A_n) = 2$                                                                                          | $\text{rank}(A_n) = 2$                                                                             | Show that $\text{rank}(A_n) = 2$ for all $n \geq 2$ . [Hint: Write each row                               |
| johnston-linear-matrix-algebra-F01802 | 177  | 0.884 | $n \geq 2$                                                                                                      | $n \geq 2$                                                                                         | Show that $\text{rank}(A_n) = 2$ for all $n \geq 2$ . [Hint: Write each row                               |
| johnston-linear-matrix-algebra-F01803 | 177  | 0.814 | $\left[\begin{array}{ll}1 & x \\ 2 & 4\end{array}\right]$                                                       | $\begin{bmatrix} 1 & x \\ 2 & 4 \end{bmatrix}$                                                     | (a) $\begin{bmatrix} 1 & x \\ 2 & 4 \end{bmatrix}$                                                        |
| johnston-linear-matrix-algebra-F01804 | 177  | 1.000 | $\left[\begin{array}{cc}x-1 & -1 \\ 1 & 1-x\end{array}\right]$                                                  | $\begin{bmatrix} x-1 & -1 \\ 1 & 1-x \end{bmatrix}$                                                | (b) $\begin{bmatrix} x-1 & -1 \\ 1 & 1-x \end{bmatrix}$                                                   |
| johnston-linear-matrix-algebra-F01805 | 177  | 1.000 | $\left[\begin{array}{ccc}1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1\end{array}\right]$                        | $\begin{bmatrix} 1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1 \end{bmatrix}$                       | $\begin{bmatrix} 1 & 2 & x \\ 2 & 3-x & 3x+1 \\ -x & -2x & -1 \end{bmatrix}$                              |
| johnston-linear-matrix-algebra-F01806 | 177  | 0.987 | $\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x\end{array}\right]$                                 | $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x \end{bmatrix}$                                | (d) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & x \end{bmatrix}$                                   |
| johnston-linear-matrix-algebra-F01807 | 177  | 0.931 | $ \text{rank}(A) -$                                                                                             | $ \text{rank}(A) -$                                                                                | of their other entries are the same) then $ \text{rank}(A) -$                                             |
| johnston-linear-matrix-algebra-F01808 | 177  | 0.967 | $\text{rank}(B) \leq 1$                                                                                         | $\text{rank}(B) \leq 1$                                                                            | $\text{rank}(B) \leq 1$ . That is, show that $\text{rank}(A)$ and $\text{rank}(B)$                        |
| johnston-linear-matrix-algebra-F01809 | 177  | 0.967 | $\text{rank}(B)$                                                                                                | $\text{rank}(B)$                                                                                   | $\text{rank}(B) \leq 1$ . That is, show that $\text{rank}(A)$ and $\text{rank}(B)$                        |
| johnston-linear-matrix-algebra-F01810 | 177  | 1.000 | $ \text{rank}(A) - \text{rank}(B)  \leq k$                                                                      | $ \text{rank}(A) - \text{rank}(B)  \leq k$                                                         | $k$ entries then $ \text{rank}(A) - \text{rank}(B)  \leq k$ .                                             |
| johnston-linear-matrix-algebra-F01811 | 177  | 1.000 | $x=1$                                                                                                           | $x = 1$                                                                                            | (a) Show that if $x = 1$ then $\text{rank}(A) = 1$ .                                                      |

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|---------------------------------------|------|---------|-----------------------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F01812 | 177  | ■ 1.000 | $x \neq 1$                                                      | $x \neq 1$                                                      | (b) Show that if $x \neq 1$ then $\text{rank}(A) = n$ .                                                                |
| johnston-linear-matrix-algebra-F01813 | 177  | ■ 1.000 | $\text{rank}(A \ B) = \text{rank}(B)$                           | $\text{rank}(AB) = \text{rank}(B)$                              | $\text{rank}(AB) = \text{rank}(B)$ . [Hint: Try to make use of The-                                                    |
| johnston-linear-matrix-algebra-F01814 | 177  | ■ 1.000 | $\text{rank}(A \ B) =$                                          | $\text{rank}(AB) =$                                             | (b) Show that if $B \in \mathcal{M}_n$ is invertible then $\text{rank}(AB) =$                                          |
| johnston-linear-matrix-algebra-F01815 | 177  | ■ 1.000 | $\text{rank}(A) \leq n / 2$                                     | $\text{rank}(A) \leq n/2$                                       | $A^2 = O$ . Show that $\text{rank}(A) \leq n/2$ .                                                                      |
| johnston-linear-matrix-algebra-F01816 | 178  | ■ 1.000 | $\mathcal{S}_1$                                                 | $\mathcal{S}_1$                                                 | 2.4.20 Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^n$ with                           |
| johnston-linear-matrix-algebra-F01817 | 178  | ■ 1.000 | $\mathcal{S}_2$                                                 | $\mathcal{S}_2$                                                 | 2.4.20 Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are subspaces of $\mathbb{R}^n$ with                           |
| johnston-linear-matrix-algebra-F01818 | 178  | ■ 1.000 | $\mathcal{S}_1 \subseteq \mathcal{S}_2$                         | $\mathcal{S}_1 \subseteq \mathcal{S}_2$                         | the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ . Show that $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$ . |
| johnston-linear-matrix-algebra-F01819 | 178  | ■ 1.000 | $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$                  | $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$                  | the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ . Show that $\dim(\mathcal{S}_1) \leq \dim(\mathcal{S}_2)$ . |
| johnston-linear-matrix-algebra-F01820 | 178  | ■ 1.000 | $\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$                     | $\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$                     | the property that $\mathcal{S}_1 \subseteq \mathcal{S}_2$ and $\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$ . Show       |
| johnston-linear-matrix-algebra-F01821 | 178  | ■ 1.000 | $\mathcal{S}_1 = \mathcal{S}_2$                                 | $\mathcal{S}_1 = \mathcal{S}_2$                                 | that $\mathcal{S}_1 = \mathcal{S}_2$ .                                                                                 |
| johnston-linear-matrix-algebra-F01822 | 178  | ■ 0.893 | $-1, 2, 3$                                                      | $-1, 2, 3$                                                      | (b) Why does removing $(-1, 2, 3)$ from $B$ not result in a                                                            |
| johnston-linear-matrix-algebra-F01823 | 178  | ■ 1.000 | $\mathbf{v} \in \text{range}(A)$                                | $\mathbf{v} \in \text{range}(A)$                                | (a) Show that if $\mathbf{v} \in \text{range}(A)$ and $\mathbf{w} \in \text{null}(A^T)$ then $\mathbf{v}$              |
| johnston-linear-matrix-algebra-F01824 | 178  | ■ 1.000 | $\mathbf{w} \in \text{null}(A^T)$                               | $\mathbf{w} \in \text{null}(A^T)$                               | (a) Show that if $\mathbf{v} \in \text{range}(A)$ and $\mathbf{w} \in \text{null}(A^T)$ then $\mathbf{v}$              |
| johnston-linear-matrix-algebra-F01825 | 178  | ■ 0.999 | $\mathbf{w} \in \text{range}(A^T)$                              | $\mathbf{w} \in \text{range}(A^T)$                              | (b) Show that if $\mathbf{v} \in \text{null}(A)$ and $\mathbf{w} \in \text{range}(A^T)$ then $\mathbf{v}$              |
| johnston-linear-matrix-algebra-F01826 | 178  | ■ 1.000 | $\text{null}(A) = \text{null}(B)$                               | $\text{null}(A) = \text{null}(B)$                               | $\text{null}(A) = \text{null}(B)$ .                                                                                    |
| johnston-linear-matrix-algebra-F01827 | 178  | ■ 1.000 | $\text{range}(A^T) = \text{range}(B^T)$                         | $\text{range}(A^T) = \text{range}(B^T)$                         | $\text{range}(A^T) = \text{range}(B^T)$ .                                                                              |
| johnston-linear-matrix-algebra-F01828 | 178  | ■ 0.981 | $\text{nullity}(AB) \leq \text{nullity}(A) + \text{nullity}(B)$ | $\text{nullity}(AB) \leq \text{nullity}(A) + \text{nullity}(B)$ | (a) Show that $\text{nullity}(AB) \leq \text{nullity}(A) + \text{nullity}(B)$ .                                        |
| johnston-linear-matrix-algebra-F01829 | 178  | ■ 1.000 | $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$      | $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$      | (b) Show that $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$ .                                             |
| johnston-linear-matrix-algebra-F01830 | 178  | ■ 1.000 | $A = \mathbf{vw}^T$                                             | $A = \mathbf{vw}^T$                                             | such that $A = \mathbf{vw}^T$ .                                                                                        |
| johnston-linear-matrix-algebra-F01831 | 178  | ■ 1.000 | $m \geq n$                                                      | $m \geq n$                                                      | (a) Suppose $m \geq n$ and $\text{rank}(A) = n$ . Show that there                                                      |
| johnston-linear-matrix-algebra-F01832 | 178  | ■ 1.000 | $B \in \mathcal{M}_{n, m}$                                      | $B \in \mathcal{M}_{n, m}$                                      | exists a matrix $B \in \mathcal{M}_{n, m}$ such that $BA = I_n$ .                                                      |
| johnston-linear-matrix-algebra-F01833 | 178  | ■ 1.000 | $BA = I_n$                                                      | $BA = I_n$                                                      | exists a matrix $B \in \mathcal{M}_{n, m}$ such that $BA = I_n$ .                                                      |
| johnston-linear-matrix-algebra-F01834 | 178  | ■ 0.999 | $n \geq m$                                                      | $n \geq m$                                                      | (b) Suppose $n \geq m$ and $\text{rank}(A) = m$ . Show that there                                                      |
| johnston-linear-matrix-algebra-F01835 | 178  | ■ 0.999 | $\text{rank}(A) = m$                                            | $\text{rank}(A) = m$                                            | (b) Suppose $n \geq m$ and $\text{rank}(A) = m$ . Show that there                                                      |
| johnston-linear-matrix-algebra-F01836 | 178  | ■ 1.000 | $C \in \mathcal{M}_{n, m}$                                      | $C \in \mathcal{M}_{n, m}$                                      | exists a matrix $C \in \mathcal{M}_{n, m}$ such that $AC = I_m$ .                                                      |
| johnston-linear-matrix-algebra-F01837 | 178  | ■ 1.000 | $AC = I_m$                                                      | $AC = I_m$                                                      | exists a matrix $C \in \mathcal{M}_{n, m}$ such that $AC = I_m$ .                                                      |
| johnston-linear-matrix-algebra-F01838 | 178  | ■ 1.000 | $R \in \mathcal{M}_{m, n}$                                      | $R \in \mathcal{M}_{m, n}$                                      | concept of the reduced row echelon form $R \in \mathcal{M}_{m, n}$ of a matrix $A \in \mathcal{M}_{m, n}$ ,            |
| johnston-linear-matrix-algebra-F01839 | 178  | ■ 1.000 | $A^{-1} \in \mathcal{M}_n$                                      | $A^{-1} \in \mathcal{M}_n$                                      | $A^{-1} \in \mathcal{M}_n$ with the property that $AA^{-1} = A^{-1}A = I$ , and we think of $A^{-1}$ as a              |
| johnston-linear-matrix-algebra-F01840 | 179  | ■ 0.896 | $(\mathbf{x} = \mathbf{0})$                                     | $(\mathbf{x} = \mathbf{0})$                                     | g) The linear system $A\mathbf{x} = \mathbf{0}$ has a unique solution ( $\mathbf{x} = \mathbf{0}$ ).                   |

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| johnston-linear-matrix-algebra-F01841 | 179  | 1.000 | $P \in \mathcal{M}_m$                                                                                | $P \in \mathcal{M}_m$                                                                      | e) There is an invertible matrix $P \in \mathcal{M}_m$ such that $A = PB$ .                                                                                                                   |
| johnston-linear-matrix-algebra-F01842 | 179  | 1.000 | $A=P \ B$                                                                                            | $A = PB$                                                                                   | e) There is an invertible matrix $P \in \mathcal{M}_m$ such that $A = PB$ .                                                                                                                   |
| johnston-linear-matrix-algebra-F01843 | 179  | 1.000 | $A=P \ R$                                                                                            | $A = PR$                                                                                   | exists an invertible matrix $P \in \mathcal{M}_m$ such that $A = PR$ .                                                                                                                        |
| johnston-linear-matrix-algebra-F01844 | 180  | 0.984 | $A=\left[\begin{array}{ll}1 & 1 \\ 1 & 1\end{array}\right]$                                          | $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$                                         | *(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$                                                                                |
| johnston-linear-matrix-algebra-F01845 | 180  | 0.984 | $B=\left[\begin{array}{ll}1 & 1 \\ 1 & 2\end{array}\right]$                                          | $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$                                         | *(a) $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$                                                                                |
| johnston-linear-matrix-algebra-F01846 | 180  | 1.000 | $A=\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 2 & 3\end{array}\right]$                                 | $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$                                 | (b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$                                                                 |
| johnston-linear-matrix-algebra-F01847 | 180  | 1.000 | $B=\left[\begin{array}{lll}0 & 1 & 2 \\ 1 & 1 & 1\end{array}\right]$                                 | $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$                                 | (b) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$                                                                 |
| johnston-linear-matrix-algebra-F01848 | 180  | 0.993 | $A=\left[\begin{array}{ccc}1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0\end{array}\right]$                        | $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$                   | *(c) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$                                     |
| johnston-linear-matrix-algebra-F01849 | 180  | 0.993 | $B=\left[\begin{array}{ccc}3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1\end{array}\right]$                    | $B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$                    | *(c) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 3 & 1 \end{bmatrix}$                                     |
| johnston-linear-matrix-algebra-F01850 | 180  | 1.000 | $A=\left[\begin{array}{ccc}1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4\end{array}\right]$                  | $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$                  | (d) $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$                                   |
| johnston-linear-matrix-algebra-F01851 | 180  | 1.000 | $B=\left[\begin{array}{ccc}1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1\end{array}\right]$                  | $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$                  | (d) $A = \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 2 \\ 2 & -1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 2 & 0 \\ 2 & 4 & -1 \end{bmatrix}$                                   |
| johnston-linear-matrix-algebra-F01852 | 180  | 0.946 | $A=\left[\begin{array}{ccc}1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3\end{array}\right]$ | $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$ | *(e) $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F01853 | 180  | 0.946 | $B=\left[\begin{array}{ccc}2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0\end{array}\right]$  | $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$  | *(e) $A = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & -3 \\ 1 & -2 & -1 \\ 1 & 1 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 & 3 \\ 0 & -1 & 2 \\ -1 & -1 & 1 \\ -1 & 0 & 0 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F01854 | 180  | 1.000 | $A=\left[\begin{array}{llll}1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0\end{array}\right]$       | $A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$        | (f) $A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$               |
| johnston-linear-matrix-algebra-F01855 | 180  | 1.000 | $B=\left[\begin{array}{llll}1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1\end{array}\right]$       | $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$        | (f) $A = \begin{bmatrix} 1 & 1 & 4 & 3 \\ 1 & 2 & 3 & 1 \\ 3 & 3 & 3 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 2 \\ 4 & 2 & 0 & 1 \end{bmatrix}$               |
| johnston-linear-matrix-algebra-F01856 | 180  | 1.000 | $R \ \mathbf{x}=\mathbf{b}$                                                                          | $R\mathbf{x} = \mathbf{b}$                                                                 | linear systems $A\mathbf{x} = \mathbf{b}$ and $R\mathbf{x} = \mathbf{b}$ have the same                                                                                                        |

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|---------------------------------------|------|-------|---------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F01857 | 180  | 1.000 | $A \mathbf{v}_j = B \mathbf{v}_j$                       | $A \mathbf{v}_j = B \mathbf{v}_j$                       | basis of $\mathbb{R}^n$ . Show that $A = B$ if and only if $A \mathbf{v}_j = B \mathbf{v}_j$ for   |
| johnston-linear-matrix-algebra-F01858 | 180  | 0.961 | $4 \times 4$                                            | $4 \times 4$                                            | <b>2.5.4</b> Suppose $A$ is a $4 \times 4$ matrix, $B$ is a $4 \times 3$ matrix, and               |
| johnston-linear-matrix-algebra-F01859 | 180  | 0.867 | $A = B C$                                               | $A = B C$                                               | $C$ is a $3 \times 4$ matrix such that $A = B C$ . Prove that $A$ is not                           |
| johnston-linear-matrix-algebra-F01860 | 180  | 1.000 | $\operatorname{null}(A) = \operatorname{null}(A^T A)$   | $\operatorname{null}(A) = \operatorname{null}(A^T A)$   | (a) Show that $\operatorname{null}(A) = \operatorname{null}(A^T A)$ .                              |
| johnston-linear-matrix-algebra-F01861 | 180  | 1.000 | $\mathbf{x} \in \operatorname{null}(A^T A)$             | $\mathbf{x} \in \operatorname{null}(A^T A)$             | [Hint: If $\mathbf{x} \in \operatorname{null}(A^T A)$ then compute $\ A \mathbf{x}\ ^2$ .]         |
| johnston-linear-matrix-algebra-F01862 | 180  | 1.000 | $\ A \mathbf{x}\ ^2$                                    | $\ A \mathbf{x}\ ^2$                                    | [Hint: If $\mathbf{x} \in \operatorname{null}(A^T A)$ then compute $\ A \mathbf{x}\ ^2$ .]         |
| johnston-linear-matrix-algebra-F01863 | 180  | 1.000 | $\operatorname{rank}(A^T A) = \operatorname{rank}(A)$   | $\operatorname{rank}(A^T A) = \operatorname{rank}(A)$   | (b) Show that $\operatorname{rank}(A^T A) = \operatorname{rank}(A)$ .                              |
| johnston-linear-matrix-algebra-F01864 | 180  | 0.975 | $\operatorname{range}(A) = \operatorname{range}(A A^T)$ | $\operatorname{range}(A) = \operatorname{range}(A A^T)$ | (b) Show that $\operatorname{range}(A) = \operatorname{range}(A A^T)$ .                            |
| johnston-linear-matrix-algebra-F01865 | 180  | 0.998 | $\operatorname{rank}(A A^T) = \operatorname{rank}(A)$   | $\operatorname{rank}(A A^T) = \operatorname{rank}(A)$   | (d) Show that $\operatorname{rank}(A A^T) = \operatorname{rank}(A)$ .                              |
| johnston-linear-matrix-algebra-F01866 | 180  | 0.994 | $\operatorname{range}(A) = \operatorname{range}(A^T A)$ | $\operatorname{range}(A) = \operatorname{range}(A^T A)$ | essarily the case that $\operatorname{range}(A) = \operatorname{range}(A^T A)$ or                  |
| johnston-linear-matrix-algebra-F01867 | 180  | 1.000 | $\operatorname{null}(A) = \operatorname{null}(A A^T)$   | $\operatorname{null}(A) = \operatorname{null}(A A^T)$   | $\operatorname{null}(A) = \operatorname{null}(A A^T)$ .                                            |
| johnston-linear-matrix-algebra-F01868 | 181  | 1.000 | $a(b+c) = a b + a c$                                    | $a(b+c) = ab + ac$                                      | $a(b+c) = ab + ac$ for all $a, b$ , and $c$ in the field).                                         |
| johnston-linear-matrix-algebra-F01869 | 181  | 1.000 | $a + i b$                                               | $a + ib$                                                | numbers (i.e., numbers of the form $a + ib$ , where $a, b \in \mathbb{R}$ and $i^2 = -1$ , which   |
| johnston-linear-matrix-algebra-F01870 | 181  | 1.000 | $i^2 = -1$                                              | $i^2 = -1$                                              | numbers (i.e., numbers of the form $a + ib$ , where $a, b \in \mathbb{R}$ and $i^2 = -1$ , which   |
| johnston-linear-matrix-algebra-F01871 | 181  | 0.996 | $\mathbb{Z}_2$                                          | $\mathbb{Z}_2$                                          | For now, we work with the set $\mathbb{Z}_2$ of numbers under arithmetic mod 2. That               |
| johnston-linear-matrix-algebra-F01872 | 181  | 1.000 | $\{0, 1\}$                                              | $\{0, 1\}$                                              | is, $\mathbb{Z}_2$ is simply the set $\{0, 1\}$ , but with the understanding that addition and     |
| johnston-linear-matrix-algebra-F01873 | 181  | 0.978 | $1 + 1 = 0$                                             | $1 + 1 = 0$                                             | tion that $1 + 1 = 0$ instead of $1 + 1 = 2$ (since there is no “2” in $\mathbb{Z}_2$ ). Phrased   |
| johnston-linear-matrix-algebra-F01874 | 181  | 0.978 | $1 + 1 = 2$                                             | $1 + 1 = 2$                                             | tion that $1 + 1 = 0$ instead of $1 + 1 = 2$ (since there is no “2” in $\mathbb{Z}_2$ ). Phrased   |
| johnston-linear-matrix-algebra-F01875 | 181  | 1.000 | $\mathbb{R}$                                            | $\mathbb{R}$                                            | linear systems with entries from $\mathbb{R}$ .                                                    |
| johnston-linear-matrix-algebra-F01876 | 182  | 0.996 | $1 = 0 - 1$                                             | $1 = 0 - 1$                                             | that $1 = 0 - 1$ , which                                                                           |
| johnston-linear-matrix-algebra-F01877 | 182  | 1.000 | $\mathbb{Z}_2^n$                                        | $\mathbb{Z}_2^n$                                        | The notation $\mathbb{Z}_2^n$                                                                      |
| johnston-linear-matrix-algebra-F01878 | 182  | 1.000 | $(x_1, x_2, x_3, x_4, x_5) =$                           | $(x_1, x_2, x_3, x_4, x_5) =$                           | It follows that the unique solution of this linear system is $(x_1, x_2, x_3, x_4, x_5) =$         |
| johnston-linear-matrix-algebra-F01879 | 182  | 1.000 | $2^n$                                                   | $2^n$                                                   | solutions of a linear system over $\mathbb{Z}_2$ in $n$ variables are the $2^n$ vectors consisting |
| johnston-linear-matrix-algebra-F01880 | 183  | 1.000 | $0 R_j$                                                 | $0 R_j$                                                 | operation, since $0 R_j$                                                                           |
| johnston-linear-matrix-algebra-F01881 | 183  | 1.000 | $1 R_j$                                                 | $1 R_j$                                                 | operation and $1 R_j$                                                                              |
| johnston-linear-matrix-algebra-F01882 | 183  | 0.980 | $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$                       | $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$                       | particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$ , and $x_1 = x_3$ (recall that $x_3 = -x_3$          |
| johnston-linear-matrix-algebra-F01883 | 183  | 0.980 | $x_1 = x_3$                                             | $x_1 = x_3$                                             | particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$ , and $x_1 = x_3$ (recall that $x_3 = -x_3$          |
| johnston-linear-matrix-algebra-F01884 | 183  | 0.980 | $x_3 = -x_3$                                            | $x_3 = -x_3$                                            | particular, $x_5 = 1, x_4 = 1, x_2 = 1 - x_3$ , and $x_1 = x_3$ (recall that $x_3 = -x_3$          |

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|-----------------------------------------|------|---------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-augmented-F01885 | 183  | ■ 1.000 | $x_{\{3\}}=0$                                                                           | $x_3 = 0$                                                               | $x_3 = 0$ or $x_3 = 1$ into the equations for $x_1$ and $x_2$ that we wrote down above.                                                                    |
| johnston-linear-matrix-augmented-F01886 | 183  | ■ 1.000 | $x_{\{3\}}=1$                                                                           | $x_3 = 1$                                                               | $x_3 = 0$ or $x_3 = 1$ into the equations for $x_1$ and $x_2$ that we wrote down above.                                                                    |
| johnston-linear-matrix-augmented-F01887 | 183  | ■ 1.000 | $2^{\{k\}}$                                                                             | $2^k$                                                                   | $2^k$ solutions, where $k$ is a non-negative integer (equal to the                                                                                         |
| johnston-linear-matrix-augmented-F01888 | 184  | ■ 0.999 | $\mathbf{v} \in \mathbb{Z}_2^{\{16\}}$                                                  | $\mathbf{v} \in \mathbb{Z}_2^{16}$                                      | <b>Figure 2.27:</b> We use a vector $\mathbf{v} \in \mathbb{Z}_2^{16}$ to encode the states of the $4 \times 4$ “Lights Out”                               |
| johnston-linear-matrix-augmented-F01889 | 184  | ■ 1.000 | $\mathbf{v}_{\{\mathrm{s}\}}, \mathbf{v}_{\{\mathrm{e}\}} \in \mathbb{Z}_2^{\{16\}}$    | $\mathbf{v}_s, \mathbf{v}_e \in \mathbb{Z}_2^{16}$                      | More generally, we use $\mathbf{v}_s, \mathbf{v}_e \in \mathbb{Z}_2^{16}$ to represent the vectors that encode the                                         |
| johnston-linear-matrix-augmented-F01890 | 184  | ■ 0.994 | $\mathbf{a}_{\{1\}}, \mathbf{a}_{\{2\}}, \ldots, \mathbf{a}_{\{16\}}$                   | $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$                    | If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$ denote the 16 different vectors that are potentially added to $\mathbf{v}_s$                       |
| johnston-linear-matrix-augmented-F01891 | 184  | ■ 0.994 | $\mathbf{v}_{\{\mathrm{s}\}}$                                                           | $\mathbf{v}_s$                                                          | If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_{16}$ denote the 16 different vectors that are potentially added to $\mathbf{v}_s$                       |
| johnston-linear-matrix-augmented-F01892 | 184  | ■ 1.000 | $\mathbf{x} \in \mathbb{Z}_2^{\{16\}}$                                                  | $\mathbf{x} \in \mathbb{Z}_2^{16}$                                      | when we press the 16 different buttons, and we let $\mathbf{x} \in \mathbb{Z}_2^{16}$ denote the vector                                                    |
| johnston-linear-matrix-augmented-F01893 | 184  | ■ 1.000 | $x_{\{j\}}=1$                                                                           | $x_j = 1$                                                               | that encodes which buttons we press (i.e., $x_j = 1$ if we press the $j$ -th button                                                                        |
| johnston-linear-matrix-augmented-F01894 | 185  | ■ 0.937 | $x_{\{j\}}=0$                                                                           | $x_j = 0$                                                               | and $x_j = 0$ otherwise) then our goal is to find $x_1, x_2, \dots, x_{16}$ so that                                                                        |
| johnston-linear-matrix-augmented-F01895 | 185  | ■ 0.937 | $x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{16\}}$                                              | $x_1, x_2, \dots, x_{16}$                                               | and $x_j = 0$ otherwise) then our goal is to find $x_1, x_2, \dots, x_{16}$ so that                                                                        |
| johnston-linear-matrix-augmented-F01896 | 185  | ■ 0.919 | $A \mathbf{x} = \mathbf{v}_{\{\mathrm{e}\}} - \mathbf{v}_{\{\mathrm{s}\}}$              | $A\mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$                             | matrix equation $A\mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$ , where $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_{16}]$ , which we can |
| johnston-linear-matrix-augmented-F01897 | 185  | ■ 0.919 | $A = [\mathbf{a}_{\{1\}} \mid \mathbf{a}_{\{2\}} \mid \cdots \mid \mathbf{a}_{\{16\}}]$ | $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_{16}]$ | matrix equation $A\mathbf{x} = \mathbf{v}_e - \mathbf{v}_s$ , where $A = [\mathbf{a}_1 \mid \mathbf{a}_2 \mid \cdots \mid \mathbf{a}_{16}]$ , which we can |
| johnston-linear-matrix-augmented-F01898 | 185  | ■ 0.973 | $\mathbf{v}_{\{\mathrm{e}\}} - \mathbf{v}_{\{\mathrm{s}\}} \in \mathbb{Z}_2^{\{16\}}$   | $\mathbf{v}_e - \mathbf{v}_s \in \mathbb{Z}_2^{16}$                     | tion (2.A.1) and $\mathbf{v}_e - \mathbf{v}_s \in \mathbb{Z}_2^{16}$ is the vector with every entry equal to 1.                                            |
| johnston-linear-matrix-augmented-F01899 | 186  | ■ 0.999 | $[A \mid \mathbf{v}_{\{\mathrm{e}\}} - \mathbf{v}_{\{\mathrm{s}\}}]$                    | $[A \mid \mathbf{v}_e - \mathbf{v}_s]$                                  | The reduced row echelon form of the augmented matrix $[A \mid \mathbf{v}_e - \mathbf{v}_s]$ is                                                             |
| johnston-linear-matrix-augmented-F01900 | 186  | ■ 1.000 | $2^{\{4\}}=16$                                                                          | $2^4 = 16$                                                              | which tells us that this linear system has 4 free variables and thus $2^4 = 16$                                                                            |
| johnston-linear-matrix-augmented-F01901 | 186  | ■ 1.000 | $x_{\{13\}}, x_{\{14\}}, x_{\{15\}}$                                                    | $x_{13}, x_{14}, x_{15}$                                                | the free variables $x_{13}, x_{14}, x_{15}$ , and $x_{16}$ . The simplest solution arises from                                                             |
| johnston-linear-matrix-augmented-F01902 | 186  | ■ 1.000 | $x_{\{16\}}$                                                                            | $x_{16}$                                                                | the free variables $x_{13}, x_{14}, x_{15}$ , and $x_{16}$ . The simplest solution arises from                                                             |
| johnston-linear-matrix-augmented-F01903 | 186  | ■ 0.992 | $x_{\{13\}}=0, x_{\{14\}}=1, x_{\{15\}}=0, x_{\{16\}}=0$                                | $x_{13} = 0, x_{14} = 1, x_{15} = 0, x_{16} = 0$                        | choosing $x_{13} = 0, x_{14} = 1, x_{15} = 0, x_{16} = 0$ , which gives                                                                                    |
| johnston-linear-matrix-augmented-F01904 | 187  | ■ 0.992 | $2^{\{16\}}=65536$                                                                      | $2^{16} = 65536$                                                        | of that grid cannot be turned into the all-“on” configuration (of the $2^{16} = 65536$                                                                     |
| johnston-linear-matrix-augmented-F01905 | 187  | ■ 1.000 | $65536 / 16=4096$                                                                       | $65536/16 = 4096$                                                       | starting configurations, only $65536/16 = 4096$ are solvable). For example, of                                                                             |
| johnston-linear-matrix-augmented-F01906 | 188  | ■ 0.994 | $\mathbf{v}_{\{\mathrm{s}\}}=(0,0,0,0,1,1,1,1,0)$                                       | $\mathbf{v}_s = (0, 0, 0, 0, 1, 1, 1, 1, 0)$                            | We represent the starting state via the vector $\mathbf{v}_s = (0, 0, 0, 0, 1, 1, 1, 1, 0)$                                                                |
| johnston-linear-matrix-augmented-F01907 | 188  | ■ 0.994 | $\mathbf{v}_{\{\mathrm{e}\}}=(1,1,1,1,1,1,1,1,1)$                                       | $\mathbf{v}_e = (1, 1, 1, 1, 1, 1, 1, 1, 1)$                            | and the target state via the vector $\mathbf{v}_e = (1, 1, 1, 1, 1, 1, 1, 1, 1)$ . If we let                                                               |
| johnston-linear-matrix-augmented-F01908 | 188  | ■ 0.916 | $9 \times 9$                                                                            | $9 \times 9$                                                            | A be the $9 \times 9$ binary matrix whose $j$ -th column (for each $1 \leq j \leq 9$ )                                                                     |
| johnston-linear-matrix-augmented-F01909 | 188  | ■ 0.916 | $1 \leq j \leq 9$                                                                       | $1 \leq j \leq 9$                                                       | A be the $9 \times 9$ binary matrix whose $j$ -th column (for each $1 \leq j \leq 9$ )                                                                     |
| johnston-linear-matrix-augmented-F01910 | 188  | ■ 1.000 | $[I \mid \mathbf{x}]$                                                                   | $[I \mid \mathbf{x}]$                                                   | echelon form of this augmented matrix has the form $[I \mid \mathbf{x}]$ , where                                                                           |
| johnston-linear-matrix-augmented-F01911 | 188  | ■ 1.000 | $\mathbf{x}=(1,1,0,0,0,0,1,1,0)$                                                        | $\mathbf{x} = (1, 1, 0, 0, 0, 0, 1, 1, 0)$                              | $\mathbf{x} = (1, 1, 0, 0, 0, 0, 1, 1, 0)$ . It follows that this linear system has a unique                                                               |

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|---------------------------------------|------|---------|----------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F01912 | 189  | ■ 1.000 | $\mathbb{Z}_2^k$                                   | $\mathbb{Z}_2^k$                                   | lives in $\mathbb{Z}_2^k$ , whereas                                                                                          |
| johnston-linear-matrix-algebra-F01913 | 189  | ■ 1.000 | $A \mathbf{x} = \mathbf{1}$                        | $A\mathbf{x} = \mathbf{1}$                         | <i>Proof.</i> Recall that the linear system we want to solve has the form $A\mathbf{x} = \mathbf{1}$ ,                       |
| johnston-linear-matrix-algebra-F01914 | 189  | ■ 0.516 | $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{Z}_2^n$ | $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{Z}_2^n$ | where $\mathbf{1} = (1, 1, \dots, 1) \in \mathbb{Z}_2^n$ is the vector with every entry equal to 1. In other                 |
| johnston-linear-matrix-algebra-F01915 | 189  | ■ 1.000 | $\mathbf{1} \in \text{range}(A)$                   | $\mathbf{1} \in \text{range}(A)$                   | words, we want to show that $\mathbf{1} \in \text{range}(A)$ . Furthermore, $A$ has the following                            |
| johnston-linear-matrix-algebra-F01916 | 189  | ■ 1.000 | $A^T = A$                                          | $A^T = A$                                          | • $A$ is symmetric (i.e., $A^T = A$ ), since button $i$ is a neighbor of button $j$ if                                       |
| johnston-linear-matrix-algebra-F01917 | 189  | ■ 1.000 | $\mathbf{y} \in \text{null}(A)$                    | $\mathbf{y} \in \text{null}(A)$                    | steps. The first step is to show that every $\mathbf{y} \in \text{null}(A)$ (i.e., every $\mathbf{y} \in \mathbb{Z}_2^n$ for |
| johnston-linear-matrix-algebra-F01918 | 189  | ■ 1.000 | $\mathbf{y} \in \mathbb{Z}_2^n$                    | $\mathbf{y} \in \mathbb{Z}_2^n$                    | steps. The first step is to show that every $\mathbf{y} \in \text{null}(A)$ (i.e., every $\mathbf{y} \in \mathbb{Z}_2^n$ for |
| johnston-linear-matrix-algebra-F01919 | 189  | ■ 1.000 | $A \mathbf{y} = \mathbf{0}$                        | $A\mathbf{y} = \mathbf{0}$                         | which $A\mathbf{y} = \mathbf{0}$ ) has an even number of entries equal to 1 (in other words, it                              |
| johnston-linear-matrix-algebra-F01920 | 189  | ■ 1.000 | $y_j = 0$                                          | $y_j = 0$                                          | deleting every row and column $j$ for which $y_j = 0$ . For example, if                                                      |
| johnston-linear-matrix-algebra-F01921 | 189  | ■ 0.808 | $B^T = B$                                          | $B^T = B$                                          | Then $B^T = B$ as well and it also has all of its diagonal entries equal to 1.                                               |
| johnston-linear-matrix-algebra-F01922 | 189  | ■ 1.000 | $B \mathbf{1} = \mathbf{0}$                        | $B\mathbf{1} = \mathbf{0}$                         | Furthermore, the fact that $A\mathbf{y} = \mathbf{0}$ tells us that $B\mathbf{1} = \mathbf{0}$ . However, the entries of     |
| johnston-linear-matrix-algebra-F01923 | 189  | ■ 0.839 | $B \mathbf{1}$                                     | $B\mathbf{1}$                                      | $B\mathbf{1}$ just count the number of 1s in each row of $B$ , so every row of $B$ contains an                               |
| johnston-linear-matrix-algebra-F01924 | 190  | ■ 0.874 | $\mathbf{1}$                                       | $\mathbf{1}$                                       | all of the rows of $R$ results in the all-ones vector $\mathbf{1}$ . It follows that $\mathbf{1} \in \text{range}(A)$ ,      |
| johnston-linear-matrix-algebra-F01925 | 190  | ■ 0.982 | $\mathbb{Z}_3$                                     | $\mathbb{Z}_3$                                     | modulus larger than 2. For example, $\mathbb{Z}_3$ denotes the set $\{0, 1, 2\}$ under arith-                                |
| johnston-linear-matrix-algebra-F01926 | 190  | ■ 0.982 | $\{0, 1, 2\}$                                      | $\{0, 1, 2\}$                                      | modulus larger than 2. For example, $\mathbb{Z}_3$ denotes the set $\{0, 1, 2\}$ under arith-                                |
| johnston-linear-matrix-algebra-F01927 | 190  | ■ 0.999 | $\bmod 3$                                          | $\text{mod}3$                                      | metic mod 3. That is, we perform addition and multiplication in the usual ways,                                              |
| johnston-linear-matrix-algebra-F01928 | 190  | ■ 1.000 | $2+2=4$                                            | $2 + 2 = 4$                                        | of having $2 + 2 = 4$ , we have $2 + 2 = 1$ since the addition “rolls over” back to                                          |
| johnston-linear-matrix-algebra-F01929 | 190  | ■ 1.000 | $2+2=1$                                            | $2 + 2 = 1$                                        | of having $2 + 2 = 4$ , we have $2 + 2 = 1$ since the addition “rolls over” back to                                          |
| johnston-linear-matrix-algebra-F01930 | 190  | ■ 0.919 | $4: 00$                                            | $4 : 00$                                           | it will be 4:00, since $9 + 7 = 16 = 12 + 4$ , and the extra 12 is irrelevant (it                                            |
| johnston-linear-matrix-algebra-F01931 | 190  | ■ 0.919 | $9+7=16=12+4$                                      | $9 + 7 = 16 = 12 + 4$                              | it will be 4:00, since $9 + 7 = 16 = 12 + 4$ , and the extra 12 is irrelevant (it                                            |
| johnston-linear-matrix-algebra-F01932 | 190  | ■ 0.924 | $\bmod -12$                                        | $\text{mod} - 12$                                  | displayed on the clock). For example, in mod-12 arithmetic, $9 + 7 = 4$ (see                                                 |
| johnston-linear-matrix-algebra-F01933 | 190  | ■ 0.924 | $9+7=4$                                            | $9 + 7 = 4$                                        | displayed on the clock). For example, in mod-12 arithmetic, $9 + 7 = 4$ (see                                                 |
| johnston-linear-matrix-algebra-F01934 | 191  | ■ 1.000 | $0-1=2, 0-2=1$                                     | $0 - 1 = 2, 0 - 2 = 1$                             | $0 - 1 = 2, 0 - 2 = 1$ ,                                                                                                     |
| johnston-linear-matrix-algebra-F01935 | 191  | ■ 1.000 | $1-2=2$                                            | $1 - 2 = 2$                                        | $1 - 2 = 2$ , and so on                                                                                                      |
| johnston-linear-matrix-algebra-F01936 | 191  | ■ 1.000 | $\mathbb{Z}_p$                                     | $\mathbb{Z}_p$                                     | systems over $\mathbb{Z}_p$ (i.e., using just the numbers $0, 1, 2, \dots, p - 1$ ) when $p$ is any                          |
| johnston-linear-matrix-algebra-F01937 | 191  | ■ 1.000 | $0, 1, 2, \dots, p-1$                              | $0, 1, 2, \dots, p - 1$                            | systems over $\mathbb{Z}_p$ (i.e., using just the numbers $0, 1, 2, \dots, p - 1$ ) when $p$ is any                          |
| johnston-linear-matrix-algebra-F01938 | 191  | ■ 1.000 | $(w, x, y, z) =$                                   | $(w, x, y, z) =$                                   | It follows that the unique solution of this linear system is $(w, x, y, z) =$                                                |
| johnston-linear-matrix-algebra-F01939 | 192  | ■ 1.000 | $3^4=81$                                           | $3^4 = 81$                                         | striped red ovals), for a total of $3^4 = 81$ cards in the deck. Some examples                                               |

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|---------------------------------------|---------------|---------------|-------------------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------|---------------|-----------------|----------------|---|-----|---------|-------|-----|---|-------|----------|---------|-----|---|--------|------|--------|-------|
| johnston-linear-matrix-aligned-F01940 | 192           | 1.000         | $\mathbb{Z}_3^4$                                      | $\mathbb{Z}_3^4$                                      | the 81 cards as a vector in $\mathbb{Z}_3^4$ by encoding each of the properties as one                                                                                                                                                                                                                                                                                                                   |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01941 | 192           | 1.000         | $\mathbf{v}=(c, p, d, n)$                             | $\mathbf{v}=(c,p,d,n)$                                | Specifically, we can write the cards as vectors of the form $\mathbf{v}=(c,p,d,n)$ ,                                                                                                                                                                                                                                                                                                                     |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01942 | 192           | 1.000         | $c, p, d$                                             | $c,p,d$                                               | where $c, p, d$ , and $n$ encode the color, shape, shading, and number of                                                                                                                                                                                                                                                                                                                                |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01943 | 193           | 0.516         | $(d)$                                                 | $(d)$                                                 | <table><tr><th>Value</th><th>Color (<math>c</math>)</th><th>Shape (<math>p</math>)</th><th>Shading (<math>d</math>)</th><th>Number (<math>n</math>)</th></tr><tr><td>0</td><td>red</td><td>diamond</td><td>solid</td><td>one</td></tr><tr><td>1</td><td>green</td><td>squiggle</td><td>striped</td><td>two</td></tr><tr><td>2</td><td>purple</td><td>oval</td><td>hollow</td><td>three</td></tr></table> | Value | Color ( $c$ ) | Shape ( $p$ ) | Shading ( $d$ ) | Number ( $n$ ) | 0 | red | diamond | solid | one | 1 | green | squiggle | striped | two | 2 | purple | oval | hollow | three |
| Value                                 | Color ( $c$ ) | Shape ( $p$ ) | Shading ( $d$ )                                       | Number ( $n$ )                                        |                                                                                                                                                                                                                                                                                                                                                                                                          |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| 0                                     | red           | diamond       | solid                                                 | one                                                   |                                                                                                                                                                                                                                                                                                                                                                                                          |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| 1                                     | green         | squiggle      | striped                                               | two                                                   |                                                                                                                                                                                                                                                                                                                                                                                                          |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| 2                                     | purple        | oval          | hollow                                                | three                                                 |                                                                                                                                                                                                                                                                                                                                                                                                          |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01944 | 193           | 1.000         | $\mathbf{v}=(c, p, d, n)=(0,2,1,0)$                   | $\mathbf{v}=(c,p,d,n)=(0,2,1,0)$                      | by the vector $\mathbf{v}=(c,p,d,n)=(0,2,1,0)$ .                                                                                                                                                                                                                                                                                                                                                         |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01945 | 193           | 1.000         | $\mathbf{v}_1, \mathbf{v}_2$                          | $\mathbf{v}_1, \mathbf{v}_2$                          | to the condition that their associated vectors $\mathbf{v}_1, \mathbf{v}_2$ , and $\mathbf{v}_3$ are such that                                                                                                                                                                                                                                                                                           |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01946 | 193           | 1.000         | $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$   | $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$   | $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$ . To see why this is the case, just notice that the only ways                                                                                                                                                                                                                                                                                        |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01947 | 193           | 1.000         | $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$       | $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$       | set with them. This is because, given $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{Z}_3^4$ , there is a unique                                                                                                                                                                                                                                                                                                |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01948 | 193           | 1.000         | $\mathbf{v}_3 \in \mathbb{Z}_3^4$                     | $\mathbf{v}_3 \in \mathbb{Z}_3^4$                     | $\mathbf{v}_3 \in \mathbb{Z}_3^4$ such that $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$ (in particular, $\mathbf{v}_3=-\mathbf{v}_1-\mathbf{v}_2$ ).                                                                                                                                                                                                                                             |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01949 | 193           | 1.000         | $\mathbf{v}_3=-\mathbf{v}_1-\mathbf{v}_2$             | $\mathbf{v}_3=-\mathbf{v}_1-\mathbf{v}_2$             | $\mathbf{v}_3 \in \mathbb{Z}_3^4$ such that $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$ (in particular, $\mathbf{v}_3=-\mathbf{v}_1-\mathbf{v}_2$ ).                                                                                                                                                                                                                                             |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01950 | 193           | 1.000         | $26 \times 3=78$                                      | $26 \times 3=78$                                      | • If 26 sets (and thus $26 \times 3=78$ cards) are removed from the deck,                                                                                                                                                                                                                                                                                                                                |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01951 | 193           | 1.000         | $\mathbf{v}_1-\mathbf{v}_2=\mathbf{v}_2-\mathbf{v}_3$ | $\mathbf{v}_1-\mathbf{v}_2=\mathbf{v}_2-\mathbf{v}_3$ | • Since $\mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3=\mathbf{0}$ is equivalent to $\mathbf{v}_1-\mathbf{v}_2=\mathbf{v}_2-\mathbf{v}_3$ in $\mathbb{Z}_3^4$ ,                                                                                                                                                                                                                                                 |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01952 | 193           | 0.999         | $(x, 2, y, x)$                                        | $(x,2,y,x)$                                           | the form $(x,2,y,x)$ is displayed below:                                                                                                                                                                                                                                                                                                                                                                 |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01953 | 194           | 1.000         | $\mathbb{Z}_5$                                        | $\mathbb{Z}_5$                                        | exist in $\mathbb{Z}_5$ —the only                                                                                                                                                                                                                                                                                                                                                                        |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01954 | 194           | 1.000         | $2x=1$                                                | $2x=1$                                                | in this case we run into issues like the equation $2x=1$ not having a solution.                                                                                                                                                                                                                                                                                                                          |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01955 | 194           | 0.982         | $\{0,1,2,\ldots,p-1\}$                                | $\{0,1,2,\ldots,p-1\}$                                | of $\{0,1,2,\ldots,p-1\}$ , so in particular we can always “divide” by any scalar—                                                                                                                                                                                                                                                                                                                       |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01956 | 194           | 1.000         | $y \in \mathbb{Z}_p$                                  | $y \in \mathbb{Z}_p$                                  | given any $y \in \mathbb{Z}_p$ , we can find an $x \in \mathbb{Z}_p$ such that $xy=1$ . For example, the                                                                                                                                                                                                                                                                                                 |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01957 | 194           | 1.000         | $x \in \mathbb{Z}_p$                                  | $x \in \mathbb{Z}_p$                                  | given any $y \in \mathbb{Z}_p$ , we can find an $x \in \mathbb{Z}_p$ such that $xy=1$ . For example, the                                                                                                                                                                                                                                                                                                 |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01958 | 194           | 1.000         | $xy=1$                                                | $xy=1$                                                | given any $y \in \mathbb{Z}_p$ , we can find an $x \in \mathbb{Z}_p$ such that $xy=1$ . For example, the                                                                                                                                                                                                                                                                                                 |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01959 | 194           | 1.000         | $2x=3$                                                | $2x=3$                                                | For example, to solve the equation $2x=3$ in $\mathbb{Z}_5$ , instead of dividing by 2 we                                                                                                                                                                                                                                                                                                                |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01960 | 195           | 0.982         | $2 \cdot 3=1$                                         | $2 \cdot 3=1$                                         | multiply by 3, since the $\mathbb{Z}_5$ multiplication table above shows us that $2 \cdot 3=1$ .                                                                                                                                                                                                                                                                                                         |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01961 | 195           | 1.000         | $x=4$                                                 | $x=4$                                                 | Multiplying the equation $2x=3$ through by 3 then gives us $x=4$ .                                                                                                                                                                                                                                                                                                                                       |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01962 | 195           | 0.998         | $w, x$                                                | $w,x$                                                 | It follows that $y$ is a free variable and $w, x$ , and $z$ are leading variables                                                                                                                                                                                                                                                                                                                        |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01963 | 195           | 1.000         | $z=2, x=1-y$                                          | $z=2, x=1-y$                                          | satisfying $z=2, x=1-y$ , and $w=3-2y$ . By letting $y$ range over the five                                                                                                                                                                                                                                                                                                                              |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |
| johnston-linear-matrix-aligned-F01964 | 195           | 1.000         | $w=3-2y$                                              | $w=3-2y$                                              | satisfying $z=2, x=1-y$ , and $w=3-2y$ . By letting $y$ range over the five                                                                                                                                                                                                                                                                                                                              |       |               |               |                 |                |   |     |         |       |     |   |       |          |         |     |   |        |      |        |       |

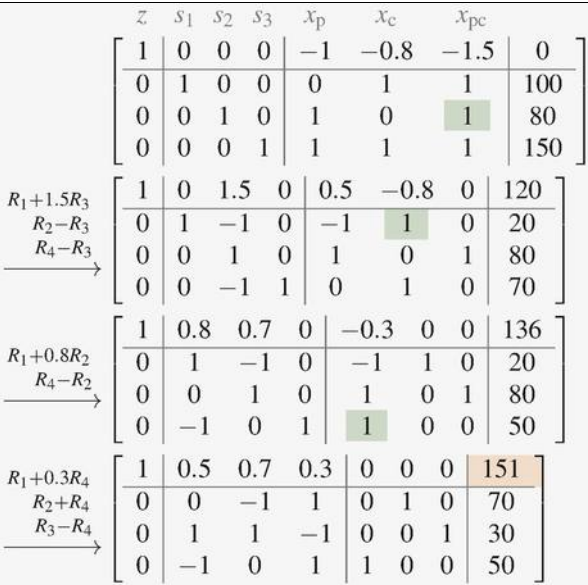
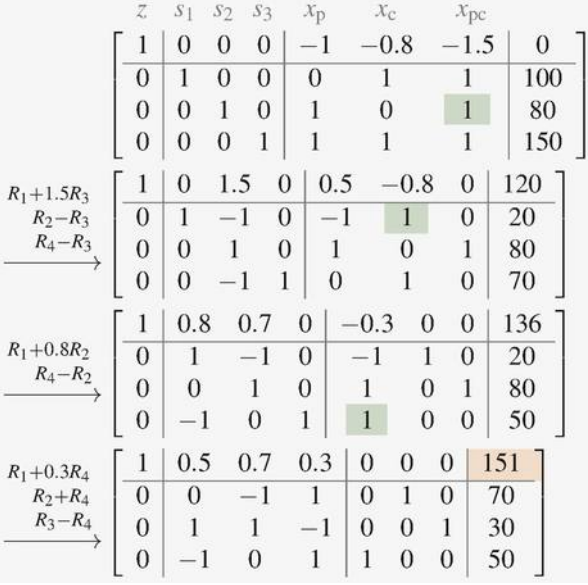
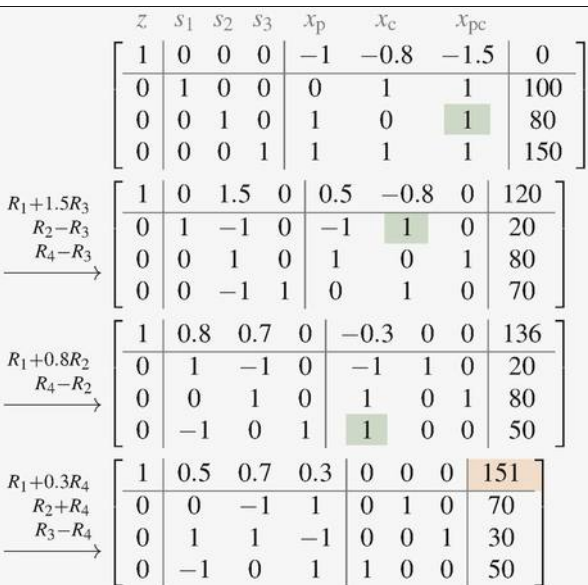
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| johnston-linear-matrix-algebra-F01965 | 195  | ■ 1.000 | $p=2$                                                                               | $p = 2$                                                                             | We already answered this question when $p = 2$ : recall that, in this case, the                                                 |
| johnston-linear-matrix-algebra-F01966 | 195  | ■ 1.000 | $p^{\{k\}}$                                                                         | $p^k$                                                                               | solutions or exactly $p^k$ solutions, where $k$ is a non-negative integer (equal                                                |
| johnston-linear-matrix-algebra-F01967 | 196  | ■ 0.863 | $v, w, x, y, z$                                                                     | $v, w, x, y, z$                                                                     | <b>*2.A.7</b> Find integers $v, w, x, y, z$ (or show that none exist)                                                           |
| johnston-linear-matrix-algebra-F01968 | 197  | ■ 0.999 | $k \times k$                                                                        | $k \times k$                                                                        | <b>**2.A.10</b> Suppose that $B$ is a symmetric $k \times k$ matrix, all                                                        |
| johnston-linear-matrix-algebra-F01969 | 197  | ■ 0.988 | $x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}} \in \mathbb{R}$                            | $x_1, x_2, \dots, x_n \in \mathbb{R}$                                               | linear in the variables. We typically use $x_1, x_2, \dots, x_n \in \mathbb{R}$ as the variables, so                            |
| johnston-linear-matrix-algebra-F01970 | 198  | ■ 1.000 | $\mathbf{x} \geq \mathbf{0}$                                                        | $\mathbf{x} \geq \mathbf{0}$                                                        | and the vector inequalities are meant entrywise (e.g., $\mathbf{x} \geq \mathbf{0}$ means that each                             |
| johnston-linear-matrix-algebra-F01971 | 198  | ■ 0.781 | $A \mathbf{x} \leq \mathbf{b}$                                                      | $A \mathbf{x} \leq \mathbf{b}$                                                      | entry of $\mathbf{x}$ is non-negative, and $A \mathbf{x} \leq \mathbf{b}$ means that each entry of $A \mathbf{x}$ is no larger  |
| johnston-linear-matrix-algebra-F01972 | 198  | ■ 1.000 | $A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ | $A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ | in the following form, where $A \in \mathcal{M}_{m,n}, \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and |
| johnston-linear-matrix-algebra-F01973 | 198  | ■ 0.988 | $\mathbf{c} \cdot \mathbf{x}$                                                       | $\mathbf{c} \cdot \mathbf{x}$                                                       | or minimized (i.e., $\mathbf{c} \cdot \mathbf{x}$ ), and the <b>optimal value</b> is the maximal or minimal                     |
| johnston-linear-matrix-algebra-F01974 | 198  | ■ 1.000 | $-(\mathbf{c} \cdot \mathbf{x})$                                                    | $-(\mathbf{c} \cdot \mathbf{x})$                                                    | minimizing $\mathbf{c} \cdot \mathbf{x}$ is the same as maximizing $-(\mathbf{c} \cdot \mathbf{x})$ and then multiplying the    |
| johnston-linear-matrix-algebra-F01975 | 199  | ■ 0.699 | $A, \mathbf{b}$                                                                     | $A, \mathbf{b}$                                                                     | If we wanted to, we could then specify exactly what $A, \mathbf{b}$ , and $\mathbf{c}$ are in this                              |
| johnston-linear-matrix-algebra-F01976 | 199  | ■ 0.699 | $\mathbf{c}$                                                                        | $\mathbf{c}$                                                                        | If we wanted to, we could then specify exactly what $A, \mathbf{b}$ , and $\mathbf{c}$ are in this                              |
| johnston-linear-matrix-algebra-F01977 | 199  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(1,2)$                                           | $(x_1, x_2) = (1, 2)$                                                               | $(x_1, x_2) = (1, 2)$ , but the original minimization problem has optimal value                                                 |
| johnston-linear-matrix-algebra-F01978 | 199  | ■ 1.000 | $-x_{\{1\}}-2 x_{\{2\}}=-5$                                                         | $-x_1 - 2x_2 = -5$                                                                  | $-x_1 - 2x_2 = -5$ , whereas its standard form has optimal value $x_1 + 2x_2 = 5$ .                                             |
| johnston-linear-matrix-algebra-F01979 | 199  | ■ 1.000 | $x_{\{1\}}+2 x_{\{2\}}=5$                                                           | $x_1 + 2x_2 = 5$                                                                    | $-x_1 - 2x_2 = -5$ , whereas its standard form has optimal value $x_1 + 2x_2 = 5$ .                                             |
| johnston-linear-matrix-algebra-F01980 | 199  | ■ 1.000 | $\geq$                                                                              | $\geq$                                                                              | Equality constraints and “ $\geq$ ” constraints can both be converted into “ $\leq$ ” con-                                      |
| johnston-linear-matrix-algebra-F01981 | 199  | ■ 1.000 | $\leq$                                                                              | $\leq$                                                                              | Equality constraints and “ $\geq$ ” constraints can both be converted into “ $\leq$ ” con-                                      |
| johnston-linear-matrix-algebra-F01982 | 199  | ■ 1.000 | $x=a$                                                                               | $x = a$                                                                             | equality constraints, recall that $x = a$ is equivalent to the pair of inequalities                                             |
| johnston-linear-matrix-algebra-F01983 | 199  | ■ 1.000 | $x \leq a$                                                                          | $x \leq a$                                                                          | $x \leq a$ and $x \geq a$ , which in turn is equivalent to the pair of inequalities $x \leq a$ and                              |
| johnston-linear-matrix-algebra-F01984 | 199  | ■ 1.000 | $x \geq a$                                                                          | $x \geq a$                                                                          | $x \leq a$ and $x \geq a$ , which in turn is equivalent to the pair of inequalities $x \leq a$ and                              |
| johnston-linear-matrix-algebra-F01985 | 199  | ■ 1.000 | $-x \leq -a$                                                                        | $-x \leq -a$                                                                        | $-x \leq -a$ , so one equality constraint can be converted into two “ $\leq$ ” constraints.                                     |
| johnston-linear-matrix-algebra-F01986 | 199  | ■ 0.936 | $3 x_{\{1\}}-4 x_{\{2\}} \geq 1$                                                    | $3x_1 - 4x_2 \geq 1$                                                                | We multiply the constraint $3x_1 - 4x_2 \geq 1$ by $-1$ to obtain the equivalent                                                |
| johnston-linear-matrix-algebra-F01987 | 199  | ■ 1.000 | $-3 x_{\{1\}}+4 x_{\{2\}} \leq-1$                                                   | $-3x_1 + 4x_2 \leq -1$                                                              | constraints $-3x_1 + 4x_2 \leq -1$ , and we split the equality constraint $2x_1 +$                                              |
| johnston-linear-matrix-algebra-F01988 | 199  | ■ 1.000 | $2 x_{\{1\}}+$                                                                      | $2x_1 +$                                                                            | constraints $-3x_1 + 4x_2 \leq -1$ , and we split the equality constraint $2x_1 +$                                              |
| johnston-linear-matrix-algebra-F01989 | 199  | ■ 1.000 | $x_{\{2\}}=5$                                                                       | $x_2 = 5$                                                                           | $x_2 = 5$ into the equivalent pair of inequality constraints $2x_1 + x_2 \leq 5$ and                                            |
| johnston-linear-matrix-algebra-F01990 | 199  | ■ 1.000 | $2 x_{\{1\}}+x_{\{2\}} \leq 5$                                                      | $2x_1 + x_2 \leq 5$                                                                 | $x_2 = 5$ into the equivalent pair of inequality constraints $2x_1 + x_2 \leq 5$ and                                            |
| johnston-linear-matrix-algebra-F01991 | 199  | ■ 1.000 | $-2 x_{\{1\}}-x_{\{2\}} \leq-5$                                                     | $-2x_1 - x_2 \leq -5$                                                               | $-2x_1 - x_2 \leq -5$ . After making these changes, the linear program has the                                                  |
| johnston-linear-matrix-algebra-F01992 | 200  | ■ 0.900 | $x_{\{1\}}+3\left(x_{\{2\}}^{+}-x_{\{2\}}^{-}\right) \leq 4$                        | $x_1 + 3\left(x_2^{+} - x_2^{-}\right) \leq 4$                                      | $x_1 + 3(x_2^{+} - x_2^{-}) \leq 4$ ,                                                                                           |

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| johnston-linear-matrix-algebra-F01993 | 200  | 1.000 | $x_{\{1\}}+3 x_{\{2\}}^{\{+\}}-3 x_{\{2\}}^{\{-\}} \leq 4$                         | $x_1+3x_2^+-3x_2^- \leq 4$                                 | $x_1+3x_2^+-3x_2^- \leq 4$ , so                                                                                       |
| johnston-linear-matrix-algebra-F01994 | 200  | 1.000 | $x=x^{\{+\}}-x^{\{-\}}$                                                            | $x=x^+-x^-$                                                | since every real number $x$ can be written in the form $x=x^+-x^-$ , where $x^+ \geq 0$                               |
| johnston-linear-matrix-algebra-F01995 | 200  | 1.000 | $x^{\{+\}} \geq 0$                                                                 | $x^+ \geq 0$                                               | since every real number $x$ can be written in the form $x=x^+-x^-$ , where $x^+ \geq 0$                               |
| johnston-linear-matrix-algebra-F01996 | 200  | 1.000 | $x^{\{-\}} \geq 0$                                                                 | $x^- \geq 0$                                               | and $x^- \geq 0$ . We illustrate how this fact can be used to write even more linear                                  |
| johnston-linear-matrix-algebra-F01997 | 200  | 1.000 | $x_{\{2\}} \geq 0$                                                                 | $x_2 \geq 0$                                               | that there is no $x_2 \geq 0$ constraint. To rewrite this linear program in an                                        |
| johnston-linear-matrix-algebra-F01998 | 200  | 1.000 | $x_{\{2\}}=x_{\{2\}}^{\{+\}}-x_{\{2\}}^{\{-\}}$                                    | $x_2=x_2^+-x_2^-$                                          | equivalent form with all non-negative variables, we write $x_2=x_2^+-x_2^-$ ,                                         |
| johnston-linear-matrix-algebra-F01999 | 200  | 1.000 | $x_{\{2\}}^{\{+\}}, x_{\{2\}}^{\{-\}} \geq 0$                                      | $x_2^+, x_2^- \geq 0$                                      | where $x_2^+, x_2^- \geq 0$ , and then we replace all instances of $x_2$ in the linear                                |
| johnston-linear-matrix-algebra-F02000 | 200  | 1.000 | $x_{\{2\}}^{\{+\}}-x_{\{2\}}^{\{-\}}$                                              | $x_2^+-x_2^-$                                              | program with $x_2^+-x_2^-$ :                                                                                          |
| johnston-linear-matrix-algebra-F02001 | 201  | 0.989 | $x_{\{1\}}^{\{+\}}-x_{\{1\}}^{\{-\}}$                                              | $x_1^+-x_1^-$                                              | the unconstrained variable $x_1$ by $x_1^+-x_1^-$ , where $x_1^+, x_1^- \geq 0$ :                                     |
| johnston-linear-matrix-algebra-F02002 | 201  | 0.989 | $x_{\{1\}}^{\{+\}}, x_{\{1\}}^{\{-\}} \geq 0$                                      | $x_1^+, x_1^- \geq 0$                                      | the unconstrained variable $x_1$ by $x_1^+-x_1^-$ , where $x_1^+, x_1^- \geq 0$ :                                     |
| johnston-linear-matrix-algebra-F02003 | 201  | 1.000 | $\mathbf{x}=\left(x_{\{1\}}, x_{\{2\}}\right)$                                     | $\mathbf{x}=(x_1, x_2)$                                    | region $\mathbf{x}=(x_1, x_2)$ is a subset of $\mathbb{R}^2$ . The constraints $x_1 \geq 0$ and $x_2 \geq 0$ are sim- |
| johnston-linear-matrix-algebra-F02004 | 201  | 1.000 | $x_{\{1\}} \geq 0$                                                                 | $x_1 \geq 0$                                               | region $\mathbf{x}=(x_1, x_2)$ is a subset of $\mathbb{R}^2$ . The constraints $x_1 \geq 0$ and $x_2 \geq 0$ are sim- |
| johnston-linear-matrix-algebra-F02005 | 201  | 1.000 | $x_{\{2\}} \leq-x_{\{1\}}+3$                                                       | $x_2 \leq-x_1+3$                                           | into the forms $x_2 \leq-x_1+3$ (i.e., everything on and below the line $x_2=-x_1+3$ )                                |
| johnston-linear-matrix-algebra-F02006 | 201  | 1.000 | $x_{\{2\}}=-x_{\{1\}}+3$                                                           | $x_2=-x_1+3$                                               | into the forms $x_2 \leq-x_1+3$ (i.e., everything on and below the line $x_2=-x_1+3$ )                                |
| johnston-linear-matrix-algebra-F02007 | 202  | 1.000 | $f\left(x_{\{1\}}, x_{\{2\}}\right)=x_{\{1\}}+2 x_{\{2\}}$                         | $f\left(x_1, x_2\right)=x_1+2 x_2$                         | $f\left(x_1, x_2\right)=x_1+2 x_2$ .                                                                                  |
| johnston-linear-matrix-algebra-F02008 | 202  | 1.000 | $x_{\{2\}} \leq x_{\{1\}}+1$                                                       | $x_2 \leq x_1+1$                                           | and $x_2 \leq x_1+1$ (i.e., everything on and below the line $x_2=x_1+1$ ), respectively.                             |
| johnston-linear-matrix-algebra-F02009 | 202  | 1.000 | $x_{\{2\}}=x_{\{1\}}+1$                                                            | $x_2=x_1+1$                                                | and $x_2 \leq x_1+1$ (i.e., everything on and below the line $x_2=x_1+1$ ), respectively.                             |
| johnston-linear-matrix-algebra-F02010 | 202  | 1.000 | $z=x_{\{1\}}+2 x_{\{2\}}$                                                          | $z=x_1+2 x_2$                                              | region. To this end, we define $z=x_1+2 x_2$ , so that our goal is to maximize $z$ .                                  |
| johnston-linear-matrix-algebra-F02011 | 202  | 1.000 | $x_{\{2\}}=-x_{\{1\}} / 2+z / 2$                                                   | $x_2=-x_1 / 2+z / 2$                                       | We can rearrange this equation to get $x_2=-x_1 / 2+z / 2$ , which is a line with                                     |
| johnston-linear-matrix-algebra-F02012 | 202  | 1.000 | $-1 / 2$                                                                           | $-1 / 2$                                                   | slope $-1 / 2$ and y-intercept $z / 2$ . We want to find the largest possible $z$ such that                           |
| johnston-linear-matrix-algebra-F02013 | 202  | 1.000 | $z / 2$                                                                            | $z / 2$                                                    | slope $-1 / 2$ and y-intercept $z / 2$ . We want to find the largest possible $z$ such that                           |
| johnston-linear-matrix-algebra-F02014 | 202  | 1.000 | $z=5$                                                                              | $z=5$                                                      | in Figure 2.33(b). We see that the largest such value of $z$ is $z=5$ (corresponding                                  |
| johnston-linear-matrix-algebra-F02015 | 202  | 1.000 | $\mathbf{x}=\left(x_{\{1\}}, x_{\{2\}}\right)=(1,2)$                               | $\mathbf{x}=\left(x_1, x_2\right)=(1,2)$                   | it is attained at the vector $\mathbf{x}=\left(x_1, x_2\right)=(1,2)$ .                                               |
| johnston-linear-matrix-algebra-F02016 | 203  | 0.999 | $x_{\{1\}}=x_{\{2\}}=0$                                                            | $x_1=x_2=0$                                                | $x_1=x_2=0$ ).                                                                                                        |
| johnston-linear-matrix-algebra-F02017 | 203  | 0.980 | $\left(x_{\{1\}}, x_{\{2\}}\right)$                                                | $\left(x_1, x_2\right)$                                    | is unbounded because we can choose $x_2=x_1+1$ so that $\left(x_1, x_2\right)$ is a feasible                          |
| johnston-linear-matrix-algebra-F02018 | 203  | 1.000 | $x_{\{1\}}+2 x_{\{2\}}=x_{\{1\}}+2\left(x_{\{1\}}+1\right)=3 x_{\{1\}}+2$          | $x_1+2 x_2=x_1+2\left(x_1+1\right)=3 x_1+2$                | $x_1+2 x_2=x_1+2\left(x_1+1\right)=3 x_1+2$ , which can be made as large as we like                                   |
| johnston-linear-matrix-algebra-F02019 | 203  | 1.000 | $\left(x_{\{1\}}-x_{\{2\}}\right)+\left(-x_{\{1\}}+x_{\{2\}}\right)=0 \leq-1-1=-2$ | $\left(x_1-x_2\right)+\left(-x_1+x_2\right)=0 \leq-1-1=-2$ | $\left(x_1-x_2\right)+\left(-x_1+x_2\right)=0 \leq-1-1=-2$ , which is not true no matter what                         |

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| johnston-linear-matrix-algebra-F02020 | 203  | 1.000 | $\infty$                                                                                                                                                                                      | $\infty$                                                        | optimal value is $\infty$ , and if it is infeasible then we might say that its optimal value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02021 | 203  | 1.000 | $-\infty$                                                                                                                                                                                     | $-\infty$                                                       | is $-\infty$ . Similarly, if a minimization problem is unbounded then we sometimes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| johnston-linear-matrix-algebra-F02022 | 204  | 1.000 | $x_2 - x_3$                                                                                                                                                                                   | $x_2 - x_3$                                                     | by $x_2 - x_3$ where $x_2, x_3 \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F02023 | 204  | 1.000 | $x_2, x_3 \geq 0$                                                                                                                                                                             | $x_2, x_3 \geq 0$                                               | by $x_2 - x_3$ where $x_2, x_3 \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F02024 | 204  | 1.000 | $\mathbf{b} \geq \mathbf{0}$                                                                                                                                                                  | $\mathbf{b} \geq \mathbf{0}$                                    | <b>Step 1.5: Hope that <math>\mathbf{b} \geq \mathbf{0}</math>.</b> If $\mathbf{b}$ (the vector containing the coefficients on the right-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02025 | 204  | 1.000 | $\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$                                                                                                                                                      | $\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$                        | fix it later—for this particular example we just note that $\mathbf{b} = (1, 2, 3) \geq \mathbf{0}$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| johnston-linear-matrix-algebra-F02026 | 204  | 1.000 | $x \leq 5$                                                                                                                                                                                    | $x \leq 5$                                                      | the inequality $x \leq 5$ is equivalent to the pair of constraints $s + x = 5, s \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02027 | 204  | 1.000 | $s + x = 5, s \geq 0$                                                                                                                                                                         | $s + x = 5, s \geq 0$                                           | the inequality $x \leq 5$ is equivalent to the pair of constraints $s + x = 5, s \geq 0$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02028 | 204  | 1.000 | $s$                                                                                                                                                                                           | $s$                                                             | The new variable $s$ is called a <b>slack variable</b> , and it just measures the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F02029 | 204  | 0.997 | $z = \mathbf{c} \cdot \mathbf{x}$                                                                                                                                                             | $z = \mathbf{c} \cdot \mathbf{x}$                               | $z = \mathbf{c} \cdot \mathbf{x}$ ), and we rephrase the linear program as maximizing over $z$ with                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02030 | 204  | 1.000 | $z - \mathbf{c} \cdot \mathbf{x} = 0$                                                                                                                                                         | $z - \mathbf{c} \cdot \mathbf{x} = 0$                           | the additional constraint that $z - \mathbf{c} \cdot \mathbf{x} = 0$ . Making this change to the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02031 | 205  | 1.000 | $z - \mathbf{c}^T \mathbf{x} = 0$                                                                                                                                                             | $z - \mathbf{c}^T \mathbf{x} = 0$                               | In particular, the top row simply says that $z - \mathbf{c}^T \mathbf{x} = 0$ and the bottom                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02032 | 205  | 1.000 | $[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$                                                                                                                                                  | $[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$                    | block row $[\mathbf{0} \mid I \mid A \mid \mathbf{b}]$ represents the equality constraints involving slack                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02033 | 205  | 1.000 | $z, s_1, s_2$                                                                                                                                                                                 | $z, s_1, s_2$                                                   | it directly. In particular, the newly-introduced variables $z, s_1, s_2$ , and $s_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02034 | 205  | 1.000 | $s_3$                                                                                                                                                                                         | $s_3$                                                           | it directly. In particular, the newly-introduced variables $z, s_1, s_2$ , and $s_3$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02035 | 206  | 1.000 | $z = 0$                                                                                                                                                                                       | $z = 0$                                                         | gives us a value of $z = 0$ in the objective function, and we would like to                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F02036 | 206  | 1.000 | $z = 2x_1 + x_2 - x_3$                                                                                                                                                                        | $z = 2x_1 + x_2 - x_3$                                          | To increase the value of the objective function $z = 2x_1 + x_2 - x_3$ , we                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F02037 | 206  | 1.000 | $x_2 = x_3 = 0$                                                                                                                                                                               | $x_2 = x_3 = 0$                                                 | Since we are leaving $x_2 = x_3 = 0$ alone for now, the constraints simply                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02038 | 206  | 1.000 | $x_1 = 2$                                                                                                                                                                                     | $x_1 = 2$                                                       | the second one, so we wish to set $x_1 = 2$ and update the other vari-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-F02039 | 206  | 0.643 | $\begin{array}{l} z \text{ \& } s_1 \text{ \& } s_2 \text{ \& } s_3 \text{ \& } x_1 \text{ \& } \\ x_2 \text{ \& } x_3 \end{array}$                                                           | $z \quad s_1 \quad s_2 \quad s_3 \quad x_1 \quad x_2 \quad x_3$ | <div> <math display="block">  \begin{array}{c}  z \quad s_1 \quad s_2 \quad s_3 \quad x_1 \quad x_2 \quad x_3 \\  \left[ \begin{array}{cccc ccc} 1 &amp; 0 &amp; 0 &amp; 0 &amp; -2 &amp; -1 &amp; 1 &amp; 0 \\ 0 &amp; 1 &amp; 0 &amp; 0 &amp; -3 &amp; -2 &amp; 2 &amp; 1 \\ 0 &amp; 0 &amp; 1 &amp; 0 &amp; 1 &amp; -1 &amp; 1 &amp; 2 \\ 0 &amp; 0 &amp; 0 &amp; 1 &amp; -1 &amp; 3 &amp; -3 &amp; 3 \end{array} \right] \\  \begin{array}{l} R_1 + 2R_3 \\ R_2 + 3R_3 \\ R_4 + R_3 \end{array} \longrightarrow \left[ \begin{array}{cccc ccc} 1 &amp; 0 &amp; 2 &amp; 0 &amp; 0 &amp; -3 &amp; 3 &amp; 4 \\ 0 &amp; 1 &amp; 3 &amp; 0 &amp; 0 &amp; -5 &amp; 5 &amp; 7 \\ 0 &amp; 0 &amp; 1 &amp; 0 &amp; 1 &amp; -1 &amp; 1 &amp; 2 \\ 0 &amp; 0 &amp; 1 &amp; 1 &amp; 0 &amp; 2 &amp; -2 &amp; 5 \end{array} \right]  \end{array}  </math> <div> <div>new “leading” entry</div> </div> </div> |
| johnston-linear-matrix-algebra-F02040 | 206  | —     | $\left[ \begin{array}{c ccc ccc} 1 & 0 & 0 & 0 & -2 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & -2 & 2 & 1 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 0 & 1 & -1 & 3 & -3 & 3 \end{array} \right]$ |                                                                 | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Identifier                            | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                | Rendered                                                 | Image                                                                                                                     |
|---------------------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02041 | 206  | —       | $\xrightarrow{\begin{smallmatrix} R_1+2R_3 \\ R_2+3R_3 \\ R_4+R_3 \end{smallmatrix}} \left[ \begin{array}{cccc cccc} 1 & 0 & 2 & 0 & 0 & -3 & 3 & 4 \\ 0 & 1 & 3 & 0 & 0 & -5 & 5 & 7 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 0 & \uparrow & 2 & -2 & 5 \end{array} \right] .$ |                                                          |                                                                                                                           |
| johnston-linear-matrix-algebra-F02042 | 206  | ■ 1.000 | $s_{\{1\}}, s_{\{3\}}$                                                                                                                                                                                                                                                                      | $s_1, s_3$                                               | the equality constraints directly from it. We just have to think of $s_1, s_3$ ,                                          |
| johnston-linear-matrix-algebra-F02043 | 206  | ■ 1.000 | $s_{\{2\}}, x_{\{2\}}$                                                                                                                                                                                                                                                                      | $s_2, x_2$                                               | columns) and $s_2, x_2$ , and $x_3$ as the free variables. Setting this new set of                                        |
| johnston-linear-matrix-algebra-F02044 | 206  | ■ 1.000 | $z=4$                                                                                                                                                                                                                                                                                       | $z = 4$                                                  | of $z = 4$ in the objective function is better than the value of $z = 0$ that we                                          |
| johnston-linear-matrix-algebra-F02045 | 207  | ■ 1.000 | $z=23 / 2-(7 / 2) s_{\{2\}}-(3 / 2) s_{\{3\}}$                                                                                                                                                                                                                                              | $z = 23/2 - (7/2)s_2 - (3/2)s_3$                         | tells us that $z = 23/2 - (7/2)s_2 - (3/2)s_3$ . Since $s_2, s_3 \geq 0$ we conclude                                      |
| johnston-linear-matrix-algebra-F02046 | 207  | ■ 1.000 | $s_{\{2\}}, s_{\{3\}} \geq 0$                                                                                                                                                                                                                                                               | $s_2, s_3 \geq 0$                                        | tells us that $z = 23/2 - (7/2)s_2 - (3/2)s_3$ . Since $s_2, s_3 \geq 0$ we conclude                                      |
| johnston-linear-matrix-algebra-F02047 | 207  | ■ 1.000 | $z=23 / 2$                                                                                                                                                                                                                                                                                  | $z = 23/2$                                               | that $z = 23/2$ is the optimal value of this linear program, and it is attained                                           |
| johnston-linear-matrix-algebra-F02048 | 207  | ■ 0.636 | $x_{\{1\}}=9 / 2$                                                                                                                                                                                                                                                                           | $x_1 = 9/2$                                              | when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$ ).                                           |
| johnston-linear-matrix-algebra-F02049 | 207  | ■ 0.636 | $x_{\{2\}}=5 / 2$                                                                                                                                                                                                                                                                           | $x_2 = 5/2$                                              | when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$ ).                                           |
| johnston-linear-matrix-algebra-F02050 | 207  | ■ 0.636 | $s_{\{1\}}=39 / 2$                                                                                                                                                                                                                                                                          | $s_1 = 39/2$                                             | when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$ ).                                           |
| johnston-linear-matrix-algebra-F02051 | 207  | ■ 0.636 | $s_{\{2\}}=s_{\{3\}}=x_{\{3\}}=0$                                                                                                                                                                                                                                                           | $s_2 = s_3 = x_3 = 0$                                    | when $x_1 = 9/2$ and $x_2 = 5/2$ (and $s_1 = 39/2$ and $s_2 = s_3 = x_3 = 0$ ).                                           |
| johnston-linear-matrix-algebra-F02052 | 207  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(0,0,0)$                                                                                                                                                                                                                                      | $(x_1, x_2, x_3) = (0, 0, 0)$                            | from $(x_1, x_2, x_3) = (0, 0, 0)$ to $(x_1, x_2, x_3) = (2, 0, 0)$ and then to $(x_1, x_2, x_3) =$                       |
| johnston-linear-matrix-algebra-F02053 | 207  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(2,0,0)$                                                                                                                                                                                                                                      | $(x_1, x_2, x_3) = (2, 0, 0)$                            | from $(x_1, x_2, x_3) = (0, 0, 0)$ to $(x_1, x_2, x_3) = (2, 0, 0)$ and then to $(x_1, x_2, x_3) =$                       |
| johnston-linear-matrix-algebra-F02054 | 207  | ■ 0.755 | $(9 / 2,5 / 2,0)$                                                                                                                                                                                                                                                                           | $(9/2, 5/2, 0)$                                          | $(9/2, 5/2, 0)$ , as illustrated in Figure 2.35.                                                                          |
| johnston-linear-matrix-algebra-F02055 | 208  | ■ 1.000 | $\binom{m}{n}$                                                                                                                                                                                                                                                                              | $\binom{m}{n}$                                           | have up to $\binom{m}{n}$                                                                                                 |
| johnston-linear-matrix-algebra-F02056 | 208  | ■ 0.940 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(0,0)$                                                                                                                                                                                                                                                   | $(x_1, x_2) = (0, 0)$                                    | with a value of $z = 0$ in the objective function at $(x_1, x_2) = (0, 0)$ and then moves                                 |
| johnston-linear-matrix-algebra-F02057 | 208  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(2,0)$                                                                                                                                                                                                                                                   | $(x_1, x_2) = (2, 0)$                                    | to $(x_1, x_2) = (2, 0)$ to get a value of $z = 4$ , and finally to $(x_1, x_2) = (9/2, 5/2)$ to get the                  |
| johnston-linear-matrix-algebra-F02058 | 208  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(9 / 2,5 / 2)$                                                                                                                                                                                                                                           | $(x_1, x_2) = (9/2, 5/2)$                                | to $(x_1, x_2) = (2, 0)$ to get a value of $z = 4$ , and finally to $(x_1, x_2) = (9/2, 5/2)$ to get the                  |
| johnston-linear-matrix-algebra-F02059 | 208  | ■ 1.000 | $x_{\{\mathrm{p}\}}, x_{\{\mathrm{c}\}}$                                                                                                                                                                                                                                                    | $x_{\mathrm{p}}, x_{\mathrm{c}}$                         | We start by formulating this problem as a linear program. Let $x_{\mathrm{p}}, x_{\mathrm{c}}$ ,                          |
| johnston-linear-matrix-algebra-F02060 | 208  | ■ 1.000 | $x_{\{\mathrm{pc}\}}$                                                                                                                                                                                                                                                                       | $x_{\mathrm{pc}}$                                        | and $x_{\mathrm{pc}}$ denote the number of dozen peanut butter, chocolate chip, and                                       |
| johnston-linear-matrix-algebra-F02061 | 209  | ■ 1.000 | $x_{\{\mathrm{p}\}}, x_{\{\mathrm{c}\}}, x_{\{\mathrm{pc}\}} \geq 0$                                                                                                                                                                                                                        | $x_{\mathrm{p}}, x_{\mathrm{c}}, x_{\mathrm{pc}} \geq 0$ | and each of the variables is non-negative as well (i.e., $x_{\mathrm{p}}, x_{\mathrm{c}}, x_{\mathrm{pc}} \geq 0$ ) since |



| Identifier                               | Page | Conf.   | LaTeX source                                                        | Rendered                                                     | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| johnston-linear-matrix-algebra<br>F02065 | 210  | ■ 0.976 | $x_{\mathrm{c}}$                                                    | $x_c$                                                        |  <p>Augmented matrix for <math>x_c</math> with columns <math>z, s_1, s_2, s_3, x_p, x_c, x_{pc}</math> and rows <math>R_1, R_2, R_3, R_4</math>. The matrix is transformed through three steps: 1) <math>R_1 + 1.5R_3, R_2 - R_3, R_4 - R_3</math>; 2) <math>R_1 + 0.8R_2, R_4 - R_2</math>; 3) <math>R_1 + 0.3R_4, R_2 + R_4, R_3 - R_4</math>. The final matrix has a leading 1 in the <math>x_c</math> column.</p>                                                             |
| johnston-linear-matrix-algebra<br>F02066 | 210  | ■ 0.976 | $\left[\begin{array}{l} 1 \\ \hline 0 \\ 0 \\ 0 \end{array}\right]$ | $\left[\begin{array}{c} 1 \\ 0 \\ 0 \\ 0 \end{array}\right]$ |  <p>Augmented matrix for the column vector <math>\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}</math> with columns <math>z, s_1, s_2, s_3, x_p, x_c, x_{pc}</math> and rows <math>R_1, R_2, R_3, R_4</math>. The matrix is transformed through three steps: 1) <math>R_1 + 1.5R_3, R_2 - R_3, R_4 - R_3</math>; 2) <math>R_1 + 0.8R_2, R_4 - R_2</math>; 3) <math>R_1 + 0.3R_4, R_2 + R_4, R_3 - R_4</math>. The final matrix has a leading 1 in the <math>z</math> column.</p> |
| johnston-linear-matrix-algebra<br>F02067 | 210  | ■ 0.976 | $R_{\{1\}} + 1.5 R_{\{3\}}$                                         | $R_1 + 1.5R_3$                                               |  <p>Augmented matrix for the row operation <math>R_1 + 1.5R_3</math> with columns <math>z, s_1, s_2, s_3, x_p, x_c, x_{pc}</math> and rows <math>R_1, R_2, R_3, R_4</math>. The matrix is transformed through three steps: 1) <math>R_1 + 1.5R_3, R_2 - R_3, R_4 - R_3</math>; 2) <math>R_1 + 0.8R_2, R_4 - R_2</math>; 3) <math>R_1 + 0.3R_4, R_2 + R_4, R_3 - R_4</math>. The final matrix has a leading 1 in the <math>z</math> column.</p>                                  |



| Identifier                         | Page | Conf.   | LaTeX source                                        | Rendered                         | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|------|---------|-----------------------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-augm F02071 | 210  | ■ 0.994 | $\$ 151$                                            | $\$151$                          | most money that the bakery can make per week) is \$151. Furthermore, it                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| johnston-linear-matrix-augm F02072 | 210  | ■ 1.000 | $x_{\mathrm{p}}=50$                                 | $x_{\mathrm{p}} = 50$            | is attained when they bake $x_{\mathrm{p}} = 50$ dozen peanut butter cookies, $x_{\mathrm{c}} = 70$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| johnston-linear-matrix-augm F02073 | 210  | ■ 1.000 | $x_{\mathrm{c}}=70$                                 | $x_{\mathrm{c}} = 70$            | is attained when they bake $x_{\mathrm{p}} = 50$ dozen peanut butter cookies, $x_{\mathrm{c}} = 70$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| johnston-linear-matrix-augm F02074 | 210  | ■ 1.000 | $x_{\mathrm{pc}}=30$                                | $x_{\mathrm{pc}} = 30$           | dozen chocolate chip cookies, and $x_{\mathrm{pc}} = 30$ dozen peanut butter chocolate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| johnston-linear-matrix-augm F02075 | 211  | ■ 0.703 | $\left(x_1, x_2\right)=(0.8,1.6)$                   | $(x_1, x_2) = (0.8, 1.6)$        | is 5.6, which is attained at the vector $(x_1, x_2) = (0.8, 1.6)$ . If we try to find                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| johnston-linear-matrix-augm F02076 | 211  | ■ 1.000 | $\left(x_1, x_2\right)=(0,1)$                       | $(x_1, x_2) = (0, 1)$            | these values down to $(x_1, x_2) = (0, 1)$ , we run into a big problem—this                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-augm F02077 | 211  | ■ 1.000 | $-2 x_1+x_2 \leq 0$                                 | $-2x_1 + x_2 \leq 0$             | vector is not even feasible (it violates the constraint $-2x_1 + x_2 \leq 0$ )!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-augm F02078 | 211  | ■ 1.000 | $x_1=0.8$                                           | $x_1 = 0.8$                      | Of the four possible ways of rounding each of $x_1 = 0.8$ and $x_2 = 1.6$ up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| johnston-linear-matrix-augm F02079 | 211  | ■ 1.000 | $x_2=1.6$                                           | $x_2 = 1.6$                      | Of the four possible ways of rounding each of $x_1 = 0.8$ and $x_2 = 1.6$ up                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| johnston-linear-matrix-augm F02080 | 211  | ■ 1.000 | $\left(x_1, x_2\right)=(1,1)$                       | $(x_1, x_2) = (1, 1)$            | or down, only one results in a feasible vector: $(x_1, x_2) = (1, 1)$ . However,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-augm F02081 | 211  | ■ 1.000 | $x_1+3 x_2=4$                                       | $x_1 + 3x_2 = 4$                 | this vector is not optimal—it gives a value of $x_1 + 3x_2 = 4$ in the objec-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| johnston-linear-matrix-augm F02082 | 211  | ■ 0.997 | $\mathbf{b} \not\geq \mathbf{0}$                    | $\mathbf{b} \not\geq \mathbf{0}$ | which is only feasible if $\mathbf{b} \geq \mathbf{0}$ . To illustrate the problem with having $\mathbf{b} \not\geq \mathbf{0}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-augm F02083 | 212  | ■ 1.000 | $-y$                                                | $-y$                             | maximize $-y$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-augm F02084 | 212  | ■ 1.000 | $y \geq 0$                                          | $y \geq 0$                       | To construct this feasibility LP we introduce yet another variable $y \geq 0$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-augm F02085 | 212  | ■ 1.000 | $y=2$                                               | $y = 2$                          | $y = 2$ (in general, we set each $x_j = 0$ and then set $y$ large enough to satisfy all                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| johnston-linear-matrix-augm F02086 | 212  | ■ 1.000 | $\left(x_1, x_2, y\right)=\left(x_1, x_2, 0\right)$ | $(x_1, x_2, y) = (x_1, x_2, 0)$  | linear program (2.B.10) is feasible, since $(x_1, x_2, y) = (x_1, x_2, 0)$ is a feasible                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-augm F02087 | 212  | ■ 1.000 | $\left(x_1, x_2, y\right)=(0,0,2)$                  | $(x_1, x_2, y) = (0, 0, 2)$      | “obvious” feasible vector $(x_1, x_2, y) = (0, 0, 2)$ that we noted earlier, after                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| johnston-linear-matrix-augm F02088 | 212  | ■ 1.000 | $-2 x_1+x_2-y \leq-2$                               | $-2x_1 + x_2 - y \leq -2$        | corresponding to the “ $-2x_1 + x_2 - y \leq -2$ ” constraint (since this one                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| johnston-linear-matrix-augm F02089 | 213  | ■ 0.915 | $(1 \quad 2) \mathbf{R}_4$                          | $(1/2)R_4$                       | <div><div><div><div><div>1</div><div>0</div><div>0</div><div>1</div><div>−2</div><div>1</div><div>0</div><div>−2</div></div><div><div>0</div><div>1</div><div>0</div><div>0</div><div>1</div><div>1</div><div>0</div><div>2</div></div><div><div>0</div><div>0</div><div>1</div><div>−1</div><div>1</div><div>−2</div><div>0</div><div>1</div></div><div><div>0</div><div>0</div><div>0</div><div>−1</div><div>2</div><div>−1</div><div>1</div><div>2</div></div></div><div><div><div>(1/2)R<sub>4</sub></div><div>R<sub>1</sub>+2R<sub>4</sub></div><div>R<sub>2</sub>−R<sub>4</sub></div><div>R<sub>3</sub>−R<sub>4</sub></div></div><div>→</div><div><div>1</div><div>0</div><div>0</div><div>0</div><div>0</div><div>0</div><div>1</div><div>0</div></div><div><div>0</div><div>1</div><div>0</div><div>1/2</div><div>0</div><div>3/2</div><div>−1/2</div><div>1</div></div><div><div>0</div><div>0</div><div>1</div><div>−1/2</div><div>0</div><div>−3/2</div><div>−1/2</div><div>0</div></div><div><div>0</div><div>0</div><div>0</div><div>−1/2</div><div>1</div><div>−1/2</div><div>1/2</div><div>1</div></div></div></div></div> |
| johnston-linear-matrix-augm F02090 | 213  | ■ 0.915 | $\mathbf{R}_1+2 \mathbf{R}_4$                       | $R_1 + 2R_4$                     | <div><div><div><div><div>1</div><div>0</div><div>0</div><div>1</div><div>−2</div><div>1</div><div>0</div><div>−2</div></div><div><div>0</div><div>1</div><div>0</div><div>0</div><div>1</div><div>1</div><div>0</div><div>2</div></div><div><div>0</div><div>0</div><div>1</div><div>−1</div><div>1</div><div>−2</div><div>0</div><div>1</div></div><div><div>0</div><div>0</div><div>0</div><div>−1</div><div>2</div><div>−1</div><div>1</div><div>2</div></div></div><div><div><div>(1/2)R<sub>4</sub></div><div>R<sub>1</sub>+2R<sub>4</sub></div><div>R<sub>2</sub>−R<sub>4</sub></div><div>R<sub>3</sub>−R<sub>4</sub></div></div><div>→</div><div><div>1</div><div>0</div><div>0</div><div>0</div><div>0</div><div>0</div><div>1</div><div>0</div></div><div><div>0</div><div>1</div><div>0</div><div>1/2</div><div>0</div><div>3/2</div><div>−1/2</div><div>1</div></div><div><div>0</div><div>0</div><div>1</div><div>−1/2</div><div>0</div><div>−3/2</div><div>−1/2</div><div>0</div></div><div><div>0</div><div>0</div><div>0</div><div>−1/2</div><div>1</div><div>−1/2</div><div>1/2</div><div>1</div></div></div></div></div> |

| Identifier                               | Page | Conf. | LaTeX source                                                         | Rendered                                                             | Image                                                                               |
|------------------------------------------|------|-------|----------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra<br>F02091 | 213  | 0.915 | $R_{\{2\}}-R_{\{4\}}$                                                | $R_2 - R_4$                                                          |                                                                                     |
| johnston-linear-matrix-algebra<br>F02092 | 213  | 0.915 | $R_{\{3\}}-R_{\{4\}}$                                                | $R_3 - R_4$                                                          |                                                                                     |
| johnston-linear-matrix-algebra<br>F02093 | 213  | 0.982 | L P                                                                  | $LP$                                                                 | value of this feasibility $LP$ is 0, so the original linear program (2.B.10) is     |
| johnston-linear-matrix-algebra<br>F02094 | 213  | 1.000 | $x_{\{1\}}=1$                                                        | $x_1 = 1$                                                            | feasible. In particular, since this final tableau has $x_1 = 1$ and $x_2 = 0$ , we  |
| johnston-linear-matrix-algebra<br>F02095 | 213  | 1.000 | $x_{\{2\}}=0$                                                        | $x_2 = 0$                                                            | feasible. In particular, since this final tableau has $x_1 = 1$ and $x_2 = 0$ , we  |
| johnston-linear-matrix-algebra<br>F02096 | 213  | 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(1,0)$                            | $(x_1, x_2) = (1, 0)$                                                | conclude that $(x_1, x_2) = (1, 0)$ is a feasible vector of that original LP. It is |
| johnston-linear-matrix-algebra<br>F02097 | 213  | 1.000 | $-y<0$                                                               | $-y < 0$                                                             | If we had instead found an optimal value of $-y < 0$ in the above example,          |
| johnston-linear-matrix-algebra<br>F02098 | 213  | 0.494 | $\left.\begin{array}{l} 0 \\ 0 \\ 2 \\ -1 \\ -2 \end{array}\right\}$ | $\left.\begin{array}{l} 0 \\ 0 \\ 2 \\ -1 \\ -2 \end{array}\right\}$ |                                                                                     |
| johnston-linear-matrix-algebra<br>F02099 | 213  | 0.494 | $R_{\{1\}}+R_{\{2\}}$                                                | $R_1 + R_2$                                                          |                                                                                     |

| Identifier                            | Page | Conf. | LaTeX source                                                                                                                                                                                                                                | Rendered                                                                                                                                                            | Image                                                                                                                                                                                  |
|---------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02100 | 214  | 1.000 | $\xrightarrow{\substack{(1/3)R_4 \\ R_1+R_4 \\ R_2-R_4}}$                                                                                                                                                                                   | $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{bmatrix}$                   | $\begin{bmatrix} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{bmatrix}$                    |
| johnston-linear-matrix-algebra-F02101 | 214  | —     | $\left[\begin{array}{c ccc cc} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{array}\right]$                                                                        | $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{bmatrix}$                   | —                                                                                                                                                                                      |
| johnston-linear-matrix-algebra-F02102 | 214  | 1.000 | $\left[\begin{array}{c ccc cc} 1 & 5 & / & 3 & 0 & 1 & / & 3 & 0 & 0 & 8 & / & 3 \\ 0 & 1 & / & 3 & 0 & -1 & / & 3 & 1 & 0 & 4 & / & 3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 2 & / & 3 & 0 & 1 & / & 3 & 0 & 1 & 2 & / & 3 \end{array}\right]$ | $\begin{bmatrix} 1 & 5/3 & 0 & 1/3 & 0 & 0 & 8/3 \\ 0 & 1/3 & 0 & -1/3 & 1 & 0 & 4/3 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2/3 & 0 & 1/3 & 0 & 1 & 2/3 \end{bmatrix}$ | $\begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 3 & 2 \end{bmatrix}$                                      |
| johnston-linear-matrix-algebra-F02103 | 214  | 1.000 | $\left(x_1, x_2\right)=(4 / 3, 2 / 3)$                                                                                                                                                                                                      | $(x_1, x_2) = (4/3, 2/3)$                                                                                                                                           | attained at the vector $(x_1, x_2) = (4/3, 2/3)$ . We can visualize the feasible                                                                                                       |
| johnston-linear-matrix-algebra-F02104 | 215  | 1.000 | $y_1, y_2 \geq 0$                                                                                                                                                                                                                           | $y_1, y_2 \geq 0$                                                                                                                                                   | We need $y_1, y_2 \geq 0$                                                                                                                                                              |
| johnston-linear-matrix-algebra-F02105 | 215  | 0.998 | $\left(x_1+x_2\right) \leq 3$                                                                                                                                                                                                               | $(x_1+x_2 \leq 3)$                                                                                                                                                  | But if we now add the first constraint $(x_1+x_2 \leq 3)$ again, we similarly see that                                                                                                 |
| johnston-linear-matrix-algebra-F02106 | 215  | 1.000 | $y_1$                                                                                                                                                                                                                                       | $y_1$                                                                                                                                                               | A bit more generally, if we add up $y_1$ times the first constraint and $y_2$ times                                                                                                    |
| johnston-linear-matrix-algebra-F02107 | 215  | 1.000 | $y_2$                                                                                                                                                                                                                                       | $y_2$                                                                                                                                                               | A bit more generally, if we add up $y_1$ times the first constraint and $y_2$ times                                                                                                    |
| johnston-linear-matrix-algebra-F02108 | 215  | 1.000 | $3 y_1+y_2$                                                                                                                                                                                                                                 | $3 y_1+y_2$                                                                                                                                                         | then $3 y_1+y_2$ must also be an upper bound on $x_1+2 x_2$ (i.e., the objective                                                                                                       |
| johnston-linear-matrix-algebra-F02109 | 215  | 1.000 | $x_1+2 x_2$                                                                                                                                                                                                                                 | $x_1+2 x_2$                                                                                                                                                         | then $3 y_1+y_2$ must also be an upper bound on $x_1+2 x_2$ (i.e., the objective                                                                                                       |
| johnston-linear-matrix-algebra-F02110 | 215  | 1.000 | $y_1=3 / 2$                                                                                                                                                                                                                                 | $y_1=3 / 2$                                                                                                                                                         | function of the linear program). It is straightforward to check that $y_1=3 / 2$ ,                                                                                                     |
| johnston-linear-matrix-algebra-F02111 | 215  | 1.000 | $y_2=1 / 2$                                                                                                                                                                                                                                 | $y_2=1 / 2$                                                                                                                                                         | $y_2=1 / 2$ works and establishes the bound $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$ , just                                                                                                 |
| johnston-linear-matrix-algebra-F02112 | 215  | 1.000 | $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$                                                                                                                                                                                                         | $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$                                                                                                                                 | $y_2=1 / 2$ works and establishes the bound $x_1+2 x_2 \leq 3(3 / 2)+(1 / 2)=5$ , just                                                                                                 |
| johnston-linear-matrix-algebra-F02113 | 215  | 0.997 | $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$                                                                                                                                            | $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$                                                                    | Suppose $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$ are |
| johnston-linear-matrix-algebra-F02114 | 215  | 0.997 | $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$                                                                                                                                                                                  | $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$                                                                                                          | Suppose $A \in \mathcal{M}_{m, n}(\mathbb{R}), \mathbf{b} \in \mathbb{R}^m, \mathbf{c} \in \mathbb{R}^n$ are fixed, and $\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$ are |
| johnston-linear-matrix-algebra-F02115 | 216  | 1.000 | $\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}$                                                                                                                                                                                  | $\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}$                                                                                                          | $\mathbf{y} \cdot(A \mathbf{x})=\mathbf{y}^T A \mathbf{x}$                                                                                                                             |
| johnston-linear-matrix-algebra-F02116 | 216  | 1.000 | $\left(A^T \mathbf{y}\right) \cdot \mathbf{x}$                                                                                                                                                                                              | $\left(A^T \mathbf{y}\right) \cdot \mathbf{x}$                                                                                                                      | $\left(A^T \mathbf{y}\right) \cdot \mathbf{x}$ . This is a                                                                                                                             |
| johnston-linear-matrix-algebra-F02117 | 216  | 1.000 | $\mathbf{y} \in \mathbb{R}^m$                                                                                                                                                                                                               | $\mathbf{y} \in \mathbb{R}^m$                                                                                                                                       | If $\mathbf{x} \in \mathbb{R}^n$ is a feasible vector of a (primal) linear program and $\mathbf{y} \in \mathbb{R}^m$ is a                                                              |

| Identifier                                                                     | Page                                                                 | Conf. | LaTeX source                                                                                                        | Rendered                                                                                              | Image                                                                                                                                                                                                                                                                                                               |        |      |                                                                                |                                                                      |
|--------------------------------------------------------------------------------|----------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------|--------------------------------------------------------------------------------|----------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02118                                          | 216                                                                  | 1.000 | $\mathbf{y} \cdot (A \mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$                                                  | $\mathbf{y} \cdot (A\mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$                                     | $\mathbf{y} \cdot (A\mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$ . However, by using the fact that $A^T \mathbf{y} \geq \mathbf{c}$ , a similar argument                                                                                                                                                           |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02119                                          | 216                                                                  | 1.000 | $A^T \mathbf{y} \geq \mathbf{c}$                                                                                    | $A^T \mathbf{y} \geq \mathbf{c}$                                                                      | $\mathbf{y} \cdot (A\mathbf{x}) \leq \mathbf{y} \cdot \mathbf{b}$ . However, by using the fact that $A^T \mathbf{y} \geq \mathbf{c}$ , a similar argument                                                                                                                                                           |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02120                                          | 216                                                                  | 1.000 | $\mathbf{y} \cdot (A \mathbf{x}) = \left( A^T \mathbf{y} \right) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$ | $\mathbf{y} \cdot (A\mathbf{x}) = (A^T \mathbf{y}) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$ | shows that $\mathbf{y} \cdot (A\mathbf{x}) = (A^T \mathbf{y}) \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{x}$ . Stringing these inequalities together                                                                                                                                                            |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02121                                          | 216                                                                  | 1.000 | $\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{y}$                        | $\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{y}$          | shows that $\mathbf{c} \cdot \mathbf{x} \leq \mathbf{y} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{y}$ , as desired.                                                                                                                                                                                               |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02122                                          | 216                                                                  | 1.000 | $\mathbf{c} \cdot \mathbf{x} = \mathbf{b} \cdot \mathbf{y}$                                                         | $\mathbf{c} \cdot \mathbf{x} = \mathbf{b} \cdot \mathbf{y}$                                           | problems, respectively, for which $\mathbf{c} \cdot \mathbf{x} = \mathbf{b} \cdot \mathbf{y}$ , then they must be optimal. In                                                                                                                                                                                       |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02123                                          | 216                                                                  | 0.984 | $\mathbf{c} \cdot \mathbf{x} = 5$                                                                                   | $\mathbf{c} \cdot \mathbf{x} = 5$                                                                     | program that gives $\mathbf{c} \cdot \mathbf{x} = 5$ , so 5 is a lower bound on the optimal value of                                                                                                                                                                                                                |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02124                                          | 217                                                                  | 1.000 | $(y_1, y_2) = (3 / 2, 1 / 2)$                                                                                       | $(y_1, y_2) = (3/2, 1/2)$                                                                             | It is straightforward to verify that $(y_1, y_2) = (3/2, 1/2)$ is a feasible vector                                                                                                                                                                                                                                 |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02125                                          | 217                                                                  | 0.679 | $\mathbf{b} \cdot \mathbf{y} = 5$                                                                                   | $\mathbf{b} \cdot \mathbf{y} = 5$                                                                     | of the dual program, and it gives $\mathbf{b} \cdot \mathbf{y} = 5$ , so 5 is an upper bound on the                                                                                                                                                                                                                 |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02126                                          | 217                                                                  | 1.000 | $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$                                                                       | $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$                                                         | It does not take long to see that $(x_1, x_2, x_3, x_4, x_5) = (1, 1, 1, 1, 1)$ is a                                                                                                                                                                                                                                |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02127                                          | 217                                                                  | 0.998 | $x_2 + x_3$                                                                                                         | $x_2 + x_3$                                                                                           | of $x_2 + x_3$ and $x_4 + x_5$ in order for the first and fourth constraints to still be                                                                                                                                                                                                                            |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02128                                          | 217                                                                  | 0.998 | $x_4 + x_5$                                                                                                         | $x_4 + x_5$                                                                                           | of $x_2 + x_3$ and $x_4 + x_5$ in order for the first and fourth constraints to still be                                                                                                                                                                                                                            |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02129                                          | 218                                                                  | 1.000 | $\mathbf{x}_*$                                                                                                      | $\mathbf{x}_*$                                                                                        | $\mathbf{x}_*$ and $\mathbf{y}_*$ are                                                                                                                                                                                                                                                                               |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02130                                          | 218                                                                  | 1.000 | $\mathbf{y}_*$                                                                                                      | $\mathbf{y}_*$                                                                                        | $\mathbf{x}_*$ and $\mathbf{y}_*$ are                                                                                                                                                                                                                                                                               |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02131                                          | 218                                                                  | 1.000 | $(y_1, y_2, y_3, y_4, y_5) = (1, 1, 1, 1, 1) / 3$                                                                   | $(y_1, y_2, y_3, y_4, y_5) = (1, 1, 1, 1, 1)/3$                                                       | that the vector $(y_1, y_2, y_3, y_4, y_5) = (1, 1, 1, 1, 1)/3$ works. Since we have                                                                                                                                                                                                                                |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02132                                          | 218                                                                  | 1.000 | $\mathbf{x}_* \in \mathbb{R}^n$                                                                                     | $\mathbf{x}_* \in \mathbb{R}^n$                                                                       | If $\mathbf{x}_* \in \mathbb{R}^n$ is a feasible vector of a (primal) linear program satisfying                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02133                                          | 218                                                                  | 1.000 | $\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$                                                    | $\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$                                      | $\mathbf{c} \cdot \mathbf{x}_* \geq \mathbf{c} \cdot \mathbf{x}$ for all feasible vectors $\mathbf{x} \in \mathbb{R}^n$ , then there exists a feasible                                                                                                                                                              |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02134                                          | 218                                                                  | 1.000 | $\mathbf{y}_* \in \mathbb{R}^m$                                                                                     | $\mathbf{y}_* \in \mathbb{R}^m$                                                                       | vector $\mathbf{y}_* \in \mathbb{R}^m$ of its dual problem such that                                                                                                                                                                                                                                                |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02135                                          | 218                                                                  | 0.993 | $\mathbf{b} \cdot \mathbf{y}$                                                                                       | $\mathbf{b} \cdot \mathbf{y}$                                                                         | problem by some quantity $\mathbf{b} \cdot \mathbf{y}$ . A similar argument shows that if a dual problem                                                                                                                                                                                                            |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02136                                          | 219                                                                  | 0.877 | $\quad y$                                                                                                           | $y$                                                                                                   | <table><tr><th>Primal</th><th>Dual</th></tr><tr><td>maximize: <math>x_1 + 2x_2</math><br/>subject to: <math>-x_1 + x_2 \leq 1</math><br/><math>x_1, x_2 \geq 0</math></td><td>minimize: <math>y</math><br/>subject to: <math>-y \geq 1</math><br/><math>y \geq 2</math><br/><math>y \geq 0</math></td></tr></table> | Primal | Dual | maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |
| Primal                                                                         | Dual                                                                 |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02137                                          | 219                                                                  | 0.877 | $-x_1 + x_2 \leq 1$                                                                                                 | $-x_1 + x_2 \leq 1$                                                                                   | <table><tr><th>Primal</th><th>Dual</th></tr><tr><td>maximize: <math>x_1 + 2x_2</math><br/>subject to: <math>-x_1 + x_2 \leq 1</math><br/><math>x_1, x_2 \geq 0</math></td><td>minimize: <math>y</math><br/>subject to: <math>-y \geq 1</math><br/><math>y \geq 2</math><br/><math>y \geq 0</math></td></tr></table> | Primal | Dual | maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |
| Primal                                                                         | Dual                                                                 |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| johnston-linear-matrix-algebra-F02138                                          | 219                                                                  | 0.877 | $\quad -y \geq 1$                                                                                                   | $-y \geq 1$                                                                                           | <table><tr><th>Primal</th><th>Dual</th></tr><tr><td>maximize: <math>x_1 + 2x_2</math><br/>subject to: <math>-x_1 + x_2 \leq 1</math><br/><math>x_1, x_2 \geq 0</math></td><td>minimize: <math>y</math><br/>subject to: <math>-y \geq 1</math><br/><math>y \geq 2</math><br/><math>y \geq 0</math></td></tr></table> | Primal | Dual | maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |
| Primal                                                                         | Dual                                                                 |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |
| maximize: $x_1 + 2x_2$<br>subject to: $-x_1 + x_2 \leq 1$<br>$x_1, x_2 \geq 0$ | minimize: $y$<br>subject to: $-y \geq 1$<br>$y \geq 2$<br>$y \geq 0$ |       |                                                                                                                     |                                                                                                       |                                                                                                                                                                                                                                                                                                                     |        |      |                                                                                |                                                                      |

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| johnston-linear-matrix-algebra<br>F02139 | 219  | 0.877 | $x_{\{1\}}, \quad \text{\quad} x_{\{2\}} \geq 0$ | $x_1, \quad x_2 \geq 0$ | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>-x_1 + x_2 \leq 1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>y</math><br/> subject to: <math>-y \geq 1</math><br/> <math>y \geq 2</math><br/> <math>y \geq 0</math> </div> </div>                                                                            |
| johnston-linear-matrix-algebra<br>F02140 | 219  | 0.877 | $y \geq 2$                                       | $y \geq 2$              | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>-x_1 + x_2 \leq 1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>y</math><br/> subject to: <math>-y \geq 1</math><br/> <math>y \geq 2</math><br/> <math>y \geq 0</math> </div> </div>                                                                            |
| johnston-linear-matrix-algebra<br>F02141 | 219  | 1.000 | $-y \geq 1$                                      | $-y \geq 1$             | possible that $-y \geq 1$ and $y \geq 0$ simultaneously).                                                                                                                                                                                                                                                                                                                                              |
| johnston-linear-matrix-algebra<br>F02142 | 219  | 0.987 | $-y_{\{1\}}-y_{\{2\}}$                           | $-y_1 - y_2$            | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02143 | 219  | 0.987 | $x_{\{1\}}-x_{\{2\}} \leq -1$                    | $x_1 - x_2 \leq -1$     | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02144 | 219  | 0.987 | $\text{\quad} y_{\{1\}}-y_{\{2\}} \geq 1$        | $y_1 - y_2 \geq 1$      | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02145 | 219  | 0.987 | $-x_{\{1\}}+x_{\{2\}} \leq -1$                   | $-x_1 + x_2 \leq -1$    | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02146 | 219  | 0.987 | $-y_{\{1\}}+y_{\{2\}} \geq 2$                    | $-y_1 + y_2 \geq 2$     | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02147 | 219  | 0.987 | $y_{\{1\}}, \quad \text{\quad} y_{\{2\}} \geq 0$ | $y_1, \quad y_2 \geq 0$ | <div> <div> <b>Primal</b><br/> maximize: <math>x_1 + 2x_2</math><br/> subject to: <math>x_1 - x_2 \leq -1</math><br/> <math>-x_1 + x_2 \leq -1</math><br/> <math>x_1, \quad x_2 \geq 0</math> </div> <div> <b>Dual</b><br/> minimize: <math>-y_1 - y_2</math><br/> subject to: <math>y_1 - y_2 \geq 1</math><br/> <math>-y_1 + y_2 \geq 2</math><br/> <math>y_1, \quad y_2 \geq 0</math> </div> </div> |
| johnston-linear-matrix-algebra<br>F02148 | 219  | 1.000 | $0 \geq 3$                                       | $0 \geq 3$              | (false) inequality $0 \geq 3$ .                                                                                                                                                                                                                                                                                                                                                                        |
| johnston-linear-matrix-algebra<br>F02149 | 220  | 1.000 | $y_{\{1\}}-y_{\{2\}}$                            | $y_1 - y_2$             | but rather always appear together in the form $y_1 - y_2$ . We can thus replace                                                                                                                                                                                                                                                                                                                        |

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| johnston-linear-matrix-algebra-F02150 | 220  | ■ 1.000 | $y_{\{*\}}$                                                                        | $y_*$                                                                              | $y_1 - y_2$ with the single (potentially negative) variable $y_*$ , which results in the                                                                                                           |
| johnston-linear-matrix-algebra-F02151 | 220  | ■ 0.852 | $\geq 0$                                                                           | $\geq 0$                                                                           | unconstrained (rather than $\geq 0$ ). This makes a fair amount of sense intuitively—                                                                                                              |
| johnston-linear-matrix-algebra-F02152 | 221  | —       | $\begin{aligned} &\geq \\ &\leq \\ &\leq \end{aligned}$                            | $\geq$<br>$\leq$<br>$\leq$                                                         | —                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02153 | 221  | —       | $\begin{aligned} &\leq 0 \\ &\text{unc} \\ &\geq 0 \end{aligned}$                  | $\leq 0$<br><b>unc</b><br>$\geq 0$                                                 | —                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02154 | 221  | —       | $\begin{aligned} &\geq 0 \\ &\text{unconstrained} \\ &\leq 0 \end{aligned}$        | $\geq 0$<br><b>unconstrained</b><br>$\leq 0$                                       | —                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02155 | 221  | —       | $\begin{aligned} &\geq \\ &= \\ &\leq \end{aligned}$                               | $\geq$<br>$=$<br>$\leq$                                                            | —                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02156 | 221  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(0,-3)$                                         | $(x_1, x_2) = (0, -3)$                                                             | a while. It does not take long to find the vector $(x_1, x_2) = (0, -3)$ , which                                                                                                                   |
| johnston-linear-matrix-algebra-F02157 | 221  | ■ 0.999 | $\mathbf{c} \cdot \mathbf{x}=-3$                                                   | $\mathbf{c} \cdot \mathbf{x} = -3$                                                 | is feasible for the primal problem and has $\mathbf{c} \cdot \mathbf{x} = -3$ , and the vector                                                                                                     |
| johnston-linear-matrix-algebra-F02158 | 221  | ■ 0.965 | $\left(y_{\{1\}}, y_{\{2\}}\right)=(1,0)$                                          | $(y_1, y_2) = (1, 0)$                                                              | $(y_1, y_2) = (1, 0)$ , which is feasible for the dual problem and has $\mathbf{b} \cdot \mathbf{y} = -3$ .                                                                                        |
| johnston-linear-matrix-algebra-F02159 | 221  | ■ 0.965 | $\mathbf{b} \cdot \mathbf{y}=-3$                                                   | $\mathbf{b} \cdot \mathbf{y} = -3$                                                 | $(y_1, y_2) = (1, 0)$ , which is feasible for the dual problem and has $\mathbf{b} \cdot \mathbf{y} = -3$ .                                                                                        |
| johnston-linear-matrix-algebra-F02160 | 222  | ■ 1.000 | $\mathbf{x} \leq A^{-1} \mathbf{b}$                                                | $\mathbf{x} \leq A^{-1} \mathbf{b}$                                                | *(a) If $A \mathbf{x} \leq \mathbf{b}$ and $A$ is invertible, then $\mathbf{x} \leq A^{-1} \mathbf{b}$ .                                                                                           |
| johnston-linear-matrix-algebra-F02161 | 222  | ■ 1.000 | $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$                              | $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$                              | (b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$ .                                |
| johnston-linear-matrix-algebra-F02162 | 222  | ■ 1.000 | $\mathbf{x} \geq \mathbf{y}$                                                       | $\mathbf{x} \geq \mathbf{y}$                                                       | (b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$ .                                |
| johnston-linear-matrix-algebra-F02163 | 222  | ■ 1.000 | $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$                     | $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$                     | (b) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\mathbf{x} \geq \mathbf{y}$ then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$ .                                |
| johnston-linear-matrix-algebra-F02164 | 222  | ■ 1.000 | $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$  | $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$  | *(c) If $\mathbf{c}, \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{c} \geq \mathbf{0}$ and $\mathbf{x} \geq \mathbf{y}$ , then $\mathbf{c} \cdot \mathbf{x} \geq \mathbf{c} \cdot \mathbf{y}$ . |
| johnston-linear-matrix-algebra-F02165 | 223  | ■ 1.000 | $0 < c \in \mathbb{R}$                                                             | $0 < c \in \mathbb{R}$                                                             | **2.B.8 Suppose $0 < c \in \mathbb{R}$ and consider the linear pro-                                                                                                                                |
| johnston-linear-matrix-algebra-F02166 | 223  | ■ 1.000 | $\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$                   | $\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$                   | (a) Suppose $\mathbf{x}_1, \dots, \mathbf{x}_k, \mathbf{y} \in \mathbb{R}^n$ . Construct a linear                                                                                                  |
| johnston-linear-matrix-algebra-F02167 | 223  | ■ 1.000 | $\mathbf{y} \in$                                                                   | $\mathbf{y} \in$                                                                   | program that determines whether or not $\mathbf{y} \in$                                                                                                                                            |
| johnston-linear-matrix-algebra-F02168 | 223  | ■ 1.000 | $\operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$                | $\operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$                | $\operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$ .                                                                                                                              |
| johnston-linear-matrix-algebra-F02169 | 223  | ■ 0.827 | $\mathbf{y} \in \operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$ | $\mathbf{y} \in \operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$ | show that $\mathbf{y} \in \operatorname{span}\left(\mathbf{x}_1, \dots, \mathbf{x}_k\right)$ if and only if $\mathbf{y}$ is                                                                        |
| johnston-linear-matrix-algebra-F02170 | 223  | ■ 1.000 | $\mathbf{x}_1, \dots, \mathbf{x}_k$                                                | $\mathbf{x}_1, \dots, \mathbf{x}_k$                                                | orthogonal to every vector that $\mathbf{x}_1, \dots, \mathbf{x}_k$ are orthog-                                                                                                                    |
| johnston-linear-matrix-algebra-F02171 | 223  | ■ 1.000 | $\mathbf{x}_1 \cdot \mathbf{z}=\dots=\mathbf{x}_k \cdot \mathbf{z}=0$              | $\mathbf{x}_1 \cdot \mathbf{z}=\dots=\mathbf{x}_k \cdot \mathbf{z}=0$              | onal to (i.e., $\mathbf{x}_1 \cdot \mathbf{z}=\dots=\mathbf{x}_k \cdot \mathbf{z}=0$ implies $\mathbf{y} \cdot \mathbf{z}=0$ ).                                                                    |
| johnston-linear-matrix-algebra-F02172 | 223  | ■ 1.000 | $\mathbf{y} \cdot \mathbf{z}=0$                                                    | $\mathbf{y} \cdot \mathbf{z}=0$                                                    | onal to (i.e., $\mathbf{x}_1 \cdot \mathbf{z}=\dots=\mathbf{x}_k \cdot \mathbf{z}=0$ implies $\mathbf{y} \cdot \mathbf{z}=0$ ).                                                                    |
| johnston-linear-matrix-algebra-F02173 | 223  | ■ 1.000 | $\ A \mathbf{x}-\mathbf{b}\ _1$                                                    | $\ A \mathbf{x}-\mathbf{b}\ _1$                                                    | that makes $\ A \mathbf{x}-\mathbf{b}\ _1$ as small as possible.                                                                                                                                   |
| johnston-linear-matrix-algebra-F02174 | 223  | ■ 1.000 | $ a  \leq b$                                                                       | $ a  \leq b$                                                                       | [Hint: $ a  \leq b$ if and only if $-b \leq a \leq b$ .]                                                                                                                                           |
| johnston-linear-matrix-algebra-F02175 | 223  | ■ 1.000 | $-b \leq a \leq b$                                                                 | $-b \leq a \leq b$                                                                 | [Hint: $ a  \leq b$ if and only if $-b \leq a \leq b$ .]                                                                                                                                           |

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| johnston-linear-matrix-algebra-F02176 | 223  | ■ 1.000 | $\ A \mathbf{x} - \mathbf{b}\ _{\infty}$        | $\ A\mathbf{x} - \mathbf{b}\ _{\infty}$         | that makes $\ A\mathbf{x} - \mathbf{b}\ _{\infty}$ as small as possible.                                                 |
| johnston-linear-matrix-algebra-F02177 | 223  | ■ 1.000 | $a_{1,j} + a_{2,j} + \cdots + a_{n,j} = 1$      | $a_{1,j} + a_{2,j} + \cdots + a_{n,j} = 1$      | (i.e., $a_{1,j} + a_{2,j} + \cdots + a_{n,j} = 1$ for each $1 \leq j \leq n$ ). Show                                     |
| johnston-linear-matrix-algebra-F02178 | 223  | ■ 1.000 | $A \mathbf{x} = \mathbf{x}$                     | $A\mathbf{x} = \mathbf{x}$                      | for which $A\mathbf{x} = \mathbf{x}$ and $\mathbf{x} \geq \mathbf{0}$ .                                                  |
| johnston-linear-matrix-algebra-F02179 | 223  | ■ 1.000 | $A^T \mathbf{y} = \mathbf{y} \dots$             | $A^T \mathbf{y} = \mathbf{y} \dots$             | “obvious” vector $\mathbf{y} \in \mathbb{R}^n$ for which $A^T \mathbf{y} = \mathbf{y}$ ... what is it?]                  |
| johnston-linear-matrix-algebra-F02180 | 223  | ■ 1.000 | $A = C R$                                       | $A = CR$                                        | $A \in \mathcal{M}_{m,n}$ , then we see that if we can write $A = CR$ for some $C \in \mathcal{M}_{m,r}$ and             |
| johnston-linear-matrix-algebra-F02181 | 223  | ■ 1.000 | $C \in \mathcal{M}_{m,r}$                       | $C \in \mathcal{M}_{m,r}$                       | $A \in \mathcal{M}_{m,n}$ , then we see that if we can write $A = CR$ for some $C \in \mathcal{M}_{m,r}$ and             |
| johnston-linear-matrix-algebra-F02182 | 223  | ■ 1.000 | $R \in \mathcal{M}_{r,n}$                       | $R \in \mathcal{M}_{r,n}$                       | $R \in \mathcal{M}_{r,n}$ , then we must have                                                                            |
| johnston-linear-matrix-algebra-F02183 | 224  | ■ 1.000 | $\text{rank}(A) \leq r$                         | $\text{rank}(A) \leq r$                         | <i>Proof.</i> We already showed that $\text{rank}(A) \leq r$ , so to complete the proof we just                          |
| johnston-linear-matrix-algebra-F02184 | 224  | ■ 1.000 | $A = P \hat{R}$                                 | $A = P\hat{R}$                                  | recall from Theorem 2.5.3 that we can write $A = P\hat{R}$ , where $P$ is some invertible                                |
| johnston-linear-matrix-algebra-F02185 | 224  | ■ 1.000 | $\hat{R}$                                       | $\hat{R}$                                       | matrix and $\hat{R}$ is the reduced row echelon form of $A$ . If $r = \text{rank}(A)$ then the first                     |
| johnston-linear-matrix-algebra-F02186 | 225  | ■ 1.000 | $C^T$                                           | $C^T$                                           | We just write $C^T$                                                                                                      |
| johnston-linear-matrix-algebra-F02187 | 225  | ■ 1.000 | $\hat{R} = EA$                                  | $\hat{R} = EA$                                  | It follows that $\hat{R} = EA$ (and equivalently, $A = P\hat{R}$ , where $P = E^{-1}$ ), where                           |
| johnston-linear-matrix-algebra-F02188 | 225  | ■ 1.000 | $P = E^{-1}$                                    | $P = E^{-1}$                                    | It follows that $\hat{R} = EA$ (and equivalently, $A = P\hat{R}$ , where $P = E^{-1}$ ), where                           |
| johnston-linear-matrix-algebra-F02189 | 225  | ■ 1.000 | $\text{rank}(A) = 2$                            | $\text{rank}(A) = 2$                            | Since $\hat{R}$ has two non-zero rows, it follows that $\text{rank}(A) = 2$ . The matrices                               |
| johnston-linear-matrix-algebra-F02190 | 225  | ■ 1.000 | $m r$                                           | $mr$                                            | rank- $r$ matrix $A \in \mathcal{M}_{m,n}$ , we can just keep track of the $mr$ entries of $C$ and $rn$                  |
| johnston-linear-matrix-algebra-F02191 | 225  | ■ 1.000 | $r n$                                           | $rn$                                            | rank- $r$ matrix $A \in \mathcal{M}_{m,n}$ , we can just keep track of the $mr$ entries of $C$ and $rn$                  |
| johnston-linear-matrix-algebra-F02192 | 225  | ■ 0.998 | $11 \times 11 = 121$                            | $11 \times 11 = 121$                            | we could list all $11 \times 11 = 121$ of its entries. However, a more economical way                                    |
| johnston-linear-matrix-algebra-F02193 | 225  | ■ 1.000 | $(11 \times 2) + (2 \times 11) = 44$            | $(11 \times 2) + (2 \times 11) = 44$            | track of $(11 \times 2) + (2 \times 11) = 44$ numbers instead of the 121 numbers required                                |
| johnston-linear-matrix-algebra-F02194 | 225  | ■ 1.000 | $r=1$                                           | $r = 1$                                         | In the most extreme case when $r = 1$ , Theorem 2.C.1 says that a non-zero                                               |
| johnston-linear-matrix-algebra-F02195 | 226  | ■ 1.000 | $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ | $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ | vectors $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ and $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ with         |
| johnston-linear-matrix-algebra-F02196 | 226  | ■ 1.000 | $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ | $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ | vectors $\{\mathbf{v}_j\}_{j=1}^r \subset \mathbb{R}^m$ and $\{\mathbf{w}_j\}_{j=1}^r \subset \mathbb{R}^n$ with         |
| johnston-linear-matrix-algebra-F02197 | 226  | ■ 1.000 | $\{\mathbf{v}_i\}_{i=1}^r$                      | $\{\mathbf{v}_i\}_{i=1}^r$                      | is exactly $r = \text{rank}(A)$ . Furthermore, the sets $\{\mathbf{v}_i\}_{i=1}^r$ and $\{\mathbf{w}_i\}_{i=1}^r$ can be |
| johnston-linear-matrix-algebra-F02198 | 226  | ■ 1.000 | $\{\mathbf{w}_i\}_{i=1}^r$                      | $\{\mathbf{w}_i\}_{i=1}^r$                      | is exactly $r = \text{rank}(A)$ . Furthermore, the sets $\{\mathbf{v}_i\}_{i=1}^r$ and $\{\mathbf{w}_i\}_{i=1}^r$ can be |
| johnston-linear-matrix-algebra-F02199 | 226  | ■ 1.000 | $R \in$                                         | $R \in$                                         | <i>Proof.</i> Use the rank decomposition to write $A = CR$ , where $C \in \mathcal{M}_{m,r}$ and $R \in$                 |
| johnston-linear-matrix-algebra-F02200 | 226  | ■ 1.000 | $\mathcal{M}_{r,n}$                             | $\mathcal{M}_{r,n}$                             | $\mathcal{M}_{r,n}$ , and then write $C$ and $R$ in terms of their columns and rows, respectively:                       |
| johnston-linear-matrix-algebra-F02201 | 226  | ■ 1.000 | $\mathbf{v}_j \mathbf{w}_j^T$                   | $\mathbf{v}_j \mathbf{w}_j^T$                   | $\mathbf{v}_j \mathbf{w}_j^T$ have rank 1, so Theorem 2.4.11(a) tells us that                                            |
| johnston-linear-matrix-algebra-F02202 | 226  | ■ 1.000 | $\{\mathbf{v}_j\}_{j=1}^r$                      | $\{\mathbf{v}_j\}_{j=1}^r$                      | ward way: the vectors $\{\mathbf{v}_j\}_{j=1}^r$ and $\{\mathbf{w}_j\}_{j=1}^r$ are just the columns of the matrices     |

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| johnston-linear-matrix-algebra-F02203 | 226  | 1.000 | $\left\{\mathbf{w}_j\right\}_{j=1}^r$                                                                                              | $\{\mathbf{w}_j\}_{j=1}^r$                                                                                                           | ward way: the vectors $\{\mathbf{v}_j\}_{j=1}^r$ and $\{\mathbf{w}_j\}_{j=1}^r$ are just the columns of the matrices                    |
| johnston-linear-matrix-algebra-F02204 | 228  | 0.998 | $\operatorname{rank}(B) \leq \operatorname{rank}(A)$                                                                               | $\operatorname{rank}(B) \leq \operatorname{rank}(A)$                                                                                 | Suppose $A \in \mathcal{M}_{m,n}$ . If $B$ is a submatrix of $A$ then $\operatorname{rank}(B) \leq \operatorname{rank}(A)$ .            |
| johnston-linear-matrix-algebra-F02205 | 228  | 1.000 | $C \in \mathcal{M}_{m,\ell}$                                                                                                       | $C \in \mathcal{M}_{m,\ell}$                                                                                                         | If a submatrix $C \in \mathcal{M}_{m,\ell}$ is created by erasing some columns of $A$ , then                                            |
| johnston-linear-matrix-algebra-F02206 | 228  | 1.000 | $\operatorname{rank}(C) \leq \operatorname{rank}(A)$                                                                               | $\operatorname{rank}(C) \leq \operatorname{rank}(A)$                                                                                 | $\operatorname{rank}(C) \leq \operatorname{rank}(A)$ since $\operatorname{range}(C)$ is the span of the columns of $C$ , which          |
| johnston-linear-matrix-algebra-F02207 | 228  | 1.000 | $\operatorname{range}(C)$                                                                                                          | $\operatorname{range}(C)$                                                                                                            | $\operatorname{rank}(C) \leq \operatorname{rank}(A)$ since $\operatorname{range}(C)$ is the span of the columns of $C$ , which          |
| johnston-linear-matrix-algebra-F02208 | 228  | 0.972 | $\operatorname{rank}(C)$                                                                                                           | $\operatorname{rank}(C)$                                                                                                             | On the other hand, we can also express $\operatorname{rank}(C)$ in terms of its rows                                                    |
| johnston-linear-matrix-algebra-F02209 | 229  | 1.000 | $\operatorname{rank}(C)=\operatorname{rank}\left(C^T\right)$                                                                       | $\operatorname{rank}(C) = \operatorname{rank}\left(C^T\right)$                                                                       | $\operatorname{rank}(C) = \operatorname{rank}(C^T)$ .                                                                                   |
| johnston-linear-matrix-algebra-F02210 | 229  | 1.000 | $\operatorname{rank}(B)=3$                                                                                                         | $\operatorname{rank}(B) = 3$                                                                                                         | In fact, $\operatorname{rank}(B) = 3$                                                                                                   |
| johnston-linear-matrix-algebra-F02211 | 229  | 0.998 | $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_m$ :                                                                               | $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_m$ :                                                                                 | $\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_m$ :                                                                                    |
| johnston-linear-matrix-algebra-F02212 | 229  | 1.000 | $B \in \mathcal{M}_{k,\ell}$                                                                                                       | $B \in \mathcal{M}_{k,\ell}$                                                                                                         | If a submatrix $B \in \mathcal{M}_{k,\ell}$ is created by erasing some rows of $C$ , then $\operatorname{rank}(B) \leq$                 |
| johnston-linear-matrix-algebra-F02213 | 229  | 1.000 | $\operatorname{rank}(B) \leq$                                                                                                      | $\operatorname{rank}(B) \leq$                                                                                                        | If a submatrix $B \in \mathcal{M}_{k,\ell}$ is created by erasing some rows of $C$ , then $\operatorname{rank}(B) \leq$                 |
| johnston-linear-matrix-algebra-F02214 | 229  | 0.991 | $\operatorname{range}\left(B^T\right)$                                                                                             | $\operatorname{range}\left(B^T\right)$                                                                                               | $\operatorname{rank}(C)$ since $\operatorname{range}(B^T)$ is the span of the rows of $B$ , which are a subset of                       |
| johnston-linear-matrix-algebra-F02215 | 229  | 1.000 | $A=\left[\begin{array}{lllll}1 & 1 & 0 & 0 & 1 \\1 & 0 & 1 & 0 & 1 \\1 & 0 & 0 & 1 & 1\end{array}\right]$                          | $A = \left[\begin{array}{ccccc}1 & 1 & 0 & 0 & 1 \\1 & 0 & 1 & 0 & 1 \\1 & 0 & 0 & 1 & 1\end{array}\right]$                          | a) $A = \left[\begin{array}{ccccc}1 & 1 & 0 & 0 & 1 \\1 & 0 & 1 & 0 & 1 \\1 & 0 & 0 & 1 & 1\end{array}\right]$                          |
| johnston-linear-matrix-algebra-F02216 | 229  | 1.000 | $B=\left[\begin{array}{ccccc}1 & 1 & 1 & 1 & 1 \\1 & 2 & 4 & 8 & 16 \\1 & 2 & 5 & 14 & 41 \\1 & 3 & 9 & 27 & 81\end{array}\right]$ | $B = \left[\begin{array}{ccccc}1 & 1 & 1 & 1 & 1 \\1 & 2 & 4 & 8 & 16 \\1 & 2 & 5 & 14 & 41 \\1 & 3 & 9 & 27 & 81\end{array}\right]$ | b) $B = \left[\begin{array}{ccccc}1 & 1 & 1 & 1 & 1 \\1 & 2 & 4 & 8 & 16 \\1 & 2 & 5 & 14 & 41 \\1 & 3 & 9 & 27 & 81\end{array}\right]$ |
| johnston-linear-matrix-algebra-F02217 | 229  | 0.937 | $\operatorname{rank}(A) \geq 3$                                                                                                    | $\operatorname{rank}(A) \geq 3$                                                                                                      | conclude that $\operatorname{rank}(A) \geq 3$ (in fact, since $A$ is $3 \times 5$ , its rank cannot                                     |
| johnston-linear-matrix-algebra-F02218 | 229  | 0.937 | $3 \times 5$                                                                                                                       | $3 \times 5$                                                                                                                         | conclude that $\operatorname{rank}(A) \geq 3$ (in fact, since $A$ is $3 \times 5$ , its rank cannot                                     |
| johnston-linear-matrix-algebra-F02219 | 229  | 0.992 | $\operatorname{rank}(B) \geq 3$                                                                                                    | $\operatorname{rank}(B) \geq 3$                                                                                                      | it has rank 3. It follows that $\operatorname{rank}(B) \geq 3$ .                                                                        |
| johnston-linear-matrix-algebra-F02220 | 229  | 1.000 | $r \times r$                                                                                                                       | $r \times r$                                                                                                                         | there exists an invertible $r \times r$ submatrix of $A$ .                                                                              |
| johnston-linear-matrix-algebra-F02221 | 229  | 0.999 | $r \leq \operatorname{rank}(A)$                                                                                                    | $r \leq \operatorname{rank}(A)$                                                                                                      | <i>Proof.</i> Theorem 2.C.3 tells us that $r \leq \operatorname{rank}(A)$ , so we just need to show that                                |
| johnston-linear-matrix-algebra-F02222 | 230  | 1.000 | $\operatorname{rank}(C)=r$                                                                                                         | $\operatorname{rank}(C) = r$                                                                                                         | columns (in the same order that they appear in $A$ ). Then $\operatorname{rank}(C) = r$ as well,                                        |
| johnston-linear-matrix-algebra-F02223 | 230  | 0.999 | $\operatorname{rank}\left(C^T\right)=r$                                                                                            | $\operatorname{rank}\left(C^T\right) = r$                                                                                            | so $\operatorname{rank}(C^T) = r$ , so there is some set of $r$ rows of $C$ that span $\operatorname{range}(C^T)$ . Let                 |
| johnston-linear-matrix-algebra-F02224 | 230  | 0.999 | $\operatorname{range}\left(C^T\right)$                                                                                             | $\operatorname{range}\left(C^T\right)$                                                                                               | so $\operatorname{rank}(C^T) = r$ , so there is some set of $r$ rows of $C$ that span $\operatorname{range}(C^T)$ . Let                 |
| johnston-linear-matrix-algebra-F02225 | 230  | 1.000 | $B \in \mathcal{M}_r$                                                                                                              | $B \in \mathcal{M}_r$                                                                                                                | $B \in \mathcal{M}_r$ be the submatrix consisting of these rows. Then $\operatorname{rank}(B) = r$ as well,                             |
| johnston-linear-matrix-algebra-F02226 | 230  | 1.000 | $\operatorname{rank}(B)=r$                                                                                                         | $\operatorname{rank}(B) = r$                                                                                                         | $B \in \mathcal{M}_r$ be the submatrix consisting of these rows. Then $\operatorname{rank}(B) = r$ as well,                             |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02227 | 230  | 1.000 | $\left[\begin{array}{lll}1 & 2 & 1 \\ 3 & 2 & 1\end{array}\right]$                                            | $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$                                           | (b) $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$                                           |
| johnston-linear-matrix-algebra-F02228 | 230  | 1.000 | $\left[\begin{array}{ccc}1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12\end{array}\right]$               | $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}$              | $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}$                  |
| johnston-linear-matrix-algebra-F02229 | 230  | 1.000 | $\left[\begin{array}{llll}1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1\end{array}\right]$ | $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$ | $\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$     |
| johnston-linear-matrix-algebra-F02230 | 230  | 0.644 | $\operatorname{rank}(A) \leq 3$                                                                               | $\operatorname{rank}(A) \leq 3$                                                                  | $\operatorname{rank}(A) \leq 3$ .                                                                    |
| johnston-linear-matrix-algebra-F02231 | 230  | 0.999 | $\operatorname{rank}(A) \geq 4$                                                                               | $\operatorname{rank}(A) \geq 4$                                                                  | $\operatorname{rank}(A) \geq 4$ .                                                                    |
| johnston-linear-matrix-algebra-F02232 | 230  | 0.994 | $(n-1) \times (n-1)$                                                                                          | $(n-1) \times (n-1)$                                                                             | (e) If $A \in \mathcal{M}_n$ has an $(n-1) \times (n-1)$ submatrix equal                             |
| johnston-linear-matrix-algebra-F02233 | 230  | 0.998 | $\operatorname{rank}(A) \leq 1$                                                                               | $\operatorname{rank}(A) \leq 1$                                                                  | to the zero matrix then $\operatorname{rank}(A) \leq 1$ .                                            |
| johnston-linear-matrix-algebra-F02234 | 230  | 0.999 | $P \in \mathcal{M}_r$                                                                                         | $P \in \mathcal{M}_r$                                                                            | and $R \in \mathcal{M}_{r,n}$ . Let $P \in \mathcal{M}_r$ be any invertible matrix, and              |
| johnston-linear-matrix-algebra-F02235 | 230  | 1.000 | $\tilde{C} = CP^{-1}$                                                                                         | $\tilde{C} = CP^{-1}$                                                                            | define $\tilde{C} = CP^{-1}$ and $\tilde{R} = PR$ . Show that $A = \tilde{C}\tilde{R}$ is also a     |
| johnston-linear-matrix-algebra-F02236 | 230  | 1.000 | $\tilde{R} = PR$                                                                                              | $\tilde{R} = PR$                                                                                 | define $\tilde{C} = CP^{-1}$ and $\tilde{R} = PR$ . Show that $A = \tilde{C}\tilde{R}$ is also a     |
| johnston-linear-matrix-algebra-F02237 | 230  | 1.000 | $A = \tilde{C}\tilde{R}$                                                                                      | $A = \tilde{C}\tilde{R}$                                                                         | define $\tilde{C} = CP^{-1}$ and $\tilde{R} = PR$ . Show that $A = \tilde{C}\tilde{R}$ is also a     |
| johnston-linear-matrix-algebra-F02238 | 230  | 0.994 | $\operatorname{rank}(A) \leq z$                                                                               | $\operatorname{rank}(A) \leq z$                                                                  | that $\operatorname{rank}(A) \leq z$ .                                                               |
| johnston-linear-matrix-algebra-F02239 | 230  | 1.000 | $A = PDQ$                                                                                                     | $A = PDQ$                                                                                        | the form $A = PDQ$ , where $P \in \mathcal{M}_m$ and $Q \in \mathcal{M}_n$ are                       |
| johnston-linear-matrix-algebra-F02240 | 230  | 1.000 | $Q \in \mathcal{M}_n$                                                                                         | $Q \in \mathcal{M}_n$                                                                            | the form $A = PDQ$ , where $P \in \mathcal{M}_m$ and $Q \in \mathcal{M}_n$ are                       |
| johnston-linear-matrix-algebra-F02241 | 230  | 1.000 | $A = PBQ$                                                                                                     | $A = PBQ$                                                                                        | $A = PBQ$ if and only if $\operatorname{rank}(A) = \operatorname{rank}(B)$ .                         |
| johnston-linear-matrix-algebra-F02242 | 230  | 1.000 | $\operatorname{rank}(A) = \operatorname{rank}(B)$                                                             | $\operatorname{rank}(A) = \operatorname{rank}(B)$                                                | $A = PBQ$ if and only if $\operatorname{rank}(A) = \operatorname{rank}(B)$ .                         |
| johnston-linear-matrix-algebra-F02243 | 230  | 1.000 | $a_0, a_1, \dots, a_m$                                                                                        | $a_0, a_1, \dots, a_m$                                                                           | 2.C.9 Suppose $a_0, a_1, \dots, a_m$ are distinct real numbers and                                   |
| johnston-linear-matrix-algebra-F02244 | 230  | 1.000 | $V \in \mathcal{M}_{m+1, n+1}$                                                                                | $V \in \mathcal{M}_{m+1, n+1}$                                                                   | let $V \in \mathcal{M}_{m+1, n+1}$ be the matrix                                                     |
| johnston-linear-matrix-algebra-F02245 | 230  | 0.934 | $\operatorname{rank}(V) = \min\{m+1, n+1\}$                                                                   | $\operatorname{rank}(V) = \min\{m+1, n+1\}$                                                      | Show that $\operatorname{rank}(V) = \min\{m+1, n+1\}$ (i.e., $V$ has full                            |
| johnston-linear-matrix-algebra-F02246 | 231  | 1.000 | $U$                                                                                                           | $U$                                                                                              | we refer to it as $U$ from now on. Similarly, the matrix $P$ that we constructed                     |
| johnston-linear-matrix-algebra-F02247 | 231  | 1.000 | $L$                                                                                                           | $L$                                                                                              | happened to be lower triangular, so we refer to it as $L$ from now on. Being able                    |
| johnston-linear-matrix-algebra-F02248 | 231  | 1.000 | $\mathbf{LU}$                                                                                                 | <b>LU</b>                                                                                        | Suppose $A \in \mathcal{M}_{m,n}$ . An <b>LU decomposition</b> of $A$ is any factorization of        |
| johnston-linear-matrix-algebra-F02249 | 231  | 1.000 | $A = LU$                                                                                                      | $A = LU$                                                                                         | the form $A = LU$ , where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is |
| johnston-linear-matrix-algebra-F02250 | 231  | 1.000 | $L \in \mathcal{M}_m$                                                                                         | $L \in \mathcal{M}_m$                                                                            | the form $A = LU$ , where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is |
| johnston-linear-matrix-algebra-F02251 | 231  | 1.000 | $U \in \mathcal{M}_{m, n}$                                                                                    | $U \in \mathcal{M}_{m,n}$                                                                        | the form $A = LU$ , where $L \in \mathcal{M}_m$ is lower triangular and $U \in \mathcal{M}_{m,n}$ is |

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| johnston-linear-matrix-algebra-F02252 | 231  | 1.000 | $U=R$                                                                                               | $U = R$                                                                                  | 1) The matrix $U = R$ from Equation (2.D.1) was upper triangular just as a                                                           |
| johnston-linear-matrix-algebra-F02253 | 231  | 1.000 | $L=P$                                                                                               | $L = P$                                                                                  | result of being chosen to be a row echelon form of $A$ , but why was $L = P$                                                         |
| johnston-linear-matrix-algebra-F02254 | 232  | 1.000 | $L=E^{-1}$                                                                                          | $L = E^{-1}$                                                                             | form of $A$ , and setting $L = E^{-1}$ .                                                                                             |
| johnston-linear-matrix-algebra-F02255 | 232  | 1.000 | $j<i$                                                                                               | $j < i$                                                                                  | operations $R_i + cR_j$ with $j < i$ preserve lower triangularity, so $E$ must be lower                                              |
| johnston-linear-matrix-algebra-F02256 | 232  | 0.873 | $A=\left[\begin{array}{cccc}2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8\end{array}\right]$ | $A = \begin{bmatrix} 2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8 \end{bmatrix}$ | Compute an LU decomposition of the matrix $A = \begin{bmatrix} 2 & 1 & -2 & 1 \\ -4 & -4 & 3 & 0 \\ 2 & -5 & -2 & 8 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra-F02257 | 233  | 1.000 | $L\ D^{-1}$                                                                                         | $LD^{-1}$                                                                                | entries as $L$ then $LD^{-1}$ is a lower triangular matrix with ones on its                                                          |
| johnston-linear-matrix-algebra-F02258 | 233  | 1.000 | $D\ U$                                                                                              | $DU$                                                                                     | diagonal and $DU$ is an upper triangular matrix, so $A = (LD^{-1})(DU)$ is a                                                         |
| johnston-linear-matrix-algebra-F02259 | 233  | 1.000 | $A=\left(L\ D^{-1}\right)\left(D\ U\right)$                                                         | $A = (LD^{-1})(DU)$                                                                      | diagonal and $DU$ is an upper triangular matrix, so $A = (LD^{-1})(DU)$ is a                                                         |
| johnston-linear-matrix-algebra-F02260 | 234  | 0.976 | $R_{\{2\}}-\mathbf{2}R_{\{1\}},\ R_{\{3\}}-\mathbf{3}R_{\{1\}}$                                     | $R_2 - \mathbf{2}R_1, R_3 - \mathbf{3}R_1$                                               | $R_2 - \mathbf{2}R_1, R_3 - \mathbf{3}R_1$ ,                                                                                         |
| johnston-linear-matrix-algebra-F02261 | 234  | 0.871 | $R_{\{3\}}-\mathbf{4}R_{\{2\}}$                                                                     | $R_3 - \mathbf{4}R_2$                                                                    | and $R_3 - \mathbf{4}R_2$ .                                                                                                          |
| johnston-linear-matrix-algebra-F02262 | 234  | 1.000 | $\ell_{i,\ j}$                                                                                      | $\ell_{i,j}$                                                                             | Here, $\ell_{i,j}$ refers to                                                                                                         |
| johnston-linear-matrix-algebra-F02263 | 234  | 1.000 | $j=1$                                                                                               | $j = 1$                                                                                  | If $j = 1$ then the sum                                                                                                              |
| johnston-linear-matrix-algebra-F02264 | 234  | 0.882 | $\mathbf{a}_{\{1\}}^T=\mathbf{u}_{\{1\}}^T$                                                         | $\mathbf{a}_1^T = \mathbf{u}_1^T$                                                        | (so $\mathbf{a}_1^T = \mathbf{u}_1^T$ ).                                                                                             |
| johnston-linear-matrix-algebra-F02265 | 234  | 1.000 | $n(n+1)\ / \ 2$                                                                                     | $n(n+1)/2$                                                                               | specified by the $n(n+1)/2$ potentially non-zero entries in each of $L$ and                                                          |
| johnston-linear-matrix-algebra-F02266 | 234  | 1.000 | $n(n+1)=n^{\{2\}}+n$                                                                                | $n(n+1) = n^2 + n$                                                                       | $U$ , for a total of $n(n+1) = n^2 + n$ numbers, so it seems reasonable that we                                                      |
| johnston-linear-matrix-algebra-F02267 | 234  | 1.000 | $n^{\{2\}}$                                                                                         | $n^2$                                                                                    | can freely choose $n$ of those numbers while still recovering the $n^2$ entries                                                      |
| johnston-linear-matrix-algebra-F02268 | 234  | 1.000 | $[U\ \mid\ E]\left(R_{\{2\}}+2\ R_{\{1\}},\ R_{\{3\}}-R_{\{1\}}\right).$                            | $[U \mid E] (R_2 + 2R_1, R_3 - R_1$                                                      | that we used to put $[A \mid I]$ into row echelon form $[U \mid E] (R_2 + 2R_1, R_3 - R_1,$                                          |
| johnston-linear-matrix-algebra-F02269 | 234  | 1.000 | $R_{\{3\}}-3\ R_{\{2\}}$                                                                            | $R_3 - 3R_2$                                                                             | and $R_3 - 3R_2$ ). This is not a coincidence—roughly speaking, it is a result of                                                    |
| johnston-linear-matrix-algebra-F02270 | 234  | 1.000 | $A,\ U\ \in\ \mathcal{M}_{\{m\},\ n}$                                                               | $A, U \in \mathcal{M}_{m,n}$                                                             | Suppose $A, U \in \mathcal{M}_{m,n}$ . A unit lower triangular matrix $L \in \mathcal{M}_m$ is such that                             |
| johnston-linear-matrix-algebra-F02271 | 234  | 1.000 | $\mathbf{a}_{\{i\}}^T$                                                                              | $\mathbf{a}_i^T$                                                                         | In particular, if we let $\mathbf{a}_i^T$ and $\mathbf{u}_i^T$ denote the $i$ -th rows of $A$ and $U$ , respectively,                |
| johnston-linear-matrix-algebra-F02272 | 234  | 1.000 | $\mathbf{u}_{\{i\}}^T$                                                                              | $\mathbf{u}_i^T$                                                                         | In particular, if we let $\mathbf{a}_i^T$ and $\mathbf{u}_i^T$ denote the $i$ -th rows of $A$ and $U$ , respectively,                |
| johnston-linear-matrix-algebra-F02273 | 234  | 1.000 | $R_{\{i\}}+\ell_{i,\ j}\ R_{\{j\}}$                                                                 | $R_i + \ell_{i,j}R_j$                                                                    | tions of the form $R_i + \ell_{i,j}R_j$ (with $j < i$ ) that converts $U$ into $A$ . For example,                                    |
| johnston-linear-matrix-algebra-F02274 | 235  | 1.000 | $R_{\{i\}}-\ell_{i,\ j}\ R_{\{j\}}$                                                                 | $R_i - \ell_{i,j}R_j$                                                                    | is $R_i - \ell_{i,j}R_j$ .                                                                                                           |
| johnston-linear-matrix-algebra-F02275 | 235  | 1.000 | $R_{\{2\}}-2\ R_{\{1\}},\ R_{\{3\}}+3\ R_{\{1\}}$                                                   | $R_2 - 2R_1, R_3 + 3R_1$                                                                 | $R_2 - 2R_1, R_3 + 3R_1$ ,                                                                                                           |
| johnston-linear-matrix-algebra-F02276 | 235  | 1.000 | $R_{\{4\}}+R_{\{1\}}$                                                                               | $R_4 + R_1$                                                                              | and $R_4 + R_1$ , so the                                                                                                             |
| johnston-linear-matrix-algebra-F02277 | 235  | 1.000 | $\mathbf{a}_{\{m\}}^T$                                                                              | $\mathbf{a}_m^T$                                                                         | which is exactly $\mathbf{a}_m^T$ , via Equation (2.D.2). Similar sequences of row operations                                        |

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| johnston-linear-matrix-algebra-F02278 | 235  | 1.000 | $R_{m-1}$                                                   | $R_{m-1}$                                            | that involves adding a multiple of $R_{m-1}$ , then all row operations that involve                                                                                               |
| johnston-linear-matrix-algebra-F02279 | 235  | 1.000 | $R_{m-2}$                                                   | $R_{m-2}$                                            | adding a multiple of $R_{m-2}$ , and so on. In particular, $U$ is transformed into $A$ via                                                                                        |
| johnston-linear-matrix-algebra-F02280 | 236  | 1.000 | $i > j$                                                     | $i > j$                                              | operations of the form $R_i + cR_j$ with $i > j$ .                                                                                                                                |
| johnston-linear-matrix-algebra-F02281 | 236  | 1.000 | $L=I$                                                       | $L = I$                                              | just choose $L = I$ and $U = A$ ) but requires a swap row operation (or an operation                                                                                              |
| johnston-linear-matrix-algebra-F02282 | 236  | 1.000 | $U=A$                                                       | $U = A$                                              | just choose $L = I$ and $U = A$ ) but requires a swap row operation (or an operation                                                                                              |
| johnston-linear-matrix-algebra-F02283 | 236  | 1.000 | $i < j$                                                     | $i < j$                                              | of the form $R_i + cR_j$ for some $i < j$ instead of $i > j$ ) to put it in row echelon                                                                                           |
| johnston-linear-matrix-algebra-F02284 | 236  | 1.000 | $L \ U \ \mathbf{x}=\mathbf{b}$                             | $LU\mathbf{x} = \mathbf{b}$                          | the coefficient matrix of the linear system $A\mathbf{x} = \mathbf{b}$ . Then $LU\mathbf{x} = \mathbf{b}$ , which is a                                                            |
| johnston-linear-matrix-algebra-F02285 | 236  | 1.000 | $L \ \mathbf{y}=\mathbf{b}$                                 | $L\mathbf{y} = \mathbf{b}$                           | <ul style="list-style-type: none"> <li>First, solve the linear system <math>L\mathbf{y} = \mathbf{b}</math> for the vector <math>\mathbf{y}</math>. This linear system</li> </ul> |
| johnston-linear-matrix-algebra-F02286 | 236  | 1.000 | $U \ \mathbf{x}=\mathbf{y}$                                 | $U\mathbf{x} = \mathbf{y}$                           | <ul style="list-style-type: none"> <li>Next, solve the linear system <math>U\mathbf{x} = \mathbf{y}</math> for the vector <math>\mathbf{x}</math>. This linear system</li> </ul>  |
| johnston-linear-matrix-algebra-F02287 | 237  | 1.000 | $A \in$                                                     | $A \in$                                              | shape of the linear systems saves us a lot of work in the end. In particular, if $A \in$                                                                                          |
| johnston-linear-matrix-algebra-F02288 | 237  | 1.000 | $2n^3 / 3$                                                  | $2n^3/3$                                             | elimination requires roughly $2n^3/3$ operations (i.e., additions, subtractions,                                                                                                  |
| johnston-linear-matrix-algebra-F02289 | 237  | 1.000 | $L \ \mathbf{y} =$                                          | $L\mathbf{y} =$                                      | multiplications, and/or divisions), whereas each of the triangular systems $L\mathbf{y} =$                                                                                        |
| johnston-linear-matrix-algebra-F02290 | 237  | 1.000 | $2n^2$                                                      | $2n^2$                                               | these two triangular systems via roughly $2n^2$ operations is considerably quicker                                                                                                |
| johnston-linear-matrix-algebra-F02291 | 237  | 1.000 | $\mathbf{y}=(0, 2, -6, -15)$                                | $\mathbf{y} = (0, 2, -6, -15)$                       | so $\mathbf{y} = (0, 2, -6, -15)$ . Next, we solve the upper triangular system $U\mathbf{x} = \mathbf{y}$ :                                                                       |
| johnston-linear-matrix-algebra-F02292 | 238  | 0.951 | $\mathbf{x}=(w, x, y, z)=(22, -9, 3, 5)$                    | $\mathbf{x} = (w, x, y, z) = (22, -9, 3, 5)$         | $\mathbf{x} = (w, x, y, z) = (22, -9, 3, 5)$ .                                                                                                                                    |
| johnston-linear-matrix-algebra-F02293 | 238  | 0.593 | $\{(0, 3), (1, -1), (2, -3), (3, 9)\}$                      | $\{(0, 3), (1, -1), (2, -3), (3, 9)\}$               | a) $\{(0, 3), (1, -1), (2, -3), (3, 9)\}$                                                                                                                                         |
| johnston-linear-matrix-algebra-F02294 | 238  | 1.000 | $\{(0, -1), (1, -1), (2, -1), (3, 5)\}$                     | $\{(0, -1), (1, -1), (2, -1), (3, 5)\}$              | b) $\{(0, -1), (1, -1), (2, -1), (3, 5)\}$                                                                                                                                        |
| johnston-linear-matrix-algebra-F02295 | 238  | 1.000 | $p(x)=c_3x^3+c_2x^2+c_1x+c_0$                               | $p(x) = c_3x^3 + c_2x^2 + c_1x + c_0$                | a) If the polynomial that we seek is $p(x) = c_3x^3 + c_2x^2 + c_1x + c_0$ , then                                                                                                 |
| johnston-linear-matrix-algebra-F02296 | 238  | 1.000 | $p(0)=3, p(1)=0, p(2)=1$                                    | $p(0) = 3, p(1) = 0, p(2) = 1$                       | the equations $p(0) = 3, p(1) = 0, p(2) = 1$ , and $p(3) = 18$ can be                                                                                                             |
| johnston-linear-matrix-algebra-F02297 | 238  | 1.000 | $p(3)=18$                                                   | $p(3) = 18$                                          | the equations $p(0) = 3, p(1) = 0, p(2) = 1$ , and $p(3) = 18$ can be                                                                                                             |
| johnston-linear-matrix-algebra-F02298 | 239  | 1.000 | $L \ \mathbf{y}=\mathbf{b}=(3, -1, 3, 9)$                   | $L\mathbf{y} = \mathbf{b} = (3, -1, 3, 9)$           | Then solving the lower triangular linear system $L\mathbf{y} = \mathbf{b} = (3, -1, 3, 9)$                                                                                        |
| johnston-linear-matrix-algebra-F02299 | 239  | 1.000 | $\mathbf{y}=(3, -4, 2, 12)$                                 | $\mathbf{y} = (3, -4, 2, 12)$                        | gives $\mathbf{y} = (3, -4, 2, 12)$ , and solving the upper triangular linear sys-                                                                                                |
| johnston-linear-matrix-algebra-F02300 | 239  | 1.000 | $U \ \mathbf{x}=\mathbf{y}=(3, -4, 2, 12)$                  | $U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$          | tem $U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$ gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$ .                                                                      |
| johnston-linear-matrix-algebra-F02301 | 239  | 1.000 | $\mathbf{x}=\left(c_0, c_1, c_2, c_3\right)=(3, -1, -5, 2)$ | $\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$ | tem $U\mathbf{x} = \mathbf{y} = (3, -4, 2, 12)$ gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (3, -1, -5, 2)$ .                                                                      |
| johnston-linear-matrix-algebra-F02302 | 239  | 1.000 | $x=0, 1, 2$                                                 | $x = 0, 1, 2$                                        | straightforward to verify by plugging $x = 0, 1, 2$ and 3 into it, and it                                                                                                         |
| johnston-linear-matrix-algebra-F02303 | 239  | 1.000 | $L \ \mathbf{y}=\mathbf{b}=(-1, -1, -1, 5)$                 | $L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$         | ear system $L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$ , which gives $\mathbf{y} = (-1, 0, 0, 6)$ .                                                                              |
| johnston-linear-matrix-algebra-F02304 | 239  | 1.000 | $\mathbf{y}=(-1, 0, 0, 6)$                                  | $\mathbf{y} = (-1, 0, 0, 6)$                         | ear system $L\mathbf{y} = \mathbf{b} = (-1, -1, -1, 5)$ , which gives $\mathbf{y} = (-1, 0, 0, 6)$ .                                                                              |

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| johnston-linear-matrix-algebra-F02305 | 239  | ■ 1.000 | $U \, \mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$                    | $U\mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$                       | Solving the upper triangular linear system $U\mathbf{x} = \mathbf{y} = (-1, 0, 0, 6)$                                                             |
| johnston-linear-matrix-algebra-F02306 | 239  | ■ 1.000 | $\mathbf{x} = \left( c_0, c_1, c_2, c_3 \right) = (-1, 2, -3, 1)$ | $\mathbf{x} = (c_0, c_1, c_2, c_3) = (-1, 2, -3, 1)$             | then gives $\mathbf{x} = (c_0, c_1, c_2, c_3) = (-1, 2, -3, 1)$ , so the (unique) degree-                                                         |
| johnston-linear-matrix-algebra-F02307 | 239  | ■ 0.817 | $(0, -1), (1, -1), (2, -1)$                                       | $(0, -1), (1, -1), (2, -1)$                                      | goes through the four points $(0, -1), (1, -1), (2, -1)$ , and $(3, 5)$ :                                                                         |
| johnston-linear-matrix-algebra-F02308 | 239  | ■ 0.817 | $(3, 5)$                                                          | $(3, 5)$                                                         | goes through the four points $(0, -1), (1, -1), (2, -1)$ , and $(3, 5)$ :                                                                         |
| johnston-linear-matrix-algebra-F02309 | 240  | ■ 1.000 | $\mathbf{x} = A^{-1} \mathbf{b}, \mathbf{y} = A^{-1} \mathbf{c}$  | $\mathbf{x} = A^{-1} \mathbf{b}, \mathbf{y} = A^{-1} \mathbf{c}$ | $\mathbf{x} = A^{-1} \mathbf{b}$ , $\mathbf{y} = A^{-1} \mathbf{c}$ , and $\mathbf{z} = A^{-1} \mathbf{d}$ . Both this approach and the one based |
| johnston-linear-matrix-algebra-F02310 | 240  | ■ 1.000 | $\mathbf{z} = A^{-1} \mathbf{d}$                                  | $\mathbf{z} = A^{-1} \mathbf{d}$                                 | $\mathbf{x} = A^{-1} \mathbf{b}$ , $\mathbf{y} = A^{-1} \mathbf{c}$ , and $\mathbf{z} = A^{-1} \mathbf{d}$ . Both this approach and the one based |
| johnston-linear-matrix-algebra-F02311 | 240  | ■ 1.000 | $2n^3$                                                            | $2n^3$                                                           | quires roughly $2n^3$ operations (additions, subtractions, multiplica-                                                                            |
| johnston-linear-matrix-algebra-F02312 | 241  | ■ 1.000 | $A = E^{-1} R$                                                    | $A = E^{-1} R$                                                   | until it is in row echelon form $[R \mid E]$ and noting that $A = E^{-1} R$ :                                                                     |
| johnston-linear-matrix-algebra-F02313 | 241  | ■ 1.000 | $E^{-1}$                                                          | $E^{-1}$                                                         | decomposition). On the other hand, $E^{-1}$ is not lower triangular, but its rows                                                                 |
| johnston-linear-matrix-algebra-F02314 | 242  | ■ 1.000 | $P = I$                                                           | $P = I$                                                          | $P = I$ then we get                                                                                                                               |
| johnston-linear-matrix-algebra-F02315 | 242  | ■ 0.793 | $0)$                                                              | $0)$                                                             | 0). In this case, we can write                                                                                                                    |
| johnston-linear-matrix-algebra-F02316 | 242  | ■ 0.999 | $A = E^{-1} U = P L U$                                            | $A = E^{-1} U = PLU$                                             | It follows that $A = E^{-1} U = PLU$ , where $P$ is a permutation matrix, $L$ is                                                                  |
| johnston-linear-matrix-algebra-F02317 | 242  | ■ 1.000 | $A = P L U$                                                       | $A = PLU$                                                        | the form $A = PLU$ , where $P \in \mathcal{M}_m$ is a permutation matrix, $L \in \mathcal{M}_m$ is                                                |
| johnston-linear-matrix-algebra-F02318 | 242  | ■ 1.000 | $P \in \mathcal{M}_n$                                             | $P \in \mathcal{M}_n$                                            | matrices $P \in \mathcal{M}_n$ are special for the fact that are not only invertible, but in                                                      |
| johnston-linear-matrix-algebra-F02319 | 242  | ■ 1.000 | $P^{-1} = P^T$                                                    | $P^{-1} = P^T$                                                   | fact their inverse is simply their transpose: $P^{-1} = P^T$ . We prove this fact in                                                              |
| johnston-linear-matrix-algebra-F02320 | 242  | ■ 1.000 | $[U \mid E]$                                                      | $[U \mid E]$                                                     | If Gaussian elimination row-reduces $[A \mid I]$ to $[U \mid E]$ , where $U$ is a row                                                             |
| johnston-linear-matrix-algebra-F02321 | 242  | ■ 1.000 | $E^{-1} = P L$                                                    | $E^{-1} = PL$                                                    | triangular matrix $L$ such that $E^{-1} = PL$ . In particular,                                                                                    |
| johnston-linear-matrix-algebra-F02322 | 243  | ■ 1.000 | $P_* \in \mathcal{M}_m$                                           | $P_* \in \mathcal{M}_m$                                          | into lower triangular form. That is, we can find a permutation matrix $P_* \in \mathcal{M}_m$                                                     |
| johnston-linear-matrix-algebra-F02323 | 243  | ■ 1.000 | $L_* \in \mathcal{M}_m$                                           | $L_* \in \mathcal{M}_m$                                          | and a lower triangular matrix $L_* \in \mathcal{M}_m$ such that $E = L_* P_*$ . Taking the inverse                                                |
| johnston-linear-matrix-algebra-F02324 | 243  | ■ 1.000 | $E = L_* P_*$                                                     | $E = L_* P_*$                                                    | and a lower triangular matrix $L_* \in \mathcal{M}_m$ such that $E = L_* P_*$ . Taking the inverse                                                |
| johnston-linear-matrix-algebra-F02325 | 243  | ■ 1.000 | $P_*^{-1} = P_*^T$                                                | $P_*^{-1} = P_*^T$                                               | Since $P_*^{-1} = P_*^T$ is also a permutation matrix, and $L_*^{-1}$ is also lower triangular,                                                   |
| johnston-linear-matrix-algebra-F02326 | 243  | ■ 1.000 | $L_*^{-1}$                                                        | $L_*^{-1}$                                                       | Since $P_*^{-1} = P_*^T$ is also a permutation matrix, and $L_*^{-1}$ is also lower triangular,                                                   |
| johnston-linear-matrix-algebra-F02327 | 243  | ■ 1.000 | $P = P_*^{-1}$                                                    | $P = P_*^{-1}$                                                   | it follows that we can set $P = P_*^{-1}$ and $L = L_*^{-1}$ to complete the proof.                                                               |
| johnston-linear-matrix-algebra-F02328 | 243  | ■ 1.000 | $L = L_*^{-1}$                                                    | $L = L_*^{-1}$                                                   | it follows that we can set $P = P_*^{-1}$ and $L = L_*^{-1}$ to complete the proof.                                                               |
| johnston-linear-matrix-algebra-F02329 | 243  | ■ 1.000 | $E P$                                                             | $EP$                                                             | The reason this that works is simply that $EP$ is lower triangular (which is                                                                      |
| johnston-linear-matrix-algebra-F02330 | 243  | ■ 1.000 | $P^{-1} E^{-1}$                                                   | $P^{-1} E^{-1}$                                                  | tion), so its inverse $P^{-1} E^{-1}$ is lower triangular too. If we define $L = P^{-1} E^{-1}$                                                   |
| johnston-linear-matrix-algebra-F02331 | 243  | ■ 1.000 | $L = P^{-1} E^{-1}$                                               | $L = P^{-1} E^{-1}$                                              | tion), so its inverse $P^{-1} E^{-1}$ is lower triangular too. If we define $L = P^{-1} E^{-1}$                                                   |

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| johnston-linear-matrix-algebra-F02332 | 243  | ■ 0.999 | $A=\left[\begin{array}{cccc}1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1\end{array}\right]$ | $A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$ | Construct a PLU decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra-F02333 | 243  | ■ 1.000 | $P^T$                                                                                                           | $P^T$                                                                                                | We then choose $P^T$ to have ones in its entries corresponding to the topmost                                                                      |
| johnston-linear-matrix-algebra-F02334 | 244  | ■ 1.000 | $\mathbf{c}=P^T \mathbf{b}$                                                                                     | $\mathbf{c} = P^T \mathbf{b}$                                                                        | so $\mathbf{c} = P^T \mathbf{b}$ means                                                                                                             |
| johnston-linear-matrix-algebra-F02335 | 244  | ■ 1.000 | $P \mathbf{c}=\mathbf{b}$                                                                                       | $P\mathbf{c} = \mathbf{b}$                                                                           | $P\mathbf{c} = \mathbf{b}$ .                                                                                                                       |
| johnston-linear-matrix-algebra-F02336 | 244  | ■ 1.000 | $PLU \mathbf{x}=\mathbf{b}$                                                                                     | $PLU\mathbf{x} = \mathbf{b}$                                                                         | decomposition $A = PLU$ , then $PLU\mathbf{x} = \mathbf{b}$ is a linear system that we can solve                                                   |
| johnston-linear-matrix-algebra-F02337 | 244  | ■ 1.000 | $L \mathbf{y}=\mathbf{c}$                                                                                       | $L\mathbf{y} = \mathbf{c}$                                                                           | <ul style="list-style-type: none"> <li>Solve the linear system <math>L\mathbf{y} = \mathbf{c}</math> for <math>\mathbf{y}</math>.</li> </ul>       |
| johnston-linear-matrix-algebra-F02338 | 245  | ■ 1.000 | $\mathbf{b}=(0,1,2,3)$                                                                                          | $\mathbf{b} = (0, 1, 2, 3)$                                                                          | to solve the linear system $A\mathbf{x} = \mathbf{b}$ , where $\mathbf{b} = (0, 1, 2, 3)$ .                                                        |
| johnston-linear-matrix-algebra-F02339 | 245  | ■ 1.000 | $P, L$                                                                                                          | $P, L$                                                                                               | We let $P, L$ , and $U$ be the matrices that we computed in Example 2.D.5                                                                          |
| johnston-linear-matrix-algebra-F02340 | 245  | ■ 1.000 | $\mathbf{y}=(0,2,1,3/2)$                                                                                        | $\mathbf{y} = (0, 2, 1, 3/2)$                                                                        | so $\mathbf{y} = (0, 2, 1, 3/2)$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F02341 | 245  | ■ 1.000 | $\mathbf{x}=(-3,2,0,1)$                                                                                         | $\mathbf{x} = (-3, 2, 0, 1)$                                                                         | so the unique solution of this linear system is $\mathbf{x} = (-3, 2, 0, 1)$ .                                                                     |
| johnston-linear-matrix-algebra-F02342 | 245  | ■ 0.840 | $1 \leq k \leq r$                                                                                               | $1 \leq k \leq r$                                                                                    | invertible for all $1 \leq k \leq r$ then $A$ has a unit LU decomposition.                                                                         |
| johnston-linear-matrix-algebra-F02343 | 245  | ■ 0.835 | $a_{1,1}$                                                                                                       | $a_{1,1}$                                                                                            | result is trivial, since any $1 \times 1$ matrix $A = [a_{1,1}]$ can be written in the form                                                        |
| johnston-linear-matrix-algebra-F02344 | 245  | ■ 1.000 | $L=[1]$                                                                                                         | $L = [1]$                                                                                            | $A = LU$ with $L = [1]$ and $U = [a_{1,1}]$ .                                                                                                      |
| johnston-linear-matrix-algebra-F02345 | 245  | ■ 1.000 | $U=\left[a_{1,1}\right]$                                                                                        | $U = [a_{1,1}]$                                                                                      | $A = LU$ with $L = [1]$ and $U = [a_{1,1}]$ .                                                                                                      |
| johnston-linear-matrix-algebra-F02346 | 246  | ■ 1.000 | $LU=\left(LD^{-1}\right)(DU)$                                                                                   | $LU = (LD^{-1})(DU)$                                                                                 | $LU = (LD^{-1})(DU)$                                                                                                                               |
| johnston-linear-matrix-algebra-F02347 | 246  | ■ 0.743 | $B \in \mathcal{M}_{n-1}$                                                                                       | $B \in \mathcal{M}_{n-1}$                                                                            | Since $B \in \mathcal{M}_{n-1}$ , we know by the inductive hypothesis that it has a unit LU                                                        |
| johnston-linear-matrix-algebra-F02348 | 246  | ■ 1.000 | $B=LU$                                                                                                          | $B = LU$                                                                                             | decomposition $B = LU$ . Our goal is to extend this to a unit LU decomposition                                                                     |
| johnston-linear-matrix-algebra-F02349 | 246  | ■ 1.000 | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$                                                                   | $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$                                                        | of $A$ . To this end, we want to find column vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^{n-1}$ and a scalar                                    |
| johnston-linear-matrix-algebra-F02350 | 246  | ■ 1.000 | $d \in \mathbb{R}$                                                                                              | $d \in \mathbb{R}$                                                                                   | $d \in \mathbb{R}$ such that                                                                                                                       |
| johnston-linear-matrix-algebra-F02351 | 246  | ■ 1.000 | $c=d+\mathbf{x}^T \mathbf{y}$                                                                                   | $c = d + \mathbf{x}^T \mathbf{y}$                                                                    | chose $L$ and $U$ in the first place. The bottom-right block requires $c = d + \mathbf{x}^T \mathbf{y}$ ,                                          |
| johnston-linear-matrix-algebra-F02352 | 246  | ■ 1.000 | $d=c-\mathbf{x}^T \mathbf{y}$                                                                                   | $d = c - \mathbf{x}^T \mathbf{y}$                                                                    | which can be satisfied just by choosing $d = c - \mathbf{x}^T \mathbf{y}$ . The top-right block                                                    |
| johnston-linear-matrix-algebra-F02353 | 246  | ■ 0.989 | $L \mathbf{y}=\mathbf{v}$                                                                                       | $L\mathbf{y} = \mathbf{v}$                                                                           | requires us to find a $\mathbf{y}$ such that $L\mathbf{y} = \mathbf{v}$ , which can be be done straightforwardly                                   |
| johnston-linear-matrix-algebra-F02354 | 246  | ■ 1.000 | $\mathbf{y}=L^{-1} \mathbf{v}$                                                                                  | $\mathbf{y} = L^{-1} \mathbf{v}$                                                                     | $\mathbf{y} = L^{-1} \mathbf{v}$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F02355 | 246  | ■ 1.000 | $\mathbf{w}^T=\mathbf{x}^T U$                                                                                   | $\mathbf{w}^T = \mathbf{x}^T U$                                                                      | block is correct: $\mathbf{w}^T = \mathbf{x}^T U$ . To this end, we split into two separate cases:                                                 |
| johnston-linear-matrix-algebra-F02356 | 246  | ■ 0.999 | $U=BL^{-1}$                                                                                                     | $U = BL^{-1}$                                                                                        | <b>Case 1:</b> $B$ is invertible. In this case, $B = LU$ , so $U = BL^{-1}$ is also invertible, so                                                 |

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| johnston-linear-matrix-algebra-F02357 | 246  | 1.000 | $\mathbf{x}=\left(U^{-1}\right)^T \mathbf{w}$                                                                                                                                                                                                                                | $\mathbf{x}=\left(U^{-1}\right)^T \mathbf{w}$                                                                                                                                                                                                                                | we can simply choose $\mathbf{x}=\left(U^{-1}\right)^T \mathbf{w}$ .                                                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F02358 | 246  | 1.000 | $\operatorname{rank}(A)=$                                                                                                                                                                                                                                                    | $\operatorname{rank}(A)=$                                                                                                                                                                                                                                                    | (by one of the hypotheses of the theorem), we conclude that $\operatorname{rank}(A)=$                                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02359 | 246  | 0.987 | $\mathbf{w}^T$                                                                                                                                                                                                                                                               | $\mathbf{w}^T$                                                                                                                                                                                                                                                               | $\operatorname{rank}(B)=r$ . It follows that $\mathbf{w}^T$ must be a linear combination of the rows                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F02360 | 246  | 1.000 | $\operatorname{rank}(A)>r$                                                                                                                                                                                                                                                   | $\operatorname{rank}(A)>r$                                                                                                                                                                                                                                                   | which in turn would imply $\operatorname{rank}(A)>r$ .                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02361 | 246  | 1.000 | $\mathbf{x}^T$                                                                                                                                                                                                                                                               | $\mathbf{x}^T$                                                                                                                                                                                                                                                               | equation $\mathbf{w}^T=\mathbf{x}^T U$ says—in particular, we choose the entries of $\mathbf{x}^T$ to                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02362 | 246  | 0.997 | $d, \mathbf{x}$                                                                                                                                                                                                                                                              | $d, \mathbf{x}$                                                                                                                                                                                                                                                              | Now that we know how to construct each of $d, \mathbf{x}$ , and $\mathbf{y}$ , the inductive step and                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02363 | 246  | 0.523 | $A=\left[\begin{array}{ccc} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right]$                                                                                                                                                                                       | $A=\left[\begin{array}{ccc} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right]$                                                                                                                                                                                       | Show that the matrix $A=\left[\begin{array}{ccc} 3 & -2 & -1 \\ 3 & -1 & 4 \\ 1 & 5 & 2 \end{array}\right]$ has a unit LU decomposition.                                                                                                                                         |
| johnston-linear-matrix-algebra-F02364 | 247  | 0.852 | $\left[\begin{array}{ll} 1 & 2 \\ 2 & 5 \end{array}\right]$                                                                                                                                                                                                                  | $\left[\begin{array}{ll} 1 & 2 \\ 2 & 5 \end{array}\right]$                                                                                                                                                                                                                  | *(a) $\left[\begin{array}{ll} 1 & 2 \\ 2 & 5 \end{array}\right]$                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02365 | 247  | 1.000 | $\left[\begin{array}{cc} 2 & -1 \\ 4 & 1 \end{array}\right]$                                                                                                                                                                                                                 | $\left[\begin{array}{cc} 2 & -1 \\ 4 & 1 \end{array}\right]$                                                                                                                                                                                                                 | (b) $\left[\begin{array}{cc} 2 & -1 \\ 4 & 1 \end{array}\right]$                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02366 | 247  | 0.995 | $\left[\begin{array}{ccc} 3 & 1 & 2 \\ -3 & -3 & -1 \end{array}\right]$                                                                                                                                                                                                      | $\left[\begin{array}{ccc} 3 & 1 & 2 \\ -3 & -3 & -1 \end{array}\right]$                                                                                                                                                                                                      | *(c) $\left[\begin{array}{ccc} 3 & 1 & 2 \\ -3 & -3 & -1 \end{array}\right]$                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02367 | 247  | 1.000 | $\left[\begin{array}{ccc} 1 & 2 & -1 \\ -1 & -3 & -2 \\ 3 & 5 & -8 \end{array}\right]$                                                                                                                                                                                       | $\left[\begin{array}{ccc} 1 & 2 & -1 \\ -1 & -3 & -2 \\ 3 & 5 & -8 \end{array}\right]$                                                                                                                                                                                       | (f) $\left[\begin{array}{ccc} 1 & 2 & -1 \\ -1 & -3 & -2 \\ 3 & 5 & -8 \end{array}\right]$                                                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02368 | 247  | 0.546 | $\left[\begin{array}{ccc} 1 & -4 & 5 \\ 3 & -9 & 8 \\ -2 & 5 & -2 \end{array}\right]$                                                                                                                                                                                        | $\left[\begin{array}{ccc} 1 & -4 & 5 \\ 3 & -9 & 8 \\ -2 & 5 & -2 \end{array}\right]$                                                                                                                                                                                        | *(g) $\left[\begin{array}{ccc} 1 & -4 & 5 \\ 3 & -9 & 8 \\ -2 & 5 & -2 \end{array}\right]$                                                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02369 | 247  | 1.000 | $\left[\begin{array}{cccc} 2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10 \end{array}\right]$                                                                                                                                                                        | $\left[\begin{array}{cccc} 2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10 \end{array}\right]$                                                                                                                                                                        | (h) $\left[\begin{array}{cccc} 2 & -1 & 4 & 3 \\ -4 & 4 & -7 & -6 \\ 6 & -7 & 12 & 10 \end{array}\right]$                                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02370 | 247  | 0.947 | $\left[\begin{array}{ll} 1 & 0 \\ 2 & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 3 \\ 0 & 1 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=\left[\begin{array}{l} 1 \\ -1 \end{array}\right]$                                                       | $\left[\begin{array}{ll} 1 & 0 \\ 2 & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 3 \\ 0 & 1 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=\left[\begin{array}{l} 1 \\ -1 \end{array}\right]$                                                       | *(a) $\left[\begin{array}{ll} 1 & 0 \\ 2 & 1 \end{array}\right]\left[\begin{array}{ll} 2 & 3 \\ 0 & 1 \end{array}\right]\left[\begin{array}{l} x \\ y \end{array}\right]=\left[\begin{array}{l} 1 \\ -1 \end{array}\right]$                                                      |
| johnston-linear-matrix-algebra-F02371 | 247  | 0.518 | $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{array}\right]\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \\ z \end{array}\right]=\left[\begin{array}{l} 3 \\ 1 \\ 2 \end{array}\right]$ | $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{array}\right]\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \\ z \end{array}\right]=\left[\begin{array}{l} 3 \\ 1 \\ 2 \end{array}\right]$ | (b) $\left[\begin{array}{ccc} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{array}\right]\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]\left[\begin{array}{l} x \\ y \\ z \end{array}\right]=\left[\begin{array}{l} 3 \\ 1 \\ 2 \end{array}\right]$ |

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| johnston-linear-matrix-algebra-F02372 | 247  | 0.556 | $\left[\begin{array}{lll}0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0\end{array}\right]\left[\begin{array}{l}0 \\ 1 \\ 0\end{array}\right]\left[\begin{array}{l}1 \\ 0 \\ 0\end{array}\right]\left[\begin{array}{l}1 \\ 2 \\ 3\end{array}\right]\left[\begin{array}{l}x \\ y \\ z\end{array}\right]=\left[\begin{array}{l}1 \\ 2 \\ 3\end{array}\right]$ | $\left[\begin{array}{ccc}0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0\end{array}\right]\left[\begin{array}{ccc}1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1\end{array}\right]\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1\end{array}\right]\left[\begin{array}{c}x \\ y \\ z\end{array}\right]=\left[\begin{array}{c}1 \\ 2 \\ 3\end{array}\right]$ | <b>(c)</b> $\left[\begin{array}{ccc}0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0\end{array}\right]\left[\begin{array}{ccc}1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1\end{array}\right]\left[\begin{array}{ccc}1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1\end{array}\right]\left[\begin{array}{c}x \\ y \\ z\end{array}\right]=\left[\begin{array}{c}1 \\ 2 \\ 3\end{array}\right]$ |
| johnston-linear-matrix-algebra-F02373 | 247  | 0.917 | $\left[\begin{array}{llll}1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1\end{array}\right]\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0\end{array}\right]\left[\begin{array}{c}w \\ x \\ y \\ z\end{array}\right]=\left[\begin{array}{c}1 \\ 1 \\ 1 \\ 1\end{array}\right]$                        | $\left[\begin{array}{cccc}1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1\end{array}\right]\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0\end{array}\right]\left[\begin{array}{c}w \\ x \\ y \\ z\end{array}\right]=\left[\begin{array}{c}1 \\ 1 \\ 1 \\ 1\end{array}\right]$                         | <b>(d)</b> $\left[\begin{array}{cccc}1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 \\ 1 & 1 & 3 & 1\end{array}\right]\left[\begin{array}{ccc}2 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0\end{array}\right]\left[\begin{array}{c}w \\ x \\ y \\ z\end{array}\right]=\left[\begin{array}{c}1 \\ 1 \\ 1 \\ 1\end{array}\right]$                         |
| johnston-linear-matrix-algebra-F02374 | 247  | 0.661 | $\left[\begin{array}{ll}0 & 2 \\ 1 & 3\end{array}\right]$                                                                                                                                                                                                                                                                                       | $\left[\begin{array}{cc}0 & 2 \\ 1 & 3\end{array}\right]$                                                                                                                                                                                                                                                                                        | <b>(a)</b> $\left[\begin{array}{cc}0 & 2 \\ 1 & 3\end{array}\right]$                                                                                                                                                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02375 | 247  | 0.950 | $\left[\begin{array}{lll}1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3\end{array}\right]$                                                                                                                                                                                                                                                                 | $\left[\begin{array}{ccc}1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3\end{array}\right]$                                                                                                                                                                                                                                                                  | <b>(c)</b> $\left[\begin{array}{ccc}1 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3\end{array}\right]$                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02376 | 247  | 0.999 | $\left[\begin{array}{llll}0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4\end{array}\right]$                                                                                                                                                                                                                                                    | $\left[\begin{array}{cccc}0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4\end{array}\right]$                                                                                                                                                                                                                                                     | <b>(d)</b> $\left[\begin{array}{cccc}0 & 2 & 3 & 3 \\ 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4\end{array}\right]$                                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02377 | 248  | 0.584 | $\left[\begin{array}{ll}1 & 3 \\ 3 & 2\end{array}\right]$                                                                                                                                                                                                                                                                                       | $\left[\begin{array}{cc}1 & 3 \\ 3 & 2\end{array}\right]$                                                                                                                                                                                                                                                                                        | <b>(a)</b> $\left[\begin{array}{cc}1 & 3 \\ 3 & 2\end{array}\right]$                                                                                                                                                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02378 | 248  | 1.000 | $\left[\begin{array}{cc}0 & 1 \\ 1 & -2\end{array}\right]$                                                                                                                                                                                                                                                                                      | $\left[\begin{array}{cc}0 & 1 \\ 1 & -2\end{array}\right]$                                                                                                                                                                                                                                                                                       | <b>(b)</b> $\left[\begin{array}{cc}0 & 1 \\ 1 & -2\end{array}\right]$                                                                                                                                                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02379 | 248  | 0.997 | $\left[\begin{array}{llll}4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2\end{array}\right]$                                                                                                                                                                                                                                   | $\left[\begin{array}{cccc}4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2\end{array}\right]$                                                                                                                                                                                                                                    | <b>(d)</b> $\left[\begin{array}{cccc}4 & 5 & 5 & 0 \\ 3 & 4 & 3 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 3 & 4 & 2\end{array}\right]$                                                                                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F02380 | 248  | 1.000 | $A=L D U$                                                                                                                                                                                                                                                                                                                                       | $A=LDU$                                                                                                                                                                                                                                                                                                                                          | is any factorization of the form $A=LDU$ , where $L$ and $U$                                                                                                                                                                                                                                                                                                |
| johnston-linear-matrix-algebra-F02381 | 248  | 1.000 | $A=L D L^T$                                                                                                                                                                                                                                                                                                                                     | $A=LDL^T$                                                                                                                                                                                                                                                                                                                                        | then it is unique and has the form $A=LDL^T$ .                                                                                                                                                                                                                                                                                                              |
| johnston-linear-matrix-algebra-F02382 | 248  | 0.994 | $A=\left[\begin{array}{ll}0 & 0 \\ 1 & 1\end{array}\right]$                                                                                                                                                                                                                                                                                     | $A=\left[\begin{array}{cc}0 & 0 \\ 1 & 1\end{array}\right]$                                                                                                                                                                                                                                                                                      | <b>**2.D.13</b> Consider the matrix $A=\left[\begin{array}{cc}0 & 0 \\ 1 & 1\end{array}\right]$ .                                                                                                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02383 | 249  | 1.000 | $A \mathbf{e}_1=(2,1)$                                                                                                                                                                                                                                                                                                                          | $A\mathbf{e}_1=(2,1)$                                                                                                                                                                                                                                                                                                                            | $A\mathbf{e}_1=(2,1)$ , which is                                                                                                                                                                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02384 | 249  | 1.000 | $A \mathbf{e}_2=(1,2)$                                                                                                                                                                                                                                                                                                                          | $A\mathbf{e}_2=(1,2)$                                                                                                                                                                                                                                                                                                                            | $A\mathbf{e}_2=(1,2)$ , which is                                                                                                                                                                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02385 | 249  | 1.000 | $A=\left[\begin{array}{ll}2 & 1 \\ 1 & 2\end{array}\right]$                                                                                                                                                                                                                                                                                     | $A=\left[\begin{array}{cc}2 & 1 \\ 1 & 2\end{array}\right]$                                                                                                                                                                                                                                                                                      | <b>Figure 3.1:</b> The matrix $A=\left[\begin{array}{cc}2 & 1 \\ 1 & 2\end{array}\right]$ acting as a linear transformation on $\mathbb{R}^2$ .                                                                                                                                                                                                             |
| johnston-linear-matrix-algebra-F02386 | 249  | 1.000 | $\mathbf{v}_1=(1,1)$                                                                                                                                                                                                                                                                                                                            | $\mathbf{v}_1=(1,1)$                                                                                                                                                                                                                                                                                                                             | and instead consider how it acts on the input grid defined by $\mathbf{v}_1=(1,1)$ and                                                                                                                                                                                                                                                                      |
| johnston-linear-matrix-algebra-F02387 | 249  | 1.000 | $\mathbf{v}_2=(-1,1)$                                                                                                                                                                                                                                                                                                                           | $\mathbf{v}_2=(-1,1)$                                                                                                                                                                                                                                                                                                                            | $\mathbf{v}_2=(-1,1)$ , then this matrix transforms $\mathbb{R}^2$ in the manner shown in Figure 3.2,                                                                                                                                                                                                                                                       |

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|-----------------------------------------|------|-------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-augmented-F02388 | 250  | 1.000 | $(1,0)$                                                                           | $(1,0)$                                                                               | away either of $(1,0)$                                                                                                                              |
| johnston-linear-matrix-augmented-F02389 | 250  | 0.993 | $\left\{\mathbf{v}_1, \mathbf{v}_2\right\}$                                       | $\{\mathbf{v}_1, \mathbf{v}_2\}$                                                      | different input grid), we investigate how to find a particular basis ( $\{\mathbf{v}_1, \mathbf{v}_2\}$ in                                          |
| johnston-linear-matrix-augmented-F02390 | 250  | 1.000 | $\mathbb{R}^2=\operatorname{span}((1,0),(0,1),(1,1))$                             | $\mathbb{R}^2 = \operatorname{span}((1,0),(0,1),(1,1))$                               | For example, it is the case that $\mathbb{R}^2 = \operatorname{span}((1,0),(0,1),(1,1))$ , so every                                                 |
| johnston-linear-matrix-augmented-F02391 | 250  | 0.985 | $(1,0),(0,1)$                                                                     | $(1,0),(0,1)$                                                                         | vector $(x,y) \in \mathbb{R}^2$ can be written as a linear combination of $(1,0),(0,1)$ , and                                                       |
| johnston-linear-matrix-augmented-F02392 | 250  | 1.000 | $\{(1,0),(0,1),(1,1)\}$                                                           | $\{(1,0),(0,1),(1,1)\}$                                                               | that the set $\{(1,0),(0,1),(1,1)\}$ is not linearly independent (and thus not a                                                                    |
| johnston-linear-matrix-augmented-F02393 | 250  | 1.000 | $(1,1)$                                                                           | $(1,1)$                                                                               | basis). If we throw away the vector $(1,1)$ then we arrive at the standard basis                                                                    |
| johnston-linear-matrix-augmented-F02394 | 250  | 0.975 | $\{(1,0),(0,1)\}$                                                                 | $\{(1,0),(0,1)\}$                                                                     | $\{(1,0),(0,1)\}$ , and linear combinations of the standard basis vectors <i>are</i> unique                                                         |
| johnston-linear-matrix-augmented-F02395 | 251  | 0.998 | $(2,1)=2(1,0)+(0,1)$                                                              | $(2,1) = 2(1,0) + (0,1)$                                                              | standard basis vectors is simply $(2,1) = 2(1,0) + (0,1)$ .                                                                                         |
| johnston-linear-matrix-augmented-F02396 | 252  | 1.000 | $\mathbb{R}^n, B=\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right\}$ | $\mathbb{R}^n, B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$               | Suppose $\mathcal{S}$ is a subspace of $\mathbb{R}^n$ , $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis of $\mathcal{S}$ , and |
| johnston-linear-matrix-augmented-F02397 | 252  | 1.000 | $\left\{\mathbf{e}_1, \mathbf{e}_2\right\}$                                       | $\{\mathbf{e}_1, \mathbf{e}_2\}$                                                      | that order does not matter for bases (e.g., $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are the                           |
| johnston-linear-matrix-augmented-F02398 | 252  | 1.000 | $\left\{\mathbf{e}_2, \mathbf{e}_1\right\}$                                       | $\{\mathbf{e}_2, \mathbf{e}_1\}$                                                      | that order does not matter for bases (e.g., $\{\mathbf{e}_1, \mathbf{e}_2\}$ and $\{\mathbf{e}_2, \mathbf{e}_1\}$ are the                           |
| johnston-linear-matrix-augmented-F02399 | 252  | 0.992 | $\left(c_1, c_2\right)$                                                           | $(c_1, c_2)$                                                                          | $(c_1, c_2)$ and $(c_2, c_1)$ are different vectors). In other words, we want to                                                                    |
| johnston-linear-matrix-augmented-F02400 | 252  | 0.992 | $\left(c_2, c_1\right)$                                                           | $(c_2, c_1)$                                                                          | $(c_1, c_2)$ and $(c_2, c_1)$ are different vectors). In other words, we want to                                                                    |
| johnston-linear-matrix-augmented-F02401 | 252  | 1.000 | $\mathcal{S}=\mathbb{R}^n$                                                        | $\mathcal{S} = \mathbb{R}^n$                                                          | When working with the standard basis of $\mathbb{R}^n$ (i.e., when $\mathcal{S} = \mathbb{R}^n$ and $B =$                                           |
| johnston-linear-matrix-augmented-F02402 | 252  | 0.999 | $c_1, c_2, \ldots, c_n$                                                           | $c_1, c_2, \dots, c_n$                                                                | to the standard basis. This is because the unique scalars $c_1, c_2, \dots, c_n$ for which                                                          |
| johnston-linear-matrix-augmented-F02403 | 252  | 1.000 | $\mathbf{v}=\left(c_1, c_2, \ldots, c_n\right)=\left[\mathbf{v}\right]_B$         | $\mathbf{v} = (c_1, c_2, \dots, c_n) = [\mathbf{v}]_B$                                | $\mathbf{v} = (c_1, c_2, \dots, c_n) = [\mathbf{v}]_B$ . The true utility of coordinates and coordinate vectors                                     |
| johnston-linear-matrix-augmented-F02404 | 252  | 1.000 | $\mathbf{v}=(5,1), B=\{(2,1),(1,2)\}, \mathcal{S}=\mathbb{R}^2$                   | $\mathbf{v} = (5,1), B = \{(2,1),(1,2)\}, \mathcal{S} = \mathbb{R}^2$                 | a) $\mathbf{v} = (5,1), B = \{(2,1),(1,2)\}, \mathcal{S} = \mathbb{R}^2$                                                                            |
| johnston-linear-matrix-augmented-F02405 | 252  | 1.000 | $\mathbf{v}=(5,4,3), B=\{(1,2,1),(2,1,1)\}, \mathcal{S}=\operatorname{span}(B)$   | $\mathbf{v} = (5,4,3), B = \{(1,2,1),(2,1,1)\}, \mathcal{S} = \operatorname{span}(B)$ | b) $\mathbf{v} = (5,4,3), B = \{(1,2,1),(2,1,1)\}, \mathcal{S} = \operatorname{span}(B)$                                                            |
| johnston-linear-matrix-augmented-F02406 | 253  | 1.000 | $(5,1)$                                                                           | $(5,1)$                                                                               | columns in a matrix with $(5,1)$ as the augmented right-hand side:                                                                                  |
| johnston-linear-matrix-augmented-F02407 | 253  | 1.000 | $c_1=3, c_2=-1$                                                                   | $c_1 = 3, c_2 = -1$                                                                   | see that the unique solution is $c_1 = 3, c_2 = -1$ , so $[\mathbf{v}]_B = (3, -1)$ :                                                               |
| johnston-linear-matrix-augmented-F02408 | 253  | 1.000 | $[\mathbf{v}]_B=(3,-1)$                                                           | $[\mathbf{v}]_B = (3, -1)$                                                            | see that the unique solution is $c_1 = 3, c_2 = -1$ , so $[\mathbf{v}]_B = (3, -1)$ :                                                               |
| johnston-linear-matrix-augmented-F02409 | 253  | 0.997 | $c_1=1, c_2=2$                                                                    | $c_1 = 1, c_2 = 2$                                                                    | see that the unique solution is $c_1 = 1, c_2 = 2$ , so $[\mathbf{v}]_B = (1, 2)$ :                                                                 |
| johnston-linear-matrix-augmented-F02410 | 253  | 0.997 | $[\mathbf{v}]_B=(1,2)$                                                            | $[\mathbf{v}]_B = (1, 2)$                                                             | see that the unique solution is $c_1 = 1, c_2 = 2$ , so $[\mathbf{v}]_B = (1, 2)$ :                                                                 |
| johnston-linear-matrix-augmented-F02411 | 253  | 1.000 | $\left[\mathbf{v}_j\right]_B=\mathbf{e}_j$                                        | $[\mathbf{v}]_B = \mathbf{e}_j$                                                       | It is worth noting that if $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a basis then $[\mathbf{v}]_B = \mathbf{e}_j$ (the           |
| johnston-linear-matrix-augmented-F02412 | 253  | 1.000 | $1 \leq j \leq k$                                                                 | $1 \leq j \leq k$                                                                     | $j$ -th standard basis vector) for all $1 \leq j \leq k$ , since the unique linear combination                                                      |
| johnston-linear-matrix-augmented-F02413 | 253  | 1.000 | $\mathbf{v}_j$                                                                    | $\mathbf{v}_j$                                                                        | of $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ resulting in $\mathbf{v}_j$ is quite trivial:                                              |
| johnston-linear-matrix-augmented-F02414 | 253  | 1.000 | $\mathbf{e}_1, \mathbf{v}_2$                                                      | $\mathbf{e}_1, \mathbf{v}_2$                                                          | so that $\mathbf{v}_1$ looks like $\mathbf{e}_1$ , $\mathbf{v}_2$ looks like $\mathbf{e}_2$ , and so on (see Figure 3.4). It is worth               |

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| johnston-linear-matrix-algebra-F02415 | 254  | 0.844 | $\mathcal{S}=\operatorname{span}((1,2,1),(2,1,1))$        | $\mathcal{S} = \operatorname{span}((1,2,1),(2,1,1))$            | $\mathbb{R}^3$ , it also lives in the 2-dimensional plane $\mathcal{S} = \operatorname{span}((1,2,1),(2,1,1))$ and |
| johnston-linear-matrix-algebra-F02416 | 254  | 1.000 | $\mathcal{S}=\mathbb{R}^2$                                | $\mathcal{S} = \mathbb{R}^2$                                    | $\mathcal{S} = \mathbb{R}^2$ and the basis is $B = \{(2,1),(1,2)\}$ , as in Example 3.1.1(a).                      |
| johnston-linear-matrix-algebra-F02417 | 254  | 1.000 | $B=\{(2,1),(1,2)\}$                                       | $B = \{(2,1),(1,2)\}$                                           | $\mathcal{S} = \mathbb{R}^2$ and the basis is $B = \{(2,1),(1,2)\}$ , as in Example 3.1.1(a).                      |
| johnston-linear-matrix-algebra-F02418 | 254  | 1.000 | $\mathbb{R}^{85}$                                         | $\mathbb{R}^{85}$                                               | of $\mathbb{R}^{85}$ (for example), then we can use coordinate vectors to describe vectors in                      |
| johnston-linear-matrix-algebra-F02419 | 254  | 1.000 | $[\mathbf{v}]_B$                                          | $[\mathbf{v}]_B$                                                | the coordinate vector $[\mathbf{v}]_B$ of the vector $\mathbf{v} = (2,1,-3,1,2) \in \operatorname{range}(A)$ :     |
| johnston-linear-matrix-algebra-F02420 | 254  | 1.000 | $\mathbf{v}=(2,1,-3,1,2) \in \operatorname{range}(A)$     | $\mathbf{v} = (2,1,-3,1,2) \in \operatorname{range}(A)$         | the coordinate vector $[\mathbf{v}]_B$ of the vector $\mathbf{v} = (2,1,-3,1,2) \in \operatorname{range}(A)$ :     |
| johnston-linear-matrix-algebra-F02421 | 255  | 1.000 | $c_1=2$                                                   | $c_1 = 2$                                                       | It follows that $c_1 = 2$ and $c_2 = -3$ , so $[\mathbf{v}]_B = (2,-3)$ .                                          |
| johnston-linear-matrix-algebra-F02422 | 255  | 1.000 | $c_2=-3$                                                  | $c_2 = -3$                                                      | It follows that $c_1 = 2$ and $c_2 = -3$ , so $[\mathbf{v}]_B = (2,-3)$ .                                          |
| johnston-linear-matrix-algebra-F02423 | 255  | 1.000 | $[\mathbf{v}]_B=(2,-3)$                                   | $[\mathbf{v}]_B = (2,-3)$                                       | It follows that $c_1 = 2$ and $c_2 = -3$ , so $[\mathbf{v}]_B = (2,-3)$ .                                          |
| johnston-linear-matrix-algebra-F02424 | 255  | 0.991 | $\mathbf{v}=(2,1,-3,1,2)$                                 | $\mathbf{v} = (2,1,-3,1,2)$                                     | 2-dimensional (i.e., its rank is 2), so representing the vector $\mathbf{v} = (2,1,-3,1,2)$                        |
| johnston-linear-matrix-algebra-F02425 | 255  | 1.000 | $[\mathbf{v}+\mathbf{w}]_B=[\mathbf{v}]_B+[\mathbf{w}]_B$ | $[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$ | a) $[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$ , and                                           |
| johnston-linear-matrix-algebra-F02426 | 255  | 0.997 | $[c \mathbf{v}]_B=c[\mathbf{v}]_B$                        | $[c\mathbf{v}]_B = c[\mathbf{v}]_B$                             | b) $[c\mathbf{v}]_B = c[\mathbf{v}]_B$ .                                                                           |
| johnston-linear-matrix-algebra-F02427 | 255  | 1.000 | $B=\{(1,2,0,2,1),(0,1,1,1,0)\}$                           | $B = \{(1,2,0,2,1),(0,1,1,1,0)\}$                               | Let $B = \{(1,2,0,2,1),(0,1,1,1,0)\}$ be a basis of the subspace                                                   |
| johnston-linear-matrix-algebra-F02428 | 255  | 1.000 | $\mathbf{w}=(1,0,-2,0,1)$                                 | $\mathbf{w} = (1,0,-2,0,1)$                                     | and suppose $\mathbf{v} = (2,1,-3,1,2)$ and $\mathbf{w} = (1,0,-2,0,1)$ . Compute the fol-                         |
| johnston-linear-matrix-algebra-F02429 | 255  | 1.000 | $[2 \mathbf{v}-5 \mathbf{w}]_B$                           | $[2\mathbf{v} - 5\mathbf{w}]_B$                                 | a) $[2\mathbf{v} - 5\mathbf{w}]_B$                                                                                 |
| johnston-linear-matrix-algebra-F02430 | 255  | 1.000 | $2[\mathbf{v}]_B-5[\mathbf{w}]_B$                         | $2[\mathbf{v}]_B - 5[\mathbf{w}]_B$                             | b) $2[\mathbf{v}]_B - 5[\mathbf{w}]_B$                                                                             |
| johnston-linear-matrix-algebra-F02431 | 255  | 1.000 | $2 \mathbf{v}-5 \mathbf{w}$                               | $2\mathbf{v} - 5\mathbf{w}$                                     | Next, we can compute the coordinate vector of $2\mathbf{v} - 5\mathbf{w}$ in the same                              |
| johnston-linear-matrix-algebra-F02432 | 256  | 1.000 | $\mathbf{v}_1=(1,0.002)$                                  | $\mathbf{v}_1 = (1,0.002)$                                      | if $\mathbf{v}_1 = (1,0.002)$ and                                                                                  |
| johnston-linear-matrix-algebra-F02433 | 256  | 1.000 | $\mathbf{v}_2=(1,0.001)$                                  | $\mathbf{v}_2 = (1,0.001)$                                      | $\mathbf{v}_2 = (1,0.001)$ then                                                                                    |
| johnston-linear-matrix-algebra-F02434 | 256  | 1.000 | $[(-1,1)]_B=$                                             | $[(-1,1)]_B =$                                                  | $[(-1,1)]_B =$                                                                                                     |
| johnston-linear-matrix-algebra-F02435 | 256  | 1.000 | $[2 \mathbf{v}-5 \mathbf{w}]_B=(-1,4)$                    | $[2\mathbf{v} - 5\mathbf{w}]_B = (-1,4)$                        | so $[2\mathbf{v} - 5\mathbf{w}]_B = (-1,4)$ .                                                                      |
| johnston-linear-matrix-algebra-F02436 | 256  | 1.000 | $[\mathbf{w}]_B$                                          | $[\mathbf{w}]_B$                                                | just need to compute $[\mathbf{w}]_B$ :                                                                            |
| johnston-linear-matrix-algebra-F02437 | 256  | 1.000 | $[\mathbf{w}]_B=(1,-2)$                                   | $[\mathbf{w}]_B = (1,-2)$                                       | It follows that $[\mathbf{w}]_B = (1,-2)$ , so we see that                                                         |
| johnston-linear-matrix-algebra-F02438 | 256  | 1.000 | $\mathbf{v}_1=(1,0.2)$                                    | $\mathbf{v}_1 = (1,0.2)$                                        | this, suppose $\mathbf{v}_1 = (1,0.2)$ and $\mathbf{v}_2 = (1,0.1)$ , and consider the basis $B =$                 |
| johnston-linear-matrix-algebra-F02439 | 256  | 1.000 | $\mathbf{v}_2=(1,0.1)$                                    | $\mathbf{v}_2 = (1,0.1)$                                        | this, suppose $\mathbf{v}_1 = (1,0.2)$ and $\mathbf{v}_2 = (1,0.1)$ , and consider the basis $B =$                 |
| johnston-linear-matrix-algebra-F02440 | 257  | 1.000 | $\mathbf{v} \neq \mathbf{w} \in B$                        | $\mathbf{v} \neq \mathbf{w} \in B$                              | a) $\mathbf{v} \cdot \mathbf{w} = 0$ for all $\mathbf{v} \neq \mathbf{w} \in B$ , and                              |
| johnston-linear-matrix-algebra-F02441 | 257  | 1.000 | $\mathbf{v} \in B$                                        | $\mathbf{v} \in B$                                              | b) $\ \mathbf{v}\  = 1$ for all $\mathbf{v} \in B$ .                                                               |

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| johnston-linear-matrix-algebra-F02442 | 257  | ■ 0.998 | $\ \mathbf{v}\ =\left\ [\mathbf{v}]_B\right\ $                                               | $\ \mathbf{v}\  = \ \mathbf{v}\ _B$                                            | If $B$ is an orthonormal basis then $\ \mathbf{v}\  = \ \mathbf{v}\ _B$ , so $[\mathbf{v}]_B$ cannot be large                                                |
| johnston-linear-matrix-algebra-F02443 | 257  | ■ 1.000 | $B=\{(1,1),(1,-1)\}$                                                                         | $B = \{(1,1), (1,-1)\}$                                                        | Let $B = \{(1,1), (1,-1)\}$ and $C = \{(3,1), (0,1)\}$ be bases of $\mathbb{R}^2$ , and                                                                      |
| johnston-linear-matrix-algebra-F02444 | 257  | ■ 1.000 | $C=\{(3,1),(0,1)\}$                                                                          | $C = \{(3,1), (0,1)\}$                                                         | Let $B = \{(1,1), (1,-1)\}$ and $C = \{(3,1), (0,1)\}$ be bases of $\mathbb{R}^2$ , and                                                                      |
| johnston-linear-matrix-algebra-F02445 | 257  | ■ 1.000 | $[\mathbf{v}]_B=(5,1)$                                                                       | $[\mathbf{v}]_B = (5,1)$                                                       | suppose that $[\mathbf{v}]_B = (5,1)$ . Compute $[\mathbf{v}]_C$ .                                                                                           |
| johnston-linear-matrix-algebra-F02446 | 257  | ■ 1.000 | $[\mathbf{v}]_C$                                                                             | $[\mathbf{v}]_C$                                                               | suppose that $[\mathbf{v}]_B = (5,1)$ . Compute $[\mathbf{v}]_C$ .                                                                                           |
| johnston-linear-matrix-algebra-F02447 | 257  | ■ 1.000 | $\mathbf{v}=5(1,1)+(1,-1)=(6,4)$                                                             | $\mathbf{v} = 5(1,1) + (1,-1) = (6,4)$                                         | Since $[\mathbf{v}]_B = (5,1)$ we know that $\mathbf{v} = 5(1,1) + (1,-1) = (6,4)$ . Next,                                                                   |
| johnston-linear-matrix-algebra-F02448 | 257  | ■ 1.000 | $\mathbf{v}=(6,4)$                                                                           | $\mathbf{v} = (6,4)$                                                           | we simply represent $\mathbf{v} = (6,4)$ in the basis $C$ by finding scalars $c_1$ and $c_2$                                                                 |
| johnston-linear-matrix-algebra-F02449 | 257  | ■ 1.000 | $(6,4)=c_1(3,1)+c_2(0,1)$                                                                    | $(6,4) = c_1(3,1) + c_2(0,1)$                                                  | so that $(6,4) = c_1(3,1) + c_2(0,1)$ :                                                                                                                      |
| johnston-linear-matrix-algebra-F02450 | 257  | ■ 1.000 | $[\mathbf{v}]_C=(2,2)$                                                                       | $[\mathbf{v}]_C = (2,2)$                                                       | It follows that $c_1 = 2$ and $c_2 = 2$ , so we conclude that $[\mathbf{v}]_C = (2,2)$ , as                                                                  |
| johnston-linear-matrix-algebra-F02451 | 258  | ■ 1.000 | $P_{C \leftarrow B}$                                                                         | $P_{C \leftarrow B}$                                                           | notation $P_{C \leftarrow B}$                                                                                                                                |
| johnston-linear-matrix-algebra-F02452 | 258  | ■ 1.000 | $P_{B \rightarrow C}$                                                                        | $P_{B \rightarrow C}$                                                          | instead of $P_{B \rightarrow C}$ , but                                                                                                                       |
| johnston-linear-matrix-algebra-F02453 | 258  | ■ 1.000 | $\left[\mathbf{v}_1\right]_C,\left[\mathbf{v}_2\right]_C,\ldots,\left[\mathbf{v}_k\right]_C$ | $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C, \dots, [\mathbf{v}_k]_C$                  | whose columns are the coordinate vectors $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C, \dots, [\mathbf{v}_k]_C$ :                                                     |
| johnston-linear-matrix-algebra-F02454 | 259  | ■ 1.000 | $\mathcal{S}=\operatorname{span}(C)$                                                         | $\mathcal{S} = \operatorname{span}(C)$                                         | that $\mathcal{S} = \operatorname{span}(C)$ .                                                                                                                |
| johnston-linear-matrix-algebra-F02455 | 259  | ■ 1.000 | $P_{C \leftarrow B}[\mathbf{v}]_B=[\mathbf{v}]_C$                                            | $P_{C \leftarrow B}[\mathbf{v}]_B = [\mathbf{v}]_C$                            | a) $P_{C \leftarrow B}[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$ , and                                                            |
| johnston-linear-matrix-algebra-F02456 | 259  | ■ 1.000 | $\left(P_{C \leftarrow B}\right)^{-1}=P_{B \leftarrow C}$                                    | $(P_{C \leftarrow B})^{-1} = P_{B \leftarrow C}$                               | b) $P_{C \leftarrow B}$ is invertible and $(P_{C \leftarrow B})^{-1} = P_{B \leftarrow C}$ .                                                                 |
| johnston-linear-matrix-algebra-F02457 | 259  | ■ 1.000 | $\mathbf{v}=c_1 \mathbf{v}_1+c_2 \mathbf{v}_2+\cdots+c_k \mathbf{v}_k$                       | $\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$ | see that property (a) holds, suppose $\mathbf{v} \in \mathcal{S}$ and write $\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k$ , |
| johnston-linear-matrix-algebra-F02458 | 259  | ■ 1.000 | $[\mathbf{v}]_B=\left(c_1, c_2, \ldots, c_k\right)$                                          | $[\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$                                      | so that $[\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$ . We can then directly compute                                                                             |
| johnston-linear-matrix-algebra-F02459 | 259  | ■ 1.000 | $P \in \mathcal{M}_k$                                                                        | $P \in \mathcal{M}_k$                                                          | suppose $P \in \mathcal{M}_k$ is any matrix for which $P[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$ . For every                    |
| johnston-linear-matrix-algebra-F02460 | 259  | ■ 1.000 | $P[\mathbf{v}]_B=[\mathbf{v}]_C$                                                             | $P[\mathbf{v}]_B = [\mathbf{v}]_C$                                             | suppose $P \in \mathcal{M}_k$ is any matrix for which $P[\mathbf{v}]_B = [\mathbf{v}]_C$ for all $\mathbf{v} \in \mathcal{S}$ . For every                    |
| johnston-linear-matrix-algebra-F02461 | 259  | ■ 0.999 | $\mathbf{v}=\mathbf{v}_j$                                                                    | $\mathbf{v} = \mathbf{v}_j$                                                    | $1 \leq j \leq k$ , if $\mathbf{v} = \mathbf{v}_j$ then we see that $[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$ (the $j$ -th standard basis           |
| johnston-linear-matrix-algebra-F02462 | 259  | ■ 0.999 | $[\mathbf{v}]_B=\left[\mathbf{v}_j\right]_B=\mathbf{e}_j$                                    | $[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$                             | $1 \leq j \leq k$ , if $\mathbf{v} = \mathbf{v}_j$ then we see that $[\mathbf{v}]_B = [\mathbf{v}_j]_B = \mathbf{e}_j$ (the $j$ -th standard basis           |
| johnston-linear-matrix-algebra-F02463 | 259  | ■ 1.000 | $P[\mathbf{v}]_B=P \mathbf{e}_j$                                                             | $P[\mathbf{v}]_B = P \mathbf{e}_j$                                             | vector), so $P[\mathbf{v}]_B = P \mathbf{e}_j$ is the $j$ -th column of $P$ . On the other hand, it is also the                                              |
| johnston-linear-matrix-algebra-F02464 | 259  | ■ 1.000 | $P[\mathbf{v}]_B=[\mathbf{v}]_C=\left[\mathbf{v}_j\right]_C$                                 | $P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$                          | case that $P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$ . The $j$ -th column of $P$ thus equals $[\mathbf{v}_j]_C$ for each                          |
| johnston-linear-matrix-algebra-F02465 | 259  | ■ 1.000 | $\left[\mathbf{v}_j\right]_C$                                                                | $[\mathbf{v}_j]_C$                                                             | case that $P[\mathbf{v}]_B = [\mathbf{v}]_C = [\mathbf{v}_j]_C$ . The $j$ -th column of $P$ thus equals $[\mathbf{v}_j]_C$ for each                          |
| johnston-linear-matrix-algebra-F02466 | 259  | ■ 1.000 | $P=P_{C \leftarrow B}$                                                                       | $P = P_{C \leftarrow B}$                                                       | $1 \leq j \leq k$ , so $P = P_{C \leftarrow B}$ .                                                                                                            |
| johnston-linear-matrix-algebra-F02467 | 259  | ■ 1.000 | $P_{B \leftarrow C} P_{C \leftarrow B}$                                                      | $P_{B \leftarrow C} P_{C \leftarrow B}$                                        | It follows that the $j$ -th column of $P_{B \leftarrow C} P_{C \leftarrow B}$ equals $\mathbf{e}_j$ for each $1 \leq j \leq k$ ,                             |
| johnston-linear-matrix-algebra-F02468 | 259  | ■ 1.000 | $P_{B \leftarrow C} P_{C \leftarrow B}=I$                                                    | $P_{B \leftarrow C} P_{C \leftarrow B} = I$                                    | so $P_{B \leftarrow C} P_{C \leftarrow B} = I$ . Theorem 2.2.7 then tells us that $P_{C \leftarrow B}$ is invertible with                                    |

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| johnston-linear-matrix-algebra-F02469 | 259  | ■ 0.999 | $[\mathbf{v}]_B =$                                              | $[\mathbf{v}]_B =$                                       | Equation (3.1.1) we used the change-of-basis matrix $P_{C \leftarrow B}$ to change $[\mathbf{v}]_B =$                                              |
| johnston-linear-matrix-algebra-F02470 | 259  | ■ 1.000 | $[\mathbf{w}]_B = (-1, 4)$                                      | $[\mathbf{w}]_B = (-1, 4)$                               | have $[\mathbf{w}]_B = (-1, 4)$ then we can re-use that same change-of-basis matrix to                                                             |
| johnston-linear-matrix-algebra-F02471 | 259  | ■ 1.000 | $B = \left\{ \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \right\}$ | $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$       | Then $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ are bases of $S = \text{span}(B)$ . |
| johnston-linear-matrix-algebra-F02472 | 259  | ■ 1.000 | $C = \left\{ \mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3 \right\}$ | $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$       | Then $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $C = \{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ are bases of $S = \text{span}(B)$ . |
| johnston-linear-matrix-algebra-F02473 | 259  | ■ 1.000 | $[\mathbf{v}]_B = (1, 2, 3)$                                    | $[\mathbf{v}]_B = (1, 2, 3)$                             | b) Compute $[\mathbf{v}]_C$ if $[\mathbf{v}]_B = (1, 2, 3)$ .                                                                                      |
| johnston-linear-matrix-algebra-F02474 | 260  | ■ 1.000 | $\mathbf{v}_1 =$                                                | $\mathbf{v}_1 =$                                         | $\mathbf{v}_1 =$                                                                                                                                   |
| johnston-linear-matrix-algebra-F02475 | 260  | ■ 0.604 | $c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + c_3 \mathbf{w}_3$        | $c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + c_3 \mathbf{w}_3$ | $c_1 \mathbf{w}_1 + c_2 \mathbf{w}_2 + c_3 \mathbf{w}_3$ .                                                                                         |
| johnston-linear-matrix-algebra-F02476 | 260  | ■ 1.000 | $[\mathbf{w}]_C$                                                | $[\mathbf{w}]_C$                                         | c) Compute $[\mathbf{w}]_C$ if $[\mathbf{w}]_B = (2, 0, -1)$ .                                                                                     |
| johnston-linear-matrix-algebra-F02477 | 260  | ■ 1.000 | $[\mathbf{w}]_B = (2, 0, -1)$                                   | $[\mathbf{w}]_B = (2, 0, -1)$                            | c) Compute $[\mathbf{w}]_C$ if $[\mathbf{w}]_B = (2, 0, -1)$ .                                                                                     |
| johnston-linear-matrix-algebra-F02478 | 260  | ■ 1.000 | $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C$                            | $[\mathbf{v}_1]_C, [\mathbf{v}_2]_C$                     | a) We must first compute $[\mathbf{v}_1]_C$ , $[\mathbf{v}_2]_C$ , and $[\mathbf{v}_3]_C$ (i.e., we must write                                     |
| johnston-linear-matrix-algebra-F02479 | 260  | ■ 1.000 | $[\mathbf{v}_3]_C$                                              | $[\mathbf{v}_3]_C$                                       | a) We must first compute $[\mathbf{v}_1]_C$ , $[\mathbf{v}_2]_C$ , and $[\mathbf{v}_3]_C$ (i.e., we must write                                     |
| johnston-linear-matrix-algebra-F02480 | 260  | ■ 1.000 | $\mathbf{w}_1, \mathbf{w}_2$                                    | $\mathbf{w}_1, \mathbf{w}_2$                             | $\mathbf{v}_1$ , $\mathbf{v}_2$ , and $\mathbf{v}_3$ each as a linear combination of $\mathbf{w}_1$ , $\mathbf{w}_2$ , and $\mathbf{w}_3$ ). For   |
| johnston-linear-matrix-algebra-F02481 | 260  | ■ 1.000 | $\mathbf{w}_3$                                                  | $\mathbf{w}_3$                                           | $\mathbf{v}_1$ , $\mathbf{v}_2$ , and $\mathbf{v}_3$ each as a linear combination of $\mathbf{w}_1$ , $\mathbf{w}_2$ , and $\mathbf{w}_3$ ). For   |
| johnston-linear-matrix-algebra-F02482 | 260  | ■ 1.000 | $c_1 = 1, c_2 = -1, c_3 = 1$                                    | $c_1 = 1, c_2 = -1, c_3 = 1$                             | that $c_1 = 1, c_2 = -1, c_3 = 1$ , so $[\mathbf{v}_1]_C = (1, -1, 1)$ . Similar compu-                                                            |
| johnston-linear-matrix-algebra-F02483 | 260  | ■ 1.000 | $[\mathbf{v}_1]_C = (1, -1, 1)$                                 | $[\mathbf{v}_1]_C = (1, -1, 1)$                          | that $c_1 = 1, c_2 = -1, c_3 = 1$ , so $[\mathbf{v}_1]_C = (1, -1, 1)$ . Similar compu-                                                            |
| johnston-linear-matrix-algebra-F02484 | 260  | ■ 1.000 | $[\mathbf{v}_2]_C = (0, -1, 1)$                                 | $[\mathbf{v}_2]_C = (0, -1, 1)$                          | tations show that $[\mathbf{v}_2]_C = (0, -1, 1)$ and $[\mathbf{v}_3]_C = (0, -2, 1)$ , so                                                         |
| johnston-linear-matrix-algebra-F02485 | 260  | ■ 1.000 | $[\mathbf{v}_3]_C = (0, -2, 1)$                                 | $[\mathbf{v}_3]_C = (0, -2, 1)$                          | tations show that $[\mathbf{v}_2]_C = (0, -1, 1)$ and $[\mathbf{v}_3]_C = (0, -2, 1)$ , so                                                         |
| johnston-linear-matrix-algebra-F02486 | 260  | ■ 1.000 | $\mathbb{R}^k$                                                  | $\mathbb{R}^k$                                           | different ways of making the subspace $S$ look like $\mathbb{R}^k$ , and the change-of-basis                                                       |
| johnston-linear-matrix-algebra-F02487 | 260  | ■ 1.000 | $P_{B \leftarrow C}$                                            | $P_{B \leftarrow C}$                                     | matrices $P_{C \leftarrow B}$ and $P_{B \leftarrow C}$ are linear transformations that convert these two                                           |
| johnston-linear-matrix-algebra-F02488 | 261  | ■ 1.000 | $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$                     | $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$              | transformation $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$ (Theorem 1.4.1):                                                                        |
| johnston-linear-matrix-algebra-F02489 | 261  | ■ 1.000 | $[\mathbf{v}_j]_B$                                              | $[\mathbf{v}_j]_B$                                       | transformation that sends $[\mathbf{v}_j]_B$ to $[\mathbf{v}_j]_C$ for all $1 \leq j \leq k$ .                                                     |
| johnston-linear-matrix-algebra-F02490 | 261  | ■ 1.000 | $E = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$              | $E = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$       | Let $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and $E = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ be bases of $\mathbb{R}^3$ , where   |
| johnston-linear-matrix-algebra-F02491 | 261  | ■ 1.000 | $P_{E \leftarrow B}$                                            | $P_{E \leftarrow B}$                                     | Compute $P_{E \leftarrow B}$ and $P_{B \leftarrow E}$ .                                                                                            |
| johnston-linear-matrix-algebra-F02492 | 261  | ■ 1.000 | $P_{B \leftarrow E}$                                            | $P_{B \leftarrow E}$                                     | Compute $P_{E \leftarrow B}$ and $P_{B \leftarrow E}$ .                                                                                            |
| johnston-linear-matrix-algebra-F02493 | 261  | ■ 0.998 | $[\mathbf{v}]_E = \mathbf{v}$                                   | $[\mathbf{v}]_E = \mathbf{v}$                            | already been done for us (i.e., recall that $[\mathbf{v}]_E = \mathbf{v}$ for all vectors $\mathbf{v}$ ):                                          |
| johnston-linear-matrix-algebra-F02494 | 262  | ■ 0.950 | $m \neq n$                                                      | $m \neq n$                                               | $m \neq n$ ), even with                                                                                                                            |
| johnston-linear-matrix-algebra-F02495 | 262  | ■ 1.000 | $[\mathbf{e}_1]_B, [\mathbf{e}_2]_B$                            | $[\mathbf{e}_1]_B, [\mathbf{e}_2]_B$                     | To construct $P_{B \leftarrow E}$ , we could compute each of $[\mathbf{e}_1]_B$ , $[\mathbf{e}_2]_B$ , and $[\mathbf{e}_3]_B$                      |

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| johnston-linear-matrix-a4602<br>F02496 | 262  | 1.000 | $\left[\mathbf{e}_3\right]_B$                                          | $[\mathbf{e}_3]_B$                                        | To construct $P_{B \leftarrow E}$ , we could compute each of $[\mathbf{e}_1]_B$ , $[\mathbf{e}_2]_B$ , and $[\mathbf{e}_3]_B$                                    |
| johnston-linear-matrix-a4602<br>F02497 | 262  | 1.000 | $P_{B \leftarrow E} = P_{E \leftarrow B}^{-1}$                         | $P_{B \leftarrow E} = P_{E \leftarrow B}^{-1}$            | $P_{B \leftarrow E} = P_{E \leftarrow B}^{-1}$ , so we can just invert the change of basis matrix that we                                                        |
| johnston-linear-matrix-a4602<br>F02498 | 262  | 1.000 | $B = \left\{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \right\}$ | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ | Suppose $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis of $\mathbb{R}^n$ and $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear       |
| johnston-linear-matrix-a4602<br>F02499 | 262  | 1.000 | $[T]_B \in \mathcal{M}_n$                                              | $[T]_B \in \mathcal{M}_n$                                 | transformation. Then there exists a unique matrix $[T]_B \in \mathcal{M}_n$ for which                                                                            |
| johnston-linear-matrix-a4602<br>F02500 | 262  | 1.000 | $[T]_B$                                                                | $[T]_B$                                                   | the coordinate vector of the input and then multiply by the matrix $[T]_B$ (see                                                                                  |
| johnston-linear-matrix-a4602<br>F02501 | 262  | 1.000 | $B = \left\{ \mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n \right\}$ | $B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ | we get if we use the standard basis $B = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ .                                                                  |
| johnston-linear-matrix-a4602<br>F02502 | 262  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$                      | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$         | rem 1.4.1 (just replace the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$        |
| johnston-linear-matrix-a4602<br>F02503 | 262  | 1.000 | $S, T : \mathbb{R}^n \rightarrow \mathbb{R}^n$                         | $S, T : \mathbb{R}^n \rightarrow \mathbb{R}^n$            | $S, T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations then $[S \circ T]_B = [S]_B [T]_B$ , which is                                          |
| johnston-linear-matrix-a4602<br>F02504 | 262  | 1.000 | $[S \circ T]_B = [S]_B [T]_B$                                          | $[S \circ T]_B = [S]_B [T]_B$                             | $S, T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ are linear transformations then $[S \circ T]_B = [S]_B [T]_B$ , which is                                          |
| johnston-linear-matrix-a4602<br>F02505 | 263  | 1.000 | $[T(\mathbf{v})]_B$                                                    | $[T(\mathbf{v})]_B$                                       | $[\mathbf{v}]_B$ to $[T(\mathbf{v})]_B$ . In other words, Theorem 3.1.4 says that we can first compute $T(\mathbf{v})$                                           |
| johnston-linear-matrix-a4602<br>F02506 | 263  | 0.995 | $T(x, y) = (2x + y, x + 2y)$                                           | $T(x, y) = (2x + y, x + 2y)$                              | Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (2x + y, x + 2y)$ . Compute $[T]_B$                                                     |
| johnston-linear-matrix-a4602<br>F02507 | 263  | 1.000 | $B = \left\{ \mathbf{e}_1, \mathbf{e}_2 \right\}$                      | $B = \{\mathbf{e}_1, \mathbf{e}_2\}$                      | a) $B = \{\mathbf{e}_1, \mathbf{e}_2\}$                                                                                                                          |
| johnston-linear-matrix-a4602<br>F02508 | 263  | 1.000 | $T(1, 1)$                                                              | $T(1, 1)$                                                 | b) First, we need to compute $T(1, 1)$ and $T(1, -1)$ :                                                                                                          |
| johnston-linear-matrix-a4602<br>F02509 | 263  | 1.000 | $T(1, -1)$                                                             | $T(1, -1)$                                                | b) First, we need to compute $T(1, 1)$ and $T(1, -1)$ :                                                                                                          |
| johnston-linear-matrix-a4602<br>F02510 | 265  | 1.000 | $P_{C \leftarrow B} [T]_B P_{B \leftarrow C}$                          | $P_{C \leftarrow B} [T]_B P_{B \leftarrow C}$             | <i>Proof.</i> We simply multiply the matrix $P_{C \leftarrow B} [T]_B P_{B \leftarrow C}$ on the right by an arbi-                                               |
| johnston-linear-matrix-a4602<br>F02511 | 265  | 1.000 | $[T]_C$                                                                | $[T]_C$                                                   | However, we know from Theorem 3.1.4 that $[T]_C$ is the <i>unique</i> matrix for which                                                                           |
| johnston-linear-matrix-a4602<br>F02512 | 265  | 1.000 | $[T]_C [\mathbf{v}]_C = [T(\mathbf{v})]_C$                             | $[T]_C [\mathbf{v}]_C = [T(\mathbf{v})]_C$                | $[T]_C [\mathbf{v}]_C = [T(\mathbf{v})]_C$ for all $\mathbf{v} \in \mathbb{R}^n$ , so it follows that $P_{C \leftarrow B} [T]_B P_{B \leftarrow C} = [T]_C$ , as |
| johnston-linear-matrix-a4602<br>F02513 | 265  | 1.000 | $P_{C \leftarrow B} [T]_B P_{B \leftarrow C} = [T]_C$                  | $P_{C \leftarrow B} [T]_B P_{B \leftarrow C} = [T]_C$     | $[T]_C [\mathbf{v}]_C = [T(\mathbf{v})]_C$ for all $\mathbf{v} \in \mathbb{R}^n$ , so it follows that $P_{C \leftarrow B} [T]_B P_{B \leftarrow C} = [T]_C$ , as |
| johnston-linear-matrix-a4602<br>F02514 | 265  | 1.000 | $[T(\mathbf{v})]_C$                                                    | $[T(\mathbf{v})]_C$                                       | are numerous different ways of computing $[T(\mathbf{v})]_C$ from $\mathbf{v}$ , such as:                                                                        |
| johnston-linear-matrix-a4602<br>F02515 | 265  | 1.000 | $B = \{(1, 2, 3), (2, -1, 0), (-1, 1, 2)\}$                            | $B = \{(1, 2, 3), (2, -1, 0), (-1, 1, 2)\}$               | Let $B = \{(1, 2, 3), (2, -1, 0), (-1, 1, 2)\}$ be a basis of $\mathbb{R}^3$ and let $T : \mathbb{R}^3 \rightarrow$                                              |
| johnston-linear-matrix-a4602<br>F02516 | 265  | 1.000 | $T : \mathbb{R}^3 \rightarrow$                                         | $T : \mathbb{R}^3 \rightarrow$                            | Let $B = \{(1, 2, 3), (2, -1, 0), (-1, 1, 2)\}$ be a basis of $\mathbb{R}^3$ and let $T : \mathbb{R}^3 \rightarrow$                                              |
| johnston-linear-matrix-a4602<br>F02517 | 266  | 1.000 | $[T]_E$                                                                | $[T]_E$                                                   | written $[T]$ as $[T]_E$ .                                                                                                                                       |
| johnston-linear-matrix-a4602<br>F02518 | 266  | 1.000 | $[T]_B = P_{B \leftarrow E} [T] P_{E \leftarrow B}$                    | $[T]_B = P_{B \leftarrow E} [T] P_{E \leftarrow B}$       | Theorem 3.1.5 tells us that $[T]_B = P_{B \leftarrow E} [T] P_{E \leftarrow B}$ . Well, we recall from                                                           |
| johnston-linear-matrix-a4602<br>F02519 | 266  | 1.000 | $A = [T]_C$                                                            | $A = [T]_C$                                               | <i>common</i> linear transformation $T$ such that $A = [T]_C$ and $B = [T]_D$ ?                                                                                  |
| johnston-linear-matrix-a4602<br>F02520 | 266  | 1.000 | $B = [T]_D$                                                            | $B = [T]_D$                                               | <i>common</i> linear transformation $T$ such that $A = [T]_C$ and $B = [T]_D$ ?                                                                                  |
| johnston-linear-matrix-a4602<br>F02521 | 266  | 1.000 | $B = P_{D \leftarrow C} A P_{C \leftarrow D}$                          | $B = P_{D \leftarrow C} A P_{C \leftarrow D}$             | we know that $A = [T]_C$ and $B = [T]_D$ if and only if $B = P_{D \leftarrow C} A P_{C \leftarrow D}$ . However,                                                 |
| johnston-linear-matrix-a4602<br>F02522 | 266  | 1.000 | $P_{C \leftarrow D} = P_{D \leftarrow C}^{-1}$                         | $P_{C \leftarrow D} = P_{D \leftarrow C}^{-1}$            | we also know from Theorem 3.1.3(b) that $P_{C \leftarrow D} = P_{D \leftarrow C}^{-1}$ , so we have                                                              |

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| johnston-linear-matrix-algebra-F02523 | 267  | 1.000 | $A=P^{-1}BP$                                                                                                 | $A = PBP^{-1}$                                                                                                       | $A = PBP^{-1}$ . However,                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02524 | 267  | 1.000 | $B=P^{-1}AP$                                                                                                 | $B = P^{-1}AP$                                                                                                       | since $B = P^{-1}AP$ .                                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F02525 | 267  | 1.000 | $B=PAP^{-1}$                                                                                                 | $B = PAP^{-1}$                                                                                                       | is an invertible matrix $P \in \mathcal{M}_n$ such that $B = PAP^{-1}$ . This property is important                                                                                                       |
| johnston-linear-matrix-algebra-F02526 | 267  | 1.000 | $P=\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$                                                            | $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$                                                                  | are similar, since straightforward calculation shows that if $P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ then                                                                                     |
| johnston-linear-matrix-algebra-F02527 | 267  | 1.000 | $A=\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$                                                             | $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$                                                                   | Show that the matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ are not similar.                                                         |
| johnston-linear-matrix-algebra-F02528 | 267  | 1.000 | $B=\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                                                             | $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$                                                                   | Show that the matrices $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ are not similar.                                                         |
| johnston-linear-matrix-algebra-F02529 | 268  | 1.000 | $\operatorname{tr}(A)$                                                                                       | $\operatorname{tr}(A)$                                                                                               | Suppose $A \in \mathcal{M}_n$ is a square matrix. Then the <b>trace</b> of $A$ , denoted by $\operatorname{tr}(A)$ ,                                                                                      |
| johnston-linear-matrix-algebra-F02530 | 268  | 1.000 | $\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}$                                       | $\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}$                                               | b) $\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}$                                                                                                                                 |
| johnston-linear-matrix-algebra-F02531 | 268  | 0.884 | $\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right)=2+5=7$                           | $\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right) = 2 + 5 = 7$                             | a) $\operatorname{tr}\left(\begin{bmatrix} 2 & 2 \\ 4 & 5 \end{bmatrix}\right) = 2 + 5 = 7.$                                                                                                              |
| johnston-linear-matrix-algebra-F02532 | 268  | 1.000 | $\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right)=3-2-1=0$ | $\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right) = 3 - 2 - 1 = 0$ | b) $\operatorname{tr}\left(\begin{bmatrix} 3 & 6 & -3 \\ 0 & -2 & 3 \\ 2 & 4 & -1 \end{bmatrix}\right) = 3 - 2 - 1 = 0.$                                                                                  |
| johnston-linear-matrix-algebra-F02533 | 268  | 1.000 | $\operatorname{tr}(A+B)=\operatorname{tr}(A)+\operatorname{tr}(B)$                                           | $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$                                               | a) $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$                                                                                                                                 |
| johnston-linear-matrix-algebra-F02534 | 268  | 1.000 | $\operatorname{tr}(cA)=c\operatorname{tr}(A)$                                                                | $\operatorname{tr}(cA) = c\operatorname{tr}(A)$                                                                      | b) $\operatorname{tr}(cA) = c\operatorname{tr}(A)$                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02535 | 268  | 1.000 | $\operatorname{tr}(AB)=\operatorname{tr}(BA)$                                                                | $\operatorname{tr}(AB) = \operatorname{tr}(BA)$                                                                      | c) $\operatorname{tr}(AB) = \operatorname{tr}(BA)$                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02536 | 269  | 1.000 | $[AB]_{i,i}$                                                                                                 | $[AB]_{i,i}$                                                                                                         | notation $[AB]_{i,i}$                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02537 | 269  | 1.000 | $(i,i)$                                                                                                      | $(i,i)$                                                                                                              | means the $(i,i)$ -entry                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02538 | 269  | 1.000 | $a_{i,j}b_{j,i}=b_{j,i}a_{i,j}$                                                                              | $a_{i,j}b_{j,i} = b_{j,i}a_{i,j}$                                                                                    | where the central step is accomplished by noting that $a_{i,j}b_{j,i} = b_{j,i}a_{i,j}$ and the                                                                                                           |
| johnston-linear-matrix-algebra-F02539 | 269  | 1.000 | $B=\begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$                                                           | $B = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$                                                                 | Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$ . Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$ .                       |
| johnston-linear-matrix-algebra-F02540 | 269  | 1.000 | $\operatorname{tr}(AB)$                                                                                      | $\operatorname{tr}(AB)$                                                                                              | Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$ . Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$ .                       |
| johnston-linear-matrix-algebra-F02541 | 269  | 1.000 | $\operatorname{tr}(BA)$                                                                                      | $\operatorname{tr}(BA)$                                                                                              | Suppose $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$ . Compute $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$ .                       |
| johnston-linear-matrix-algebra-F02542 | 269  | 0.997 | $\operatorname{tr}(A)=\operatorname{tr}(B)$                                                                  | $\operatorname{tr}(A) = \operatorname{tr}(B)$                                                                        |  If $A \in \mathcal{M}_n$ and $B \in \mathcal{M}_n$ are similar then $\operatorname{tr}(A) = \operatorname{tr}(B)$ . |

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| johnston-linear-matrix-algebra-F02543 | 270  | 1.000 | $B=\left[\begin{array}{ll}1 & 2 \\ 3 & 5\end{array}\right]$             | $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$                                | Show that the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ are not similar. |
| johnston-linear-matrix-algebra-F02544 | 270  | 0.991 | $\operatorname{rank}(A)=\operatorname{rank}(B)=2$                       | $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$                             | The rank does not help us here, since $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$ . On the                                              |
| johnston-linear-matrix-algebra-F02545 | 270  | 1.000 | $\operatorname{tr}(A)=5$                                                | $\operatorname{tr}(A) = 5$                                                        | other hand, $\operatorname{tr}(A) = 5$ and $\operatorname{tr}(B) = 6$ , so we know that $A$ and $B$ are not                                       |
| johnston-linear-matrix-algebra-F02546 | 270  | 1.000 | $\operatorname{tr}(B)=6$                                                | $\operatorname{tr}(B) = 6$                                                        | other hand, $\operatorname{tr}(A) = 5$ and $\operatorname{tr}(B) = 6$ , so we know that $A$ and $B$ are not                                       |
| johnston-linear-matrix-algebra-F02547 | 270  | 0.995 | $\operatorname{tr}(A \, B \, C)=\operatorname{tr}(C \, A \, B)$         | $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$                                 | and similarly that $\operatorname{tr}(ABC) = \operatorname{tr}(CAB)$ . However, the other similar-looking                                         |
| johnston-linear-matrix-algebra-F02548 | 270  | 0.856 | $\operatorname{tr}(A \, B \, C)=\operatorname{tr}(A \, C \, B)$         | $\operatorname{tr}(ABC) = \operatorname{tr}(ACB)$                                 | equalities like $\operatorname{tr}(ABC) = \operatorname{tr}(ACB)$ do <i>not</i> hold in general. For example, if                                  |
| johnston-linear-matrix-algebra-F02549 | 270  | 1.000 | $\operatorname{tr}(A \, B \, C)=1$                                      | $\operatorname{tr}(ABC) = 1$                                                      | then direct calculation reveals that $\operatorname{tr}(ABC) = 1$ , but $\operatorname{tr}(ACB) = 0$ .                                            |
| johnston-linear-matrix-algebra-F02550 | 270  | 1.000 | $\operatorname{tr}(A \, C \, B)=0$                                      | $\operatorname{tr}(ACB) = 0$                                                      | then direct calculation reveals that $\operatorname{tr}(ABC) = 1$ , but $\operatorname{tr}(ACB) = 0$ .                                            |
| johnston-linear-matrix-algebra-F02551 | 270  | 1.000 | $A, \, B, \, C, \, D \in \mathcal{M}_n$                                 | $A, B, C, D \in \mathcal{M}_n$                                                    | if $A, B, C, D \in \mathcal{M}_n$ then                                                                                                            |
| johnston-linear-matrix-algebra-F02552 | 270  | 1.000 | $\operatorname{tr}(A \, B \, C \, D)$                                   | $\operatorname{tr}(ABCD)$                                                         | However, $\operatorname{tr}(ABCD)$ does not necessarily equal the trace of any of the other 20                                                    |
| johnston-linear-matrix-algebra-F02553 | 270  | 0.848 | $A, \, B, \, C$                                                         | $A, B, C$                                                                         | possible products of $A, B, C$ , and $D$ .                                                                                                        |
| johnston-linear-matrix-algebra-F02554 | 270  | 0.999 | $\mathbf{v}=(6,1), \, B=\{(3,0),(0,2)\}, \, \mathcal{S}=\mathbb{R}^2$   | $\mathbf{v} = (6, 1), B = \{(3, 0), (0, 2)\}, \mathcal{S} = \mathbb{R}^2$         | *(a) $\mathbf{v} = (6, 1), B = \{(3, 0), (0, 2)\}, \mathcal{S} = \mathbb{R}^2$                                                                    |
| johnston-linear-matrix-algebra-F02555 | 270  | 0.991 | $\mathbf{v}=(4,2), \, B=\{(1,1),(1,-1)\}, \, \mathcal{S}=\mathbb{R}^2$  | $\mathbf{v} = (4, 2), B = \{(1, 1), (1, -1)\}, \mathcal{S} = \mathbb{R}^2$        | (b) $\mathbf{v} = (4, 2), B = \{(1, 1), (1, -1)\}, \mathcal{S} = \mathbb{R}^2$                                                                    |
| johnston-linear-matrix-algebra-F02556 | 270  | 0.987 | $\mathbf{v}=(2,3,1), \, B=\{(1,0,-1),(1,2,1)\}, \, \mathcal{S}$         | $\mathbf{v} = (2, 3, 1), B = \{(1, 0, -1), (1, 2, 1)\}, \mathcal{S}$              | *(c) $\mathbf{v} = (2, 3, 1), B = \{(1, 0, -1), (1, 2, 1)\}, \mathcal{S}$ is the                                                                  |
| johnston-linear-matrix-algebra-F02557 | 270  | 1.000 | $x-y+z=0$                                                               | $x - y + z = 0$                                                                   | plane in $\mathbb{R}^3$ with equation $x - y + z = 0$ .                                                                                           |
| johnston-linear-matrix-algebra-F02558 | 270  | 1.000 | $\mathbf{v}=(1,1,1), \, B=\{(1,2,3),(-1,0,1),(2,2,1)\}, \, \mathcal{S}$ | $\mathbf{v} = (1, 1, 1), B = \{(1, 2, 3), (-1, 0, 1), (2, 2, 1)\}, \mathcal{S} =$ | (d) $\mathbf{v} = (1, 1, 1), B = \{(1, 2, 3), (-1, 0, 1), (2, 2, 1)\}, \mathcal{S} =$                                                             |
| johnston-linear-matrix-algebra-F02559 | 270  | 1.000 | $\mathbf{v}=(2,3)$                                                      | $\mathbf{v} = (2, 3)$                                                             | 3.1.2 Suppose $\mathbf{v} = (2, 3)$ and $\mathbf{w} = (2, -1)$ .                                                                                  |
| johnston-linear-matrix-algebra-F02560 | 270  | 1.000 | $\mathbf{w}=(2,-1)$                                                     | $\mathbf{w} = (2, -1)$                                                            | 3.1.2 Suppose $\mathbf{v} = (2, 3)$ and $\mathbf{w} = (2, -1)$ .                                                                                  |
| johnston-linear-matrix-algebra-F02561 | 270  | 0.999 | $[\mathbf{v}]_B=(1,0)$                                                  | $[\mathbf{v}]_B = (1, 0)$                                                         | *(a) Find a basis $B$ of $\mathbb{R}^2$ such that $[\mathbf{v}]_B = (1, 0)$ and                                                                   |
| johnston-linear-matrix-algebra-F02562 | 270  | 0.992 | $[\mathbf{w}]_B=(0,1)$                                                  | $[\mathbf{w}]_B = (0, 1)$                                                         | $[\mathbf{w}]_B = (0, 1)$ .                                                                                                                       |
| johnston-linear-matrix-algebra-F02563 | 270  | 0.998 | $[\mathbf{w}]_B=(-1,2)$                                                 | $[\mathbf{w}]_B = (-1, 2)$                                                        | $[\mathbf{w}]_B = (-1, 2)$ .                                                                                                                      |
| johnston-linear-matrix-algebra-F02564 | 270  | 1.000 | $\mathbf{v}=(1,5,6,8,-9)$                                               | $\mathbf{v} = (1, 5, 6, 8, -9)$                                                   | and $\mathbf{v} = (1, 5, 6, 8, -9)$ , then $\mathbf{v} \in \operatorname{span}(B)$ , and compute $[\mathbf{v}]_B$ .                               |
| johnston-linear-matrix-algebra-F02565 | 270  | 1.000 | $\mathbf{v} \in \operatorname{span}(B)$                                 | $\mathbf{v} \in \operatorname{span}(B)$                                           | and $\mathbf{v} = (1, 5, 6, 8, -9)$ , then $\mathbf{v} \in \operatorname{span}(B)$ , and compute $[\mathbf{v}]_B$ .                               |
| johnston-linear-matrix-algebra-F02566 | 270  | 0.995 | $B=\{(1,2),(3,4)\}, \, C=\{(1,0),(0,1)\}$                               | $B = \{(1, 2), (3, 4)\}, C = \{(1, 0), (0, 1)\}$                                  | *(a) $B = \{(1, 2), (3, 4)\}, C = \{(1, 0), (0, 1)\}$                                                                                             |
| johnston-linear-matrix-algebra-F02567 | 270  | 1.000 | $B=\{(1,0),(0,1)\}, \, C=\{(2,1),(-4,3)\}$                              | $B = \{(1, 0), (0, 1)\}, C = \{(2, 1), (-4, 3)\}$                                 | (b) $B = \{(1, 0), (0, 1)\}, C = \{(2, 1), (-4, 3)\}$                                                                                             |
| johnston-linear-matrix-algebra-F02568 | 270  | 0.964 | $B=\{(1,2),(3,4)\}, \, C=\{(2,1),(-4,3)\}$                              | $B = \{(1, 2), (3, 4)\}, C = \{(2, 1), (-4, 3)\}$                                 | *(c) $B = \{(1, 2), (3, 4)\}, C = \{(2, 1), (-4, 3)\}$                                                                                            |
| johnston-linear-matrix-algebra-F02569 | 270  | 1.000 | $B=\{(4,0,0),(0,4,0),(1,4,1)\}$                                         | $B = \{(4, 0, 0), (0, 4, 0), (1, 4, 1)\}$                                         | (d) $B = \{(4, 0, 0), (0, 4, 0), (1, 4, 1)\},$                                                                                                    |
| johnston-linear-matrix-algebra-F02570 | 270  | 1.000 | $C=\{(1,2,3),(1,0,1),(-1,1,2)\}$                                        | $C = \{(1, 2, 3), (1, 0, 1), (-1, 1, 2)\}$                                        | $C = \{(1, 2, 3), (1, 0, 1), (-1, 1, 2)\}$                                                                                                        |

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| johnston-linear-matrix-algebra-F02571 | 270  | ■ 0.925 | $\set{(2,3),(1,-1)}$                                                                                                                                                                                                                                                                                                                                                                                                                                     | $\{(2,3),(1,-1)\}$                                                                                                                                                                                                                                                                             | $\{(2,3),(1,-1)\}$ :                                                                                                                                                                                                                                                                                |
| johnston-linear-matrix-algebra-F02572 | 270  | ■ 0.996 | $T\left(v_{\{1\}}, v_{\{2\}}\right)=\left(v_{\{1\}}-v_{\{2\}}, v_{\{1\}}+3 v_{\{2\}}\right)$                                                                                                                                                                                                                                                                                                                                                             | $T\left(v_1, v_2\right)=\left(v_1-v_2, v_1+3 v_2\right)$                                                                                                                                                                                                                                       | *(a) $T\left(v_1, v_2\right)=\left(v_1-v_2, v_1+3 v_2\right)$ .                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02573 | 270  | ■ 1.000 | $T\left(v_{\{1\}}, v_{\{2\}}\right)=\left(v_{\{1\}}+2 v_{\{2\}}, v_{\{1\}}+2 v_{\{2\}}\right)$                                                                                                                                                                                                                                                                                                                                                           | $T\left(v_1, v_2\right)=\left(v_1+2 v_2, v_1+2 v_2\right)$                                                                                                                                                                                                                                     | (b) $T\left(v_1, v_2\right)=\left(v_1+2 v_2, v_1+2 v_2\right)$ .                                                                                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F02574 | 270  | ■ 1.000 | $T\left(v_{\{1\}}, v_{\{2\}}\right)=\left(-v_{\{1\}}+2 v_{\{2\}}, 3 v_{\{1\}}\right)$                                                                                                                                                                                                                                                                                                                                                                    | $T\left(v_1, v_2\right)=\left(-v_1+2 v_2, 3 v_1\right)$                                                                                                                                                                                                                                        | *(c) $T\left(v_1, v_2\right)=\left(-v_1+2 v_2, 3 v_1\right)$ .                                                                                                                                                                                                                                      |
| johnston-linear-matrix-algebra-F02575 | 270  | ■ 1.000 | $T\left(v_{\{1\}}, v_{\{2\}}\right)=\left(4 v_{\{1\}}+3 v_{\{2\}}, 5 v_{\{1\}}+v_{\{2\}}\right)$                                                                                                                                                                                                                                                                                                                                                         | $T\left(v_1, v_2\right)=\left(4 v_1+3 v_2, 5 v_1+v_2\right)$                                                                                                                                                                                                                                   | (d) $T\left(v_1, v_2\right)=\left(4 v_1+3 v_2, 5 v_1+v_2\right)$ .                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02576 | 271  | ■ 1.000 | $B=\set{(3,1,2),(0,-1,-1),(2,-1,0)}$                                                                                                                                                                                                                                                                                                                                                                                                                     | $B=\{(3,1,2),(0,-1,-1),(2,-1,0)\}$                                                                                                                                                                                                                                                             | to the basis $B=\{(3,1,2),(0,-1,-1),(2,-1,0)\}$ :                                                                                                                                                                                                                                                   |
| johnston-linear-matrix-algebra-F02577 | 271  | ■ 0.595 | $T\left(v_{\{1\}}, v_{\{2\}}, v_{\{3\}}\right)=$                                                                                                                                                                                                                                                                                                                                                                                                         | $T\left(v_1, v_2, v_3\right)=$                                                                                                                                                                                                                                                                 | *(a) $T\left(v_1, v_2, v_3\right)=$                                                                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02578 | 271  | ■ 0.802 | $\begin{aligned} & \& A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \& -1 \\ & \& -1 \& 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 2 & -1 \\ -1 & 0 \end{array}\right] \\ & \& \text { (b) } A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \& A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 1 & -2 \\ -2 & 4 \end{array}\right] \end{aligned}$ | $A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 2 & -1 \\ -1 & 0 \end{array}\right]$<br>(b) $A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 1 & -2 \\ -2 & 4 \end{array}\right]$ | *(a) $A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 2 & -1 \\ -1 & 0 \end{array}\right]$<br>(b) $A=\left[\begin{array}{cc} 1 & -1 \\ -1 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 1 & -2 \\ -2 & 4 \end{array}\right]$ |
| johnston-linear-matrix-algebra-F02579 | 271  | ■ 0.816 | $A=\left[\begin{array}{lll} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                      | $A=\left[\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{array}\right]$                                                                                                                                                                                                            | *(c) $A=\left[\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{array}\right] \text { and } B=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right]$                                                                                                           |
| johnston-linear-matrix-algebra-F02580 | 271  | ■ 0.816 | $B=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                     | $B=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right]$                                                                                                                                                                                                           | *(c) $A=\left[\begin{array}{ccc} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{array}\right] \text { and } B=\left[\begin{array}{ccc} 1 & -1 & 0 \\ 3 & 1 & 2 \\ 1 & 1 & 1 \end{array}\right]$                                                                                                           |
| johnston-linear-matrix-algebra-F02581 | 271  | ■ 1.000 | $A=\left[\begin{array}{ccc} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                     | $A=\left[\begin{array}{ccc} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{array}\right]$                                                                                                                                                                                                           | (d) $A=\left[\begin{array}{ccc} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{ccc} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{array}\right]$                                                                                                          |
| johnston-linear-matrix-algebra-F02582 | 271  | ■ 1.000 | $B=\left[\begin{array}{ccc} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                    | $B=\left[\begin{array}{ccc} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{array}\right]$                                                                                                                                                                                                          | (d) $A=\left[\begin{array}{ccc} 2 & 0 & 1 \\ 1 & 1 & 3 \\ -1 & 2 & 1 \end{array}\right] \text { and } B=\left[\begin{array}{ccc} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 0 & 0 & 2 \end{array}\right]$                                                                                                          |
| johnston-linear-matrix-algebra-F02583 | 271  | ■ 0.997 | $A=\left[\begin{array}{ll} 1 & 0 \\ 0 & 2 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                            | $A=\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \end{array}\right]$                                                                                                                                                                                                                                  | *(a) $A=\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 3 & -2 \\ 1 & 0 \end{array}\right]$                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02584 | 271  | ■ 0.997 | $B=\left[\begin{array}{cc} 3 & -2 \\ 1 & 0 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                           | $B=\left[\begin{array}{cc} 3 & -2 \\ 1 & 0 \end{array}\right]$                                                                                                                                                                                                                                 | *(a) $A=\left[\begin{array}{cc} 1 & 0 \\ 0 & 2 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 3 & -2 \\ 1 & 0 \end{array}\right]$                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02585 | 271  | ■ 1.000 | $A=\left[\begin{array}{ll} 1 & 2 \\ 0 & 3 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                            | $A=\left[\begin{array}{cc} 1 & 2 \\ 0 & 3 \end{array}\right]$                                                                                                                                                                                                                                  | (b) $A=\left[\begin{array}{cc} 1 & 2 \\ 0 & 3 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 2 & 0 \\ 1 & 2 \end{array}\right]$                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02586 | 271  | ■ 1.000 | $B=\left[\begin{array}{ll} 2 & 0 \\ 1 & 2 \end{array}\right]$                                                                                                                                                                                                                                                                                                                                                                                            | $B=\left[\begin{array}{cc} 2 & 0 \\ 1 & 2 \end{array}\right]$                                                                                                                                                                                                                                  | (b) $A=\left[\begin{array}{cc} 1 & 2 \\ 0 & 3 \end{array}\right] \text { and } B=\left[\begin{array}{cc} 2 & 0 \\ 1 & 2 \end{array}\right]$                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02587 | 271  | ■ 0.995 | $P_{\{B \rightarrow B\}}=I$                                                                                                                                                                                                                                                                                                                                                                                                                              | $P_{B \leftarrow B}=I$                                                                                                                                                                                                                                                                         | *(a) If $B$ is a basis of a subspace $\mathcal{S}$ of $\mathbb{R}^n$ then $P_{B \leftarrow B}=I$ .                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02588 | 271  | ■ 1.000 | $\operatorname{tr}(A B)=\operatorname{tr}(A) \operatorname{tr}(B)$                                                                                                                                                                                                                                                                                                                                                                                       | $\operatorname{tr}(A B)=\operatorname{tr}(A) \operatorname{tr}(B)$                                                                                                                                                                                                                             | $\operatorname{tr}(A B)=\operatorname{tr}(A) \operatorname{tr}(B)$ .                                                                                                                                                                                                                                |

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| johnston-linear-matrix-algebra-F02589 | 271  | ■ 1.000 | $\operatorname{tr}(A^T)=\operatorname{tr}(A)$                                             | $\operatorname{tr}(A^T) = \operatorname{tr}(A)$                                           | <b>(h)</b> If $A$ is a square matrix then $\operatorname{tr}(A^T) = \operatorname{tr}(A)$ .                                            |
| johnston-linear-matrix-algebra-F02590 | 271  | ■ 1.000 | $C=\left\{\mathbf{w}_1, \ldots, \mathbf{w}_m\right\} \subset \mathcal{S}$                 | $C = \{\mathbf{w}_1, \dots, \mathbf{w}_m\} \subset \mathcal{S}$                           | basis $B$ . Let $C = \{\mathbf{w}_1, \dots, \mathbf{w}_m\} \subset \mathcal{S}$ be a set of vectors and                                |
| johnston-linear-matrix-algebra-F02591 | 271  | ■ 1.000 | $D=\left\{\left[\mathbf{w}_1\right]_B, \ldots,\left[\mathbf{w}_m\right]_B\right\}$        | $D = \{[\mathbf{w}_1]_B, \dots, [\mathbf{w}_m]_B\}$                                       | let $D = \{[\mathbf{w}_1]_B, \dots, [\mathbf{w}_m]_B\}$ be the set of corresponding co-                                                |
| johnston-linear-matrix-algebra-F02592 | 271  | ■ 1.000 | $P=P_B \rightarrow C$                                                                     | $P = P_{B \leftarrow C}$                                                                  | a basis $C$ of $\mathcal{S}$ such that $P = P_{B \leftarrow C}$ .                                                                      |
| johnston-linear-matrix-algebra-F02593 | 271  | ■ 0.997 | $B, C$                                                                                    | $B, C$                                                                                    | <b>**3.1.14</b> Let $\mathcal{S}$ be a subspace of $\mathbb{R}^n$ with bases $B, C$ , and $E$ .                                        |
| johnston-linear-matrix-algebra-F02594 | 271  | ■ 1.000 | $P_{C \rightarrow B}=P_{C \rightarrow E} P_{E \rightarrow B}$                             | $P_{C \leftarrow B} = P_{C \leftarrow E} P_{E \leftarrow B}$                              | <b>(a)</b> Show that $P_{C \leftarrow B} = P_{C \leftarrow E} P_{E \leftarrow B}$ .                                                    |
| johnston-linear-matrix-algebra-F02595 | 271  | ■ 1.000 | $\left[P_{E \rightarrow C} \mid P_{E \rightarrow B}\right]$                               | $[P_{E \leftarrow C} \mid P_{E \leftarrow B}]$                                            | $[P_{E \leftarrow C} \mid P_{E \leftarrow B}]$ is $[I \mid P_{C \leftarrow B}]$ .                                                      |
| johnston-linear-matrix-algebra-F02596 | 271  | ■ 1.000 | $\left[I \mid P_{C \rightarrow B}\right]$                                                 | $[I \mid P_{C \leftarrow B}]$                                                             | $[P_{E \leftarrow C} \mid P_{E \leftarrow B}]$ is $[I \mid P_{C \leftarrow B}]$ .                                                      |
| johnston-linear-matrix-algebra-F02597 | 271  | ■ 0.998 | $\operatorname{nullity}(A)=\operatorname{nullity}(B)$                                     | $\operatorname{nullity}(A) = \operatorname{nullity}(B)$                                   | similar, then $\operatorname{nullity}(A) = \operatorname{nullity}(B)$ .                                                                |
| johnston-linear-matrix-algebra-F02598 | 272  | ■ 0.999 | $f: \mathcal{M}_n \rightarrow \mathbb{R}$                                                 | $f: \mathcal{M}_n \rightarrow \mathbb{R}$                                                 | <b>**3.1.20</b> Show that the only linear function $f: \mathcal{M}_n \rightarrow \mathbb{R}$                                           |
| johnston-linear-matrix-algebra-F02599 | 272  | ■ 1.000 | $f(A B C)=f(A C B)$                                                                       | $f(ABC) = f(ACB)$                                                                         | with the property that $f(ABC) = f(ACB)$ for all $A, B, C \in$                                                                         |
| johnston-linear-matrix-algebra-F02600 | 272  | ■ 1.000 | $A, B, C \in$                                                                             | $A, B, C \in$                                                                             | with the property that $f(ABC) = f(ACB)$ for all $A, B, C \in$                                                                         |
| johnston-linear-matrix-algebra-F02601 | 272  | ■ 1.000 | $f(A)=0$                                                                                  | $f(A) = 0$                                                                                | $f(A) = 0$ for all $A \in \mathcal{M}_n$ .                                                                                             |
| johnston-linear-matrix-algebra-F02602 | 272  | ■ 1.000 | $f\left(\mathbf{e}_i \mathbf{e}_j^T\right)$                                               | $f\left(\mathbf{e}_i \mathbf{e}_j^T\right)$                                               | [Hint: Compute $f(\mathbf{e}_i \mathbf{e}_j^T)$ in multiple different ways.]                                                           |
| johnston-linear-matrix-algebra-F02603 | 272  | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w}=\left[\mathbf{v}\right]_B \cdot\left[\mathbf{w}\right]_B$    | $\mathbf{v} \cdot \mathbf{w} = [\mathbf{v}]_B \cdot [\mathbf{w}]_B$                       | Remark 3.1.2) then $\mathbf{v} \cdot \mathbf{w} = [\mathbf{v}]_B \cdot [\mathbf{w}]_B$ .                                               |
| johnston-linear-matrix-algebra-F02604 | 272  | ■ 1.000 | $B \subset \mathbb{R}^n$                                                                  | $B \subset \mathbb{R}^n$                                                                  | <b>**3.1.22</b> Suppose $B \subset \mathbb{R}^n$ is a set consisting of non-zero                                                       |
| johnston-linear-matrix-algebra-F02605 | 272  | ■ 1.000 | $\left\{R^{\theta}\left(\mathbf{e}_1\right), R^{\theta}\left(\mathbf{e}_2\right)\right\}$ | $\left\{R^{\theta}\left(\mathbf{e}_1\right), R^{\theta}\left(\mathbf{e}_2\right)\right\}$ | <b>(a)</b> Show that $\left\{R^{\theta}\left(\mathbf{e}_1\right), R^{\theta}\left(\mathbf{e}_2\right)\right\}$ is an orthonormal basis |
| johnston-linear-matrix-algebra-F02606 | 272  | ■ 1.000 | $A \mathbf{x} \in \mathbb{R}^n$                                                           | $A \mathbf{x} \in \mathbb{R}^n$                                                           | linear transformation that sends $\mathbf{x} \in \mathbb{R}^n$ to $A \mathbf{x} \in \mathbb{R}^n$ . That linear transformation         |
| johnston-linear-matrix-algebra-F02607 | 272  | ■ 1.000 | $A: A \mathbf{e}_1, A \mathbf{e}_2, \ldots, A \mathbf{e}_n$                               | $A: A \mathbf{e}_1, A \mathbf{e}_2, \dots, A \mathbf{e}_n$                                | equal to the columns of $A$ : $A \mathbf{e}_1, A \mathbf{e}_2, \dots, A \mathbf{e}_n$ (see Figure 3.10).                               |
| johnston-linear-matrix-algebra-F02608 | 272  | ■ 1.000 | $\operatorname{det}(A)$                                                                   | $\det(A)$                                                                                 | The determinant of $A$ , which we denote by $\det(A)$ , is the area (or volume,                                                        |
| johnston-linear-matrix-algebra-F02609 | 273  | ■ 1.000 | $\operatorname{det}(A)=2$                                                                 | $\det(A) = 2$                                                                             | area of the unit square (i.e., $\det(A) = 2$ ), then it also doubles the area of any                                                   |
| johnston-linear-matrix-algebra-F02610 | 273  | ■ 0.534 | $A(\operatorname{det}(A))=2.46$                                                           | $A(\det(A) = 2.46$                                                                        | amount by the matrix $A$ ( $\det(A) = 2.46$ in this case).                                                                             |
| johnston-linear-matrix-algebra-F02611 | 273  | ■ 1.000 | $\operatorname{det}(I)=1$                                                                 | $\det(I) = 1$                                                                             | $\det(I) = 1$ . Secondly, since every $A \in \mathcal{M}_n$ expands space by a factor of $\det(A)$ ,                                   |
| johnston-linear-matrix-algebra-F02612 | 273  | ■ 1.000 | $\operatorname{det}(B)$                                                                   | $\det(B)$                                                                                 | and similarly each $B \in \mathcal{M}_n$ expands space by a factor of $\det(B)$ , it must be the                                       |
| johnston-linear-matrix-algebra-F02613 | 273  | ■ 1.000 | $n=2$                                                                                     | $n = 2$                                                                                   | when $n = 2$ in Figure 3.13.                                                                                                           |
| johnston-linear-matrix-algebra-F02614 | 274  | ■ 1.000 | $\mathbf{a}_1$                                                                            | $\mathbf{a}_1$                                                                            | parallel to the side $\mathbf{a}_1$ .                                                                                                  |
| johnston-linear-matrix-algebra-F02615 | 274  | ■ 1.000 | $A=\left[\mathbf{a}_1 \mid \mathbf{a}_2\right]$                                           | $A = [\mathbf{a}_1 \mid \mathbf{a}_2]$                                                    | <b>Figure 3.12:</b> Scaling one of the columns of $A = [\mathbf{a}_1 \mid \mathbf{a}_2]$ scales the area of the                        |
| johnston-linear-matrix-algebra-F02616 | 274  | ■ 1.000 | $2 \mathbf{a}_2$                                                                          | $2 \mathbf{a}_2$                                                                          | amount. Here, the parallelogram with sides $\mathbf{a}_1$ and $2 \mathbf{a}_2$ has twice the area of the                               |
| johnston-linear-matrix-algebra-F02617 | 274  | ■ 1.000 | $\mathbf{a}_2$                                                                            | $\mathbf{a}_2$                                                                            | parallelogram with sides $\mathbf{a}_1$ and $\mathbf{a}_2$ .                                                                           |

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| johnston-linear-matrix-algebra-F02618 | 274  | ■ 0.998 | $\operatorname{det} : \mathcal{M}_n \rightarrow \mathbb{R}$                                                                                                                      | $\det : \mathcal{M}_n \rightarrow \mathbb{R}$                                                                               | The <b>determinant</b> is the (unique!) function $\det : \mathcal{M}_n \rightarrow \mathbb{R}$ that satisfies the                                       |
| johnston-linear-matrix-algebra-F02619 | 274  | ■ 1.000 | $\operatorname{det}(A B)=\operatorname{det}(A) \operatorname{det}(B)$                                                                                                            | $\det(AB) = \det(A) \det(B)$                                                                                                | b) $\det(AB) = \det(A) \det(B)$ for all $A, B \in \mathcal{M}_n$ , and                                                                                  |
| johnston-linear-matrix-algebra-F02620 | 274  | ■ 1.000 | $\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^n$                                                                                      | $\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^n$                                 | c) for all $c \in \mathbb{R}$ and all $\mathbf{v}, \mathbf{w}, \mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^n$ , it is the case that |
| johnston-linear-matrix-algebra-F02621 | 275  | ■ 1.000 | $\operatorname{det}(A) \neq 0$                                                                                                                                                   | $\det(A) \neq 0$                                                                                                            | It follows that $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$ . This makes sense geomet-                                                              |
| johnston-linear-matrix-algebra-F02622 | 275  | ■ 1.000 | $\operatorname{det}\left(A^{-1}\right)=1 / \operatorname{det}(A)$                                                                                                                | $\det\left(A^{-1}\right) = 1/\det(A)$                                                                                       | It follows that $\det(A) \neq 0$ and $\det(A^{-1}) = 1/\det(A)$ . This makes sense geomet-                                                              |
| johnston-linear-matrix-algebra-F02623 | 275  | ■ 1.000 | $\operatorname{det}(A)=0$                                                                                                                                                        | $\det(A) = 0$                                                                                                               | $A$ is not invertible then $\det(A) = 0$ (see Figure 3.14).                                                                                             |
| johnston-linear-matrix-algebra-F02624 | 275  | ■ 1.000 | $\operatorname{det}\left(P^{-1} A P\right)=0$                                                                                                                                    | $\det\left(P^{-1} A P\right) = 0$                                                                                           | so our goal now is to show that $\det(P^{-1}AP) = 0$ . By recalling that the first                                                                      |
| johnston-linear-matrix-algebra-F02625 | 275  | ■ 0.817 | $P \mathbf{e}_1=\mathbf{x}$                                                                                                                                                      | $P\mathbf{e}_1 = \mathbf{x}$                                                                                                | column of $P$ is $\mathbf{x}$ (i.e., $P\mathbf{e}_1 = \mathbf{x}$ ), we see that                                                                        |
| johnston-linear-matrix-algebra-F02626 | 275  | ■ 1.000 | $P^{-1} A P$                                                                                                                                                                     | $P^{-1}AP$                                                                                                                  | In other words, the first column of $P^{-1}AP$ is the zero vector $\mathbf{0}$ , so our goal                                                            |
| johnston-linear-matrix-algebra-F02627 | 276  | ■ 1.000 | $\operatorname{det}\left(P^{-1} A P\right)=\operatorname{det}(A)$                                                                                                                | $\det\left(P^{-1} A P\right) = \det(A)$                                                                                     | that $\det(P^{-1}AP) = \det(A)$ for every invertible matrix $P$ . In other words, the                                                                   |
| johnston-linear-matrix-algebra-F02628 | 276  | ■ 0.965 | $\operatorname{det}(A)=\operatorname{det}(B)$                                                                                                                                    | $\det(A) = \det(B)$                                                                                                         | ! If $A \in \mathcal{M}_n$ and $B \in \mathcal{M}_n$ are similar then $\det(A) = \det(B)$ .                                                             |
| johnston-linear-matrix-algebra-F02629 | 276  | ■ 1.000 | $\operatorname{det}(c A)=c^n \operatorname{det}(A)$                                                                                                                              | $\det(cA) = c^n \det(A)$                                                                                                    | a) $\det(cA) = c^n \det(A)$ , and                                                                                                                       |
| johnston-linear-matrix-algebra-F02630 | 276  | ■ 1.000 | $\operatorname{det}\left(A^T\right)=\operatorname{det}(A)$                                                                                                                       | $\det\left(A^T\right) = \det(A)$                                                                                            | b) $\det(A^T) = \det(A)$ .                                                                                                                              |
| johnston-linear-matrix-algebra-F02631 | 276  | ■ 1.000 | $c^n$                                                                                                                                                                            | $c^n$                                                                                                                       | resulting volume by $c^n$ (rather than just by $c$ , which is what we might mistakenly                                                                  |
| johnston-linear-matrix-algebra-F02632 | 276  | ■ 1.000 | $2^2=4$                                                                                                                                                                          | $2^2 = 4$                                                                                                                   | determinant) by a factor of $2^2 = 4$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F02633 | 277  | ■ 0.998 | $A, B, C \in \mathcal{M}_3$                                                                                                                                                      | $A, B, C \in \mathcal{M}_3$                                                                                                 | Suppose that $A, B, C \in \mathcal{M}_3$ are matrices with $\det(A) = 2$ , $\det(B) = 3$ , and                                                          |
| johnston-linear-matrix-algebra-F02634 | 277  | ■ 0.998 | $\operatorname{det}(A)=2, \operatorname{det}(B)=3$                                                                                                                               | $\det(A) = 2, \det(B) = 3$                                                                                                  | Suppose that $A, B, C \in \mathcal{M}_3$ are matrices with $\det(A) = 2$ , $\det(B) = 3$ , and                                                          |
| johnston-linear-matrix-algebra-F02635 | 277  | ■ 1.000 | $\operatorname{det}(C)=5$                                                                                                                                                        | $\det(C) = 5$                                                                                                               | $\det(C) = 5$ . Compute each of the following determinants:                                                                                             |
| johnston-linear-matrix-algebra-F02636 | 277  | ■ 1.000 | $\operatorname{det}(A B)$                                                                                                                                                        | $\det(AB)$                                                                                                                  | a) $\det(AB)$                                                                                                                                           |
| johnston-linear-matrix-algebra-F02637 | 277  | ■ 1.000 | $\operatorname{det}\left(A^2 C^T B^{-1}\right)$                                                                                                                                  | $\det\left(A^2 C^T B^{-1}\right)$                                                                                           | b) $\det(A^2C^TB^{-1})$                                                                                                                                 |
| johnston-linear-matrix-algebra-F02638 | 277  | ■ 1.000 | $\operatorname{det}\left(3 A B^{-2} C^2\right)$                                                                                                                                  | $\det\left(3 A B^{-2} C^2\right)$                                                                                           | c) $\det(3AB^{-2}C^2)$                                                                                                                                  |
| johnston-linear-matrix-algebra-F02639 | 277  | ■ 1.000 | $\operatorname{det}\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right)$                                                                                                    | $\det\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right)$                                                             | d) $\det\left(2 A^{-3} B^{-2}\left(C^T B\right)^4(C / 5)^2\right)$                                                                                      |
| johnston-linear-matrix-algebra-F02640 | 277  | ■ 1.000 | $\operatorname{det}(A B)=\operatorname{det}(A) \operatorname{det}(B)=2 \cdot 3=6$                                                                                                | $\det(AB) = \det(A) \det(B) = 2 \cdot 3 = 6$                                                                                | a) $\det(AB) = \det(A) \det(B) = 2 \cdot 3 = 6$ .                                                                                                       |
| johnston-linear-matrix-algebra-F02641 | 277  | ■ 1.000 | $\operatorname{det}\left(A^2 C^T B^{-1}\right)=\operatorname{det}(A)^2 \operatorname{det}\left(C^T\right) \operatorname{det}\left(B^{-1}\right)=2^2 \cdot 5 \cdot(1 / 3)=20 / 3$ | $\det\left(A^2 C^T B^{-1}\right) = \det(A)^2 \det\left(C^T\right) \det\left(B^{-1}\right) = 2^2 \cdot 5 \cdot (1/3) = 20/3$ | b) $\det(A^2C^TB^{-1}) = \det(A)^2 \det(C^T) \det(B^{-1}) = 2^2 \cdot 5 \cdot (1/3) = 20/3$ .                                                           |
| johnston-linear-matrix-algebra-F02642 | 277  | ■ 1.000 | $\operatorname{det}\left(3 A B^{-2} C^2\right)=3^3 \operatorname{det}(A) \operatorname{det}(B)^{-2} \operatorname{det}(C)^2=27 \cdot 2 \cdot(1 / 9) \cdot 25=$                   | $\det\left(3 A B^{-2} C^2\right) = 3^3 \det(A) \det(B)^{-2} \det(C)^2 = 27 \cdot 2 \cdot (1/9) \cdot 25 =$                  | c) $\det(3AB^{-2}C^2) = 3^3 \det(A) \det(B)^{-2} \det(C)^2 = 27 \cdot 2 \cdot (1/9) \cdot 25 =$                                                         |

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| johnston-linear-matrix-algebra-F02643 | 278  | ■ 1.000 | $c \mathbf{e}_{\{i\}}$                                                                          | $c\mathbf{e}_i$                                | by adding $c\mathbf{e}_i$ to the $j$ -th column of $I$ , so we can use multilinearity (defining           |
| johnston-linear-matrix-algebra-F02644 | 278  | ■ 0.523 | $1+c \cdot 0=1$                                                                                 | $1 + c \cdot 0 = 1$                            | determinant is 0. This elementary matrix thus has determinant $1 + c \cdot 0 = 1$ .                       |
| johnston-linear-matrix-algebra-F02645 | 278  | ■ 1.000 | $R_{\{2\}}+\frac{1}{2} R_{\{1\}}$                                                               | $R_2 + \frac{1}{2}R_1$                         | <b>Figure 3.17:</b> The elementary matrix corresponding to the row operation $R_2 + \frac{1}{2}R_1$       |
| johnston-linear-matrix-algebra-F02646 | 279  | ■ 1.000 | $\mathbf{e}_{\{i\}}+\mathbf{e}_{\{j\}}$                                                         | $\mathbf{e}_i + \mathbf{e}_j$                  | equal $\mathbf{e}_i + \mathbf{e}_j$ . Since this matrix has two equal columns, it is not invertible, so   |
| johnston-linear-matrix-algebra-F02647 | 280  | ■ 1.000 | $f(x)$                                                                                          | $f(x)$                                         | is defined as the area under the graph of $f(x)$ from $x = a$ to $x = b$ :                                |
| johnston-linear-matrix-algebra-F02648 | 280  | ■ 1.000 | $x=b$                                                                                           | $x = b$                                        | is defined as the area under the graph of $f(x)$ from $x = a$ to $x = b$ :                                |
| johnston-linear-matrix-algebra-F02649 | 280  | ■ 1.000 | $f(x) \geq 0$                                                                                   | $f(x) \geq 0$                                  | However, this geometric definition is only accurate if $f(x) \geq 0$ for all                              |
| johnston-linear-matrix-algebra-F02650 | 281  | ■ 1.000 | $c_{\{1\}} R_{\{i_{\{1\}}\}}, c_{\{2\}} R_{\{i_{\{2\}}\}}, \ldots, c_{\{k\}} R_{\{i_{\{k\}}\}}$ | $c_1R_{i_1}, c_2R_{i_2}, \dots, c_kR_{i_k}$    | via $k$ “multiplication” row operations $c_1R_{i_1}, c_2R_{i_2}, \dots, c_kR_{i_k}$ as well as $s$        |
| johnston-linear-matrix-algebra-F02651 | 281  | ■ 1.000 | $E_{\{m\}} \cdots E_{\{2\}} E_{\{1\}} A=I$                                                      | $E_m \cdots E_2E_1A = I$                       | <i>Proof.</i> Recall that if $A$ is invertible then we can write $E_m \cdots E_2E_1A = I$ , where         |
| johnston-linear-matrix-algebra-F02652 | 281  | ■ 0.999 | $E_{\{1\}}, E_{\{2\}}, \ldots, E_{\{m\}}$                                                       | $E_1, E_2, \dots, E_m$                         | $E_1, E_2, \dots, E_m$ are the elementary matrices corresponding to the row operations                    |
| johnston-linear-matrix-algebra-F02653 | 282  | ■ 1.000 | $1 / (-1)^s=(-1)^s$                                                                             | $1/(-1)^s = (-1)^s$                            | since $1/(-1)^s = (-1)^s$ .                                                                               |
| johnston-linear-matrix-algebra-F02654 | 282  | ■ 1.000 | $\left[\begin{array}{ll} 0 & 2 \\ 3 & 4 \end{array}\right]$                                     | $\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$ | b) $\begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}$                                                         |
| johnston-linear-matrix-algebra-F02655 | 282  | ■ 1.000 | $\frac{1}{2} R_{\{1\}}$                                                                         | $\frac{1}{2}R_1$                               | the only “multiplication” row operation that we performed was $\frac{1}{2}R_1$ ,                          |
| johnston-linear-matrix-algebra-F02656 | 282  | ■ 1.000 | $1 /(1 / 2)=2$                                                                                  | $1/(1/2) = 2$                                  | we conclude that the determinant of this matrix is $1/(1/2) = 2$ .                                        |
| johnston-linear-matrix-algebra-F02657 | 282  | ■ 1.000 | $s=1$                                                                                           | $s = 1$                                        | Since we performed $s = 1$ “swap” row operation, as well as the                                           |
| johnston-linear-matrix-algebra-F02658 | 282  | ■ 0.999 | $\frac{1}{3} R_{\{1\}}$                                                                         | $\frac{1}{3}R_1$                               | two “multiplication” row operations $\frac{1}{3}R_1$ and $\frac{1}{2}R_2$ , it follows that               |
| johnston-linear-matrix-algebra-F02659 | 282  | ■ 0.997 | $(-1) /((1 / 2)(1 / 3))=-6$                                                                     | $(-1)/((1/2)(1/3)) = -6$                       | the determinant of this matrix is $(-1)/((1/2)(1/3)) = -6$ .                                              |
| johnston-linear-matrix-algebra-F02660 | 283  | ■ 1.000 | $\frac{1}{2} R_{\{3\}}$                                                                         | $\frac{1}{2}R_3$                               | “multiplication” row operation that we performed was $\frac{1}{2}R_3$ , we con-                           |
| johnston-linear-matrix-algebra-F02661 | 283  | ■ 1.000 | $a_{\{1,1\}}, a_{\{2,2\}}, \ldots, a_{\{n, n\}} \neq 0$                                         | $a_{1,1}, a_{2,2}, \dots, a_{n,n} \neq 0$      | $a_{1,1}, a_{2,2}, \dots, a_{n,n} \neq 0$ . Then we can construct a sequence of row operations that       |
| johnston-linear-matrix-algebra-F02662 | 283  | ■ 1.000 | $a_{\{n, n\}}$                                                                                  | $a_{n,n}$                                      | row-reduces $A$ to $I$ . First, divide the $n$ -th row of $A$ by $a_{n,n}$ and then, for each             |
| johnston-linear-matrix-algebra-F02663 | 283  | ■ 1.000 | $1 \leq i \leq n-1$                                                                             | $1 \leq i \leq n - 1$                          | $1 \leq i \leq n - 1$ , subtract $a_{i,n}R_n$ from $R_i$ :                                                |
| johnston-linear-matrix-algebra-F02664 | 283  | ■ 1.000 | $a_{\{i, n\}} R_{\{n\}}$                                                                        | $a_{i,n}R_n$                                   | $1 \leq i \leq n - 1$ , subtract $a_{i,n}R_n$ from $R_i$ :                                                |
| johnston-linear-matrix-algebra-F02665 | 283  | ■ 1.000 | $R_{\{i\}}$                                                                                     | $R_i$                                          | $1 \leq i \leq n - 1$ , subtract $a_{i,n}R_n$ from $R_i$ :                                                |
| johnston-linear-matrix-algebra-F02666 | 283  | ■ 1.000 | $a_{\{n-1, n-1\}}$                                                                              | $a_{n-1,n-1}$                                  | $(n - 1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n - 2$ , subtract $a_{i,n-1}R_{n-1}$ |
| johnston-linear-matrix-algebra-F02667 | 283  | ■ 1.000 | $1 \leq i \leq n-2$                                                                             | $1 \leq i \leq n - 2$                          | $(n - 1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n - 2$ , subtract $a_{i,n-1}R_{n-1}$ |
| johnston-linear-matrix-algebra-F02668 | 283  | ■ 1.000 | $a_{\{i, n-1\}} R_{\{n-1\}}$                                                                    | $a_{i,n-1}R_{n-1}$                             | $(n - 1)$ -th row by $a_{n-1,n-1}$ and then, for each $1 \leq i \leq n - 2$ , subtract $a_{i,n-1}R_{n-1}$ |

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| johnston-linear-matrix-algebra-F02669 | 284  | 0.894 | $a_{1,1} a_{2,2} \cdots a_{n,n} =$                                                                                  | $a_{1,1} a_{2,2} \cdots a_{n,n} =$                                                                                  | On the other hand, if $A$ has a diagonal entry equal to 0, then $a_{1,1} a_{2,2} \cdots a_{n,n} =$                                                |
| johnston-linear-matrix-algebra-F02670 | 284  | 1.000 | $a_{i,i} = 0$                                                                                                       | $a_{i,i} = 0$                                                                                                       | is not invertible). To see this, suppose that the $a_{i,i} = 0$ (and furthermore, $i$ is                                                          |
| johnston-linear-matrix-algebra-F02671 | 284  | 0.958 | $r_{1,1}, r_{2,2}, \ldots, r_{n,n}$                                                                                 | $r_{1,1}, r_{2,2}, \ldots, r_{n,n}$                                                                                 | where $r_{1,1}, r_{2,2}, \ldots, r_{n,n}$ are the diagonal entries of the row echelon form $R$ .                                                  |
| johnston-linear-matrix-algebra-F02672 | 284  | 1.000 | $\left[ \begin{array}{cccc} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right]$ | $\left[ \begin{array}{cccc} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right]$ | d) $\left[ \begin{array}{cccc} 0 & 2 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ -2 & 0 & 1 & 1 \end{array} \right]$                            |
| johnston-linear-matrix-algebra-F02673 | 284  | 1.000 | $2 \cdot 1 = 2$                                                                                                     | $2 \cdot 1 = 2$                                                                                                     | $2 \cdot 1 = 2$ .                                                                                                                                 |
| johnston-linear-matrix-algebra-F02674 | 285  | 1.000 | $(1/3) R_1$                                                                                                         | $(1/3) R_1$                                                                                                         | that we applied was $(1/3) R_1$ , we conclude that the determinant of                                                                             |
| johnston-linear-matrix-algebra-F02675 | 285  | 0.999 | $(1 \cdot (-2) \cdot 1) / (1/3) = -6$                                                                               | $(1 \cdot (-2) \cdot 1) / (1/3) = -6$                                                                               | the original matrix is $(1 \cdot (-2) \cdot 1) / (1/3) = -6$ .                                                                                    |
| johnston-linear-matrix-algebra-F02676 | 285  | 1.000 | $(-1) \cdot 1 \cdot 2 \cdot (-7/2) \cdot (5/7) = 5$                                                                 | $(-1) \cdot 1 \cdot 2 \cdot (-7/2) \cdot (5/7) = 5$                                                                 | original matrix is $(-1) \cdot 1 \cdot 2 \cdot (-7/2) \cdot (5/7) = 5$ .                                                                          |
| johnston-linear-matrix-algebra-F02677 | 285  | 0.435 | $[a]$                                                                                                               | $[a]$                                                                                                               | of a $1 \times 1$ matrix $[a]$                                                                                                                    |
| johnston-linear-matrix-algebra-F02678 | 286  | 1.000 | $B = \left[ \begin{array}{cc} 0 & 2 \\ 3 & 5 \end{array} \right]$                                                   | $B = \left[ \begin{array}{cc} 0 & 2 \\ 3 & 5 \end{array} \right]$                                                   | Show that the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix}$ are not similar. |
| johnston-linear-matrix-algebra-F02679 | 286  | 0.966 | $\operatorname{tr}(A) = \operatorname{tr}(B) = 5$                                                                   | $\operatorname{tr}(A) = \operatorname{tr}(B) = 5$                                                                   | similar, since $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$ and $\operatorname{tr}(A) = \operatorname{tr}(B) = 5$ . However,             |
| johnston-linear-matrix-algebra-F02680 | 286  | 1.000 | $\operatorname{det}(A) \neq \operatorname{det}(B)$                                                                  | $\operatorname{det}(A) \neq \operatorname{det}(B)$                                                                  | Since $\operatorname{det}(A) \neq \operatorname{det}(B)$ , we conclude that $A$ and $B$ are not similar.                                          |
| johnston-linear-matrix-algebra-F02681 | 288  | 1.000 | $a e i + b f g + c d h - a f h - b d i - c e g$                                                                     | $a e i + b f g + c d h - a f h - b d i - c e g$                                                                     | Determinant: $a e i + b f g + c d h - a f h - b d i - c e g$                                                                                      |
| johnston-linear-matrix-algebra-F02682 | 288  | 1.000 | $\left[ \begin{array}{ccc} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{array} \right]$                                | $\left[ \begin{array}{ccc} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{array} \right]$                                | d) $\left[ \begin{array}{ccc} 0 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & 0 \end{array} \right]$                                                           |
| johnston-linear-matrix-algebra-F02683 | 289  | 1.000 | $m_{i,j}$                                                                                                           | $m_{i,j}$                                                                                                           | We typically use $m_{i,j}$                                                                                                                        |
| johnston-linear-matrix-algebra-F02684 | 289  | 1.000 | $c_{i,j}$                                                                                                           | $c_{i,j}$                                                                                                           | $c_{i,j}$ to denote the                                                                                                                           |
| johnston-linear-matrix-algebra-F02685 | 289  | 0.898 | $(\mathbf{i}, \mathbf{j})$                                                                                          | $(\mathbf{i}, \mathbf{j})$                                                                                          | a) the <b>(i,j)-minor</b> of $A$ is the determinant of the $(n-1) \times (n-1)$                                                                   |
| johnston-linear-matrix-algebra-F02686 | 289  | 0.843 | $\mathbf{i}, \mathbf{j}$                                                                                            | $\mathbf{i}, \mathbf{j}$                                                                                            | b) the <b>(i,j)-cofactor</b> of $A$ is the quantity $(-1)^{i+j} m_{i,j}$ , where $m_{i,j}$ is the                                                 |
| johnston-linear-matrix-algebra-F02687 | 289  | 0.843 | $(-1)^{i+j} m_{i,j}$                                                                                                | $(-1)^{i+j} m_{i,j}$                                                                                                | b) the <b>(i,j)-cofactor</b> of $A$ is the quantity $(-1)^{i+j} m_{i,j}$ , where $m_{i,j}$ is the                                                 |
| johnston-linear-matrix-algebra-F02688 | 289  | 0.860 | $(-1)^{i+j}$                                                                                                        | $(-1)^{i+j}$                                                                                                        | minor $m_{i,j}$ by $(-1)^{i+j}$ . In this case, the $(1,2)$ -cofactor of $A$ is                                                                   |
| johnston-linear-matrix-algebra-F02689 | 289  | 1.000 | $c_{1,2} = -m_{1,2} = -(-6) = 6$                                                                                    | $c_{1,2} = -m_{1,2} = -(-6) = 6$                                                                                    | $c_{1,2} = -m_{1,2} = -(-6) = 6$ that we did earlier).                                                                                            |
| johnston-linear-matrix-algebra-F02690 | 290  | 0.793 | $(3, 2)$                                                                                                            | $(3, 2)$                                                                                                            | $(3, 2)$ -cofactors have their signs switched (we have highlighted these                                                                          |
| johnston-linear-matrix-algebra-F02691 | 291  | 1.000 | $\{c_{i,j}\}_{i,j=1}^n$                                                                                             | $\{c_{i,j}\}_{i,j=1}^n$                                                                                             | Suppose $A \in \mathcal{M}_n$ has cofactors $\{c_{i,j}\}_{i,j=1}^n$ . Then                                                                        |
| johnston-linear-matrix-algebra-F02692 | 293  | 1.000 | $c_{1,1}$                                                                                                           | $c_{1,1}$                                                                                                           | $c_{1,1}$ . To see why the first equality holds, notice that if the matrix                                                                        |

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| johnston-linear-matrix-algebra-F02693 | 294  | ■ 1.000 | $i-1$                                                                                                                                                         | $i-1$                                                                                                                                           | for the $i$ -th term in the sum, we perform $i-1$ swap operations: first we swap                                                                    |
| johnston-linear-matrix-algebra-F02694 | 294  | ■ 1.000 | $i-2$                                                                                                                                                         | $i-2$                                                                                                                                           | rows $i$ and $i-1$ , then $i-1$ and $i-2$ , then $i-2$ and $i-3$ , and so on until rows                                                             |
| johnston-linear-matrix-algebra-F02695 | 294  | ■ 1.000 | $i-3$                                                                                                                                                         | $i-3$                                                                                                                                           | rows $i$ and $i-1$ , then $i-1$ and $i-2$ , then $i-2$ and $i-3$ , and so on until rows                                                             |
| johnston-linear-matrix-algebra-F02696 | 294  | ■ 0.992 | $(-1)^{i-1}$                                                                                                                                                  | $(-1)^{i-1}$                                                                                                                                    | 2 and 1. These swaps introduce a sign of $(-1)^{i-1}$ , which explains why the $c_{i,1}$                                                            |
| johnston-linear-matrix-algebra-F02697 | 294  | ■ 0.992 | $c_{i,1}$                                                                                                                                                     | $c_{i,1}$                                                                                                                                       | 2 and 1. These swaps introduce a sign of $(-1)^{i-1}$ , which explains why the $c_{i,1}$                                                            |
| johnston-linear-matrix-algebra-F02698 | 294  | ■ 1.000 | $\operatorname{det}(A)=\operatorname{det}(\left(A^T\right))$                                                                                                  | $\det(A)=\det(A^T)$                                                                                                                             | columns), recall from Theorem 3.2.2(b) that $\det(A)=\det(A^T)$ , and a cofactor                                                                    |
| johnston-linear-matrix-algebra-F02699 | 296  | ■ 1.000 | $f(I)=2$                                                                                                                                                      | $f(I)=2$                                                                                                                                        | require that $f(I)=2$ instead of $f(I)=1$ :                                                                                                         |
| johnston-linear-matrix-algebra-F02700 | 296  | ■ 1.000 | $f(I)=1$                                                                                                                                                      | $f(I)=1$                                                                                                                                        | require that $f(I)=2$ instead of $f(I)=1$ :                                                                                                         |
| johnston-linear-matrix-algebra-F02701 | 296  | ■ 1.000 | $f(A\ B)=f(A)\ f(B)$                                                                                                                                          | $f(AB)=f(A)f(B)$                                                                                                                                | b) $f(AB)=f(A)f(B)$ for all $A,B\in\mathcal{M}_n$ .                                                                                                 |
| johnston-linear-matrix-algebra-F02702 | 296  | ■ 1.000 | $A=B=I$                                                                                                                                                       | $A=B=I$                                                                                                                                         | properties (a) and (b) described above. To see why, simply let $A=B=I$                                                                              |
| johnston-linear-matrix-algebra-F02703 | 296  | ■ 0.998 | $\left[\begin{array}{ll}1 & 2 \\ 3 & 5\end{array}\right]$                                                                                                     | $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$                                                                                                  | (b) $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$                                                                                                  |
| johnston-linear-matrix-algebra-F02704 | 296  | ■ 1.000 | $\left[\begin{array}{ccc}1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1\end{array}\right]$                                                                             | $\begin{bmatrix} 1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1 \end{bmatrix}$                                                                           | (d) $\begin{bmatrix} 1 & 3 & 0 \\ 0 & -2 & 2 \\ -1 & 0 & 1 \end{bmatrix}$                                                                           |
| johnston-linear-matrix-algebra-F02705 | 296  | ■ 0.832 | $\left[\begin{array}{ccc}2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7\end{array}\right]$                                                                              | $\begin{bmatrix} 2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7 \end{bmatrix}$                                                                            | *(e) $\begin{bmatrix} 2 & 0 & 1 \\ 5 & 2 & 8 \\ 3 & -2 & 7 \end{bmatrix}$                                                                           |
| johnston-linear-matrix-algebra-F02706 | 296  | ■ 1.000 | $\left[\begin{array}{ccc}3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3\end{array}\right]$                                                                              | $\begin{bmatrix} 3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$                                                                            | $\begin{bmatrix} 3 & 2 & -6 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$                                                                                |
| johnston-linear-matrix-algebra-F02707 | 296  | ■ 1.000 | $\left[\begin{array}{cccc}1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2\end{array}\right]$                                               | $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$                                              | $\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & -3 \\ 4 & 3 & 1 & 2 \\ 0 & 0 & 2 & -2 \end{bmatrix}$                                                  |
| johnston-linear-matrix-algebra-F02708 | 297  | ■ 0.994 | $\left[\begin{array}{cccc}3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6\end{array}\right]$                                              | $\begin{bmatrix} 3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6 \end{bmatrix}$                                             | *(a) $\begin{bmatrix} 3 & 1 & 6 & 4 \\ -2 & 5 & 3 & 3 \\ 6 & -2 & 4 & 1 \\ 3 & 4 & -2 & 6 \end{bmatrix}$                                            |
| johnston-linear-matrix-algebra-F02709 | 297  | ■ 1.000 | $\left[\begin{array}{ccccc}-2 & 6 & -1 & 6 & -1 \\ -1 & 3 & -1 & 4 & 3 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2\end{array}\right]$ | $\begin{bmatrix} -2 & 6 & -1 & 6 & -1 \\ 5 & 4 & -1 & 3 & -1 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2 \end{bmatrix}$ | (b) $\begin{bmatrix} -2 & 6 & -1 & 6 & -1 \\ 5 & 4 & -1 & 3 & -1 \\ 4 & 3 & 4 & -2 & 1 \\ 6 & 6 & 5 & 4 & 2 \\ -1 & 2 & -2 & -1 & -2 \end{bmatrix}$ |

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| johnston-linear-matrix-algebra-F02710 | 297  | 0.999 | $\left[\begin{array}{cccccc}2 & 2 & 5 & -1 & 1 & -1 \\ 3 & 0 & 0 & 1 & 1 & 1 \\ 1 & 2 & 5 & -1 & 0 & 3 \\ 4 & 0 & 5 & 5 & -1 & 3 \\ 5 & -1 & 5 & 2 & 0 & -1 \\ 3 & 0 & 3 & 3 & 4 & 3\end{array}\right]$ | <div><div>*(c)</div><div><math display="block">\begin{bmatrix} 2 &amp; 2 &amp; 5 &amp; -1 &amp; 1 &amp; -1 \\ 1 &amp; 3 &amp; 0 &amp; 0 &amp; 1 &amp; 1 \\ 1 &amp; 2 &amp; 5 &amp; -1 &amp; 0 &amp; 3 \\ 4 &amp; 0 &amp; 5 &amp; 5 &amp; -1 &amp; 3 \\ 5 &amp; -1 &amp; 5 &amp; 2 &amp; 0 &amp; -1 \\ 3 &amp; 0 &amp; 3 &amp; 3 &amp; 4 &amp; 3 \end{bmatrix}</math></div></div>     |                                                                                                                                                                                                                                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02711 | 297  | 1.000 | $\left[\begin{array}{cccccc}-2 & 3 & 2 & 5 & 2 & 4 \\ 2 & -1 & -3 & 2 & 0 & 6 \\ -2 & 2 & -2 & -2 & -3 & 3 \\ 3 & 3 & 2 & 2 & -3 & 1 \\ 1 & 6 & 0 & -1 & -3 & -2\end{array}\right]$                     | <div><div></div><div><math display="block">\begin{bmatrix} -2 &amp; 3 &amp; 2 &amp; 5 &amp; 2 &amp; 4 \\ -1 &amp; 2 &amp; -1 &amp; -3 &amp; 2 &amp; 0 \\ 6 &amp; -2 &amp; 3 &amp; 5 &amp; 4 &amp; -1 \\ -2 &amp; 2 &amp; -2 &amp; -2 &amp; -3 &amp; 3 \\ 3 &amp; 3 &amp; 2 &amp; 2 &amp; -3 &amp; 1 \\ 1 &amp; 6 &amp; 0 &amp; -1 &amp; -3 &amp; -2 \end{bmatrix}</math></div></div> | <div><div>(d)</div><div><math display="block">\begin{bmatrix} -2 &amp; 3 &amp; 2 &amp; 5 &amp; 2 &amp; 4 \\ -1 &amp; 2 &amp; -1 &amp; -3 &amp; 2 &amp; 0 \\ 6 &amp; -2 &amp; 3 &amp; 5 &amp; 4 &amp; -1 \\ -2 &amp; 2 &amp; -2 &amp; -2 &amp; -3 &amp; 3 \\ 3 &amp; 3 &amp; 2 &amp; 2 &amp; -3 &amp; 1 \\ 1 &amp; 6 &amp; 0 &amp; -1 &amp; -3 &amp; -2 \end{bmatrix}</math></div></div> |
| johnston-linear-matrix-algebra-F02712 | 297  | 1.000 | A, B, C \in \mathcal{M}_5                                                                                                                                                                               | A, B, C \in \mathcal{M}_5                                                                                                                                                                                                                                                                                                                                                            | 3.2.3 Suppose that A, B, C \in \mathcal{M}_5 are such that det(A) = 2,                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02713 | 297  | 1.000 | det(B)=3                                                                                                                                                                                                | det(B) = 3                                                                                                                                                                                                                                                                                                                                                                           | det(B) = 3, and det(C) = 0. Compute the determinant of                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02714 | 297  | 1.000 | det(C)=0                                                                                                                                                                                                | det(C) = 0                                                                                                                                                                                                                                                                                                                                                                           | det(B) = 3, and det(C) = 0. Compute the determinant of                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02715 | 297  | 0.904 | A B^T                                                                                                                                                                                                   | AB^T                                                                                                                                                                                                                                                                                                                                                                                 | *(a) AB^T                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-F02716 | 297  | 1.000 | A^2 B                                                                                                                                                                                                   | A^2B                                                                                                                                                                                                                                                                                                                                                                                 | *(c) A^2B                                                                                                                                                                                                                                                                                                                                                                               |
| johnston-linear-matrix-algebra-F02717 | 297  | 1.000 | 2 A^2                                                                                                                                                                                                   | 2A^2                                                                                                                                                                                                                                                                                                                                                                                 | (d) 2A^2                                                                                                                                                                                                                                                                                                                                                                                |
| johnston-linear-matrix-algebra-F02718 | 297  | 1.000 | A^{-1} B^3                                                                                                                                                                                              | A^{-1}B^3                                                                                                                                                                                                                                                                                                                                                                            | *(e) A^{-1}B^3                                                                                                                                                                                                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F02719 | 297  | 1.000 | A^3 B^{-2} A^{-4} B^2 A                                                                                                                                                                                 | A^3B^{-2}A^{-4}B^2A                                                                                                                                                                                                                                                                                                                                                                  | (f) A^3B^{-2}A^{-4}B^2A                                                                                                                                                                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F02720 | 297  | 1.000 | 3 A^2 \left( B^T B / 2 \right)^{-1}                                                                                                                                                                     | 3A^2 \left( B^T B / 2 \right)^{-1}                                                                                                                                                                                                                                                                                                                                                   | *(g) 3A^2(B^TB/2)^{-1}                                                                                                                                                                                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F02721 | 297  | 1.000 | A^{64} B^{-14} C^9 A^{-6} B^{13}                                                                                                                                                                        | A^{64}B^{-14}C^9A^{-6}B^{13}                                                                                                                                                                                                                                                                                                                                                         | (h) A^{64}B^{-14}C^9A^{-6}B^{13}                                                                                                                                                                                                                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02722 | 297  | 1.000 | det(A)=6                                                                                                                                                                                                | det(A) = 6                                                                                                                                                                                                                                                                                                                                                                           | 3.2.4 Suppose that A \in \mathcal{M}_4 is a matrix with det(A) = 6.                                                                                                                                                                                                                                                                                                                     |
| johnston-linear-matrix-algebra-F02723 | 297  | 1.000 | 3 R_3                                                                                                                                                                                                   | 3R_3                                                                                                                                                                                                                                                                                                                                                                                 | (b) 3R_3                                                                                                                                                                                                                                                                                                                                                                                |
| johnston-linear-matrix-algebra-F02724 | 297  | 0.961 | R_{\{1\}} \leftrightharpoons R_{\{4\}}                                                                                                                                                                  | R_1 \leftrightarrow R_4                                                                                                                                                                                                                                                                                                                                                              | *(c) R_1 \leftrightarrow R_4                                                                                                                                                                                                                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02725 | 297  | 1.000 | R_{\{2\}}-2 R_{\{1\}}, R_{\{3\}}-4 R_{\{1\}}                                                                                                                                                            | R_2 - 2R_1, R_3 - 4R_1                                                                                                                                                                                                                                                                                                                                                               | (d) R_2 - 2R_1, R_3 - 4R_1                                                                                                                                                                                                                                                                                                                                                              |
| johnston-linear-matrix-algebra-F02726 | 297  | 0.991 | 2 R_{\{1\}}, R_{\{2\}} \leftrightharpoons R_{\{4\}}                                                                                                                                                     | 2R_1, R_2 \leftrightarrow R_4                                                                                                                                                                                                                                                                                                                                                        | *(e) 2R_1, R_2 \leftrightarrow R_4                                                                                                                                                                                                                                                                                                                                                      |
| johnston-linear-matrix-algebra-F02727 | 297  | 1.000 | R_{\{1\}} \leftrightharpoons R_{\{2\}}, R_{\{2\}} \leftrightharpoons R_{\{3\}}                                                                                                                          | R_1 \leftrightarrow R_2, R_2 \leftrightarrow R_3                                                                                                                                                                                                                                                                                                                                     | (f) R_1 \leftrightarrow R_2, R_2 \leftrightarrow R_3                                                                                                                                                                                                                                                                                                                                    |
| johnston-linear-matrix-algebra-F02728 | 297  | 1.000 | 2 R_{\{2\}}, 3 R_{\{3\}}, 4 R_{\{4\}}                                                                                                                                                                   | 2R_2, 3R_3, 4R_4                                                                                                                                                                                                                                                                                                                                                                     | *(g) 2R_2, 3R_3, 4R_4                                                                                                                                                                                                                                                                                                                                                                   |
| johnston-linear-matrix-algebra-F02729 | 297  | 1.000 | R_{\{1\}} \leftrightharpoons R_{\{3\}}, 2 R_{\{2\}}, R_{\{3\}}-3 R_{\{1\}}, R_{\{4\}}-7 R_{\{2\}}, 3 R_{\{4\}}                                                                                          | R_1 \leftrightarrow R_3, 2R_2, R_3 - 3R_1, R_4 - 7R_2, 3R_4                                                                                                                                                                                                                                                                                                                          | (h) R_1 \leftrightarrow R_3, 2R_2, R_3 - 3R_1, R_4 - 7R_2, 3R_4                                                                                                                                                                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02730 | 297  | 0.979 | $\left[\begin{array}{lll}g & h & i \\ d & e & f \\ a & b & c\end{array}\right]$                                                                                                                         | <div><div></div><div><math display="block">\begin{bmatrix} g &amp; h &amp; i \\ d &amp; e &amp; f \\ a &amp; b &amp; c \end{bmatrix}</math></div></div>                                                                                                                                                                                                                              | <div><div></div><div><math display="block">\begin{bmatrix} g &amp; h &amp; i \\ d &amp; e &amp; f \\ a &amp; b &amp; c \end{bmatrix}</math></div></div>                                                                                                                                                                                                                                 |

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| johnston-linear-matrix-algebra-F02731 | 297  | 1.000 | $\left[\begin{array}{ccc}a & b & c \\ 2d & 2e & 2f \\ 3g & 3h & 3i\end{array}\right]$                                                                                | $\begin{bmatrix} a & b & c \\ 2d & 2e & 2f \\ 3g & 3h & 3i \end{bmatrix}$         | <b>(b)</b> $\begin{bmatrix} a & b & c \\ 2d & 2e & 2f \\ 3g & 3h & 3i \end{bmatrix}$                      |
| johnston-linear-matrix-algebra-F02732 | 297  | 0.566 | $\left[\begin{array}{ccc}a & b+a & 2c \\ d & e+d & 2f \\ g & h+g & 2i\end{array}\right]$                                                                             | $\begin{bmatrix} a & b+a & 2c \\ d & e+d & 2f \\ g & h+g & 2i \end{bmatrix}$      | $\begin{bmatrix} a & b+a & 2c \\ d & e+d & 2f \\ g & h+g & 2i \end{bmatrix}$                              |
| johnston-linear-matrix-algebra-F02733 | 297  | 1.000 | $\left[\begin{array}{ccc}a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i\end{array}\right]$                                                                              | $\begin{bmatrix} a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i \end{bmatrix}$       | <b>(d)</b> $\begin{bmatrix} a & 2b & 3c \\ 2d & 4e & 6f \\ 3g & 6h & 9i \end{bmatrix}$                    |
| johnston-linear-matrix-algebra-F02734 | 297  | 0.997 | $\operatorname{det}(-A)=$                                                                                                                                            | $\det(-A) =$                                                                      | <b>*(a)</b> For all square matrices $A$ , it is true that $\det(-A) =$                                    |
| johnston-linear-matrix-algebra-F02735 | 297  | 1.000 | $-\operatorname{det}(A)$                                                                                                                                             | $-\det(A)$                                                                        | $-\det(A)$ .                                                                                              |
| johnston-linear-matrix-algebra-F02736 | 297  | 1.000 | $\operatorname{det}(A\ B)=\operatorname{det}(B\ A)$                                                                                                                  | $\det(AB) = \det(BA)$                                                             | $\det(AB) = \det(BA)$ .                                                                                   |
| johnston-linear-matrix-algebra-F02737 | 297  | 1.000 | $\operatorname{det}(A+B)=\operatorname{det}(A)+\operatorname{det}(B)$                                                                                                | $\det(A+B) = \det(A) + \det(B)$                                                   | $\det(A+B) = \det(A) + \det(B)$ .                                                                         |
| johnston-linear-matrix-algebra-F02738 | 297  | 0.952 | $\operatorname{det}(A)=3$                                                                                                                                            | $\det(A) = 3$                                                                     | <b>(d)</b> If $A$ is a $3 \times 3$ matrix with $\det(A) = 3$ , then $\operatorname{rank}(A) =$           |
| johnston-linear-matrix-algebra-F02739 | 297  | 1.000 | $\operatorname{det}(A)=1$                                                                                                                                            | $\det(A) = 1$                                                                     | form is $I$ , then $\det(A) = 1$ .                                                                        |
| johnston-linear-matrix-algebra-F02740 | 297  | 1.000 | $B_{\{n\}}$                                                                                                                                                          | $B_n$                                                                             | <b>3.2.7</b> Let $B_n$ be the $n \times n$ “backwards identity” matrix                                    |
| johnston-linear-matrix-algebra-F02741 | 297  | 1.000 | $\operatorname{det}\left(B_{\{n\}}\right)$                                                                                                                           | $\det(B_n)$                                                                       | Find a formula for $\det(B_n)$ in terms of $n$ .                                                          |
| johnston-linear-matrix-algebra-F02742 | 297  | 0.977 | $\left[P_{\{\mathbf{u}\}}\right]$                                                                                                                                    | $[P_{\mathbf{u}}]$                                                                | <b>*3.2.10</b> Let $[P_{\mathbf{u}}]$ be the standard matrix of a projection                              |
| johnston-linear-matrix-algebra-F02743 | 297  | 1.000 | $\operatorname{det}\left(\left(\left[P_{\{\mathbf{u}\}}\right]\right)\right)$                                                                                        | $\det([P_{\mathbf{u}}])$                                                          | onto a line (see Section 1.4.2). Compute $\det([P_{\mathbf{u}}])$ .                                       |
| johnston-linear-matrix-algebra-F02744 | 297  | 1.000 | $\operatorname{det}\left(A^{\{r\}}\right)=\left(\operatorname{det}(A)\right)^{\{r\}}$                                                                                | $\det(A^r) = (\det(A))^r$                                                         | $\det(A^r) = (\det(A))^r$ .                                                                               |
| johnston-linear-matrix-algebra-F02745 | 297  | 1.000 | $A^{\{T\}}=-A$                                                                                                                                                       | $A^T = -A$                                                                        | $A^T = -A$ . Show that skew-symmetric matrices are not in-                                                |
| johnston-linear-matrix-algebra-F02746 | 298  | 1.000 | $A^{\{R\}}$                                                                                                                                                          | $A^R$                                                                             | <b>3.2.14</b> Given a matrix $A \in \mathcal{M}_n$ , let $A^R$ denote the matrix                          |
| johnston-linear-matrix-algebra-F02747 | 298  | 1.000 | $\operatorname{det}\left(A^{\{R\}}\right)$                                                                                                                           | $\det(A^R)$                                                                       | Find a formula for $\det(A^R)$ in terms of $\det(A)$ and $n$ . [Hint:                                     |
| johnston-linear-matrix-algebra-F02748 | 298  | 1.000 | $\operatorname{det}\left(A_{\{n\}}\right)$                                                                                                                           | $\det(A_n)$                                                                       | <b>(a)</b> Use computer software to compute $\det(A_n)$ when                                              |
| johnston-linear-matrix-algebra-F02749 | 298  | 1.000 | $n=2,3,4,5$                                                                                                                                                          | $n = 2, 3, 4, 5$                                                                  | $n = 2, 3, 4, 5$ .                                                                                        |
| johnston-linear-matrix-algebra-F02750 | 298  | 0.971 | $(*)$                                                                                                                                                                | $(*)$                                                                             | where asterisks $(*)$ denote blocks whose values are                                                      |
| johnston-linear-matrix-algebra-F02751 | 298  | 1.000 | $\prod_{j=1}^{\{n\}}$                                                                                                                                                | $\prod_{j=1}^n$                                                                   | [Side note: The notation $\prod_{j=1}^n$ refers to a product                                              |
| johnston-linear-matrix-algebra-F02752 | 298  | 1.000 | $A \in \mathcal{M}_{\{m\}}, B \in \mathcal{M}_{\{m, n\}}, C \in \mathcal{M}_{\{m, m\}}$                                                                              | $A \in \mathcal{M}_m, B \in \mathcal{M}_{m,n}, C \in \mathcal{M}_{n,m}$           | <b>3.2.17</b> Suppose $A \in \mathcal{M}_m, B \in \mathcal{M}_{m,n}, C \in \mathcal{M}_{n,m}$ , and       |
| johnston-linear-matrix-algebra-F02753 | 298  | 0.922 | $\operatorname{det}\left(I_{\{n\}}+\mathbf{w}^{\{T\}}\ \mathbf{w}^{\{T\}}\right)=1+\mathbf{w}^{\{T\}}\ \mathbf{w}^{\{T\}}$                                           | $\det(I_n + \mathbf{w}\mathbf{w}^T) = 1 + \mathbf{w}^T \mathbf{w}$                | <b>(b)</b> Use part (a) to show that $\det(I_n + \mathbf{w}\mathbf{w}^T) = 1 + \mathbf{w}^T \mathbf{w}$ . |
| johnston-linear-matrix-algebra-F02754 | 298  | 0.827 | $\operatorname{det}\left(A+\mathbf{w}^{\{T\}}\ A^{-1}\ \mathbf{w}^{\{T\}}\right)=\left(1+\mathbf{w}^{\{T\}}\ A^{-1}\ \mathbf{w}^{\{T\}}\right)\operatorname{det}(A)$ | $\det(A + \mathbf{w}\mathbf{w}^T) = (1 + \mathbf{w}^T A^{-1} \mathbf{w}) \det(A)$ | <b>(c)</b> Show that $\det(A + \mathbf{w}\mathbf{w}^T) = (1 + \mathbf{w}^T A^{-1} \mathbf{w}) \det(A)$ .  |
| johnston-linear-matrix-algebra-F02755 | 298  | 0.971 | $A+\mathbf{w}^{\{T\}}\ A^{-1}\ \mathbf{w}^{\{T\}}=A\left(I_{\{n\}}+A^{-1}\ \mathbf{w}^{\{T\}}\ \mathbf{w}^{\{T\}}\right)$                                            | $A + \mathbf{w}\mathbf{w}^T = A(I_n + A^{-1} \mathbf{w}\mathbf{w}^T)$             | [Hint: Write $A + \mathbf{w}\mathbf{w}^T = A(I_n + A^{-1} \mathbf{w}\mathbf{w}^T)$ .]                     |

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| johnston-linear-matrix-algebra-F02756 | 298  | ■ 1.000 | $(-1)^{n+1}$                                       | $(-1)^{n+1}$                                       | equal to $(-1)^{n+1}$ .                                                                                             |
| johnston-linear-matrix-algebra-F02757 | 298  | ■ 1.000 | $f: \mathcal{M}_n \rightarrow$                     | $f: \mathcal{M}_n \rightarrow$                     | <b>3.2.22</b> Show that there does not exist a function $f: \mathcal{M}_n \rightarrow$                              |
| johnston-linear-matrix-algebra-F02758 | 298  | ■ 1.000 | $f(A+cB)=$                                         | $f(A+cB)=$                                         | c) for all $c \in \mathbb{R}$ and $A, B \in \mathcal{M}_n$ , we have $f(A+cB)=$                                     |
| johnston-linear-matrix-algebra-F02759 | 298  | ■ 1.000 | $f(A)+cf(B)$                                       | $f(A)+cf(B)$                                       | $f(A)+cf(B)$ .                                                                                                      |
| johnston-linear-matrix-algebra-F02760 | 299  | ■ 1.000 | $\lambda$                                          | $\lambda$                                          | to $\lambda$ . It does not                                                                                          |
| johnston-linear-matrix-algebra-F02761 | 299  | ■ 1.000 | $A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$            | $A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$            | diagonal then $A \mathbf{e}_j = a_{j,j} \mathbf{e}_j$ for all $1 \leq j \leq n$ .                                   |
| johnston-linear-matrix-algebra-F02762 | 299  | ■ 1.000 | $a_{1,1}, a_{2,2}, \dots, a_{n,n}$                 | $a_{1,1}, a_{2,2}, \dots, a_{n,n}$                 | $A$ 's diagonal entries $a_{1,1}, a_{2,2}, \dots, a_{n,n}$ .                                                        |
| johnston-linear-matrix-algebra-F02763 | 300  | ■ 1.000 | $A \mathbf{0} = \lambda \mathbf{0}$                | $A \mathbf{0} = \lambda \mathbf{0}$                | that $A \mathbf{0} = \lambda \mathbf{0}$ for all                                                                    |
| johnston-linear-matrix-algebra-F02764 | 300  | ■ 1.000 | $A \mathbf{v} = 3 \mathbf{v}$                      | $A \mathbf{v} = 3 \mathbf{v}$                      | Since $A \mathbf{v} = 3 \mathbf{v}$ , the eigenvalue corresponding to $\mathbf{v}$ is $\lambda = 3$ . The effect of |
| johnston-linear-matrix-algebra-F02765 | 300  | ■ 1.000 | $\lambda = 3$                                      | $\lambda = 3$                                      | Since $A \mathbf{v} = 3 \mathbf{v}$ , the eigenvalue corresponding to $\mathbf{v}$ is $\lambda = 3$ . The effect of |
| johnston-linear-matrix-algebra-F02766 | 300  | ■ 1.000 | $\lambda = 1$                                      | $\lambda = 1$                                      | $\lambda = 1$ is an eigenvalue of $A$ , and find an eigenvector that it corresponds to.                             |
| johnston-linear-matrix-algebra-F02767 | 300  | ■ 1.000 | $A \mathbf{v} = 1 \mathbf{v}$                      | $A \mathbf{v} = 1 \mathbf{v}$                      | We want to find a non-zero vector $\mathbf{v} \in \mathbb{R}^2$ such that $A \mathbf{v} = 1 \mathbf{v}$ . This is a |
| johnston-linear-matrix-algebra-F02768 | 300  | ■ 1.000 | $v_1 = -v_2$                                       | $v_1 = -v_2$                                       | $v_1 = -v_2$ . In other words, the eigenvectors are the (non-zero) vectors of                                       |
| johnston-linear-matrix-algebra-F02769 | 300  | ■ 1.000 | $\mathbf{v} = (-v_2, v_2)$                         | $\mathbf{v} = (-v_2, v_2)$                         | the form $\mathbf{v} = (-v_2, v_2)$ . We (arbitrarily) choose $v_2 = 1$ so that we get                              |
| johnston-linear-matrix-algebra-F02770 | 300  | ■ 1.000 | $v_2 = 1$                                          | $v_2 = 1$                                          | the form $\mathbf{v} = (-v_2, v_2)$ . We (arbitrarily) choose $v_2 = 1$ so that we get                              |
| johnston-linear-matrix-algebra-F02771 | 300  | ■ 1.000 | $\mathbf{v} = (-1, 1)$                             | $\mathbf{v} = (-1, 1)$                             | $\mathbf{v} = (-1, 1)$ as an eigenvector of $A$ with corresponding eigenvalue $\lambda = 1$ .                       |
| johnston-linear-matrix-algebra-F02772 | 301  | ■ 1.000 | $A \mathbf{v} - \lambda \mathbf{v}$                | $A \mathbf{v} - \lambda \mathbf{v}$                | factoring $A \mathbf{v} - \lambda \mathbf{v}$ . It is                                                               |
| johnston-linear-matrix-algebra-F02773 | 301  | ■ 1.000 | $(A - \lambda) \mathbf{v} = \mathbf{0}$            | $(A - \lambda) \mathbf{v} = \mathbf{0}$            | $(A - \lambda) \mathbf{v} = \mathbf{0}$ , but this                                                                  |
| johnston-linear-matrix-algebra-F02774 | 301  | ■ 0.955 | $A \mathbf{v} = \lambda \mathbf{v}$                | $A \mathbf{v} = \lambda \mathbf{v}$                | only if there is a non-zero vector $\mathbf{v}$ such that $A \mathbf{v} = \lambda \mathbf{v}$ . By moving both      |
| johnston-linear-matrix-algebra-F02775 | 301  | ■ 1.000 | $(A - \lambda I) \mathbf{v} = \mathbf{0}$          | $(A - \lambda I) \mathbf{v} = \mathbf{0}$          | $(A - \lambda I) \mathbf{v} = \mathbf{0}$ has a non-zero solution. By Theorem 2.2.4, this is equiva-                |
| johnston-linear-matrix-algebra-F02776 | 301  | ■ 1.000 | $A - \lambda I$                                    | $A - \lambda I$                                    | lent to $A - \lambda I$ not being invertible. We have thus arrived at the following                                 |
| johnston-linear-matrix-algebra-F02777 | 301  | ■ 1.000 | $\det(A - \lambda I)$                              | $\det(A - \lambda I)$                              | is to compute $\det(A - \lambda I)$ via one of the explicit formulas for the                                        |
| johnston-linear-matrix-algebra-F02778 | 301  | ■ 1.000 | $\det(A - \lambda I) = 0$                          | $\det(A - \lambda I) = 0$                          | is not invertible if and only if $\det(A - \lambda I) = 0$ , by Theorem 3.2.1). We                                  |
| johnston-linear-matrix-algebra-F02779 | 301  | ■ 0.998 | $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ | $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ | Compute all of the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$ .                   |
| johnston-linear-matrix-algebra-F02780 | 302  | ■ 1.000 | $\lambda = -1$                                     | $\lambda = -1$                                     | so the eigenvalues of $A$ are $\lambda = -1$ and $\lambda = 6$ .                                                    |
| johnston-linear-matrix-algebra-F02781 | 302  | ■ 1.000 | $\lambda = 6$                                      | $\lambda = 6$                                      | so the eigenvalues of $A$ are $\lambda = -1$ and $\lambda = 6$ .                                                    |
| johnston-linear-matrix-algebra-F02782 | 302  | ■ 1.000 | $\lambda^2 - 5\lambda - 6$                         | $\lambda^2 - 5\lambda - 6$                         | As an alternative to factoring the polynomial $\lambda^2 - 5\lambda - 6$ , we could have                            |

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| johnston-linear-matrix-algebra-F02783 | 302  | 1.000 | $a \lambda^2 + b \lambda + c = 0$                          | $a \lambda^2 + b \lambda + c = 0$             | solutions of $a \lambda^2 + b \lambda + c = 0$ are                                                           |
| johnston-linear-matrix-algebra-F02784 | 302  | 1.000 | $\lambda = (5 \pm 7) / 2$                                  | $\lambda = (5 \pm 7) / 2$                     | In this case, the quadratic formula gives $\lambda = (5 \pm 7) / 2$ , which of course                        |
| johnston-linear-matrix-algebra-F02785 | 302  | 1.000 | $\mathbf{v} \in \text{null}(A - \lambda I)$                | $\mathbf{v} \in \text{null}(A - \lambda I)$   | sponding eigenvalue $\lambda$ if and only if $\mathbf{v} \in \text{null}(A - \lambda I)$ .                   |
| johnston-linear-matrix-algebra-F02786 | 302  | 1.000 | $(A + I) \mathbf{v} = \mathbf{0}$                          | $(A + I) \mathbf{v} = \mathbf{0}$             | systems $(A + I) \mathbf{v} = \mathbf{0}$ and $(A - 6I) \mathbf{v} = \mathbf{0}$ , respectively:             |
| johnston-linear-matrix-algebra-F02787 | 302  | 1.000 | $(A - 6I) \mathbf{v} = \mathbf{0}$                         | $(A - 6I) \mathbf{v} = \mathbf{0}$            | systems $(A + I) \mathbf{v} = \mathbf{0}$ and $(A - 6I) \mathbf{v} = \mathbf{0}$ , respectively:             |
| johnston-linear-matrix-algebra-F02788 | 302  | 0.998 | $(A - \lambda I) \mathbf{v} =$                             | $(A - \lambda I) \mathbf{v} =$                | $\lambda = -1$ : In this case, we want to solve the linear system $(A - \lambda I) \mathbf{v} =$             |
| johnston-linear-matrix-algebra-F02789 | 302  | 1.000 | $\text{left}(-v_2, v_2) = v_2(-1, 1)$                      | $(-v_2, v_2) = v_2(-1, 1)$                    | $(-v_2, v_2) = v_2(-1, 1)$ .                                                                                 |
| johnston-linear-matrix-algebra-F02790 | 303  | 0.999 | $(2 / 5, 1)$                                               | $(2/5, 1)$                                    | By multiplying $(2/5, 1)$                                                                                    |
| johnston-linear-matrix-algebra-F02791 | 303  | 1.000 | $v_1 = 2 v_2 / 5$                                          | $v_1 = 2 v_2 / 5$                             | ing variable with $v_1 = 2 v_2 / 5$ , so the eigenvectors with corre-                                        |
| johnston-linear-matrix-algebra-F02792 | 303  | 0.999 | $\mathbf{v} = \text{left}(2 v_2 / 5, v_2) = v_2(2 / 5, 1)$ | $\mathbf{v} = (2 v_2 / 5, v_2) = v_2(2/5, 1)$ | $\mathbf{v} = (2 v_2 / 5, v_2) = v_2(2/5, 1)$ .                                                              |
| johnston-linear-matrix-algebra-F02793 | 304  | 1.000 | $(2 - \lambda)$                                            | $(2 - \lambda)$                               | $(2 - \lambda)$ , since this                                                                                 |
| johnston-linear-matrix-algebra-F02794 | 304  | 0.816 | $\mathbf{v}_1 = (-1, 1)$                                   | $\mathbf{v}_1 = (-1, 1)$                      | $\mathbf{v}_1 = (-1, 1)$ and $\mathbf{v}_2 = (2/5, 1)$ , which have corresponding eigenvalues $-1$ and $6$ , |
| johnston-linear-matrix-algebra-F02795 | 304  | 0.816 | $\mathbf{v}_2 = (2 / 5, 1)$                                | $\mathbf{v}_2 = (2/5, 1)$                     | $\mathbf{v}_1 = (-1, 1)$ and $\mathbf{v}_2 = (2/5, 1)$ , which have corresponding eigenvalues $-1$ and $6$ , |
| johnston-linear-matrix-algebra-F02796 | 304  | 1.000 | $\lambda = -2, \lambda = 2$                                | $\lambda = -2, \lambda = 2$                   | gives us $\lambda = -2, \lambda = 2$ , and $\lambda = 4$ . To find eigenvectors corresponding                |
| johnston-linear-matrix-algebra-F02797 | 304  | 1.000 | $\lambda = 4$                                              | $\lambda = 4$                                 | gives us $\lambda = -2, \lambda = 2$ , and $\lambda = 4$ . To find eigenvectors corresponding                |
| johnston-linear-matrix-algebra-F02798 | 304  | 1.000 | $\lambda = -2$                                             | $\lambda = -2$                                | $\lambda = -2$ : In this case, the linear system is $(A + 2I) \mathbf{v} = \mathbf{0}$ , which we can        |
| johnston-linear-matrix-algebra-F02799 | 304  | 1.000 | $(A + 2I) \mathbf{v} = \mathbf{0}$                         | $(A + 2I) \mathbf{v} = \mathbf{0}$            | $\lambda = -2$ : In this case, the linear system is $(A + 2I) \mathbf{v} = \mathbf{0}$ , which we can        |
| johnston-linear-matrix-algebra-F02800 | 304  | 1.000 | $v_3 = 0$                                                  | $v_3 = 0$                                     | ing variables with $v_3 = 0$ and $v_1 = -v_2$ . The eigenvectors cor-                                        |
| johnston-linear-matrix-algebra-F02801 | 304  | 0.954 | $(-1, 1, 0)$                                               | $(-1, 1, 0)$                                  | multiples of $(-1, 1, 0)$ .                                                                                  |
| johnston-linear-matrix-algebra-F02802 | 304  | 1.000 | $\lambda = 2$                                              | $\lambda = 2$                                 | $\lambda = 2$ : We work through this eigenvector calculation a bit more quickly.                             |
| johnston-linear-matrix-algebra-F02803 | 305  | 0.706 | $\mathbf{3} \times \mathbf{3}$                             | $\mathbf{3} \times \mathbf{3}$                | <b>Ugly <math>3 \times 3</math> Matrix</b>                                                                   |
| johnston-linear-matrix-algebra-F02804 | 305  | 1.000 | $v_1 = 0$                                                  | $v_1 = 0$                                     | ables with $v_1 = 0$ and $v_2 = -v_3$ . The eigenvectors correspond-                                         |
| johnston-linear-matrix-algebra-F02805 | 305  | 1.000 | $v_2 = -v_3$                                               | $v_2 = -v_3$                                  | ables with $v_1 = 0$ and $v_2 = -v_3$ . The eigenvectors correspond-                                         |
| johnston-linear-matrix-algebra-F02806 | 305  | 1.000 | $(A - 4I) \mathbf{v} = \mathbf{0}$                         | $(A - 4I) \mathbf{v} = \mathbf{0}$            | $\lambda = 4$ : In this case, the linear system is $(A - 4I) \mathbf{v} = \mathbf{0}$ . Since the reduced    |
| johnston-linear-matrix-algebra-F02807 | 305  | 1.000 | $A - 4I$                                                   | $A - 4I$                                      | row echelon form of $A - 4I$ is                                                                              |
| johnston-linear-matrix-algebra-F02808 | 305  | 1.000 | $v_1 = v_2$                                                | $v_1 = v_2$                                   | ables. Furthermore, $v_1 = v_2$ and $v_3 = 0$ , so the eigenvectors                                          |

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| johnston-linear-matrix-algebra<br>F02809 | 305  | 1.000 | $A=\left[\begin{array}{ccc}1 & 2 & 3 \\ 1 & -2 & 1 \\ 2 & 1 & 2\end{array}\right]$ | $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$  | Compute the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & 1 \\ 3 & 2 & 1 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra<br>F02810 | 305  | 1.000 | $-\lambda^3+16\lambda+24=0$                                                        | $-\lambda^3 + 16\lambda + 24 = 0$                                         | degree polynomial equations) like $-\lambda^3 + 16\lambda + 24 = 0$ are covered in                               |
| johnston-linear-matrix-algebra<br>F02811 | 306  | 0.996 | $-(-2)^3+16(-2)+$                                                                  | $-(-2)^3 + 16(-2) +$                                                      | only one of them that is actually a solution (indeed, $-(-2)^3 + 16(-2) +$                                       |
| johnston-linear-matrix-algebra<br>F02812 | 306  | 1.000 | $24=8-32+24=0$                                                                     | $24 = 8 - 32 + 24 = 0$                                                    | $24 = 8 - 32 + 24 = 0$ ).                                                                                        |
| johnston-linear-matrix-algebra<br>F02813 | 306  | 1.000 | $-\lambda^3+16\lambda+$                                                            | $-\lambda^3 + 16\lambda +$                                                | use polynomial long division to obtain the result of dividing $-\lambda^3 + 16\lambda +$                         |
| johnston-linear-matrix-algebra<br>F02814 | 306  | 1.000 | $\lambda+2$                                                                        | $\lambda + 2$                                                             | 24 by $\lambda + 2$ (which we know is a factor since $\lambda = -2$ is a root of the cubic                       |
| johnston-linear-matrix-algebra<br>F02815 | 306  | 0.990 | $-\lambda^3+16\lambda+24=(\lambda+2)(-\lambda^2+2\lambda+12)$                      | $-\lambda^3 + 16\lambda + 24 = (\lambda + 2)(-\lambda^2 + 2\lambda + 12)$ | It follows that $-\lambda^3 + 16\lambda + 24 = (\lambda + 2)(-\lambda^2 + 2\lambda + 12)$ , so the roots of      |
| johnston-linear-matrix-algebra<br>F02816 | 306  | 1.000 | $-\lambda^3+16\lambda+24$                                                          | $-\lambda^3 + 16\lambda + 24$                                             | $-\lambda^3 + 16\lambda + 24$ (i.e., the eigenvalues of $A$ ) are $\lambda = -2$ as well as the two              |
| johnston-linear-matrix-algebra<br>F02817 | 306  | 1.000 | $-\lambda^2+2\lambda+12$                                                           | $-\lambda^2 + 2\lambda + 12$                                              | roots of $-\lambda^2 + 2\lambda + 12$ , which we can compute via the quadratic formula:                          |
| johnston-linear-matrix-algebra<br>F02818 | 306  | 0.998 | $\lambda=-2, \lambda=1+\sqrt{13}$                                                  | $\lambda = -2, \lambda = 1 + \sqrt{13}$                                   | The eigenvalues of $A$ are thus $\lambda = -2, \lambda = 1 + \sqrt{13}$ , and $\lambda = 1 -$                    |
| johnston-linear-matrix-algebra<br>F02819 | 306  | 0.998 | $\lambda=1-$                                                                       | $\lambda = 1 -$                                                           | The eigenvalues of $A$ are thus $\lambda = -2, \lambda = 1 + \sqrt{13}$ , and $\lambda = 1 -$                    |
| johnston-linear-matrix-algebra<br>F02820 | 306  | 1.000 | $\sqrt{13}$                                                                        | $\sqrt{13}$                                                               | $\sqrt{13}$ . If we were feeling adventurous, we could also find the eigenvectors                                |
| johnston-linear-matrix-algebra<br>F02821 | 306  | 1.000 | $A-$                                                                               | $A -$                                                                     | corresponding to these eigenvalues by solving the linear system $(A -$                                           |
| johnston-linear-matrix-algebra<br>F02822 | 306  | 1.000 | $\lambda I)\mathbf{v}=\mathbf{0}$                                                  | $\lambda I)\mathbf{v} = \mathbf{0}$                                       | $\lambda I)\mathbf{v} = \mathbf{0}$ for each of these three choices of $\lambda$ . However, it is perhaps a      |
| johnston-linear-matrix-algebra<br>F02823 | 306  | 0.921 | $5\times 5$                                                                        | $5 \times 5$                                                              | be <i>impossible</i> to do so if the matrix is $5 \times 5$ or larger (see Remark A.2.1). For                    |
| johnston-linear-matrix-algebra<br>F02824 | 307  | 1.000 | $\operatorname{det}(\lambda I-A)$                                                  | $\det(\lambda I - A)$                                                     | $\det(\lambda I - A)$ . These two                                                                                |
| johnston-linear-matrix-algebra<br>F02825 | 307  | 1.000 | $p_A: \mathbb{R} \rightarrow \mathbb{R}$                                           | $p_A : \mathbb{R} \rightarrow \mathbb{R}$                                 | Suppose $A$ is a square matrix. The function $p_A : \mathbb{R} \rightarrow \mathbb{R}$ defined by                |
| johnston-linear-matrix-algebra<br>F02826 | 307  | 1.000 | $(a_{1,1}-\lambda)(a_{2,2}-\lambda)\cdots(a_{n,n}-\lambda)$                        | $(a_{1,1} - \lambda)(a_{2,2} - \lambda) \cdots (a_{n,n} - \lambda)$       | product of the diagonal entries, $(a_{1,1} - \lambda)(a_{2,2} - \lambda) \cdots (a_{n,n} - \lambda)$ , which has |
| johnston-linear-matrix-algebra<br>F02827 | 307  | 1.000 | $p_A$                                                                              | $p_A$                                                                     | A bit more explicitly, if the characteristic polynomial of $A$ is $p_A$ and $\lambda_0$ is a                     |
| johnston-linear-matrix-algebra<br>F02828 | 307  | 1.000 | $\lambda_{\{0\}}$                                                                  | $\lambda_0$                                                               | A bit more explicitly, if the characteristic polynomial of $A$ is $p_A$ and $\lambda_0$ is a                     |
| johnston-linear-matrix-algebra<br>F02829 | 307  | 1.000 | $(\lambda_{\{0\}}-\lambda)$                                                        | $(\lambda_0 - \lambda)$                                                   | of the term $(\lambda_0 - \lambda)$ in the factored form of $p_A(\lambda)$ . We now work through an              |
| johnston-linear-matrix-algebra<br>F02830 | 307  | 1.000 | $p_A(\lambda)$                                                                     | $p_A(\lambda)$                                                            | of the term $(\lambda_0 - \lambda)$ in the factored form of $p_A(\lambda)$ . We now work through an              |
| johnston-linear-matrix-algebra<br>F02831 | 308  | 1.000 | $\sqrt{-1}=i$                                                                      | $\sqrt{-1} = i$                                                           | $\sqrt{-1} = i$ . See                                                                                            |
| johnston-linear-matrix-algebra<br>F02832 | 309  | 1.000 | $a+bi$                                                                             | $a + bi$                                                                  | conjugate of $a + bi$                                                                                            |
| johnston-linear-matrix-algebra<br>F02833 | 309  | 1.000 | $\overline{a+bi}=a-bi$                                                             | $\overline{a + bi} = a - bi$                                              | $\overline{a + bi} = a - bi$ .                                                                                   |
| johnston-linear-matrix-algebra<br>F02834 | 309  | 1.000 | $\mathcal{M}_{\{m,n\}}(X)$                                                         | $\mathcal{M}_{m,n}(X)$                                                    | $\mathcal{M}_{m,n}(X)$ is the set of                                                                             |

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| johnston-linear-matrix-algebra-F02835 | 309  | 1.000 | $\mathcal{M}_n(X)$                                                                                   | $\mathcal{M}_n(X)$                                                                                      | use $\mathcal{M}_n(X)$ to                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02836 | 309  | 0.999 | $\overline{-1+2i}=-1-2i$                                                                             | $\overline{-1+2i} = -1 - 2i$                                                                            | complex conjugates of each other: $\overline{-1+2i} = -1 - 2i$ . This always happens for                                                                                             |
| johnston-linear-matrix-algebra-F02837 | 309  | 1.000 | $p_A(\lambda)=0$                                                                                     | $p_A(\lambda) = 0$                                                                                      | coefficients, so if $p_A(\lambda) = 0$ then $p_A(\bar{\lambda}) = \overline{p_A(\lambda)} = 0$ as well.                                                                              |
| johnston-linear-matrix-algebra-F02838 | 309  | 1.000 | $p_A(\bar{\lambda})=\overline{p_A(\lambda)}=0$                                                       | $p_A(\bar{\lambda}) = \overline{p_A(\lambda)} = 0$                                                      | coefficients, so if $p_A(\lambda) = 0$ then $p_A(\bar{\lambda}) = \overline{p_A(\lambda)} = 0$ as well.                                                                              |
| johnston-linear-matrix-algebra-F02839 | 309  | 1.000 | $\mathcal{M}_{m,n}(\mathbb{R})$                                                                      | $\mathcal{M}_{m,n}(\mathbb{R})$                                                                         | matrix can have, we use the notation $\mathcal{M}_{m,n}(\mathbb{R})$ to denote the set of $m \times n$                                                                               |
| johnston-linear-matrix-algebra-F02840 | 309  | 1.000 | $\mathcal{M}_{m,n}(\mathbb{C})$                                                                      | $\mathcal{M}_{m,n}(\mathbb{C})$                                                                         | matrices with real entries and $\mathcal{M}_{m,n}(\mathbb{C})$ to denote the set of $m \times n$ matrices                                                                            |
| johnston-linear-matrix-algebra-F02841 | 309  | 1.000 | $p_A(\lambda)=p_B(\lambda)$                                                                          | $p_A(\lambda) = p_B(\lambda)$                                                                           | polynomials are the same: $p_A(\lambda) = p_B(\lambda)$ .                                                                                                                            |
| johnston-linear-matrix-algebra-F02842 | 310  | 1.000 | $\lambda^n$                                                                                          | $\lambda^n$                                                                                             | The coefficient of $\lambda^n$                                                                                                                                                       |
| johnston-linear-matrix-algebra-F02843 | 310  | 0.990 | $(-1)^n$                                                                                             | $(-1)^n$                                                                                                | $(-1)^n$ .                                                                                                                                                                           |
| johnston-linear-matrix-algebra-F02844 | 310  | 1.000 | $A=\left[\begin{array}{ccc}4 & 0 & 2 \\ -2 & 6 & 2 \\ 0 & 4\end{array}\right]$                       | $A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$                                | Show that $A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$ are not similar. |
| johnston-linear-matrix-algebra-F02845 | 310  | 1.000 | $B=\left[\begin{array}{ccc}8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8\end{array}\right]$               | $B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$                            | Show that $A = \begin{bmatrix} 4 & 0 & 2 \\ -2 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 8 & -5 & -5 \\ 5 & -2 & -5 \\ -5 & 5 & 8 \end{bmatrix}$ are not similar. |
| johnston-linear-matrix-algebra-F02846 | 310  | 0.975 | $\operatorname{rank}(A)=\operatorname{rank}(B)=3,$<br>$\operatorname{tr}(A)=\operatorname{tr}(B)=14$ | $\operatorname{rank}(A) = \operatorname{rank}(B) = 3, \operatorname{tr}(A) = \operatorname{tr}(B) = 14$ | $\operatorname{rank}(A) = \operatorname{rank}(B) = 3,$ $\operatorname{tr}(A) = \operatorname{tr}(B) = 14,$ and $\det(A) = \det(B) = 72$ .                                            |
| johnston-linear-matrix-algebra-F02847 | 310  | 0.975 | $\operatorname{det}(A)=\operatorname{det}(B)=72$                                                     | $\det(A) = \det(B) = 72$                                                                                | $\operatorname{rank}(A) = \operatorname{rank}(B) = 3,$ $\operatorname{tr}(A) = \operatorname{tr}(B) = 14,$ and $\det(A) = \det(B) = 72$ .                                            |
| johnston-linear-matrix-algebra-F02848 | 310  | 1.000 | $p_A(\lambda) \neq p_B(\lambda)$                                                                     | $p_A(\lambda) \neq p_B(\lambda)$                                                                        | Since $p_A(\lambda) \neq p_B(\lambda)$ , we conclude that $A$ and $B$ are not similar.                                                                                               |
| johnston-linear-matrix-algebra-F02849 | 310  | 1.000 | $\lambda^2$                                                                                          | $\lambda^2$                                                                                             | Notice in the above example that the coefficient of $\lambda^2$ in the characteristic                                                                                                |
| johnston-linear-matrix-algebra-F02850 | 310  | 1.000 | $A \in \mathcal{M}_n(\mathbb{C})$                                                                    | $A \in \mathcal{M}_n(\mathbb{C})$                                                                       | Denote the $n$ eigenvalues of $A \in \mathcal{M}_n(\mathbb{C})$ (listed according to algebraic                                                                                       |
| johnston-linear-matrix-algebra-F02851 | 310  | 1.000 | $\lambda_1, \lambda_2, \dots, \lambda_n$                                                             | $\lambda_1, \lambda_2, \dots, \lambda_n$                                                                | multiplicity) by $\lambda_1, \lambda_2, \dots, \lambda_n$ , and its characteristic polynomial by                                                                                     |
| johnston-linear-matrix-algebra-F02852 | 310  | 1.000 | $c_0=\operatorname{det}(A)=\lambda_1 \lambda_2 \cdots \lambda_n$                                     | $c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$                                                  | <i>Proof.</i> We start by proving the claim that $c_0 = \det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$ . To see                                                                     |
| johnston-linear-matrix-algebra-F02853 | 311  | 1.000 | $\lambda=0$                                                                                          | $\lambda = 0$                                                                                           | $\lambda = 0$ into the above formulas shows that                                                                                                                                     |
| johnston-linear-matrix-algebra-F02854 | 311  | 1.000 | $(-1)^{n-1} c_{n-1}=\lambda_1+\lambda_2+\cdots+\lambda_n$                                            | $(-1)^{n-1} c_{n-1} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$                                       | To prove the claim that $(-1)^{n-1} c_{n-1} = \lambda_1 + \lambda_2 + \cdots + \lambda_n$ , we simply mul-                                                                           |
| johnston-linear-matrix-algebra-F02855 | 311  | 1.000 | $\lambda^{n-1}$                                                                                      | $\lambda^{n-1}$                                                                                         | and observe that the coefficient of $\lambda^{n-1}$ on the right-hand side (i.e., $c_{n-1}$ ) is                                                                                     |
| johnston-linear-matrix-algebra-F02856 | 311  | 1.000 | $c_{n-1}$                                                                                            | $c_{n-1}$                                                                                               | and observe that the coefficient of $\lambda^{n-1}$ on the right-hand side (i.e., $c_{n-1}$ ) is                                                                                     |
| johnston-linear-matrix-algebra-F02857 | 311  | 1.000 | $(-1)^{n-1}(\lambda_1+\lambda_2+\cdots+\lambda_n)$                                                   | $(-1)^{n-1}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$                                                | $(-1)^{n-1}(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$ .                                                                                                                           |
| johnston-linear-matrix-algebra-F02858 | 311  | 1.000 | $(-1)^{n-1} c_{n-1}=\operatorname{tr}(A)$                                                            | $(-1)^{n-1} c_{n-1} = \operatorname{tr}(A)$                                                             | On the other hand, we can show that $(-1)^{n-1} c_{n-1} = \operatorname{tr}(A)$ by considering                                                                                       |

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| johnston-linear-matrix-algebra-F02859 | 311  | ■ 0.997 | $n-1$                                                                                      | $n - 1$                                                                       | term of the form $\lambda^{n-1}$ is the first one, since it is the only one containing $n - 1$                                                                                             |
| johnston-linear-matrix-algebra-F02860 | 312  | ■ 1.000 | $2 \times 3 = 6$                                                                           | $2 \times 3 = 6$                                                              | $2 \times 3 = 6$ .                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F02861 | 313  | ■ 1.000 | $A \in \mathcal{M}_{m,n}(\mathbb{C})$                                                      | $A \in \mathcal{M}_{m,n}(\mathbb{C})$                                         | $A \in \mathcal{M}_{m,n}(\mathbb{C})$ means                                                                                                                                                |
| johnston-linear-matrix-algebra-F02862 | 313  | ■ 0.742 | $\lambda=1,1$                                                                              | $\lambda = 1, 1$                                                              | It follows that $A$ has eigenvalues $\lambda = 1, 1$ , and $2$ , so its trace is $\text{tr}(A) =$                                                                                          |
| johnston-linear-matrix-algebra-F02863 | 313  | ■ 0.742 | $\text{operatorname{tr}}(A)=$                                                              | $\text{tr}(A) =$                                                              | It follows that $A$ has eigenvalues $\lambda = 1, 1$ , and $2$ , so its trace is $\text{tr}(A) =$                                                                                          |
| johnston-linear-matrix-algebra-F02864 | 313  | ■ 1.000 | $1+1+2=4$                                                                                  | $1 + 1 + 2 = 4$                                                               | $1 + 1 + 2 = 4$ and its determinant is $\det(A) = 1 \times 1 \times 2 = 2$ , which agree                                                                                                   |
| johnston-linear-matrix-algebra-F02865 | 313  | ■ 1.000 | $\text{operatorname{det}}(A)=1 \times 1 \times 2=2$                                        | $\det(A) = 1 \times 1 \times 2 = 2$                                           | $1 + 1 + 2 = 4$ and its determinant is $\det(A) = 1 \times 1 \times 2 = 2$ , which agree                                                                                                   |
| johnston-linear-matrix-algebra-F02866 | 313  | ■ 1.000 | $A^{\{*\}}$                                                                                | $A^*$                                                                         | construct its <b>conjugate transpose</b> $A^*$ (instead of its standard transpose $A^T$ ),                                                                                                 |
| johnston-linear-matrix-algebra-F02867 | 313  | ■ 1.000 | $\left[\begin{array}{cc}2i & 1-3i \\ -3 & 5+2i\end{array}\right]$                          | $\left[\begin{array}{cc}2i & 1-3i \\ -3 & 5+2i\end{array}\right]$             | a) $\left[\begin{array}{cc}2i & 1-3i \\ -3 & 5+2i\end{array}\right]$                                                                                                                       |
| johnston-linear-matrix-algebra-F02868 | 313  | ■ 1.000 | $\left[\begin{array}{cc}0 & 2+3i \\ 2-3i & 4\end{array}\right]$                            | $\left[\begin{array}{cc}0 & 2+3i \\ 2-3i & 4\end{array}\right]$               | b) $\left[\begin{array}{cc}0 & 2+3i \\ 2-3i & 4\end{array}\right]$                                                                                                                         |
| johnston-linear-matrix-algebra-F02869 | 313  | ■ 1.000 | $\left[\begin{array}{cc}i & 1-i \\ -2 & -2-i \\ 1+i & 2+3i\end{array}\right]$              | $\left[\begin{array}{cc}i & 1-i \\ -2 & -2-i \\ 1+i & 2+3i\end{array}\right]$ | d) $\left[\begin{array}{cc}i & 1-i \\ -2 & -2-i \\ 1+i & 2+3i\end{array}\right]$                                                                                                           |
| johnston-linear-matrix-algebra-F02870 | 314  | ■ 1.000 | $\mathbf{v} \cdot (A \mathbf{w}) = \left(A^{\{*\}} \mathbf{v}\right) \cdot \mathbf{w}$     | $\mathbf{v} \cdot (A\mathbf{w}) = (A^*\mathbf{v}) \cdot \mathbf{w}$           | $\mathbf{v} \cdot (A\mathbf{w}) = (A^*\mathbf{v}) \cdot \mathbf{w}$                                                                                                                        |
| johnston-linear-matrix-algebra-F02871 | 314  | ■ 1.000 | $\mathbf{v}^{\{*\}} A \mathbf{v}$                                                          | $\mathbf{v}^* A \mathbf{v}$                                                   | Since $\mathbf{v}^* A \mathbf{v}$ is a scalar,                                                                                                                                             |
| johnston-linear-matrix-algebra-F02872 | 314  | ■ 1.000 | $\overline{z}z =  z ^2$                                                                    | $\bar{z}z =  z ^2$                                                            | $\bar{z}z =  z ^2$ for all $z \in \mathbb{C}$ .                                                                                                                                            |
| johnston-linear-matrix-algebra-F02873 | 314  | ■ 1.000 | $z \in \mathbb{C}$                                                                         | $z \in \mathbb{C}$                                                            | $\bar{z}z =  z ^2$ for all $z \in \mathbb{C}$ .                                                                                                                                            |
| johnston-linear-matrix-algebra-F02874 | 314  | ■ 1.000 | $A^{\{*\}}=A$                                                                              | $A^* = A$                                                                     | In the special case when a matrix $A \in \mathcal{M}_n(\mathbb{C})$ is such that $A^* = A$ , like in                                                                                       |
| johnston-linear-matrix-algebra-F02875 | 314  | ■ 1.000 | $\mathbf{v} \in \mathbb{C}^n$                                                              | $\mathbf{v} \in \mathbb{C}^n$                                                 | <i>Proof.</i> Suppose that $\mathbf{v} \in \mathbb{C}^n$ is an eigenvector of $A$ with corresponding eigen-                                                                                |
| johnston-linear-matrix-algebra-F02876 | 314  | ■ 1.000 | $\lambda \in \mathbb{C}$                                                                   | $\lambda \in \mathbb{C}$                                                      | value $\lambda \in \mathbb{C}$ . Then computing $\mathbf{v}^* A \mathbf{v}$ shows that                                                                                                     |
| johnston-linear-matrix-algebra-F02877 | 315  | ■ 1.000 | $\lambda \mathbf{v}^{\{*\}} \mathbf{v} = \overline{\lambda} \mathbf{v}^{\{*\}} \mathbf{v}$ | $\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v}$     | $\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda = \bar{\lambda}$ , which shows that $\lambda \in \mathbb{R}$ . |
| johnston-linear-matrix-algebra-F02878 | 315  | ■ 1.000 | $\mathbf{v}^{\{*\}} \mathbf{v}$                                                            | $\mathbf{v}^* \mathbf{v}$                                                     | $\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda = \bar{\lambda}$ , which shows that $\lambda \in \mathbb{R}$ . |
| johnston-linear-matrix-algebra-F02879 | 315  | ■ 1.000 | $\lambda = \overline{\lambda}$                                                             | $\lambda = \bar{\lambda}$                                                     | $\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda = \bar{\lambda}$ , which shows that $\lambda \in \mathbb{R}$ . |
| johnston-linear-matrix-algebra-F02880 | 315  | ■ 1.000 | $\lambda \in \mathbb{R}$                                                                   | $\lambda \in \mathbb{R}$                                                      | $\lambda \mathbf{v}^* \mathbf{v} = \bar{\lambda} \mathbf{v}^* \mathbf{v}$ by $\mathbf{v}^* \mathbf{v}$ to see that $\lambda = \bar{\lambda}$ , which shows that $\lambda \in \mathbb{R}$ . |
| johnston-linear-matrix-algebra-F02881 | 315  | ■ 1.000 | $\left[\begin{array}{cc}2 & 1+i \\ 1-i & 3\end{array}\right]$                              | $\left[\begin{array}{cc}2 & 1+i \\ 1-i & 3\end{array}\right]$                 | a) $\left[\begin{array}{cc}2 & 1+i \\ 1-i & 3\end{array}\right]$                                                                                                                           |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02882 | 315  | 1.000 | $\left[\begin{array}{ccc}0 & 1 & -1 \\ 1 & -3 & 2 \\ -1 & 2 & 1\end{array}\right]$    | $\begin{bmatrix} 0 & 1 & -1 \\ 1 & -3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$    | b) $\begin{bmatrix} 0 & 1 & -1 \\ 1 & -3 & 2 \\ -1 & 2 & 1 \end{bmatrix}$                                                                                                        |
| johnston-linear-matrix-algebra-F02883 | 315  | 1.000 | $-\lambda^3-2\lambda^2+9\lambda-2$                                                    | $-\lambda^3-2\lambda^2+9\lambda-2$                                        | polynomial long division to divide $-\lambda^3-2\lambda^2+9\lambda-2$ by $\lambda-2$ :                                                                                           |
| johnston-linear-matrix-algebra-F02884 | 315  | 1.000 | $\lambda-2$                                                                           | $\lambda-2$                                                               | polynomial long division to divide $-\lambda^3-2\lambda^2+9\lambda-2$ by $\lambda-2$ :                                                                                           |
| johnston-linear-matrix-algebra-F02885 | 315  | 0.972 | $-\lambda^3-2\lambda^2+9\lambda-2=(\lambda-2)\left(-\lambda^2-4\lambda+1\right)$      | $-\lambda^3-2\lambda^2+9\lambda-2=(\lambda-2)(-\lambda^2-4\lambda+1)$     | It follows that $-\lambda^3-2\lambda^2+9\lambda-2=(\lambda-2)(-\lambda^2-4\lambda+1)$ , so                                                                                       |
| johnston-linear-matrix-algebra-F02886 | 315  | 1.000 | $-\lambda^2-4\lambda+1$                                                               | $-\lambda^2-4\lambda+1$                                                   | are $\lambda=2$ as well as the two roots of $-\lambda^2-4\lambda+1$ , which can be                                                                                               |
| johnston-linear-matrix-algebra-F02887 | 316  | 1.000 | $\operatorname{null}(A-\lambda I)$                                                    | $\operatorname{null}(A-\lambda I)$                                        | $\operatorname{null}(A-\lambda I)$ .                                                                                                                                             |
| johnston-linear-matrix-algebra-F02888 | 316  | 0.943 | $(-1,1,0),(0,1,-1)$                                                                   | $(-1,1,0),(0,1,-1)$                                                       | the non-zero multiples of $(-1,1,0)$ , $(0,1,-1)$ , and $(1,1,0)$ , respectively. It fol-                                                                                        |
| johnston-linear-matrix-algebra-F02889 | 316  | 0.943 | $(1,1,0)$                                                                             | $(1,1,0)$                                                                 | the non-zero multiples of $(-1,1,0)$ , $(0,1,-1)$ , and $(1,1,0)$ , respectively. It fol-                                                                                        |
| johnston-linear-matrix-algebra-F02890 | 316  | 1.000 | $\{(-1,1,0)\},\{(0,1,-1)\}$                                                           | $\{(-1,1,0)\},\{(0,1,-1)\}$                                               | lows that $\{(-1,1,0)\},\{(0,1,-1)\}$ , and $\{(1,1,0)\}$ are bases of the eigenspaces                                                                                           |
| johnston-linear-matrix-algebra-F02891 | 316  | 1.000 | $\{(1,1,0)\}$                                                                         | $\{(1,1,0)\}$                                                             | lows that $\{(-1,1,0)\},\{(0,1,-1)\}$ , and $\{(1,1,0)\}$ are bases of the eigenspaces                                                                                           |
| johnston-linear-matrix-algebra-F02892 | 316  | 1.000 | $A=\left[\begin{array}{ccc}2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1\end{array}\right]$ | $A=\begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ | Compute bases of the eigenspaces of the matrix $A=\begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ .                                                       |
| johnston-linear-matrix-algebra-F02893 | 317  | 0.925 | $(A-\lambda I)$                                                                       | $(A-\lambda I)$                                                           | $\operatorname{nullity}(A-\lambda I)$ .                                                                                                                                          |
| johnston-linear-matrix-algebra-F02894 | 317  | 0.999 | $(A-2I)\mathbf{v}=\mathbf{0}$                                                         | $(A-2I)\mathbf{v}=\mathbf{0}$                                             | by solving the linear system $(A-2I)\mathbf{v}=\mathbf{0}$ as follows:                                                                                                           |
| johnston-linear-matrix-algebra-F02895 | 317  | 1.000 | $v_1=3v_2$                                                                            | $v_1=3v_2$                                                                | variables with $v_3=0$ and $v_1=3v_2$ . The eigenvectors corre-                                                                                                                  |
| johnston-linear-matrix-algebra-F02896 | 317  | 1.000 | $v_2(3,1,0)$                                                                          | $v_2(3,1,0)$                                                              | sponding to the eigenvalue $\lambda=2$ thus have the form $v_2(3,1,0)$ ,                                                                                                         |
| johnston-linear-matrix-algebra-F02897 | 317  | 1.000 | $\{(3,1,0)\}$                                                                         | $\{(3,1,0)\}$                                                             | so $\{(3,1,0)\}$ is a basis of this eigenspace.                                                                                                                                  |
| johnston-linear-matrix-algebra-F02898 | 317  | 1.000 | $v_1=v_3$                                                                             | $v_1=v_3$                                                                 | variable with $v_1=v_3$ . The eigenvectors corresponding to the                                                                                                                  |
| johnston-linear-matrix-algebra-F02899 | 317  | 1.000 | $v_2(0,1,0)+v_3(1,0,1)$                                                               | $v_2(0,1,0)+v_3(1,0,1)$                                                   | eigenvalue $\lambda=-1$ thus have the form $v_2(0,1,0)+v_3(1,0,1)$ ,                                                                                                             |
| johnston-linear-matrix-algebra-F02900 | 317  | 1.000 | $\{(0,1,0),(1,0,1)\}$                                                                 | $\{(0,1,0),(1,0,1)\}$                                                     | so $\{(0,1,0),(1,0,1)\}$ is a basis of this eigenspace.                                                                                                                          |
| johnston-linear-matrix-algebra-F02901 | 318  | 1.000 | $(B-\lambda I)$                                                                       | $(B-\lambda I)$                                                           | multiplicity of $\lambda$ as an eigenvalue of $A$ (and $\operatorname{nullity}(B-\lambda I)$ is its geometric                                                                    |
| johnston-linear-matrix-algebra-F02902 | 318  | 1.000 | $B=\left[\begin{array}{ccc}2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1\end{array}\right]$   | $B=\begin{bmatrix} 2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$   | Show that $A=\begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ and $B=\begin{bmatrix} 2 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$ are not similar. |
| johnston-linear-matrix-algebra-F02903 | 318  | 1.000 | $(B+I)\mathbf{v}=\mathbf{0}$                                                          | $(B+I)\mathbf{v}=\mathbf{0}$                                              | multiplicity as an eigenvalue of $B$ , we solve the linear system $(B+I)\mathbf{v}=\mathbf{0}$                                                                                   |
| johnston-linear-matrix-algebra-F02904 | 318  | 1.000 | $v_1=-v_3$                                                                            | $v_1=-v_3$                                                                | with $v_1=-v_3$ and $v_2=0$ . The eigenvectors corresponding to $\lambda=-1$                                                                                                     |
| johnston-linear-matrix-algebra-F02905 | 318  | 1.000 | $v_2=0$                                                                               | $v_2=0$                                                                   | with $v_1=-v_3$ and $v_2=0$ . The eigenvectors corresponding to $\lambda=-1$                                                                                                     |

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|---------------------------------------|------|-------|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F02986 | 318  | 1.000 | $\mathbf{v}_3(-1,0,1)$                                                             | $v_3(-1,0,1)$                                                                      | as an eigenvalue of $B$ are thus of the form $v_3(-1,0,1)$ , so $\{(-1,0,1)\}$ is                                                 |
| johnston-linear-matrix-algebra-F02987 | 318  | 1.000 | $\{(-1,0,1)\}$                                                                     | $\{(-1,0,1)\}$                                                                     | as an eigenvalue of $B$ are thus of the form $v_3(-1,0,1)$ , so $\{(-1,0,1)\}$ is                                                 |
| johnston-linear-matrix-algebra-F02988 | 319  | 1.000 | $\lambda_1$                                                                        | $\lambda_1$                                                                        | eigenvalue $\lambda_1$                                                                                                            |
| johnston-linear-matrix-algebra-F02989 | 319  | 0.999 | $\left\{\mathbf{v}_1, \ldots, \mathbf{v}_k\right\}$                                | $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$                                           | $k$ , so that there exists a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of the eigenspace $\text{nullity}(A - \lambda_1 I)$ . |
| johnston-linear-matrix-algebra-F02910 | 319  | 0.999 | $\operatorname{nullity}(A - \lambda_1 I)$                                          | $\text{nullity}(A - \lambda_1 I)$                                                  | $k$ , so that there exists a basis $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ of the eigenspace $\text{nullity}(A - \lambda_1 I)$ . |
| johnston-linear-matrix-algebra-F02911 | 319  | 1.000 | $V \in \mathcal{M}_{n,n-k}$                                                        | $V \in \mathcal{M}_{n,n-k}$                                                        | $V \in \mathcal{M}_{n,n-k}$ ; they are just chosen to make $P$ invertible).                                                       |
| johnston-linear-matrix-algebra-F02912 | 319  | 1.000 | $B \in \mathcal{M}_{k,n-k}$                                                        | $B = \mathcal{M}_{k,n-k}$                                                          | where $B = \mathcal{M}_{k,n-k}$ and $C \in \mathcal{M}_{n-k,n-k}$ are ugly matrices whose entries we do                           |
| johnston-linear-matrix-algebra-F02913 | 319  | 1.000 | $C \in \mathcal{M}_{n-k,n-k}$                                                      | $C \in \mathcal{M}_{n-k,n-k}$                                                      | where $B = \mathcal{M}_{k,n-k}$ and $C \in \mathcal{M}_{n-k,n-k}$ are ugly matrices whose entries we do                           |
| johnston-linear-matrix-algebra-F02914 | 319  | 1.000 | $(\lambda_1 - \lambda)^k$                                                          | $(\lambda_1 - \lambda)^k$                                                          | The characteristic polynomial of $A$ thus has $(\lambda_1 - \lambda)^k$ as a factor, so the alge-                                 |
| johnston-linear-matrix-algebra-F02915 | 320  | 0.614 | $\{(0,0,1,2)\}$                                                                    | $\{(0,0,1,2)\}$                                                                    | has a 1-dimensional eigenspace with $\{(0,0,1,2)\}$ as a basis, but this is                                                       |
| johnston-linear-matrix-algebra-F02916 | 320  | 1.000 | $\lambda = -3$                                                                     | $\lambda = -3$                                                                     | For $\lambda = -3$ , all we know is that its geometric multiplicity is between                                                    |
| johnston-linear-matrix-algebra-F02917 | 320  | 1.000 | $(A+3I)\mathbf{v}=\mathbf{0}$                                                      | $(A+3I)\mathbf{v}=\mathbf{0}$                                                      | this end, we solve the linear system $(A+3I)\mathbf{v}=\mathbf{0}$ to find a basis of this                                        |
| johnston-linear-matrix-algebra-F02918 | 320  | 1.000 | $\{(0,1,0,0)\}$                                                                    | $\{(0,1,0,0)\}$                                                                    | ear system and see that $\{(0,1,0,0)\}$ is a basis of the eigenspace, so the                                                      |
| johnston-linear-matrix-algebra-F02919 | 321  | 0.997 | $(a_{2,2}, \ldots, a_{n,n})$                                                       | $a_{2,2}, \ldots, a_{n,n})$                                                        | $a_{2,2}, \ldots, a_{n,n})$ .                                                                                                     |
| johnston-linear-matrix-algebra-F02920 | 321  | 0.996 | $\left[\begin{array}{lll} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]$  | $\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]$  | a) $\left[\begin{array}{ccc} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{array}\right]$                                              |
| johnston-linear-matrix-algebra-F02921 | 321  | 1.000 | $\left[\begin{array}{ccc} 7 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{array}\right]$ | $\left[\begin{array}{ccc} 7 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{array}\right]$ | b) $\left[\begin{array}{ccc} 7 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -3 \end{array}\right]$                                             |
| johnston-linear-matrix-algebra-F02922 | 321  | 1.000 | $\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]$  | $\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]$  | c) $\left[\begin{array}{ccc} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{array}\right]$                                              |
| johnston-linear-matrix-algebra-F02923 | 322  | 0.852 | $\{(1,0,0)\}$                                                                      | $\{(1,0,0)\}$                                                                      | One basis of this null space is thus $\{(1,0,0)\}$ .                                                                              |
| johnston-linear-matrix-algebra-F02924 | 322  | 1.000 | $A - \lambda I =$                                                                  | $A - \lambda I =$                                                                  | $\lambda = 4$ : We want to find the null space of the matrix $A - \lambda I =$                                                    |
| johnston-linear-matrix-algebra-F02925 | 322  | 0.954 | $\{(2,3,0)\}$                                                                      | $\{(2,3,0)\}$                                                                      | One basis of this null space is thus $\{(2,3,0)\}$ .                                                                              |
| johnston-linear-matrix-algebra-F02926 | 322  | 1.000 | $A - 6I$                                                                           | $A - 6I$                                                                           | $A - 6I$ :                                                                                                                        |
| johnston-linear-matrix-algebra-F02927 | 322  | 0.813 | $\{(16,25,10)\}$                                                                   | $\{(16,25,10)\}$                                                                   | One basis of this null space is thus $\{(16,25,10)\}$ .                                                                           |
| johnston-linear-matrix-algebra-F02928 | 322  | 0.938 | $[\mathbf{v}], \left\{\mathbf{e}_1\right\}, \left\{\mathbf{e}_2\right\}$           | $[\mathbf{v}], \{\mathbf{e}_1\}, \{\mathbf{e}_2\}$                                 | that bases of the corresponding eigenspaces are $[\mathbf{v}], \{\mathbf{e}_1\}, \{\mathbf{e}_2\}$ , and                          |
| johnston-linear-matrix-algebra-F02929 | 322  | 1.000 | $\left\{\mathbf{e}_3\right\}$                                                      | $\{\mathbf{e}_3\}$                                                                 | $\{\mathbf{e}_3\}$ , respectively.                                                                                                |

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| johnston-linear-matrix-algebra<br>F02930 | 322  | ■ 0.957 | $A=\left[\begin{array}{ll}1 & 5 \\ 4 & 2\end{array}\right], \mathbf{v}=\left[\begin{array}{l}1 \\ 1\end{array}\right]$                                                                   | $A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$                                                                 | *(a) $A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ .                                                                 |
| johnston-linear-matrix-algebra<br>F02931 | 322  | ■ 1.000 | $A=\left[\begin{array}{cc}1 & -1 \\ 3 & 6\end{array}\right], \mathbf{v}=\left[\begin{array}{c}-2 \\ 5+\sqrt{13}\end{array}\right]$                                                       | $A = \begin{bmatrix} 1 & -1 \\ 3 & 6 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -2 \\ 5 + \sqrt{13} \end{bmatrix}$                                                   | (b) $A = \begin{bmatrix} 1 & -1 \\ 3 & 6 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -2 \\ 5 + \sqrt{13} \end{bmatrix}$ .                                                    |
| johnston-linear-matrix-algebra<br>F02932 | 322  | ■ 0.921 | $A=\left[\begin{array}{ccc}4 & 1 & 1 \\ 4 & 1 & 1\end{array}\right], \mathbf{v}=\left[\begin{array}{c}-4 \\ 7 \\ 9\end{array}\right]$                                                    | $A = \begin{bmatrix} 4 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -4 \\ 7 \\ 9 \end{bmatrix}$                                     | *(c) $A = \begin{bmatrix} 4 & 1 & 1 \\ 3 & 3 & -1 \\ 4 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} -4 \\ 7 \\ 9 \end{bmatrix}$ .                                     |
| johnston-linear-matrix-algebra<br>F02933 | 322  | ■ 1.000 | $A=\left[\begin{array}{ccc}2 & -1 & -1 \\ 3 & 2 & 4\end{array}\right], \mathbf{v}=\left[\begin{array}{c}2 \\ -1+i \\ -1-3i\end{array}\right]$                                            | $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ 3 & 2 & 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1 + i \\ -1 - 3i \end{bmatrix}$                         | (d) $A = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & 0 \\ 3 & 2 & 4 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ -1 + i \\ -1 - 3i \end{bmatrix}$ .                          |
| johnston-linear-matrix-algebra<br>F02934 | 323  | ■ 0.884 | $A=\left[\begin{array}{llll}3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0\end{array}\right], \mathbf{v}=\left[\begin{array}{c}13 \\ 14 \\ 6 \\ -27\end{array}\right]$ | $A = \begin{bmatrix} 3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 13 \\ 14 \\ 6 \\ -27 \end{bmatrix}$ | *(e) $A = \begin{bmatrix} 3 & 1 & 5 & 5 \\ 3 & 2 & 2 & 5 \\ 3 & 3 & 5 & 5 \\ 2 & 5 & 2 & 0 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 13 \\ 14 \\ 6 \\ -27 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra<br>F02935 | 323  | ■ 0.754 | $\lambda=-3, A=\left[\begin{array}{ll}1 & 5 \\ 4 & 2\end{array}\right]$                                                                                                                  | $\lambda = -3, A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$                                                                                                      | *(a) $\lambda = -3, A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$ .                                                                                                      |
| johnston-linear-matrix-algebra<br>F02936 | 323  | ■ 1.000 | $\lambda=\sqrt{2}, A=\left[\begin{array}{ll}0 & 1 \\ 2 & 0\end{array}\right]$                                                                                                            | $\lambda = \sqrt{2}, A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$                                                                                                | (b) $\lambda = \sqrt{2}, A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ .                                                                                                 |
| johnston-linear-matrix-algebra<br>F02937 | 323  | ■ 0.985 | $\lambda=2, A=\left[\begin{array}{ccc}0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0\end{array}\right]$                                                                                         | $\lambda = 2, A = \begin{bmatrix} 0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0 \end{bmatrix}$                                                                              | *(c) $\lambda = 2, A = \begin{bmatrix} 0 & 3 & -1 \\ 2 & -1 & -1 \\ -2 & 3 & 0 \end{bmatrix}$ .                                                                              |
| johnston-linear-matrix-algebra<br>F02938 | 323  | ■ 1.000 | $\lambda=2+i, A=\left[\begin{array}{ccc}-1 & 0 & -1 \\ 2 & 1 & 2 \\ 2 & -2 & 3\end{array}\right]$                                                                                        | $\lambda = 2 + i, A = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix}$                                                                           | (d) $\lambda = 2 + i, A = \begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 1 \\ 2 & -2 & 3 \end{bmatrix}$ .                                                                            |
| johnston-linear-matrix-algebra<br>F02939 | 323  | ■ 0.969 | $\lambda=-1, A=\left[\begin{array}{cccc}3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0\end{array}\right]$                                                             | $\lambda = -1, A = \begin{bmatrix} 3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 \end{bmatrix}$                                                   | *(e) $\lambda = -1, A = \begin{bmatrix} 3 & 2 & 3 & 2 \\ 0 & 0 & 3 & -1 \\ 4 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 \end{bmatrix}$ .                                                   |
| johnston-linear-matrix-algebra<br>F02940 | 323  | ■ 0.976 | $\left[\begin{array}{cc}1 & 2 \\ -1 & -2\end{array}\right]$                                                                                                                              | $\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$                                                                                                                      | *(a) $\begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$                                                                                                                        |
| johnston-linear-matrix-algebra<br>F02941 | 323  | ■ 0.916 | $\left[\begin{array}{ll}0 & 1 \\ 0 & 0\end{array}\right]$                                                                                                                                | $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$                                                                                                                        | *(c) $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$                                                                                                                          |
| johnston-linear-matrix-algebra<br>F02942 | 323  | ■ 1.000 | $\left[\begin{array}{cc}2 & 1 \\ 3 & -1\end{array}\right]$                                                                                                                               | $\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$                                                                                                                       | (d) $\begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$                                                                                                                          |

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| johnston-linear-matrix-algebra<br>F02943 | 323  | 0.599 | $\left[\begin{array}{ccc}3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7\end{array}\right]$                                                                                          | $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$                                                                                        | *(e) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}$                                                                                        |
| johnston-linear-matrix-algebra<br>F02944 | 323  | 0.992 | $\left[\begin{array}{lll}2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2\end{array}\right]$                                                                                           | $\begin{bmatrix} 2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$                                                                                         | (f) $\begin{bmatrix} 2 & 3 & 0 \\ 3 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}$                                                                                          |
| johnston-linear-matrix-algebra<br>F02945 | 323  | 0.981 | $\left[\begin{array}{cccc}2 & 1 & 0 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2\end{array}\right]$                                                           | $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$                                                          | *(g) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & -3 & 2 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$                                                          |
| johnston-linear-matrix-algebra<br>F02946 | 323  | 1.000 | $\left[\begin{array}{cccc}2 & 1 & -1 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2\end{array}\right]$                                                           | $\begin{bmatrix} 2 & 1 & -1 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$                                                          | (h) $\begin{bmatrix} 2 & 1 & -1 & -1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$                                                           |
| johnston-linear-matrix-algebra<br>F02947 | 323  | 0.999 | $\left[\begin{array}{cccc}19 & 10 & 5 & 22 \\ 9 & 17 & 5 & 19 \\ -8 & -10 & 2 & -18 \\ -11 & -10 & -5 & -14\end{array}\right]$                                            | $\begin{bmatrix} 19 & 10 & 5 & 22 \\ 9 & 17 & 5 & 19 \\ -8 & -10 & 2 & -18 \\ -11 & -10 & -5 & -14 \end{bmatrix}$                                           | *(a) $\begin{bmatrix} 19 & 10 & 5 & 22 \\ 9 & 17 & 5 & 19 \\ -8 & -10 & 2 & -18 \\ -11 & -10 & -5 & -14 \end{bmatrix}$                                           |
| johnston-linear-matrix-algebra<br>F02948 | 323  | 1.000 | $\left[\begin{array}{cccc}12 & -3 & -6 & -3 \\ 1 & 16 & 2 & 1 \\ -3 & -3 & 9 & -3 \\ -1 & -1 & -2 & 14\end{array}\right]$                                                 | $\begin{bmatrix} 12 & -3 & -6 & -3 \\ 1 & 16 & 2 & 1 \\ -3 & -3 & 9 & -3 \\ -1 & -1 & -2 & 14 \end{bmatrix}$                                                | (b) $\begin{bmatrix} 12 & -3 & -6 & -3 \\ 1 & 16 & 2 & 1 \\ -3 & -3 & 9 & -3 \\ -1 & -1 & -2 & 14 \end{bmatrix}$                                                 |
| johnston-linear-matrix-algebra<br>F02949 | 323  | 1.000 | $\left[\begin{array}{ccccc}6 & -4 & 4 & 0 & 4 \\ 0 & 10 & -6 & 0 & -6 \\ 16 & 20 & -20 & -8 & -26 \\ -8 & -8 & 11 & 10 & 11 \\ -16 & -20 & 24 & 8 & 30\end{array}\right]$ | $\begin{bmatrix} 6 & -4 & 4 & 0 & 4 \\ 0 & 10 & -6 & 0 & -6 \\ 16 & 20 & -20 & -8 & -26 \\ -8 & -8 & 11 & 10 & 11 \\ -16 & -20 & 24 & 8 & 30 \end{bmatrix}$ | *(c) $\begin{bmatrix} 6 & -4 & 4 & 0 & 4 \\ 0 & 10 & -6 & 0 & -6 \\ 16 & 20 & -20 & -8 & -26 \\ -8 & -8 & 11 & 10 & 11 \\ -16 & -20 & 24 & 8 & 30 \end{bmatrix}$ |
| johnston-linear-matrix-algebra<br>F02950 | 323  | 1.000 | $\operatorname{det}(A-\lambda I)=\operatorname{det}(\lambda I-A)$                                                                                                         | $\det(A-\lambda I)=\det(\lambda I-A)$                                                                                                                       | *(e) If $A \in \mathcal{M}_n$ then $\det(A-\lambda I)=\det(\lambda I-A)$ .                                                                                       |
| johnston-linear-matrix-algebra<br>F02951 | 323  | 0.991 | $A \in \mathcal{M}_n(\mathbb{R})$                                                                                                                                         | $A \in \mathcal{M}_n(\mathbb{R})$                                                                                                                           | *(g) Every diagonal matrix $A \in \mathcal{M}_n(\mathbb{R})$ is symmetric.                                                                                       |
| johnston-linear-matrix-algebra<br>F02952 | 323  | 1.000 | $k \in \mathbb{R}$                                                                                                                                                        | $k \in \mathbb{R}$                                                                                                                                          | *3.3.6 Let $k \in \mathbb{R}$ and consider the following matrix:                                                                                                 |
| johnston-linear-matrix-algebra<br>F02953 | 323  | 0.999 | $A=\left[\begin{array}{ccc}1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9\end{array}\right]$                                                                                       | $A=\begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$                                                                                     | 3.3.7 Let $A=\begin{bmatrix} 1 & 0 & 0 \\ -8 & 4 & -5 \\ 8 & 0 & 9 \end{bmatrix}$ .                                                                              |
| johnston-linear-matrix-algebra<br>F02954 | 323  | 1.000 | $A^2-3A+I$                                                                                                                                                                | $A^2-3A+I$                                                                                                                                                  | (c) Find all eigenvalues of $A^2-3A+I$ .                                                                                                                         |
| johnston-linear-matrix-algebra<br>F02955 | 323  | 0.998 | $A^2-$                                                                                                                                                                    | $A^2-$                                                                                                                                                      | [Hint: You can explicitly calculate the matrix $A^2-$                                                                                                            |
| johnston-linear-matrix-algebra<br>F02956 | 323  | 1.000 | $3A+I$                                                                                                                                                                    | $3A+I$                                                                                                                                                      | $3A+I$ and find its eigenvalues, but there is a much                                                                                                             |
| johnston-linear-matrix-algebra<br>F02957 | 323  | 1.000 | $\overline{\mathbf{v}}$                                                                                                                                                   | $\overline{\mathbf{v}}$                                                                                                                                     | value $\lambda$ then $\overline{\mathbf{v}}$ is an eigenvector of $A$ corresponding to the                                                                       |
| johnston-linear-matrix-algebra<br>F02958 | 323  | 1.000 | $\overline{\lambda}$                                                                                                                                                      | $\lambda$                                                                                                                                                   | eigenvalue $\overline{\lambda}$ .                                                                                                                                |
| johnston-linear-matrix-algebra<br>F02959 | 324  | 0.928 | $A \in \mathcal{M}_2(\mathbb{C})$                                                                                                                                         | $A \in \mathcal{M}_2(\mathbb{C})$                                                                                                                           | **3.3.10 Suppose $A \in \mathcal{M}_2(\mathbb{C})$ . Find a formula for the                                                                                      |

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| johnston-linear-matrix-algebra-F02960 | 324  | 1.000 | $A^4=I$                                                                                 | $A^4 = I$                                                                               | $A^4 = I$ . Find all possible eigenvalues of $A$ .                                                                           |
| johnston-linear-matrix-algebra-F02961 | 324  | 1.000 | $A^*=-A$                                                                                | $A^* = -A$                                                                              | if $A^* = -A$ . Show that the eigenvalues of skew-Hermitian                                                                  |
| johnston-linear-matrix-algebra-F02962 | 324  | 0.991 | $b \in \mathbb{R}$                                                                      | $b \in \mathbb{R}$                                                                      | matrices are imaginary (i.e., of the form $bi$ for some $b \in \mathbb{R}$ ).                                                |
| johnston-linear-matrix-algebra-F02963 | 324  | 0.955 | $\mathbf{v}^* \mathbf{w}$                                                               | $\mathbf{v}^* \mathbf{w}$                                                               | which equals $\mathbf{v}^* \mathbf{w}$ if $\mathbf{v}$ and $\mathbf{w}$ are column vectors.                                  |
| johnston-linear-matrix-algebra-F02964 | 324  | 1.000 | $\mathbf{v} \in \mathbb{C}^m$                                                           | $\mathbf{v} \in \mathbb{C}^m$                                                           | (a) Show that $\mathbf{v} \cdot (A\mathbf{w}) = (A^* \mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and |
| johnston-linear-matrix-algebra-F02965 | 324  | 1.000 | $\mathbf{w} \in \mathbb{C}^n$                                                           | $\mathbf{w} \in \mathbb{C}^n$                                                           | $\mathbf{w} \in \mathbb{C}^n$ .                                                                                              |
| johnston-linear-matrix-algebra-F02966 | 324  | 1.000 | $B \in \mathcal{M}_{n,m}(\mathbb{C})$                                                   | $B \in \mathcal{M}_{n,m}(\mathbb{C})$                                                   | (b) Show that if $B \in \mathcal{M}_{n,m}(\mathbb{C})$ is such that $\mathbf{v} \cdot (A\mathbf{w}) =$                       |
| johnston-linear-matrix-algebra-F02967 | 324  | 1.000 | $\mathbf{v} \cdot (A\mathbf{w}) =$                                                      | $\mathbf{v} \cdot (A\mathbf{w}) =$                                                      | (b) Show that if $B \in \mathcal{M}_{n,m}(\mathbb{C})$ is such that $\mathbf{v} \cdot (A\mathbf{w}) =$                       |
| johnston-linear-matrix-algebra-F02968 | 324  | 1.000 | $(B\mathbf{v}) \cdot \mathbf{w}$                                                        | $(B\mathbf{v}) \cdot \mathbf{w}$                                                        | $(B\mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and $\mathbf{w} \in \mathbb{C}^n$ , then $B = A^*$ .  |
| johnston-linear-matrix-algebra-F02969 | 324  | 1.000 | $B=A^*$                                                                                 | $B = A^*$                                                                               | $(B\mathbf{v}) \cdot \mathbf{w}$ for all $\mathbf{v} \in \mathbb{C}^m$ and $\mathbf{w} \in \mathbb{C}^n$ , then $B = A^*$ .  |
| johnston-linear-matrix-algebra-F02970 | 324  | 1.000 | $P\mathbf{v}$                                                                           | $P\mathbf{v}$                                                                           | eigenvector of $B$ if and only if $P\mathbf{v}$ is an eigenvector of $A$ .                                                   |
| johnston-linear-matrix-algebra-F02971 | 324  | 1.000 | $\lambda \neq 0$                                                                        | $\lambda \neq 0$                                                                        | (a) Show that if $\lambda \neq 0$ is an eigenvalue of $AB$ then it is                                                        |
| johnston-linear-matrix-algebra-F02972 | 324  | 0.999 | $BAB\mathbf{v}$                                                                         | $BAB\mathbf{v}$                                                                         | tor of $AB$ corresponding to $\lambda$ and simplify $BAB\mathbf{v}$ .]                                                       |
| johnston-linear-matrix-algebra-F02973 | 324  | 1.000 | $\mu \neq \bar{\lambda}$                                                                | $\mu \neq \bar{\lambda}$                                                                | to an eigenvalue $\mu \neq \bar{\lambda}$ of $A^*$ . Show that $\mathbf{v} \cdot \mathbf{w} = 0$ .                           |
| johnston-linear-matrix-algebra-F02974 | 324  | 0.997 | $\lambda_1, \lambda_2, \dots$                                                           | $\lambda_1, \lambda_2, \dots$                                                           | <b>**3.3.22</b> Suppose $A \in \mathcal{M}_n(\mathbb{C})$ has eigenvalues $\lambda_1, \lambda_2, \dots$ ,                    |
| johnston-linear-matrix-algebra-F02975 | 324  | 0.992 | $\lambda_n$                                                                             | $\lambda_n$                                                                             | $\lambda_n$ with corresponding unit (i.e., length 1) eigenvectors $\mathbf{v}_1$ ,                                           |
| johnston-linear-matrix-algebra-F02976 | 324  | 1.000 | $\mathbf{v}_2, \dots, \mathbf{v}_n$                                                     | $\mathbf{v}_2, \dots, \mathbf{v}_n$                                                     | $\mathbf{v}_2, \dots, \mathbf{v}_n$ . Let $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$ .                                  |
| johnston-linear-matrix-algebra-F02977 | 324  | 1.000 | $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$                                         | $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$                                         | $\mathbf{v}_2, \dots, \mathbf{v}_n$ . Let $B = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^*$ .                                  |
| johnston-linear-matrix-algebra-F02978 | 324  | 1.000 | $0, \lambda_2, \lambda_3, \dots, \lambda_n$                                             | $0, \lambda_2, \lambda_3, \dots, \lambda_n$                                             | (b) Show that $B$ has eigenvalues $0, \lambda_2, \lambda_3, \dots, \lambda_n$ . [Side                                        |
| johnston-linear-matrix-algebra-F02979 | 324  | 1.000 | $\lambda_1 \neq \lambda_2$                                                              | $\lambda_1 \neq \lambda_2$                                                              | note: If it helps, you can assume that $\lambda_1 \neq \lambda_2$ ,                                                          |
| johnston-linear-matrix-algebra-F02980 | 324  | 0.999 | $\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$                             | $\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$                             | $\lambda_1 \neq \lambda_3, \dots, \lambda_1 \neq \lambda_n$ .] [Hint: Use Exercise 3.3.21.]                                  |
| johnston-linear-matrix-algebra-F02981 | 324  | 1.000 | $(-1)^n \lambda^n$                                                                      | $(-1)^n \lambda^n$                                                                      | nomial (with leading term $(-1)^n \lambda^n$ ) of our choosing.                                                              |
| johnston-linear-matrix-algebra-F02982 | 324  | 1.000 | $p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$                       | $p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$                       | is $p_C(\lambda) = -(\lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0)$ .                                                       |
| johnston-linear-matrix-algebra-F02983 | 324  | 1.000 | $p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$ | $p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$ | is $p_C(\lambda) = (-1)^n (\lambda^n + a_{n-1} \lambda^{n-1} + \dots + a_1 \lambda + a_0)$ .                                 |
| johnston-linear-matrix-algebra-F02984 | 324  | 1.000 | $p_C$                                                                                   | $p_C$                                                                                   | of the polynomial $p_C$ . Although we have used polynomials                                                                  |
| johnston-linear-matrix-algebra-F02985 | 325  | 0.987 | $v_i$                                                                                   | $v_i$                                                                                   | that its largest entry $v_i$ has $v_i = 1$ .]                                                                                |
| johnston-linear-matrix-algebra-F02986 | 325  | 0.987 | $v_i = 1$                                                                               | $v_i = 1$                                                                               | that its largest entry $v_i$ has $v_i = 1$ .]                                                                                |
| johnston-linear-matrix-algebra-F02987 | 325  | 1.000 | $A = P D P^{-1}$                                                                        | $A = P D P^{-1}$                                                                        | $D \in \mathcal{M}_n$ and an invertible matrix $P \in \mathcal{M}_n$ such that $A = P D P^{-1}$ .                            |
| johnston-linear-matrix-algebra-F02988 | 325  | 1.000 | $D^k$                                                                                   | $D^k$                                                                                   | Since $D$ is diagonal, $D^k$ can be computed entry-wise, so we can compute                                                   |

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| johnston-linear-matrix-algebra-F02989 | 325  | 1.000 | $P^{-1}$                                                                                                                                  | $P^{-1}$                                                                                                                                 | right by $P^{-1}$ .                                                                                                                                                            |
| johnston-linear-matrix-algebra-F02990 | 326  | 0.999 | $P \ D$                                                                                                                                   | $PD$                                                                                                                                     | $PD$ in terms of scaled                                                                                                                                                        |
| johnston-linear-matrix-algebra-F02991 | 326  | 1.000 | $P, D \in \mathcal{M}_n$                                                                                                                  | $P, D \in \mathcal{M}_n$                                                                                                                 | Let $A \in \mathcal{M}_n$ and suppose $P, D \in \mathcal{M}_n$ are such that $P$ is invertible and $D$ is                                                                      |
| johnston-linear-matrix-algebra-F02992 | 326  | 1.000 | $A \ P=P \ D$                                                                                                                             | $AP = PD$                                                                                                                                | it is equivalent to $AP = PD$ . If we write $P$ in terms of its columns $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$                                                      |
| johnston-linear-matrix-algebra-F02993 | 326  | 1.000 | $A \ P$                                                                                                                                   | $AP$                                                                                                                                     | Since $AP = PD$ if and only if the columns of $AP$ equal the columns of $PD$                                                                                                   |
| johnston-linear-matrix-algebra-F02994 | 326  | 1.000 | $A \ \mathbf{v}_j=d_{j,j} \ \mathbf{v}_j$                                                                                                 | $A\mathbf{v}_j = d_{j,j}\mathbf{v}_j$                                                                                                    | in the same order, we conclude that $AP = PD$ if and only if $A\mathbf{v}_j = d_{j,j}\mathbf{v}_j$ for all                                                                     |
| johnston-linear-matrix-algebra-F02995 | 326  | 1.000 | $d_{1,1}, d_{2,2}, \ldots, d_{n,n}$                                                                                                       | $d_{1,1}, d_{2,2}, \dots, d_{n,n}$                                                                                                       | of $P$ ) are eigenvectors of $A$ with corresponding eigenvalues $d_{1,1}, d_{2,2}, \dots, d_{n,n}$ ,                                                                           |
| johnston-linear-matrix-algebra-F02996 | 326  | 1.000 | $\lambda_1=-1$                                                                                                                            | $\lambda_1 = -1$                                                                                                                         | ues $\lambda_1 = -1$ and $\lambda_2 = 6$ corresponding to the eigenvectors $\mathbf{v}_1 = (-1, 1)$                                                                            |
| johnston-linear-matrix-algebra-F02997 | 326  | 1.000 | $\lambda_2=6$                                                                                                                             | $\lambda_2 = 6$                                                                                                                          | ues $\lambda_1 = -1$ and $\lambda_2 = 6$ corresponding to the eigenvectors $\mathbf{v}_1 = (-1, 1)$                                                                            |
| johnston-linear-matrix-algebra-F02998 | 326  | 0.982 | $\mathbf{v}_2=(2,5)$                                                                                                                      | $\mathbf{v}_2 = (2, 5)$                                                                                                                  | and $\mathbf{v}_2 = (2, 5)$ , , respectively. We thus stick these eigenvalues along the                                                                                        |
| johnston-linear-matrix-algebra-F02999 | 327  | 1.000 | $\mathbf{v}_2=(0,1,0)$                                                                                                                    | $\mathbf{v}_2 = (0, 1, 0)$                                                                                                               | $\mathbf{v}_2 = (0, 1, 0)$ and                                                                                                                                                 |
| johnston-linear-matrix-algebra-F03000 | 327  | 0.998 | $\mathbf{v}_3=(1,0,1)$                                                                                                                    | $\mathbf{v}_3 = (1, 0, 1)$                                                                                                               | $\mathbf{v}_3 = (1, 0, 1)$ , or                                                                                                                                                |
| johnston-linear-matrix-algebra-F03001 | 327  | 1.000 | $\mathbf{v}_2=(1,0,1)$                                                                                                                    | $\mathbf{v}_2 = (1, 0, 1)$                                                                                                               | $\mathbf{v}_2 = (1, 0, 1)$ and                                                                                                                                                 |
| johnston-linear-matrix-algebra-F03002 | 327  | 1.000 | $\mathbf{v}_3=(0,1,0)$                                                                                                                    | $\mathbf{v}_3 = (0, 1, 0)$                                                                                                               | $\mathbf{v}_3 = (0, 1, 0)$ . Either                                                                                                                                            |
| johnston-linear-matrix-algebra-F03003 | 327  | 1.000 | $P \ D \ P^{-1}$                                                                                                                          | $PDP^{-1}$                                                                                                                               | let's multiply out $PDP^{-1}$ and see that it does indeed equal $A$ :                                                                                                          |
| johnston-linear-matrix-algebra-F03004 | 327  | 1.000 | $B=\left\{\mathbf{v}_1, \mathbf{v}_2\right\}=\{(-1,1),(2,5)\}$                                                                            | $B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-1, 1), (2, 5)\}$                                                                               | transformation in the basis $B = \{\mathbf{v}_1, \mathbf{v}_2\} = \{(-1, 1), (2, 5)\}$ consisting of                                                                           |
| johnston-linear-matrix-algebra-F03005 | 327  | 1.000 | $[T]_B=D$                                                                                                                                 | $[T]_B = D$                                                                                                                              | eigenvectors of $A$ then $[T]_B = D$ is diagonal:                                                                                                                              |
| johnston-linear-matrix-algebra-F03006 | 327  | 1.000 | $\lambda_1=2$                                                                                                                             | $\lambda_1 = 2$                                                                                                                          | according to algebraic multiplicity) $\lambda_1 = 2$ and $\lambda_2 = \lambda_3 = -1$ , and we                                                                                 |
| johnston-linear-matrix-algebra-F03007 | 327  | 1.000 | $\lambda_2=\lambda_3=-1$                                                                                                                  | $\lambda_2 = \lambda_3 = -1$                                                                                                             | according to algebraic multiplicity) $\lambda_1 = 2$ and $\lambda_2 = \lambda_3 = -1$ , and we                                                                                 |
| johnston-linear-matrix-algebra-F03008 | 327  | 1.000 | $\mathbf{v}_1=(3,1,0), \mathbf{v}_2=(0,1,0)$                                                                                              | $\mathbf{v}_1 = (3, 1, 0), \mathbf{v}_2 = (0, 1, 0)$                                                                                     | from the corresponding eigenvectors $\mathbf{v}_1 = (3, 1, 0)$ , $\mathbf{v}_2 = (0, 1, 0)$ , and                                                                              |
| johnston-linear-matrix-algebra-F03009 | 328  | 1.000 | $B_1, B_2, \ldots, B_m$                                                                                                                   | $B_1, B_2, \dots, B_m$                                                                                                                   | Suppose $A$ is a square matrix. If $B_1, B_2, \dots, B_m$ are bases of eigenspaces                                                                                             |
| johnston-linear-matrix-algebra-F03010 | 328  | 1.000 | $\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \ldots, \mathbf{v}_m \in \operatorname{span}(B_m)$ | $\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \dots, \mathbf{v}_m \in \operatorname{span}(B_m)$ | <i>Proof.</i> First, let $\mathbf{v}_1 \in \operatorname{span}(B_1), \mathbf{v}_2 \in \operatorname{span}(B_2), \dots, \mathbf{v}_m \in \operatorname{span}(B_m)$ be arbitrary |
| johnston-linear-matrix-algebra-F03011 | 329  | 1.000 | $\gamma_j$                                                                                                                                | $\gamma_j$                                                                                                                               | Here, $\gamma_j$ is the                                                                                                                                                        |
| johnston-linear-matrix-algebra-F03012 | 329  | 1.000 | $B_j$                                                                                                                                     | $B_j$                                                                                                                                    | basis $B_j$ .                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F03013 | 329  | 1.000 | $\lambda_1, \lambda_2, \ldots, \lambda_m$                                                                                                 | $\lambda_1, \lambda_2, \dots, \lambda_m$                                                                                                 | of $A$ corresponding to eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ , respectively). We first show                                                                    |
| johnston-linear-matrix-algebra-F03014 | 329  | 1.000 | $1 \leq \ell \leq m$                                                                                                                      | $1 \leq \ell \leq m$                                                                                                                     | a) If $1 \leq \ell \leq m$ is such that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\ell\}$ is linearly dependent, then at                                                |
| johnston-linear-matrix-algebra-F03015 | 329  | 1.000 | $\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell\right\}$                                                                      | $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\ell\}$                                                                                 | a) If $1 \leq \ell \leq m$ is such that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_\ell\}$ is linearly dependent, then at                                                |

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| johnston-linear-matrix-algebra-F03016 | 329  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$                           | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$                           | least one of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$ is the zero vector.                                                             |
| johnston-linear-matrix-algebra-F03017 | 329  | 1.000 | $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$              | $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$              | b) If $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$ then $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$ . |
| johnston-linear-matrix-algebra-F03018 | 329  | 1.000 | $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$              | $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$              | b) If $\mathbf{v}_1 + \mathbf{v}_2 + \cdots + \mathbf{v}_m = \mathbf{0}$ then $\mathbf{v}_1 = \mathbf{v}_2 = \cdots = \mathbf{v}_m = \mathbf{0}$ . |
| johnston-linear-matrix-algebra-F03019 | 329  | 1.000 | $k \leq \ell$                                                                   | $k \leq \ell$                                                                   | $k \leq \ell$ be the smallest integer such that $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is linearly dependent                       |
| johnston-linear-matrix-algebra-F03020 | 329  | 0.934 | $c_1, c_2, \ldots, c_{k-1}$                                                     | $c_1, c_2, \ldots, c_{k-1}$                                                     | then there exist scalars $c_1, c_2, \ldots, c_{k-1}$ such that                                                                                     |
| johnston-linear-matrix-algebra-F03021 | 329  | 1.000 | $\lambda_k$                                                                     | $\lambda_k$                                                                     | Also, multiplying both sides of Equation (3.4.1) by $\lambda_k$ gives                                                                              |
| johnston-linear-matrix-algebra-F03022 | 329  | 1.000 | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}$                      | $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}$                      | linearly dependent, $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{k-1}\}$ is linearly independent, which implies                              |
| johnston-linear-matrix-algebra-F03023 | 329  | 1.000 | $c_j(\lambda_j - \lambda_k) = 0$                                                | $c_j(\lambda_j - \lambda_k) = 0$                                                | $c_j(\lambda_j - \lambda_k) = 0$ for all $1 \leq j \leq k-1$ . Since $\lambda_1, \lambda_2, \ldots, \lambda_m$ are distinct, this is               |
| johnston-linear-matrix-algebra-F03024 | 329  | 1.000 | $1 \leq j \leq k-1$                                                             | $1 \leq j \leq k-1$                                                             | $c_j(\lambda_j - \lambda_k) = 0$ for all $1 \leq j \leq k-1$ . Since $\lambda_1, \lambda_2, \ldots, \lambda_m$ are distinct, this is               |
| johnston-linear-matrix-algebra-F03025 | 329  | 1.000 | $c_1 = c_2 = \cdots = c_{k-1} = 0$                                              | $c_1 = c_2 = \cdots = c_{k-1} = 0$                                              | only possible if $c_1 = c_2 = \cdots = c_{k-1} = 0$ , so $\mathbf{v}_k = \mathbf{0}$ , which proves fact (a).                                      |
| johnston-linear-matrix-algebra-F03026 | 329  | 1.000 | $\mathbf{v}_k = \mathbf{0}$                                                     | $\mathbf{v}_k = \mathbf{0}$                                                     | only possible if $c_1 = c_2 = \cdots = c_{k-1} = 0$ , so $\mathbf{v}_k = \mathbf{0}$ , which proves fact (a).                                      |
| johnston-linear-matrix-algebra-F03027 | 329  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$                              | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$                              | then at least one of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ is the zero vector. Suppose that $\mathbf{v}_m = \mathbf{0}$ (but a        |
| johnston-linear-matrix-algebra-F03028 | 329  | 1.000 | $\mathbf{v}_m = \mathbf{0}$                                                     | $\mathbf{v}_m = \mathbf{0}$                                                     | then at least one of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ is the zero vector. Suppose that $\mathbf{v}_m = \mathbf{0}$ (but a        |
| johnston-linear-matrix-algebra-F03029 | 329  | 0.958 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{m-1}$                          | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{m-1}$                          | similar argument works if it's one of $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_{m-1}$ that equals $\mathbf{0}$ ). Then                      |
| johnston-linear-matrix-algebra-F03030 | 329  | 1.000 | $B_1 \cup B_2 \cup \cdots \cup B_m$                                             | $B_1 \cup B_2 \cup \cdots \cup B_m$                                             | To now show that $B_1 \cup B_2 \cup \cdots \cup B_m$ is linearly independent, we give names                                                        |
| johnston-linear-matrix-algebra-F03031 | 329  | 1.000 | $B_j = \{\mathbf{v}_{j,1}, \mathbf{v}_{j,2}, \ldots, \mathbf{v}_{j,\gamma_j}\}$ | $B_j = \{\mathbf{v}_{j,1}, \mathbf{v}_{j,2}, \ldots, \mathbf{v}_{j,\gamma_j}\}$ | to the vectors in each of these bases via $B_j = \{\mathbf{v}_{j,1}, \mathbf{v}_{j,2}, \ldots, \mathbf{v}_{j,\gamma_j}\}$ for each                 |
| johnston-linear-matrix-algebra-F03032 | 329  | 1.000 | $1 \leq j \leq m$                                                               | $1 \leq j \leq m$                                                               | $1 \leq j \leq m$ . Now suppose that some linear combination of these vectors equals                                                               |
| johnston-linear-matrix-algebra-F03033 | 329  | 1.000 | $c_{j,1} = c_{j,2} = \cdots = c_{j,\gamma_j} = 0$                               | $c_{j,1} = c_{j,2} = \cdots = c_{j,\gamma_j} = 0$                               | tells us that $c_{j,1} = c_{j,2} = \cdots = c_{j,\gamma_j} = 0$ for all $1 \leq j \leq m$ . It follows that every                                  |
| johnston-linear-matrix-algebra-F03034 | 330  | 1.000 | $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$                             | $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$                             | Show that the matrix $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$ cannot be diagonalized.                                                   |
| johnston-linear-matrix-algebra-F03035 | 330  | 1.000 | $\mathbf{v} = v_2(1, 1)$                                                        | $\mathbf{v} = v_2(1, 1)$                                                        | $v_2$ , so all eigenvectors corresponding to $\lambda = 2$ are of the form $\mathbf{v} = v_2(1, 1)$ .                                              |
| johnston-linear-matrix-algebra-F03036 | 331  | 1.000 | $D, P \in \mathcal{M}_n(\mathbb{R})$                                            | $D, P \in \mathcal{M}_n(\mathbb{R})$                                            | onalized as $A = PDP^{-1}$ via <i>real</i> matrices $D, P \in \mathcal{M}_n(\mathbb{R})$ then we say that $A$                                      |
| johnston-linear-matrix-algebra-F03037 | 331  | 1.000 | $D, P \in \mathcal{M}_n(\mathbb{C})$                                            | $D, P \in \mathcal{M}_n(\mathbb{C})$                                            | $D, P \in \mathcal{M}_n(\mathbb{C})$ then we say that $A$ is <b>diagonalizable over</b> $\mathbb{C}$ . If we just say                              |
| johnston-linear-matrix-algebra-F03038 | 331  | 1.000 | $\mathbb{C}$                                                                    | $\mathbb{C}$                                                                    | $D, P \in \mathcal{M}_n(\mathbb{C})$ then we say that $A$ is <b>diagonalizable over</b> $\mathbb{C}$ . If we just say                              |
| johnston-linear-matrix-algebra-F03039 | 331  | 1.000 | $B_1, B_2, \ldots, B_m \subseteq B$                                             | $B_1, B_2, \ldots, B_m \subseteq B$                                             | to their corresponding eigenvalues. Specifically, let $B_1, B_2, \ldots, B_m \subseteq B$ be                                                       |
| johnston-linear-matrix-algebra-F03040 | 331  | 0.995 | $B_1, B_2, \ldots$                                                              | $B_1, B_2, \ldots$                                                              | and note that the fact that $\lambda_1, \lambda_2, \ldots, \lambda_m$ are distinct ensures that $B_1, B_2, \ldots$ ,                               |
| johnston-linear-matrix-algebra-F03041 | 331  | 1.000 | $B_m$                                                                           | $B_m$                                                                           | $B_m$ are disjoint (after all, no non-zero vector can be scaled by two different                                                                   |
| johnston-linear-matrix-algebra-F03042 | 331  | 0.999 | $\lambda_j$                                                                     | $\lambda_j$                                                                     | of $\lambda_j$ is at least $ B_j $ . Then the fact that $n =  B  =  B_1  + \cdots +  B_m $ implies that                                            |

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|-------------------------------------------|------|-------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-<br>F03043 | 331  | 0.999 | $\left B_{\{j\}}\right $                                                                               | $ B_j $                                                                                      | of $\lambda_j$ is at least $ B_j $ . Then the fact that $n =  B  =  B_1  + \cdots +  B_m $ implies that                 |
| johnston-linear-matrix-algebra-<br>F03044 | 331  | 0.999 | $n= B =\left B_{\{1\}}\right +\cdots+\left B_{\{m\}}\right $                                           | $n =  B  =  B_1  + \cdots +  B_m $                                                           | of $\lambda_j$ is at least $ B_j $ . Then the fact that $n =  B  =  B_1  + \cdots +  B_m $ implies that                 |
| johnston-linear-matrix-algebra-<br>F03045 | 332  | 1.000 | $\left B_{\{1\}}\right ,\left B_{\{2\}}\right ,\ldots,\left B_{\{m\}}\right $                          | $ B_1 ,  B_2 , \dots,  B_m $                                                                 | the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ , respectively. Since $ B_1 ,  B_2 , \dots,  B_m $ are         |
| johnston-linear-matrix-algebra-<br>F03046 | 332  | 0.998 | $\left B_{\{1\}}\right +\left B_{\{2\}}\right +\cdots+\left B_{\{m\}}\right =n$                        | $ B_1  +  B_2  + \cdots +  B_m  = n$                                                         | $ B_1  +  B_2  + \cdots +  B_m  = n$ . If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$ , then $ B  = n$            |
| johnston-linear-matrix-algebra-<br>F03047 | 332  | 0.998 | $B=B_{\{1\}} \cup B_{\{2\}} \cup \cdots \cup B_{\{m\}}$                                                | $B = B_1 \cup B_2 \cup \cdots \cup B_m$                                                      | $ B_1  +  B_2  + \cdots +  B_m  = n$ . If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$ , then $ B  = n$            |
| johnston-linear-matrix-algebra-<br>F03048 | 332  | 0.998 | $ B =n$                                                                                                | $ B  = n$                                                                                    | $ B_1  +  B_2  + \cdots +  B_m  = n$ . If we define $B = B_1 \cup B_2 \cup \cdots \cup B_m$ , then $ B  = n$            |
| johnston-linear-matrix-algebra-<br>F03049 | 332  | 0.979 | $1+2=3=n$                                                                                              | $1 + 2 = 3 = n$                                                                              | plicities of its eigenvalues were 1 and 2, and $1 + 2 = 3 = n$ .                                                        |
| johnston-linear-matrix-algebra-<br>F03050 | 332  | 0.793 | $\operatorname{not} A=\left[\begin{array}{ccc}5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2\end{array}\right]$ | $\operatorname{not} A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$ | Determine whether or not $A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$ can be diagonalized. |
| johnston-linear-matrix-algebra-<br>F03051 | 333  | 1.000 | $a \lambda^2+b \lambda+c$                                                                              | $a\lambda^2 + b\lambda + c$                                                                  | $a\lambda^2 + b\lambda + c$ are                                                                                         |
| johnston-linear-matrix-algebra-<br>F03052 | 333  | 0.997 | $1+$                                                                                                   | $1+$                                                                                         | Since there are $n$ distinct eigenvalues, their geometric multiplies thus sum to $1 +$                                  |
| johnston-linear-matrix-algebra-<br>F03053 | 333  | 0.997 | $1+\cdots+1=n$                                                                                         | $1 + \cdots + 1 = n$                                                                         | $1 + \cdots + 1 = n$ . It then follows from Theorem 3.4.3 that $A$ is diagonalizable.                                   |
| johnston-linear-matrix-algebra-<br>F03054 | 333  | 0.972 | $\operatorname{not} A=\left[\begin{array}{ll}1 & 1 \\ 1 & 0\end{array}\right]$                         | $\operatorname{not} A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$                        | Determine whether or not $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is diagonalizable.                          |
| johnston-linear-matrix-algebra-<br>F03055 | 334  | 1.000 | $1+\varepsilon$                                                                                        | $1 + \varepsilon$                                                                            | $1 + \varepsilon$ results in its eigenvalues being                                                                      |
| johnston-linear-matrix-algebra-<br>F03056 | 334  | 1.000 | $\varepsilon \notin\{0,4\}$                                                                            | $\varepsilon \notin \{0,4\}$                                                                 | which are distinct as long as $\varepsilon \notin \{0,4\}$ . It follows that nearby matrices                            |
| johnston-linear-matrix-algebra-<br>F03057 | 335  | 1.000 | $Q^{\ast}$                                                                                             | $Q^*$                                                                                        | Recall that $Q^*$ is the                                                                                                |
| johnston-linear-matrix-algebra-<br>F03058 | 335  | 1.000 | $Q$                                                                                                    | $Q$                                                                                          | of $Q$ , which was                                                                                                      |
| johnston-linear-matrix-algebra-<br>F03059 | 335  | 1.000 | $A^{\{500\}}$                                                                                          | $A^{500}$                                                                                    | it is much simpler to compute $A^{500}$ by performing this diagonalization and then                                     |
| johnston-linear-matrix-algebra-<br>F03060 | 335  | 1.000 | $k=500$                                                                                                | $k = 500$                                                                                    | plugging $k = 500$ into that formula, rather than multiplying $A$ by itself 499                                         |
| johnston-linear-matrix-algebra-<br>F03061 | 335  | 1.000 | $P, \, Q, \, D \, \in \, \mathcal{M}_{\{n\}}$                                                          | $P, Q, D \in \mathcal{M}_n$                                                                  | Suppose $P, Q, D \in \mathcal{M}_n$ are such that $D$ is diagonal with diagonal entries                                 |
| johnston-linear-matrix-algebra-<br>F03062 | 335  | 0.999 | $d_{\{1\}}, \, d_{\{2\}}, \, \ldots, \, d_{\{n\}}$                                                     | $d_1, d_2, \dots, d_n$                                                                       | $d_1, d_2, \dots, d_n$ , in that order. Write $P$ and $Q$ in terms of their columns:                                    |
| johnston-linear-matrix-algebra-<br>F03063 | 336  | 1.000 | $\mathbf{p}_{\{1\}} \, \mathbf{q}_{\{1\}}^{\ast}$                                                      | $\mathbf{p}_1 \mathbf{q}_1^*$                                                                | like $\mathbf{p}_1 \mathbf{q}_1^*$ . A column                                                                           |
| johnston-linear-matrix-algebra-<br>F03064 | 336  | 1.000 | $(-1)^{\{k\}}$                                                                                         | $(-1)^k$                                                                                     | by $(-1)^k$ here if we                                                                                                  |
| johnston-linear-matrix-algebra-<br>F03065 | 336  | 1.000 | $\mathbf{p}_{\{j\}}$                                                                                   | $\mathbf{p}_j$                                                                               | where $\mathbf{p}_j$ is the $j$ -th column of $P$ and $\mathbf{q}_j^*$ is the $j$ -th row of $P^{-1}$ . Furthermore, in |
| johnston-linear-matrix-algebra-<br>F03066 | 336  | 1.000 | $\mathbf{q}_{\{j\}}^{\ast}$                                                                            | $\mathbf{q}_j^*$                                                                             | where $\mathbf{p}_j$ is the $j$ -th column of $P$ and $\mathbf{q}_j^*$ is the $j$ -th row of $P^{-1}$ . Furthermore, in |
| johnston-linear-matrix-algebra-<br>F03067 | 336  | 1.000 | $A^{\{k\}}=P \, D^{\{k\}} \, P^{-1}$                                                                   | $A^k = PD^kP^{-1}$                                                                           | this case we have $A^k = PD^kP^{-1}$ , from which the above theorem implies                                             |

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| johnston-linear-matrix-algebra<br>F03068 | 336  | ■ 0.996 | $\mathbf{p}_j \mathbf{q}_j^*$                                                                  | $\mathbf{p}_j \mathbf{q}_j^*$                                                                  | $\mathbf{p}_j$ is a column vector and each $\mathbf{q}_j^*$ is a row vector, the products $\mathbf{p}_j \mathbf{q}_j^*$ are rank-1                   |
| johnston-linear-matrix-algebra<br>F03069 | 337  | ■ 1.000 | $F_3 = F_2 + F_1 = 1 + 1 = 2$                                                                  | $F_3 = F_2 + F_1 = 1 + 1 = 2$                                                                  | $F_3 = F_2 + F_1 = 1 + 1 = 2,$                                                                                                                       |
| johnston-linear-matrix-algebra<br>F03070 | 337  | ■ 0.999 | $F_4 = F_3 + F_2 = 2 + 1 = 3$                                                                  | $F_4 = F_3 + F_2 = 2 + 1 = 3$                                                                  | $F_4 = F_3 + F_2 = 2 + 1 = 3,$                                                                                                                       |
| johnston-linear-matrix-algebra<br>F03071 | 337  | ■ 1.000 | $\phi = 1.61803 \dots$                                                                         | $\phi = 1.61803 \dots$                                                                         | starts $\phi = 1.61803 \dots$                                                                                                                        |
| johnston-linear-matrix-algebra<br>F03072 | 337  | ■ 1.000 | $\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$ | $\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$ | $\mathbf{p}_1 \mathbf{q}_1^*, \mathbf{p}_2 \mathbf{q}_2^*, \dots, \mathbf{p}_n \mathbf{q}_n^*$ (we have not yet discussed linear combinations of ma- |
| johnston-linear-matrix-algebra<br>F03073 | 337  | ■ 0.988 | $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$                                                       | $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$                                                       | The first few terms of this sequence are $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, \dots$ . Find                                                             |
| johnston-linear-matrix-algebra<br>F03074 | 337  | ■ 1.000 | $F_n$                                                                                          | $F_n$                                                                                          | an explicit formula for $F_n$ that does not depend on the previous terms in                                                                          |
| johnston-linear-matrix-algebra<br>F03075 | 337  | ■ 1.000 | $F_{n+1} = F_n + F_{n-1}$                                                                      | $F_{n+1} = F_n + F_{n-1}$                                                                      | terms of matrix multiplication—the fact that $F_{n+1} = F_n + F_{n-1}$ is equivalent                                                                 |
| johnston-linear-matrix-algebra<br>F03076 | 337  | ■ 1.000 | $\lambda^2 - \lambda - 1$                                                                      | $\lambda^2 - \lambda - 1$                                                                      | recall from Example 3.4.5 that its characteristic polynomial is $\lambda^2 - \lambda - 1,$                                                           |
| johnston-linear-matrix-algebra<br>F03077 | 337  | ■ 1.000 | $\phi = (1 + \sqrt{5}) / 2$                                                                    | $\phi = (1 + \sqrt{5}) / 2$                                                                    | For convenience, we define $\phi = (1 + \sqrt{5}) / 2,$ so that the eigenvalues of $A$                                                               |
| johnston-linear-matrix-algebra<br>F03078 | 337  | ■ 0.999 | $1 - \phi$                                                                                     | $1 - \phi$                                                                                     | are $\phi$ and $1 - \phi$ .                                                                                                                          |
| johnston-linear-matrix-algebra<br>F03079 | 337  | ■ 1.000 | $\lambda = \phi$                                                                               | $\lambda = \phi$                                                                               | Next, we find the corresponding eigenvectors. For $\lambda = \phi,$ we find a                                                                        |
| johnston-linear-matrix-algebra<br>F03080 | 338  | ■ 1.000 | $1 - (1 - \phi)(-\phi) =$                                                                      | $1 - (1 - \phi)(-\phi) =$                                                                      | $1 - (1 - \phi)(-\phi) =$                                                                                                                            |
| johnston-linear-matrix-algebra<br>F03081 | 338  | ■ 0.628 | $\phi^2 + \phi + 1 = 0$                                                                        | $\phi^2 + \phi + 1 = 0$                                                                        | $-\phi^2 + \phi + 1 = 0,$ since                                                                                                                      |
| johnston-linear-matrix-algebra<br>F03082 | 338  | ■ 1.000 | $A^n = P D^n P^{-1}$                                                                           | $A^n = P D^n P^{-1}$                                                                           | then $A^n = P D^n P^{-1}.$                                                                                                                           |
| johnston-linear-matrix-algebra<br>F03083 | 338  | ■ 0.935 | $(F_n)$                                                                                        | $(F_n)$                                                                                        | an integer $(F_n)$ for all                                                                                                                           |
| johnston-linear-matrix-algebra<br>F03084 | 338  | ■ 0.830 | $\text{null}(A - \phi I)$                                                                      | $\text{null}(A - \phi I)$                                                                      | basis of $\text{null}(A - \phi I)$ as follows:                                                                                                       |
| johnston-linear-matrix-algebra<br>F03085 | 338  | ■ 1.000 | $\{(\phi, 1)\}$                                                                                | $\{(\phi, 1)\}$                                                                                | One basis of this null space is thus $\{(\phi, 1)\}.$ On the other hand, for $\lambda =$                                                             |
| johnston-linear-matrix-algebra<br>F03086 | 338  | ■ 1.000 | $\lambda =$                                                                                    | $\lambda =$                                                                                    | One basis of this null space is thus $\{(\phi, 1)\}.$ On the other hand, for $\lambda =$                                                             |
| johnston-linear-matrix-algebra<br>F03087 | 338  | ■ 0.950 | $\text{null}(A - (1 - \phi) I)$                                                                | $\text{null}(A - (1 - \phi) I)$                                                                | $1 - \phi,$ we find a basis of $\text{null}(A - (1 - \phi) I):$                                                                                      |
| johnston-linear-matrix-algebra<br>F03088 | 338  | ■ 0.997 | $\{(1 - \phi, 1)\}$                                                                            | $\{(1 - \phi, 1)\}$                                                                            | It follows that one basis of this null space is $\{(1 - \phi, 1)\}.$                                                                                 |
| johnston-linear-matrix-algebra<br>F03089 | 339  | ■ 0.986 | $\lambda_3 = \lambda_4 = 0$                                                                    | $\lambda_3 = \lambda_4 = 0$                                                                    | $\lambda_3 = \lambda_4 = 0$ might                                                                                                                    |
| johnston-linear-matrix-algebra<br>F03090 | 339  | ■ 1.000 | $[P D^k P^{-1}]_{1,4}$                                                                         | $[P D^k P^{-1}]_{1,4}$                                                                         | $[P D^k P^{-1}]_{1,4}$ is the dot                                                                                                                    |
| johnston-linear-matrix-algebra<br>F03091 | 339  | ■ 1.000 | $D^k P^{-1}$                                                                                   | $D^k P^{-1}$                                                                                   | row of $D^k P^{-1}.$                                                                                                                                 |
| johnston-linear-matrix-algebra<br>F03092 | 339  | ■ 0.978 | $A \in \mathcal{M}_5$                                                                          | $A \in \mathcal{M}_5$                                                                          | $A \in \mathcal{M}_5$ of this graph—the matrix whose $(i, j)$ -entry equals 1 if there is                                                            |
| johnston-linear-matrix-algebra<br>F03093 | 339  | ■ 1.000 | $[A^k]_{1,4}$                                                                                  | $[A^k]_{1,4}$                                                                                  | equal to $[A^k]_{1,4}.$                                                                                                                              |
| johnston-linear-matrix-algebra<br>F03094 | 339  | ■ 1.000 | $A^k =$                                                                                        | $A^k =$                                                                                        | so we now have a complete diagonalization of $A.$ It follows that $A^k =$                                                                            |

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| johnston-linear-matrix-algebra-F03095 | 339  | 1.000 | $P^{-1} D^k P$                                             | $PD^k P^{-1}$                                        | $PD^k P^{-1}$ for all integers $k \geq 0$ , so the formula that we desire is simply the                       |
| johnston-linear-matrix-algebra-F03096 | 339  | 1.000 | $\left[A^k\right]_{1,4} = \left[P^{-1} D^k P\right]_{1,4}$ | $[A^k]_{1,4} = [PD^k P^{-1}]_{1,4}$                  | save some effort by recalling that $[A^k]_{1,4} = [PD^k P^{-1}]_{1,4}$ is the dot product                     |
| johnston-linear-matrix-algebra-F03097 | 340  | 1.000 | $r \in \mathbb{C}$                                         | $r \in \mathbb{C}$                                   | if $r \in \mathbb{C}$ too.                                                                                    |
| johnston-linear-matrix-algebra-F03098 | 340  | 1.000 | $P^{-1} D^k P$                                             | $PD^k$                                               | of the 1st row of $PD^k$ with the 4th column of $P^{-1}$ . That is, the number of                             |
| johnston-linear-matrix-algebra-F03099 | 340  | 0.991 | $D^r$                                                      | $D^r$                                                | where $D^r$ is obtained by raising each of its diagonal entries to the $r$ -th                                |
| johnston-linear-matrix-algebra-F03100 | 340  | 1.000 | $A^r$                                                      | $A^r$                                                | we know that $A^r$ does not change if we choose a different diagonalization                                   |
| johnston-linear-matrix-algebra-F03101 | 340  | 1.000 | $(-1)^{1/2} = i$                                           | $(-1)^{1/2} = i$                                     | an exponentiation will be complex in general, since we have $(-1)^{1/2} = i$ ,                                |
| johnston-linear-matrix-algebra-F03102 | 340  | 1.000 | $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$        | $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$  | Use Definition 3.4.2 to compute the following powers of $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$ . |
| johnston-linear-matrix-algebra-F03103 | 340  | 1.000 | $A^{1/2}$                                                  | $A^{1/2}$                                            | a) $A^{1/2}$                                                                                                  |
| johnston-linear-matrix-algebra-F03104 | 340  | 1.000 | $A^\pi$                                                    | $A^\pi$                                              | c) $A^\pi$                                                                                                    |
| johnston-linear-matrix-algebra-F03105 | 340  | 0.710 | $(4, 1)$                                                   | $(4, 1)$                                             | vectors $(4, 1)$ and $(1, 1)$ , respectively. We can thus diagonalize $A$ as                                  |
| johnston-linear-matrix-algebra-F03106 | 341  | 1.000 | $r = 1/2$                                                  | $r = 1/2$                                            | If we plug $r = 1/2$                                                                                          |
| johnston-linear-matrix-algebra-F03107 | 341  | 1.000 | $r = -1$                                                   | $r = -1$                                             | instead find an explicit formula for $A^r$ and then plug $r = -1$ into                                        |
| johnston-linear-matrix-algebra-F03108 | 341  | 1.000 | $r = \pi$                                                  | $r = \pi$                                            | c) Plugging $r = \pi$ into the formula that we found in part (b) gives                                        |
| johnston-linear-matrix-algebra-F03109 | 342  | 1.000 | $\lambda_j^{-1}$                                           | $\lambda_j^{-1}$                                     | only if each $\lambda_j^{-1}$                                                                                 |
| johnston-linear-matrix-algebra-F03110 | 342  | 1.000 | $B^2 = A$                                                  | $B^2 = A$                                            | a matrix $B \in \mathcal{M}_n$ such that $B^2 = A$ . Well, if $A = PDP^{-1}$ is a diagonalization of          |
| johnston-linear-matrix-algebra-F03111 | 342  | 1.000 | $B^k = A$                                                  | $B^k = A$                                            | $A \in \mathcal{M}_n$ if $B^k = A$ , and the same argument as above shows that $B = A^{1/k}$ is one           |
| johnston-linear-matrix-algebra-F03112 | 342  | 1.000 | $B = A^{1/k}$                                              | $B = A^{1/k}$                                        | $A \in \mathcal{M}_n$ if $B^k = A$ , and the same argument as above shows that $B = A^{1/k}$ is one           |
| johnston-linear-matrix-algebra-F03113 | 342  | 1.000 | $A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$       | $A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$ | Find a cube root of the matrix $A = \begin{bmatrix} -2 & 6 \\ 3 & -5 \end{bmatrix}$ .                         |
| johnston-linear-matrix-algebra-F03114 | 342  | 0.719 | $(1, -1)$                                                  | $(1, -1)$                                            | $(2, 1)$ and $(1, -1)$ , respectively. We can thus diagonalize $A$ as follows:                                |
| johnston-linear-matrix-algebra-F03115 | 343  | 1.000 | $A^{1/2}, B$                                               | $A^{1/2}, B$                                         | like $A^{1/2}$ , $B$ really is a                                                                              |
| johnston-linear-matrix-algebra-F03116 | 343  | 1.000 | $B^3 = A$                                                  | $B^3 = A$                                            | $B^3 = A$ :                                                                                                   |
| johnston-linear-matrix-algebra-F03117 | 343  | 1.000 | $-A^{1/2}$                                                 | $-A^{1/2}$                                           | is one, and $-A^{1/2}$ is another one. To find two more square roots of $A$ ,                                 |
| johnston-linear-matrix-algebra-F03118 | 343  | 0.978 | $-B$                                                       | $-B$                                                 | as well as $-B$ .                                                                                             |
| johnston-linear-matrix-algebra-F03119 | 344  | 0.531 | $F_{\mathbf{u}}^2 = I$                                     | $F_{\mathbf{u}}^2 = I$                               | unit vector $\mathbf{u}$ , then $F_{\mathbf{u}}^2 = I$ :                                                      |

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| johnston-linear-matrix-algebra-F03120 | 344  | 0.990 | $\left[F_{\mathbf{u}}\right]=2\mathbf{u}\mathbf{u}^T-I$ | $[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I$  | Since the standard matrix of $F_{\mathbf{u}}$ is $[F_{\mathbf{u}}] = 2\mathbf{u}\mathbf{u}^T - I$ , every matrix of the form            |
| johnston-linear-matrix-algebra-F03121 | 344  | 0.446 | $2\mathbf{u}\mathbf{u}^T-I$                             | $2\mathbf{u}\mathbf{u}^T - I$                     | $2\mathbf{u}\mathbf{u}^T - I$ is a square root of $I$ (i.e., $(2\mathbf{u}\mathbf{u}^T - I)^2 = I$ for all unit vectors $\mathbf{u}$ ). |
| johnston-linear-matrix-algebra-F03122 | 344  | 0.446 | $\left(2\mathbf{u}\mathbf{u}^T-I\right)^2=I$            | $(2\mathbf{u}\mathbf{u}^T - I)^2 = I$             | $2\mathbf{u}\mathbf{u}^T - I$ is a square root of $I$ (i.e., $(2\mathbf{u}\mathbf{u}^T - I)^2 = I$ for all unit vectors $\mathbf{u}$ ). |
| johnston-linear-matrix-algebra-F03123 | 344  | 1.000 | $k>1$                                                   | $k > 1$                                           | Suppose $A \in \mathcal{M}_n(\mathbb{C})$ is diagonalizable and let $k > 1$ be an integer.                                              |
| johnston-linear-matrix-algebra-F03124 | 344  | 1.000 | $k^n$                                                   | $k^n$                                             | a) If $A$ 's eigenvalues are distinct, it has exactly $k^n$ different $k$ -th roots.                                                    |
| johnston-linear-matrix-algebra-F03125 | 344  | 1.000 | $A^{r+s}=A^rA^s$                                        | $A^{r+s} = A^r A^s$                               | matrices with non-negative real eigenvalues. That is, $A^{r+s} = A^r A^s$ and $(A^r)^s =$                                               |
| johnston-linear-matrix-algebra-F03126 | 344  | 1.000 | $\left(A^r\right)^s=$                                   | $(A^r)^s =$                                       | matrices with non-negative real eigenvalues. That is, $A^{r+s} = A^r A^s$ and $(A^r)^s =$                                               |
| johnston-linear-matrix-algebra-F03127 | 344  | 1.000 | $A^{rs}$                                                | $A^{rs}$                                          | $A^{rs}$ for all real numbers $r$ and $s$ , as long as $A$ has non-negative real eigenvalues                                            |
| johnston-linear-matrix-algebra-F03128 | 345  | 1.000 | $p(D)$                                                  | $p(D)$                                            | where $p(D)$ is obtained by applying $p$ to each diagonal entry of $D$ .                                                                |
| johnston-linear-matrix-algebra-F03129 | 345  | 1.000 | $p(x)=c_kx^k+\cdots+c_2x^2+c_1x+c_0$                    | $p(x) = c_k x^k + \cdots + c_2 x^2 + c_1 x + c_0$ | <i>Proof.</i> We just write $p(x) = c_k x^k + \cdots + c_2 x^2 + c_1 x + c_0$ , and then make use of                                    |
| johnston-linear-matrix-algebra-F03130 | 345  | 1.000 | $\left(PDP^{-1}\right)^r=PD^rP^{-1}$                    | $(PDP^{-1})^r = PD^r P^{-1}$                      | the fact that $(PDP^{-1})^r = PD^r P^{-1}$ for all $r$ :                                                                                |
| johnston-linear-matrix-algebra-F03131 | 345  | 1.000 | $p(\lambda)$                                            | $p(\lambda)$                                      | $\mathcal{M}_n$ has an eigenvalue $\lambda$ with corresponding eigenvector $\mathbf{v}$ , then $p(\lambda)$ is an                       |
| johnston-linear-matrix-algebra-F03132 | 345  | 0.979 | $p(A)$                                                  | $p(A)$                                            | eigenvalue of $p(A)$ corresponding to the same eigenvector $\mathbf{v}$ . In fact, this is true                                         |
| johnston-linear-matrix-algebra-F03133 | 345  | 1.000 | $p(x)=x^3-3x^2+2x-4$                                    | $p(x) = x^3 - 3x^2 + 2x - 4$                      | $p(x) = x^3 - 3x^2 + 2x - 4$ and $A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$ .                                                  |
| johnston-linear-matrix-algebra-F03134 | 346  | 0.991 | $f(A)$                                                  | $f(A)$                                            | Don't worry— $f(A)$ is                                                                                                                  |
| johnston-linear-matrix-algebra-F03135 | 346  | 1.000 | $f(D)$                                                  | $f(D)$                                            | where $f(D)$ is obtained by applying $f$ to each of the diagonal entries of $D$ .                                                       |
| johnston-linear-matrix-algebra-F03136 | 346  | 1.000 | $e^A$                                                   | $e^A$                                             | a) $e^A$                                                                                                                                |
| johnston-linear-matrix-algebra-F03137 | 346  | 1.000 | $\sin(A)$                                               | $\sin(A)$                                         | b) $\sin(A)$                                                                                                                            |
| johnston-linear-matrix-algebra-F03138 | 347  | 1.000 | $\sin(x)$                                               | $\sin(x)$                                         | function $\sin(x)$ instead of $e^x$ to the diagonal entries of $D$ , gives                                                              |
| johnston-linear-matrix-algebra-F03139 | 347  | 1.000 | $e^x$                                                   | $e^x$                                             | function $\sin(x)$ instead of $e^x$ to the diagonal entries of $D$ , gives                                                              |
| johnston-linear-matrix-algebra-F03140 | 347  | 0.996 | $f(\lambda)$                                            | $f(\lambda)$                                      | $f(\lambda)$ must be an eigenvalue of $f(A)$ corresponding to the same eigenvector $\mathbf{v}$ .                                       |
| johnston-linear-matrix-algebra-F03141 | 347  | 1.000 | $(a,b)$                                                 | $(a,b)$                                           | variable—on some open interval $(a,b)$ :                                                                                                |
| johnston-linear-matrix-algebra-F03142 | 348  | 1.000 | $f(A)=e^A$                                              | $f(A) = e^A$                                      | and polynomials) is the matrix exponential $f(A) = e^A$ . The following theorem                                                         |
| johnston-linear-matrix-algebra-F03143 | 348  | 1.000 | $e^0=1$                                                 | $e^0 = 1$                                         | analogous to the facts that, for real numbers, $e^0 = 1$ and $1/e^x = e^{-x}$ .                                                         |
| johnston-linear-matrix-algebra-F03144 | 348  | 1.000 | $1/e^x=e^{-x}$                                          | $1/e^x = e^{-x}$                                  | analogous to the facts that, for real numbers, $e^0 = 1$ and $1/e^x = e^{-x}$ .                                                         |
| johnston-linear-matrix-algebra-F03145 | 348  | 1.000 | $e^0=I$                                                 | $e^O = I$                                         | a) $e^O = I$ , and                                                                                                                      |

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| johnston-linear-matrix-algebra-F03146 | 348  | 1.000 | $\left(e^A\right)^{-1}=e^{-A}$                                                                                                            | $(e^A)^{-1} = e^{-A}$                                                                                                                                                                                                                    | b) $e^A$ is invertible and $(e^A)^{-1} = e^{-A}$ .                                                                                           |
| johnston-linear-matrix-algebra-F03147 | 348  | 1.000 | $e^{-A}$                                                                                                                                  | $e^{-A}$                                                                                                                                                                                                                                 | Similarly, for part (b) we just multiply $e^A$ by $e^{-A}$ and work with their diagonal                                                      |
| johnston-linear-matrix-algebra-F03148 | 349  | 1.000 | $e^{A+B}=e^A e^B$                                                                                                                         | $e^{A+B} = e^A e^B$                                                                                                                                                                                                                      | Note, however, that it is typically not the case that $e^{A+B} = e^A e^B$ (even                                                              |
| johnston-linear-matrix-algebra-F03149 | 349  | 1.000 | $e^{x+y}=e^x e^y$                                                                                                                         | $e^{x+y} = e^x e^y$                                                                                                                                                                                                                      | though $e^{x+y} = e^x e^y$ for all real numbers $x$ and $y$ ). The reason that this property                                                 |
| johnston-linear-matrix-algebra-F03150 | 349  | 1.000 | $\operatorname{det}\left(e^A\right)=e^{\operatorname{tr}(A)}$                                                                             | $\det(e^A) = e^{\operatorname{tr}(A)}$                                                                                                                                                                                                   | If $A \in \mathcal{M}_n$ is diagonalizable then $\det(e^A) = e^{\operatorname{tr}(A)}$ .                                                     |
| johnston-linear-matrix-algebra-F03151 | 349  | 1.000 | $e^{\lambda_1}, e^{\lambda_2}, \ldots, e^{\lambda_n}$                                                                                     | $e^{\lambda_1}, e^{\lambda_2}, \ldots, e^{\lambda_n}$                                                                                                                                                                                    | that the eigenvalues of $e^A$ are $e^{\lambda_1}, e^{\lambda_2}, \ldots, e^{\lambda_n}$ :                                                    |
| johnston-linear-matrix-algebra-F03152 | 350  | 0.214 | $\left(\mathbf{c}\right)\left[\begin{array}{l} 1 \\ 1 \\ 0 \end{array}\right]$                                                            | $(\mathbf{c}) \begin{bmatrix} 1 & \\ 1 & 1 \\ 0 & \end{bmatrix}$                                                                                                                                                                         | $(\mathbf{a}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$<br>$(\mathbf{c}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F03153 | 350  | 0.643 | $\left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right] \text{ (d) } \left[\begin{array}{l} -2 \\ -3 \end{array}\right]$              | $\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$<br>(d) $\begin{bmatrix} -2 & 6 \\ -3 & 7 \end{bmatrix}$                                                                                                                                  | (b) $\begin{bmatrix} 2 & 1 \\ -1 & 2 \\ -2 & 6 \\ -3 & 7 \end{bmatrix}$<br>(d) $\begin{bmatrix} -2 & 6 \\ -3 & 7 \end{bmatrix}$              |
| johnston-linear-matrix-algebra-F03154 | 350  | 0.395 | $\left[\begin{array}{lll} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right]$                                                         | $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$                                                                                                                                                                      | (c) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$                                                                      |
| johnston-linear-matrix-algebra-F03155 | 350  | 1.000 | $\left[\begin{array}{ccc} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{array}\right]$                                                      | $\begin{bmatrix} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{bmatrix}$                                                                                                                                                                   | (f) $\begin{bmatrix} 5 & 0 & -3 \\ -3 & 2 & 3 \\ 6 & 0 & -4 \end{bmatrix}$                                                                   |
| johnston-linear-matrix-algebra-F03156 | 350  | 0.705 | $\left[\begin{array}{ccc} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{array}\right]$                                                        | $\begin{bmatrix} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$                                                                                                                                                                     | (g) $\begin{bmatrix} 3 & 0 & 1 \\ 0 & -1 & 2 \\ 0 & 0 & 2 \end{bmatrix}$                                                                     |
| johnston-linear-matrix-algebra-F03157 | 350  | 1.000 | $\left[\begin{array}{ccc} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{array}\right]$                                                      | $\begin{bmatrix} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{bmatrix}$                                                                                                                                                                   | (h) $\begin{bmatrix} -3 & 4 & 5 \\ -3 & 5 & 3 \\ 1 & -2 & 1 \end{bmatrix}$                                                                   |
| johnston-linear-matrix-algebra-F03158 | 350  | 0.262 | $\left(\mathbf{c}\right)\left[\begin{array}{c} 1 \\ 1 \\ 1 \end{array}\right] \left[\begin{array}{cc} 2 & 1 \\ -1 & 2 \end{array}\right]$ | $(\mathbf{a}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$<br>$(\mathbf{c}) \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$ | (a) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}$<br>(c) $\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$                           |
| johnston-linear-matrix-algebra-F03159 | 350  | 1.000 | $\left[\begin{array}{cc} 2+2i & 1-i \\ 2 & 1-i \end{array}\right]$                                                                        | $\begin{bmatrix} 2+2i & 1-i \\ 2 & 1-i \end{bmatrix}$                                                                                                                                                                                    | (b) $\begin{bmatrix} 2+2i & 1-i \\ 2 & 1-i \end{bmatrix}$                                                                                    |
| johnston-linear-matrix-algebra-F03160 | 350  | 1.000 | $\left[\begin{array}{cc} 1 & 1 \\ -9 & -5 \end{array}\right]$                                                                             | $\begin{bmatrix} 1 & 1 \\ -9 & -5 \end{bmatrix}$                                                                                                                                                                                         | (d) $\begin{bmatrix} 1 & 1 \\ -9 & -5 \end{bmatrix}$                                                                                         |

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| johnston-linear-matrix-algebra-F03161 | 350  | 0.741 | $\left[\begin{array}{ccc}2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0\end{array}\right]$       | $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$       | <b>*(e)</b> $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$                 |
| johnston-linear-matrix-algebra-F03162 | 350  | 1.000 | $\left[\begin{array}{ccc}-2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4\end{array}\right]$  | $\begin{bmatrix} -2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4 \end{bmatrix}$  | <b>(f)</b> $\begin{bmatrix} -2 & -3 & -2 \\ 4 & 6 & 4 \\ -4 & -5 & -4 \end{bmatrix}$             |
| johnston-linear-matrix-algebra-F03163 | 350  | 0.932 | $\left[\begin{array}{ccc}1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i\end{array}\right]$ | $\begin{bmatrix} 1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i \end{bmatrix}$ | <b>*(g)</b> $\begin{bmatrix} 1+i & -1 & 0 \\ 1 & -1+i & 0 \\ 2 & -2 & i \end{bmatrix}$           |
| johnston-linear-matrix-algebra-F03164 | 350  | 1.000 | $\left[\begin{array}{ccc}2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1\end{array}\right]$     | $\begin{bmatrix} 2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1 \end{bmatrix}$     | <b>(h)</b> $\begin{bmatrix} 2 & 4 & -2 \\ 1 & 3 & -1 \\ 3 & -1 & 1 \end{bmatrix}$                |
| johnston-linear-matrix-algebra-F03165 | 350  | 1.000 | $\sqrt{A}$                                                                             | $\sqrt{A}$                                                                 | <b>(c)</b> $\sqrt{A}$ , Exercise 3.4.1(d).                                                       |
| johnston-linear-matrix-algebra-F03166 | 351  | 1.000 | $L_{\{n\}}$                                                                            | $L_n$                                                                      | Find an explicit formula for $L_n$ . [Hint: This sequence satis-                                 |
| johnston-linear-matrix-algebra-F03167 | 351  | 1.000 | $P_{\{n\}}$                                                                            | $P_n$                                                                      | Find an explicit formula for $P_n$ .                                                             |
| johnston-linear-matrix-algebra-F03168 | 351  | 1.000 | $F_{\{n+1\}} F_{\{n-1\}} - F_{\{n\}}^2 = (-1)^n$                                       | $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$                                          | <b>(b)</b> Show that $F_{n+1}F_{n-1} - F_n^2 = (-1)^n$ for all $n \geq 1$ .                      |
| johnston-linear-matrix-algebra-F03169 | 351  | 1.000 | $F_{\{m+n\}} = F_{\{m+1\}} F_{\{n\}} + F_{\{m\}} F_{\{n-1\}}$                          | $F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$                                        | <b>(c)</b> Show that $F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$ for all $m, n \geq 1$ .                 |
| johnston-linear-matrix-algebra-F03170 | 351  | 1.000 | $m, n \geq 1$                                                                          | $m, n \geq 1$                                                              | <b>(c)</b> Show that $F_{m+n} = F_{m+1}F_n + F_mF_{n-1}$ for all $m, n \geq 1$ .                 |
| johnston-linear-matrix-algebra-F03171 | 351  | 0.996 | $A^{\{m\}} A^{\{n\}} = A^{\{m+n\}}$                                                    | $A^m A^n = A^{m+n}$                                                        | [Hint: $A^m A^n = A^{m+n}$ for all square matrices $A$ .]                                        |
| johnston-linear-matrix-algebra-F03172 | 351  | 1.000 | $A = \mathbf{xy}^T$                                                                    | $A = \mathbf{xy}^T$                                                        | <b>3.4.13</b> Suppose $A = \mathbf{xy}^T$ for some non-zero column vec-                          |
| johnston-linear-matrix-algebra-F03173 | 351  | 0.954 | $\mathbf{x} \cdot \mathbf{y}$                                                          | $\mathbf{x} \cdot \mathbf{y}$                                              | equal to $\mathbf{x} \cdot \mathbf{y}$ , with corresponding eigenvector $\mathbf{x}$ .           |
| johnston-linear-matrix-algebra-F03174 | 351  | 1.000 | $\mathbf{x} \cdot \mathbf{y} \neq 0$                                                   | $\mathbf{x} \cdot \mathbf{y} \neq 0$                                       | <b>(b)</b> Show that $A$ is diagonalizable if and only if $\mathbf{x} \cdot \mathbf{y} \neq 0$ . |
| johnston-linear-matrix-algebra-F03175 | 351  | 1.000 | $A \neq B$                                                                             | $A \neq B$                                                                 | able then it might be the case that $A \neq B$ .                                                 |
| johnston-linear-matrix-algebra-F03176 | 351  | 1.000 | $P_{\{1\}} D_{\{1\}} P_{\{1\}}^{-1} = P_{\{2\}} D_{\{2\}} P_{\{2\}}^{-1}$              | $P_1 D_1 P_1^{-1} = P_2 D_2 P_2^{-1}$                                      | inition 3.4.2. That is, show that if $P_1 D_1 P_1^{-1} = P_2 D_2 P_2^{-1}$ are                   |
| johnston-linear-matrix-algebra-F03177 | 351  | 1.000 | $P_{\{1\}} D_{\{1\}}^r P_{\{1\}}^{-1} = P_{\{2\}} D_{\{2\}}^r P_{\{2\}}^{-1}$          | $P_1 D_1^r P_1^{-1} = P_2 D_2^r P_2^{-1}$                                  | two diagonalizations of a matrix then $P_1 D_1^r P_1^{-1} = P_2 D_2^r P_2^{-1}$                  |
| johnston-linear-matrix-algebra-F03178 | 351  | 1.000 | $P_{\{1\}} f(D_{\{1\}}) P_{\{1\}}^{-1} =$                                              | $P_1 f(D_1) P_1^{-1} =$                                                    | are two diagonalizations of a matrix then $P_1 f(D_1) P_1^{-1} =$                                |
| johnston-linear-matrix-algebra-F03179 | 351  | 1.000 | $P_{\{2\}} f(D_{\{2\}}) P_{\{2\}}^{-1}$                                                | $P_2 f(D_2) P_2^{-1}$                                                      | $P_2 f(D_2) P_2^{-1}$ for all functions $f$ .                                                    |
| johnston-linear-matrix-algebra-F03180 | 351  | 1.000 | $\lambda^{\{k\}}$                                                                      | $\lambda^k$                                                                | tor of $A^k$ with eigenvalue $\lambda^k$ .                                                       |
| johnston-linear-matrix-algebra-F03181 | 351  | 1.000 | $(A^{\{r\}})^{\{s\}} = A^{\{rs\}}$                                                     | $(A^r)^s = A^{rs}$                                                         | <b>(b)</b> $(A^r)^s = A^{rs}$ .                                                                  |
| johnston-linear-matrix-algebra-F03182 | 351  | 0.999 | $e^{\left(A^{\{T\}}\right)} = e^{\{A\}}^{\{T\}}$                                       | $e^{(A^T)} = (e^A)^T$                                                      | $e^{(A^T)} = (e^A)^T$ .                                                                          |
| johnston-linear-matrix-algebra-F03183 | 351  | 1.000 | $e^{\{A+B\}} =$                                                                        | $e^{A+B} =$                                                                | [Be careful: remember that it is <i>not</i> the case that $e^{A+B} =$                            |
| johnston-linear-matrix-algebra-F03184 | 351  | 1.000 | $e^{\{A\}} e^{\{B\}}$                                                                  | $e^A e^B$                                                                  | $e^A e^B$ in general.]                                                                           |
| johnston-linear-matrix-algebra-F03185 | 351  | 1.000 | $\sin^{\{2\}}(A) + \cos^{\{2\}}(A) = I$                                                | $\sin^2(A) + \cos^2(A) = I$                                                | $\sin^2(A) + \cos^2(A) = I$ .                                                                    |

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|---------------------------------------|------|-------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03186 | 351  | 1.000 | $p(\lambda)=\lambda^n+a_{n-1}\lambda^{n-1}+\cdots+a_1\lambda+a_0$     | $p(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$ | <b>matrix</b> of the polynomial $p(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$ is the matrix                                      |
| johnston-linear-matrix-algebra-F03187 | 351  | 1.000 | $C=V^{-1}DV$                                                          | $C = VDV^{-1}$                                                              | That is, show that $C = VDV^{-1}$ , where $D$ is the diag-                                                                                                     |
| johnston-linear-matrix-algebra-F03188 | 351  | 1.000 | $A=P^{-1}DP$                                                          | $A = PDP^{-1} \in \mathcal{M}_2$                                            | <b>Figure 3.25:</b> If $A = PDP^{-1} \in \mathcal{M}_2$ has eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ with corresponding                                  |
| johnston-linear-matrix-algebra-F03189 | 352  | 1.000 | $\lambda_1, \lambda_2$                                                | $\lambda_2$                                                                 | eigenvalues $\lambda_1$ and $\lambda_2$ , respectively, then its diagonalization skews the basis $\{\mathbf{v}_1, \mathbf{v}_2\}$                              |
| johnston-linear-matrix-algebra-F03190 | 352  | 1.000 | $N$                                                                   | $N$                                                                         | does not hold for non-diagonalizable matrices, since $N$ has no non-zero                                                                                       |
| johnston-linear-matrix-algebra-F03191 | 353  | 1.000 | $A, B \in \mathcal{M}_n(\mathbb{C})$                                  | $A, B \in \mathcal{M}_n(\mathbb{C})$                                        | Suppose $A, B \in \mathcal{M}_n(\mathbb{C})$ are diagonalizable. Then $A$ and $B$ are similar if                                                               |
| johnston-linear-matrix-algebra-F03192 | 354  | 1.000 | $B=Q^{-1}AQ$                                                          | $B = QDQ^{-1}$                                                              | Multiplying the equation $B = QDQ^{-1}$ on the left by $Q^{-1}$ and on the right by $Q$                                                                        |
| johnston-linear-matrix-algebra-F03193 | 354  | 1.000 | $Q^{-1}BQ=D$                                                          | $Q^{-1}$                                                                    | Multiplying the equation $B = QDQ^{-1}$ on the left by $Q^{-1}$ and on the right by $Q$                                                                        |
| johnston-linear-matrix-algebra-F03194 | 354  | 1.000 | $Q^{-1}BQ=D$                                                          | $Q^{-1}BQ = D$                                                              | shows that $Q^{-1}BQ = D$ . Substituting this into the equation $A = PDP^{-1}$ shows                                                                           |
| johnston-linear-matrix-algebra-F03195 | 354  | 0.983 | $(PQ^{-1})^{-1}=QP^{-1}$                                              | $(PQ^{-1})^{-1} = QP^{-1}$                                                  | $(PQ^{-1})^{-1} = QP^{-1}$ .                                                                                                                                   |
| johnston-linear-matrix-algebra-F03196 | 354  | 0.999 | $B=\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$                      | $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$                          | <b>(a)</b> $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$                                           |
| johnston-linear-matrix-algebra-F03197 | 354  | 1.000 | $B=\begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$                      | $B = \begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$                          | <b>(b)</b> $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 3 & 0 \end{bmatrix}$                                           |
| johnston-linear-matrix-algebra-F03198 | 354  | 0.996 | $A=\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$                      | $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$                          | <b>(c)</b> $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$                                          |
| johnston-linear-matrix-algebra-F03199 | 354  | 0.996 | $B=\begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$                     | $B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$                         | <b>(c)</b> $A = \begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 3 & 5 \end{bmatrix}$                                          |
| johnston-linear-matrix-algebra-F03200 | 354  | 1.000 | $A=\begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$                     | $A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$                         | <b>(d)</b> $A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$                                         |
| johnston-linear-matrix-algebra-F03201 | 354  | 1.000 | $B=\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$                     | $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$                         | <b>(d)</b> $A = \begin{bmatrix} 2 & 1 \\ -3 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$                                         |
| johnston-linear-matrix-algebra-F03202 | 354  | 0.980 | $A=\begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$                      | $A = \begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$                          | <b>(e)</b> $A = \begin{bmatrix} 3 & 3 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$                                          |
| johnston-linear-matrix-algebra-F03203 | 354  | 0.999 | $A=\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ | $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$     | <b>(f)</b> $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F03204 | 354  | 0.999 | $B=\begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$ | $B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$     | <b>(f)</b> $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 3 \\ 8 & 9 & 4 \\ 7 & 6 & 5 \end{bmatrix}$ |

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| johnston-linear-matrix-algebra-F03206 | 354  | 0.962 | $A=\left[\begin{array}{ccc}3 & 1 & -1 \\ 0 & 2 & 1 \\ 0 & 2\end{array}\right]$      | $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ | *(g) $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F03207 | 354  | 0.962 | $B=\left[\begin{array}{ccc}3 & 1 & 2 \\ -2 & 3 & 0 \\ -1 & 1\end{array}\right]$     | $B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$ | *(g) $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 2 & 1 \\ -1 & 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 & 2 \\ -2 & 3 & 0 \\ 1 & -1 & 1 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F03208 | 354  | 0.736 | $B \in \mathcal{M}_3$                                                               | $B \in \mathcal{M}_3$                                                     | (f) If $A \in \mathcal{M}_3$ and $B \in \mathcal{M}_3$ each have eigenvalues 1, 2,                                                                           |
| johnston-linear-matrix-algebra-F03209 | 355  | 1.000 | $A = P B$                                                                           | $AP = PB$                                                                 | [Hint: Rewrite $A = PBP^{-1}$ as $AP = PB$ and solve a                                                                                                       |
| johnston-linear-matrix-algebra-F03210 | 355  | 1.000 | $A^2=N$                                                                             | $A^2 = N$                                                                 | not exist a matrix $A \in \mathcal{M}_2(\mathbb{C})$ such that $A^2 = N$ .                                                                                   |
| johnston-linear-matrix-algebra-F03211 | 355  | 1.000 | $c_{\{i, j\}}=(-1)^{i+j} m_{\{i, j\}}$                                              | $c_{i,j} = (-1)^{i+j} m_{i,j}$                                            | matrix $A \in \mathcal{M}_n$ is the quantity $c_{i,j} = (-1)^{i+j} m_{i,j}$ , where $m_{i,j}$ is the determinant                                             |
| johnston-linear-matrix-algebra-F03212 | 355  | 0.993 | $\operatorname{cof}(A)$                                                             | $\operatorname{cof}(A)$                                                   | The <b>cofactor matrix</b> of $A \in \mathcal{M}_n$ , denoted by $\operatorname{cof}(A)$ , is the matrix whose                                               |
| johnston-linear-matrix-algebra-F03213 | 355  | 1.000 | $\left[\begin{array}{cc}2 & 3 \\ -1 & 4\end{array}\right]$                          | $\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$                           | a) $\begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$                                                                                                           |
| johnston-linear-matrix-algebra-F03214 | 356  | 1.000 | $\operatorname{cof}(A)^T$                                                           | $\operatorname{cof}(A)^T$                                                 | The matrix $\operatorname{cof}(A)^T$ is                                                                                                                      |
| johnston-linear-matrix-algebra-F03215 | 356  | 0.998 | $\operatorname{adj}(A)$                                                             | $\operatorname{adj}(A)$                                                   | $\operatorname{adj}(A)$ .                                                                                                                                    |
| johnston-linear-matrix-algebra-F03216 | 356  | 0.980 | $A \operatorname{cof}(A)^T$                                                         | $A \operatorname{cof}(A)^T$                                               | the product $A \operatorname{cof}(A)^T$ straight from the definition of matrix multiplication, we                                                            |
| johnston-linear-matrix-algebra-F03217 | 356  | 0.660 | $\operatorname{product} A \operatorname{cof}(A)^T$                                  | <b>product</b> $A \operatorname{cof}(A)^T$                                | On the other hand, if $i \neq j$ then the $(i, j)$ -entry of the product $A \operatorname{cof}(A)^T$ is                                                      |
| johnston-linear-matrix-algebra-F03218 | 357  | 1.000 | $\left[A \operatorname{cof}(A)^T\right]_{i, j}$                                     | $[A \operatorname{cof}(A)^T]_{i,j}$                                       | which is exactly the quantity $[A \operatorname{cof}(A)^T]_{i,j}$ from Equation (3.A.1). Since $B$                                                           |
| johnston-linear-matrix-algebra-F03219 | 357  | 1.000 | $0=\operatorname{det}(B)=$                                                          | $0 = \det(B) =$                                                           | has two identical rows, it is not invertible, so we conclude that $0 = \det(B) =$                                                                            |
| johnston-linear-matrix-algebra-F03220 | 357  | 1.000 | $A \operatorname{cof}(A)^T=\operatorname{det}(A) I$                                 | $A \operatorname{cof}(A)^T = \det(A) I$                                   | $A \operatorname{cof}(A)^T = \det(A) I$ . By dividing both sides by $\det(A)$ , we see that                                                                  |
| johnston-linear-matrix-algebra-F03221 | 357  | 1.000 | $\operatorname{det}(A)=a d-b c$                                                     | $\det(A) = ad - bc$                                                       | Since $\det(A) = ad - bc$ , it follows that                                                                                                                  |
| johnston-linear-matrix-algebra-F03222 | 358  | 1.000 | $A_{\{j\}}$                                                                         | $A_j$                                                                     | Suppose $A \in \mathcal{M}_n$ is invertible and $\mathbf{b} \in \mathbb{R}^n$ . Define $A_j$ to be the matrix that                                           |
| johnston-linear-matrix-algebra-F03223 | 359  | 1.000 | $A^{-1} \mathbf{b}$                                                                 | $A^{-1} \mathbf{b}$                                                       | the entries of $A^{-1} \mathbf{b}$ , which we do via Theorem 3.A.1:                                                                                          |
| johnston-linear-matrix-algebra-F03224 | 359  | 1.000 | $c_{\{1, j\}}, c_{\{2, j\}}, \ldots, c_{\{n, j\}}$                                  | $c_{1,j}, c_{2,j}, \dots, c_{n,j}$                                        | where $c_{1,j}, c_{2,j}, \dots, c_{n,j}$ are cofactors of $A$ and the final equality above follows                                                           |
| johnston-linear-matrix-algebra-F03225 | 359  | 1.000 | $c_{\{1, j\}} b_1+c_{\{2, j\}} b_2+$                                                | $c_{1,j} b_1 + c_{2,j} b_2 +$                                             | from the definition of matrix multiplication. Well, the quantity $c_{1,j} b_1 + c_{2,j} b_2 +$                                                               |
| johnston-linear-matrix-algebra-F03226 | 359  | 1.000 | $\cdots+c_{\{n, j\}} b_n$                                                           | $\dots + c_{n,j} b_n$                                                     | $\dots + c_{n,j} b_n$ above on the right is exactly the cofactor expansion of $A_j$ along its                                                                |
| johnston-linear-matrix-algebra-F03227 | 359  | 1.000 | $c_{\{1, j\}} b_1+$                                                                 | $c_{1,j} b_1 +$                                                           | It follows from the cofactor expansion theorem (Theorem 3.2.8) that $c_{1,j} b_1 +$                                                                          |
| johnston-linear-matrix-algebra-F03228 | 359  | 1.000 | $c_{\{2, j\}} b_2+\cdots+c_{\{n, j\}} b_n=\operatorname{det}\left(A_{\{j\}}\right)$ | $c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$                           | $c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$ , so $x_j = \det(A_j) / \det(A)$ , as desired.                                                               |
| johnston-linear-matrix-algebra-F03229 | 359  | 1.000 | $x_{\{j\}}=\operatorname{det}\left(A_{\{j\}}\right) / \operatorname{det}(A)$        | $x_j = \det(A_j) / \det(A)$                                               | $c_{2,j} b_2 + \dots + c_{n,j} b_n = \det(A_j)$ , so $x_j = \det(A_j) / \det(A)$ , as desired.                                                               |
| johnston-linear-matrix-algebra-F03230 | 359  | 1.000 | $A, A_{\{1\}}, A_{\{2\}}$                                                           | $A, A_1, A_2$                                                             | We start by writing out the four matrices $A, A_1, A_2$ , and $A_3$ whose                                                                                    |

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| johnston-linear-matrix-algebra-F03231 | 359  | ■ 1.000 | $A_{\{3\}}$                                          | $A_3$                                                | We start by writing out the four matrices $A, A_1, A_2$ , and $A_3$ whose                            |
| johnston-linear-matrix-algebra-F03232 | 361  | ■ 1.000 | $\sigma:\{1,2,\ldots,n\}\rightarrow$                 | $\sigma:\{1,2,\ldots,n\}\rightarrow$                 | $\sigma:\{1,2,\ldots,n\}\rightarrow$                                                                 |
| johnston-linear-matrix-algebra-F03233 | 361  | ■ 0.999 | $\{1,2,\ldots,n\}$                                   | $\{1,2,\ldots,n\}$                                   | $\{1,2,\ldots,n\}$ means                                                                             |
| johnston-linear-matrix-algebra-F03234 | 361  | ■ 1.000 | $\sigma$                                             | $\sigma$                                             | that $\sigma$ is a function                                                                          |
| johnston-linear-matrix-algebra-F03235 | 361  | ■ 1.000 | $1,2,\ldots,n$                                       | $1,2,\ldots,n$                                       | that takes $1,2,\ldots,n$                                                                            |
| johnston-linear-matrix-algebra-F03236 | 361  | ■ 1.000 | $\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$ | $\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$ | A function $\sigma:\{1,2,\ldots,n\}\rightarrow\{1,2,\ldots,n\}$ is called a <b>permutation</b> if    |
| johnston-linear-matrix-algebra-F03237 | 361  | ■ 0.996 | $\sigma(i)\neq\sigma(j)$                             | $\sigma(i)\neq\sigma(j)$                             | $\sigma(i)\neq\sigma(j)$ whenever $i\neq j$ .                                                        |
| johnston-linear-matrix-algebra-F03238 | 361  | ■ 1.000 | $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$               | $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$               | For example, the function $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $\sigma(1)=2$ ,          |
| johnston-linear-matrix-algebra-F03239 | 361  | ■ 1.000 | $\sigma(1)=2$                                        | $\sigma(1)=2$                                        | For example, the function $\sigma:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $\sigma(1)=2$ ,          |
| johnston-linear-matrix-algebra-F03240 | 361  | ■ 0.973 | $\sigma(2)=3$                                        | $\sigma(2)=3$                                        | $\sigma(2)=3$ , and $\sigma(3)=1$ is a permutation (it just shuffles 1, 2, and 3 around),            |
| johnston-linear-matrix-algebra-F03241 | 361  | ■ 0.973 | $\sigma(3)=1$                                        | $\sigma(3)=1$                                        | $\sigma(2)=3$ , and $\sigma(3)=1$ is a permutation (it just shuffles 1, 2, and 3 around),            |
| johnston-linear-matrix-algebra-F03242 | 361  | ■ 1.000 | $f:\{1,2,3\}\rightarrow\{1,2,3\}$                    | $f:\{1,2,3\}\rightarrow\{1,2,3\}$                    | but the function $f:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $f(1)=2$ , $f(2)=2$ , and              |
| johnston-linear-matrix-algebra-F03243 | 361  | ■ 1.000 | $f(1)=2, f(2)=2$                                     | $f(1)=2, f(2)=2$                                     | but the function $f:\{1,2,3\}\rightarrow\{1,2,3\}$ defined by $f(1)=2$ , $f(2)=2$ , and              |
| johnston-linear-matrix-algebra-F03244 | 361  | ■ 1.000 | $f(3)=1$                                             | $f(3)=1$                                             | $f(3)=1$ is not (since $f(1)=f(2)$ ).                                                                |
| johnston-linear-matrix-algebra-F03245 | 361  | ■ 1.000 | $f(1)=f(2)$                                          | $f(1)=f(2)$                                          | $f(3)=1$ is not (since $f(1)=f(2)$ ).                                                                |
| johnston-linear-matrix-algebra-F03246 | 361  | ■ 1.000 | $(\sigma(1)\sigma(2)\cdots\sigma(n))$                | $(\sigma(1)\sigma(2)\cdots\sigma(n))$                | parentheses: $(\sigma(1)\sigma(2)\cdots\sigma(n))$ . For example, the permutation discussed          |
| johnston-linear-matrix-algebra-F03247 | 361  | ■ 1.000 | $\sigma(1)=2, \sigma(2)=3$                           | $\sigma(1)=2, \sigma(2)=3$                           | earlier with $\sigma(1)=2$ , $\sigma(2)=3$ , and $\sigma(3)=1$ is described succinctly by its        |
| johnston-linear-matrix-algebra-F03248 | 361  | ■ 0.699 | $\sigma=(4132)$                                      | $\sigma=(4132)$                                      | via its one-line notation $\sigma=(4\ 1\ 3\ 2)$ then we know that it acts on $\{1,2,3,4\}$           |
| johnston-linear-matrix-algebra-F03249 | 361  | ■ 1.000 | $\sigma(1)=4, \sigma(2)=1, \sigma(3)=3$              | $\sigma(1)=4, \sigma(2)=1, \sigma(3)=3$              | via $\sigma(1)=4$ , $\sigma(2)=1$ , $\sigma(3)=3$ , and $\sigma(4)=2$ .                              |
| johnston-linear-matrix-algebra-F03250 | 361  | ■ 1.000 | $\sigma(4)=2$                                        | $\sigma(4)=2$                                        | via $\sigma(1)=4$ , $\sigma(2)=1$ , $\sigma(3)=3$ , and $\sigma(4)=2$ .                              |
| johnston-linear-matrix-algebra-F03251 | 361  | ■ 1.000 | $n!=n\cdot(n-1)\cdot(n-2)\cdots 2\cdot 1$            | $n!=n\cdot(n-1)\cdot(n-2)\cdots 2\cdot 1$            | There are exactly $n!=n\cdot(n-1)\cdot(n-2)\cdots 2\cdot 1$ permutations $\sigma$ that act           |
| johnston-linear-matrix-algebra-F03252 | 361  | ■ 1.000 | $\sigma(1)$                                          | $\sigma(1)$                                          | on the set $\{1,2,\ldots,n\}$ . To see this, notice that $\sigma(1)$ can be any of $n$ different     |
| johnston-linear-matrix-algebra-F03253 | 361  | ■ 1.000 | $\sigma(2)$                                          | $\sigma(2)$                                          | values, $\sigma(2)$ can be any of the $n-1$ remaining values (since it cannot be the                 |
| johnston-linear-matrix-algebra-F03254 | 361  | ■ 0.970 | $\sigma(3)$                                          | $\sigma(3)$                                          | same as $\sigma(1)$ ), $\sigma(3)$ can be any of the $n-2$ remaining values not equal to $\sigma(1)$ |
| johnston-linear-matrix-algebra-F03255 | 361  | ■ 0.970 | $n-2$                                                | $n-2$                                                | same as $\sigma(1)$ ), $\sigma(3)$ can be any of the $n-2$ remaining values not equal to $\sigma(1)$ |
| johnston-linear-matrix-algebra-F03256 | 361  | ■ 1.000 | $S_{\{n\}}$                                          | $S_n$                                                | is denoted by $S_n$ .                                                                                |
| johnston-linear-matrix-algebra-F03257 | 361  | ■ 1.000 | $S_{\{2\}}$                                          | $S_2$                                                | a) $S_2$                                                                                             |
| johnston-linear-matrix-algebra-F03258 | 361  | ■ 1.000 | $S_{\{3\}}$                                          | $S_3$                                                | b) $S_3$                                                                                             |

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| johnston-linear-matrix-algebra-F03259 | 361  | 0.772 | $2!=2$                                    | $2! = 2$                                  | a) There are only $2! = 2$ permutations that act on the set $\{1, 2\}$ , and                                       |
| johnston-linear-matrix-algebra-F03260 | 361  | 0.772 | $\{1, 2\}$                                | $\{1, 2\}$                                | a) There are only $2! = 2$ permutations that act on the set $\{1, 2\}$ , and                                       |
| johnston-linear-matrix-algebra-F03261 | 361  | 0.984 | $3!=6$                                    | $3! = 6$                                  | b) There are $3! = 6$ permutations that act on the set $\{1, 2, 3\}$ , and they                                    |
| johnston-linear-matrix-algebra-F03262 | 361  | 0.984 | $\{1, 2, 3\}$                             | $\{1, 2, 3\}$                             | b) There are $3! = 6$ permutations that act on the set $\{1, 2, 3\}$ , and they                                    |
| johnston-linear-matrix-algebra-F03263 | 361  | 1.000 | $\sigma, \tau \in S_n$                    | $\sigma, \tau \in S_n$                    | just like we do with linear transformations. Specifically, if $\sigma, \tau \in S_n$ then we                       |
| johnston-linear-matrix-algebra-F03264 | 361  | 1.000 | $\sigma \circ \tau$                       | $\sigma \circ \tau$                       | define their <b>composition</b> $\sigma \circ \tau$ by                                                             |
| johnston-linear-matrix-algebra-F03265 | 361  | 0.600 | $\iota$                                   | $\iota$                                   | Similarly, the <b>identity permutation</b> , which we denote by $\iota$ , is the one that does                     |
| johnston-linear-matrix-algebra-F03266 | 361  | 0.644 | $\iota = (1\ 2\ \cdots\ n)$               | $\iota = (1\ 2\ \cdots\ n)$               | nothing at all: $\iota = (1\ 2\ \cdots\ n)$ , so $\iota(j) = j$ for all $1 \leq j \leq n$ , and the <b>inverse</b> |
| johnston-linear-matrix-algebra-F03267 | 361  | 0.644 | $\iota(j) = j$                            | $\iota(j) = j$                            | nothing at all: $\iota = (1\ 2\ \cdots\ n)$ , so $\iota(j) = j$ for all $1 \leq j \leq n$ , and the <b>inverse</b> |
| johnston-linear-matrix-algebra-F03268 | 361  | 1.000 | $\sigma^{-1}$                             | $\sigma^{-1}$                             | $\sigma^{-1}$ of a permutation $\sigma$ is the one that “undoes” it:                                               |
| johnston-linear-matrix-algebra-F03269 | 362  | 1.000 | $\sigma(\tau(1))$                         | $\sigma(\tau(1))$                         | by writing $\sigma(\tau(1))$ ,                                                                                     |
| johnston-linear-matrix-algebra-F03270 | 362  | 1.000 | $\sigma(\tau(2)), \dots, \sigma(\tau(5))$ | $\sigma(\tau(2)), \dots, \sigma(\tau(5))$ | $\sigma(\tau(2)), \dots, \sigma(\tau(5))$ in                                                                       |
| johnston-linear-matrix-algebra-F03271 | 362  | 1.000 | $\sigma(j) = k$                           | $\sigma(j) = k$                           | If $\sigma(j) = k$ then                                                                                            |
| johnston-linear-matrix-algebra-F03272 | 362  | 1.000 | $\sigma^{-1}(k) = j$                      | $\sigma^{-1}(k) = j$                      | $\sigma^{-1}(k) = j$ .                                                                                             |
| johnston-linear-matrix-algebra-F03273 | 362  | 0.306 | $\sigma = (32514)$                        | $\sigma = (32514)$                        | Suppose $\sigma = (3\ 2\ 5\ 1\ 4)$ and $\tau = (5\ 1\ 4\ 2\ 3)$ . Compute the one-line                             |
| johnston-linear-matrix-algebra-F03274 | 362  | 0.306 | $\tau = (5142)$                           | $\tau = (5142)$                           | Suppose $\sigma = (3\ 2\ 5\ 1\ 4)$ and $\tau = (5\ 1\ 4\ 2\ 3)$ . Compute the one-line                             |
| johnston-linear-matrix-algebra-F03275 | 362  | 1.000 | $\tau \circ \sigma$                       | $\tau \circ \sigma$                       | b) $\tau \circ \sigma$                                                                                             |
| johnston-linear-matrix-algebra-F03276 | 362  | 1.000 | $\tau^{-1}$                               | $\tau^{-1}$                               | d) $\tau^{-1}$                                                                                                     |
| johnston-linear-matrix-algebra-F03277 | 362  | 1.000 | $\sigma(\tau(j))$                         | $\sigma(\tau(j))$                         | a) We just directly compute $\sigma(\tau(j))$ for each $1 \leq j \leq 5$ :                                         |
| johnston-linear-matrix-algebra-F03278 | 362  | 1.000 | $1 \leq j \leq 5$                         | $1 \leq j \leq 5$                         | a) We just directly compute $\sigma(\tau(j))$ for each $1 \leq j \leq 5$ :                                         |
| johnston-linear-matrix-algebra-F03279 | 362  | 1.000 | $\sigma \circ \tau = (43125)$             | $\sigma \circ \tau = (43125)$             | We thus conclude that $\sigma \circ \tau = (4\ 3\ 1\ 2\ 5)$ .                                                      |
| johnston-linear-matrix-algebra-F03280 | 362  | 1.000 | $\tau(\sigma(j))$                         | $\tau(\sigma(j))$                         | b) Again, we just directly compute $\tau(\sigma(j))$ for each $1 \leq j \leq 5$ :                                  |
| johnston-linear-matrix-algebra-F03281 | 362  | 0.999 | $\tau \circ \sigma = (41352)$             | $\tau \circ \sigma = (41352)$             | It follows that $\tau \circ \sigma = (4\ 1\ 3\ 5\ 2)$ .                                                            |
| johnston-linear-matrix-algebra-F03282 | 362  | 0.994 | $\sigma^{-1} = (42153)$                   | $\sigma^{-1} = (42153)$                   | notation, so $\sigma^{-1} = (4\ 2\ 1\ 5\ 3)$ .                                                                     |
| johnston-linear-matrix-algebra-F03283 | 362  | 0.637 | $\tau$                                    | $\tau$                                    | d) The numbers 1, 2, 3, 4, 5 appear in positions 2, 4, 5, 3, 1 in $\tau$ ’s one-                                   |
| johnston-linear-matrix-algebra-F03284 | 362  | 1.000 | $\tau^{-1} = (24531)$                     | $\tau^{-1} = (24531)$                     | line notation, so $\tau^{-1} = (2\ 4\ 5\ 3\ 1)$ .                                                                  |
| johnston-linear-matrix-algebra-F03285 | 362  | 1.000 | $P_{\sigma}$                              | $P_{\sigma}$                              | In other words, the permutation matrix $P_{\sigma}$ is the standard matrix of the                                  |

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|------------------------------------------|------|---------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra<br>F03286 | 362  | ■ 1.000 | $\mathbf{e}_{\{\sigma(j)\}}$                                                           | $\mathbf{e}_{\sigma(j)}$                                                 | linear transformation that sends $\mathbf{e}_j$ to $\mathbf{e}_{\sigma(j)}$ for each $1 \leq j \leq n$ . Geometrically,                                            |
| johnston-linear-matrix-algebra<br>F03287 | 362  | ■ 1.000 | $\sigma=(3421)$                                                                        | $\sigma = (3421)$                                                        | a) $\sigma = (3\ 4\ 2\ 1)$                                                                                                                                         |
| johnston-linear-matrix-algebra<br>F03288 | 362  | ■ 1.000 | $\sigma=(25134)$                                                                       | $\sigma = (25134)$                                                       | b) $\sigma = (2\ 5\ 1\ 3\ 4)$                                                                                                                                      |
| johnston-linear-matrix-algebra<br>F03289 | 362  | ■ 0.956 | $\imath \in S_6$                                                                       | $\imath \in S_6$                                                         | c) $\imath \in S_6$                                                                                                                                                |
| johnston-linear-matrix-algebra<br>F03290 | 363  | ■ 1.000 | $\sigma=\left(\begin{array}{lll}2 & 3 & 1\end{array}\right)$                           | $\sigma = \left( \begin{array}{ccc} 2 & 3 & 1 \end{array} \right)$       | <b>Figure 3.26:</b> If $\sigma = (2\ 3\ 1)$ then the permutation matrix $P_\sigma$ sends $\mathbf{e}_1$ to $\mathbf{e}_2$ , $\mathbf{e}_2$ to $\mathbf{e}_3$ , and |
| johnston-linear-matrix-algebra<br>F03291 | 363  | ■ 1.000 | $\mathbf{e}_{\{2\}}, \mathbf{e}_{\{2\}}$                                               | $\mathbf{e}_2, \mathbf{e}_2$                                             | <b>Figure 3.26:</b> If $\sigma = (2\ 3\ 1)$ then the permutation matrix $P_\sigma$ sends $\mathbf{e}_1$ to $\mathbf{e}_2$ , $\mathbf{e}_2$ to $\mathbf{e}_3$ , and |
| johnston-linear-matrix-algebra<br>F03292 | 363  | ■ 1.000 | $\mathbf{e}_{\{3\}}, \mathbf{e}_{\{4\}}, \mathbf{e}_{\{2\}}$                           | $\mathbf{e}_3, \mathbf{e}_4, \mathbf{e}_2$                               | a) As described by Definition 3.A.3, we just place $\mathbf{e}_3$ , $\mathbf{e}_4$ , $\mathbf{e}_2$ , and $\mathbf{e}_1$ as                                        |
| johnston-linear-matrix-algebra<br>F03293 | 363  | ■ 1.000 | $\mathbf{e}_{\{2\}}, \mathbf{e}_{\{5\}}, \mathbf{e}_{\{1\}}, \mathbf{e}_{\{3\}}$       | $\mathbf{e}_2, \mathbf{e}_5, \mathbf{e}_1, \mathbf{e}_3$                 | b) Similarly, for this permutation we place $\mathbf{e}_2$ , $\mathbf{e}_5$ , $\mathbf{e}_1$ , $\mathbf{e}_3$ , and $\mathbf{e}_4$ as                              |
| johnston-linear-matrix-algebra<br>F03294 | 363  | ■ 1.000 | $\mathbf{e}_{\{4\}}$                                                                   | $\mathbf{e}_4$                                                           | b) Similarly, for this permutation we place $\mathbf{e}_2$ , $\mathbf{e}_5$ , $\mathbf{e}_1$ , $\mathbf{e}_3$ , and $\mathbf{e}_4$ as                              |
| johnston-linear-matrix-algebra<br>F03295 | 363  | ■ 1.000 | $\{1,2,3,4,5,6\}$                                                                      | $\{1,2,3,4,5,6\}$                                                        | c) For the identity permutation on $\{1,2,3,4,5,6\}$ , we place $\mathbf{e}_1, \mathbf{e}_2, \dots$ ,                                                              |
| johnston-linear-matrix-algebra<br>F03296 | 363  | ■ 1.000 | $\mathbf{e}_{\{1\}}, \mathbf{e}_{\{2\}}, \ldots$                                       | $\mathbf{e}_1, \mathbf{e}_2, \dots$                                      | c) For the identity permutation on $\{1,2,3,4,5,6\}$ , we place $\mathbf{e}_1, \mathbf{e}_2, \dots$ ,                                                              |
| johnston-linear-matrix-algebra<br>F03297 | 363  | ■ 0.969 | $\mathbf{e}_{\{6\}}$                                                                   | $\mathbf{e}_6$                                                           | $\mathbf{e}_6$ as columns into a matrix. Doing so gives us exactly the $6 \times 6$                                                                                |
| johnston-linear-matrix-algebra<br>F03298 | 363  | ■ 0.969 | $6 \times 6$                                                                           | $6 \times 6$                                                             | $\mathbf{e}_6$ as columns into a matrix. Doing so gives us exactly the $6 \times 6$                                                                                |
| johnston-linear-matrix-algebra<br>F03299 | 363  | ■ 0.999 | $S_6$                                                                                  | $S_6$                                                                    | associated with the identity permutation in $S_6$ was the identity matrix in $\mathcal{M}_6$ .                                                                     |
| johnston-linear-matrix-algebra<br>F03300 | 363  | ■ 0.999 | $\mathcal{M}_6$                                                                        | $\mathcal{M}_6$                                                          | associated with the identity permutation in $S_6$ was the identity matrix in $\mathcal{M}_6$ .                                                                     |
| johnston-linear-matrix-algebra<br>F03301 | 363  | ■ 1.000 | $P_{\{\sigma\}}, P_{\{\tau\}} \in \mathcal{M}_n$                                       | $P_\sigma, P_\tau \in \mathcal{M}_n$                                     | $P_\sigma, P_\tau \in \mathcal{M}_n$ . Then                                                                                                                        |
| johnston-linear-matrix-algebra<br>F03302 | 363  | ■ 1.000 | $P_{\{\sigma\}} P_{\{\tau\}}=P_{\{\sigma \circ \tau\}}$                                | $P_\sigma P_\tau = P_{\sigma \circ \tau}$                                | a) $P_\sigma P_\tau = P_{\sigma \circ \tau}$                                                                                                                       |
| johnston-linear-matrix-algebra<br>F03303 | 363  | ■ 1.000 | $P_{\{\sigma\}}^{-1}=P_{\{\sigma\}}^T=P_{\{\sigma^{-1}\}}$                             | $P_\sigma^{-1} = P_\sigma^T = P_{\sigma^{-1}}$                           | b) $P_\sigma$ is invertible, and $P_\sigma^{-1} = P_\sigma^T = P_{\sigma^{-1}}$ .                                                                                  |
| johnston-linear-matrix-algebra<br>F03304 | 364  | ■ 1.000 | $P_{\{\sigma \circ \tau\}}$                                                            | $P_{\sigma \circ \tau}$                                                  | Recall that $P_{\sigma \circ \tau}$ is                                                                                                                             |
| johnston-linear-matrix-algebra<br>F03305 | 364  | ■ 0.843 | $\mathbf{e}_{\{(\sigma \circ \tau)(j)\}}$                                              | $\mathbf{e}_{(\sigma \circ \tau)(j)}$                                    | $\mathbf{e}_{(\sigma \circ \tau)(j)}$ .                                                                                                                            |
| johnston-linear-matrix-algebra<br>F03306 | 364  | ■ 1.000 | $P_{\{\sigma\}} P_{\{\tau\}}$                                                          | $P_\sigma P_\tau$                                                        | In particular, this means that, for all $1 \leq j \leq n$ , the $j$ -th column of $P_\sigma P_\tau$ is                                                             |
| johnston-linear-matrix-algebra<br>F03307 | 364  | ■ 0.618 | $P_{\{\sigma\}} P_{\{\sigma^{-1}\}}=P_{\{\sigma \circ \sigma^{-1}\}}=P_{\{\imath\}}=I$ | $P_\sigma P_{\sigma^{-1}} = P_{\sigma \circ \sigma^{-1}} = P_\imath = I$ | For part (b), we first note that part (a) tells us that $P_\sigma P_{\sigma^{-1}} = P_{\sigma \circ \sigma^{-1}} = P_\imath = I$ ,                                 |
| johnston-linear-matrix-algebra<br>F03308 | 364  | ■ 1.000 | $P_{\{\sigma\}}^{-1}=P_{\{\sigma^{-1}\}}$                                              | $P_\sigma^{-1} = P_{\sigma^{-1}}$                                        | so $P_\sigma^{-1} = P_{\sigma^{-1}}$ , as claimed. All that remains is to show that $P_\sigma^T = P_\sigma^{-1}$ . To this                                         |
| johnston-linear-matrix-algebra<br>F03309 | 364  | ■ 1.000 | $P_{\{\sigma\}}^T=P_{\{\sigma\}}^{-1}$                                                 | $P_\sigma^T = P_\sigma^{-1}$                                             | so $P_\sigma^{-1} = P_{\sigma^{-1}}$ , as claimed. All that remains is to show that $P_\sigma^T = P_\sigma^{-1}$ . To this                                         |
| johnston-linear-matrix-algebra<br>F03310 | 364  | ■ 1.000 | $P_{\{\sigma\}}^T P_{\{\sigma\}}$                                                      | $P_\sigma^T P_\sigma$                                                    | end, we compute the entries of $P_\sigma^T P_\sigma$ :                                                                                                             |
| johnston-linear-matrix-algebra<br>F03311 | 364  | ■ 1.000 | $\sigma(i)=\sigma(j)$                                                                  | $\sigma(i) = \sigma(j)$                                                  | If $i = j$ then $\sigma(i) = \sigma(j)$ , so the above dot product equals 1 (i.e., the diagonal                                                                    |
| johnston-linear-matrix-algebra<br>F03312 | 364  | ■ 0.998 | $\mathbf{e}_{\{\sigma(i)\}} \cdot \mathbf{e}_{\{\sigma(j)\}}=0$                        | $\mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)} = 0$                | $\mathbf{e}_{\sigma(i)} \cdot \mathbf{e}_{\sigma(j)} = 0$ (i.e., the off-diagonal entries of $P_\sigma^T P_\sigma$ all equal 0). It follows                        |

| Identifier                            | Page                         | Conf.                      | LaTeX source                                                                                                               | Rendered                                                                                                            | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
|---------------------------------------|------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------|---------------------------|-------------|---|---------------------------|-------------|------|----------------------------|-------------|------|----------------------------|-------------|---|---------------------------|-------------|---|---------------------------|-------------|------|----------------------------|
| johnston-linear-matrix-algebra-F03313 | 364                          | 1.000                      | $P_{\sigma}^T P_{\sigma}=I$                                                                                                | $P_{\sigma}^T P_{\sigma} = I$                                                                                       | that $P_{\sigma}^T P_{\sigma} = I$ , so $P_{\sigma}^T = P_{\sigma}^{-1}$ , which completes the proof.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03314 | 364                          | 1.000                      | $\operatorname{sgn}(\sigma)$                                                                                               | $\operatorname{sgn}(\sigma)$                                                                                        | The <b>sign of a permutation</b> $\sigma$ , denoted by $\operatorname{sgn}(\sigma)$ , is the quantity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03315 | 364                          | 0.998                      | $\sigma=\left(\begin{array}{llll}3 & 4 & 2 & 1\end{array}\right)$                                                          | $\sigma = \left(\begin{array}{cccc}3 & 4 & 2 & 1\end{array}\right)$                                                 | a) $\sigma = (3\ 4\ 2\ 1)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03316 | 364                          | 0.804                      | $\sigma=\left(\begin{array}{lllll}2 & 5 & 1 & 3 & 4\end{array}\right)$                                                     | $\sigma = \left(\begin{array}{ccccc}2 & 5 & 1 & 3 & 4\end{array}\right)$                                            | b) $\sigma = (2\ 5\ 1\ 3\ 4)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03317 | 365                          | 1.000                      | $(-1)^s$                                                                                                                   | $(-1)^s$                                                                                                            | The sign of a permutation equals $(-1)^s$ , where $s$ is the number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03318 | 365                          | 1.000                      | $\operatorname{sgn}(\sigma)=(-1)^3=-1$                                                                                     | $\operatorname{sgn}(\sigma) = (-1)^3 = -1$                                                                          | Since we swapped 3 times, we conclude that $\operatorname{sgn}(\sigma) = (-1)^3 = -1$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03319 | 365                          | 1.000                      | $\operatorname{sgn}(\sigma)=(-1)^4=1$                                                                                      | $\operatorname{sgn}(\sigma) = (-1)^4 = 1$                                                                           | Since we swapped 4 times, we conclude that $\operatorname{sgn}(\sigma) = (-1)^4 = 1$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03320 | 365                          | 0.936                      | $\operatorname{det}(A)=\sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)a_{\sigma(1),1}a_{\sigma(2),2}\cdots a_{\sigma(n),n}$ | $\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$ | Suppose $A \in \mathcal{M}_n$ . Then $\det(A) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03321 | 365                          | 0.860                      | $\operatorname{det}(A)=a_{1,1}a_{2,2}-a_{2,1}a_{1,2}$                                                                      | $\det(A) = a_{1,1} a_{2,2} - a_{2,1} a_{1,2}$                                                                       | $-1$ . The above theorem thus tells us that $\det(A) = a_{1,1} a_{2,2} - a_{2,1} a_{1,2}$ , which is                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03322 | 365                          | 0.998                      | $n=3$                                                                                                                      | $n = 3$                                                                                                             | Similarly, when $n = 3$ , there are six permutations in $S_3$ , which we now                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03323 | 366                          | 0.789                      | $a_{1,1}a_{2,2}a_{3,3}$                                                                                                    | $a_{1,1} a_{2,2} a_{3,3}$                                                                                           | <table><thead><tr><th><math>\sigma</math></th><th><math>\operatorname{sgn}(\sigma)</math></th><th>contribution to <math>\det(A)</math></th></tr></thead><tbody><tr><td><math>(1\ 2\ 3)</math></td><td>1</td><td><math>a_{1,1} a_{2,2} a_{3,3}</math></td></tr><tr><td><math>(1\ 3\ 2)</math></td><td><math>-1</math></td><td><math>-a_{1,1} a_{3,2} a_{2,3}</math></td></tr><tr><td><math>(2\ 1\ 3)</math></td><td><math>-1</math></td><td><math>-a_{2,1} a_{1,2} a_{3,3}</math></td></tr><tr><td><math>(2\ 3\ 1)</math></td><td>1</td><td><math>a_{2,1} a_{3,2} a_{1,3}</math></td></tr><tr><td><math>(3\ 1\ 2)</math></td><td>1</td><td><math>a_{3,1} a_{1,2} a_{2,3}</math></td></tr><tr><td><math>(3\ 2\ 1)</math></td><td><math>-1</math></td><td><math>-a_{3,1} a_{2,2} a_{1,3}</math></td></tr></tbody></table> | $\sigma$ | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$ | $(1\ 2\ 3)$ | 1 | $a_{1,1} a_{2,2} a_{3,3}$ | $(1\ 3\ 2)$ | $-1$ | $-a_{1,1} a_{3,2} a_{2,3}$ | $(2\ 1\ 3)$ | $-1$ | $-a_{2,1} a_{1,2} a_{3,3}$ | $(2\ 3\ 1)$ | 1 | $a_{2,1} a_{3,2} a_{1,3}$ | $(3\ 1\ 2)$ | 1 | $a_{3,1} a_{1,2} a_{2,3}$ | $(3\ 2\ 1)$ | $-1$ | $-a_{3,1} a_{2,2} a_{1,3}$ |
| $\sigma$                              | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 2\ 3)$                           | 1                            | $a_{1,1} a_{2,2} a_{3,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 3\ 2)$                           | $-1$                         | $-a_{1,1} a_{3,2} a_{2,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 1\ 3)$                           | $-1$                         | $-a_{2,1} a_{1,2} a_{3,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 3\ 1)$                           | 1                            | $a_{2,1} a_{3,2} a_{1,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 1\ 2)$                           | 1                            | $a_{3,1} a_{1,2} a_{2,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 2\ 1)$                           | $-1$                         | $-a_{3,1} a_{2,2} a_{1,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03324 | 366                          | 0.789                      | $-a_{1,1}a_{3,2}a_{2,3}$                                                                                                   | $-a_{1,1} a_{3,2} a_{2,3}$                                                                                          | <table><thead><tr><th><math>\sigma</math></th><th><math>\operatorname{sgn}(\sigma)</math></th><th>contribution to <math>\det(A)</math></th></tr></thead><tbody><tr><td><math>(1\ 2\ 3)</math></td><td>1</td><td><math>a_{1,1} a_{2,2} a_{3,3}</math></td></tr><tr><td><math>(1\ 3\ 2)</math></td><td><math>-1</math></td><td><math>-a_{1,1} a_{3,2} a_{2,3}</math></td></tr><tr><td><math>(2\ 1\ 3)</math></td><td><math>-1</math></td><td><math>-a_{2,1} a_{1,2} a_{3,3}</math></td></tr><tr><td><math>(2\ 3\ 1)</math></td><td>1</td><td><math>a_{2,1} a_{3,2} a_{1,3}</math></td></tr><tr><td><math>(3\ 1\ 2)</math></td><td>1</td><td><math>a_{3,1} a_{1,2} a_{2,3}</math></td></tr><tr><td><math>(3\ 2\ 1)</math></td><td><math>-1</math></td><td><math>-a_{3,1} a_{2,2} a_{1,3}</math></td></tr></tbody></table> | $\sigma$ | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$ | $(1\ 2\ 3)$ | 1 | $a_{1,1} a_{2,2} a_{3,3}$ | $(1\ 3\ 2)$ | $-1$ | $-a_{1,1} a_{3,2} a_{2,3}$ | $(2\ 1\ 3)$ | $-1$ | $-a_{2,1} a_{1,2} a_{3,3}$ | $(2\ 3\ 1)$ | 1 | $a_{2,1} a_{3,2} a_{1,3}$ | $(3\ 1\ 2)$ | 1 | $a_{3,1} a_{1,2} a_{2,3}$ | $(3\ 2\ 1)$ | $-1$ | $-a_{3,1} a_{2,2} a_{1,3}$ |
| $\sigma$                              | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 2\ 3)$                           | 1                            | $a_{1,1} a_{2,2} a_{3,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 3\ 2)$                           | $-1$                         | $-a_{1,1} a_{3,2} a_{2,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 1\ 3)$                           | $-1$                         | $-a_{2,1} a_{1,2} a_{3,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 3\ 1)$                           | 1                            | $a_{2,1} a_{3,2} a_{1,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 1\ 2)$                           | 1                            | $a_{3,1} a_{1,2} a_{2,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 2\ 1)$                           | $-1$                         | $-a_{3,1} a_{2,2} a_{1,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| johnston-linear-matrix-algebra-F03325 | 366                          | 0.789                      | $-a_{2,1}a_{1,2}a_{3,3}$                                                                                                   | $-a_{2,1} a_{1,2} a_{3,3}$                                                                                          | <table><thead><tr><th><math>\sigma</math></th><th><math>\operatorname{sgn}(\sigma)</math></th><th>contribution to <math>\det(A)</math></th></tr></thead><tbody><tr><td><math>(1\ 2\ 3)</math></td><td>1</td><td><math>a_{1,1} a_{2,2} a_{3,3}</math></td></tr><tr><td><math>(1\ 3\ 2)</math></td><td><math>-1</math></td><td><math>-a_{1,1} a_{3,2} a_{2,3}</math></td></tr><tr><td><math>(2\ 1\ 3)</math></td><td><math>-1</math></td><td><math>-a_{2,1} a_{1,2} a_{3,3}</math></td></tr><tr><td><math>(2\ 3\ 1)</math></td><td>1</td><td><math>a_{2,1} a_{3,2} a_{1,3}</math></td></tr><tr><td><math>(3\ 1\ 2)</math></td><td>1</td><td><math>a_{3,1} a_{1,2} a_{2,3}</math></td></tr><tr><td><math>(3\ 2\ 1)</math></td><td><math>-1</math></td><td><math>-a_{3,1} a_{2,2} a_{1,3}</math></td></tr></tbody></table> | $\sigma$ | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$ | $(1\ 2\ 3)$ | 1 | $a_{1,1} a_{2,2} a_{3,3}$ | $(1\ 3\ 2)$ | $-1$ | $-a_{1,1} a_{3,2} a_{2,3}$ | $(2\ 1\ 3)$ | $-1$ | $-a_{2,1} a_{1,2} a_{3,3}$ | $(2\ 3\ 1)$ | 1 | $a_{2,1} a_{3,2} a_{1,3}$ | $(3\ 1\ 2)$ | 1 | $a_{3,1} a_{1,2} a_{2,3}$ | $(3\ 2\ 1)$ | $-1$ | $-a_{3,1} a_{2,2} a_{1,3}$ |
| $\sigma$                              | $\operatorname{sgn}(\sigma)$ | contribution to $\det(A)$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 2\ 3)$                           | 1                            | $a_{1,1} a_{2,2} a_{3,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(1\ 3\ 2)$                           | $-1$                         | $-a_{1,1} a_{3,2} a_{2,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 1\ 3)$                           | $-1$                         | $-a_{2,1} a_{1,2} a_{3,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(2\ 3\ 1)$                           | 1                            | $a_{2,1} a_{3,2} a_{1,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 1\ 2)$                           | 1                            | $a_{3,1} a_{1,2} a_{2,3}$  |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |
| $(3\ 2\ 1)$                           | $-1$                         | $-a_{3,1} a_{2,2} a_{1,3}$ |                                                                                                                            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                              |                           |             |   |                           |             |      |                            |             |      |                            |             |   |                           |             |   |                           |             |      |                            |



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|----------------------------------------------------|------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03344              | 367  | 1.000 | $\cdots < j_m \leq n$                                                                                                                                 | $\cdots < j_m \leq n$                                                                                                                                 | $\cdots < j_m \leq n$ , which completes the proof.                                                                                                    |
| johnston-linear-matrix-algebra-F03345              | 367  | 0.813 | $\left(j_1, j_2, \ldots, j_n\right)$                                                                                                                  | $(j_1, j_2, \ldots, j_n)$                                                                                                                             | ply says that $\det(AB) = \det(A) \det(B)$ , since the only tuple $(j_1, j_2, \ldots, j_n)$ with                                                      |
| johnston-linear-matrix-algebra-F03346              | 367  | 1.000 | $1 \leq j_1 < j_2 < \cdots < j_n \leq n$                                                                                                              | $1 \leq j_1 < j_2 < \cdots < j_n \leq n$                                                                                                              | $1 \leq j_1 < j_2 < \cdots < j_n \leq n$ is $(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$ . At the other extreme,                                     |
| johnston-linear-matrix-algebra-F03347              | 367  | 1.000 | $\left(j_1, j_2, \ldots, j_n\right)=(1,2, \ldots, n)$                                                                                                 | $(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$                                                                                                         | $1 \leq j_1 < j_2 < \cdots < j_n \leq n$ is $(j_1, j_2, \ldots, j_n) = (1, 2, \ldots, n)$ . At the other extreme,                                     |
| johnston-linear-matrix-algebra-F03348              | 368  | 1.000 | $m=1$                                                                                                                                                 | $m=1$                                                                                                                                                 | if $m=1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and                                         |
| johnston-linear-matrix-algebra-F03349              | 368  | 1.000 | $A \in \mathcal{M}_{1, n}$                                                                                                                            | $A \in \mathcal{M}_{1,n}$                                                                                                                             | if $m=1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and                                         |
| johnston-linear-matrix-algebra-F03350              | 368  | 1.000 | $B \in \mathcal{M}_{n, 1}$                                                                                                                            | $B \in \mathcal{M}_{n,1}$                                                                                                                             | if $m=1$ then $A \in \mathcal{M}_{1,n}$ is a row vector and $B \in \mathcal{M}_{n,1}$ is a column vector, and                                         |
| johnston-linear-matrix-algebra-F03351              | 368  | 0.998 | $\left[\begin{array}{lll} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right]$                                                                     | $\left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right]$                                                                     | *(g) $\left[\begin{array}{ccc} 0 & 0 & 1 \\ 0 & 2 & 3 \\ 4 & 5 & 6 \end{array}\right]$                                                                |
| johnston-linear-matrix-algebra-F03352              | 369  | 0.995 | $\left[\begin{array}{llll} 3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{array}\right]$                                       | $\left[\begin{array}{cccc} 3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{array}\right]$                                       | $\left[\begin{array}{cccc} 3 & 2 & 4 & 3 \\ 1 & 1 & 3 & 2 \\ 4 & 0 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{array}\right]$                                       |
| johnston-linear-matrix-algebra-F03353              | 369  | 1.000 | $\left[\begin{array}{lllll} 6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6 \end{array}\right]$ | $\left[\begin{array}{ccccc} 6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6 \end{array}\right]$ | $\left[\begin{array}{ccccc} 6 & 2 & 1 & 0 & 4 \\ 4 & 2 & 0 & 4 & 6 \\ 0 & 6 & 3 & 4 & 4 \\ 0 & 3 & 6 & 0 & 3 \\ 2 & 6 & 1 & 6 & 6 \end{array}\right]$ |
| johnston-linear-matrix-algebra-F03354              | 369  | 0.264 | $\circ$                                                                                                                                               | $\circ$                                                                                                                                               | *(a) $(1\ 2\ 3) \circ (3\ 2\ 1)$                                                                                                                      |
| johnston-linear-matrix-algebra-F03355 [conf 0.087] | 369  | 0.087 | $\circ(2331)$                                                                                                                                         | $\circ(2331)$                                                                                                                                         | (b) $(3\ 1\ 2) \circ (2\ 3\ 1)$                                                                                                                       |
| johnston-linear-matrix-algebra-F03356 [conf 0.079] | 369  | 0.079 | $\circ(42.331)$                                                                                                                                       | $\circ(42.331)$                                                                                                                                       | *(c) $(1\ 3\ 2\ 4) \circ (4\ 2\ 3\ 1)$                                                                                                                |
| johnston-linear-matrix-algebra-F03357              | 369  | 0.649 | $(24113) \circ (4132)$                                                                                                                                | $(24113) \circ (4132)$                                                                                                                                | (d) $(2\ 4\ 1\ 3) \circ (4\ 1\ 3\ 2)$                                                                                                                 |
| johnston-linear-matrix-algebra-F03358              | 369  | 0.792 | $(241536) \circ (361524)$                                                                                                                             | $(241536) \circ (361524)$                                                                                                                             | (h) $(2\ 4\ 1\ 5\ 3\ 6) \circ (3\ 6\ 1\ 5\ 2\ 4)$                                                                                                     |
| johnston-linear-matrix-algebra-F03359              | 369  | 0.999 | $\operatorname{cof}(I)=I$                                                                                                                             | $\operatorname{cof}(I) = I$                                                                                                                           | *(c) $\operatorname{cof}(I) = I$                                                                                                                      |
| johnston-linear-matrix-algebra-F03360              | 369  | 1.000 | $\operatorname{cof}(A)=\operatorname{cof}(B)$                                                                                                         | $\operatorname{cof}(A) = \operatorname{cof}(B)$                                                                                                       | (d) If $A, B \in \mathcal{M}_n$ are such that $\operatorname{cof}(A) = \operatorname{cof}(B)$ then                                                    |
| johnston-linear-matrix-algebra-F03361              | 369  | 1.000 | $\operatorname{rank}(\operatorname{cof}(A))$                                                                                                          | $\operatorname{rank}(\operatorname{cof}(A))$                                                                                                          | $\operatorname{rank}(\operatorname{cof}(A))$ .                                                                                                        |
| johnston-linear-matrix-algebra-F03362              | 369  | 0.998 | $\imath^{-1}=\imath$                                                                                                                                  | $\imath^{-1} = \imath$                                                                                                                                | $\imath^{-1} = \imath$ .                                                                                                                              |
| johnston-linear-matrix-algebra-F03363              | 369  | 1.000 | $\sigma \circ$                                                                                                                                        | $\sigma \circ$                                                                                                                                        | (h) Composition of permutations is commutative: $\sigma \circ$                                                                                        |
| johnston-linear-matrix-algebra-F03364              | 369  | 1.000 | $\tau=\tau \circ \sigma$                                                                                                                              | $\tau = \tau \circ \sigma$                                                                                                                            | $\tau = \tau \circ \sigma$ for all $\sigma, \tau \in S_n$ .                                                                                           |
| johnston-linear-matrix-algebra-F03365              | 369  | 1.000 | $S_5$                                                                                                                                                 | $S_5$                                                                                                                                                 | *(i) There are exactly 120 permutations in $S_5$ .                                                                                                    |
| johnston-linear-matrix-algebra-F03366              | 369  | 0.998 | $\operatorname{cof}(A\ B)=\operatorname{cof}(A)\operatorname{cof}(B)$                                                                                 | $\operatorname{cof}(AB) = \operatorname{cof}(A) \operatorname{cof}(B)$                                                                                | $\operatorname{cof}(AB) = \operatorname{cof}(A) \operatorname{cof}(B)$ .                                                                              |
| johnston-linear-matrix-algebra-F03367              | 369  | 1.000 | $\operatorname{cof}\left(A^k\right)=(\operatorname{cof}(A))^k$                                                                                        | $\operatorname{cof}\left(A^k\right) = (\operatorname{cof}(A))^k$                                                                                      | Show that $\operatorname{cof}(A^k) = (\operatorname{cof}(A))^k$ . [Side note: This result is                                                          |

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|---------------------------------------|------|-------|-----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03368 | 369  | 1.000 | $\operatorname{cof}(cA)=$                                                                     | $\operatorname{cof}(cA) =$                                                                          | <b>3.A.9</b> Suppose $A \in \mathcal{M}_n$ and $c \in \mathbb{R}$ . Show that $\operatorname{cof}(cA) =$                                     |
| johnston-linear-matrix-algebra-F03369 | 369  | 1.000 | $c^{\{n-1\}} \operatorname{cof}(A)$                                                           | $c^{n-1} \operatorname{cof}(A)$                                                                     | $c^{n-1} \operatorname{cof}(A)$ .                                                                                                            |
| johnston-linear-matrix-algebra-F03370 | 369  | 1.000 | $\operatorname{det}(\operatorname{cof}(A))=(\operatorname{det}(A))^{\{n-1\}}$                 | $\det(\operatorname{cof}(A)) = (\det(A))^{n-1}$                                                     | $\det(\operatorname{cof}(A)) = (\det(A))^{n-1}$ .                                                                                            |
| johnston-linear-matrix-algebra-F03371 | 369  | 1.000 | $\operatorname{cof}(\operatorname{cof}(A))=(\operatorname{det}(A))^{\{n-2\}}A$                | $\operatorname{cof}(\operatorname{cof}(A)) = (\det(A))^{n-2}A$                                      | $\operatorname{cof}(\operatorname{cof}(A)) = (\det(A))^{n-2}A$ .                                                                             |
| johnston-linear-matrix-algebra-F03372 | 369  | 1.000 | $\operatorname{cof}(\operatorname{cof}(A))=0$                                                 | $\operatorname{cof}(\operatorname{cof}(A)) = O$                                                     | vertible, and it shows in particular that $\operatorname{cof}(\operatorname{cof}(A)) = O$                                                    |
| johnston-linear-matrix-algebra-F03373 | 369  | 1.000 | $A \in \mathcal{M}_n, A \mathbf{x} = \mathbf{b}$                                              | $A \in \mathcal{M}_n, A\mathbf{x} = \mathbf{b}$                                                     | of Cramer’s rule (Theorem 3.A.2). Let $A \in \mathcal{M}_n, A\mathbf{x} = \mathbf{b}$ ,                                                      |
| johnston-linear-matrix-algebra-F03374 | 369  | 1.000 | $A^{-1}A_{\{j\}}$                                                                             | $A^{-1}A_j$                                                                                         | (a) Compute the columns of the matrix $A^{-1}A_j$ .                                                                                          |
| johnston-linear-matrix-algebra-F03375 | 369  | 1.000 | $\operatorname{det}(A^{-1}A_{\{j\}})$                                                         | $\det(A^{-1}A_j)$                                                                                   | (b) Compute $\det(A^{-1}A_j)$ in two different ways: by using                                                                                |
| johnston-linear-matrix-algebra-F03376 | 369  | 1.000 | $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \operatorname{sgn}(\tau)$ | $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$ | (a) Show that $\operatorname{sgn}(\sigma \circ \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau)$ .                          |
| johnston-linear-matrix-algebra-F03377 | 369  | 1.000 | $m > n$                                                                                       | $m > n$                                                                                             | $\det(AB)$ if $m > n$ ?                                                                                                                      |
| johnston-linear-matrix-algebra-F03378 | 370  | 1.000 | $\operatorname{per}(A)$                                                                       | $\operatorname{per}(A)$                                                                             | $\operatorname{per}(A)$ , is the quantity defined by                                                                                         |
| johnston-linear-matrix-algebra-F03379 | 370  | 1.000 | $\left[ \begin{array}{cc} 1 & 3 \\ 0 & -2 \end{array} \right]$                                | $\left[ \begin{array}{cc} 1 & 3 \\ 0 & -2 \end{array} \right]$                                      | (b) $\left[ \begin{array}{cc} 1 & 3 \\ 0 & -2 \end{array} \right]$                                                                           |
| johnston-linear-matrix-algebra-F03380 | 370  | 1.000 | $\left[ \begin{array}{ccc} 1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2 \end{array} \right]$          | $\left[ \begin{array}{ccc} 1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2 \end{array} \right]$                | (f) $\left[ \begin{array}{ccc} 1 & 1 & 3 \\ -4 & 2 & 1 \\ 3 & 1 & 2 \end{array} \right]$                                                     |
| johnston-linear-matrix-algebra-F03381 | 370  | 1.000 | $\operatorname{per}(A^T) = \operatorname{per}(A)$                                             | $\operatorname{per}(A^T) = \operatorname{per}(A)$                                                   | satisfies $\operatorname{per}(A^T) = \operatorname{per}(A)$ for all $A \in \mathcal{M}_n$ .                                                  |
| johnston-linear-matrix-algebra-F03382 | 370  | 1.000 | $\operatorname{per}(A) = a_{1,1} a_{2,2} \cdots a_{n,n}$                                      | $\operatorname{per}(A) = a_{1,1} a_{2,2} \cdots a_{n,n}$                                            | $\operatorname{per}(A) = a_{1,1} a_{2,2} \cdots a_{n,n}$ .                                                                                   |
| johnston-linear-matrix-algebra-F03383 | 370  | 0.999 | $\operatorname{per}(AB) =$                                                                    | $\operatorname{per}(AB) =$                                                                          | Exercise 3.A.16) does <i>not</i> satisfy the property $\operatorname{per}(AB) =$                                                             |
| johnston-linear-matrix-algebra-F03384 | 370  | 1.000 | $\operatorname{per}(A) \operatorname{per}(B)$                                                 | $\operatorname{per}(A) \operatorname{per}(B)$                                                       | $\operatorname{per}(A) \operatorname{per}(B)$ .                                                                                              |
| johnston-linear-matrix-algebra-F03385 | 370  | 1.000 | $\mathbf{v}_0 \in \mathbb{R}^n$                                                               | $\mathbf{v}_0 \in \mathbb{R}^n$                                                                     | starting vector. That is, if we start with a randomly-chosen vector $\mathbf{v}_0 \in \mathbb{R}^n$ , we                                     |
| johnston-linear-matrix-algebra-F03386 | 371  | 0.997 | $A = \left[ \begin{array}{cc} 4 & 1 \\ 2 & 5 \end{array} \right]$                             | $A = \left[ \begin{array}{cc} 4 & 1 \\ 2 & 5 \end{array} \right]$                                   | <b>Figure 3.27:</b> Repeatedly applying the matrix (linear transformation) $A = \left[ \begin{array}{cc} 4 & 1 \\ 2 & 5 \end{array} \right]$ |
| johnston-linear-matrix-algebra-F03387 | 371  | 1.000 | $\mathbf{v} = (1, 2) / \sqrt{5}$                                                              | $\mathbf{v} = (1, 2) / \sqrt{5}$                                                                    | squished closer and closer to the line in the direction of $\mathbf{v} = (1, 2) / \sqrt{5}$ , which is a                                     |
| johnston-linear-matrix-algebra-F03388 | 371  | 1.000 | $\mathbf{v}_{k-1}$                                                                            | $\mathbf{v}_{k-1}$                                                                                  | The idea here is that multiplying $\mathbf{v}_{k-1}$ by $A$ skews it a bit closer toward the                                                 |
| johnston-linear-matrix-algebra-F03389 | 371  | 1.000 | $\mathbf{v}_k^T A \mathbf{v}_k$                                                               | $\mathbf{v}_k^T A \mathbf{v}_k$                                                                     | We claim that, as $k$ gets large, $\mathbf{v}_k^T A \mathbf{v}_k$ typically becomes a better and better                                      |
| johnston-linear-matrix-algebra-F03390 | 371  | 1.000 | $\mathbf{v}_0$                                                                                | $\mathbf{v}_0$                                                                                      | We start by picking $\mathbf{v}_0$ arbitrarily. If we select $\mathbf{v}_0 = \mathbf{e}_1$ then power                                        |
| johnston-linear-matrix-algebra-F03391 | 371  | 1.000 | $\mathbf{v}_0 = \mathbf{e}_1$                                                                 | $\mathbf{v}_0 = \mathbf{e}_1$                                                                       | We start by picking $\mathbf{v}_0$ arbitrarily. If we select $\mathbf{v}_0 = \mathbf{e}_1$ then power                                        |

| Identifier                            | Page                | Conf.                   | LaTeX source                                                           | Rendered                                                        | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
|---------------------------------------|---------------------|-------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------|-------------------------|----------------------------------------------------------|---|---|---|---------------------------------------|---|-------------|------|-------------|---|-------------|------|-------------|---|-------------|------|-------------|---|-------------|------|-------------|---|-------------|------|-------------|
| johnston-linear-matrix-algebra-F03392 | 372                 | 0.546                   | $A \mathbf{v}_{k-1}$                                                   | $A\mathbf{v}_{k-1}$                                             | <table><tr><th><math>k</math></th><th><math>A\mathbf{v}_{k-1}</math></th><th><math>\ A\mathbf{v}_{k-1}\ </math></th><th><math>\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ </math></th></tr><tr><td>0</td><td>–</td><td>–</td><td><math>\mathbf{v}_0 = \mathbf{e}_1 = (1,0)</math></td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table> | $k$ | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $ | 0 | – | – | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ | 1 | (1.00,5.00) | 5.10 | (0.20,0.98) | 2 | (2.16,4.90) | 5.36 | (0.40,0.92) | 3 | (2.23,5.68) | 6.10 | (0.37,0.93) | 4 | (2.23,5.55) | 5.98 | (0.37,0.93) | 5 | (2.23,5.57) | 6.00 | (0.37,0.93) |
| $k$                                   | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $               |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 0                                     | –                   | –                       | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$                                  |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 1                                     | (1.00,5.00)         | 5.10                    | (0.20,0.98)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 2                                     | (2.16,4.90)         | 5.36                    | (0.40,0.92)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 3                                     | (2.23,5.68)         | 6.10                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 4                                     | (2.23,5.55)         | 5.98                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 5                                     | (2.23,5.57)         | 6.00                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03393 | 372                 | 0.546                   | $\left\ A \mathbf{v}_{k-1}\right\ $                                    | $\ A\mathbf{v}_{k-1}\ $                                         | <table><tr><th><math>k</math></th><th><math>A\mathbf{v}_{k-1}</math></th><th><math>\ A\mathbf{v}_{k-1}\ </math></th><th><math>\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ </math></th></tr><tr><td>0</td><td>–</td><td>–</td><td><math>\mathbf{v}_0 = \mathbf{e}_1 = (1,0)</math></td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table> | $k$ | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $ | 0 | – | – | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ | 1 | (1.00,5.00) | 5.10 | (0.20,0.98) | 2 | (2.16,4.90) | 5.36 | (0.40,0.92) | 3 | (2.23,5.68) | 6.10 | (0.37,0.93) | 4 | (2.23,5.55) | 5.98 | (0.37,0.93) | 5 | (2.23,5.57) | 6.00 | (0.37,0.93) |
| $k$                                   | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $               |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 0                                     | –                   | –                       | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$                                  |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 1                                     | (1.00,5.00)         | 5.10                    | (0.20,0.98)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 2                                     | (2.16,4.90)         | 5.36                    | (0.40,0.92)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 3                                     | (2.23,5.68)         | 6.10                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 4                                     | (2.23,5.55)         | 5.98                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 5                                     | (2.23,5.57)         | 6.00                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03394 | 372                 | 0.546                   | $\mathbf{v}_{k}=A \mathbf{v}_{k-1} /\left\ A \mathbf{v}_{k-1}\right\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $        | <table><tr><th><math>k</math></th><th><math>A\mathbf{v}_{k-1}</math></th><th><math>\ A\mathbf{v}_{k-1}\ </math></th><th><math>\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ </math></th></tr><tr><td>0</td><td>–</td><td>–</td><td><math>\mathbf{v}_0 = \mathbf{e}_1 = (1,0)</math></td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table> | $k$ | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $ | 0 | – | – | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ | 1 | (1.00,5.00) | 5.10 | (0.20,0.98) | 2 | (2.16,4.90) | 5.36 | (0.40,0.92) | 3 | (2.23,5.68) | 6.10 | (0.37,0.93) | 4 | (2.23,5.55) | 5.98 | (0.37,0.93) | 5 | (2.23,5.57) | 6.00 | (0.37,0.93) |
| $k$                                   | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $               |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 0                                     | –                   | –                       | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$                                  |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 1                                     | (1.00,5.00)         | 5.10                    | (0.20,0.98)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 2                                     | (2.16,4.90)         | 5.36                    | (0.40,0.92)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 3                                     | (2.23,5.68)         | 6.10                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 4                                     | (2.23,5.55)         | 5.98                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 5                                     | (2.23,5.57)         | 6.00                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03395 | 372                 | 0.546                   | $\mathbf{v}_{\{0\}}=\mathbf{e}_{\{1\}}=(1,0)$                          | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$                           | <table><tr><th><math>k</math></th><th><math>A\mathbf{v}_{k-1}</math></th><th><math>\ A\mathbf{v}_{k-1}\ </math></th><th><math>\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ </math></th></tr><tr><td>0</td><td>–</td><td>–</td><td><math>\mathbf{v}_0 = \mathbf{e}_1 = (1,0)</math></td></tr><tr><td>1</td><td>(1.00,5.00)</td><td>5.10</td><td>(0.20,0.98)</td></tr><tr><td>2</td><td>(2.16,4.90)</td><td>5.36</td><td>(0.40,0.92)</td></tr><tr><td>3</td><td>(2.23,5.68)</td><td>6.10</td><td>(0.37,0.93)</td></tr><tr><td>4</td><td>(2.23,5.55)</td><td>5.98</td><td>(0.37,0.93)</td></tr><tr><td>5</td><td>(2.23,5.57)</td><td>6.00</td><td>(0.37,0.93)</td></tr></table> | $k$ | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $ | 0 | – | – | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$ | 1 | (1.00,5.00) | 5.10 | (0.20,0.98) | 2 | (2.16,4.90) | 5.36 | (0.40,0.92) | 3 | (2.23,5.68) | 6.10 | (0.37,0.93) | 4 | (2.23,5.55) | 5.98 | (0.37,0.93) | 5 | (2.23,5.57) | 6.00 | (0.37,0.93) |
| $k$                                   | $A\mathbf{v}_{k-1}$ | $\ A\mathbf{v}_{k-1}\ $ | $\mathbf{v}_k = A\mathbf{v}_{k-1}/\ A\mathbf{v}_{k-1}\ $               |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 0                                     | –                   | –                       | $\mathbf{v}_0 = \mathbf{e}_1 = (1,0)$                                  |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 1                                     | (1.00,5.00)         | 5.10                    | (0.20,0.98)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 2                                     | (2.16,4.90)         | 5.36                    | (0.40,0.92)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 3                                     | (2.23,5.68)         | 6.10                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 4                                     | (2.23,5.55)         | 5.98                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| 5                                     | (2.23,5.57)         | 6.00                    | (0.37,0.93)                                                            |                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03396 | 372                 | 0.997                   | $\mathbf{v}_{\{6\}}, \mathbf{v}_{\{7\}}, \mathbf{v}_{\{8\}}$           | $\mathbf{v}_6, \mathbf{v}_7, \mathbf{v}_8$                      | We could of course keep going and compute $\mathbf{v}_6, \mathbf{v}_7, \mathbf{v}_8$ , and so on,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03397 | 372                 | 1.000                   | $\mathbf{v} \approx \mathbf{v}_{\{5\}} \approx (0.37,0.93)$            | $\mathbf{v} \approx \mathbf{v}_5 \approx (0.37,0.93)$           | with corresponding unit eigenvector $\mathbf{v} \approx \mathbf{v}_5 \approx (0.37,0.93)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03398 | 372                 | 0.942                   | $\mathbf{v}=(2,5) / \sqrt{29} \approx (0.37,0.93)$                     | $\mathbf{v} = (2,5)/\sqrt{29} \approx (0.37,0.93)$              | $\mathbf{v} = (2,5)/\sqrt{29} \approx (0.37,0.93)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03399 | 372                 | 1.000                   | $\mathbf{v}_{\{0\}}, \mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots$   | $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \ldots$              | Also, we can visualize the vectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2, \ldots$ that we computed via                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03400 | 373                 | 1.000                   | $\mathbf{v}_{\{7\}} \approx (-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$  | $\mathbf{v}_7 \approx (-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$ | close to $\mathbf{v}_7 \approx (-0.26,-0.41,0.49,0.38,0.49,-0.38,-0.10)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03401 | 373                 | 1.000                   | $\lambda=-4.92$                                                        | $\lambda = -4.92$                                               | located near $\lambda = -4.92$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03402 | 373                 | 0.999                   | $s-1$                                                                  | $s - 1$                                                         | multiplications and $s - 1$ additions, rather than $n$ multiplications and $n - 1$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03403 | 374                 | 1.000                   | $p_{\{C\}}(\lambda)=(-1)^n p(\lambda)$                                 | $p_C(\lambda) = (-1)^n p(\lambda)$                              | $p_C(\lambda) = (-1)^n p(\lambda)$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |
| johnston-linear-matrix-algebra-F03404 | 374                 | 1.000                   | $C \mathbf{v}_{\{k\}}$                                                 | $C\mathbf{v}_k$                                                 | $C\mathbf{v}_k$ only requires $n$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                     |                         |                                                          |   |   |   |                                       |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |   |             |      |             |

| Identifier                            | Page                                                                    | Conf.                           | LaTeX source                                                            | Rendered                                                                | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
|---------------------------------------|-------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------|---------------------------------|---|---------------------------------|------|---|---------------------------------|------|---|---------------------------------|------|---|---------------------------------|------|--|----------|--|----|---------------------------------|------|----|---------------------------------|------|
| johnston-linear-matrix-algebra-F03405 | 374                                                                     | 0.976                           | $2 \times 2, 3 \times 3$                                                | $2 \times 2, 3 \times 3$                                                | polynomial. While this method works fine for small (e.g., $2 \times 2, 3 \times 3$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03406 | 374                                                                     | 1.000                           | $p(x)=\lambda^n+a_{n-1} \lambda^{n-1}+\cdots+a_1 \lambda+a_0$           | $p(x)=\lambda^n+a_{n-1}\lambda^{n-1}+\cdots+a_1\lambda+a_0$             | Specifically, to find the roots of $p(x)=\lambda^n+a_{n-1}\lambda^{n-1}+\cdots+a_1\lambda+a_0$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03407 | 374                                                                     | 0.999                           | $2 \times n-1$                                                          | $2n-1$                                                                  | to this task since $C$ is so sparse (it only has $2n-1$ non-zero entries, despite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03408 | 374                                                                     | 1.000                           | $p(x)=x^6-2x^5+2x^4-3x^3+x^2+x-2$                                       | $p(x)=x^6-2x^5+2x^4-3x^3+x^2+x-2$                                       | nomial $p(x)=x^6-2x^5+2x^4-3x^3+x^2+x-2$ , we first construct its                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03409 | 374                                                                     | 1.000                           | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | <table><tr><th><math>k</math></th><th><math>\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ </math></th><th><math>\mathbf{v}_k^T C \mathbf{v}_k</math></th></tr><tr><td>0</td><td>(1.00,0.00,0.00,0.00,0.00,0.00)</td><td>0.00</td></tr><tr><td>1</td><td>(0.00,0.00,0.00,0.00,0.00,1.00)</td><td>2.00</td></tr><tr><td>2</td><td>(0.00,0.00,0.00,0.00,0.45,0.89)</td><td>1.20</td></tr><tr><td>3</td><td>(0.00,0.00,0.00,0.33,0.67,0.67)</td><td>1.33</td></tr><tr><td></td><td><math>\vdots</math></td><td></td></tr><tr><td>26</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.69</td></tr><tr><td>27</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.68</td></tr></table> | $k$ | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | $\mathbf{v}_k^T C \mathbf{v}_k$ | 0 | (1.00,0.00,0.00,0.00,0.00,0.00) | 0.00 | 1 | (0.00,0.00,0.00,0.00,0.00,1.00) | 2.00 | 2 | (0.00,0.00,0.00,0.00,0.45,0.89) | 1.20 | 3 | (0.00,0.00,0.00,0.33,0.67,0.67) | 1.33 |  | $\vdots$ |  | 26 | (0.06,0.10,0.17,0.28,0.48,0.81) | 1.69 | 27 | (0.06,0.10,0.17,0.28,0.48,0.81) | 1.68 |
| $k$                                   | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | $\mathbf{v}_k^T C \mathbf{v}_k$ |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 0                                     | (1.00,0.00,0.00,0.00,0.00,0.00)                                         | 0.00                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 1                                     | (0.00,0.00,0.00,0.00,0.00,1.00)                                         | 2.00                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 2                                     | (0.00,0.00,0.00,0.00,0.45,0.89)                                         | 1.20                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 3                                     | (0.00,0.00,0.00,0.33,0.67,0.67)                                         | 1.33                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
|                                       | $\vdots$                                                                |                                 |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 26                                    | (0.06,0.10,0.17,0.28,0.48,0.81)                                         | 1.69                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 27                                    | (0.06,0.10,0.17,0.28,0.48,0.81)                                         | 1.68                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03410 | 374                                                                     | 1.000                           | $\mathbf{v}_k^T C \mathbf{v}_k$                                         | $\mathbf{v}_k^T C \mathbf{v}_k$                                         | <table><tr><th><math>k</math></th><th><math>\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ </math></th><th><math>\mathbf{v}_k^T C \mathbf{v}_k</math></th></tr><tr><td>0</td><td>(1.00,0.00,0.00,0.00,0.00,0.00)</td><td>0.00</td></tr><tr><td>1</td><td>(0.00,0.00,0.00,0.00,0.00,1.00)</td><td>2.00</td></tr><tr><td>2</td><td>(0.00,0.00,0.00,0.00,0.45,0.89)</td><td>1.20</td></tr><tr><td>3</td><td>(0.00,0.00,0.00,0.33,0.67,0.67)</td><td>1.33</td></tr><tr><td></td><td><math>\vdots</math></td><td></td></tr><tr><td>26</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.69</td></tr><tr><td>27</td><td>(0.06,0.10,0.17,0.28,0.48,0.81)</td><td>1.68</td></tr></table> | $k$ | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | $\mathbf{v}_k^T C \mathbf{v}_k$ | 0 | (1.00,0.00,0.00,0.00,0.00,0.00) | 0.00 | 1 | (0.00,0.00,0.00,0.00,0.00,1.00) | 2.00 | 2 | (0.00,0.00,0.00,0.00,0.45,0.89) | 1.20 | 3 | (0.00,0.00,0.00,0.33,0.67,0.67) | 1.33 |  | $\vdots$ |  | 26 | (0.06,0.10,0.17,0.28,0.48,0.81) | 1.69 | 27 | (0.06,0.10,0.17,0.28,0.48,0.81) | 1.68 |
| $k$                                   | $\mathbf{v}_k=C \mathbf{v}_{k-1} / \left\  C \mathbf{v}_{k-1} \right\ $ | $\mathbf{v}_k^T C \mathbf{v}_k$ |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 0                                     | (1.00,0.00,0.00,0.00,0.00,0.00)                                         | 0.00                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 1                                     | (0.00,0.00,0.00,0.00,0.00,1.00)                                         | 2.00                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 2                                     | (0.00,0.00,0.00,0.00,0.45,0.89)                                         | 1.20                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 3                                     | (0.00,0.00,0.00,0.33,0.67,0.67)                                         | 1.33                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
|                                       | $\vdots$                                                                |                                 |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 26                                    | (0.06,0.10,0.17,0.28,0.48,0.81)                                         | 1.69                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| 27                                    | (0.06,0.10,0.17,0.28,0.48,0.81)                                         | 1.68                            |                                                                         |                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03411 | 374                                                                     | 0.992                           | $1.68,-0.72,-0.17 \pm 1.32 i$                                           | $1.68,-0.72,-0.17 \pm 1.32i$                                            | indeed, it is—the roots of $p$ are approximately $1.68,-0.72,-0.17 \pm 1.32i$ ,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03412 | 374                                                                     | 1.000                           | $0.69 \pm 0.67 i$                                                       | $0.69 \pm 0.67i$                                                        | and $0.69 \pm 0.67i$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03413 | 375                                                                     | 1.000                           | $(1,1) / \sqrt{2}$                                                      | $(1,1)/\sqrt{2}$                                                        | $(1,1)/\sqrt{2}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03414 | 375                                                                     | 1.000                           | $(1,3) / \sqrt{10}$                                                     | $(1,3)/\sqrt{10}$                                                       | $(1,3)/\sqrt{10}$ . Similarly,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03415 | 375                                                                     | 1.000                           | $k=1,2,3, \ldots$                                                       | $k=1,2,3, \ldots$                                                       | ting $\mathbf{v}_k=A \mathbf{v}_{k-1} / \left\  A \mathbf{v}_{k-1} \right\ $ for $k=1,2,3, \ldots$ , then absolutely nothing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03416 | 375                                                                     | 1.000                           | $\mathbf{v}_0=(0.80,0.60)$                                              | $\mathbf{v}_0=(0.80,0.60)$                                              | be $\mathbf{v}_0=(0.80,0.60)$ then power iteration does indeed converge to the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03417 | 375                                                                     | 1.000                           | $A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$          | $A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$          | Apply power iteration to the matrix $A=\left[\begin{array}{cc} 1 & 1 \\ 3 & -1 \end{array}\right]$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03418 | 376                                                                     | 1.000                           | $y=-3 x$                                                                | $y=-3x$                                                                 | $y=-3x$ are the                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03419 | 376                                                                     | 1.000                           | $A^2=4 I$                                                               | $A^2=4I$                                                                | “winning” as $A$ is applied more times. In particular, $A^2=4I$ , so applying $A$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03420 | 376                                                                     | 1.000                           | $\mathbf{v}_1=A \mathbf{v}_0 / \left\  A \mathbf{v}_0 \right\ $         | $\mathbf{v}_1=A \mathbf{v}_0 / \left\  A \mathbf{v}_0 \right\ $         | a real vector $\mathbf{v}_0$ then $\mathbf{v}_1=A \mathbf{v}_0 / \left\  A \mathbf{v}_0 \right\ $ will also be real, as will $\mathbf{v}_2, \mathbf{v}_3$ , and so                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03421 | 376                                                                     | 1.000                           | $\mathbf{v}_2, \mathbf{v}_3$                                            | $\mathbf{v}_2, \mathbf{v}_3$                                            | a real vector $\mathbf{v}_0$ then $\mathbf{v}_1=A \mathbf{v}_0 / \left\  A \mathbf{v}_0 \right\ $ will also be real, as will $\mathbf{v}_2, \mathbf{v}_3$ , and so                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |
| johnston-linear-matrix-algebra-F03422 | 376                                                                     | 1.000                           | $ \lambda = \bar{\lambda} $                                             | $ \lambda = \bar{\lambda} $                                             | complex conjugates have the same absolute value of each other: $ \lambda = \bar{\lambda} $ for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |                                                                         |                                 |   |                                 |      |   |                                 |      |   |                                 |      |   |                                 |      |  |          |  |    |                                 |      |    |                                 |      |

| Identifier                            | Page | Conf.   | LaTeX source                                                                         | Rendered                                                                             | Image                                                                                                                                                                       |
|---------------------------------------|------|---------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03423 | 376  | ■ 0.999 | $1 \pm i$                                                                            | $1 \pm i$                                                                            | has eigenvalues $1 \pm i$ , which have the same absolute value as each other:                                                                                               |
| johnston-linear-matrix-algebra-F03424 | 377  | ■ 1.000 | $\mathbf{v}_0 \in \mathbb{C}_n$                                                      | $\mathbf{v}_0 \in \mathbb{C}_n$                                                      | $\mathbf{v}_0 \in \mathbb{C}_n$ instead.                                                                                                                                    |
| johnston-linear-matrix-algebra-F03425 | 377  | ■ 1.000 | $ 1+i = 1-i =\sqrt{2}$                                                               | $ 1+i  =  1-i  = \sqrt{2}$                                                           | $ 1+i  =  1-i  = \sqrt{2}$ . Power iteration thus does not necessarily converge to                                                                                          |
| johnston-linear-matrix-algebra-F03426 | 377  | ■ 1.000 | $\left\{ \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n \right\}$                   | $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$                                | $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ with corresponding eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$ , respec-                                     |
| johnston-linear-matrix-algebra-F03427 | 377  | ■ 1.000 | $ \lambda_1  >  \lambda_j $                                                          | $ \lambda_1  >  \lambda_j $                                                          | • $ \lambda_1  >  \lambda_j $ for all $2 \leq j \leq n$ , and                                                                                                               |
| johnston-linear-matrix-algebra-F03428 | 377  | ■ 1.000 | $2 \leq j \leq n$                                                                    | $2 \leq j \leq n$                                                                    | • $ \lambda_1  >  \lambda_j $ for all $2 \leq j \leq n$ , and                                                                                                               |
| johnston-linear-matrix-algebra-F03429 | 377  | ■ 1.000 | $\mathbf{v}_0 \notin \text{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$ | $\mathbf{v}_0 \notin \text{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$ | • $\mathbf{v}_0 \notin \text{span}\{\mathbf{w}_2, \mathbf{w}_3, \dots, \mathbf{w}_n\}$ .                                                                                    |
| johnston-linear-matrix-algebra-F03430 | 377  | ■ 1.000 | $\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$              | $\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$              | If we define $\mathbf{v}_k = A \mathbf{v}_{k-1} / \ A \mathbf{v}_{k-1}\ $ for all $k \geq 1$ then $\lim_{k \rightarrow \infty} \mathbf{v}_k^* A \mathbf{v}_k = \lambda_1$ . |
| johnston-linear-matrix-algebra-F03431 | 377  | ■ 1.000 | $\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $                             | $\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $                             | for $\mathbf{v}_k$ a bit more explicitly as $\mathbf{v}_k = A^k \mathbf{v}_0 / \ A^k \mathbf{v}_0\ $ . With this in mind, we begin                                          |
| johnston-linear-matrix-algebra-F03432 | 377  | ■ 1.000 | $A^k \mathbf{v}_0$                                                                   | $A^k \mathbf{v}_0$                                                                   | by investigating what happens to $A^k \mathbf{v}_0$ as $k$ gets large.                                                                                                      |
| johnston-linear-matrix-algebra-F03433 | 377  | ■ 1.000 | $\mathbf{w}_j$                                                                       | $\mathbf{w}_j$                                                                       | Without loss of generality, we can assume that each eigenvector $\mathbf{w}_j$ is a                                                                                         |
| johnston-linear-matrix-algebra-F03434 | 377  | ■ 1.000 | $A \mathbf{w}_j = \lambda_j \mathbf{w}_j$                                            | $A \mathbf{w}_j = \lambda_j \mathbf{w}_j$                                            | Then repeatedly using the fact that $A \mathbf{w}_j = \lambda_j \mathbf{w}_j$ for each $1 \leq j \leq n$ gives                                                              |
| johnston-linear-matrix-algebra-F03435 | 377  | ■ 0.985 | $ \lambda_j / \lambda_1  < 1$                                                        | $ \lambda_j / \lambda_1  < 1$                                                        | We then notice that, since $ \lambda_1  >  \lambda_j $ for all $2 \leq j \leq n$ , we have $ \lambda_j / \lambda_1  < 1$                                                    |
| johnston-linear-matrix-algebra-F03436 | 377  | ■ 1.000 | $(\lambda_j / \lambda_1)^k \rightarrow 0$                                            | $(\lambda_j / \lambda_1)^k \rightarrow 0$                                            | and so $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $2 \leq j \leq n$ as well. For simplicity, we                                            |
| johnston-linear-matrix-algebra-F03437 | 377  | ■ 1.000 | $k \rightarrow \infty$                                                               | $k \rightarrow \infty$                                                               | and so $(\lambda_j / \lambda_1)^k \rightarrow 0$ as $k \rightarrow \infty$ for all $2 \leq j \leq n$ as well. For simplicity, we                                            |
| johnston-linear-matrix-algebra-F03438 | 377  | ■ 1.000 | $\mathbf{r}_k$                                                                       | $\mathbf{r}_k$                                                                       | then define $\mathbf{r}_k$ to be the “remainder” vector that consists of terms that become                                                                                  |
| johnston-linear-matrix-algebra-F03439 | 378  | ■ 1.000 | $ \lambda_1 ^{2k}$                                                                   | $ \lambda_1 ^{2k}$                                                                   | The $ \lambda_1 ^{2k}$ term here                                                                                                                                            |
| johnston-linear-matrix-algebra-F03440 | 378  | ■ 0.994 | $\bar{\lambda}_1^k$                                                                  | $\bar{\lambda}_1^k$                                                                  | combining $\bar{\lambda}_1^k$ from                                                                                                                                          |
| johnston-linear-matrix-algebra-F03441 | 378  | ■ 1.000 | $\mathbf{v}_k^*$                                                                     | $\mathbf{v}_k^*$                                                                     | the $\mathbf{v}_k^*$ on the left                                                                                                                                            |
| johnston-linear-matrix-algebra-F03442 | 378  | ■ 1.000 | $\lambda_1^k$                                                                        | $\lambda_1^k$                                                                        | with $\lambda_1^k$ from the $\mathbf{v}_k$                                                                                                                                  |
| johnston-linear-matrix-algebra-F03443 | 378  | ■ 1.000 | $\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$                              | $\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$                              | In particular, by using the fact that $\lim_{k \rightarrow \infty} \mathbf{r}_k = \mathbf{0}$ we can now compute the desired                                                |
| johnston-linear-matrix-algebra-F03444 | 378  | ■ 1.000 | $\mathbf{w}_1$                                                                       | $\mathbf{w}_1$                                                                       | where at the end we use the facts that $\mathbf{w}_1$ is a unit vector (twice) and that it is                                                                               |
| johnston-linear-matrix-algebra-F03445 | 378  | ■ 1.000 | $(\lambda_1 /  \lambda_1 )^k$                                                        | $(\lambda_1 /  \lambda_1 )^k$                                                        | scalar $(\lambda_1 /  \lambda_1 )^k$ can change sign (or in the case of complex matrices, bounce                                                                            |
| johnston-linear-matrix-algebra-F03446 | 378  | ■ 0.990 | $\lambda_1 /  \lambda_1  = -1$                                                       | $\lambda_1 /  \lambda_1  = -1$                                                       | being negative, and thus $\lambda_1 /  \lambda_1  = -1$ ).                                                                                                                  |
| johnston-linear-matrix-algebra-F03447 | 378  | ■ 1.000 | $ \lambda_2 / \lambda_1 $                                                            | $ \lambda_2 / \lambda_1 $                                                            | the ratio $ \lambda_2 / \lambda_1 $ of the two largest (in absolute value) eigenvalues of $A$ —the                                                                          |

| Identifier                            | Page                                                                  | Conf.                           | LaTeX source                                                                        | Rendered                                                                   | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
|---------------------------------------|-----------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------|---------------------------------|---|--------------------|-------|---|--------------------|-------|---|---------------------|-------|---|--------------------|-------|---|---------------------|-------|
| johnston-linear-matrix-algebra-F03448 | 378                                                                   | 1.000                           | $A=\left[\begin{array}{ccc}8 & -1 & 0 \\ -1 & 2 & 5 \\ 5 & -6 & \end{array}\right]$ | $A = \begin{bmatrix} 8 & -1 & 0 \\ -1 & 2 & 5 \\ 0 & 5 & -6 \end{bmatrix}$ | Apply power iteration to the matrix $A = \begin{bmatrix} 8 & -1 & 0 \\ -1 & 2 & 5 \\ 0 & 5 & -6 \end{bmatrix}$ .                                                                                                                                                                                                                                                                                                                                                                                          |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03449 | 379                                                                   | 1.000                           | $\mathbf{v}_{318}$                                                                  | $\mathbf{v}_{318}$                                                         | The vectors $\mathbf{v}_{318}$ and                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03450 | 379                                                                   | 1.000                           | $\mathbf{v}_{320}$                                                                  | $\mathbf{v}_{320}$                                                         | $\mathbf{v}_{320}$ are actually                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03451 | 379                                                                   | 1.000                           | $\mathbf{v}_{319}$                                                                  | $\mathbf{v}_{319}$                                                         | (and similarly for $\mathbf{v}_{319}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03452 | 379                                                                   | 1.000                           | $\mathbf{v}_{321}$                                                                  | $\mathbf{v}_{321}$                                                         | and $\mathbf{v}_{321}$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03453 | 379                                                                   | 0.990                           | $\lambda_3 \approx 4.19$                                                            | $\lambda_3 \approx 4.19$                                                   | $\lambda_3 \approx 4.19$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03454 | 379                                                                   | 1.000                           | $A-\lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$                                           | $A-\lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$                                  | $A-\lambda_1 \mathbf{w}_1 \mathbf{w}_1^T$ may                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03455 | 379                                                                   | 1.000                           | $k=63$                                                                              | $k=63$                                                                     | to be very, very wrong. After $k=63$ iterations, the eigenvalue estimate                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03456 | 379                                                                   | 0.803                           | $k=156$                                                                             | $k=156$                                                                    | dips back down below 8.00, after $k=156$ iterations it dips down below                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03457 | 379                                                                   | 1.000                           | $k=320$                                                                             | $k=320$                                                                    | $k=320$ :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03458 | 379                                                                   | 0.999                           | $\lambda_1 \approx -8.41$                                                           | $\lambda_1 \approx -8.41$                                                  | $\lambda_1 \approx -8.41$ and $\lambda_2 \approx 8.22$ of $A$ are so close to each other in absolute                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03459 | 379                                                                   | 0.999                           | $\lambda_2 \approx 8.22$                                                            | $\lambda_2 \approx 8.22$                                                   | $\lambda_1 \approx -8.41$ and $\lambda_2 \approx 8.22$ of $A$ are so close to each other in absolute                                                                                                                                                                                                                                                                                                                                                                                                      |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03460 | 379                                                                   | 0.942                           | $ \lambda_2/\lambda_1  \approx 0.98$                                                | $ \lambda_2/\lambda_1  \approx 0.98$                                       | value (so the ratio $ \lambda_2/\lambda_1  \approx 0.98$ is close to 1). Furthermore, the starting                                                                                                                                                                                                                                                                                                                                                                                                        |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03461 | 379                                                                   | 0.763                           | $\operatorname{span}\{(0.97,-0.22,-0.08)\}$                                         | $\operatorname{span}\{(0.97,-0.22,-0.08)\}$                                | is $\operatorname{span}\{(0.97,-0.22,-0.08)\}$ ), so it starts off by getting pulled toward                                                                                                                                                                                                                                                                                                                                                                                                               |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03462 | 379                                                                   | 1.000                           | $\mathbf{v}_k^* A \mathbf{v}_k$                                                     | $\mathbf{v}_k^* A \mathbf{v}_k$                                            | plot of the eigenvalue estimate $\mathbf{v}_k^* A \mathbf{v}_k$ , as a function of the iteration $k$ , is                                                                                                                                                                                                                                                                                                                                                                                                 |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03463 | 380                                                                   | 1.000                           | $\mathbf{w}_2$                                                                      | $\mathbf{w}_2$                                                             | Keep in mind that $\mathbf{w}_2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03464 | 380                                                                   | 0.993                           | $A=\left[\begin{array}{lll}1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1\end{array}\right]$   | $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$    | Use power iteration to find all eigenvalues of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix}$ .                                                                                                                                                                                                                                                                                                                                                                                  |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03465 | 380                                                                   | 1.000                           | $\lambda_1 \approx 8.53$                                                            | $\lambda_1 \approx 8.53$                                                   | Indeed, the dominant eigenvalue of $A$ is $\lambda_1 \approx 8.53$ with corresponding                                                                                                                                                                                                                                                                                                                                                                                                                     |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03466 | 380                                                                   | 1.000                           | $\mathbf{w}_1 \approx (0.38,0.85,0.37)$                                             | $\mathbf{w}_1 \approx (0.38,0.85,0.37)$                                    | unit eigenvector $\mathbf{w}_1 \approx (0.38,0.85,0.37)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| johnston-linear-matrix-algebra-F03467 | 380                                                                   | 1.000                           | $\mathbf{v}_k=B \mathbf{v}_{k-1} / \left\ \mathbf{v}_{k-1}\right\ $                 | $\mathbf{v}_k=B \mathbf{v}_{k-1} / \left\ B \mathbf{v}_{k-1}\right\ $      | <table><tr><th><math>k</math></th><th><math>\mathbf{v}_k=B \mathbf{v}_{k-1} / \left\ B \mathbf{v}_{k-1}\right\ </math></th><th><math>\mathbf{v}_k^T B \mathbf{v}_k</math></th></tr><tr><td>0</td><td>( 1.00,0.00, 0.00)</td><td>−0.21</td></tr><tr><td>1</td><td>(−0.10,0.57, 0.81)</td><td>−0.52</td></tr><tr><td>2</td><td>( 0.81,0.22, −0.54)</td><td>−1.81</td></tr><tr><td>3</td><td>(−0.68,0.04, 0.73)</td><td>−1.99</td></tr><tr><td>4</td><td>( 0.72,0.03, −0.69)</td><td>−2.00</td></tr></table> | $k$ | $\mathbf{v}_k=B \mathbf{v}_{k-1} / \left\ B \mathbf{v}_{k-1}\right\ $ | $\mathbf{v}_k^T B \mathbf{v}_k$ | 0 | ( 1.00,0.00, 0.00) | −0.21 | 1 | (−0.10,0.57, 0.81) | −0.52 | 2 | ( 0.81,0.22, −0.54) | −1.81 | 3 | (−0.68,0.04, 0.73) | −1.99 | 4 | ( 0.72,0.03, −0.69) | −2.00 |
| $k$                                   | $\mathbf{v}_k=B \mathbf{v}_{k-1} / \left\ B \mathbf{v}_{k-1}\right\ $ | $\mathbf{v}_k^T B \mathbf{v}_k$ |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| 0                                     | ( 1.00,0.00, 0.00)                                                    | −0.21                           |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| 1                                     | (−0.10,0.57, 0.81)                                                    | −0.52                           |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| 2                                     | ( 0.81,0.22, −0.54)                                                   | −1.81                           |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| 3                                     | (−0.68,0.04, 0.73)                                                    | −1.99                           |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |
| 4                                     | ( 0.72,0.03, −0.69)                                                   | −2.00                           |                                                                                     |                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |                                                                       |                                 |   |                    |       |   |                    |       |   |                     |       |   |                    |       |   |                     |       |

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|-------------------------------------------------------------|----------------------------------------------------------------------|---------------------------------|----------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------|---------------------|---------------------------------------------------------------|---|----------------------------------------------------------------------|-------|---|----------------------|-------|---|---------------------|-------|---|----------------------|-------|
| johnston-linear-matrix-algebra-F03468                       | 380                                                                  | 1.000                           | $\mathbf{v}_{k}^{\mathrm{T}} B \mathbf{v}_{k}$                       | $\mathbf{v}_k^T B \mathbf{v}_k$                                      | <table><tr><th><math>k</math></th><th><math>\mathbf{v}_k = B \mathbf{v}_{k-1} / \ B \mathbf{v}_{k-1}\ </math></th><th><math>\mathbf{v}_k^T B \mathbf{v}_k</math></th></tr><tr><td>0</td><td>( 1.00, 0.00, 0.00)</td><td>−0.21</td></tr><tr><td>1</td><td>(−0.10, 0.57, 0.81)</td><td>−0.52</td></tr><tr><td>2</td><td>( 0.81, 0.22, −0.54)</td><td>−1.81</td></tr><tr><td>3</td><td>(−0.68, 0.04, 0.73)</td><td>−1.99</td></tr><tr><td>4</td><td>( 0.72, 0.03, −0.69)</td><td>−2.00</td></tr></table>      | $k$      | $\mathbf{v}_k = B \mathbf{v}_{k-1} / \ B \mathbf{v}_{k-1}\ $ | $\mathbf{v}_k^T B \mathbf{v}_k$                             | 0                                                                   | ( 1.00, 0.00, 0.00) | −0.21                                                         | 1 | (−0.10, 0.57, 0.81)                                                  | −0.52 | 2 | ( 0.81, 0.22, −0.54) | −1.81 | 3 | (−0.68, 0.04, 0.73) | −1.99 | 4 | ( 0.72, 0.03, −0.69) | −2.00 |
| $k$                                                         | $\mathbf{v}_k = B \mathbf{v}_{k-1} / \ B \mathbf{v}_{k-1}\ $         | $\mathbf{v}_k^T B \mathbf{v}_k$ |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| 0                                                           | ( 1.00, 0.00, 0.00)                                                  | −0.21                           |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| 1                                                           | (−0.10, 0.57, 0.81)                                                  | −0.52                           |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| 2                                                           | ( 0.81, 0.22, −0.54)                                                 | −1.81                           |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| 3                                                           | (−0.68, 0.04, 0.73)                                                  | −1.99                           |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| 4                                                           | ( 0.72, 0.03, −0.69)                                                 | −2.00                           |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03469                       | 380                                                                  | 1.000                           | $\lambda_2 \approx -2.00$                                            | $\lambda_2 \approx -2.00$                                            | in absolute value) is thus $\lambda_2 \approx -2.00$ , with corresponding unit eigenvector                                                                                                                                                                                                                                                                                                                                                                                                                 |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03470                       | 380                                                                  | 1.000                           | $\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$                           | $\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$                           | $\mathbf{w}_2 \approx (0.72, 0.03, -0.69)$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03471                       | 381                                                                  | 1.000                           | $\lambda_3 \approx 0.47$                                             | $\lambda_3 \approx 0.47$                                             | value) is thus $\lambda_3 \approx 0.47$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03472                       | 381                                                                  | 1.000                           | $-2, (9 + \sqrt{65}) / 2 \approx 8.53$                               | $-2, (9 + \sqrt{65})/2 \approx 8.53$                                 | which indeed has roots $-2, (9 + \sqrt{65})/2 \approx 8.53$ , and $(9 - \sqrt{65})/2 \approx 0.47$ .                                                                                                                                                                                                                                                                                                                                                                                                       |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03473                       | 381                                                                  | 1.000                           | $(9 - \sqrt{65}) / 2 \approx 0.47$                                   | $(9 - \sqrt{65})/2 \approx 0.47$                                     | which indeed has roots $-2, (9 + \sqrt{65})/2 \approx 8.53$ , and $(9 - \sqrt{65})/2 \approx 0.47$ .                                                                                                                                                                                                                                                                                                                                                                                                       |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03474                       | 381                                                                  | 1.000                           | $6.25, -2.55 \pm 1.88 i$                                             | $6.25, -2.55 \pm 1.88i$                                              | has eigenvalues equal to approximately $6.25, -2.55 \pm 1.88i$ , and $-0.16$ . How-                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03475                       | 381                                                                  | 1.000                           | $-2.55 \pm 1.88 i$                                                   | $-2.55 \pm 1.88i$                                                    | that the next two eigenvalues $-2.55 \pm 1.88i$ are equal to each other in absolute                                                                                                                                                                                                                                                                                                                                                                                                                        |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03476                       | 381                                                                  | 0.985                           | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$          | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$          | <table><tr><th>Positive</th><th>Not positive</th></tr><tr><td><math>\left[\begin{array}{ll} 3 &amp; 1 \\ 2 &amp; 7 \end{array}\right]</math></td><td><math>\left[\begin{array}{ll} 2 &amp; 4 &amp; 1 \\ 8 &amp; 1 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{ll} 1 &amp; -1 \\ -2 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{lll} 4 &amp; 3 &amp; 2 \\ 1 &amp; 0 &amp; 1 \end{array}\right]</math></td></tr></table> | Positive | Not positive                                                 | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ |                     | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$ |   | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |       |   |                      |       |   |                     |       |   |                      |       |
| Positive                                                    | Not positive                                                         |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$  |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$        |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03477                       | 381                                                                  | 0.985                           | $\left[\begin{array}{lll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ | $\left[\begin{array}{lll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ | <table><tr><th>Positive</th><th>Not positive</th></tr><tr><td><math>\left[\begin{array}{ll} 3 &amp; 1 \\ 2 &amp; 7 \end{array}\right]</math></td><td><math>\left[\begin{array}{ll} 2 &amp; 4 &amp; 1 \\ 8 &amp; 1 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{ll} 1 &amp; -1 \\ -2 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{lll} 4 &amp; 3 &amp; 2 \\ 1 &amp; 0 &amp; 1 \end{array}\right]</math></td></tr></table> | Positive | Not positive                                                 | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ |                     | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$ |   | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |       |   |                      |       |   |                     |       |   |                      |       |
| Positive                                                    | Not positive                                                         |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$  |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$        |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03478                       | 381                                                                  | 0.985                           | $\left[\begin{array}{cc} 1 & -1 \\ -2 & 3 \end{array}\right]$        | $\left[\begin{array}{cc} 1 & -1 \\ -2 & 3 \end{array}\right]$        | <table><tr><th>Positive</th><th>Not positive</th></tr><tr><td><math>\left[\begin{array}{ll} 3 &amp; 1 \\ 2 &amp; 7 \end{array}\right]</math></td><td><math>\left[\begin{array}{ll} 2 &amp; 4 &amp; 1 \\ 8 &amp; 1 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{ll} 1 &amp; -1 \\ -2 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{lll} 4 &amp; 3 &amp; 2 \\ 1 &amp; 0 &amp; 1 \end{array}\right]</math></td></tr></table> | Positive | Not positive                                                 | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ |                     | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$ |   | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |       |   |                      |       |   |                     |       |   |                      |       |
| Positive                                                    | Not positive                                                         |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$  |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$        |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03479                       | 381                                                                  | 0.985                           | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ | <table><tr><th>Positive</th><th>Not positive</th></tr><tr><td><math>\left[\begin{array}{ll} 3 &amp; 1 \\ 2 &amp; 7 \end{array}\right]</math></td><td><math>\left[\begin{array}{ll} 2 &amp; 4 &amp; 1 \\ 8 &amp; 1 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{ll} 1 &amp; -1 \\ -2 &amp; 3 \end{array}\right]</math></td></tr><tr><td></td><td><math>\left[\begin{array}{lll} 4 &amp; 3 &amp; 2 \\ 1 &amp; 0 &amp; 1 \end{array}\right]</math></td></tr></table> | Positive | Not positive                                                 | $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$ |                     | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$ |   | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |       |   |                      |       |   |                     |       |   |                      |       |
| Positive                                                    | Not positive                                                         |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| $\left[\begin{array}{ll} 3 & 1 \\ 2 & 7 \end{array}\right]$ | $\left[\begin{array}{ll} 2 & 4 & 1 \\ 8 & 1 & 3 \end{array}\right]$  |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{ll} 1 & -1 \\ -2 & 3 \end{array}\right]$        |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
|                                                             | $\left[\begin{array}{lll} 4 & 3 & 2 \\ 1 & 0 & 1 \end{array}\right]$ |                                 |                                                                      |                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03480                       | 382                                                                  | 1.000                           | $[A \mathbf{v}]_j$                                                   | $[A\mathbf{v}]_j$                                                    | The notation $[A\mathbf{v}]_j$                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03481                       | 382                                                                  | 0.662                           | $\lambda_1 >  \lambda_j $                                            | $\lambda_1 >  \lambda_j $                                            | $\lambda_1 >  \lambda_j $ for all $2 \leq j \leq n$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03482                       | 382                                                                  | 1.000                           | $\lambda_j (2 \leq j \leq n)$                                        | $\lambda_j (2 \leq j \leq n)$                                        | d) Every eigenvector corresponding to an eigenvalue $\lambda_j (2 \leq j \leq n)$                                                                                                                                                                                                                                                                                                                                                                                                                          |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03483                       | 382                                                                  | 1.000                           | $\mathbf{v} \geq \mathbf{w}$                                         | $\mathbf{v} \geq \mathbf{w}$                                         | the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or                                                                                                                                                                                                                                                                                                                                                                             |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03484                       | 382                                                                  | 1.000                           | $\mathbf{v} > \mathbf{w}$                                            | $\mathbf{v} > \mathbf{w}$                                            | the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or                                                                                                                                                                                                                                                                                                                                                                             |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03485                       | 382                                                                  | 1.000                           | $v_j \geq w_j$                                                       | $v_j \geq w_j$                                                       | the proof. For brevity, we write $\mathbf{v} \geq \mathbf{w}$ and $\mathbf{v} > \mathbf{w}$ to indicate that $v_j \geq w_j$ or                                                                                                                                                                                                                                                                                                                                                                             |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |
| johnston-linear-matrix-algebra-F03486                       | 382                                                                  | 0.999                           | $v_j > w_j$                                                          | $v_j > w_j$                                                          | $v_j > w_j$ for all $j$ , respectively (just like we did in Section 2.B). In particular,                                                                                                                                                                                                                                                                                                                                                                                                                   |          |                                                              |                                                             |                                                                     |                     |                                                               |   |                                                                      |       |   |                      |       |   |                     |       |   |                      |       |

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|---------------------------------------|------|-------|-------------------------------------------------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03487 | 382  | 0.874 | $\mathbf{v} \geq \mathbf{0}$                    | $\mathbf{v} \geq \mathbf{0}$                  | this means that non-negativity of $\mathbf{v}$ can be denoted by $\mathbf{v} \geq \mathbf{0}$ , and positivity of $\mathbf{v}$              |
| johnston-linear-matrix-algebra-F03488 | 382  | 1.000 | $\mathbf{v} > \mathbf{0}$                       | $\mathbf{v} > \mathbf{0}$                     | can be denoted by $\mathbf{v} > \mathbf{0}$ . We also let $S^n \subset \mathbb{R}^n$ be the set of unit vectors with                        |
| johnston-linear-matrix-algebra-F03489 | 382  | 1.000 | $S^n \subset \mathbb{R}^n$                      | $S^n \subset \mathbb{R}^n$                    | can be denoted by $\mathbf{v} > \mathbf{0}$ . We also let $S^n \subset \mathbb{R}^n$ be the set of unit vectors with                        |
| johnston-linear-matrix-algebra-F03490 | 382  | 1.000 | $S^n$                                           | $S^n$                                         | <b>Figure 3.28:</b> The set $S^n$ is the portion of the unit circle/sphere/hypersphere that is                                              |
| johnston-linear-matrix-algebra-F03491 | 382  | 1.000 | $\mathbf{v} \in S^n$                            | $\mathbf{v} \in S^n$                          | Note that if $\mathbf{v} \in S^n$ then $A\mathbf{v} > \mathbf{0}$ , since it is computed by multiplying together                            |
| johnston-linear-matrix-algebra-F03492 | 382  | 1.000 | $A \mathbf{v} > \mathbf{0}$                     | $A\mathbf{v} > \mathbf{0}$                    | Note that if $\mathbf{v} \in S^n$ then $A\mathbf{v} > \mathbf{0}$ , since it is computed by multiplying together                            |
| johnston-linear-matrix-algebra-F03493 | 382  | 1.000 | $L: S^n \rightarrow \mathbb{R}$                 | $L: S^n \rightarrow \mathbb{R}$               | positive. We can thus define a function $L: S^n \rightarrow \mathbb{R}$ by                                                                  |
| johnston-linear-matrix-algebra-F03494 | 382  | 1.000 | $c > 0$                                         | $c > 0$                                       | In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for |
| johnston-linear-matrix-algebra-F03495 | 382  | 1.000 | $A \mathbf{v} \geq c \mathbf{v}$                | $A\mathbf{v} \geq c\mathbf{v}$                | In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for |
| johnston-linear-matrix-algebra-F03496 | 382  | 1.000 | $L(\mathbf{v}) > 0$                             | $L(\mathbf{v}) > 0$                           | In particular, there exists a $c > 0$ such that $A\mathbf{v} \geq c\mathbf{v}$ since $A\mathbf{v} > \mathbf{0}$ (so $L(\mathbf{v}) > 0$ for |
| johnston-linear-matrix-algebra-F03497 | 382  | 1.000 | $L(\mathbf{v})$                                 | $L(\mathbf{v})$                               | such $c$ (i.e., the value of $L(\mathbf{v})$ ) can be found by increasing $c$ until one of the                                              |
| johnston-linear-matrix-algebra-F03498 | 383  | 1.000 | $ [B \mathbf{w}]_i $                            | $ [B\mathbf{w}]_i $                           | clear, $ [B\mathbf{w}]_i $ is the                                                                                                           |
| johnston-linear-matrix-algebra-F03499 | 383  | 1.000 | $B \mathbf{w}$                                  | $B\mathbf{w}$                                 | the $i$ -th entry of $B\mathbf{w}$ ,                                                                                                        |
| johnston-linear-matrix-algebra-F03500 | 383  | 0.998 | $[B \mathbf{w}]_i$                              | $[B \mathbf{w}]_i$                            | while $[B \mathbf{w}]_i$ is the                                                                                                             |
| johnston-linear-matrix-algebra-F03501 | 383  | 1.000 | $B \mathbf{w} $                                 | $B \mathbf{w} $                               | $i$ -th entry of $B \mathbf{w} $ .                                                                                                          |
| johnston-linear-matrix-algebra-F03502 | 383  | 1.000 | $\mu \in \mathbb{R}$                            | $\mu \in \mathbb{R}$                          | is, there is a particular number $\mu \in \mathbb{R}$ and a particular vector $\mathbf{v}_* \in S^n$ with the                               |
| johnston-linear-matrix-algebra-F03503 | 383  | 1.000 | $\mathbf{v}_* \in S^n$                          | $\mathbf{v}_* \in S^n$                        | is, there is a particular number $\mu \in \mathbb{R}$ and a particular vector $\mathbf{v}_* \in S^n$ with the                               |
| johnston-linear-matrix-algebra-F03504 | 383  | 1.000 | $L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$      | $L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$    | property that $L(\mathbf{v}_*) = \mu \geq L(\mathbf{v})$ for all $\mathbf{v} \in S^n$ .                                                     |
| johnston-linear-matrix-algebra-F03505 | 383  | 0.998 | $A, \mathbf{v}_*, \mu$                          | $A, \mathbf{v}_*, \mu$                        | between $A, \mathbf{v}_*$ , and $\mu$ that we will use repeatedly.                                                                          |
| johnston-linear-matrix-algebra-F03506 | 383  | 0.998 | $\mu$                                           | $\mu$                                         | between $A, \mathbf{v}_*$ , and $\mu$ that we will use repeatedly.                                                                          |
| johnston-linear-matrix-algebra-F03507 | 383  | 1.000 | $L(\mathbf{v}) = \mu$                           | $L(\mathbf{v}) = \mu$                         | <b>Claim 1:</b> If $\mathbf{v} \in S^n$ and $L(\mathbf{v}) = \mu$ then $A\mathbf{v} = \mu\mathbf{v}$ . To see why this claim holds,         |
| johnston-linear-matrix-algebra-F03508 | 383  | 1.000 | $A \mathbf{v} = \mu \mathbf{v}$                 | $A\mathbf{v} = \mu\mathbf{v}$                 | <b>Claim 1:</b> If $\mathbf{v} \in S^n$ and $L(\mathbf{v}) = \mu$ then $A\mathbf{v} = \mu\mathbf{v}$ . To see why this claim holds,         |
| johnston-linear-matrix-algebra-F03509 | 383  | 1.000 | $A \mathbf{v} \geq \mu \mathbf{v}$              | $A\mathbf{v} \geq \mu\mathbf{v}$              | notice that $L(\mathbf{v}) = \mu$ implies $A\mathbf{v} \geq \mu\mathbf{v}$ and thus $A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$ . Suppose |
| johnston-linear-matrix-algebra-F03510 | 383  | 1.000 | $A \mathbf{v} - \mu \mathbf{v} \geq \mathbf{0}$ | $A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$ | notice that $L(\mathbf{v}) = \mu$ implies $A\mathbf{v} \geq \mu\mathbf{v}$ and thus $A\mathbf{v} - \mu\mathbf{v} \geq \mathbf{0}$ . Suppose |
| johnston-linear-matrix-algebra-F03511 | 383  | 0.985 | $A \mathbf{v} - \mu \mathbf{v} \neq \mathbf{0}$ | $A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$ | (for the purpose of establishing a contradiction) that $A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$ . Since                                |
| johnston-linear-matrix-algebra-F03512 | 383  | 1.000 | $A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$   | $A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$ | $A$ is positive, this implies that $A(A\mathbf{v} - \mu\mathbf{v}) > \mathbf{0}$ , and re-arranging then                                    |
| johnston-linear-matrix-algebra-F03513 | 383  | 1.000 | $A(A\mathbf{v}) > \mu A\mathbf{v}$              | $A(A\mathbf{v}) > \mu A\mathbf{v}$            | gives $A(A\mathbf{v}) > \mu A\mathbf{v}$ . It follows that                                                                                  |
| johnston-linear-matrix-algebra-F03514 | 383  | 1.000 | $\mu \geq L(A\mathbf{v}/\ A\mathbf{v}\ )$       | $\mu \geq L(A\mathbf{v}/\ A\mathbf{v}\ )$     | so $\mu \geq L(A\mathbf{v}/\ A\mathbf{v}\ )$ . It follows that our assumption that $A\mathbf{v} - \mu\mathbf{v} \neq \mathbf{0}$            |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03515 | 383  | 1.000 | $\mathbf{v}^{\wedge{*}}$                                                        | $\mathbf{v}^*$                                                             | eigenvector $\mathbf{v}^*$ . The remaining claims will help us understand this particular                                                                |
| johnston-linear-matrix-algebra-F03516 | 383  | 1.000 | $1 \leq i \leq n$                                                               | $1 \leq i \leq n$                                                          | so for each $1 \leq i \leq n$ , the sum $[\mathbf{A}\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$ has at                                |
| johnston-linear-matrix-algebra-F03517 | 383  | 1.000 | $[\mathbf{A} \mathbf{v}]_i = a_{i,1} v_1 + a_{i,2} v_2 + \cdots + a_{i,n} v_n$  | $[\mathbf{A}\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$ | so for each $1 \leq i \leq n$ , the sum $[\mathbf{A}\mathbf{v}]_i = a_{i,1}v_1 + a_{i,2}v_2 + \cdots + a_{i,n}v_n$ has at                                |
| johnston-linear-matrix-algebra-F03518 | 383  | 1.000 | $\mathbf{v} = (\mathbf{A} \mathbf{v}) / \mu > \mathbf{0}$                       | $\mathbf{v} = (\mathbf{A}\mathbf{v})/\mu > \mathbf{0}$                     | $\mathbf{v} = (\mathbf{A}\mathbf{v})/\mu > \mathbf{0}$ as well.                                                                                          |
| johnston-linear-matrix-algebra-F03519 | 383  | 1.000 | $\mathbf{v}_*$                                                                  | $\mathbf{v}_*$                                                             | Claim 3 above shows that the eigenvector $\mathbf{v}_*$ of $A$ is positive. Since we want the                                                            |
| johnston-linear-matrix-algebra-F03520 | 383  | 1.000 | $\mu = \lambda_1$                                                               | $\mu = \lambda_1$                                                          | we must prove that $\mu = \lambda_1$ . The fourth and final claim demonstrates this fact:                                                                |
| johnston-linear-matrix-algebra-F03521 | 383  | 1.000 | $B \in \mathcal{M}_n(\mathbb{R})$                                               | $B \in \mathcal{M}_n(\mathbb{R})$                                          | <b>Claim 4:</b> If $B \in \mathcal{M}_n(\mathbb{R})$ is such that $O \leq B \leq A$ and $\lambda$ is an eigenvalue of $B$                                |
| johnston-linear-matrix-algebra-F03522 | 383  | 1.000 | $O \leq B \leq A$                                                               | $O \leq B \leq A$                                                          | <b>Claim 4:</b> If $B \in \mathcal{M}_n(\mathbb{R})$ is such that $O \leq B \leq A$ and $\lambda$ is an eigenvalue of $B$                                |
| johnston-linear-matrix-algebra-F03523 | 383  | 1.000 | $ \lambda  \leq \mu$                                                            | $ \lambda  \leq \mu$                                                       | then $ \lambda  \leq \mu$ . To see why this property holds, notice that if $\mathbf{w}$ is a unit                                                        |
| johnston-linear-matrix-algebra-F03524 | 383  | 0.997 | $ \mathbf{w} $                                                                  | $ \mathbf{w} $                                                             | fact that $A$ is positive. If we let $ \mathbf{w} $ denote the vector whose entries are                                                                  |
| johnston-linear-matrix-algebra-F03525 | 383  | 0.996 | $ \mathbf{w}  = (\left w_1\right , \left w_2\right , \ldots, \left w_n\right )$ | $ \mathbf{w}  = ( w_1 ,  w_2 , \dots,  w_n )$                              | the absolute values of the entries of $\mathbf{w}$ (i.e., $ \mathbf{w}  = ( w_1 ,  w_2 , \dots,  w_n )$ )                                                |
| johnston-linear-matrix-algebra-F03526 | 383  | 1.000 | $B \mathbf{w}  \geq  \lambda  \mathbf{w} $                                      | $B \mathbf{w}  \geq  \lambda  \mathbf{w} $                                 | then the above inequalities implies $B \mathbf{w}  \geq  \lambda  \mathbf{w} $ . However, since $A - B$                                                  |
| johnston-linear-matrix-algebra-F03527 | 383  | 1.000 | $(A - B) \mathbf{w}  \geq \mathbf{0}$                                           | $(A - B) \mathbf{w}  \geq \mathbf{0}$                                      | is entry-wise non-negative, it is also the case that $(A - B) \mathbf{w}  \geq \mathbf{0}$ , so                                                          |
| johnston-linear-matrix-algebra-F03528 | 383  | 1.000 | $A \mathbf{w}  \geq B \mathbf{w}  \geq  \lambda  \mathbf{w} $                   | $A \mathbf{w}  \geq B \mathbf{w}  \geq  \lambda  \mathbf{w} $              | $A \mathbf{w}  \geq B \mathbf{w}  \geq  \lambda  \mathbf{w} $ . It follows that $L( \mathbf{w} ) \geq  \lambda $ , but since $\mu \geq L( \mathbf{w} )$  |
| johnston-linear-matrix-algebra-F03529 | 383  | 1.000 | $L( \mathbf{w} ) \geq  \lambda $                                                | $L( \mathbf{w} ) \geq  \lambda $                                           | $A \mathbf{w}  \geq B \mathbf{w}  \geq  \lambda  \mathbf{w} $ . It follows that $L( \mathbf{w} ) \geq  \lambda $ , but since $\mu \geq L( \mathbf{w} )$  |
| johnston-linear-matrix-algebra-F03530 | 383  | 1.000 | $\mu \geq L( \mathbf{w} )$                                                      | $\mu \geq L( \mathbf{w} )$                                                 | $A \mathbf{w}  \geq B \mathbf{w}  \geq  \lambda  \mathbf{w} $ . It follows that $L( \mathbf{w} ) \geq  \lambda $ , but since $\mu \geq L( \mathbf{w} )$  |
| johnston-linear-matrix-algebra-F03531 | 383  | 1.000 | $\mu \geq  \lambda $                                                            | $\mu \geq  \lambda $                                                       | we then have $\mu \geq  \lambda $ .                                                                                                                      |
| johnston-linear-matrix-algebra-F03532 | 384  | 1.000 | $\mu \leq \lambda$                                                              | $\mu \leq \lambda$                                                         | If $\mu \leq \lambda$ then we can                                                                                                                        |
| johnston-linear-matrix-algebra-F03533 | 384  | 1.000 | $\mu \geq \lambda$                                                              | $\mu \geq \lambda$                                                         | arbitrarily. If $\mu \geq \lambda$                                                                                                                       |
| johnston-linear-matrix-algebra-F03534 | 384  | 1.000 | $\mathbf{v}_1 > c\mathbf{v}$                                                    | $\mathbf{v}_1 > c\mathbf{v}$                                               | that $\mathbf{v}_1 > c\mathbf{v}$ .                                                                                                                      |
| johnston-linear-matrix-algebra-F03535 | 384  | 1.000 | $B = A$                                                                         | $B = A$                                                                    | In particular, if we choose $B = A$ in Claim 4 then we see that $\mu$ is indeed                                                                          |
| johnston-linear-matrix-algebra-F03536 | 384  | 1.000 | $\lambda = \mu$                                                                 | $\lambda = \mu$                                                            | goal is to show that this implies $\lambda = \mu$ ). Then $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ , and since $\mathbf{A}\mathbf{v} > \mathbf{0}$ (by |
| johnston-linear-matrix-algebra-F03537 | 384  | 1.000 | $\lambda > 0$                                                                   | $\lambda > 0$                                                              | Claim 2), we conclude that $\lambda > 0$ . Then there exists some real scalar $c \neq 0$                                                                 |
| johnston-linear-matrix-algebra-F03538 | 384  | 1.000 | $c\mu \geq c\lambda$                                                            | $c\mu \geq c\lambda$                                                       | such that (i) $c\mu \geq c\lambda$ , and (ii) $\mathbf{v}_1 > c\mathbf{v}$ , so                                                                          |
| johnston-linear-matrix-algebra-F03539 | 384  | 1.000 | $A(\mathbf{v}_1 - c\mathbf{v}) =$                                               | $A(\mathbf{v}_1 - c\mathbf{v}) =$                                          | ity (3.B.3) is actually an equality. Claim 1 then tells us that $A(\mathbf{v}_1 - c\mathbf{v}) =$                                                        |
| johnston-linear-matrix-algebra-F03540 | 384  | 1.000 | $\mu(\mathbf{v}_1 - c\mathbf{v})$                                               | $\mu(\mathbf{v}_1 - c\mathbf{v})$                                          | $\mu(\mathbf{v}_1 - c\mathbf{v})$ , so $\mathbf{v}_1 - c\mathbf{v}$ is an eigenvector of $A$ corresponding to the eigenvalue                             |
| johnston-linear-matrix-algebra-F03541 | 384  | 1.000 | $\mathbf{v}_1 - c\mathbf{v}$                                                    | $\mathbf{v}_1 - c\mathbf{v}$                                               | $\mu(\mathbf{v}_1 - c\mathbf{v})$ , so $\mathbf{v}_1 - c\mathbf{v}$ is an eigenvector of $A$ corresponding to the eigenvalue                             |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F03542 | 384  | 1.000 | $ \lambda  = \mu$                                                                     | $ \lambda  = \mu$                                                                     | goal is to show that if $ \lambda  = \mu$ then $\lambda = \mu$ .                                 |
| johnston-linear-matrix-algebra-F03543 | 384  | 1.000 | $b_{i,j} w_j$                                                                         | $b_{i,j} w_j$                                                                         | the terms $b_{i,j} w_j$ are non-negative real multiples of each other. Since $b_{i,j} > 0$       |
| johnston-linear-matrix-algebra-F03544 | 384  | 1.000 | $b_{i,j} > 0$                                                                         | $b_{i,j} > 0$                                                                         | the terms $b_{i,j} w_j$ are non-negative real multiples of each other. Since $b_{i,j} > 0$       |
| johnston-linear-matrix-algebra-F03545 | 385  | 1.000 | $\mathbf{v}_5 = (0.58, 0.58, 0.58)$                                                   | $\mathbf{v}_5 = (0.58, 0.58, 0.58)$                                                   | $\mathbf{v}_5 = (0.58, 0.58, 0.58)$ .                                                            |
| johnston-linear-matrix-algebra-F03546 | 385  | 1.000 | $\lambda = 7$                                                                         | $\lambda = 7$                                                                         | The calculation above suggests that $\lambda = 7$ is the dominant eigenvalue                     |
| johnston-linear-matrix-algebra-F03547 | 385  | 1.000 | $\lambda \mathbf{v}$                                                                  | $\lambda \mathbf{v}$                                                                  | is much easier to just compute $A\mathbf{v}$ and see that it equals $\lambda \mathbf{v}$ :       |
| johnston-linear-matrix-algebra-F03548 | 385  | 1.000 | $n \approx 5$                                                                         | $n \approx 5$                                                                         | web pages to be ranked ( $n \approx 5$ billion at the time of this writing) and we let $r_i$     |
| johnston-linear-matrix-algebra-F03549 | 385  | 1.000 | $r_i$                                                                                 | $r_i$                                                                                 | web pages to be ranked ( $n \approx 5$ billion at the time of this writing) and we let $r_i$     |
| johnston-linear-matrix-algebra-F03550 | 385  | 1.000 | $i, p_j$                                                                              | $i, p_j$                                                                              | denote the rank of page $i$ , $p_j$ denote the number of pages that page $j$ links to,           |
| johnston-linear-matrix-algebra-F03551 | 386  | 0.915 | $\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}$                                      | A, B, C, D                                                                            | Suppose that 5 web pages, which we call A, B, C, D, and E, link to each                          |
| johnston-linear-matrix-algebra-F03552 | 386  | 1.000 | $r_A, r_B, r_C, r_D$                                                                  | $r_A, r_B, r_C, r_D$                                                                  | system (3.B.4) for determining the ranks $r_A, r_B, r_C, r_D$ , and $r_E$ of these               |
| johnston-linear-matrix-algebra-F03553 | 386  | 1.000 | $r_E$                                                                                 | $r_E$                                                                                 | system (3.B.4) for determining the ranks $r_A, r_B, r_C, r_D$ , and $r_E$ of these               |
| johnston-linear-matrix-algebra-F03554 | 386  | 0.999 | $(r_A, r_B, r_C, r_D, r_E) =$                                                         | $(r_A, r_B, r_C, r_D, r_E) =$                                                         | tions of this linear system are exactly the multiples of $(r_A, r_B, r_C, r_D, r_E) =$           |
| johnston-linear-matrix-algebra-F03555 | 386  | 1.000 | $r_A > r_B = r_C > r_E > r_D$                                                         | $r_A > r_B = r_C > r_E > r_D$                                                         | ular, we have $r_A > r_B = r_C > r_E > r_D$ as the resulting ranking of how                      |
| johnston-linear-matrix-algebra-F03556 | 386  | 1.000 | $1 / p_j$                                                                             | $1/p_j$                                                                               | $1/p_j$ , where $p_j$ is the total number of edges coming out of vertex $j$ . For                |
| johnston-linear-matrix-algebra-F03557 | 386  | 1.000 | $p_j$                                                                                 | $p_j$                                                                                 | $1/p_j$ , where $p_j$ is the total number of edges coming out of vertex $j$ . For                |
| johnston-linear-matrix-algebra-F03558 | 387  | 1.000 | $A \mathbf{r} = \mathbf{r}$                                                           | $A\mathbf{r} = \mathbf{r}$                                                            | pages then simply has the form $A\mathbf{r} = \mathbf{r}$ , where $A$ is the adjacency matrix    |
| johnston-linear-matrix-algebra-F03559 | 387  | 1.000 | $\mathbf{r} = (r_A, r_B, r_C, r_D, r_E)$                                              | $\mathbf{r} = (r_A, r_B, r_C, r_D, r_E)$                                              | It is straightforward to verify that if $\mathbf{r} = (r_A, r_B, r_C, r_D, r_E)$ then the linear |
| johnston-linear-matrix-algebra-F03560 | 388  | 1.000 | $\lambda_2 = -0.77$                                                                   | $\lambda_2 = -0.77$                                                                   | $\lambda_2 = -0.77$ that is                                                                      |
| johnston-linear-matrix-algebra-F03561 | 388  | 0.562 | $\lambda_1 = 1 \mathrm{in}$                                                           | $\lambda_1 = 1$ in                                                                    | eigenvalue $\lambda_1 = 1$ in                                                                    |
| johnston-linear-matrix-algebra-F03562 | 388  | 0.998 | $(-1 \pm i \sqrt{3}) / 2$                                                             | $(-1 \pm i\sqrt{3})/2$                                                                | has eigenvalues 1 and $(-1 \pm i\sqrt{3})/2$ , each of which have absolute value 1.              |
| johnston-linear-matrix-algebra-F03563 | 388  | 1.000 | $A + \varepsilon J$                                                                   | $A + \varepsilon J$                                                                   | it is typically applied to $A + \varepsilon J$ instead, where $J$ is the matrix of appropriate   |
| johnston-linear-matrix-algebra-F03564 | 388  | 1.000 | $J$                                                                                   | $J$                                                                                   | it is typically applied to $A + \varepsilon J$ instead, where $J$ is the matrix of appropriate   |
| johnston-linear-matrix-algebra-F03565 | 388  | 1.000 | $\varepsilon$                                                                         | $\varepsilon$                                                                         | size with every entry equal to 1 and $\varepsilon$ is some very small (but positive) real        |
| johnston-linear-matrix-algebra-F03566 | 389  | 1.000 | $\left[ \begin{array}{cc} 3 & -1 \\ -4 & 5 \end{array} \right]$                       | $\left[ \begin{array}{cc} 3 & -1 \\ -4 & 5 \end{array} \right]$                       | (b) $\left[ \begin{array}{cc} 3 & -1 \\ -4 & 5 \end{array} \right]$                              |
| johnston-linear-matrix-algebra-F03567 | 389  | 0.877 | $\left[ \begin{array}{ccc} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{array} \right]$ | $\left[ \begin{array}{ccc} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{array} \right]$ | *(c) $\left[ \begin{array}{ccc} -2 & 1 & 1 \\ 4 & 1 & 2 \\ 1 & 0 & -2 \end{array} \right]$       |

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| johnston-linear-matrix-algebra-F03568 | 389  | 1.000 | $\left[\begin{array}{ccc}3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2\end{array}\right]$                             | $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2 \end{bmatrix}$                            | (d) $\begin{bmatrix} 3 & -1 & 2 \\ 0 & 4 & 1 \\ -4 & 1 & 2 \end{bmatrix}$                                  |
| johnston-linear-matrix-algebra-F03569 | 389  | 0.497 | $\left[\begin{array}{llll}3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2\end{array}\right]$ | $\begin{bmatrix} 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}$ | *(e) $\begin{bmatrix} 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}$      |
| johnston-linear-matrix-algebra-F03570 | 389  | 0.999 | $\left[\begin{array}{llll}0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4\end{array}\right]$ | $\begin{bmatrix} 0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4 \end{bmatrix}$ | (f) $\begin{bmatrix} 0 & 5 & 5 & 2 \\ 5 & 4 & 7 & 2 \\ 5 & 4 & 0 & 6 \\ 7 & 5 & 5 & 4 \end{bmatrix}$       |
| johnston-linear-matrix-algebra-F03571 | 389  | 1.000 | $\left[\begin{array}{cc}-2 & 4 \\ 5 & 5\end{array}\right]$                                                    | $\begin{bmatrix} -2 & 4 \\ 5 & 5 \end{bmatrix}$                                                  | (b) $\begin{bmatrix} -2 & 4 \\ 5 & 5 \end{bmatrix}$                                                        |
| johnston-linear-matrix-algebra-F03572 | 389  | 1.000 | $\left[\begin{array}{ccc}-2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0\end{array}\right]$                             | $\begin{bmatrix} -2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0 \end{bmatrix}$                            | (d) $\begin{bmatrix} -2 & 3 & -1 \\ 5 & 2 & 3 \\ 3 & 3 & 0 \end{bmatrix}$                                  |
| johnston-linear-matrix-algebra-F03573 | 389  | 0.990 | $A \in \mathcal{M}_3(\mathbb{R})$                                                                             | $A \in \mathcal{M}_3(\mathbb{R})$                                                                | (b) If $A \in \mathcal{M}_3(\mathbb{R})$ has eigenvalues 3, 2, and 1, then                                 |
| johnston-linear-matrix-algebra-F03574 | 389  | 1.000 | $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$                                                                    | $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$                                                       | $A \in \mathcal{M}_n(\mathbb{R})$ then the sequence of vectors $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2,$ |
| johnston-linear-matrix-algebra-F03575 | 389  | 1.000 | $\mathbf{v}_0 = \mathbf{e}_2$                                                                                 | $\mathbf{v}_0 = \mathbf{e}_2$                                                                    | with $\mathbf{v}_0 = \mathbf{e}_2$ instead of $\mathbf{v}_0 = \mathbf{e}_1$ .                              |
| johnston-linear-matrix-algebra-F03576 | 389  | 0.415 | $**3$                                                                                                         | $**3$                                                                                            | **3.B.6 Suppose $A \in \mathcal{M}_n$ satisfies the hypotheses of The-                                     |
| johnston-linear-matrix-algebra-F03577 | 389  | 1.000 | $A \in \mathcal{M}_2(\mathbb{R})$                                                                             | $A \in \mathcal{M}_2(\mathbb{R})$                                                                | 3.B.7 Show that if $A \in \mathcal{M}_2(\mathbb{R})$ is positive then all of its                           |
| johnston-linear-matrix-algebra-F03578 | 390  | 0.495 | $\left[\begin{array}{ll}0 & 1 \\ 1 & 1\end{array}\right]$                                                     | $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$                                                   | *(a) $\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$                                                        |
| johnston-linear-matrix-algebra-F03579 | 390  | 1.000 | $\left[\begin{array}{cc}2 & 1 \\ 1 & -1\end{array}\right]$                                                    | $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$                                                  | (b) $\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$                                                        |
| johnston-linear-matrix-algebra-F03580 | 390  | 0.534 | $\left[\begin{array}{cc}i & i \\ i & i\end{array}\right]$                                                     | $\begin{bmatrix} i & i \\ i & i \end{bmatrix}$                                                   | *(c) $\begin{bmatrix} i & i \\ i & i \end{bmatrix}$                                                        |
| johnston-linear-matrix-algebra-F03581 | 390  | 0.798 | $\left[\begin{array}{ll}\sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i\end{array}\right]$                 | $\begin{bmatrix} \sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i \end{bmatrix}$               | (d) $\begin{bmatrix} \sqrt{5}+i & \sqrt{5}-i \\ \sqrt{5}-i & \sqrt{5}+i \end{bmatrix}$                     |
| johnston-linear-matrix-algebra-F03582 | 390  | 0.809 | $*(\mathbf{e})\left[\begin{array}{lll}0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0\end{array}\right]$                  | *(e) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$                         | *(e) $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$                                   |
| johnston-linear-matrix-algebra-F03583 | 390  | 1.000 | $\left[\begin{array}{ccc}0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1\end{array}\right]$                              | $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$                             | (f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$                                   |
| johnston-linear-matrix-algebra-F03584 | 390  | 1.000 | $\left[A^k\right]_{i,j}>0$                                                                                    | $[A^k]_{i,j} > 0$                                                                                | $k$ (that may depend on $i$ and $j$ ) such that $[A^k]_{i,j} > 0$ .                                        |
| johnston-linear-matrix-algebra-F03585 | 390  | 0.972 | $\left[\begin{array}{cc}-1 & 1 \\ 1 & 1\end{array}\right]$                                                    | $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$                                                  | *(a) $\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$                                                       |

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| johnston-linear-matrix-algebra-F03586 | 390  | 1.000 | $\left[\begin{array}{ll}1 & 1 \\ 0 & 1\end{array}\right]$                         | $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$                        | (b) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$                                                                                                       |
| johnston-linear-matrix-algebra-F03587 | 390  | 0.817 | $\left[\begin{array}{ll}0 & 1 \\ 2 & 3\end{array}\right]$                         | $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$                        | *(c) $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$                                                                                                      |
| johnston-linear-matrix-algebra-F03588 | 390  | 1.000 | $\left[\begin{array}{cc}-2 & 1 \\ 1 & 3\end{array}\right]$                        | $\begin{bmatrix} -2 & 1 \\ 1 & 3 \end{bmatrix}$                       | (d) $\begin{bmatrix} -2 & 1 \\ 1 & 3 \end{bmatrix}$                                                                                                      |
| johnston-linear-matrix-algebra-F03589 | 390  | 1.000 | $\left[\begin{array}{ccc}1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1\end{array}\right]$ | $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ | $\begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$                                                                                    |
| johnston-linear-matrix-algebra-F03590 | 390  | 0.997 | $\left[\begin{array}{lll}0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0\end{array}\right]$   | $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$   | (f) $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$                                                                                  |
| johnston-linear-matrix-algebra-F03591 | 390  | 1.000 | $A=\left[\begin{array}{cc}-3 & -2 \\ 4 & 1\end{array}\right]$                     | $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$                  | Compute bases of the eigenspaces of the matrix $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$ .                                                    |
| johnston-linear-matrix-algebra-F03592 | 391  | 1.000 | $4+(1-i)(-2-2i)=$                                                                 | $4 + (1 - i)(-2 - 2i) =$                                              | $4 + (1 - i)(-2 - 2i) =$                                                                                                                                 |
| johnston-linear-matrix-algebra-F03593 | 391  | 1.000 | $4+(-2+2i-2i-2)=$                                                                 | $4 + (-2 + 2i - 2i - 2) =$                                            | $4 + (-2 + 2i - 2i - 2) =$                                                                                                                               |
| johnston-linear-matrix-algebra-F03594 | 391  | 0.935 | $4-4=0$                                                                           | $4 - 4 = 0$                                                           | $4 - 4 = 0.$                                                                                                                                             |
| johnston-linear-matrix-algebra-F03595 | 391  | 1.000 | $\bar{B}=B$                                                                       | $B = B$                                                               | real entries, so $\overline{B} = B$ .                                                                                                                    |
| johnston-linear-matrix-algebra-F03596 | 391  | 1.000 | $\lambda_{\pm}=-1\pm 2i$                                                          | $\lambda_{\pm} = -1 \pm 2i$                                           | We showed in Example 3.3.8 that $A$ has eigenvalues $\lambda_{\pm} = -1 \pm 2i$ .                                                                        |
| johnston-linear-matrix-algebra-F03597 | 391  | 1.000 | $\mathbf{v}_{\pm}$                                                                | $\mathbf{v}_{\pm}$                                                    | We thus want to find the non-zero vectors $\mathbf{v}_{\pm}$ satisfying the linear systems                                                               |
| johnston-linear-matrix-algebra-F03598 | 391  | 0.998 | $(A-(-1\pm 2i)I)\mathbf{v}_{\pm}=\mathbf{0}$                                      | $(A - (-1 \pm 2i)I)\mathbf{v}_{\pm} = \mathbf{0}$                     | $(A - (-1 \pm 2i)I)\mathbf{v}_{\pm} = \mathbf{0}$ . We start by solving the “+” system:                                                                  |
| johnston-linear-matrix-algebra-F03599 | 391  | 0.999 | $\lambda_{+}=-1+2i$                                                               | $\lambda_{+} = -1 + 2i$                                               | It follows that one basis of the eigenspace corresponding to $\lambda_{+} = -1 + 2i$                                                                     |
| johnston-linear-matrix-algebra-F03600 | 391  | 1.000 | $\mathbf{v}_{+}=(1,-1-i)$                                                         | $\mathbf{v}_{+} = (1, -1 - i)$                                        | consists of the single vector $\mathbf{v}_{+} = (1, -1 - i)$ . A similar calculation shows                                                               |
| johnston-linear-matrix-algebra-F03601 | 391  | 1.000 | $\lambda_{-}=-1-2i$                                                               | $\lambda_{-} = -1 - 2i$                                               | that one basis of the eigenspace corresponding to $\lambda_{-} = -1 - 2i$ consists                                                                       |
| johnston-linear-matrix-algebra-F03602 | 391  | 1.000 | $\mathbf{v}_{-}=(1,-1+i)$                                                         | $\mathbf{v}_{-} = (1, -1 + i)$                                        | of the vector $\mathbf{v}_{-} = (1, -1 + i)$ .                                                                                                           |
| johnston-linear-matrix-algebra-F03603 | 391  | 1.000 | $\overline{-1+2i}=$                                                               | $\overline{-1 + 2i} =$                                                | this case, its eigenvalues are complex conjugates of each other, since $\overline{-1 + 2i} =$                                                            |
| johnston-linear-matrix-algebra-F03604 | 391  | 1.000 | $-1-2i$                                                                           | $-1 - 2i$                                                             | $-1 - 2i$ , and its eigenvectors are also complex conjugates of each other, since                                                                        |
| johnston-linear-matrix-algebra-F03605 | 391  | 1.000 | $\overline{(1,-1-i)}=(1,-1+i)$                                                    | $\overline{(1, -1 - i)} = (1, -1 + i)$                                | $\overline{(1, -1 - i)} = (1, -1 + i)$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F03606 | 391  | 1.000 | $B\mathbf{v}=\lambda\mathbf{v}$                                                   | $B\mathbf{v} = \lambda\mathbf{v}$                                     | is an eigenvector of $B$ with corresponding eigenvalue $\lambda$ . Then $B\mathbf{v} = \lambda\mathbf{v}$ , which                                        |
| johnston-linear-matrix-algebra-F03607 | 391  | 1.000 | $B\overline{\mathbf{v}}$                                                          | $B\overline{\mathbf{v}}$                                              | we can use to compute $B\overline{\mathbf{v}}$ as follows:                                                                                               |
| johnston-linear-matrix-algebra-F03608 | 391  | 1.000 | $B\overline{\mathbf{v}}=\bar{\lambda}\overline{\mathbf{v}}$                       | $B\overline{\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}$         | Since $B\overline{\mathbf{v}} = \bar{\lambda}\overline{\mathbf{v}}$ , we conclude that $\overline{\mathbf{v}}$ is an eigenvector of $B$ with correspond- |
| johnston-linear-matrix-algebra-F03609 | 391  | 0.735 | $\mathbf{2} \times \mathbf{2}$                                                    | $\mathbf{2} \times \mathbf{2}$                                        | <b>3.C.1 Geometric Interpretation for <math>2 \times 2</math> Matrices</b>                                                                               |

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| johnston-linear-matrix-algebra-F03610 | 391  | 1.000 | $\lambda=r e^{i \theta}$                                                | $\lambda = r e^{i \theta}$                                                | $\lambda = r e^{i \theta}$ , which means that $\lambda$ is a distance of $r$ from the origin in the complex                                                                                                |
| johnston-linear-matrix-algebra-F03611 | 391  | 1.000 | $r=\sqrt{a^2+b^2}$                                                      | $r = \sqrt{a^2 + b^2}$                                                    | particular, $r$ and $\theta$ can be computed via $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$ if                                                                                                     |
| johnston-linear-matrix-algebra-F03612 | 391  | 1.000 | $\theta=\arctan (b / a)$                                                | $\theta = \arctan(b/a)$                                                   | particular, $r$ and $\theta$ can be computed via $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$ if                                                                                                     |
| johnston-linear-matrix-algebra-F03613 | 391  | 1.000 | $a>0$                                                                   | $a > 0$                                                                   | $a > 0$ (if $a \leq 0$ then we have to be more careful when computing $\theta$ to make                                                                                                                     |
| johnston-linear-matrix-algebra-F03614 | 391  | 1.000 | $a \leq 0$                                                              | $a \leq 0$                                                                | $a > 0$ (if $a \leq 0$ then we have to be more careful when computing $\theta$ to make                                                                                                                     |
| johnston-linear-matrix-algebra-F03615 | 391  | 1.000 | $e^{i \theta}=\cos (\theta)+i \sin (\theta)$                            | $e^{i \theta} = \cos(\theta) + i \sin(\theta)$                            | $e^{i \theta} = \cos(\theta) + i \sin(\theta)$                                                                                                                                                             |
| johnston-linear-matrix-algebra-F03616 | 391  | 1.000 | $\operatorname{Re}(\mathbf{v})$                                         | $\operatorname{Re}(\mathbf{v})$                                           | Second, we use $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ to denote the real and imaginary parts of                                                                               |
| johnston-linear-matrix-algebra-F03617 | 391  | 1.000 | $\operatorname{Im}(\mathbf{v})$                                         | $\operatorname{Im}(\mathbf{v})$                                           | Second, we use $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ to denote the real and imaginary parts of                                                                               |
| johnston-linear-matrix-algebra-F03618 | 391  | 1.000 | $\mathbf{v}=\mathbf{x}+i \mathbf{y}$                                    | $\mathbf{v} = \mathbf{x} + i \mathbf{y}$                                  | a vector, respectively. That is, if $\mathbf{v} = \mathbf{x} + i \mathbf{y}$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ then $\operatorname{Re}(\mathbf{v}) = \mathbf{x}$                          |
| johnston-linear-matrix-algebra-F03619 | 391  | 1.000 | $\operatorname{Re}(\mathbf{v})=\mathbf{x}$                              | $\operatorname{Re}(\mathbf{v}) = \mathbf{x}$                              | a vector, respectively. That is, if $\mathbf{v} = \mathbf{x} + i \mathbf{y}$ for some $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ then $\operatorname{Re}(\mathbf{v}) = \mathbf{x}$                          |
| johnston-linear-matrix-algebra-F03620 | 391  | 1.000 | $\operatorname{Im}(\mathbf{v})=\mathbf{y}$                              | $\operatorname{Im}(\mathbf{v}) = \mathbf{y}$                              | and $\operatorname{Im}(\mathbf{v}) = \mathbf{y}$ . Also, by recalling that the complex conjugate of $\mathbf{v}$ is $\bar{\mathbf{v}} =$                                                                   |
| johnston-linear-matrix-algebra-F03621 | 391  | 1.000 | $\overline{\mathbf{v}}=$                                                | $\bar{\mathbf{v}} =$                                                      | and $\operatorname{Im}(\mathbf{v}) = \mathbf{y}$ . Also, by recalling that the complex conjugate of $\mathbf{v}$ is $\bar{\mathbf{v}} =$                                                                   |
| johnston-linear-matrix-algebra-F03622 | 391  | 1.000 | $\operatorname{Re}(\mathbf{v})-i \operatorname{Im}(\mathbf{v})$         | $\operatorname{Re}(\mathbf{v}) - i \operatorname{Im}(\mathbf{v})$         | $\operatorname{Re}(\mathbf{v}) - i \operatorname{Im}(\mathbf{v})$ , we can solve for $\operatorname{Re}(\mathbf{v})$ and $\operatorname{Im}(\mathbf{v})$ in terms of $\mathbf{v}$ and $\bar{\mathbf{v}}$ : |
| johnston-linear-matrix-algebra-F03623 | 392  | 1.000 | $a+i b=r e^{i \theta}$                                                  | $a + i b = r e^{i \theta}$                                                | <b>Figure 3.29:</b> Every complex number $a + i b$ can be written in the form $a + i b = r e^{i \theta}$ ,                                                                                                 |
| johnston-linear-matrix-algebra-F03624 | 392  | 1.000 | $[\mathbf{x}]_B$                                                        | $[\mathbf{x}]_B$                                                          | Third, recall from Section 3.1 that if $B$ is a basis of a subspace then $[\mathbf{x}]_B$                                                                                                                  |
| johnston-linear-matrix-algebra-F03625 | 392  | 0.712 | $B=\{\mathbf{v}, \mathbf{w}\}$                                          | $B = \{\mathbf{v}, \mathbf{w}\}$                                          | if $B = \{\mathbf{v}, \mathbf{w}\}$ is a basis of a 2-dimensional subspace (i.e., a plane) and $\mathbf{x} =$                                                                                              |
| johnston-linear-matrix-algebra-F03626 | 392  | 1.000 | $c \mathbf{v}+d \mathbf{w}$                                             | $c \mathbf{v} + d \mathbf{w}$                                             | $c \mathbf{v} + d \mathbf{w}$ , then $[\mathbf{x}]_B = (c, d)$ .                                                                                                                                           |
| johnston-linear-matrix-algebra-F03627 | 392  | 1.000 | $[\mathbf{x}]_B=(c, d)$                                                 | $[\mathbf{x}]_B = (c, d)$                                                 | $c \mathbf{v} + d \mathbf{w}$ , then $[\mathbf{x}]_B = (c, d)$ .                                                                                                                                           |
| johnston-linear-matrix-algebra-F03628 | 392  | 1.000 | $\pi$                                                                   | $\pi$                                                                     | an integer multiple of $\pi$ , then the upcoming theorem can be thought of as the                                                                                                                          |
| johnston-linear-matrix-algebra-F03629 | 392  | 0.999 | $r e^{i \theta} \in \mathbb{C}$                                         | $r e^{i \theta} \in \mathbb{C}$                                           | Suppose $A \in \mathcal{M}_2(\mathbb{R})$ has a complex (non-real) eigenvalue $r e^{i \theta} \in \mathbb{C}$ with                                                                                         |
| johnston-linear-matrix-algebra-F03630 | 392  | 1.000 | $\mathbf{v} \in \mathbb{C}^2$                                           | $\mathbf{v} \in \mathbb{C}^2$                                             | corresponding eigenvector $\mathbf{v} \in \mathbb{C}^2$ . If we let $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ then                                                         |
| johnston-linear-matrix-algebra-F03631 | 392  | 1.000 | $Q=[\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ | $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ | corresponding eigenvector $\mathbf{v} \in \mathbb{C}^2$ . If we let $Q = [\operatorname{Re}(\mathbf{v}) \mid -\operatorname{Im}(\mathbf{v})]$ then                                                         |
| johnston-linear-matrix-algebra-F03632 | 392  | 0.999 | $B=\{\operatorname{Re}(\mathbf{v}),-\operatorname{Im}(\mathbf{v})\}$    | $B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\}$   | $B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\}$ .                                                                                                                                  |
| johnston-linear-matrix-algebra-F03633 | 392  | 1.000 | $\lambda=-1+2 i$                                                        | $\lambda = -1 + 2 i =$                                                    | from Example 3.3.8, we recall that one of its eigenvalues is $\lambda = -1 + 2 i =$                                                                                                                        |
| johnston-linear-matrix-algebra-F03634 | 392  | 1.000 | $\sqrt{5} e^{i \theta}$                                                 | $\sqrt{5} e^{i \theta}$                                                   | $\sqrt{5} e^{i \theta}$ , where $\theta \approx \pm(0.6476) \pi$ . In Example 3.C.1, we found that one cor-                                                                                                |
| johnston-linear-matrix-algebra-F03635 | 392  | 1.000 | $\theta \approx \pm(0.6476) \pi$                                        | $\theta \approx \pm(0.6476) \pi$                                          | $\sqrt{5} e^{i \theta}$ , where $\theta \approx \pm(0.6476) \pi$ . In Example 3.C.1, we found that one cor-                                                                                                |
| johnston-linear-matrix-algebra-F03636 | 392  | 1.000 | $\mathbf{v}=(1,-1-i)$                                                   | $\mathbf{v} = (1, -1 - i)$                                                | responding eigenvector is $\mathbf{v} = (1, -1 - i)$ . It follows that this matrix can be                                                                                                                  |

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| johnston-linear-matrix-algebra-F03637 | 393  | ■ 1.000 | $r=\sqrt{5}$                                                                                                  | $r = \sqrt{5}$                                                                                | a factor of $r = \sqrt{5}$ ,                                                                                                                 |
| johnston-linear-matrix-algebra-F03638 | 393  | ■ 1.000 | $B=\{\operatorname{Re}(\mathbf{v}),-\operatorname{Im}(\mathbf{v})\}=\{(1,-1),(0,1)\}$                         | $B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$ | viewed as rotating $\mathbb{R}^2$ in the basis $B = \{\operatorname{Re}(\mathbf{v}), -\operatorname{Im}(\mathbf{v})\} = \{(1, -1), (0, 1)\}$ |
| johnston-linear-matrix-algebra-F03639 | 393  | ■ 1.000 | $\theta \approx$                                                                                              | $\theta \approx$                                                                              | counter-clockwise by approximately 0.6476 half-rotations (i.e., an angle of $\theta \approx$                                                 |
| johnston-linear-matrix-algebra-F03640 | 393  | ■ 0.936 | $(0.6476) \pi$                                                                                                | $(0.6476)\pi$                                                                                 | $(0.6476)\pi$ and stretching it by a factor of $r = \sqrt{5} \approx 2.2361$ , as in Figure 3.30.                                            |
| johnston-linear-matrix-algebra-F03641 | 393  | ■ 0.936 | $r=\sqrt{5} \approx 2.2361$                                                                                   | $r = \sqrt{5} \approx 2.2361$                                                                 | $(0.6476)\pi$ and stretching it by a factor of $r = \sqrt{5} \approx 2.2361$ , as in Figure 3.30.                                            |
| johnston-linear-matrix-algebra-F03642 | 393  | ■ 0.561 | $A=\left[\begin{array}{rl}-3 & -2 \\ 4 & 1\end{array}\right]$                                                 | $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$                                          | <b>Figure 3.30:</b> A visualization of the geometric effect of the matrix $A = \begin{bmatrix} -3 & -2 \\ 4 & 1 \end{bmatrix}$ from          |
| johnston-linear-matrix-algebra-F03643 | 393  | ■ 1.000 | $r e^{i \theta}$                                                                                              | $r e^{i \theta}$                                                                              | <i>Proof Theorem 3.C.1.</i> First, we note that since the eigenvalue $r e^{i \theta}$ is not real,                                           |
| johnston-linear-matrix-algebra-F03644 | 393  | ■ 1.000 | $r e^{-i \theta}$                                                                                             | $r e^{-i \theta}$                                                                             | it does not equal its complex conjugate $r e^{-i \theta}$ , so $A$ is diagonalizable via $A =$                                               |
| johnston-linear-matrix-algebra-F03645 | 393  | ■ 1.000 | $H \in \mathcal{M}_2(\mathbb{C})$                                                                             | $H \in \mathcal{M}_2(\mathbb{C})$                                                             | with real entries). To this end, let $H \in \mathcal{M}_2(\mathbb{C})$ be the matrix                                                         |
| johnston-linear-matrix-algebra-F03646 | 394  | ■ 0.913 | $Q H=P$                                                                                                       | $QH = P$                                                                                      | $QH = P$ , so                                                                                                                                |
| johnston-linear-matrix-algebra-F03647 | 394  | ■ 0.947 | $H P^{-1}=Q^{-1}$                                                                                             | $HP^{-1} = Q^{-1}$                                                                            | $HP^{-1} = Q^{-1}$ .                                                                                                                         |
| johnston-linear-matrix-algebra-F03648 | 394  | ■ 0.927 | $\sin (-\theta)=-\sin (\theta)$                                                                               | $\sin(-\theta) = -\sin(\theta)$                                                               | $\sin(-\theta) = -\sin(\theta)$                                                                                                              |
| johnston-linear-matrix-algebra-F03649 | 394  | ■ 1.000 | $e^{-i \theta}=\cos (\theta)-i \sin (\theta)$                                                                 | $e^{-i \theta} = \cos(\theta) - i \sin(\theta)$                                               | $e^{-i \theta} = \cos(\theta) - i \sin(\theta)$ .                                                                                            |
| johnston-linear-matrix-algebra-F03650 | 394  | ■ 1.000 | $Q=P H^{-1}$                                                                                                  | $Q = PH^{-1}$                                                                                 | which tells us that $Q$ is invertible since it implies $Q = PH^{-1}$ , which is the                                                          |
| johnston-linear-matrix-algebra-F03651 | 394  | ■ 1.000 | $D=P^{-1} A P$                                                                                                | $D = P^{-1}AP$                                                                                | relationship between $P$ and $Q$ to transform $D = P^{-1}AP$ into $Q^{-1}AQ$ and see                                                         |
| johnston-linear-matrix-algebra-F03652 | 394  | ■ 1.000 | $Q^{-1} A Q$                                                                                                  | $Q^{-1}AQ$                                                                                    | relationship between $P$ and $Q$ to transform $D = P^{-1}AP$ into $Q^{-1}AQ$ and see                                                         |
| johnston-linear-matrix-algebra-F03653 | 394  | ■ 1.000 | $H D H^{-1}$                                                                                                  | $HDH^{-1}$                                                                                    | theorem. In particular, we now compute $HDH^{-1}$ in two different ways. First,                                                              |
| johnston-linear-matrix-algebra-F03654 | 394  | ■ 1.000 | $H D H^{-1}=r\left[R^{\theta}\right]$                                                                         | $HDH^{-1} = r \left[R^{\theta}\right]$                                                        | so if we can also show that $HDH^{-1} = r \left[R^{\theta}\right]$ then we will be done. Equivalently,                                       |
| johnston-linear-matrix-algebra-F03655 | 394  | ■ 1.000 | $H D=r\left[R^{\theta}\right] H$                                                                              | $HD = r \left[R^{\theta}\right] H$                                                            | we want to show that $HD = r \left[R^{\theta}\right] H$ , which can be done simply by multiplying                                            |
| johnston-linear-matrix-algebra-F03656 | 394  | ■ 1.000 | $H$                                                                                                           | $H$                                                                                           | The matrix $H$ from Equation 3.C.1 is quite remarkable for the fact that it                                                                  |
| johnston-linear-matrix-algebra-F03657 | 394  | ■ 1.000 | $\lambda_{\pm}=-1 \pm$                                                                                        | $\lambda_{\pm} = -1 \pm$                                                                      | We start by recalling that the eigenvalues of this matrix are $\lambda_{\pm} = -1 \pm$                                                       |
| johnston-linear-matrix-algebra-F03658 | 394  | ■ 1.000 | $2 i$                                                                                                         | $2i$                                                                                          | $2i$ and a pair of corresponding eigenvectors is given by $\mathbf{v}_{\pm} = (1, -1 \mp i)$ .                                               |
| johnston-linear-matrix-algebra-F03659 | 394  | ■ 1.000 | $\mathbf{v}_{\pm}=(1,-1 \mp i)$                                                                               | $\mathbf{v}_{\pm} = (1, -1 \mp i)$                                                            | $2i$ and a pair of corresponding eigenvectors is given by $\mathbf{v}_{\pm} = (1, -1 \mp i)$ .                                               |
| johnston-linear-matrix-algebra-F03660 | 395  | ■ 1.000 | $\left[\operatorname{Re}\left(\mathbf{v}_{+}\right) \mid-\operatorname{Im}\left(\mathbf{v}_{+}\right)\right]$ | $[\operatorname{Re}(\mathbf{v}_{+}) \mid -\operatorname{Im}(\mathbf{v}_{+})]$                 | which is exactly $[\operatorname{Re}(\mathbf{v}_{+}) \mid -\operatorname{Im}(\mathbf{v}_{+})]$ , as we would hope (we want our               |
| johnston-linear-matrix-algebra-F03661 | 395  | ■ 1.000 | $\sqrt{5}\left[R^{\theta}\right]$                                                                             | $\sqrt{5} \left[R^{\theta}\right]$                                                            | (i.e., it equals $\sqrt{5} \left[R^{\theta}\right]$ , where $\theta = \arccos(-1/\sqrt{5}) \approx (0.6476)\pi$ ).                           |
| johnston-linear-matrix-algebra-F03662 | 395  | ■ 1.000 | $\theta=\arccos (-1 / \sqrt{5}) \approx(0.6476) \pi$                                                          | $\theta = \arccos(-1/\sqrt{5}) \approx (0.6476)\pi$                                           | (i.e., it equals $\sqrt{5} \left[R^{\theta}\right]$ , where $\theta = \arccos(-1/\sqrt{5}) \approx (0.6476)\pi$ ).                           |
| johnston-linear-matrix-algebra-F03663 | 395  | ■ 1.000 | $\operatorname{diag}(a, b, c, \ldots)$                                                                        | $\operatorname{diag}(a, b, c, \dots)$                                                         | Before stating the theorem, we note that we use the notation $\operatorname{diag}(a, b, c, \dots)$                                           |

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| johnston-linear-matrix-algebra-F03664 | 395  | ■ 0.999 | $a, b, c, \ldots$                                                        | $a, b, c, \dots$                                                         | to denote the diagonal matrix with diagonal entries $a, b, c, \dots$ . In fact, we even                                           |
| johnston-linear-matrix-algebra-F03665 | 395  | ■ 1.000 | $A \in \mathcal{M}_n(\mathbb{R})$                                        | $A \in \mathcal{M}_n(\mathbb{R})$                                        | Suppose $A \in \mathcal{M}_n(\mathbb{R})$ has $\ell$ complex conjugate eigenvalue pairs and $m$                                   |
| johnston-linear-matrix-algebra-F03666 | 395  | ■ 1.000 | $2 \ell$                                                                 | $2\ell$                                                                  | eigenvalues and $2\ell$                                                                                                           |
| johnston-linear-matrix-algebra-F03667 | 395  | ■ 0.951 | $2 \ell + m = n$                                                         | $2\ell + m = n$                                                          | $2\ell + m = n$ .                                                                                                                 |
| johnston-linear-matrix-algebra-F03668 | 395  | ■ 1.000 | $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$ | $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$ | with $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$ , and $\mathbf{v}_j \in \mathbb{C}^n$ for all $j$ . |
| johnston-linear-matrix-algebra-F03669 | 395  | ■ 1.000 | $\mathbf{v}_j \in \mathbb{C}^n$                                          | $\mathbf{v}_j \in \mathbb{C}^n$                                          | with $r_j, \lambda_j, \theta_j \in \mathbb{R}, \mathbf{w}_j \in \mathbb{R}^n$ , and $\mathbf{v}_j \in \mathbb{C}^n$ for all $j$ . |
| johnston-linear-matrix-algebra-F03670 | 395  | ■ 1.000 | $A = Q B Q^{-1}$                                                         | $A = Q B Q^{-1}$                                                         | Then $A = Q B Q^{-1}$ , where $B, Q \in \mathcal{M}_n(\mathbb{R})$ are defined by                                                 |
| johnston-linear-matrix-algebra-F03671 | 395  | ■ 1.000 | $B, Q \in \mathcal{M}_n(\mathbb{R})$                                     | $B, Q \in \mathcal{M}_n(\mathbb{R})$                                     | Then $A = Q B Q^{-1}$ , where $B, Q \in \mathcal{M}_n(\mathbb{R})$ are defined by                                                 |
| johnston-linear-matrix-algebra-F03672 | 395  | ■ 1.000 | $P, D \in \mathcal{M}_n(\mathbb{C})$                                     | $P, D \in \mathcal{M}_n(\mathbb{C})$                                     | as $A = P D P^{-1}$ via some $P, D \in \mathcal{M}_n(\mathbb{C})$ , but not via a <i>real</i> $P$ and $D$ (in other               |
| johnston-linear-matrix-algebra-F03673 | 396  | ■ 1.000 | $\tilde{H}$                                                              | $\tilde{H}$                                                              | simply because $\tilde{H}$ is                                                                                                     |
| johnston-linear-matrix-algebra-F03674 | 396  | ■ 0.993 | $(1 \pm i) / 2$                                                          | $(1 \pm i)/2$                                                            | has distinct eigenvalues $2$ and $(1 \pm i)/2$ and is thus diagonalizable over $\mathbb{C}$ , but                                 |
| johnston-linear-matrix-algebra-F03675 | 396  | ■ 1.000 | $1 \leq j \leq \ell$                                                     | $1 \leq j \leq \ell$                                                     | for all $1 \leq j \leq \ell$ , and we also define                                                                                 |
| johnston-linear-matrix-algebra-F03676 | 396  | ■ 1.000 | $Q_j H = P_j$                                                            | $Q_j H = P_j$                                                            | $Q_j H = P_j$ for all $1 \leq j \leq \ell$ , so block matrix multiplication then shows that                                       |
| johnston-linear-matrix-algebra-F03677 | 396  | ■ 1.000 | $H D_j =$                                                                | $H D_j =$                                                                | Similarly, we also showed in the proof of Theorem 3.C.1 that $H D_j =$                                                            |
| johnston-linear-matrix-algebra-F03678 | 396  | ■ 1.000 | $r_j [R^{\theta_j}] H$                                                   | $r_j [R^{\theta_j}] H$                                                   | $r_j [R^{\theta_j}] H$ for all $1 \leq j \leq \ell$ , so block matrix multiplication shows that                                   |
| johnston-linear-matrix-algebra-F03679 | 396  | ■ 0.999 | $\tilde{H} D \tilde{H}^{-1} = B$                                         | $\tilde{H} D \tilde{H}^{-1} = B$                                         | It follows that $\tilde{H} D \tilde{H}^{-1} = B$ , so                                                                             |
| johnston-linear-matrix-algebra-F03680 | 397  | ■ 1.000 | $-\operatorname{Im}(\mathbf{v})$                                         | $-\operatorname{Im}(\mathbf{v})$                                         | $-\operatorname{Im}(\mathbf{v})$ .                                                                                                |
| johnston-linear-matrix-algebra-F03681 | 397  | ■ 1.000 | $p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$              | $p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$              | • Its characteristic polynomial is $p_A(\lambda) = -\lambda^3 + 3\lambda^2 - (5/2)\lambda + 1$ .                                  |
| johnston-linear-matrix-algebra-F03682 | 397  | ■ 0.672 | $(1+i) / 2, (1-i) / 2$                                                   | $(1+i)/2, (1-i)/2$                                                       | • Its eigenvalues are $(1+i)/2, (1-i)/2$ , and $2$ .                                                                              |
| johnston-linear-matrix-algebra-F03683 | 397  | ■ 1.000 | $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$                  | $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$                  | are $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$ , and $\mathbf{w} = (0, 1, 2)$ .                                      |
| johnston-linear-matrix-algebra-F03684 | 397  | ■ 1.000 | $\mathbf{w} = (0, 1, 2)$                                                 | $\mathbf{w} = (0, 1, 2)$                                                 | are $\mathbf{v} = (1, i, 1), \bar{\mathbf{v}} = (1, -i, 1)$ , and $\mathbf{w} = (0, 1, 2)$ .                                      |
| johnston-linear-matrix-algebra-F03685 | 397  | ■ 0.980 | $(1+i) / 2$                                                              | $(1+i)/2$                                                                | $A$ and write it in polar form. We choose the eigenvalue $(1+i)/2$ , so $r =$                                                     |
| johnston-linear-matrix-algebra-F03686 | 397  | ■ 0.980 | $r =$                                                                    | $r =$                                                                    | $A$ and write it in polar form. We choose the eigenvalue $(1+i)/2$ , so $r =$                                                     |
| johnston-linear-matrix-algebra-F03687 | 397  | ■ 1.000 | $\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$                                  | $\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$                                  | $\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$ and $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$                                        |
| johnston-linear-matrix-algebra-F03688 | 397  | ■ 1.000 | $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$                           | $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$                           | $\sqrt{(1/2)^2 + (1/2)^2} = 1/\sqrt{2}$ and $\theta = \arctan((1/2)/(1/2)) = \arctan(1) =$                                        |
| johnston-linear-matrix-algebra-F03689 | 397  | ■ 1.000 | $(1+i) / 2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$                              | $(1+i)/2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$                                | $\pi/4$ . The polar form of this eigenvalue is thus $(1+i)/2 = \frac{1}{\sqrt{2}} e^{i\pi/4}$ .                                   |
| johnston-linear-matrix-algebra-F03690 | 397  | ■ 0.539 | $(1+$                                                                    | $(1+$                                                                    | If we denote the eigenvectors corresponding to the eigenvalues $(1+$                                                              |

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| johnston-linear-matrix-algebra-F03691 | 397  | 1.000 | $i) / 2$                                                                                                 | $i)/2$                                                                                                   | $i)/2$ and 2 by $\mathbf{v} = (1, i, 1)$ and $\mathbf{w} = (0, 1, 2)$ , respectively, it then follows                                                                      |
| johnston-linear-matrix-algebra-F03692 | 397  | 1.000 | $\mathbf{v}=(1, i, 1)$                                                                                   | $\mathbf{v} = (1, i, 1)$                                                                                 | $i)/2$ and 2 by $\mathbf{v} = (1, i, 1)$ and $\mathbf{w} = (0, 1, 2)$ , respectively, it then follows                                                                      |
| johnston-linear-matrix-algebra-F03693 | 397  | 0.920 | $(0,1,2), A$                                                                                             | $(0, 1, 2), A$                                                                                           | $(0, 1, 2), A$ stretches by a factor of 2 (this is nothing new—this is exactly                                                                                             |
| johnston-linear-matrix-algebra-F03694 | 397  | 1.000 | $\theta=\pi / 4$                                                                                         | $\theta = \pi/4$                                                                                         | then $A$ acts on $\mathcal{S}$ like a rotation by an angle of $\theta = \pi/4$ composed with a                                                                             |
| johnston-linear-matrix-algebra-F03695 | 398  | 1.000 | $r=1 / \sqrt{2}$                                                                                         | $r = 1/\sqrt{2}$                                                                                         | scaling by a factor of $r = 1/\sqrt{2}$ , as displayed below:                                                                                                              |
| johnston-linear-matrix-algebra-F03696 | 398  | 0.996 | $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$                                                    | $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$                                                    | recall that $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$ since a rotating $\mathbb{R}^2$ by an angle of $k \theta$ is equivalent                                  |
| johnston-linear-matrix-algebra-F03697 | 398  | 0.996 | $k \theta$                                                                                               | $k \theta$                                                                                               | recall that $\left[R^{\theta}\right]^k=\left[R^{k \theta}\right]$ since a rotating $\mathbb{R}^2$ by an angle of $k \theta$ is equivalent                                  |
| johnston-linear-matrix-algebra-F03698 | 398  | 1.000 | $A^{40}$                                                                                                 | $A^{40}$                                                                                                 | to compute $A^{40}$ for the matrix $A$ from Example 3.C.3 via its diagonalization, we                                                                                      |
| johnston-linear-matrix-algebra-F03699 | 398  | 1.000 | $((1+i) / 2)^{40}$                                                                                       | $((1+i) / 2)^{40}$                                                                                       | would have to compute complex numbers like $((1+i) / 2)^{40}$ , which is perhaps                                                                                           |
| johnston-linear-matrix-algebra-F03700 | 398  | 1.000 | $\left.\left((1+i) / 2\right)^{40}=1 / 2^{20}\right)$                                                    | $((1+i) / 2)^{40}=1 / 2^{20})$                                                                           | this particular case, the answer is in fact real: $((1+i) / 2)^{40}=1 / 2^{20})$ .                                                                                         |
| johnston-linear-matrix-algebra-F03701 | 399  | 1.000 | $\left[R^{40 \pi / 4}\right]$                                                                            | $\left[R^{40 \pi / 4}\right]$                                                                            | $\left[R^{40 \pi / 4}\right]$ is simply the                                                                                                                                |
| johnston-linear-matrix-algebra-F03702 | 399  | 1.000 | $\theta=40 \pi / 4=10 \pi$                                                                               | $\theta=40 \pi / 4=10 \pi$                                                                               | $\theta=40 \pi / 4=10 \pi$ is                                                                                                                                              |
| johnston-linear-matrix-algebra-F03703 | 399  | 0.996 | $r, \theta \in \mathbb{R}$                                                                               | $r, \theta \in \mathbb{R}$                                                                               | That is, find values for $r, \theta \in \mathbb{R}$ and a matrix $Q \in \mathcal{M}_2(\mathbb{R})$                                                                         |
| johnston-linear-matrix-algebra-F03704 | 399  | 0.996 | $Q \in \mathcal{M}_2(\mathbb{R})$                                                                        | $Q \in \mathcal{M}_2(\mathbb{R})$                                                                        | That is, find values for $r, \theta \in \mathbb{R}$ and a matrix $Q \in \mathcal{M}_2(\mathbb{R})$                                                                         |
| johnston-linear-matrix-algebra-F03705 | 399  | 1.000 | $Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$                                                          | $Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$                                                          | such that the given matrix has the form $Q\left(r\left[R^{\theta}\right]\right) Q^{-1}$ .                                                                                  |
| johnston-linear-matrix-algebra-F03706 | 399  | 0.725 | $\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$                      | $\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$                      | $\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$<br>$\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$ |
| johnston-linear-matrix-algebra-F03707 | 399  | 0.725 | $\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$                      | $\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$                      | $\left(\mathbf{a}\right)\left[\begin{array}{cc} 0 & -1 \\ 1 & 0 \end{array}\right]$<br>$\left(\mathbf{c}\right)\left[\begin{array}{cc} 2 & 1 \\ -2 & 0 \end{array}\right]$ |
| johnston-linear-matrix-algebra-F03708 | 399  | 1.000 | $\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$                                             | $\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$                                             | $\left(\mathbf{b}\right)\left[\begin{array}{cc} 0 & 1 \\ -2 & 0 \end{array}\right]$                                                                                        |
| johnston-linear-matrix-algebra-F03709 | 399  | 1.000 | $\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$                               | $\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$                               | $\left(\mathbf{d}\right)\left[\begin{array}{cc} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{array}\right]$                                                                          |
| johnston-linear-matrix-algebra-F03710 | 399  | 0.970 | $\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$                      | $\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$                      | $\left(\mathbf{a}\right)\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{array}\right]$                                                                 |
| johnston-linear-matrix-algebra-F03711 | 399  | 1.000 | $\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$ | $\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$ | $\left(\mathbf{b}\right)\left[\begin{array}{ccc} 1 & -\sqrt{3} & \sqrt{3} \\ 0 & 3 & 0 \\ -\sqrt{3} & 2 & 1 \end{array}\right]$                                            |

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| johnston-linear-matrix-atgebu<br>F03712 | 399  | ■ 0.943 | $\left[\begin{array}{cccc} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{array}\right]$    | $\begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$    | *(c) $\begin{bmatrix} 1 & -2 & 2 & 1 \\ 0 & 1 & 0 & 0 \\ -1 & -1 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$   |
| johnston-linear-matrix-atgebu<br>F03713 | 399  | ■ 1.000 | $\left[\begin{array}{cccc} 4 & -3 & -1 & 2 \\ 6 & -5 & 0 & 1 \\ 4 & -5 & 1 & 0 \\ 0 & -1 & 3 & -2 \end{array}\right]$ | $\begin{bmatrix} 4 & -3 & -1 & 2 \\ 6 & -5 & 0 & 1 \\ 4 & -5 & 1 & 0 \\ 0 & -1 & 3 & -2 \end{bmatrix}$ | (d) $\begin{bmatrix} 4 & -3 & -1 & 2 \\ 6 & -5 & 0 & 1 \\ 4 & -5 & 1 & 0 \\ 0 & -1 & 3 & -2 \end{bmatrix}$ |
| johnston-linear-matrix-atgebu<br>F03714 | 399  | ■ 1.000 | $A \in \mathcal{M}_4(\mathbb{R})$                                                                                     | $A \in \mathcal{M}_4(\mathbb{R})$                                                                      | (b) Every matrix $A \in \mathcal{M}_4(\mathbb{R})$ has a real eigenvalue.                                  |
| johnston-linear-matrix-atgebu<br>F03715 | 399  | ■ 1.000 | $\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$                                                               | $\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$                                                | (f) Every rotation matrix $\left[R^{\theta}\right] \in \mathcal{M}_2(\mathbb{R})$ has complex              |
| johnston-linear-matrix-atgebu<br>F03716 | 400  | ■ 1.000 | $x_{\{0\}}$                                                                                                           | $x_0$                                                                                                  | sequences at $x_0$                                                                                         |
| johnston-linear-matrix-atgebu<br>F03717 | 400  | ■ 1.000 | $x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, \ldots$                                                                             | $x_1, x_2, x_3, \dots$                                                                                 | numbers $x_1, x_2, x_3, \dots$ that are defined in a similar linear way.                                   |
| johnston-linear-matrix-atgebu<br>F03718 | 400  | ■ 1.000 | $x_{\{n\}}$                                                                                                           | $x_n$                                                                                                  | A <b>linear recurrence relation</b> is a formula that represents a term $x_n$ in a se-                     |
| johnston-linear-matrix-atgebu<br>F03719 | 400  | ■ 1.000 | $a_{\{0\}}, a_{\{1\}}, \ldots, a_{\{k-1\}}$                                                                           | $a_0, a_1, \dots, a_{k-1}$                                                                             | the sequence. That is, it is a formula for which there exist scalars $a_0, a_1, \dots, a_{k-1}$            |
| johnston-linear-matrix-atgebu<br>F03720 | 400  | ■ 0.994 | $F_{\{0\}}, F_{\{1\}}, F_{\{2\}}, \ldots$                                                                             | $F_0, F_1, F_2, \dots$                                                                                 | For example, the Fibonacci numbers $F_0, F_1, F_2, \dots$ from Example 3.4.8                               |
| johnston-linear-matrix-atgebu<br>F03721 | 400  | ■ 0.907 | $F_{\{n\}}=F_{\{n-1\}}+F_{\{n-2\}}$                                                                                   | $F_n = F_{n-1} + F_{n-2}$                                                                              | satisfy the degree-2 linear recurrence relation $F_n = F_{n-1} + F_{n-2}$ , and the initial                |
| johnston-linear-matrix-atgebu<br>F03722 | 400  | ■ 1.000 | $F_{\{0\}}=0$                                                                                                         | $F_0 = 0$                                                                                              | conditions were the requirements that $F_0 = 0$ and $F_1 = 1$ .                                            |
| johnston-linear-matrix-atgebu<br>F03723 | 400  | ■ 1.000 | $F_{\{1\}}=1$                                                                                                         | $F_1 = 1$                                                                                              | conditions were the requirements that $F_0 = 0$ and $F_1 = 1$ .                                            |
| johnston-linear-matrix-atgebu<br>F03724 | 400  | ■ 1.000 | $x_{\{0\}}=3, x_{\{1\}}=-2$                                                                                           | $x_0 = 3, x_1 = -2$                                                                                    | if $x_0 = 3, x_1 = -2$ , and $x_2 = 8$ .                                                                   |
| johnston-linear-matrix-atgebu<br>F03725 | 400  | ■ 1.000 | $x_{\{2\}}=8$                                                                                                         | $x_2 = 8$                                                                                              | if $x_0 = 3, x_1 = -2$ , and $x_2 = 8$ .                                                                   |
| johnston-linear-matrix-atgebu<br>F03726 | 400  | ■ 1.000 | $x_{\{6\}}, x_{\{7\}}$                                                                                                | $x_6, x_7$                                                                                             | By using this same procedure to compute $x_6, x_7$ , and so on, we see that                                |
| johnston-linear-matrix-atgebu<br>F03727 | 401  | ■ 1.000 | $x_{\{n\}}=x_{\{n-1\}}+x_{\{n-2\}}$                                                                                   | $x_n = x_{n-1} + x_{n-2}$                                                                              | a) $x_n = x_{n-1} + x_{n-2}$ ,                                                                             |
| johnston-linear-matrix-atgebu<br>F03728 | 401  | ■ 1.000 | $x_{\{n\}}=4 x_{\{n-1\}}-x_{\{n-2\}}-6 x_{\{n-3\}}$                                                                   | $x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$                                                                  | b) $x_n = 4x_{n-1} - x_{n-2} - 6x_{n-3}$ , and                                                             |
| johnston-linear-matrix-atgebu<br>F03729 | 401  | ■ 1.000 | $x_{\{n\}}=x_{\{n-1\}}+4 x_{\{n-3\}}+2 x_{\{n-4\}}$                                                                   | $x_n = x_{n-1} + 4x_{n-3} + 2x_{n-4}$                                                                  | c) $x_n = x_{n-1} + 4x_{n-3} + 2x_{n-4}$ .                                                                 |
| johnston-linear-matrix-atgebu<br>F03730 | 401  | ■ 1.000 | $a_{\{1\}}=a_{\{0\}}=1$                                                                                               | $a_1 = a_0 = 1$                                                                                        | a) This is a degree-2 linear recurrence relation with $a_1 = a_0 = 1$ , so its                             |
| johnston-linear-matrix-atgebu<br>F03731 | 401  | ■ 0.877 | $a_{\{2\}}=4, a_{\{1\}}=-1$                                                                                           | $a_2 = 4, a_1 = -1$                                                                                    | b) This is a degree-3 linear recurrence relation with $a_2 = 4, a_1 = -1$ ,                                |
| johnston-linear-matrix-atgebu<br>F03732 | 401  | ■ 1.000 | $a_{\{0\}}=-6$                                                                                                        | $a_0 = -6$                                                                                             | and $a_0 = -6$ , so its companion matrix is                                                                |
| johnston-linear-matrix-atgebu<br>F03733 | 401  | ■ 0.948 | $a_{\{3\}}=1, a_{\{2\}}=0, a_{\{1\}}=4$                                                                               | $a_3 = 1, a_2 = 0, a_1 = 4$                                                                            | lation with $a_3 = 1, a_2 = 0, a_1 = 4$ , and $a_0 = 2$ , so its companion                                 |
| johnston-linear-matrix-atgebu<br>F03734 | 401  | ■ 0.948 | $a_{\{0\}}=2$                                                                                                         | $a_0 = 2$                                                                                              | lation with $a_3 = 1, a_2 = 0, a_1 = 4$ , and $a_0 = 2$ , so its companion                                 |
| johnston-linear-matrix-atgebu<br>F03735 | 401  | ■ 0.999 | $x_{\{0\}}, x_{\{1\}}, x_{\{2\}}, \ldots$                                                                             | $x_0, x_1, x_2, \dots$                                                                                 | Let $x_0, x_1, x_2, \dots$ be a sequence satisfying a linear recurrence rela-                              |
| johnston-linear-matrix-atgebu<br>F03736 | 401  | ■ 1.000 | $n=0,1,2, \ldots$                                                                                                     | $n = 0, 1, 2, \dots$                                                                                   | tion with companion matrix $C$ . For each $n = 0, 1, 2, \dots$ , define $\mathbf{x}_n =$                   |

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| johnston-linear-matrix-attrib F03737 | 401  | 1.000 | $\mathbf{x}_n =$                                                                      | $\mathbf{x}_n =$                                                                      | tion with companion matrix $C$ . For each $n = 0, 1, 2, \dots$ , define $\mathbf{x}_n =$                                 |
| johnston-linear-matrix-attrib F03738 | 401  | 1.000 | $\left(x_n, x_{n+1}, \dots, x_{n+k-1}\right) \in \mathbb{R}^k$                        | $(x_n, x_{n+1}, \dots, x_{n+k-1}) \in \mathbb{R}^k$                                   | $(x_n, x_{n+1}, \dots, x_{n+k-1}) \in \mathbb{R}^k$ . Then                                                               |
| johnston-linear-matrix-attrib F03739 | 402  | 1.000 | $\mathbf{x}_0 = \left(x_0, x_1, \dots, x_{k-1}\right)$                                | $\mathbf{x}_0 = (x_0, x_1, \dots, x_{k-1})$                                           | $\mathbf{x}_0 = (x_0, x_1, \dots, x_{k-1})$                                                                              |
| johnston-linear-matrix-attrib F03740 | 402  | 1.000 | $n = k$                                                                               | $n = k$                                                                               | where we used the given recurrence relation (with $n = k$ ) to simplify the bottom                                       |
| johnston-linear-matrix-attrib F03741 | 402  | 1.000 | $C^n \mathbf{x}_0$                                                                    | $C^n \mathbf{x}_0$                                                                    | In particular, since $x_n$ equals the first entry of $C^n \mathbf{x}_0$ (or the last entry of                            |
| johnston-linear-matrix-attrib F03742 | 402  | 1.000 | $C^{n-k+1} \mathbf{x}_0$                                                              | $C^{n-k+1} \mathbf{x}_0$                                                              | $C^{n-k+1} \mathbf{x}_0$ ), any explicit formula for the powers of $C$ immediately gives us an                           |
| johnston-linear-matrix-attrib F03743 | 403  | 1.000 | $3^n - 2^n + 3(-1)^n$                                                                 | $3^n - 2^n + 3(-1)^n$                                                                 | and the top entry of this matrix-vector product is $3^n - 2^n + 3(-1)^n$ . We                                            |
| johnston-linear-matrix-attrib F03744 | 403  | 1.000 | $x_n = 3^n - 2^n + 3(-1)^n$                                                           | $x_n = 3^n - 2^n + 3(-1)^n$                                                           | thus conclude that $x_n = 3^n - 2^n + 3(-1)^n$ for all $n$ .                                                             |
| johnston-linear-matrix-attrib F03745 | 403  | 1.000 | $p(\lambda) = (-1)^k (\lambda^k - a_{k-1} \lambda^{k-1} - \dots - a_1 \lambda - a_0)$ | $p(\lambda) = (-1)^k (\lambda^k - a_{k-1} \lambda^{k-1} - \dots - a_1 \lambda - a_0)$ | is simply $p(\lambda) = (-1)^k (\lambda^k - a_{k-1} \lambda^{k-1} - \dots - a_1 \lambda - a_0)$ . Since the leading      |
| johnston-linear-matrix-attrib F03746 | 404  | 1.000 | $x_{n-j}$                                                                             | $x_{n-j}$                                                                             | $x_{n-j}$ with $\lambda^{k-j}$ .                                                                                         |
| johnston-linear-matrix-attrib F03747 | 404  | 1.000 | $\lambda^{k-j}$                                                                       | $\lambda^{k-j}$                                                                       | $x_{n-j}$ with $\lambda^{k-j}$ .                                                                                         |
| johnston-linear-matrix-attrib F03748 | 404  | 1.000 | $c_0, c_1, \dots, c_{k-1}$                                                            | $c_0, c_1, \dots, c_{k-1}$                                                            | choose $c_0, c_1, \dots, c_{k-1}$                                                                                        |
| johnston-linear-matrix-attrib F03749 | 404  | 1.000 | $n = 0$                                                                               | $n = 0$                                                                               | plugging $n = 0$ into                                                                                                    |
| johnston-linear-matrix-attrib F03750 | 404  | 1.000 | $\lambda_0, \lambda_1, \dots, \lambda_{k-1}$                                          | $\lambda_0, \lambda_1, \dots, \lambda_{k-1}$                                          | whose characteristic polynomial has distinct roots $\lambda_0, \lambda_1, \dots, \lambda_{k-1}$ . Let                    |
| johnston-linear-matrix-attrib F03751 | 404  | 1.000 | $p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \dots -$    | $p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \dots -$    | root of the characteristic polynomial $p(\lambda) = \lambda^k - a_{k-1} \lambda^{k-1} - a_{k-2} \lambda^{k-2} - \dots -$ |
| johnston-linear-matrix-attrib F03752 | 404  | 1.000 | $a_1 \lambda - a_0$                                                                   | $a_1 \lambda - a_0$                                                                   | $a_1 \lambda - a_0$ , so there is never a term of the form $0^n$ in the explicit formula (3.D.2)                         |
| johnston-linear-matrix-attrib F03753 | 404  | 1.000 | $0^n$                                                                                 | $0^n$                                                                                 | $a_1 \lambda - a_0$ , so there is never a term of the form $0^n$ in the explicit formula (3.D.2)                         |
| johnston-linear-matrix-attrib F03754 | 404  | 1.000 | $a_0 = 0$                                                                             | $a_0 = 0$                                                                             | is simply that it would imply $a_0 = 0$ , which means that the recurrence relation                                       |
| johnston-linear-matrix-attrib F03755 | 404  | 1.000 | $k - 1$                                                                               | $k - 1$                                                                               | actually has degree $k - 1$ (or less), so $p$ is not actually its characteristic polyno-                                 |
| johnston-linear-matrix-attrib F03756 | 405  | 1.000 | $\lambda_0 = 1, \lambda_1 = 2$                                                        | $\lambda_0 = 1, \lambda_1 = 2$                                                        | $\lambda_0 = 1, \lambda_1 = 2$ , and                                                                                     |
| johnston-linear-matrix-attrib F03757 | 405  | 1.000 | $\lambda_2 = 5$                                                                       | $\lambda_2 = 5$                                                                       | $\lambda_2 = 5$ , while the                                                                                              |
| johnston-linear-matrix-attrib F03758 | 405  | 1.000 | $x_0 = 1, x_1 = 4$                                                                    | $x_0 = 1, x_1 = 4$                                                                    | $x_0 = 1, x_1 = 4$ , and                                                                                                 |
| johnston-linear-matrix-attrib F03759 | 405  | 1.000 | $x_2 = 22$                                                                            | $x_2 = 22$                                                                            | $x_2 = 22$ .                                                                                                             |
| johnston-linear-matrix-attrib F03760 | 405  | 1.000 | $x_k$                                                                                 | $x_k$                                                                                 | which is exactly the claimed formula for $x_k$ . Repeating this argument for $x_{k+1}$ ,                                 |
| johnston-linear-matrix-attrib F03761 | 405  | 1.000 | $x_{k+1}$                                                                             | $x_{k+1}$                                                                             | which is exactly the claimed formula for $x_k$ . Repeating this argument for $x_{k+1}$ ,                                 |
| johnston-linear-matrix-attrib F03762 | 405  | 1.000 | $x_{k+2}$                                                                             | $x_{k+2}$                                                                             | $x_{k+2}$ , and so on shows that the result holds for all $n$ .                                                          |
| johnston-linear-matrix-attrib F03763 | 405  | 1.000 | $(c_0, c_1, c_2) = (1, -1, 1)$                                                        | $(c_0, c_1, c_2) = (1, -1, 1)$                                                        | The unique solution is $(c_0, c_1, c_2) = (1, -1, 1)$ , so we conclude that                                              |
| johnston-linear-matrix-attrib F03764 | 405  | 1.000 | $\mathbf{x}_n = C^n \mathbf{x}_0$                                                     | $\mathbf{x}_n = C^n \mathbf{x}_0$                                                     | $\mathbf{x}_n = C^n \mathbf{x}_0$ , where $C$ is the companion matrix, and then diagonalize $C$ . In                     |

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| johnston-linear-matrix-algebra-F03765 | 406  | ■ 1.000 | $V \mathbf{c} = \mathbf{x}_0$                                                                    | $V\mathbf{c} = \mathbf{x}_0$                                                                     | that if $V\mathbf{c} = \mathbf{x}_0$ then                                                                               |
| johnston-linear-matrix-algebra-F03766 | 406  | ■ 0.996 | $V^{-1} \mathbf{x}_0 = \mathbf{c}$                                                               | $V^{-1}\mathbf{x}_0 = \mathbf{c}$                                                                | $V^{-1}\mathbf{x}_0 = \mathbf{c}$ .                                                                                     |
| johnston-linear-matrix-algebra-F03767 | 406  | ■ 1.000 | $\lambda_1 = i$                                                                                  | $\lambda_1 = i$                                                                                  | pairs ( $\lambda_1 = i$ and                                                                                             |
| johnston-linear-matrix-algebra-F03768 | 406  | ■ 1.000 | $\lambda_2 = \bar{i} = -i$                                                                       | $\lambda_2 = \bar{i} = -i$                                                                       | $\lambda_2 = \bar{i} = -i$ here), as                                                                                    |
| johnston-linear-matrix-algebra-F03769 | 406  | ■ 1.000 | $c_1 = 1 - 3i$                                                                                   | $c_1 = 1 - 3i$                                                                                   | ( $c_1 = 1 - 3i$ and                                                                                                    |
| johnston-linear-matrix-algebra-F03770 | 406  | ■ 1.000 | $c_2 = \overline{1 - 3i} = 1 + 3i$                                                               | $c_2 = 1 - 3i = 1 + 3i$                                                                          | $c_2 = \overline{1 - 3i} = 1 + 3i$                                                                                      |
| johnston-linear-matrix-algebra-F03771 | 406  | ■ 0.988 | $\mathbf{c} = (c_0, c_1, \dots, c_{k-1})$                                                        | $\mathbf{c} = (c_0, c_1, \dots, c_{k-1})$                                                        | If we let $\mathbf{c} = (c_0, c_1, \dots, c_{k-1})$ be a vector for which $V\mathbf{c} = \mathbf{x}_0$ , then Theo-     |
| johnston-linear-matrix-algebra-F03772 | 406  | ■ 1.000 | $VD^n \mathbf{c}$                                                                                | $VD^n\mathbf{c}$                                                                                 | It is straightforward to explicitly compute the top entry of $VD^n\mathbf{c}$ to then                                   |
| johnston-linear-matrix-algebra-F03773 | 406  | ■ 1.000 | $x_0 = 3, x_1 = 8$                                                                               | $x_0 = 3, x_1 = 8$                                                                               | if $x_0 = 3, x_1 = 8$ , and $x_2 = 2$ .                                                                                 |
| johnston-linear-matrix-algebra-F03774 | 406  | ■ 1.000 | $x_2 = 2$                                                                                        | $x_2 = 2$                                                                                        | if $x_0 = 3, x_1 = 8$ , and $x_2 = 2$ .                                                                                 |
| johnston-linear-matrix-algebra-F03775 | 406  | ■ 1.000 | $\lambda_0 = 2, \lambda_1 = i$                                                                   | $\lambda_0 = 2, \lambda_1 = i$                                                                   | which has roots $\lambda_0 = 2, \lambda_1 = i$ , and $\lambda_2 = -i$ . Since these roots are distinct,                 |
| johnston-linear-matrix-algebra-F03776 | 406  | ■ 1.000 | $\lambda_2 = -i$                                                                                 | $\lambda_2 = -i$                                                                                 | which has roots $\lambda_0 = 2, \lambda_1 = i$ , and $\lambda_2 = -i$ . Since these roots are distinct,                 |
| johnston-linear-matrix-algebra-F03777 | 406  | ■ 1.000 | $(c_0, c_1, c_2) = (1, 1 - 3i, 1 + 3i)$                                                          | $(c_0, c_1, c_2) = (1, 1 - 3i, 1 + 3i)$                                                          | The unique solution is $(c_0, c_1, c_2) = (1, 1 - 3i, 1 + 3i)$ , so we conclude that                                    |
| johnston-linear-matrix-algebra-F03778 | 406  | ■ 0.814 | $x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + c_2 \lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ | $x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + c_2 \lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ | $x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + c_2 \lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ for all $n \geq 0$ .   |
| johnston-linear-matrix-algebra-F03779 | 406  | ■ 0.814 | $n \geq 0$                                                                                       | $n \geq 0$                                                                                       | $x_n = c_0 \lambda_0^n + c_1 \lambda_1^n + c_2 \lambda_2^n = 2^n + (1 - 3i)i^n + (1 + 3i)(-i)^n$ for all $n \geq 0$ .   |
| johnston-linear-matrix-algebra-F03780 | 407  | ■ 0.999 | $r_0 = \dots = r_{m-1} = 1$                                                                      | $r_0 = \dots = r_{m-1} = 1$                                                                      | $r_0 = \dots = r_{m-1} = 1$ .                                                                                           |
| johnston-linear-matrix-algebra-F03781 | 407  | ■ 1.000 | $\lambda_1 = i = e^{i\pi/2}$                                                                     | $\lambda_1 = i = e^{i\pi/2}$                                                                     | we can write the complex roots $\lambda_1$ and $\lambda_2$ in their polar forms $\lambda_1 = i = e^{i\pi/2}$            |
| johnston-linear-matrix-algebra-F03782 | 407  | ■ 1.000 | $\lambda_2 = -i = e^{-i\pi/2}$                                                                   | $\lambda_2 = -i = e^{-i\pi/2}$                                                                   | and $\lambda_2 = -i = e^{-i\pi/2}$ and simplify:                                                                        |
| johnston-linear-matrix-algebra-F03783 | 407  | ■ 1.000 | $e^{i\theta} + e^{-i\theta} = 2 \cos(\theta)$                                                    | $e^{i\theta} + e^{-i\theta} = 2 \cos(\theta)$                                                    | so $e^{i\theta} + e^{-i\theta} = 2 \cos(\theta)$ and $e^{i\theta} - e^{-i\theta} = 2i \sin(\theta)$ (see Appendix A.1). |
| johnston-linear-matrix-algebra-F03784 | 407  | ■ 1.000 | $e^{i\theta} - e^{-i\theta} = 2i \sin(\theta)$                                                   | $e^{i\theta} - e^{-i\theta} = 2i \sin(\theta)$                                                   | so $e^{i\theta} + e^{-i\theta} = 2 \cos(\theta)$ and $e^{i\theta} - e^{-i\theta} = 2i \sin(\theta)$ (see Appendix A.1). |
| johnston-linear-matrix-algebra-F03785 | 407  | ■ 1.000 | $\lambda_0, \lambda_1, \dots, \lambda_{m-1}$                                                     | $\lambda_0, \lambda_1, \dots, \lambda_{m-1}$                                                     | whose characteristic polynomial has roots $\lambda_0, \lambda_1, \dots, \lambda_{m-1}$ , with multiplic-                |
| johnston-linear-matrix-algebra-F03786 | 407  | ■ 1.000 | $r_0, r_1, \dots, r_{m-1}$                                                                       | $r_0, r_1, \dots, r_{m-1}$                                                                       | ities $r_0, r_1, \dots, r_{m-1}$ , respectively. Let $q_0, q_1, \dots, q_{m-1}$ be the (necessarily                     |
| johnston-linear-matrix-algebra-F03787 | 407  | ■ 1.000 | $q_0, q_1, \dots, q_{m-1}$                                                                       | $q_0, q_1, \dots, q_{m-1}$                                                                       | ities $r_0, r_1, \dots, r_{m-1}$ , respectively. Let $q_0, q_1, \dots, q_{m-1}$ be the (necessarily                     |
| johnston-linear-matrix-algebra-F03788 | 407  | ■ 1.000 | $r_0 - 1, r_1 - 1, \dots, r_{m-1} - 1$                                                           | $r_0 - 1, r_1 - 1, \dots, r_{m-1} - 1$                                                           | unique) polynomials with degrees $r_0 - 1, r_1 - 1, \dots, r_{m-1} - 1$ , respectively,                                 |
| johnston-linear-matrix-algebra-F03789 | 408  | ■ 0.874 | $x_{n-1}$                                                                                        | $x_{n-1}$                                                                                        | $x_{n-1}$ is 0).                                                                                                        |
| johnston-linear-matrix-algebra-F03790 | 408  | ■ 1.000 | $V_j$                                                                                            | $V_j$                                                                                            | Notice that each $V_j$                                                                                                  |
| johnston-linear-matrix-algebra-F03791 | 408  | ■ 1.000 | $k \times r_j$                                                                                   | $k \times r_j$                                                                                   | has size $k \times r_j$ , so $V$                                                                                        |
| johnston-linear-matrix-algebra-F03792 | 408  | ■ 1.000 | $k \times (r_0 + \dots + r_{m-1}) =$                                                             | $k \times (r_0 + \dots + r_{m-1}) =$                                                             | $k \times (r_0 + \dots + r_{m-1}) =$                                                                                    |

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| johnston-linear-matrix-atgso-2-F03793 | 408  | 1.000 | $r_{\{j\}}=1$                                                                                      | $r_j = 1$                                                    | matrix if $r_j = 1$ (i.e., $V_j$                                                                          |
| johnston-linear-matrix-atgso-2-F03794 | 408  | 1.000 | $k \times 1$                                                                                       | $k \times 1$                                                 | is $k \times 1$ ) for all $j$ . This                                                                      |
| johnston-linear-matrix-atgso-2-F03795 | 408  | 1.000 | $x_{\{0\}}=6, \ x_{\{1\}}=0$                                                                       | $x_0 = 6, x_1 = 0$                                           | if $x_0 = 6, x_1 = 0$ , and $x_2 = 3$ .                                                                   |
| johnston-linear-matrix-atgso-2-F03796 | 408  | 1.000 | $x_{\{2\}}=3$                                                                                      | $x_2 = 3$                                                    | if $x_0 = 6, x_1 = 0$ , and $x_2 = 3$ .                                                                   |
| johnston-linear-matrix-atgso-2-F03797 | 408  | 1.000 | $\lambda_{\{0\}}=2$                                                                                | $\lambda_0 = 2$                                              | which has roots $\lambda_0 = 2$ and $\lambda_1 = -1$ , with multiplicities 1 and 2, respec-               |
| johnston-linear-matrix-atgso-2-F03798 | 408  | 1.000 | $q_{\{0\}}$                                                                                        | $q_0$                                                        | tively. The polynomials $q_0$ and $q_1$ in Theorem 3.D.3 then have degrees 0                              |
| johnston-linear-matrix-atgso-2-F03799 | 408  | 1.000 | $q_{\{1\}}$                                                                                        | $q_1$                                                        | tively. The polynomials $q_0$ and $q_1$ in Theorem 3.D.3 then have degrees 0                              |
| johnston-linear-matrix-atgso-2-F03800 | 408  | 0.979 | $q_{\{0\}}(x)=c_{\{0\}}$                                                                           | $q_0(x) = c_0$                                               | and 1, respectively, so they are of the form $q_0(x) = c_0$ and $q_1(x) = c_1 + c_2x$ .                   |
| johnston-linear-matrix-atgso-2-F03801 | 408  | 0.979 | $q_{\{1\}}(x)=c_{\{1\}}+c_{\{2\}} \ x$                                                             | $q_1(x) = c_1 + c_2x$                                        | and 1, respectively, so they are of the form $q_0(x) = c_0$ and $q_1(x) = c_1 + c_2x$ .                   |
| johnston-linear-matrix-atgso-2-F03802 | 408  | 1.000 | $c_{\{0\}}, \ c_{\{1\}}, \ c_{\{2\}}$                                                              | $c_0, c_1, c_2$                                              | Our goal is thus to find $c_0, c_1, c_2$ such that                                                        |
| johnston-linear-matrix-atgso-2-F03803 | 408  | 1.000 | $n=0, 1$                                                                                           | $n = 0, 1$                                                   | By plugging $n = 0, 1$ , and 2 into this equation, we arrive at the linear                                |
| johnston-linear-matrix-atgso-2-F03804 | 408  | 1.000 | $\left(c_{\{0\}}, \ c_{\{1\}}, \ c_{\{2\}}\right)=(1,5,-3)$                                        | $(c_0, c_1, c_2) = (1, 5, -3)$                               | which has unique solution $(c_0, c_1, c_2) = (1, 5, -3)$ . It follows that                                |
| johnston-linear-matrix-atgso-2-F03805 | 408  | 1.000 | $q_{\{0\}}, \ q_{\{1\}}, \ \ldots, \ q_{\{k-1\}}$                                                  | $q_0, q_1, \dots, q_{k-1}$                                   | <i>Proof of Theorem 3.D.3.</i> We start by showing that $q_0, q_1, \dots, q_{k-1}$ exist and              |
| johnston-linear-matrix-atgso-2-F03806 | 408  | 1.000 | $q_{\{j\}}(n)=c_{\{j, \ 0\}}+c_{\{j, \ 1\}} \ n+\cdots+c_{\{j, \ r_{\{j\}}-1\}} \ n^{r_{\{j\}}-1}$ | $q_j(n) = c_{j,0} + c_{j,1}n + \dots + c_{j,r_j-1}n^{r_j-1}$ | more explicitly as $q_j(n) = c_{j,0} + c_{j,1}n + \dots + c_{j,r_j-1}n^{r_j-1}$ for $0 \leq j < m$ . Then |
| johnston-linear-matrix-atgso-2-F03807 | 408  | 1.000 | $0 \leq j < m$                                                                                     | $0 \leq j < m$                                               | more explicitly as $q_j(n) = c_{j,0} + c_{j,1}n + \dots + c_{j,r_j-1}n^{r_j-1}$ for $0 \leq j < m$ . Then |
| johnston-linear-matrix-atgso-2-F03808 | 408  | 1.000 | $c_{\{j, \ \ell\}}$                                                                                | $c_{j,\ell}$                                                 | from $\ell = 0$ ) corresponds to $c_{j,\ell}$ .                                                           |
| johnston-linear-matrix-atgso-2-F03809 | 409  | 1.000 | $0^{\{0\}}=1$                                                                                      | $0^0 = 1$                                                    | $0^0 = 1$ , so $d_0 0^0 \lambda_j^0 = d_0$ ,                                                              |
| johnston-linear-matrix-atgso-2-F03810 | 409  | 1.000 | $d_{\{0\}} \ 0^{\{0\}} \ \lambda_{\{j\}}^{\{0\}}=d_{\{0\}}$                                        | $d_0 0^0 \lambda_j^0 = d_0$                                  | $0^0 = 1$ , so $d_0 0^0 \lambda_j^0 = d_0$ ,                                                              |
| johnston-linear-matrix-atgso-2-F03811 | 409  | 0.999 | $d_{\{0\}} \ 0^{\{\ell\}} \ \lambda_{\{j\}}^{\{0\}}=0$                                             | $d_0 0^\ell \lambda_j^0 = 0$                                 | but $d_0 0^\ell \lambda_j^0 = 0$ when                                                                     |
| johnston-linear-matrix-atgso-2-F03812 | 409  | 1.000 | $\ell>0$                                                                                           | $\ell > 0$                                                   | $\ell > 0$ .                                                                                              |
| johnston-linear-matrix-atgso-2-F03813 | 409  | 0.999 | $d_{\{0\}}, \ d_{\{1\}}, \ \ldots, \ d_{\{k-1\}}$                                                  | $d_0, d_1, \dots, d_{k-1}$                                   | suppose that some linear combination (with coefficients $d_0, d_1, \dots, d_{k-1}$ ) of                   |
| johnston-linear-matrix-atgso-2-F03814 | 409  | 1.000 | $r_{\{0\}}+r_{\{1\}}+\cdots+r_{\{m-1\}}=k$                                                         | $r_0 + r_1 + \dots + r_{m-1} = k$                            | enough). Counting multiplicity, we have thus found $r_0 + r_1 + \dots + r_{m-1} = k$                      |
| johnston-linear-matrix-atgso-2-F03815 | 409  | 1.000 | $d_{\{0\}}=d_{\{1\}}=\cdots=d_{\{k-1\}}=0$                                                         | $d_0 = d_1 = \dots = d_{k-1} = 0$                            | $d_0 = d_1 = \dots = d_{k-1} = 0$ . The rows of $V$ are thus linearly independent, so it is               |
| johnston-linear-matrix-atgso-2-F03816 | 409  | 1.000 | $x_{\{0\}}=0, \ x_{\{1\}}=3, \ x_{\{2\}}=8$                                                        | $x_0 = 0, x_1 = 3, x_2 = 8$                                  | if $x_0 = 0, x_1 = 3, x_2 = 8$ , and $x_3 = 7$ .                                                          |
| johnston-linear-matrix-atgso-2-F03817 | 409  | 1.000 | $x_{\{3\}}=7$                                                                                      | $x_3 = 7$                                                    | if $x_0 = 0, x_1 = 3, x_2 = 8$ , and $x_3 = 7$ .                                                          |
| johnston-linear-matrix-atgso-2-F03818 | 410  | 1.000 | $1^{\{n\}}=1$                                                                                      | $1^n = 1$                                                    | Since $1^n = 1$ for all $n$ ,                                                                             |
| johnston-linear-matrix-atgso-2-F03819 | 410  | 1.000 | $\left(c_{\{0\}}+c_{\{1\}} \ n+c_{\{2\}} \ n^2\right) 1^{\{n\}}$                                   | $(c_0 + c_1n + c_2n^2) 1^n$                                  | $(c_0 + c_1n + c_2n^2) 1^n$                                                                               |
| johnston-linear-matrix-atgso-2-F03820 | 410  | 1.000 | $c_{\{0\}}+c_{\{1\}} \ n+c_{\{2\}} \ n^2$                                                          | $c_0 + c_1n + c_2n^2$                                        | $c_0 + c_1n + c_2n^2$ here.                                                                               |

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| johnston-linear-matrix-atglr_u<br>F03821 | 410   | ■ 1.000       | $1-\phi \approx -0.6180$                                                                            | $1-\phi \approx -0.6180$                                           | Since $1-\phi \approx -0.6180$ ,                                                                                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03822 | 410   | ■ 1.000       | $(1-\phi)^n \rightarrow 0$                                                                          | $(1-\phi)^n \rightarrow 0$                                         | $(1-\phi)^n \rightarrow 0$ as $n \rightarrow \infty$ .                                                                                                                                                                                                                                                                                                                  |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03823 | 410   | ■ 1.000       | $n \rightarrow \infty$                                                                              | $n \rightarrow \infty$                                             | $(1-\phi)^n \rightarrow 0$ as $n \rightarrow \infty$ .                                                                                                                                                                                                                                                                                                                  |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03824 | 410   | ■ 0.996       | $\lambda_0=1$                                                                                       | $\lambda_0 = 1$                                                    | which has roots $\lambda_0 = 1$ and $\lambda_1 = 3$ , with multiplicities 3 and 1, respectively.                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03825 | 410   | ■ 0.996       | $\lambda_1=3$                                                                                       | $\lambda_1 = 3$                                                    | which has roots $\lambda_0 = 1$ and $\lambda_1 = 3$ , with multiplicities 3 and 1, respectively.                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03826 | 410   | ■ 1.000       | $q_{\{0\}}(x)=c_{\{0\}}+c_{\{1\}}\ x+c_{\{2\}}\ x^{\{2\}}$                                          | $q_0(x) = c_0 + c_1x + c_2x^2$                                     | respectively, so they are of the form $q_0(x) = c_0 + c_1x + c_2x^2$ and $q_1(x) = c_3$ .                                                                                                                                                                                                                                                                               |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03827 | 410   | ■ 1.000       | $q_{\{1\}}(x)=c_{\{3\}}$                                                                            | $q_1(x) = c_3$                                                     | respectively, so they are of the form $q_0(x) = c_0 + c_1x + c_2x^2$ and $q_1(x) = c_3$ .                                                                                                                                                                                                                                                                               |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03828 | 410   | ■ 1.000       | $c_{\{0\}},\ c_{\{1\}},\ c_{\{2\}},\ c_{\{3\}}$                                                     | $c_0, c_1, c_2, c_3$                                               | Our goal is thus to find $c_0, c_1, c_2, c_3$ such that                                                                                                                                                                                                                                                                                                                 |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03829 | 410   | ■ 1.000       | $n=0,1,2$                                                                                           | $n = 0, 1, 2$                                                      | By plugging $n = 0, 1, 2$ , and 3 into this equation, we arrive at the linear                                                                                                                                                                                                                                                                                           |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03830 | 410   | ■ 1.000       | $\left(c_{\{0\}},\ c_{\{1\}},\ c_{\{2\}},\ c_{\{3\}}\right)=(1,2,3,-1)$                             | $(c_0, c_1, c_2, c_3) = (1, 2, 3, -1)$                             | which has unique solution $(c_0, c_1, c_2, c_3) = (1, 2, 3, -1)$ . It follows that                                                                                                                                                                                                                                                                                      |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03831 | 410   | ■ 1.000       | $F_{\{0\}}=0,\ F_{\{1\}}=1$                                                                         | $F_0 = 0, F_1 = 1$                                                 | satisfy $F_0 = 0, F_1 = 1$ , and $F_n = F_{n-1} + F_{n-2}$ when $n \geq 2$ ), as well as the ratios                                                                                                                                                                                                                                                                     |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03832 | 410   | ■ 1.000       | $F_{\{n\}}\ /\ F_{\{n-1\}}$                                                                         | $F_n/F_{n-1}$                                                      | <table><tr><th><math>n</math></th><th><math>F_n</math></th><th><math>F_n/F_{n-1}</math></th></tr><tr><td>0</td><td>0</td><td>–</td></tr><tr><td>1</td><td>1</td><td>–</td></tr><tr><td>2</td><td>1</td><td>1.0000</td></tr><tr><td>3</td><td>2</td><td>2.0000</td></tr><tr><td>4</td><td>3</td><td>1.5000</td></tr><tr><td>5</td><td>5</td><td>1.6667</td></tr></table> | $n$ | $F_n$ | $F_n/F_{n-1}$ | 0 | 0 | – | 1 | 1 | – | 2 | 1 | 1.0000 | 3 | 2 | 2.0000 | 4 | 3 | 1.5000 | 5 | 5 | 1.6667 |
| $n$                                      | $F_n$ | $F_n/F_{n-1}$ |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 0                                        | 0     | –             |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 1                                        | 1     | –             |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 2                                        | 1     | 1.0000        |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 3                                        | 2     | 2.0000        |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 4                                        | 3     | 1.5000        |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| 5                                        | 5     | 1.6667        |                                                                                                     |                                                                    |                                                                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03833 | 410   | ■ 1.000       | $\phi^{\{n\}}$                                                                                      | $\phi^n$                                                           | In particular, since $\phi^n$ is significantly larger than $(1-\phi)^n$ when $n$ is large, $F_n$                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03834 | 410   | ■ 1.000       | $(1-\phi)^{\{n\}}$                                                                                  | $(1-\phi)^n$                                                       | In particular, since $\phi^n$ is significantly larger than $(1-\phi)^n$ when $n$ is large, $F_n$                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03835 | 410   | ■ 1.000       | $\phi^{\{n\}}\ /\ \sqrt{5}$                                                                         | $\phi^n/\sqrt{5}$                                                  | grows in roughly the same way that $\phi^n/\sqrt{5}$ grows, so the Fibonacci sequence                                                                                                                                                                                                                                                                                   |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03836 | 411   | ■ 1.000       | $q_{\{0\}}(n)\ \lambda_{\{0\}}^{\{n\}}$                                                             | $q_0(n)\lambda_0^n$                                                | $q_0(n)\lambda_0^n$ here is a                                                                                                                                                                                                                                                                                                                                           |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03837 | 411   | ■ 1.000       | $q$                                                                                                 | $q$                                                                | $q$ and any number $c$ with $ c  < 1$ , we have                                                                                                                                                                                                                                                                                                                         |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03838 | 411   | ■ 0.887       | $q(n+1)$                                                                                            | $q(n+1)$                                                           | The two limits above are hopefully intuitive enough $q(n+1)$ and $q(n)$ have the                                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03839 | 411   | ■ 0.887       | $q(n)$                                                                                              | $q(n)$                                                             | The two limits above are hopefully intuitive enough $q(n+1)$ and $q(n)$ have the                                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03840 | 411   | ■ 1.000       | $x_{\{8\}}=6308$                                                                                    | $x_8 = 6308$                                                       | that every term is roughly 3 times as large as the previous one (e.g., $x_8 = 6308$ ,                                                                                                                                                                                                                                                                                   |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03841 | 411   | ■ 1.000       | $x_{\{7\}}=2056$                                                                                    | $x_7 = 2056$                                                       | which is roughly 3 times as large as $x_7 = 2056$ ). This now makes sense, since                                                                                                                                                                                                                                                                                        |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03842 | 412   | ■ 1.000       | $x_{\{0\}}=0,\ x_{\{1\}}=0,\ x_{\{2\}}=1,\ x_{\{n\}}=x_{\{n-1\}}-4\ x_{\{n-2\}}+4\ x_{\{n-3\}}$     | $x_0 = 0, x_1 = 0, x_2 = 1, x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$   | a) $x_0 = 0, x_1 = 0, x_2 = 1, x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$ when $n \geq 3$ , and                                                                                                                                                                                                                                                                               |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03843 | 412   | ■ 1.000       | $x_{\{0\}}=0,\ x_{\{1\}}=1,\ x_{\{2\}}=0,\ x_{\{n\}}=2\ x_{\{n-1\}}+9\ x_{\{n-2\}}-18\ x_{\{n-3\}}$ | $x_0 = 0, x_1 = 1, x_2 = 0, x_n = 2x_{n-1} + 9x_{n-2} - 18x_{n-3}$ | b) $x_0 = 0, x_1 = 1, x_2 = 0, x_n = 2x_{n-1} + 9x_{n-2} - 18x_{n-3}$ when $n \geq 3$ .                                                                                                                                                                                                                                                                                 |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |
| johnston-linear-matrix-atglr_u<br>F03844 | 412   | ■ 1.000       | $1,2\ i$                                                                                            | $1, 2i$                                                            | which has roots 1, $2i$ , and $-2i$ . Since the absolute values of these                                                                                                                                                                                                                                                                                                |     |       |               |   |   |   |   |   |   |   |   |        |   |   |        |   |   |        |   |   |        |

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| johnston-linear-matrix-atglu-<br>F03845 | 412  | ■ 1.000 | $-2\ i$                                                                                                                             | $-2i$                                                 | which has roots $1$ , $2i$ , and $-2i$ . Since the absolute values of these                              |
| johnston-linear-matrix-atglu-<br>F03846 | 412  | ■ 0.999 | $x_{\{n\}} / x_{\{n-1\}}=1$                                                                                                         | $x_n/x_{n-1} = 1$                                     | back and forth between repeating the previous term (so $x_n/x_{n-1} = 1$ )                               |
| johnston-linear-matrix-atglu-<br>F03847 | 412  | ■ 0.979 | $x_{\{n\}} / x_{\{n-1\}} \backslash \mathrm{approx} -4$                                                                             | $x_n/x_{n-1} \approx -4$                              | and growing by roughly a factor of $-4$ (so $x_n/x_{n-1} \approx -4$ ). The                              |
| johnston-linear-matrix-atglu-<br>F03848 | 412  | ■ 1.000 | $(2\ i)^{\{2\}}=(-2\ i)^{\{2\}}=-4$                                                                                                 | $(2i)^2 = (-2i)^2 = -4$                               | polynomial satisfy $(2i)^2 = (-2i)^2 = -4$ .                                                             |
| johnston-linear-matrix-atglu-<br>F03849 | 412  | ■ 1.000 | $x_{\{n\}} / x_{\{n-1\}}$                                                                                                           | $x_n/x_{n-1}$                                         | exist—the ratio $x_n/x_{n-1}$ alternates back and forth between $0$ and                                  |
| johnston-linear-matrix-atglu-<br>F03850 | 412  | ■ 1.000 | $3^{\{2\}}=(-3)^{\{2\}}=9$                                                                                                          | $3^2 = (-3)^2 = 9$                                    | its characteristic polynomial satisfy $3^2 = (-3)^2 = 9$ .                                               |
| johnston-linear-matrix-atglu-<br>F03851 | 413  | ■ 1.000 | $x_{\{n-3\}}, x_{\{n-11\}}, x_{\{n-12\}}$                                                                                           | $x_{n-3}, x_{n-11}, x_{n-12}$                         | of $x_{n-3}, x_{n-11}, x_{n-12}$ ,                                                                       |
| johnston-linear-matrix-atglu-<br>F03852 | 413  | ■ 0.915 | $x_{\{n-45\}}, x_{\{n-54\}}$                                                                                                        | $x_{n-45}, x_{n-54}$                                  | $x_{n-45}, x_{n-54}$ is $0$ .                                                                            |
| johnston-linear-matrix-atglu-<br>F03853 | 413  | ■ 1.000 | $x_{\{5\}}=6$                                                                                                                       | $x_5 = 6$                                             | the look-and-say sequence—for example, $x_5 = 6$ since the fifth terms in                                |
| johnston-linear-matrix-atglu-<br>F03854 | 413  | ■ 1.000 | $p(\backslash \mathrm{lambda})=(\backslash \mathrm{lambda}+1)(\backslash \mathrm{lambda}-1)^{\{2\}}\ q(\backslash \mathrm{lambda})$ | $p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$ | as $p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$ , where $q(\lambda)$ is the following degree-71 |
| johnston-linear-matrix-atglu-<br>F03855 | 413  | ■ 1.000 | $q(\backslash \mathrm{lambda})$                                                                                                     | $q(\lambda)$                                          | as $p(\lambda) = (\lambda + 1)(\lambda - 1)^2q(\lambda)$ , where $q(\lambda)$ is the following degree-71 |
| johnston-linear-matrix-atglu-<br>F03856 | 414  | ■ 0.992 | $x_{\{n\}}=x_{\{n-1\}}+6\ x_{\{n-2\}}$                                                                                              | $x_n = x_{n-1} + 6x_{n-2}$                            | *(a) $x_n = x_{n-1} + 6x_{n-2}$                                                                          |
| johnston-linear-matrix-atglu-<br>F03857 | 414  | ■ 1.000 | $x_{\{n\}}=6\ x_{\{n-1\}}-9\ x_{\{n-2\}}$                                                                                           | $x_n = 6x_{n-1} - 9x_{n-2}$                           | (b) $x_n = 6x_{n-1} - 9x_{n-2}$                                                                          |
| johnston-linear-matrix-atglu-<br>F03858 | 414  | ■ 0.995 | $x_{\{n\}}=6\ x_{\{n-1\}}-4\ x_{\{n-2\}}$                                                                                           | $x_n = 6x_{n-1} - 4x_{n-2}$                           | *(c) $x_n = 6x_{n-1} - 4x_{n-2}$                                                                         |
| johnston-linear-matrix-atglu-<br>F03859 | 414  | ■ 1.000 | $x_{\{n\}}=2\ x_{\{n-1\}}-2\ x_{\{n-2\}}$                                                                                           | $x_n = 2x_{n-1} - 2x_{n-2}$                           | (d) $x_n = 2x_{n-1} - 2x_{n-2}$                                                                          |
| johnston-linear-matrix-atglu-<br>F03860 | 414  | ■ 1.000 | $x_{\{n\}}=6\ x_{\{n-1\}}-11\ x_{\{n-2\}}+6\ x_{\{n-3\}}$                                                                           | $x_n = 6x_{n-1} - 11x_{n-2} + 6x_{n-3}$               | *(e) $x_n = 6x_{n-1} - 11x_{n-2} + 6x_{n-3}$                                                             |
| johnston-linear-matrix-atglu-<br>F03861 | 414  | ■ 1.000 | $x_{\{n\}}=2\ x_{\{n-2\}}+x_{\{n-3\}}$                                                                                              | $x_n = 2x_{n-2} + x_{n-3}$                            | (f) $x_n = 2x_{n-2} + x_{n-3}$                                                                           |
| johnston-linear-matrix-atglu-<br>F03862 | 414  | ■ 0.982 | $x_{\{n\}}=6\ x_{\{n-1\}}-12\ x_{\{n-2\}}+8\ x_{\{n-3\}}$                                                                           | $x_n = 6x_{n-1} - 12x_{n-2} + 8x_{n-3}$               | *(g) $x_n = 6x_{n-1} - 12x_{n-2} + 8x_{n-3}$                                                             |
| johnston-linear-matrix-atglu-<br>F03863 | 414  | ■ 1.000 | $x_{\{n\}}=7\ x_{\{n-1\}}-8\ x_{\{n-2\}}-16\ x_{\{n-3\}}$                                                                           | $x_n = 7x_{n-1} - 8x_{n-2} - 16x_{n-3}$               | (h) $x_n = 7x_{n-1} - 8x_{n-2} - 16x_{n-3}$                                                              |
| johnston-linear-matrix-atglu-<br>F03864 | 414  | ■ 1.000 | $x_{\{n\}}=5\ x_{\{n-1\}}-4\ x_{\{n-2\}}-6\ x_{\{n-3\}}$                                                                            | $x_n = 5x_{n-1} - 4x_{n-2} - 6x_{n-3}$                | *(i) $x_n = 5x_{n-1} - 4x_{n-2} - 6x_{n-3}$                                                              |
| johnston-linear-matrix-atglu-<br>F03865 | 414  | ■ 1.000 | $x_{\{n\}}=7\ x_{\{n-1\}}-17\ x_{\{n-2\}}+15\ x_{\{n-3\}}$                                                                          | $x_n = 7x_{n-1} - 17x_{n-2} + 15x_{n-3}$              | (j) $x_n = 7x_{n-1} - 17x_{n-2} + 15x_{n-3}$                                                             |
| johnston-linear-matrix-atglu-<br>F03866 | 414  | ■ 0.999 | $x_{\{0\}}=4,\ x_{\{1\}}=-3$                                                                                                        | $x_0 = 4, x_1 = -3$                                   | *(a) $x_0 = 4, x_1 = -3$ ,                                                                               |
| johnston-linear-matrix-atglu-<br>F03867 | 414  | ■ 1.000 | $x_{\{0\}}=1,\ x_{\{1\}}=0$                                                                                                         | $x_0 = 1, x_1 = 0$                                    | (b) $x_0 = 1, x_1 = 0$ ,                                                                                 |
| johnston-linear-matrix-atglu-<br>F03868 | 414  | ■ 0.993 | $x_{\{0\}}=2,\ x_{\{1\}}=0$                                                                                                         | $x_0 = 2, x_1 = 0$                                    | *(c) $x_0 = 2, x_1 = 0$ ,                                                                                |
| johnston-linear-matrix-atglu-<br>F03869 | 414  | ■ 1.000 | $x_{\{0\}}=2,\ x_{\{1\}}=3$                                                                                                         | $x_0 = 2, x_1 = 3$                                    | (d) $x_0 = 2, x_1 = 3$ ,                                                                                 |
| johnston-linear-matrix-atglu-<br>F03870 | 414  | ■ 0.997 | $x_{\{0\}}=0,\ x_{\{1\}}=2,\ x_{\{2\}}=8$                                                                                           | $x_0 = 0, x_1 = 2, x_2 = 8$                           | *(e) $x_0 = 0, x_1 = 2, x_2 = 8$ ,                                                                       |
| johnston-linear-matrix-atglu-<br>F03871 | 414  | ■ 1.000 | $x_{\{0\}}=0,\ x_{\{1\}}=2,\ x_{\{2\}}=16$                                                                                          | $x_0 = 0, x_1 = 2, x_2 = 16$                          | (f) $x_0 = 0, x_1 = 2, x_2 = 16$ ,                                                                       |
| johnston-linear-matrix-atglu-<br>F03872 | 414  | ■ 1.000 | $x_{\{0\}}=-2,\ x_{\{1\}}=1,\ x_{\{2\}}=5$                                                                                          | $x_0 = -2, x_1 = 1, x_2 = 5$                          | *(g) $x_0 = -2, x_1 = 1, x_2 = 5$ ,                                                                      |
| johnston-linear-matrix-atglu-<br>F03873 | 415  | ■ 1.000 | $x_{\{0\}}=7,\ x_{\{1\}}=8,\ x_{\{2\}}=4$                                                                                           | $x_0 = 7, x_1 = 8, x_2 = 4$                           | (h) $x_0 = 7, x_1 = 8, x_2 = 4$ ,                                                                        |

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| johnston-linear-matrix-at&#x2D;alg&#x2D;F03874 | 415  | ■ 1.000 | $x_{\{n\}}=4\ x_{\{n-1\}}-6\ x_{\{n-2\}}+4\ x_{\{n-3\}}$                                      | $x_n = 4x_{n-1} - 6x_{n-2} + 4x_{n-3}$                      | $x_n = 4x_{n-1} - 6x_{n-2} + 4x_{n-3}$ when $n \geq 3$                                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03875 | 415  | ■ 1.000 | $x_{\{n\}}=2^{\{n\}}-3^{\{n\}}$                                                               | $x_n = 2^n - 3^n$                                           | <b>(b)</b> The sequence defined by the formula $x_n = 2^n - 3^n$                              |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03876 | 415  | ■ 0.976 | $x_{\{n\}}=n^{\{2\}}-n^{\{3\}}$                                                               | $x_n = n^2 - n^3$                                           | <b>*(c)</b> The sequence defined by the formula $x_n = n^2 - n^3$                             |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03877 | 415  | ■ 1.000 | $x_{\{n\}}=\sqrt{2\ n}-$                                                                      | $x_n = \sqrt{2n}-$                                          | <b>(d)</b> The sequence defined by the formula $x_n = \sqrt{2n}-$                             |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03878 | 415  | ■ 1.000 | $\sqrt{3\ n}$                                                                                 | $\sqrt{3n}$                                                 | $\sqrt{3n}$ satisfies a linear recurrence relation.                                           |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03879 | 415  | ■ 0.916 | $\lim_{\{n\}\rightarrow\infty} x_{\{n+1\}} / x_{\{n\}}=4$                                     | $\lim_{n\rightarrow\infty} x_{n+1}/x_n = 4$                 | 2, then $\lim_{n\rightarrow\infty} x_{n+1}/x_n = 4$ .                                         |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03880 | 415  | ■ 0.853 | $x_{\{n\}}=2^{\{n\}}+3^{\{n\}}$                                                               | $x_n = 2^n + 3^n$                                           | <b>*(a)</b> $x_n = 2^n + 3^n$                                                                 |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03881 | 415  | ■ 1.000 | $x_{\{n\}}=2^{\{n\}}-3^{\{n+1\}}$                                                             | $x_n = 2^n - 3^{n+1}$                                       | <b>(b)</b> $x_n = 2^n - 3^{n+1}$                                                              |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03882 | 415  | ■ 0.977 | $x_{\{n\}}=2^{\{n\}}+3^{\{n\}}+4^{\{n\}}$                                                     | $x_n = 2^n + 3^n + 4^n$                                     | <b>*(c)</b> $x_n = 2^n + 3^n + 4^n$                                                           |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03883 | 415  | ■ 1.000 | $x_{\{n\}}=(1+n)\ 2^{\{n\}}+3^{\{n\}}$                                                        | $x_n = (1+n)2^n + 3^n$                                      | <b>(d)</b> $x_n = (1+n)2^n + 3^n$                                                             |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03884 | 415  | ■ 0.999 | $x_{\{n\}}=\left(1+n+n^{\{2\}}\right)\ 2^{\{n\}}+3^{\{n\}}$                                   | $x_n = (1+n+n^2)\ 2^n + 3^n$                                | <b>*(e)</b> $x_n = (1+n+n^2)2^n + 3^n$                                                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03885 | 415  | ■ 1.000 | $x_{\{n\}}=7+2^{\{n\}}$                                                                       | $x_n = 7 + 2^n$                                             | <b>(f)</b> $x_n = 7 + 2^n$                                                                    |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03886 | 415  | ■ 0.975 | $x_{\{n\}}=n^{\{2\}}$                                                                         | $x_n = n^2$                                                 | <b>*(g)</b> $x_n = n^2$                                                                       |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03887 | 415  | ■ 1.000 | $x_{\{n\}}=2(1+i)^{\{n\}}+2(1-i)^{\{n\}}$                                                     | $x_n = 2(1+i)^n + 2(1-i)^n$                                 | <b>(h)</b> $x_n = 2(1+i)^n + 2(1-i)^n$                                                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03888 | 415  | ■ 0.808 | $x_{\{n\}}=(n-5)\ 3^{\{n\}}+(3+i)(1-2\ i)^{\{n\}}+(3-i)(1+2\ i)^{\{n\}}$                      | $x_n = (n-5)3^n + (3+i)(1-2i)^n + (3-i)(1+2i)^n$            | <b>*(i)</b> $x_n = (n-5)3^n + (3+i)(1-2i)^n + (3-i)(1+2i)^n$                                  |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03889 | 415  | ■ 1.000 | $x_{\{n\}}=2^{\{n\}}\ \cos\left(\pi\ n\ /\ 4\right)-2^{\{n\}}\ \sin\left(\pi\ n\ /\ 4\right)$ | $x_n = 2^n \cos(\pi n/4) - 2^n \sin(\pi n/4)$               | <b>(j)</b> $x_n = 2^n \cos(\pi n/4) - 2^n \sin(\pi n/4)$                                      |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03890 | 415  | ■ 1.000 | $x_{\{n\}}=-x_{\{n-2\}}$                                                                      | $x_n = -x_{n-2}$                                            | $x_n = -x_{n-2}$ when $n \geq 2$                                                              |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03891 | 415  | ■ 1.000 | $x_{\{0\}}=4,\ x_{\{1\}}=4$                                                                   | $x_0 = 4, x_1 = 4$                                          | <b>(b)</b> $x_0 = 4, x_1 = 4,$                                                                |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03892 | 415  | ■ 1.000 | $x_{\{0\}}=3,\ x_{\{1\}}=1,\ x_{\{2\}}=-7$                                                    | $x_0 = 3, x_1 = 1, x_2 = -7$                                | <b>*(c)</b> $x_0 = 3, x_1 = 1, x_2 = -7,$                                                     |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03893 | 415  | ■ 1.000 | $x_{\{n\}}=x_{\{n-1\}}-4\ x_{\{n-2\}}+4\ x_{\{n-3\}}$                                         | $x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$                       | $x_n = x_{n-1} - 4x_{n-2} + 4x_{n-3}$ when $n \geq 3$                                         |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03894 | 415  | ■ 1.000 | $x_{\{0\}}=-1,\ x_{\{1\}}=0,\ x_{\{2\}}=8$                                                    | $x_0 = -1, x_1 = 0, x_2 = 8$                                | <b>(d)</b> $x_0 = -1, x_1 = 0, x_2 = 8,$                                                      |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03895 | 415  | ■ 1.000 | $x_{\{n\}}=4\ x_{\{n-1\}}-8\ x_{\{n-2\}}+8\ x_{\{n-3\}}$                                      | $x_n = 4x_{n-1} - 8x_{n-2} + 8x_{n-3}$                      | $x_n = 4x_{n-1} - 8x_{n-2} + 8x_{n-3}$ when $n \geq 3$                                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03896 | 415  | ■ 1.000 | $x_{\{n\}}=\left\lfloor\left(4+\sqrt{11}\right)^{\{n\}}\right\rfloor$                         | $x_n = \left\lfloor\left(4+\sqrt{11}\right)^n\right\rfloor$ | $x_n = \lfloor(4+\sqrt{11})^n\rfloor$ , where $\lfloor\cdot\rfloor$ denotes the “floor” func- |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03897 | 415  | ■ 1.000 | $\lfloor\cdot\rfloor$                                                                         | $\lfloor\cdot\rfloor$                                       | $x_n = \lfloor(4+\sqrt{11})^n\rfloor$ , where $\lfloor\cdot\rfloor$ denotes the “floor” func- |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03898 | 415  | ■ 1.000 | $4+\sqrt{11}$                                                                                 | $4+\sqrt{11}$                                               | relation should have $4+\sqrt{11}$ as a root.]                                                |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03899 | 415  | ■ 1.000 | $x_{\{n\}}=x_{\{n-1\}}-$                                                                      | $x_n = x_{n-1}-$                                            | quence defined by the linear recurrence relation $x_n = x_{n-1}-$                             |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03900 | 415  | ■ 1.000 | $4\ x_{\{n-2\}}+4\ x_{\{n-3\}}$                                                               | $4x_{n-2} + 4x_{n-3}$                                       | $4x_{n-2} + 4x_{n-3}$ , with initial conditions $x_0 = 0, x_1 = 0$ and                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03901 | 415  | ■ 1.000 | $x_{\{0\}}=0,\ x_{\{1\}}=0$                                                                   | $x_0 = 0, x_1 = 0$                                          | $4x_{n-2} + 4x_{n-3}$ , with initial conditions $x_0 = 0, x_1 = 0$ and                        |
| johnston-linear-matrix-at&#x2D;alg&#x2D;F03902 | 415  | ■ 1.000 | $x_{\{2\}}=1$                                                                                 | $x_2 = 1$                                                   | $x_2 = 1$ , repeats every term (for example, $x_0 = x_1, x_2 = x_3,$                          |

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| johnston-linear-matrix-atglu_a_F03903 | 415  | ■ 1.000 | $x_{\{0\}}=x_{\{1\}}, \ x_{\{2\}}=x_{\{3\}}$        | $x_0 = x_1, x_2 = x_3$                          | $x_2 = 1$ , repeats every term (for example, $x_0 = x_1, x_2 = x_3$ ,                                    |
| johnston-linear-matrix-atglu_a_F03904 | 417  | ■ 1.000 | $a \ b=b \ a, \ a+b=b+a$                            | $ab = ba, a + b = b + a$                        | That is, we still have all of the properties like $ab = ba, a + b = b + a$ , and                         |
| johnston-linear-matrix-atglu_a_F03905 | 417  | ■ 1.000 | $a, \ b, \ c \ \text{\in} \ \mathbb{C}$             | $a, b, c \in \mathbb{C}$                        | $a(b+c) = ab+ac$ for all $a, b, c \in \mathbb{C}$ that we would expect “numbers” to                      |
| johnston-linear-matrix-atglu_a_F03906 | 417  | ■ 0.992 | $\varepsilon \times 0=1$                            | $\varepsilon \times 0 = 1$                      | property that $\varepsilon \times 0 = 1$ (i.e., we are thinking of $\varepsilon$ as $1/0$ much like we   |
| johnston-linear-matrix-atglu_a_F03907 | 417  | ■ 0.992 | $1 \ / \ 0$                                         | $1/0$                                           | property that $\varepsilon \times 0 = 1$ (i.e., we are thinking of $\varepsilon$ as $1/0$ much like we   |
| johnston-linear-matrix-atglu_a_F03908 | 417  | ■ 1.000 | $\sqrt{-1}$                                         | $\sqrt{-1}$                                     | think of $i$ as $\sqrt{-1}$ ), then we have                                                              |
| johnston-linear-matrix-atglu_a_F03909 | 417  | ■ 1.000 | $(a+b \ i)+(c+d \ i)=(a+c)+(b+d) \ i$               | $(a + bi) + (c + di) = (a + c) + (b + d)i$      | add up their real and imaginary parts: $(a + bi) + (c + di) = (a + c) + (b + d)i$ ,                      |
| johnston-linear-matrix-atglu_a_F03910 | 417  | ■ 1.000 | $x^{\{2\}}-2 \ x+5=0$                               | $x^2 - 2x + 5 = 0$                              | For example, the equation $x^2 - 2x + 5 = 0$ has no real solutions, but it has                           |
| johnston-linear-matrix-atglu_a_F03911 | 417  | ■ 1.000 | $x=1+2 \ i$                                         | $x = 1 + 2i$                                    | $x = 1 + 2i$ as a complex solution, since                                                                |
| johnston-linear-matrix-atglu_a_F03912 | 418  | ■ 0.923 | $a+b \ i \ \text{\in} \ \mathbb{C}$                 | $a + bi \in \mathbb{C}$                         | complex number $a + bi \in \mathbb{C}$ as the vector $(a, b) \in \mathbb{R}^2$ ), but with a multiplica- |
| johnston-linear-matrix-atglu_a_F03913 | 418  | ■ 0.923 | $(a, \ b) \ \text{\in} \ \mathbb{R}^{\{2\}}$        | $(a, b) \in \mathbb{R}^2$                       | complex number $a + bi \in \mathbb{C}$ as the vector $(a, b) \in \mathbb{R}^2$ ), but with a multiplica- |
| johnston-linear-matrix-atglu_a_F03914 | 419  | ■ 1.000 | $(a+b \ i)(a-b \ i)$                                | $(a + bi)(a - bi)$                              | $(a + bi)(a - bi)$ here                                                                                  |
| johnston-linear-matrix-atglu_a_F03915 | 419  | ■ 0.957 | $(a \ b-a \ b) \ i=0$                               | $(ab - ab)i = 0$                                | $(ab - ab)i = 0$ .                                                                                       |
| johnston-linear-matrix-atglu_a_F03916 | 419  | ■ 1.000 | $z=a+b \ i$                                         | $z = a + bi$                                    | • The magnitude of a complex number $z = a + bi$ can be written in terms                                 |
| johnston-linear-matrix-atglu_a_F03917 | 419  | ■ 1.000 | $ z ^{\{2\}}=z \ \bar{z}$                           | $ z ^2 = z\bar{z}$                              | of the product of $z$ with its complex conjugate: $ z ^2 = z\bar{z}$ , since                             |
| johnston-linear-matrix-atglu_a_F03918 | 419  | ■ 1.000 | $z= z  \ u$                                         | $z =  z u$                                      | $z =  z u$ , where $ z $ is the magnitude of $z$ and $u$ is a number on the unit circle in               |
| johnston-linear-matrix-atglu_a_F03919 | 419  | ■ 1.000 | $ z $                                               | $ z $                                           | $z =  z u$ , where $ z $ is the magnitude of $z$ and $u$ is a number on the unit circle in               |
| johnston-linear-matrix-atglu_a_F03920 | 419  | ■ 1.000 | $u$                                                 | $u$                                             | $z =  z u$ , where $ z $ is the magnitude of $z$ and $u$ is a number on the unit circle in               |
| johnston-linear-matrix-atglu_a_F03921 | 419  | ■ 1.000 | $\theta \ \text{\in} [0,2 \ \pi)$                   | $\theta \in [0, 2\pi)$                          | $\theta \in [0, 2\pi)$ means                                                                             |
| johnston-linear-matrix-atglu_a_F03922 | 419  | ■ 1.000 | $2 \ \pi$                                           | $2\pi$                                          | (inclusive) and $2\pi$                                                                                   |
| johnston-linear-matrix-atglu_a_F03923 | 419  | ■ 1.000 | $\cos \ (\theta)+i \ \sin \ (\theta)$               | $\cos(\theta) + i \sin(\theta)$                 | unit circle in the complex plane can be written in the form $\cos(\theta) + i \sin(\theta)$ , as         |
| johnston-linear-matrix-atglu_a_F03924 | 419  | ■ 1.000 | $z=$                                                | $z =$                                           | It follows that we can write every complex number in the form $z =$                                      |
| johnston-linear-matrix-atglu_a_F03925 | 419  | ■ 1.000 | $ z (\cos \ (\theta)+i \ \sin \ (\theta))$          | $ z (\cos(\theta) + i \sin(\theta))$            | $ z (\cos(\theta) + i \sin(\theta))$ . However, we can simplify this expression somewhat                 |
| johnston-linear-matrix-atglu_a_F03926 | 419  | ■ 1.000 | $\operatorname{Re}(e^{i \ \theta})=\cos \ (\theta)$ | $\operatorname{Re}(e^{i\theta}) = \cos(\theta)$ | $\operatorname{Re}(e^{i\theta}) = \cos(\theta)$ and                                                      |
| johnston-linear-matrix-atglu_a_F03927 | 419  | ■ 0.995 | $\operatorname{Im}(e^{i \ \theta})=\sin \ (\theta)$ | $\operatorname{Im}(e^{i\theta}) = \sin(\theta)$ | $\operatorname{Im}(e^{i\theta}) = \sin(\theta)$ .                                                        |
| johnston-linear-matrix-atglu_a_F03928 | 419  | ■ 1.000 | $e^{i \ \theta}$                                    | $e^{i\theta}$                                   | this formula simply as a definition—just think of $e^{i\theta}$ as a shorthand for the                   |
| johnston-linear-matrix-atglu_a_F03929 | 420  | ■ 0.980 | $ \cos \ (\theta)+i \ \sin \ (\theta) ^2=$          | $ \cos(\theta) + i \sin(\theta) ^2 =$           | $ \cos(\theta) + i \sin(\theta) ^2 =$                                                                    |
| johnston-linear-matrix-atglu_a_F03930 | 420  | ■ 1.000 | $\cos ^{\{2\}}(\theta)+\sin ^{\{2\}}(\theta)=1$     | $\cos^2(\theta) + \sin^2(\theta) = 1$           | $\cos^2(\theta) + \sin^2(\theta) = 1$ ,                                                                  |

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|-------------------------------------------|------|---------|---------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-atgskw_2<br>F03931 | 420  | ■ 0.964 | $\operatorname{sign}(b) = \pm 1$                                    | $\operatorname{sign}(b) = \pm 1$                                  | $\operatorname{sign}(b) = \pm 1$ ,                                                                                             |
| johnston-linear-matrix-atgskw_2<br>F03932 | 420  | ■ 1.000 | $b < 0$                                                             | $b < 0$                                                           | or negative. If $b < 0$                                                                                                        |
| johnston-linear-matrix-atgskw_2<br>F03933 | 420  | ■ 1.000 | $-\pi < \theta < 0$                                                 | $-\pi < \theta < 0$                                               | $-\pi < \theta < 0$ , which                                                                                                    |
| johnston-linear-matrix-atgskw_2<br>F03934 | 420  | ■ 1.000 | $[0, 2 \pi)$                                                        | $[0, 2\pi)$                                                       | interval $[0, 2\pi)$ by                                                                                                        |
| johnston-linear-matrix-atgskw_2<br>F03935 | 420  | ■ 1.000 | $e^x, \cos(x)$                                                      | $e^x, \cos(x)$                                                    | $e^x$ , $\cos(x)$ , and $\sin(x)$ have Taylor series                                                                           |
| johnston-linear-matrix-atgskw_2<br>F03936 | 420  | ■ 1.000 | $x = i \theta$                                                      | $x = i\theta$                                                     | respectively. Plugging $x = i\theta$ into the Taylor series for $e^x$ then gives                                               |
| johnston-linear-matrix-atgskw_2<br>F03937 | 420  | ■ 1.000 | $z = r e^{i \theta}$                                                | $z = r e^{i\theta}$                                               | number $z \in \mathbb{C}$ in the form $z = r e^{i\theta}$ , where $r$ is the magnitude of $z$ (i.e., $r =  z $ )               |
| johnston-linear-matrix-atgskw_2<br>F03938 | 420  | ■ 1.000 | $r =  z $                                                           | $r =  z $                                                         | number $z \in \mathbb{C}$ in the form $z = r e^{i\theta}$ , where $r$ is the magnitude of $z$ (i.e., $r =  z $ )               |
| johnston-linear-matrix-atgskw_2<br>F03939 | 420  | ■ 1.000 | $\left(r_1 e^{i \theta_1}\right) \left(r_2 e^{i \theta_2}\right) =$ | $\left(r_1 e^{i\theta_1}\right) \left(r_2 e^{i\theta_2}\right) =$ | in polar form, but multiplication is quite straightforward: $\left(r_1 e^{i\theta_1}\right) \left(r_2 e^{i\theta_2}\right) =$  |
| johnston-linear-matrix-atgskw_2<br>F03940 | 420  | ■ 1.000 | $\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$          | $\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$        | $\left(r_1 r_2\right) e^{i\left(\theta_1+\theta_2\right)}$ . We can thus think of complex numbers as stretched rotations—      |
| johnston-linear-matrix-atgskw_2<br>F03941 | 420  | ■ 1.000 | $\left(r e^{i \theta}\right)^n = r^n e^{i n \theta}$                | $\left(r e^{i\theta}\right)^n = r^n e^{i n \theta}$               | multiplying a complex number in polar form by itself gives $\left(r e^{i\theta}\right)^n = r^n e^{i n \theta}$ for             |
| johnston-linear-matrix-atgskw_2<br>F03942 | 421  | ■ 1.000 | $e^{2 \pi i}=e^{0 i}=1$                                             | $e^{2 \pi i}=e^{0 i}=1$                                           | that $e^{2 \pi i}=e^{0 i}=1$ .                                                                                                 |
| johnston-linear-matrix-atgskw_2<br>F03943 | 421  | ■ 1.000 | $z=i$                                                               | $z=i$                                                             | Compute all 3 cube roots of the complex number $z=i$ .                                                                         |
| johnston-linear-matrix-atgskw_2<br>F03944 | 421  | ■ 1.000 | $z=e^{\pi i / 2}$                                                   | $z=e^{\pi i / 2}$                                                 | We start by writing $z$ in its polar form $z=e^{\pi i / 2}$ , from which we see                                                |
| johnston-linear-matrix-atgskw_2<br>F03945 | 421  | ■ 1.000 | $e^{\pi i / 6}=(\sqrt{3}+i) / 2$                                    | $e^{\pi i / 6}=(\sqrt{3}+i) / 2$                                  | Geometrically, the cube root $e^{\pi i / 6}=(\sqrt{3}+i) / 2$ of $i$ lies on the unit                                          |
| johnston-linear-matrix-atgskw_2<br>F03946 | 421  | ■ 1.000 | $r^{1 / n} e^{i \theta / n}$                                        | $r^{1 / n} e^{i \theta / n}$                                      | $r^{1 / n} e^{i \theta / n}$ its <b>principal <math>n</math>-th root</b> , which we denote by $z^{1 / n}$ . The principal root |
| johnston-linear-matrix-atgskw_2<br>F03947 | 421  | ■ 1.000 | $z^{1 / n}$                                                         | $z^{1 / n}$                                                       | $r^{1 / n} e^{i \theta / n}$ its <b>principal <math>n</math>-th root</b> , which we denote by $z^{1 / n}$ . The principal root |
| johnston-linear-matrix-atgskw_2<br>F03948 | 422  | ■ 1.000 | $p: \mathbb{C} \rightarrow \mathbb{C}$                              | $p: \mathbb{C} \rightarrow \mathbb{C}$                            | $p: \mathbb{C} \rightarrow \mathbb{C}$ with                                                                                    |
| johnston-linear-matrix-atgskw_2<br>F03949 | 422  | ■ 1.000 | $\theta=0$                                                          | $\theta=0$                                                        | $z$ is a positive real number (i.e., $\theta=0$ ) then its principal roots are positive real                                   |
| johnston-linear-matrix-atgskw_2<br>F03950 | 422  | ■ 0.996 | $-i$                                                                | $-i$                                                              | root of $-1$ is $i$ , not $-i$ ), and we showed in Example A.1.1 that the principal                                            |
| johnston-linear-matrix-atgskw_2<br>F03951 | 422  | ■ 1.000 | $z=e^{\pi i / 2}=i$                                                 | $z=e^{\pi i / 2}=i$                                               | cube root of $z=e^{\pi i / 2}=i$ is $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$ .                                               |
| johnston-linear-matrix-atgskw_2<br>F03952 | 422  | ■ 1.000 | $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$                          | $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$                        | cube root of $z=e^{\pi i / 2}=i$ is $z^{1 / 3}=e^{\pi i / 6}=(\sqrt{3}+i) / 2$ .                                               |
| johnston-linear-matrix-atgskw_2<br>F03953 | 422  | ■ 1.000 | $p: \mathbb{R} \rightarrow \mathbb{R}$                              | $p: \mathbb{R} \rightarrow \mathbb{R}$                            | A <b>polynomial</b> is a function $p: \mathbb{R} \rightarrow \mathbb{R}$ of the form                                           |
| johnston-linear-matrix-atgskw_2<br>F03954 | 422  | ■ 1.000 | $a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$                | $a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$              | where $a_0, a_1, a_2, \ldots, a_{n-1}, a_n \in \mathbb{R}$ are constants (called the <b>coefficients</b> of $p$ ).             |
| johnston-linear-matrix-atgskw_2<br>F03955 | 422  | ■ 0.995 | $a_n \neq 0$                                                        | $a_n \neq 0$                                                      | example, the polynomial displayed above has degree $n$ , as long as $a_n \neq 0$ ).                                            |
| johnston-linear-matrix-atgskw_2<br>F03956 | 422  | ■ 1.000 | $p(x)=0$                                                            | $p(x)=0$                                                          | which are the solutions to the equation $p(x)=0$ . Geometrically, the roots of                                                 |
| johnston-linear-matrix-atgskw_2<br>F03957 | 422  | ■ 1.000 | $x=0.5$                                                             | $x=0.5$                                                           | $x=0.5$ , and $x=1.5$ .                                                                                                        |

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| johnston-linear-matrix-atgkwoe<br>F03958 | 422  | ■ 1.000 | $x=1.5$                            | $x = 1.5$              | $x = 0.5$ , and $x = 1.5$ .                                                                               |
| johnston-linear-matrix-atgkwoe<br>F03959 | 423  | ■ 1.000 | $x^{\{2\}}-6\ x+8$                 | $x^2 - 6x + 8$         | a) $x^2 - 6x + 8$ ,                                                                                       |
| johnston-linear-matrix-atgkwoe<br>F03960 | 423  | ■ 1.000 | $2\ x^{\{2\}}+8\ x+8$              | $2x^2 + 8x + 8$        | b) $2x^2 + 8x + 8$ , and                                                                                  |
| johnston-linear-matrix-atgkwoe<br>F03961 | 423  | ■ 1.000 | $x^{\{2\}}+2\ x+3$                 | $x^2 + 2x + 3$         | c) $x^2 + 2x + 3$ .                                                                                       |
| johnston-linear-matrix-atgkwoe<br>F03962 | 423  | ■ 1.000 | $a=1,\ b=-6$                       | $a = 1, b = -6$        | a) This quadratic has $a = 1$ , $b = -6$ , and $c = 8$ , so the quadratic                                 |
| johnston-linear-matrix-atgkwoe<br>F03963 | 423  | ■ 1.000 | $c=8$                              | $c = 8$                | a) This quadratic has $a = 1$ , $b = -6$ , and $c = 8$ , so the quadratic                                 |
| johnston-linear-matrix-atgkwoe<br>F03964 | 423  | ■ 1.000 | $a=2,\ b=8$                        | $a = 2, b = 8$         | b) This quadratic has $a = 2$ , $b = 8$ , and $c = 8$ , so the quadratic formula                          |
| johnston-linear-matrix-atgkwoe<br>F03965 | 423  | ■ 1.000 | $a=1,\ b=2$                        | $a = 1, b = 2$         | c) This quadratic has $a = 1$ , $b = 2$ , and $c = 3$ , so the quadratic formula                          |
| johnston-linear-matrix-atgkwoe<br>F03966 | 423  | ■ 1.000 | $c=3$                              | $c = 3$                | c) This quadratic has $a = 1$ , $b = 2$ , and $c = 3$ , so the quadratic formula                          |
| johnston-linear-matrix-atgkwoe<br>F03967 | 424  | ■ 1.000 | $b\ /\ c$                          | $b/c$                  | $b/c$ , where $b$ and $c$                                                                                 |
| johnston-linear-matrix-atgkwoe<br>F03968 | 424  | ■ 1.000 | $-1\ \pm\ i\ \sqrt{2}$             | $-1 \pm i\sqrt{2}$     | are $-1 \pm i\sqrt{2}$ . This quadratic is displayed below (notice that it does                           |
| johnston-linear-matrix-atgkwoe<br>F03969 | 424  | ■ 1.000 | $a_{\{0\}},\ a_{\{n\}}\ \neq\ 0$   | $a_0, a_n \neq 0$      | where $a_0, a_1, a_2, \dots, a_{n-1}, a_n \in \mathbb{R}$ are integers and $a_0, a_n \neq 0$ . Then every |
| johnston-linear-matrix-atgkwoe<br>F03970 | 424  | ■ 1.000 | $a_{\{0\}}$                        | $a_0$                  | rational root of $p$ has the form $b/c$ , where $b$ is a divisor of $a_0$ and $c$ is a                    |
| johnston-linear-matrix-atgkwoe<br>F03971 | 424  | ■ 1.000 | $a_{\{n\}}$                        | $a_n$                  | divisor of $a_n$ .                                                                                        |
| johnston-linear-matrix-atgkwoe<br>F03972 | 424  | ■ 1.000 | $x^{\{2\}}-2\ x-3$                 | $x^2 - 2x - 3$         | a) $x^2 - 2x - 3$ ,                                                                                       |
| johnston-linear-matrix-atgkwoe<br>F03973 | 424  | ■ 1.000 | $2\ x^{\{3\}}-9\ x^{\{2\}}-6\ x+5$ | $2x^3 - 9x^2 - 6x + 5$ | b) $2x^3 - 9x^2 - 6x + 5$ , and                                                                           |
| johnston-linear-matrix-atgkwoe<br>F03974 | 424  | ■ 1.000 | $3\ x^{\{3\}}-14\ x+4$             | $3x^3 - 14x + 4$       | c) $3x^3 - 14x + 4$ .                                                                                     |
| johnston-linear-matrix-atgkwoe<br>F03975 | 424  | ■ 1.000 | $a_{\{2\}}=1$                      | $a_2 = 1$              | a) Since $a_2 = 1$ and $a_0 = 3$ , the rational root theorem tells us that the                            |
| johnston-linear-matrix-atgkwoe<br>F03976 | 424  | ■ 1.000 | $a_{\{0\}}=3$                      | $a_0 = 3$              | a) Since $a_2 = 1$ and $a_0 = 3$ , the rational root theorem tells us that the                            |
| johnston-linear-matrix-atgkwoe<br>F03977 | 424  | ■ 0.273 | $\pm 1$                            | $\pm 1$                | only possible rational roots of this polynomial are $\pm 1$ and $\pm 3$ (i.e.,                            |
| johnston-linear-matrix-atgkwoe<br>F03978 | 424  | ■ 0.273 | $\pm 3$                            | $\pm 3$                | only possible rational roots of this polynomial are $\pm 1$ and $\pm 3$ (i.e.,                            |
| johnston-linear-matrix-atgkwoe<br>F03979 | 424  | ■ 1.000 | $x=-\ 1$                           | $x = -1$               | The rational roots of this polynomial are thus $x = -1$ and $x = 3$                                       |
| johnston-linear-matrix-atgkwoe<br>F03980 | 424  | ■ 1.000 | $x=3$                              | $x = 3$                | The rational roots of this polynomial are thus $x = -1$ and $x = 3$                                       |
| johnston-linear-matrix-atgkwoe<br>F03981 | 425  | ■ 1.000 | $a_{\{3\}}=2$                      | $a_3 = 2$              | b) Since $a_3 = 2$ and $a_0 = 5$ , the rational root theorem tells us that the                            |
| johnston-linear-matrix-atgkwoe<br>F03982 | 425  | ■ 1.000 | $a_{\{0\}}=5$                      | $a_0 = 5$              | b) Since $a_3 = 2$ and $a_0 = 5$ , the rational root theorem tells us that the                            |
| johnston-linear-matrix-atgkwoe<br>F03983 | 425  | ■ 0.967 | $x=5,\ x=1\ /\ 2$                  | $x = 5, x = 1/2$       | It follows that the rational roots of this polynomial are $x = 5, x = 1/2$ ,                              |
| johnston-linear-matrix-atgkwoe<br>F03984 | 425  | ■ 1.000 | $a_{\{3\}}=3$                      | $a_3 = 3$              | c) Since $a_3 = 3$ and $a_0 = 4$ , the rational root theorem tells us that                                |
| johnston-linear-matrix-atgkwoe<br>F03985 | 425  | ■ 1.000 | $a_{\{0\}}=4$                      | $a_0 = 4$              | c) Since $a_3 = 3$ and $a_0 = 4$ , the rational root theorem tells us that                                |

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|-----------------------------------------|------|-------|---------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-atlas-<br>F03986 | 425  | 0.880 | $\pm 4/3, \pm 2/3$                          | $\pm 4/3, \pm 2/3$                            | $\pm 4/3, \pm 2/3$ , and $\pm 1/3$ . Plugging these values into the polynomial                                          |
| johnston-linear-matrix-atlas-<br>F03987 | 425  | 0.880 | $\pm 1/3$                                   | $\pm 1/3$                                     | $\pm 4/3, \pm 2/3$ , and $\pm 1/3$ . Plugging these values into the polynomial                                          |
| johnston-linear-matrix-atlas-<br>F03988 | 425  | 1.000 | $p(x)=3x^3-14x+4$                           | $p(x) = 3x^3 - 14x + 4$                       | the cubic $p(x) = 3x^3 - 14x + 4$ . However, this cubic actually has two other                                          |
| johnston-linear-matrix-atlas-<br>F03989 | 425  | 1.000 | $\sqrt{2}, \pi$                             | $\sqrt{2}, \pi$                               | include $\sqrt{2}, \pi$ , and $e$ .                                                                                     |
| johnston-linear-matrix-atlas-<br>F03990 | 425  | 1.000 | $p(x)$                                      | $p(x)$                                        | $x = 2$ is one of its roots to try to factor $p(x)$ as                                                                  |
| johnston-linear-matrix-atlas-<br>F03991 | 425  | 0.699 | $\left(p(x)=3x^3-14x+4\right)$              | $(p(x) = 3x^3 - 14x + 4)$                     | <b>Step 1:</b> Divide the highest-degree term in the dividend ( $p(x) = 3x^3 - 14x + 4$ )                               |
| johnston-linear-matrix-atlas-<br>F03992 | 425  | 0.999 | $(x-2)$                                     | $(x - 2)$                                     | by the highest-degree term in the divisor ( $x - 2$ ). In this case, we divide                                          |
| johnston-linear-matrix-atlas-<br>F03993 | 425  | 1.000 | $3x^3$                                      | $3x^3$                                        | $3x^3$ by $x$ , getting $3x^2$ as our result:                                                                           |
| johnston-linear-matrix-atlas-<br>F03994 | 425  | 1.000 | $3x^2$                                      | $3x^2$                                        | $3x^3$ by $x$ , getting $3x^2$ as our result:                                                                           |
| johnston-linear-matrix-atlas-<br>F03995 | 426  | 0.999 | $0x^2 - \left(-6x^2\right) = 6x^2$          | $0x^2 - (-6x^2) = 6x^2$                       | $0x^2 - (-6x^2) = 6x^2$ .                                                                                               |
| johnston-linear-matrix-atlas-<br>F03996 | 426  | 0.517 | $x-2$                                       | $x - 2$                                       | <b>Step 2:</b> Multiply the divisor ( $x - 2$ ) by the quotient from Step 1 ( $3x^2$ ) and subtract                     |
| johnston-linear-matrix-atlas-<br>F03997 | 426  | 1.000 | $2\left(6x^2-14x+4\right)$                  | $2(6x^2 - 14x + 4)$                           | subtraction in Step 2 ( $6x^2 - 14x + 4$ ) as the new dividend:                                                         |
| johnston-linear-matrix-atlas-<br>F03998 | 426  | 0.933 | $\left(q(x)=3x^2+6x-2\right)$               | $(q(x) = 3x^2 + 6x - 2)$                      | polynomial written at the top of this calculation ( $q(x) = 3x^2 + 6x - 2$ ).                                           |
| johnston-linear-matrix-atlas-<br>F03999 | 426  | 1.000 | $q(x)=3x^2+6x-2$                            | $q(x) = 3x^2 + 6x - 2$                        | find the roots of $q(x) = 3x^2 + 6x - 2$ , which are                                                                    |
| johnston-linear-matrix-atlas-<br>F04000 | 426  | 0.996 | $2, -1+\sqrt{5/3}$                          | $2, -1 + \sqrt{5/3}$                          | We (finally!) have shown that the roots of $p$ are $2, -1 + \sqrt{5/3}$ , and $-1 - \sqrt{5/3}$ .                       |
| johnston-linear-matrix-atlas-<br>F04001 | 426  | 0.996 | $-1-\sqrt{5/3}$                             | $-1 - \sqrt{5/3}$                             | We (finally!) have shown that the roots of $p$ are $2, -1 + \sqrt{5/3}$ , and $-1 - \sqrt{5/3}$ .                       |
| johnston-linear-matrix-atlas-<br>F04002 | 426  | 0.997 | $p(x)=3x^4+4x^3-14x^2-11x-2$                | $p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2$        | Find all roots of the polynomial $p(x) = 3x^4 + 4x^3 - 14x^2 - 11x - 2$ .                                               |
| johnston-linear-matrix-atlas-<br>F04003 | 426  | 1.000 | $a_4=3$                                     | $a_4 = 3$                                     | Since $a_4 = 3$ and $a_0 = -2$ , the rational root theorem tells us that the                                            |
| johnston-linear-matrix-atlas-<br>F04004 | 426  | 1.000 | $a_0=-2$                                    | $a_0 = -2$                                    | Since $a_4 = 3$ and $a_0 = -2$ , the rational root theorem tells us that the                                            |
| johnston-linear-matrix-atlas-<br>F04005 | 426  | 1.000 | $x=-1/3$                                    | $x = -1/3$                                    | $x = -1/3$ are the only ones that are <i>actually</i> roots of $p$ .                                                    |
| johnston-linear-matrix-atlas-<br>F04006 | 427  | 1.000 | $x^2+3x+1$                                  | $x^2 + 3x + 1$                                | to $x^2 + 3x + 1$ :                                                                                                     |
| johnston-linear-matrix-atlas-<br>F04007 | 427  | 1.000 | $2, \frac{-1}{3}, \frac{1}{2}(-3+\sqrt{5})$ | $2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$ | It follows that the roots of $p$ are $2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$ , and $\frac{1}{2}(-3 - \sqrt{5})$ . |
| johnston-linear-matrix-atlas-<br>F04008 | 427  | 1.000 | $\frac{1}{2}(-3-\sqrt{5})$                  | $\frac{1}{2}(-3 - \sqrt{5})$                  | It follows that the roots of $p$ are $2, \frac{-1}{3}, \frac{1}{2}(-3 + \sqrt{5})$ , and $\frac{1}{2}(-3 - \sqrt{5})$ . |
| johnston-linear-matrix-atlas-<br>F04009 | 427  | 1.000 | $x-r$                                       | $x - r$                                       | If $r$ is a root of a polynomial $p$ and we divide $p(x)$ by $x - r$ via polynomial                                     |
| johnston-linear-matrix-atlas-<br>F04010 | 427  | 1.000 | $p(x)=(x-r)q(x)$                            | $p(x) = (x - r)q(x)$                          | $p(x) = (x - r)q(x)$ .                                                                                                  |
| johnston-linear-matrix-atlas-<br>F04011 | 428  | 1.000 | $p(x)=x^2+1$                                | $p(x) = x^2 + 1$                              | Some polynomials like $p(x) = x^2 + 1$ do not have any real roots at all, so the                                        |

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| johnston-linear-matrix-atqzwa<br>F04012 | 428  | ■ 0.999 | $x^{\{n\}}$                               | $x^n$                        | where $a$ is the coefficient of $x^n$ in $p$ and $r_1, r_2, \dots, r_n$ are the (potentially com- |
| johnston-linear-matrix-atqzwa<br>F04013 | 428  | ■ 0.999 | $r_{\{1\}}, r_{\{2\}}, \ldots, r_{\{n\}}$ | $r_1, r_2, \dots, r_n$       | where $a$ is the coefficient of $x^n$ in $p$ and $r_1, r_2, \dots, r_n$ are the (potentially com- |
| johnston-linear-matrix-atqzwa<br>F04014 | 428  | ■ 0.679 | $2 x^{\{2\}}+8 x+8=2(x+2)^{\{2\}}$        | $2x^2 + 8x + 8 = 2(x + 2)^2$ | For example, the root $-2$ of the polynomial $2x^2 + 8x + 8 = 2(x + 2)^2$ has                     |
| johnston-linear-matrix-atqzwa<br>F04015 | 428  | ■ 0.922 | $x+2$                                     | $x + 2$                      | multiplicity 2, since the term $x + 2$ is squared, while all of the other roots that              |
| johnston-linear-matrix-atqzwa<br>F04016 | 429  | ■ 1.000 | $x y=y x$                                 | $xy = yx$                    | properties of the real numbers, like the fact that $xy = yx$ for all real numbers $x$             |
| johnston-linear-matrix-atqzwa<br>F04017 | 429  | ■ 1.000 | $m^{\{2\}}$                               | $m^2$                        | If $m$ is an even integer then so is $m^2$ .                                                      |
| johnston-linear-matrix-atqzwa<br>F04018 | 429  | ■ 1.000 | $m=2 k$                                   | $m = 2k$                     | <i>Proof.</i> If $m$ is an even integer then there is some integer $k$ such that $m = 2k$ .       |
| johnston-linear-matrix-atqzwa<br>F04019 | 429  | ■ 1.000 | $2 k^{\{2\}}$                             | $2k^2$                       | which is also even since it is 2 times the integer $2k^2$ .                                       |
| johnston-linear-matrix-atqzwa<br>F04020 | 429  | ■ 1.000 | $m=2 k+1$                                 | $m = 2k + 1$                 | odd integer $m$ we would write $m = 2k + 1$ for some integer $k$ , and if we were                 |
| johnston-linear-matrix-atqzwa<br>F04021 | 429  | ■ 1.000 | $m=7 k$                                   | $m = 7k$                     | trying to prove a theorem about multiples of 7 then we would write $m = 7k$ .                     |
| johnston-linear-matrix-atqzwa<br>F04022 | 429  | ■ 1.000 | $m^{\{2\}}=(2 k)^{\{2\}}$                 | $m^2 = (2k)^2$               | based on that, so we then simply compute $m^2 = (2k)^2$ , and we finish the proof                 |
| johnston-linear-matrix-atqzwa<br>F04023 | 429  | ■ 1.000 | $(2 k)^{\{2\}}=2\left(2 k^{\{2\}}\right)$ | $(2k)^2 = 2(2k^2)$           | by again using the definition of “even integer”: we write $(2k)^2 = 2(2k^2)$ , which              |
| johnston-linear-matrix-atqzwa<br>F04024 | 429  | ■ 1.000 | $x^{\{2\}}+y^{\{2\}} \geq 2 x y$          | $x^2 + y^2 \geq 2xy$         | If $x$ and $y$ are real numbers then $x^2 + y^2 \geq 2xy$ .                                       |
| johnston-linear-matrix-atqzwa<br>F04025 | 429  | ■ 0.997 | $x-$                                      | $x-$                         | <i>Proof.</i> Recall that the square of every real number is non-negative, so $(x-$               |
| johnston-linear-matrix-atqzwa<br>F04026 | 429  | ■ 1.000 | $y)^{\{2\}} \geq 0$                       | $y)^2 \geq 0$                | $y)^2 \geq 0$ . However, using some algebra to expand $(x - y)^2$ shows that                      |
| johnston-linear-matrix-atqzwa<br>F04027 | 429  | ■ 1.000 | $(x-y)^{\{2\}}$                           | $(x - y)^2$                  | $y)^2 \geq 0$ . However, using some algebra to expand $(x - y)^2$ shows that                      |
| johnston-linear-matrix-atqzwa<br>F04028 | 429  | ■ 1.000 | $2 x y$                                   | $2xy$                        | Adding $2xy$ to both sides of this inequality shows that $2xy \leq x^2 + y^2$ , as desired.       |
| johnston-linear-matrix-atqzwa<br>F04029 | 429  | ■ 1.000 | $2 x y \leq x^{\{2\}}+y^{\{2\}}$          | $2xy \leq x^2 + y^2$         | Adding $2xy$ to both sides of this inequality shows that $2xy \leq x^2 + y^2$ , as desired.       |
| johnston-linear-matrix-atqzwa<br>F04030 | 430  | ■ 0.889 | $x^{\{2\}}+y^{\{2\}}-2 x y$               | $x^2 + y^2 - 2xy$            | all we did in that proof—we factored $x^2 + y^2 - 2xy$ as $(x - y)^2$ ). This same trick          |
| johnston-linear-matrix-atqzwa<br>F04031 | 430  | ■ 0.998 | $a, b, c$                                 | $a, b, c$                    | If $a, b, c$ , and $x$ are real or complex numbers such that $ax^2 + bx + c = 0$                  |
| johnston-linear-matrix-atqzwa<br>F04032 | 430  | ■ 0.998 | $a x^{\{2\}}+b x+c=0$                     | $ax^2 + bx + c = 0$          | If $a, b, c$ , and $x$ are real or complex numbers such that $ax^2 + bx + c = 0$                  |
| johnston-linear-matrix-atqzwa<br>F04033 | 430  | ■ 1.000 | $b x$                                     | $bx$                         | groups that linear $bx$ term and quadratic $ax^2$ term together:                                  |
| johnston-linear-matrix-atqzwa<br>F04034 | 430  | ■ 1.000 | $a x^{\{2\}}$                             | $ax^2$                       | groups that linear $bx$ term and quadratic $ax^2$ term together:                                  |
| johnston-linear-matrix-atqzwa<br>F04035 | 431  | ■ 1.000 | $m^{\{2\}}=2 k$                           | $m^2 = 2k$                   | writing $m^2 = 2k$ for                                                                            |
| johnston-linear-matrix-atqzwa<br>F04036 | 431  | ■ 1.000 | $Q^{\{\prime \prime\}}$                   | $Q''$                        | only if $Q''$ is                                                                                  |
| johnston-linear-matrix-atqzwa<br>F04037 | 431  | ■ 1.000 | $2\left(2 k^{\{2\}}+2 k\right)$           | $2(2k^2 + 2k)$               | which is also odd since it is 1 larger than the even integer $2(2k^2 + 2k)$ .                     |
| johnston-linear-matrix-atqzwa<br>F04038 | 431  | ■ 0.597 | $2 \times 1$                              | $2 \times 1$                 | chessboard by $2 \times 1$ dominoes. That is, we lay $2 \times 1$ dominoes on top of the          |

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| johnston-linear-matrix-attrib F04039 | 432  | ■ 0.997 | $8 \times 8$                                  | $8 \times 8$                          | $8 \times 8$ chessboard.                                                                                |
| johnston-linear-matrix-attrib F04040 | 432  | ■ 0.666 | $8 \times 8 = 64$                             | $8 \times 8 = 64$                     | remove one of the $8 \times 8 = 64$ squares from the chessboard, then the remaining                     |
| johnston-linear-matrix-attrib F04041 | 434  | ■ 1.000 | $\sqrt{2}=a / b$                              | $\sqrt{2} = a/b$                      | There do not exist integers $a$ and $b$ such that $\sqrt{2} = a/b$ . In other words,                    |
| johnston-linear-matrix-attrib F04042 | 434  | ■ 1.000 | $a / b$                                       | $a/b$                                 | form $a/b$ , where $a$                                                                                  |
| johnston-linear-matrix-attrib F04043 | 434  | ■ 1.000 | $\sqrt{2} \ b=a$                              | $\sqrt{2}b = a$                       | Multiplying the equation $\sqrt{2} = a/b$ through by $b$ shows that $\sqrt{2}b = a$ , and               |
| johnston-linear-matrix-attrib F04044 | 434  | ■ 1.000 | $2 \ b^{\{2\}}=a^{\{2\}}$                     | $2b^2 = a^2$                          | then squaring shows that $2b^2 = a^2$ , so $a^2$ is even. It follows from Theorem A.3.5                 |
| johnston-linear-matrix-attrib F04045 | 434  | ■ 1.000 | $a^{\{2\}}$                                   | $a^2$                                 | then squaring shows that $2b^2 = a^2$ , so $a^2$ is even. It follows from Theorem A.3.5                 |
| johnston-linear-matrix-attrib F04046 | 434  | ■ 1.000 | $a=2 \ k$                                     | $a = 2k$                              | that $a$ is even and can be written in the form $a = 2k$ for some integer $k$ . Since                   |
| johnston-linear-matrix-attrib F04047 | 435  | ■ 1.000 | $2 \ b^{\{2\}}=(2 \ k)^{\{2\}}=4 \ k^{\{2\}}$ | $2b^2 = (2k)^2 = 4k^2$                | Plugging $a = 2k$ into the equation $2b^2 = a^2$ shows that $2b^2 = (2k)^2 = 4k^2$ ,                    |
| johnston-linear-matrix-attrib F04048 | 435  | ■ 1.000 | $b^{\{2\}}=2 \ k^{\{2\}}$                     | $b^2 = 2k^2$                          | so $b^2 = 2k^2$ . It follows that $b^2$ is even, so using Theorem A.3.5 again shows                     |
| johnston-linear-matrix-attrib F04049 | 435  | ■ 1.000 | $b^{\{2\}}$                                   | $b^2$                                 | so $b^2 = 2k^2$ . It follows that $b^2$ is even, so using Theorem A.3.5 again shows                     |
| johnston-linear-matrix-attrib F04050 | 435  | ■ 1.000 | $1+3+5+\cdots+(2 \ n-1)=n^{\{2\}}$            | $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ | perfect squares (i.e., we have shown that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ , at                    |
| johnston-linear-matrix-attrib F04051 | 435  | ■ 1.000 | $n \ \leq 5$                                  | $n \leq 5$                            | least for $n \leq 5$ ). We expect that this pattern continues forever, but how could                    |
| johnston-linear-matrix-attrib F04052 | 435  | ■ 0.999 | $P(1), \ P(2), \ P(3), \ \ldots$              | $P(1), P(2), P(3), \dots$             | $P(1), P(2), P(3), \dots$ , indexed by the natural numbers 1, 2, 3, $\dots$ , then proof by             |
| johnston-linear-matrix-attrib F04053 | 435  | ■ 0.999 | $1, 2, 3, \ \ldots$                           | 1, 2, 3, $\dots$                      | $P(1), P(2), P(3), \dots$ , indexed by the natural numbers 1, 2, 3, $\dots$ , then proof by             |
| johnston-linear-matrix-attrib F04054 | 435  | ■ 1.000 | $P(1)$                                        | $P(1)$                                | • The <b>base case</b> : we start by showing that $P(1)$ is true.                                       |
| johnston-linear-matrix-attrib F04055 | 435  | ■ 1.000 | $P(n)$                                        | $P(n)$                                | $P(n)$ holds for a                                                                                      |
| johnston-linear-matrix-attrib F04056 | 435  | ■ 1.000 | $P(n+1)$                                      | $P(n + 1)$                            | $n$ , then $P(n + 1)$ must also be true.                                                                |
| johnston-linear-matrix-attrib F04057 | 435  | ■ 1.000 | $P(2)$                                        | $P(2)$                                | $P(2)$ : True by the inductive step: if $P(1)$ is true (it is!) then $P(2)$ is true too.                |
| johnston-linear-matrix-attrib F04058 | 435  | ■ 1.000 | $P(3)$                                        | $P(3)$                                | $P(3)$ : True by the inductive step: if $P(2)$ is true (it is!) then $P(3)$ is true too.                |
| johnston-linear-matrix-attrib F04059 | 435  | ■ 0.988 | $P(4)$                                        | $P(4)$                                | $P(4)$ : True because $P(3)$ is true. You get the idea...                                               |
| johnston-linear-matrix-attrib F04060 | 435  | ■ 1.000 | $1=1$                                         | $1 = 1$                               | that $1 = 1$ , this step is complete.                                                                   |
| johnston-linear-matrix-attrib F04061 | 436  | ■ 1.000 | $2 \ n+1$                                     | $2n + 1$                              | The trick that lets us do this is to add $2n + 1$ to both sides of $P(n)$ :                             |
| johnston-linear-matrix-attrib F04062 | 436  | ■ 0.995 | $1, 3, 5, \ \ldots, \ 2 \ n-1$                | $1, 3, 5, \dots, 2n - 1$              | of 1, 3, 5, $\dots, 2n - 1$ squares, it follows that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$ , as         |
| johnston-linear-matrix-attrib F04063 | 437  | ■ 0.992 | $\left[A_{\{i, \ j\}}\right]_{\{k, \ \ell\}}$ | $[A_{i,j}]_{k,\ell}$                  | block it is. Specifically, we use $[A_{i,j}]_{k,\ell}$ to refer to the $(k, \ell)$ -entry of the matrix |
| johnston-linear-matrix-attrib F04064 | 437  | ■ 0.992 | $(k, \ \ell)$                                 | $(k, \ell)$                           | block it is. Specifically, we use $[A_{i,j}]_{k,\ell}$ to refer to the $(k, \ell)$ -entry of the matrix |
| johnston-linear-matrix-attrib F04065 | 437  | ■ 1.000 | $A_{\{i, \ j\}}$                              | $A_{i,j}$                             | $A_{i,j}$ , which itself is the matrix that makes up the $i$ -th block row and $j$ -th block            |

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| johnston-linear-matrix-attrib F04066 | 438  | 1.000 | $R, S \in \mathcal{M}_{m,n}$          | $R, S \in \mathcal{M}_{m,n}$          | matrices $R, S \in \mathcal{M}_{m,n}$ , both of which are in reduced row echelon form, then                                           |
| johnston-linear-matrix-attrib F04067 | 438  | 1.000 | $R=S$                                 | $R = S$                               | $R = S$ .                                                                                                                             |
| johnston-linear-matrix-attrib F04068 | 438  | 1.000 | $R \neq S$                            | $R \neq S$                            | To this end, suppose for the sake of contradiction that $R \neq S$ . Let $\widetilde{R}$ and $\widetilde{S}$                          |
| johnston-linear-matrix-attrib F04069 | 438  | 1.000 | $\widetilde{R}$                       | $\widetilde{R}$                       | To this end, suppose for the sake of contradiction that $R \neq S$ . Let $\widetilde{R}$ and $\widetilde{S}$                          |
| johnston-linear-matrix-attrib F04070 | 438  | 1.000 | $\widetilde{S}$                       | $\widetilde{S}$                       | To this end, suppose for the sake of contradiction that $R \neq S$ . Let $\widetilde{R}$ and $\widetilde{S}$                          |
| johnston-linear-matrix-attrib F04071 | 439  | 1.000 | $\widetilde{\mathbf{R}}$              | $\widetilde{\mathbf{R}}$              | <b>Case 1: <math>\widetilde{\mathbf{R}}</math> has the first form.</b> Then the linear system associated with $\widetilde{R}$ has the |
| johnston-linear-matrix-attrib F04072 | 439  | 0.960 | $\mathbf{c}=\mathbf{b}$               | $\mathbf{c} = \mathbf{b}$             | the unique solution $\mathbf{b}$ , so it must also have the first form with $\mathbf{c} = \mathbf{b}$ (and                            |
| johnston-linear-matrix-attrib F04073 | 439  | 1.000 | $\widetilde{S}=\widetilde{R}$         | $\widetilde{S} = \widetilde{R}$       | thus $\widetilde{S} = \widetilde{R}$ ).                                                                                               |
| johnston-linear-matrix-attrib F04074 | 439  | 1.000 | $\widetilde{R}=\widetilde{S}$         | $\widetilde{R} = \widetilde{S}$       | We have $\widetilde{R} = \widetilde{S}$ in both cases, which contradicts the fact that we chose them                                  |
| johnston-linear-matrix-attrib F04075 | 439  | 1.000 | $I_m$                                 | $I_m$                                 | $I_m$ results in a matrix $E$ , then $EA = B$ .                                                                                       |
| johnston-linear-matrix-attrib F04076 | 439  | 1.000 | $E \ A=B$                             | $EA = B$                              | $I_m$ results in a matrix $E$ , then $EA = B$ .                                                                                       |
| johnston-linear-matrix-attrib F04077 | 439  | 0.976 | $(j, j)$                              | $(j, j)$                              | case, $E$ is diagonal with all diagonal entries equal to 1, except for the $(j, j)$ -                                                 |
| johnston-linear-matrix-attrib F04078 | 440  | 1.000 | $\mathbf{e}_j+c \ \mathbf{e}_i$       | $\mathbf{e}_j + c\mathbf{e}_i$        | of $E$ is $\mathbf{e}_j + c\mathbf{e}_i$ , not                                                                                        |
| johnston-linear-matrix-attrib F04079 | 440  | 1.000 | $\mathbf{e}_i+c \ \mathbf{e}_j$       | $\mathbf{e}_i + c\mathbf{e}_j$        | $\mathbf{e}_i + c\mathbf{e}_j$ . The reason                                                                                           |
| johnston-linear-matrix-attrib F04080 | 440  | 1.000 | $\mathbf{e}_1 \ \mathbf{a}_1^T$       | $\mathbf{e}_1 \mathbf{a}_1^T$         | $\mathbf{e}_1 \mathbf{a}_1^T$ are matrices,                                                                                           |
| johnston-linear-matrix-attrib F04081 | 440  | 1.000 | $\mathbf{a}_1^T$                      | $\mathbf{a}_1^T$                      | is $m \times 1$ and $\mathbf{a}_1^T$ is $1 \times n$ .                                                                                |
| johnston-linear-matrix-attrib F04082 | 440  | 1.000 | $\{1,2, \ldots, n\}, \ \sigma$        | $\{1, 2, \dots, n\}, \sigma$          | $\{1, 2, \dots, n\}$ , $\sigma$ is a                                                                                                  |
| johnston-linear-matrix-attrib F04083 | 440  | 1.000 | $\mathbf{e}_i \ \mathbf{a}_j^T$       | $\mathbf{e}_i \mathbf{a}_j^T$         | Since $\mathbf{e}_i \mathbf{a}_j^T$ is the matrix that contains the entries from the $j$ -th row of $A$ in its                        |
| johnston-linear-matrix-attrib F04084 | 440  | 1.000 | $A+c \ \mathbf{e}_i \ \mathbf{a}_j^T$ | $A + c\mathbf{e}_i \mathbf{a}_j^T$    | $i$ -th row, it follows that $A + c\mathbf{e}_i \mathbf{a}_j^T$ is exactly the matrix $B$ that is obtained by                         |
| johnston-linear-matrix-attrib F04085 | 441  | 1.000 | $v+c \ \mathbf{w}$                    | $v + c\mathbf{w}$                     | the $v + c\mathbf{w}$ column                                                                                                          |
| johnston-linear-matrix-attrib F04086 | 441  | 0.996 | $\sigma(j)=j$                         | $\sigma(j) = j$                       | is the identity permutation (i.e., $\sigma(j) = j$ for all $1 \leq j \leq n$ ) then                                                   |
| johnston-linear-matrix-attrib F04087 | 442  | 1.000 | $p_A^{\prime}$                        | $p_A'$                                | The notation $p_A'$                                                                                                                   |
| johnston-linear-matrix-attrib F04088 | 442  | 1.000 | $\frac{\partial}{\partial \lambda_i}$ | $\frac{\partial}{\partial \lambda_i}$ | The notation $\frac{\partial}{\partial \lambda_i}$                                                                                    |
| johnston-linear-matrix-attrib F04089 | 442  | 1.000 | $\lambda_i$                           | $\lambda_i$                           | respect to $\lambda_i$ .                                                                                                              |
| johnston-linear-matrix-attrib F04090 | 442  | 1.000 | $\mu \neq 0$                          | $\mu \neq 0$                          | because $\mu \neq 0$ (this                                                                                                            |
| johnston-linear-matrix-attrib F04091 | 442  | 1.000 | $L( \mathbf{w} )=\mu$                 | $L( \mathbf{w} ) = \mu$               | so if $ \lambda  = \mu$ then in fact we have $L( \mathbf{w} ) = \mu$ . Claims 1 and 3 then tell us that                               |
| johnston-linear-matrix-attrib F04092 | 442  | 1.000 | $A \mathbf{w} =\mu \mathbf{w} $       | $A \mathbf{w}  = \mu \mathbf{w} $     | $A \mathbf{w}  = \mu \mathbf{w} $ and $ \mathbf{w}  > \mathbf{0}$ . It follows that                                                   |

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| johnston-linear-matrix-algebra-F04093 | 442  | 1.000 | $ \mathbf{w}  > \mathbf{0}$                      | $ \mathbf{w}  > \mathbf{0}$                | $A \mathbf{w}  = \mu \mathbf{w} $ and $ \mathbf{w}  > \mathbf{0}$ . It follows that                                |
| johnston-linear-matrix-algebra-F04094 | 442  | 0.948 | $(A - B) \mathbf{w}  = \mathbf{0}$               | $(A - B) \mathbf{w}  = \mathbf{0}$         | but since $A - B$ is entrywise non-negative, this is only possible if $(A - B) \mathbf{w}  = \mathbf{0}$ .         |
| johnston-linear-matrix-algebra-F04095 | 442  | 1.000 | $A - B = 0$                                      | $A - B = O$                                | Since $ \mathbf{w}  > \mathbf{0}$ , this then implies $A - B = O$ , so $A = B$ , which completes the               |
| johnston-linear-matrix-algebra-F04096 | 442  | 1.000 | $p_A(\lambda) = \det(A -$                        | $p_A(\lambda) = \det(A -$                  | when we take the derivative of the characteristic polynomial $p_A(\lambda) = \det(A -$                             |
| johnston-linear-matrix-algebra-F04097 | 442  | 0.986 | $\lambda I)$                                     | $\lambda I)$                               | $\lambda I)$ of $A$ . To show that the algebraic multiplicity of $\mu$ is 1, it suffices to show                   |
| johnston-linear-matrix-algebra-F04098 | 442  | 1.000 | $p_A'(\mu) \neq 0$                               | $p_A'(\mu) \neq 0$                         | that $p_A'(\mu) \neq 0$ (via Theorem B.6.1, for example). To see why this is the case,                             |
| johnston-linear-matrix-algebra-F04099 | 442  | 1.000 | $f: \mathbb{R}^n \rightarrow \mathbb{R}$         | $f: \mathbb{R}^n \rightarrow \mathbb{R}$   | consider the somewhat uglier multivariable function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ defined by            |
| johnston-linear-matrix-algebra-F04100 | 442  | 0.999 | $\Lambda$                                        | $\Lambda$                                  | where $\Lambda$ is the diagonal matrix whose diagonal entries are $\lambda_1, \lambda_2, \dots, \lambda_n$ . If we |
| johnston-linear-matrix-algebra-F04101 | 442  | 1.000 | $A_{\{i\}}$                                      | $A_i$                                      | let $A_i$ denote the $(n - 1) \times (n - 1)$ matrix that is obtained by removing the $i$ -th                      |
| johnston-linear-matrix-algebra-F04102 | 442  | 1.000 | $\Lambda_{\{i\}}$                                | $\Lambda_i$                                | row and column from $A$ (and similarly for $\Lambda_i$ from $\Lambda$ ), and we let $c_{i,j}$ denote               |
| johnston-linear-matrix-algebra-F04103 | 442  | 1.000 | $A - \Lambda$                                    | $A - \Lambda$                              | the $(i, j)$ -cofactor of $A - \Lambda$ , then applying a cofactor expansion along the $i$ -th                     |
| johnston-linear-matrix-algebra-F04104 | 442  | 1.000 | $a_{\{i, 1\}}, \dots, a_{\{i, n\}}$              | $a_{i,1}, \dots, a_{i,n}$                  | Since none of $a_{i,1}, \dots, a_{i,n}$ or $c_{i,1}, \dots, c_{i,n}$ depend on $\lambda_i$ , it follows that       |
| johnston-linear-matrix-algebra-F04105 | 442  | 1.000 | $c_{\{i, 1\}}, \dots, c_{\{i, n\}}$              | $c_{i,1}, \dots, c_{i,n}$                  | Since none of $a_{i,1}, \dots, a_{i,n}$ or $c_{i,1}, \dots, c_{i,n}$ depend on $\lambda_i$ , it follows that       |
| johnston-linear-matrix-algebra-F04106 | 442  | 1.000 | $B_{\{i\}}$                                      | $B_i$                                      | If we now let $B_i$ be the matrix obtained by replacing the $i$ -th row and column                                 |
| johnston-linear-matrix-algebra-F04107 | 442  | 1.000 | $p_{B_{\{i\}}}(\lambda) = -\lambda p_A(\lambda)$ | $p_{B_i}(\lambda) = -\lambda p_A(\lambda)$ | column of zeros at the $i$ -th index), then $p_{B_i}(\lambda) = -\lambda p_A(\lambda)$ for all $i$ , so            |
| johnston-linear-matrix-algebra-F04108 | 442  | 1.000 | $0 \leq B_{\{i\}} \leq A$                        | $O \leq B_i \leq A$                        | Since $O \leq B_i \leq A$ , we know from our earlier claim that the absolute values                                |
| johnston-linear-matrix-algebra-F04109 | 442  | 1.000 | $p_{B_{\{i\}}}(\mu) > 0$                         | $p_{B_i}(\mu) > 0$                         | $p_{B_i}(\mu) > 0$ for all $i$ (if $n$ is even) or $p_{B_i}(\mu) < 0$ for all $i$ (if $n$ is odd). Either          |
| johnston-linear-matrix-algebra-F04110 | 442  | 1.000 | $p_{B_{\{i\}}}(\mu) < 0$                         | $p_{B_i}(\mu) < 0$                         | $p_{B_i}(\mu) > 0$ for all $i$ (if $n$ is even) or $p_{B_i}(\mu) < 0$ for all $i$ (if $n$ is odd). Either          |
| johnston-linear-matrix-algebra-F04111 | 443  | 1.000 | $p^{(k)}$                                        | $p^{(k)}$                                  | The notation $p^{(k)}$                                                                                             |
| johnston-linear-matrix-algebra-F04112 | 443  | 0.999 | $(fg)'(x) =$                                     | $(fg)'(x) =$                               | that $(fg)'(x) =$                                                                                                  |
| johnston-linear-matrix-algebra-F04113 | 443  | 0.715 | $f'(x)g(x) + f(x)g'(x)$                          | $f'(x)g(x) + f(x)g'(x)$                    | $f'(x)g(x) + f(x)g'(x)$ .                                                                                          |
| johnston-linear-matrix-algebra-F04114 | 443  | 1.000 | $m - 1$                                          | $m - 1$                                    | $m$ if and only if $r$ is a root of each of $p$ and its first $m - 1$ derivatives:                                 |
| johnston-linear-matrix-algebra-F04115 | 443  | 1.000 | $p, p', \dots, p^{(m-1)}$                        | $p, p', \dots, p^{(m-1)}$                  | exactly $m$ then is a root of $p, p', \dots, p^{(m-1)}$ , but not of $p^{(m)}$ .                                   |
| johnston-linear-matrix-algebra-F04116 | 443  | 1.000 | $p^{(m)}$                                        | $p^{(m)}$                                  | exactly $m$ then is a root of $p, p', \dots, p^{(m-1)}$ , but not of $p^{(m)}$ .                                   |
| johnston-linear-matrix-algebra-F04117 | 443  | 0.981 | $p(x) =$                                         | $p(x) =$                                   | Recall that the factor theorem (Theorem A.2.2) lets us factor $p$ as $p(x) =$                                      |
| johnston-linear-matrix-algebra-F04118 | 443  | 1.000 | $(x - r)^m q(x)$                                 | $(x - r)^m q(x)$                           | $(x - r)^m q(x)$ , where $q$ is some polynomial with $q(r) \neq 0$ . We begin by showing                           |
| johnston-linear-matrix-algebra-F04119 | 443  | 1.000 | $q(r) \neq 0$                                    | $q(r) \neq 0$                              | $(x - r)^m q(x)$ , where $q$ is some polynomial with $q(r) \neq 0$ . We begin by showing                           |



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| johnston-linear-matrix-algebra-F04147 | 444  | ■ 0.931 | $\infty / \infty$                                                          | $\infty / \infty$                                                          | exists and is an indeterminate form of type “0/0” or “ $\infty / \infty$ ” (along with some                     |
| johnston-linear-matrix-algebra-F04148 | 444  | ■ 1.000 | $q(x)=c_{\{k\}} x^{\{k\}}+\cdots+c_{\{1\}} x+c_{\{0\}}$                    | $q(x)=c_k x^k+\cdots+c_1 x+c_0$                                            | explicitly as $q(x)=c_k x^k+\cdots+c_1 x+c_0$ . Then using L’Hôpital’s rule a total of                          |
| johnston-linear-matrix-algebra-F04149 | 445  | ■ 1.000 | $b^{\{x\}}$                                                                | $b^x$                                                                      | derivative of $b^x$ is                                                                                          |
| johnston-linear-matrix-algebra-F04150 | 445  | ■ 1.000 | $b^{\{x\}} \ln (b)$                                                        | $b^x \ln (b)$                                                              | $b^x \ln (b)$ .                                                                                                 |
| johnston-linear-matrix-algebra-F04151 | 445  | ■ 0.996 | $c_{\{k\}} k !$                                                            | $c_k k !$                                                                  | $c_k k !$ ), whereas the                                                                                        |
| johnston-linear-matrix-algebra-F04152 | 445  | ■ 1.000 | $b=1 / c$                                                                  | $b=1 / c$                                                                  | To evaluate the other limit (B.7.1), we define $b=1 / c$ and notice that $ b >1$                                |
| johnston-linear-matrix-algebra-F04153 | 445  | ■ 1.000 | $ b >1$                                                                    | $ b >1$                                                                    | To evaluate the other limit (B.7.1), we define $b=1 / c$ and notice that $ b >1$                                |
| johnston-linear-matrix-algebra-F04154 | 446  | ■ 1.000 | $\mathbf{v}=2 \mathbf{e}_{\{3\}}$                                          | $\mathbf{v}=2 \mathbf{e}_3$                                                | (a) $\mathbf{v}=2 \mathbf{e}_3$                                                                                 |
| johnston-linear-matrix-algebra-F04155 | 446  | ■ 0.992 | $\mathbf{x}=\mathbf{e}_{\{1\}}+2 \mathbf{e}_{\{2\}}$                       | $\mathbf{x}=\mathbf{e}_1+2 \mathbf{e}_2$                                   | (c) $\mathbf{x}=\mathbf{e}_1+2 \mathbf{e}_2$                                                                    |
| johnston-linear-matrix-algebra-F04156 | 446  | ■ 1.000 | $c(1,2)=(3,4)$                                                             | $c(1,2)=(3,4)$                                                             | 1.1.12 Algebraically, we note that $c(1,2)=(3,4)$ implies                                                       |
| johnston-linear-matrix-algebra-F04157 | 446  | ■ 1.000 | $2 c=4$                                                                    | $2 c=4$                                                                    | $c=3$ and $2 c=4$ , so $c=2$ . Since $c$ cannot simulta-                                                        |
| johnston-linear-matrix-algebra-F04158 | 446  | ■ 1.000 | $c=2$                                                                      | $c=2$                                                                      | $c=3$ and $2 c=4$ , so $c=2$ . Since $c$ cannot simulta-                                                        |
| johnston-linear-matrix-algebra-F04159 | 446  | ■ 0.990 | $(3,4)$                                                                    | $(3,4)$                                                                    | we notice that $(1,2)$ and $(3,4)$ point in different di-                                                       |
| johnston-linear-matrix-algebra-F04160 | 446  | ■ 1.000 | $\mathbf{v}+\mathbf{w}=\mathbf{x}$                                         | $\mathbf{v}+\mathbf{w}=\mathbf{x}$                                         | 1.1.9 We want vectors $\mathbf{v}$ and $\mathbf{w}$ such that $\mathbf{v}+\mathbf{w}=\mathbf{x}$ and            |
| johnston-linear-matrix-algebra-F04161 | 446  | ■ 1.000 | $\mathbf{v}-\mathbf{w}=\mathbf{y}$                                         | $\mathbf{v}-\mathbf{w}=\mathbf{y}$                                         | $\mathbf{v}-\mathbf{w}=\mathbf{y}$ . Adding these two equations together                                        |
| johnston-linear-matrix-algebra-F04162 | 446  | ■ 1.000 | $2 \mathbf{v}=\mathbf{x}+\mathbf{y}$                                       | $2 \mathbf{v}=\mathbf{x}+\mathbf{y}$                                       | tells us that $2 \mathbf{v}=\mathbf{x}+\mathbf{y}$ , and subtracting the second                                 |
| johnston-linear-matrix-algebra-F04163 | 446  | ■ 1.000 | $2 \mathbf{w}=\mathbf{x}-\mathbf{y}$                                       | $2 \mathbf{w}=\mathbf{x}-\mathbf{y}$                                       | equation from the first tells us that $2 \mathbf{w}=\mathbf{x}-\mathbf{y}$ , so                                 |
| johnston-linear-matrix-algebra-F04164 | 446  | ■ 1.000 | $\mathbf{v}=(\mathbf{x}+\mathbf{y}) / 2=(2,1)$                             | $\mathbf{v}=(\mathbf{x}+\mathbf{y}) / 2=(2,1)$                             | $\mathbf{v}=(\mathbf{x}+\mathbf{y}) / 2=(2,1)$ and $\mathbf{w}=(\mathbf{x}-\mathbf{y}) / 2=(1,-3)$ .            |
| johnston-linear-matrix-algebra-F04165 | 446  | ■ 1.000 | $\mathbf{w}=(\mathbf{x}-\mathbf{y}) / 2=(1,-3)$                            | $\mathbf{w}=(\mathbf{x}-\mathbf{y}) / 2=(1,-3)$                            | $\mathbf{v}=(\mathbf{x}+\mathbf{y}) / 2=(2,1)$ and $\mathbf{w}=(\mathbf{x}-\mathbf{y}) / 2=(1,-3)$ .            |
| johnston-linear-matrix-algebra-F04166 | 446  | ■ 0.996 | $\mathbf{x}=(2,2)$                                                         | $\mathbf{x}=(2,2)$                                                         | (a) $\mathbf{x}=(2,2)$                                                                                          |
| johnston-linear-matrix-algebra-F04167 | 446  | ■ 1.000 | $\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$                            | $\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$                            | (a) $\mathbf{x}=\frac{1}{2}(\mathbf{v}-\mathbf{w})$                                                             |
| johnston-linear-matrix-algebra-F04168 | 447  | ■ 1.000 | $\ \mathbf{v}\ =5, \mathbf{u}=(3 / 5,4 / 5)$                               | $\ \mathbf{v}\ =5, \mathbf{u}=(3 / 5,4 / 5)$                               | 1.2.2 (a) $\ \mathbf{v}\ =5, \mathbf{u}=(3 / 5,4 / 5)$                                                          |
| johnston-linear-matrix-algebra-F04169 | 447  | ■ 1.000 | $\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2} / 3,-1 / 2, \sqrt{10} / 6,1 / 2)$ | $\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2} / 3,-1 / 2, \sqrt{10} / 6,1 / 2)$ | (c) $\ \mathbf{v}\ =6, \mathbf{u}=(-\sqrt{2} / 3,-1 / 2, \sqrt{10} / 6,1 / 2)$                                  |
| johnston-linear-matrix-algebra-F04170 | 447  | ■ 1.000 | $\arccos (-1 / \sqrt{2})=3 \pi / 4$                                        | $\arccos (-1 / \sqrt{2})=3 \pi / 4$                                        | (c) $\arccos (-1 / \sqrt{2})=3 \pi / 4$                                                                         |
| johnston-linear-matrix-algebra-F04171 | 447  | ■ 1.000 | $\mathbf{x}=(1,-1)$                                                        | $\mathbf{x}=(1,-1)$                                                        | $\mathbf{w}=(1,1)$ , and $\mathbf{x}=(1,-1)$ . In this case, $\mathbf{v} \cdot \mathbf{w} =$                    |
| johnston-linear-matrix-algebra-F04172 | 447  | ■ 1.000 | $\mathbf{v} \cdot \mathbf{x}=1$                                            | $\mathbf{v} \cdot \mathbf{x}=1$                                            | $\mathbf{v} \cdot \mathbf{x}=1$ , but $\mathbf{w} \neq \mathbf{x}$ .                                            |
| johnston-linear-matrix-algebra-F04173 | 447  | ■ 1.000 | $\ \mathbf{v}+\mathbf{w}\  \leq\ \mathbf{v}\ +\ \mathbf{w}\ $              | $\ \mathbf{v}+\mathbf{w}\  \leq\ \mathbf{v}\ +\ \mathbf{w}\ $              | (c) True. The triangle inequality tells us that $\ \mathbf{v}+\mathbf{w}\  \leq\ \mathbf{v}\ +\ \mathbf{w}\ $ . |
| johnston-linear-matrix-algebra-F04174 | 447  | ■ 0.942 | $\mathbf{w} \leq\ \mathbf{v}\ +\ \mathbf{w}\ $                             | $\mathbf{w} \leq\ \mathbf{v}\ +\ \mathbf{w}\ $                             | $\mathbf{w} \leq\ \mathbf{v}\ +\ \mathbf{w}\ $ .                                                                |
| johnston-linear-matrix-algebra-F04175 | 447  | ■ 0.995 | $(1,0,0)$                                                                  | $(1,0,0)$                                                                  | $(1,0,0)$ and $\mathbf{w}=(-1, \sqrt{3}, 0)$ .                                                                  |

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| johnston-linear-matrix-algebra-F04176 | 447  | ■ 0.995 | $\mathbf{w}=(-1, \sqrt{3}, 0)$                                                                                                                             | $\mathbf{w} = (-1, \sqrt{3}, 0)$                                                                         | $(1, 0, 0)$ and $\mathbf{w} = (-1, \sqrt{3}, 0)$ .                                                                   |
| johnston-linear-matrix-algebra-F04177 | 447  | ■ 1.000 | $\mathbf{v}=(2,0)$                                                                                                                                         | $\mathbf{v} = (2, 0)$                                                                                    | (g) False. A counter-example is given by $\mathbf{v} = (2, 0)$                                                       |
| johnston-linear-matrix-algebra-F04178 | 447  | ■ 1.000 | $\mathbf{w}=(0,2)$                                                                                                                                         | $\mathbf{w} = (0, 2)$                                                                                    | and $\mathbf{w} = (0, 2)$ .                                                                                          |
| johnston-linear-matrix-algebra-F04179 | 447  | ■ 1.000 | $\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\  \ \mathbf{w}\  = \cos(\pi/3)(2\sqrt{3})(2) =$<br>$\ \mathbf{w}\  = \cos(\pi/3)(2\sqrt{3})(2) =$ | $\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\  \ \mathbf{w}\  = \cos(\pi/3)(2\sqrt{3})(2) =$ | 1.2.6 Since $\mathbf{v} \cdot \mathbf{w} = \cos(\theta) \ \mathbf{v}\  \ \mathbf{w}\  = \cos(\pi/3)(2\sqrt{3})(2) =$ |
| johnston-linear-matrix-algebra-F04180 | 447  | ■ 1.000 | $2\sqrt{3}$                                                                                                                                                | $2\sqrt{3}$                                                                                              | $2\sqrt{3}$ , we find that $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$ , so                                                     |
| johnston-linear-matrix-algebra-F04181 | 447  | ■ 1.000 | $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$                                                                                                                           | $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$                                                                         | $2\sqrt{3}$ , we find that $3w_1 + \sqrt{3}w_2 = 2\sqrt{3}$ , so                                                     |
| johnston-linear-matrix-algebra-F04182 | 447  | ■ 1.000 | $w_2 = 2 - \sqrt{3}w_1$                                                                                                                                    | $w_2 = 2 - \sqrt{3}w_1$                                                                                  | $w_2 = 2 - \sqrt{3}w_1$ . On the other hand, $\ \mathbf{w}\  = 2$ ,                                                  |
| johnston-linear-matrix-algebra-F04183 | 447  | ■ 1.000 | $w_1^2 + w_2^2 = 4$                                                                                                                                        | $w_1^2 + w_2^2 = 4$                                                                                      | so $w_1^2 + w_2^2 = 4$ . Plugging in our formula for $w_2$                                                           |
| johnston-linear-matrix-algebra-F04184 | 447  | ■ 1.000 | $w_2$                                                                                                                                                      | $w_2$                                                                                                    | so $w_1^2 + w_2^2 = 4$ . Plugging in our formula for $w_2$                                                           |
| johnston-linear-matrix-algebra-F04185 | 447  | ■ 1.000 | $w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$                                                                                                                          | $w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$                                                                        | gives $w_1^2 + (2 - \sqrt{3}w_1)^2 = 4$ , and simplifying                                                            |
| johnston-linear-matrix-algebra-F04186 | 447  | ■ 1.000 | $w_1(w_1 - \sqrt{3}) = 0$                                                                                                                                  | $w_1(w_1 - \sqrt{3}) = 0$                                                                                | gives $w_1(w_1 - \sqrt{3}) = 0$ , so $w_1 = 0$ or $w_1 = \sqrt{3}$ .                                                 |
| johnston-linear-matrix-algebra-F04187 | 447  | ■ 1.000 | $w_1 = 0$                                                                                                                                                  | $w_1 = 0$                                                                                                | gives $w_1(w_1 - \sqrt{3}) = 0$ , so $w_1 = 0$ or $w_1 = \sqrt{3}$ .                                                 |
| johnston-linear-matrix-algebra-F04188 | 447  | ■ 1.000 | $w_1 = \sqrt{3}$                                                                                                                                           | $w_1 = \sqrt{3}$                                                                                         | gives $w_1(w_1 - \sqrt{3}) = 0$ , so $w_1 = 0$ or $w_1 = \sqrt{3}$ .                                                 |
| johnston-linear-matrix-algebra-F04189 | 447  | ■ 0.999 | $w_2 = 2$                                                                                                                                                  | $w_2 = 2$                                                                                                | $w_2 = 2$ , and $w_1 = \sqrt{3}$ implies $w_2 = -1$ . There                                                          |
| johnston-linear-matrix-algebra-F04190 | 447  | ■ 0.999 | $w_2 = -1$                                                                                                                                                 | $w_2 = -1$                                                                                               | $w_2 = 2$ , and $w_1 = \sqrt{3}$ implies $w_2 = -1$ . There                                                          |
| johnston-linear-matrix-algebra-F04191 | 447  | ■ 0.925 | $\mathbf{w}=(\sqrt{3},-1)$                                                                                                                                 | $\mathbf{w} = (\sqrt{3}, -1)$                                                                            | $\mathbf{w} = (\sqrt{3}, -1)$ .                                                                                      |
| johnston-linear-matrix-algebra-F04192 | 447  | ■ 1.000 | $\mathbf{v}^2$                                                                                                                                             | $\mathbf{v}^2$                                                                                           | exponent is not standard notation. Does $\mathbf{v}^2$                                                               |
| johnston-linear-matrix-algebra-F04193 | 447  | ■ 0.974 | $\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$                                                                                                           | $\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$                                                         | of the entries of $\mathbf{v}$ , or the scalar $\ \mathbf{v}\ ^2 = \mathbf{v} \cdot \mathbf{v}$ , or                 |
| johnston-linear-matrix-algebra-F04194 | 447  | ■ 0.996 | $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$                                                                                                                 | $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$                                                               | $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$ satisfies $w_1 + 2w_2 = 0$ , so                                           |
| johnston-linear-matrix-algebra-F04195 | 447  | ■ 0.996 | $w_1 + 2w_2 = 0$                                                                                                                                           | $w_1 + 2w_2 = 0$                                                                                         | $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$ satisfies $w_1 + 2w_2 = 0$ , so                                           |
| johnston-linear-matrix-algebra-F04196 | 447  | ■ 1.000 | $w_1 =$                                                                                                                                                    | $w_1 =$                                                                                                  | we can choose $w_2$ arbitrarily and set $w_1 =$                                                                      |
| johnston-linear-matrix-algebra-F04197 | 447  | ■ 1.000 | $-2w_2$                                                                                                                                                    | $-2w_2$                                                                                                  | $-2w_2$ . One possibility is $\mathbf{w} = (-2, 1)$ .                                                                |
| johnston-linear-matrix-algebra-F04198 | 447  | ■ 0.975 | $\mathbf{v} \cdot \mathbf{y} = 0$                                                                                                                          | $\mathbf{v} \cdot \mathbf{y} = 0$                                                                        | part (a) that the only solutions $\mathbf{y}$ to $\mathbf{v} \cdot \mathbf{y} = 0$ are                               |
| johnston-linear-matrix-algebra-F04199 | 447  | ■ 0.531 | $\mathbf{y} = y_2(-2, 1)$                                                                                                                                  | $\mathbf{y} = y_2(-2, 1)$                                                                                | of the form $\mathbf{y} = y_2(-2, 1)$ . However, $\mathbf{y}$ then can-                                              |
| johnston-linear-matrix-algebra-F04200 | 447  | ■ 1.000 | $\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$                                                                                                                 | $\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$                                                               | not be orthogonal to $\mathbf{w}$ , since $\mathbf{w} \cdot \mathbf{y} = y_2(4 + 1)$ ,                               |
| johnston-linear-matrix-algebra-F04201 | 447  | ■ 1.000 | $\mathbf{y} = \mathbf{0}$                                                                                                                                  | $\mathbf{y} = \mathbf{0}$                                                                                | which equals 0 if and only if $\mathbf{y} = \mathbf{0}$ . However,                                                   |
| johnston-linear-matrix-algebra-F04202 | 447  | ■ 1.000 | $(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$                                                                                                       | $(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$                                                     | $(2, 2, 2) + (-1, -2, -3) + (-3, 0, 1) = (-2, 0, 0)$ .                                                               |
| johnston-linear-matrix-algebra-F04203 | 447  | ■ 1.000 | $\overline{\overline{x}} = x$                                                                                                                              | $\overline{\overline{x}} = x$                                                                            | where we used the facts that $\overline{\overline{x}} = x$ and $\overline{\overline{x \cdot y}} =$                   |

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| johnston-linear-matrix-algebra-F04204 | 447  | 1.000 | $\overline{x \cdot y} =$                                                                                   | $\overline{x} \cdot \overline{y} =$                                                                        | where we used the facts that $\overline{\overline{x}} = x$ and $\overline{\overline{x} \cdot \overline{y}} =$                |
| johnston-linear-matrix-algebra-F04205 | 447  | 1.000 | $\overline{\overline{x}} \cdot \overline{\overline{y}}$                                                    | $\overline{x} \cdot \overline{y}$                                                                          | $\overline{x} \cdot \overline{y}$ for all $x, y \in \mathbb{C}$ .                                                            |
| johnston-linear-matrix-algebra-F04206 | 447  | 1.000 | $x, y \in \mathbb{C}$                                                                                      | $x, y \in \mathbb{C}$                                                                                      | $\overline{x} \cdot \overline{y}$ for all $x, y \in \mathbb{C}$ .                                                            |
| johnston-linear-matrix-algebra-F04207 | 447  | 1.000 | $\ \mathbf{v}\  \ \mathbf{w}\  = 0$                                                                        | $\ \mathbf{v}\  \ \mathbf{w}\  = 0$                                                                        | and $\ \mathbf{v}\  \ \mathbf{w}\  = 0$ , so equality holds in the                                                           |
| johnston-linear-matrix-algebra-F04208 | 447  | 0.992 | $ \mathbf{v} \cdot \mathbf{w}  =  (c \mathbf{w}) \cdot \mathbf{w}  =  c  \ \mathbf{w}\ ^2$                 | $ \mathbf{v} \cdot \mathbf{w}  =  (c \mathbf{w}) \cdot \mathbf{w}  =  c  \ \mathbf{w}\ ^2$                 | then $ \mathbf{v} \cdot \mathbf{w}  =  (c \mathbf{w}) \cdot \mathbf{w}  =  c  \ \mathbf{w}\ ^2$ , whereas                    |
| johnston-linear-matrix-algebra-F04209 | 447  | 1.000 | $\ \mathbf{v}\  \ \mathbf{w}\  = \ c \mathbf{w}\  \ \mathbf{w}\  =  c  \ \mathbf{w}\ ^2$                   | $\ \mathbf{v}\  \ \mathbf{w}\  = \ c \mathbf{w}\  \ \mathbf{w}\  =  c  \ \mathbf{w}\ ^2$                   | $\ \mathbf{v}\  \ \mathbf{w}\  = \ c \mathbf{w}\  \ \mathbf{w}\  =  c  \ \mathbf{w}\ ^2$ , so equality                       |
| johnston-linear-matrix-algebra-F04210 | 447  | 1.000 | $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\ $                                              | $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\ $                                              | $\mathbf{v} \cdot \mathbf{w} = \ \mathbf{v}\  \ \mathbf{w}\ $ implies (by tracing the proof                                  |
| johnston-linear-matrix-algebra-F04211 | 447  | 0.985 | $\ \mathbf{w}\  \mathbf{v} - \ \mathbf{v}\  \mathbf{w}$                                                    | $\ \mathbf{w}\  \mathbf{v} - \ \mathbf{v}\  \mathbf{w}$                                                    | $\ \mathbf{w}\  \mathbf{v} - \ \mathbf{v}\  \mathbf{w}$ is 0. However, we saw in Theo-                                       |
| johnston-linear-matrix-algebra-F04212 | 447  | 0.555 | $\ \mathbf{w}\  \mathbf{v} = \ \mathbf{v}\  \mathbf{w}$                                                    | $\ \mathbf{w}\  \mathbf{v} = \ \mathbf{v}\  \mathbf{w}$                                                    | 0, so it follows that $\ \mathbf{w}\  \mathbf{v} = \ \mathbf{v}\  \mathbf{w}$ , so $\mathbf{v}$ and $\mathbf{w}$             |
| johnston-linear-matrix-algebra-F04213 | 448  | 1.000 | $\ \mathbf{v}\  = \ \mathbf{v}\ $                                                                          | $\ \mathbf{v}\  = \ \mathbf{v}\ $                                                                          | ity just says that $\ \mathbf{v}\  = \ \mathbf{v}\ $ , which is true,                                                        |
| johnston-linear-matrix-algebra-F04214 | 448  | 1.000 | $\ \mathbf{v} + \mathbf{w}\  = \ (c \mathbf{w}) + \mathbf{w}\  =$                                          | $\ \mathbf{v} + \mathbf{w}\  = \ (c \mathbf{w}) + \mathbf{w}\  =$                                          | and if $\mathbf{v} = c \mathbf{w}$ then $\ \mathbf{v} + \mathbf{w}\  = \ (c \mathbf{w}) + \mathbf{w}\  =$                    |
| johnston-linear-matrix-algebra-F04215 | 448  | 1.000 | $\ (c + 1) \mathbf{w}\  = (c + 1) \ \mathbf{w}\  = c \ \mathbf{w}\  + \ \mathbf{w}\  =$                    | $\ (c + 1) \mathbf{w}\  = (c + 1) \ \mathbf{w}\  = c \ \mathbf{w}\  + \ \mathbf{w}\  =$                    | $\ (c + 1) \mathbf{w}\  = (c + 1) \ \mathbf{w}\  = c \ \mathbf{w}\  + \ \mathbf{w}\  =$                                      |
| johnston-linear-matrix-algebra-F04216 | 448  | 1.000 | $\ c \mathbf{w}\  + \ \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $                                      | $\ c \mathbf{w}\  + \ \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $                                      | $\ c \mathbf{w}\  + \ \mathbf{w}\  = \ \mathbf{v}\  + \ \mathbf{w}\ $ .                                                      |
| johnston-linear-matrix-algebra-F04217 | 448  | 1.000 | $\ \mathbf{v} + \mathbf{w}\  =$                                                                            | $\ \mathbf{v} + \mathbf{w}\  =$                                                                            | proof of Theorem 1.2.4 that if $\ \mathbf{v} + \mathbf{w}\  =$                                                               |
| johnston-linear-matrix-algebra-F04218 | 448  | 1.000 | $2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\  \ \mathbf{w}\ $                                          | $2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\  \ \mathbf{w}\ $                                          | $\ \mathbf{v}\  + \ \mathbf{w}\ $ then $2(\mathbf{v} \cdot \mathbf{w}) = 2\ \mathbf{v}\  \ \mathbf{w}\ $ . It then           |
| johnston-linear-matrix-algebra-F04219 | 448  | 1.000 | $c \geq 0$                                                                                                 | $c \geq 0$                                                                                                 | remains is to show that $c \geq 0$ . To this end, just                                                                       |
| johnston-linear-matrix-algebra-F04220 | 448  | 1.000 | $2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$                                                     | $2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$                                                     | with $c < 0$ then $2(\mathbf{v} \cdot \mathbf{w}) = 2c \ \mathbf{w}\ ^2$ , whereas                                           |
| johnston-linear-matrix-algebra-F04221 | 448  | 0.956 | $2\ \mathbf{v}\  \ \mathbf{w}\  = 2 c  \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$                            | $2\ \mathbf{v}\  \ \mathbf{w}\  = 2 c  \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$                            | $2\ \mathbf{v}\  \ \mathbf{w}\  = 2 c  \ \mathbf{w}\ ^2 = -2c \ \mathbf{w}\ ^2$ , which con-                                 |
| johnston-linear-matrix-algebra-F04222 | 448  | 1.000 | $\ \mathbf{x} + \mathbf{y}\  \leq \ \mathbf{x}\  +$                                                        | $\ \mathbf{x} + \mathbf{y}\  \leq \ \mathbf{x}\  +$                                                        | <b>1.2.21</b> The triangle inequality tells us that $\ \mathbf{x} + \mathbf{y}\  \leq \ \mathbf{x}\  +$                      |
| johnston-linear-matrix-algebra-F04223 | 448  | 1.000 | $\ \mathbf{y}\ $                                                                                           | $\ \mathbf{y}\ $                                                                                           | $\ \mathbf{y}\ $ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . If we choose $\mathbf{x} = \mathbf{v} - \mathbf{w}$ and |
| johnston-linear-matrix-algebra-F04224 | 448  | 1.000 | $\mathbf{x} = \mathbf{v} - \mathbf{w}$                                                                     | $\mathbf{x} = \mathbf{v} - \mathbf{w}$                                                                     | $\ \mathbf{y}\ $ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . If we choose $\mathbf{x} = \mathbf{v} - \mathbf{w}$ and |
| johnston-linear-matrix-algebra-F04225 | 448  | 1.000 | $\mathbf{y} = \mathbf{w}$                                                                                  | $\mathbf{y} = \mathbf{w}$                                                                                  | $\mathbf{y} = \mathbf{w}$ , then this says that                                                                              |
| johnston-linear-matrix-algebra-F04226 | 448  | 0.999 | $f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$ | $f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$ | <b>1.2.23</b> (a) $f(x) = \ \mathbf{v} - x \mathbf{w}\ ^2 = (\mathbf{v} - x \mathbf{w}) \cdot (\mathbf{v} - x \mathbf{w}) =$ |
| johnston-linear-matrix-algebra-F04227 | 448  | 1.000 | $x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$                                | $x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$                                | $x^2 \ \mathbf{w}\ ^2 - 2x(\mathbf{v} \cdot \mathbf{w}) + \ \mathbf{v}\ ^2$                                                  |
| johnston-linear-matrix-algebra-F04228 | 448  | 1.000 | $4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$                                    | $4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$                                    | (b) $4(\mathbf{v} \cdot \mathbf{w})^2 - 4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$                                                  |
| johnston-linear-matrix-algebra-F04229 | 448  | 1.000 | $4(\mathbf{v} \cdot \mathbf{w})^2 -$                                                                       | $4(\mathbf{v} \cdot \mathbf{w})^2 -$                                                                       | (d) By parts (b) and (c), we know that $4(\mathbf{v} \cdot \mathbf{w})^2 -$                                                  |
| johnston-linear-matrix-algebra-F04230 | 448  | 1.000 | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$                                                                | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$                                                                | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$ . By adding $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$ to both                         |
| johnston-linear-matrix-algebra-F04231 | 448  | 1.000 | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$                                                                       | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$                                                                       | $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2 \leq 0$ . By adding $4\ \mathbf{v}\ ^2 \ \mathbf{w}\ ^2$ to both                         |

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| johnston-linear-matrix-attrib F04232 | 448  | 1.000 | $\left[\begin{array}{cc} 1 & 0 \\ -2 & 3 \end{array}\right]$                                                      | $\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$                                                                                | (a) $\begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}$                                                                                   |
| johnston-linear-matrix-attrib F04233 | 448  | 1.000 | $\left[\begin{array}{cc} 3 & -2 \\ 2 & 3 \end{array}\right]$                                                      | $\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$                                                                                | (c) $\begin{bmatrix} 3 & -2 \\ 2 & 3 \end{bmatrix}$                                                                                   |
| johnston-linear-matrix-attrib F04234 | 448  | 1.000 | $A^{\{5\}}=A^{\{1\}}, A^{\{6\}}=A^{\{2\}}$                                                                        | $A^5 = A^1, A^6 = A^2$                                                                                                         | this point on (i.e., $A^5 = A^1, A^6 = A^2$ , and so on). We                                                                          |
| johnston-linear-matrix-attrib F04235 | 448  | 0.979 | $A^{\{2\}}=\left[\begin{array}{ll} 1 & 2 \\ 0 & 1 \end{array}\right]$                                             | $A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                                           | 1.3.7 (a) $A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                                        |
| johnston-linear-matrix-attrib F04236 | 448  | 1.000 | $A^{\{3\}}=\left[\begin{array}{ll} 1 & 3 \\ 0 & 1 \end{array}\right]$                                             | $A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$                                                                           | (b) $A^3 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$                                                                              |
| johnston-linear-matrix-attrib F04237 | 449  | 1.000 | $k=0,1$                                                                                                           | $k = 0, 1$                                                                                                                     | $k = 0, 1$ are clear, and we also already showed                                                                                      |
| johnston-linear-matrix-attrib F04238 | 449  | 1.000 | $k=2,3$                                                                                                           | $k = 2, 3$                                                                                                                     | the $k = 2, 3$ cases. For the inductive step, as-                                                                                     |
| johnston-linear-matrix-attrib F04239 | 449  | 0.808 | $A^{\{\ell\}}=\left[\begin{array}{cc} 1 & \ell \\ 0 & 1 \end{array}\right]$                                       | $A^\ell = \begin{bmatrix} 1 & \ell \\ 0 & 1 \end{bmatrix}$                                                                     | sume that $A^\ell = \begin{bmatrix} 1 & \ell \\ 0 & 1 \end{bmatrix}$ for some particular                                              |
| johnston-linear-matrix-attrib F04240 | 449  | 1.000 | $\left[J_n\right]_{i,j}=(1,1,\ldots,1)\cdot(1,1,\ldots,1)=n$                                                      | $[J_n]_{i,j} = (1, 1, \dots, 1) \cdot (1, 1, \dots, 1) = n$                                                                    | is, $[J_n]_{i,j} = (1, 1, \dots, 1) \cdot (1, 1, \dots, 1) = n$ for all $i, j$ ,                                                      |
| johnston-linear-matrix-attrib F04241 | 449  | 1.000 | $J_n^2=nJ_n$                                                                                                      | $J_n^2 = nJ_n$                                                                                                                 | so $J_n^2 = nJ_n$ .                                                                                                                   |
| johnston-linear-matrix-attrib F04242 | 449  | 1.000 | $A^C{}^T$                                                                                                         | $AC^T$                                                                                                                         | but some of these products (like $AC^T$ ) do not                                                                                      |
| johnston-linear-matrix-attrib F04243 | 449  | 1.000 | $A^B=\left[\begin{array}{ll ll} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{array}\right]$                                | $AB = \left[ \begin{array}{cc cc} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{array} \right]$                                          | 1.3.13 (a) $AB = \left[ \begin{array}{cc cc} 3 & 3 & 3 & 3 \\ 3 & 3 & 3 & 3 \end{array} \right]$                                      |
| johnston-linear-matrix-attrib F04244 | 449  | 0.997 | $A^B=\left[\begin{array}{rr rr} 2 & 4 & 6 & 3 \\ 6 & 8 & 7 & 4 \\ \hline 2 & 4 & 6 & 3 \end{array}\right]$        | $AB = \left[ \begin{array}{cc cc} 2 & 4 & 6 & 3 \\ 3 & 5 & 7 & 4 \\ 4 & 6 & 8 & 3 \\ \hline 2 & 4 & 6 & 3 \end{array} \right]$ | (c) $AB = \left[ \begin{array}{cc cc} 2 & 4 & 6 & 3 \\ 3 & 5 & 7 & 4 \\ 4 & 6 & 8 & 3 \\ \hline 2 & 4 & 6 & 3 \end{array} \right]$    |
| johnston-linear-matrix-attrib F04245 | 449  | 1.000 | $r=n$                                                                                                             | $r = n$                                                                                                                        | 1.3.15 (a) For $AB$ to exist, we need $r = n$ , and for $BA$ to                                                                       |
| johnston-linear-matrix-attrib F04246 | 449  | 1.000 | $p=m$                                                                                                             | $p = m$                                                                                                                        | exist, we need $p = m$ . For both products to ex-                                                                                     |
| johnston-linear-matrix-attrib F04247 | 449  | 1.000 | $m=n=r=p$                                                                                                         | $m = n = r = p$                                                                                                                | (and thus $m = n = r = p$ ).                                                                                                          |
| johnston-linear-matrix-attrib F04248 | 449  | 1.000 | $\mathbf{v} \cdot \mathbf{w}=\mathbf{v}^T \mathbf{w}$                                                             | $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$                                                                        | tors then $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$ . It follows that $\mathbf{x} \cdot (A\mathbf{y}) =$                |
| johnston-linear-matrix-attrib F04249 | 449  | 1.000 | $\mathbf{x} \cdot(A \mathbf{y})=$                                                                                 | $\mathbf{x} \cdot (A\mathbf{y}) =$                                                                                             | tors then $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T \mathbf{w}$ . It follows that $\mathbf{x} \cdot (A\mathbf{y}) =$                |
| johnston-linear-matrix-attrib F04250 | 449  | 1.000 | $\mathbf{x}^T A \mathbf{y}$                                                                                       | $\mathbf{x}^T A\mathbf{y}$                                                                                                     | $\mathbf{x}^T A\mathbf{y}$ and $(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$ . Since |
| johnston-linear-matrix-attrib F04251 | 449  | 1.000 | $\left(A^T \mathbf{x}\right) \cdot \mathbf{y}=\left(A^T \mathbf{x}\right)^T \mathbf{y}=\mathbf{x}^T A \mathbf{y}$ | $(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$                                 | $\mathbf{x}^T A\mathbf{y}$ and $(A^T \mathbf{x}) \cdot \mathbf{y} = (A^T \mathbf{x})^T \mathbf{y} = \mathbf{x}^T A\mathbf{y}$ . Since |
| johnston-linear-matrix-attrib F04252 | 449  | 1.000 | $\mathbf{x} \cdot(A \mathbf{y})=\mathbf{x}^T A \mathbf{y}$                                                        | $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^T A\mathbf{y}$                                                                    | (b) Just like in part (a), $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^T A\mathbf{y}$ and                                            |
| johnston-linear-matrix-attrib F04253 | 449  | 1.000 | $(B \mathbf{x}) \cdot \mathbf{y}=\mathbf{x}^T B^T \mathbf{y}$                                                     | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$                                                                 | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$ . We thus know that $\mathbf{x}^T A\mathbf{y} =$                       |
| johnston-linear-matrix-attrib F04254 | 449  | 1.000 | $\mathbf{x}^T A \mathbf{y}=$                                                                                      | $\mathbf{x}^T A\mathbf{y} =$                                                                                                   | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^T B^T \mathbf{y}$ . We thus know that $\mathbf{x}^T A\mathbf{y} =$                       |

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| johnston-linear-matrix-algebra-F04255 | 449  | 1.000 | $\mathbf{x}^T \mathbf{B}^T \mathbf{y}$           | $\mathbf{x}^T \mathbf{B}^T \mathbf{y}$           | $\mathbf{x}^T \mathbf{B}^T \mathbf{y}$ for all $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$ . However, if |
| johnston-linear-matrix-algebra-F04256 | 449  | 1.000 | $\mathbf{x} = \mathbf{e}_i$                      | $\mathbf{x} = \mathbf{e}_i$                      | we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$    |
| johnston-linear-matrix-algebra-F04257 | 449  | 1.000 | $\mathbf{y} = \mathbf{e}_j$                      | $\mathbf{y} = \mathbf{e}_j$                      | we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$    |
| johnston-linear-matrix-algebra-F04258 | 449  | 1.000 | $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$   | $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$   | we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^T \mathbf{A} \mathbf{y} = a_{i,j}$    |
| johnston-linear-matrix-algebra-F04259 | 449  | 1.000 | $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$ | $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$ | and $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$ , so we see that $a_{i,j} = b_{j,i}$ for                                |
| johnston-linear-matrix-algebra-F04260 | 449  | 1.000 | $a_{i,j} = b_{j,i}$                              | $a_{i,j} = b_{j,i}$                              | and $\mathbf{x}^T \mathbf{B}^T \mathbf{y} = b_{j,i}$ , so we see that $a_{i,j} = b_{j,i}$ for                                |
| johnston-linear-matrix-algebra-F04261 | 449  | 0.994 | $[\mathbf{e}_i]_k = 0$                           | $[\mathbf{e}_i]_k = 0$                           | $[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$ . It follows that                                            |
| johnston-linear-matrix-algebra-F04262 | 449  | 0.994 | $k \neq i$                                       | $k \neq i$                                       | $[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$ . It follows that                                            |
| johnston-linear-matrix-algebra-F04263 | 449  | 0.994 | $[\mathbf{e}_i]_i = 1$                           | $[\mathbf{e}_i]_i = 1$                           | $[\mathbf{e}_i]_k = 0$ if $k \neq i$ and $[\mathbf{e}_i]_i = 1$ . It follows that                                            |
| johnston-linear-matrix-algebra-F04264 | 449  | 1.000 | $\mathbf{a}_i$                                   | $\mathbf{a}_i$                                   | (c) We let $\mathbf{a}_i$ denote the $i$ -th column of $A$ and then                                                          |
| johnston-linear-matrix-algebra-F04265 | 449  | 1.000 | $(A - B)\mathbf{v} = \mathbf{0}$                 | $(A - B)\mathbf{v} = \mathbf{0}$                 | $(A - B)\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in \mathbb{R}^n$ . In particular then,                                 |
| johnston-linear-matrix-algebra-F04266 | 449  | 1.000 | $(A - B)\mathbf{e}_i = \mathbf{0}$               | $(A - B)\mathbf{e}_i = \mathbf{0}$               | $(A - B)\mathbf{e}_i = \mathbf{0}$ for all $i$ , so part (a) tells us that                                                   |
| johnston-linear-matrix-algebra-F04267 | 449  | 1.000 | $(A + B) + C$                                    | $(A + B) + C$                                    | 1.3.19 (a) We compute the $(i, j)$ -entry of both $(A + B) + C$                                                              |
| johnston-linear-matrix-algebra-F04268 | 449  | 1.000 | $A + (B + C)$                                    | $A + (B + C)$                                    | and $A + (B + C)$ :                                                                                                          |
| johnston-linear-matrix-algebra-F04269 | 449  | 1.000 | $(A + B) + C =$                                  | $(A + B) + C =$                                  | $A + (B + C)$ are all the same, so $(A + B) + C =$                                                                           |
| johnston-linear-matrix-algebra-F04270 | 449  | 1.000 | $c(A + B)$                                       | $c(A + B)$                                       | (b) We compute the $(i, j)$ -entry of both $c(A + B)$                                                                        |
| johnston-linear-matrix-algebra-F04271 | 449  | 1.000 | $cA + cB$                                        | $cA + cB$                                        | and $cA + cB$ :                                                                                                              |
| johnston-linear-matrix-algebra-F04272 | 449  | 0.987 | $cA +$                                           | $cA +$                                           | It follows that the entries of $c(A + B)$ and $cA +$                                                                         |
| johnston-linear-matrix-algebra-F04273 | 449  | 1.000 | $cB$                                             | $cB$                                             | $cB$ are all the same, so $c(A + B) = cA + cB$ .                                                                             |
| johnston-linear-matrix-algebra-F04274 | 450  | 1.000 | $(c + d)A$                                       | $(c + d)A$                                       | (c) We compute the $(i, j)$ -entry of both $(c + d)A$                                                                        |
| johnston-linear-matrix-algebra-F04275 | 450  | 1.000 | $cA + dA$                                        | $cA + dA$                                        | and $cA + dA$ :                                                                                                              |
| johnston-linear-matrix-algebra-F04276 | 450  | 1.000 | $dA$                                             | $dA$                                             | $dA$ are all the same, so $(c + d)A = cA + dA$ .                                                                             |
| johnston-linear-matrix-algebra-F04277 | 450  | 1.000 | $c(dA)$                                          | $c(dA)$                                          | (d) We compute the $(i, j)$ -entry of both $c(dA)$ and                                                                       |
| johnston-linear-matrix-algebra-F04278 | 450  | 0.936 | $(cd)A$                                          | $(cd)A$                                          | $(cd)A$ :                                                                                                                    |
| johnston-linear-matrix-algebra-F04279 | 450  | 1.000 | $(AB)C$                                          | $(AB)C$                                          | 1.3.20 (a) We compute the $(i, j)$ -entry of both $(AB)C$ and                                                                |
| johnston-linear-matrix-algebra-F04280 | 450  | 1.000 | $A(BC)$                                          | $A(BC)$                                          | $A(BC)$ :                                                                                                                    |
| johnston-linear-matrix-algebra-F04281 | 450  | 1.000 | $AC + BC$                                        | $AC + BC$                                        | and $AC + BC$ :                                                                                                              |
| johnston-linear-matrix-algebra-F04282 | 450  | 1.000 | $A(B + C) =$                                     | $A(B + C) =$                                     | $AB + AC$ are all the same, so $A(B + C) =$                                                                                  |
| johnston-linear-matrix-algebra-F04283 | 450  | 1.000 | $c(AB)$                                          | $c(AB)$                                          | (c) We compute the $(i, j)$ -entry of both $c(AB)$ and                                                                       |

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|----------------------------------------|------|-------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| johnston-linear-matrix-attached-F04284 | 450  | 1.000 | $(c\ A)\ B$                                                                                                                    | $(cA)B$                                                                                      | $(cA)B$ :                                                                                    |
| johnston-linear-matrix-attached-F04285 | 450  | 0.734 | $I_{\{m\}}\ A$                                                                                                                 | $I_m A$                                                                                      | <b>1.3.21</b> We recall that the $(i, j)$ -entry of $I_m A$ is the dot prod-                 |
| johnston-linear-matrix-attached-F04286 | 450  | 0.998 | $A$ : $a_{\{i, j\}}$                                                                                                           | $A : a_{i,j}$                                                                                | column of $A$ : $a_{i,j}$ . All entries of $I_m A$ and $A$ thus                              |
| johnston-linear-matrix-attached-F04287 | 450  | 1.000 | $A^{\{k\}}\ A^{\{\ell\}}=(A\ A\ \cdots\ A)(A\ A\ \cdots\ A)$                                                                   | $A^k A^\ell = (AA \cdots A)(AA \cdots A)$                                                    | <b>1.3.22</b> (a) Just note that $A^k A^\ell = (AA \cdots A)(AA \cdots A)$ ,                 |
| johnston-linear-matrix-attached-F04288 | 450  | 1.000 | $k+\ell$                                                                                                                       | $k + \ell$                                                                                   | the second has $\ell$ copies of $A$ , for a total of $k + \ell$                              |
| johnston-linear-matrix-attached-F04289 | 450  | 1.000 | $A^{\{k+\ell\}}$                                                                                                               | $A^{k+\ell}$                                                                                 | copies of $A$ (and is thus equal to $A^{k+\ell}$ ).                                          |
| johnston-linear-matrix-attached-F04290 | 450  | 1.000 | $k\ \ell$                                                                                                                      | $k\ell$                                                                                      | there are $\ell$ parentheses, for a total of $k\ell$ copies                                  |
| johnston-linear-matrix-attached-F04291 | 450  | 1.000 | $A^{\{k\ \ell\}}$                                                                                                              | $A^{k\ell}$                                                                                  | of $A$ (which is thus equal to $A^{k\ell}$ ).                                                |
| johnston-linear-matrix-attached-F04292 | 450  | 1.000 | $\left[\left(\left(A^{\{T\}}\right)^{\{T\}}\right)^{\{T\}}\right]_{\{i, j\}}=\left[A^{\{T\}}\right]_{\{j, i\}}=[A]_{\{i, j\}}$ | $\left[\left(A^T\right)^T\right]_{i,j} = \left[A^T\right]_{j,i} = [A]_{i,j}$                 | $[(A^T)^T]_{i,j} = [A^T]_{j,i} = [A]_{i,j}$ , so $(A^T)^T = A$ .                             |
| johnston-linear-matrix-attached-F04293 | 450  | 1.000 | $[(A+$                                                                                                                         | $[(A+$                                                                                       | (b) Compute the $(i, j)$ -entry of both sides: $[(A +$                                       |
| johnston-linear-matrix-attached-F04294 | 450  | 0.431 | $\left.\right\}^{\{T\}}\right]_{\{i, j\}}=[A+B]_{\{j, i\}}=[A]_{\{j, i\}}+[B]_{\{j, i\}}=\left[A^{\{T\}}\right]_{\{i, j\}}+$   | $\left.\right\}^T]_{i,j} = [A + B]_{j,i} = [A]_{j,i} + [B]_{j,i} = \left[A^T\right]_{i,j} +$ | $\left.\right\}^T]_{i,j} = [A + B]_{j,i} = [A]_{j,i} + [B]_{j,i} = \left[A^T\right]_{i,j} +$ |
| johnston-linear-matrix-attached-F04295 | 450  | 1.000 | $\left[B^{\{T\}}\right]_{\{i, j\}}=\left[A^{\{T\}}+B^{\{T\}}\right]_{\{i, j\}}$                                                | $\left[B^T\right]_{i,j} = \left[A^T + B^T\right]_{i,j}$                                      | $\left[B^T\right]_{i,j} = \left[A^T + B^T\right]_{i,j}$ , so $(A + B)^T = A^T + B^T$ .       |
| johnston-linear-matrix-attached-F04296 | 450  | 1.000 | $\left[\left(c\ A\right)^{\{T\}}\right]_{\{i, j\}}=[c\ A]_{\{j, i\}}=c[A]_{\{j, i\}}=c\ \left[A^{\{T\}}\right]_{\{i, j\}}=$    | $\left[(cA)^T\right]_{i,j} = [cA]_{j,i} = c[A]_{j,i} = c\left[A^T\right]_{i,j} =$            | $[(cA)^T]_{i,j} = [cA]_{j,i} = c[A]_{j,i} = c[A^T]_{i,j} =$                                  |
| johnston-linear-matrix-attached-F04297 | 450  | 0.998 | $\left[c\ A^{\{T\}}\right]_{\{i, j\}}$                                                                                         | $\left[cA^T\right]_{i,j}$                                                                    | $[cA^T]_{i,j}$ , so $(cA)^T = cA^T$ .                                                        |
| johnston-linear-matrix-attached-F04298 | 450  | 1.000 | $p\ \times\ 1$                                                                                                                 | $p \times 1$                                                                                 | as a $p \times 1$ block matrix and $B$ as a $1 \times 1$ block matrix,                       |
| johnston-linear-matrix-attached-F04299 | 450  | 1.000 | $\mathbf{a}_{\{1\}},\ \mathbf{a}_{\{2\}},\ \ldots\ \mathbf{a}_{\{n\}}$                                                         | $\mathbf{a}_1, \mathbf{a}_2, \dots \mathbf{a}_n$                                             | where $\mathbf{a}_1, \mathbf{a}_2, \dots \mathbf{a}_n$ are the columns of $A$ . Com-         |
| johnston-linear-matrix-attached-F04300 | 450  | 1.000 | $(A\ B)^{\{T\}}=$                                                                                                              | $(AB)^T =$                                                                                   | The rows of $AB$ are the columns of $(AB)^T =$                                               |
| johnston-linear-matrix-attached-F04301 | 450  | 1.000 | $\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$                                                                   | $\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$                                 | (a) $\left[\begin{array}{cc} 1 & 2 \\ 3 & -1 \end{array}\right]$                             |
| johnston-linear-matrix-attached-F04302 | 450  | 1.000 | $\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$                                                          | $\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$                        | (c) $\left[\begin{array}{ccc} 1 & 1 & 0 \\ 1 & 1 & -1 \end{array}\right]$                    |
| johnston-linear-matrix-attached-F04303 | 451  | 1.000 | $T(1,0,0)=$                                                                                                                    | $T(1,0,0) =$                                                                                 | (e) This time, we must be more clever. $T(1,0,0) =$                                          |
| johnston-linear-matrix-attached-F04304 | 451  | 1.000 | $T(0,1,0)=T(1,1,0)-T(1,0,0)=$                                                                                                  | $T(0,1,0) = T(1,1,0) - T(1,0,0) =$                                                           | column is $T(0,1,0) = T(1,1,0) - T(1,0,0) =$                                                 |
| johnston-linear-matrix-attached-F04305 | 451  | 1.000 | $(0,1,2)-(1,2,3)=(-1,-1,-1)$                                                                                                   | $(0,1,2) - (1,2,3) = (-1, -1, -1)$                                                           | $(0,1,2) - (1,2,3) = (-1, -1, -1)$ . Its third                                               |
| johnston-linear-matrix-attached-F04306 | 451  | 1.000 | $T(0,0,1)=T(1,1,1)-T(1,1,0)=$                                                                                                  | $T(0,0,1) = T(1,1,1) - T(1,1,0) =$                                                           | column is $T(0,0,1) = T(1,1,1) - T(1,1,0) =$                                                 |
| johnston-linear-matrix-attached-F04307 | 451  | 1.000 | $(0,0,1)-(0,1,2)=(0,-1,-1)$                                                                                                    | $(0,0,1) - (0,1,2) = (0, -1, -1)$                                                            | $(0,0,1) - (0,1,2) = (0, -1, -1)$ . It follows                                               |
| johnston-linear-matrix-attached-F04308 | 451  | 1.000 | $[T]=$                                                                                                                         | $[T] =$                                                                                      | (which completely determines $T$ ) is $[T] =$                                                |
| johnston-linear-matrix-attached-F04309 | 451  | 1.000 | $\left[T\left(\mathbf{e}_{\{1\}}\right)\mid T\left(\mathbf{e}_{\{2\}}\right)\right]$                                           | $[T(\mathbf{e}_1) \mid T(\mathbf{e}_2)]$                                                     | $[T(\mathbf{e}_1) \mid T(\mathbf{e}_2)]$ .                                                   |
| johnston-linear-matrix-attached-F04310 | 451  | 1.000 | $T\left(\mathbf{e}_{\{1\}}\right)=(2,1)$                                                                                       | $T(\mathbf{e}_1) = (2,1)$                                                                    | (e) False. If $T(\mathbf{e}_1) = (2,1)$ and $T(\mathbf{e}_2) = (1,3)$                        |
| johnston-linear-matrix-attached-F04311 | 451  | 1.000 | $T\left(\mathbf{e}_{\{2\}}\right)=(1,3)$                                                                                       | $T(\mathbf{e}_2) = (1,3)$                                                                    | (e) False. If $T(\mathbf{e}_1) = (2,1)$ and $T(\mathbf{e}_2) = (1,3)$                        |

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|---------------------------------------|------|-------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra-F04312 | 451  | 1.000 | $T(1,1)=T(\mathbf{e}_1+\mathbf{e}_2)=T(\mathbf{e}_1)+T(\mathbf{e}_2)=(2,1)+(1,3)=(3,4) \neq (3,3)$ | $T(1,1)=T(\mathbf{e}_1+\mathbf{e}_2)=T(\mathbf{e}_1)+T(\mathbf{e}_2)=(2,1)+(1,3)=(3,4) \neq (3,3)$ | then $T(1,1)=T(\mathbf{e}_1+\mathbf{e}_2)=T(\mathbf{e}_1)+T(\mathbf{e}_2)=(2,1)+(1,3)=(3,4) \neq (3,3)$ .   |
| johnston-linear-matrix-algebra-F04313 | 451  | 1.000 | $\mathbf{v}=(0,1,0)$                                                                               | $\mathbf{v}=(0,1,0)$                                                                               | and $\mathbf{v}=(0,1,0)$ . Then $R_{xz}^\theta(\mathbf{v})=\mathbf{v}$ , but                                |
| johnston-linear-matrix-algebra-F04314 | 451  | 1.000 | $R_{xz}^\theta(\mathbf{v})=\mathbf{v}$                                                             | $R_{xz}^\theta(\mathbf{v})=\mathbf{v}$                                                             | and $\mathbf{v}=(0,1,0)$ . Then $R_{xz}^\theta(\mathbf{v})=\mathbf{v}$ , but                                |
| johnston-linear-matrix-algebra-F04315 | 451  | 1.000 | $R_{xy}^\theta \circ R_{yz}^\theta(\mathbf{v})=R_{xy}^\theta(0,0,1)=(0,0,1) \neq \mathbf{v}$       | $(R_{xy}^\theta \circ R_{yz}^\theta)(\mathbf{v})=R_{xy}^\theta(0,0,1)=(0,0,1) \neq \mathbf{v}$     | $(R_{xy}^\theta \circ R_{yz}^\theta)(\mathbf{v})=R_{xy}^\theta(0,0,1)=(0,0,1) \neq \mathbf{v}$ .            |
| johnston-linear-matrix-algebra-F04316 | 451  | 0.623 | $R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$                                             | $R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$                                             | It follows that $R_{xy}^\theta \circ R_{yz}^\theta \neq R_{xz}^\theta$ .                                    |
| johnston-linear-matrix-algebra-F04317 | 451  | 1.000 | $T(2(1,1))=(4,2)$                                                                                  | $T(2(1,1))=(4,2)$                                                                                  | $T(2(1,1))=(4,2)$ , but $2T(1,1)=(2,2)$ .                                                                   |
| johnston-linear-matrix-algebra-F04318 | 451  | 1.000 | $2T(1,1)=(2,2)$                                                                                    | $2T(1,1)=(2,2)$                                                                                    | $T(2(1,1))=(4,2)$ , but $2T(1,1)=(2,2)$ .                                                                   |
| johnston-linear-matrix-algebra-F04319 | 451  | 1.000 | $T(0,0)=(0,-1) \neq \mathbf{0}$                                                                    | $T(0,0)=(0,-1) \neq \mathbf{0}$                                                                    | cally, $T(0,0)=(0,-1) \neq \mathbf{0}$ .                                                                    |
| johnston-linear-matrix-algebra-F04320 | 451  | 1.000 | $T(2(1,1))=(2\sqrt{2},2)$                                                                          | $T(2(1,1))=(2\sqrt{2},2)$                                                                          | cally, $T(2(1,1))=(2\sqrt{2},2)$ , but $2T(1,1)=$                                                           |
| johnston-linear-matrix-algebra-F04321 | 451  | 1.000 | $2T(1,1)=(4,2\sqrt{2})$                                                                            | $2T(1,1)=(4,2\sqrt{2})$                                                                            | cally, $T(2(1,1))=(2\sqrt{2},2)$ , but $2T(1,1)=$                                                           |
| johnston-linear-matrix-algebra-F04322 | 451  | 1.000 | $T(1,1)+T(-1,-1)=(2,2)$                                                                            | $T(1,1)+T(-1,-1)=(2,2)$                                                                            | Specifically, $T(1,1)+T(-1,-1)=(2,2)$ , but                                                                 |
| johnston-linear-matrix-algebra-F04323 | 451  | 1.000 | $T((1,1)+(-1,-1))=(0,0)$                                                                           | $T((1,1)+(-1,-1))=(0,0)$                                                                           | $T((1,1)+(-1,-1))=(0,0)$ .                                                                                  |
| johnston-linear-matrix-algebra-F04324 | 451  | 1.000 | $\frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$                                        | $\frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$                                        | is $\frac{1}{10} \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$ .                                            |
| johnston-linear-matrix-algebra-F04325 | 451  | 0.996 | $\frac{1}{5} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$                                        | $\frac{1}{5} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$                                        | is $\frac{1}{5} \begin{bmatrix} -3 & 4 \\ 4 & 3 \end{bmatrix}$ .                                            |
| johnston-linear-matrix-algebra-F04326 | 451  | 0.999 | $\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$            | $\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$            | is $\begin{bmatrix} \cos(\pi/5) & -\sin(\pi/5) \\ \sin(\pi/5) & \cos(\pi/5) \end{bmatrix}$ .                |
| johnston-linear-matrix-algebra-F04327 | 451  | 0.798 | $\mathbf{u}=\frac{1}{\sqrt{2}}(1,1)$                                                               | $\mathbf{u}=\frac{1}{\sqrt{2}}(1,1)$                                                               | direction of $\mathbf{u}=\frac{1}{\sqrt{2}}(1,1)$ and rotation are                                          |
| johnston-linear-matrix-algebra-F04328 | 451  | 1.000 | $T(\mathbf{0})=OT(\mathbf{0})=\mathbf{0}$                                                          | $T(\mathbf{0})=OT(\mathbf{0})=\mathbf{0}$                                                          | $c=0$ and $\mathbf{v}=\mathbf{0}$ then this says that $T(\mathbf{0})=OT(\mathbf{0})=\mathbf{0}$ ,           |
| johnston-linear-matrix-algebra-F04329 | 451  | 0.902 | $T(c_1\mathbf{v}_1+\dots)$                                                                         | $T(c_1\mathbf{v}_1+\dots)$                                                                         | <b>1.4.9</b> For the “if” direction, assume that $T(c_1\mathbf{v}_1+\dots)$                                 |
| johnston-linear-matrix-algebra-F04330 | 451  | 1.000 | $c_k\mathbf{v}_k=c_1T(\mathbf{v}_1)+\dots+c_kT(\mathbf{v}_k)$                                      | $c_k\mathbf{v}_k=c_1T(\mathbf{v}_1)+\dots+c_kT(\mathbf{v}_k)$                                      | $c_k\mathbf{v}_k=c_1T(\mathbf{v}_1)+\dots+c_kT(\mathbf{v}_k)$ for all $\mathbf{v}_1,\dots,\mathbf{v}_k \in$ |
| johnston-linear-matrix-algebra-F04331 | 451  | 1.000 | $\mathbf{v}_1,\dots,\mathbf{v}_k \in$                                                              | $\mathbf{v}_1,\dots,\mathbf{v}_k \in$                                                              | $c_k\mathbf{v}_k=c_1T(\mathbf{v}_1)+\dots+c_kT(\mathbf{v}_k)$ for all $\mathbf{v}_1,\dots,\mathbf{v}_k \in$ |
| johnston-linear-matrix-algebra-F04332 | 451  | 1.000 | $T(c_1\mathbf{v}_1)=c_1T(\mathbf{v}_1)$                                                            | $T(c_1\mathbf{v}_1)=c_1T(\mathbf{v}_1)$                                                            | then this says that $T(c_1\mathbf{v}_1)=c_1T(\mathbf{v}_1)$ , and if we                                     |
| johnston-linear-matrix-algebra-F04333 | 451  | 1.000 | $c_1=c_2=1$                                                                                        | $c_1=c_2=1$                                                                                        | choose $k=2$ and $c_1=c_2=1$ then it says that                                                              |
| johnston-linear-matrix-algebra-F04334 | 451  | 1.000 | $T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$                                     | $T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$                                     | $T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$ . These are the two                          |
| johnston-linear-matrix-algebra-F04335 | 451  | 1.000 | $T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$                                     | $T(\mathbf{v}_1+\mathbf{v}_2)=T(\mathbf{v}_1)+T(\mathbf{v}_2)$                                     |                                                                                                             |

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|---------------------------------------|------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-atlas-1-F04337 | 451  | 1.000 | $\left(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k\right) = \left(c_1 \mathbf{v}_1\right) + \left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right)$ | $T\left(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k\right) = T\left(c_1 \mathbf{v}_1\right) + T\left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right)$ | see that $T\left(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k\right) = T\left(c_1 \mathbf{v}_1\right) + T\left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right)$          |
| johnston-linear-matrix-atlas-1-F04338 | 451  | 1.000 | $\left(c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k\right) = c_1 \mathbf{v}_1 + \left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right)$              | $\cdots + c_k \mathbf{v}_k) = c_1 T\left(\mathbf{v}_1\right) + T\left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right) = \cdots =$                      | $\cdots + c_k \mathbf{v}_k) = c_1 T\left(\mathbf{v}_1\right) + T\left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right) = \cdots =$                                        |
| johnston-linear-matrix-atlas-1-F04339 | 451  | 1.000 | $c_1 \mathbf{v}_1 + \cdots + c_k \mathbf{v}_k = c_1 \mathbf{v}_1 + \left(c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k\right)$                           | $c_1 T\left(\mathbf{v}_1\right) + \cdots + c_k T\left(\mathbf{v}_k\right)$                                                                                 | $c_1 T\left(\mathbf{v}_1\right) + \cdots + c_k T\left(\mathbf{v}_k\right)$ for all $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ and                                 |
| johnston-linear-matrix-atlas-2-F04340 | 452  | 1.000 | $[S+T] \mathbf{v} = (S+T) \mathbf{v}$                                                                                                                   | $[S+T] \mathbf{v} = (S+T) \mathbf{v}$                                                                                                                      | for all $\mathbf{v} \in \mathbb{R}^n$ it is the case that $[S+T] \mathbf{v} = (S+T) \mathbf{v}$                                                                              |
| johnston-linear-matrix-atlas-2-F04341 | 452  | 0.977 | $T(S \mathbf{v}) = S(T \mathbf{v}) = [S] \mathbf{v} + [T] \mathbf{v} = ([S] + [T]) \mathbf{v}$                                                          | $T(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) = [S] \mathbf{v} + [T] \mathbf{v} = ([S] + [T]) \mathbf{v}$                                                 | $T(\mathbf{v}) = S(\mathbf{v}) + T(\mathbf{v}) = [S] \mathbf{v} + [T] \mathbf{v} = ([S] + [T]) \mathbf{v}$                                                                   |
| johnston-linear-matrix-atlas-2-F04342 | 452  | 1.000 | $[T] \mathbf{v} = (T) \mathbf{v}$                                                                                                                       | $[T] \mathbf{v}$                                                                                                                                           | $[T] \mathbf{v}$ . Since $[S+T] \mathbf{v} = ([S] + [T]) \mathbf{v}$ for all                                                                                                 |
| johnston-linear-matrix-atlas-2-F04343 | 452  | 1.000 | $[S+T] \mathbf{v} = ([S] + [T]) \mathbf{v}$                                                                                                             | $[S+T] \mathbf{v} = ([S] + [T]) \mathbf{v}$                                                                                                                | $[T] \mathbf{v}$ . Since $[S+T] \mathbf{v} = ([S] + [T]) \mathbf{v}$ for all                                                                                                 |
| johnston-linear-matrix-atlas-2-F04344 | 452  | 1.000 | $[cS] \mathbf{v} = (cS) \mathbf{v} = c(S \mathbf{v}) = c([S] \mathbf{v}) = (c[S]) \mathbf{v}$                                                           | $[cS] \mathbf{v} = (cS) \mathbf{v} = cS(\mathbf{v}) = c[S] \mathbf{v} = (c[S]) \mathbf{v}$                                                                 | that $[cS] \mathbf{v} = (cS) \mathbf{v} = cS(\mathbf{v}) = c[S] \mathbf{v} = (c[S]) \mathbf{v}$ .                                                                            |
| johnston-linear-matrix-atlas-2-F04345 | 452  | 1.000 | $[cS] \mathbf{v} = (c[S]) \mathbf{v}$                                                                                                                   | $[cS] \mathbf{v} = (c[S]) \mathbf{v}$                                                                                                                      | Since $[cS] \mathbf{v} = (c[S]) \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^n$ , it follows                                                                               |
| johnston-linear-matrix-atlas-2-F04346 | 452  | 1.000 | $[cS] = c[S]$                                                                                                                                           | $[cS] = c[S]$                                                                                                                                              | from Exercise 1.3.18(d) that $[cS] = c[S]$ .                                                                                                                                 |
| johnston-linear-matrix-atlas-2-F04347 | 452  | 1.000 | $160 \pi / 4 = 40 \pi$                                                                                                                                  | $160 \pi / 4 = 40 \pi$                                                                                                                                     | $160 \pi / 4 = 40 \pi$ . Since $40 \pi$ is a multiple of $2 \pi$ , this                                                                                                      |
| johnston-linear-matrix-atlas-2-F04348 | 452  | 1.000 | $40 \pi$                                                                                                                                                | $40 \pi$                                                                                                                                                   | $160 \pi / 4 = 40 \pi$ . Since $40 \pi$ is a multiple of $2 \pi$ , this                                                                                                      |
| johnston-linear-matrix-atlas-2-F04349 | 452  | 1.000 | $A^{160} = I$                                                                                                                                           | $A^{160} = I$                                                                                                                                              | $A^{160} = I$ .                                                                                                                                                              |
| johnston-linear-matrix-atlas-2-F04350 | 452  | 0.974 | $\cos(\theta) \sin(\theta) - \cos(\theta) \sin(\theta) = 0$                                                                                             | $\cos(\theta) \sin(\theta) - \cos(\theta) \sin(\theta) = 0$                                                                                                | $\cos(\theta) \sin(\theta) - \cos(\theta) \sin(\theta) = 0$ . We thus con-                                                                                                   |
| johnston-linear-matrix-atlas-2-F04351 | 452  | 1.000 | $\mathbf{u} = (1, m)$                                                                                                                                   | $\mathbf{u} = (1, m)$                                                                                                                                      | $y = mx$ is $\mathbf{u} = (1, m)$ , so we know that the standard                                                                                                             |
| johnston-linear-matrix-atlas-2-F04352 | 452  | 1.000 | $E_{i, j} = \mathbf{e}_i \mathbf{e}_j^T$                                                                                                                | $E_{i, j} = \mathbf{e}_i \mathbf{e}_j^T$                                                                                                                   | (b) First, we notice that $E_{i, j} = \mathbf{e}_i \mathbf{e}_j^T$ , so $E_{i, j}^2 =$                                                                                       |
| johnston-linear-matrix-atlas-2-F04353 | 452  | 1.000 | $E_{i, j}^2 =$                                                                                                                                          | $E_{i, j}^2 =$                                                                                                                                             | (b) First, we notice that $E_{i, j} = \mathbf{e}_i \mathbf{e}_j^T$ , so $E_{i, j}^2 =$                                                                                       |
| johnston-linear-matrix-atlas-2-F04354 | 452  | 0.978 | $\mathbf{e}_i \mathbf{e}_j^T \mathbf{e}_i \mathbf{e}_j^T = \mathbf{e}_i \left(\mathbf{e}_j \cdot \mathbf{e}_i\right) \mathbf{e}_j^T = \mathbf{0}$       | $\mathbf{e}_i \mathbf{e}_j^T \mathbf{e}_i \mathbf{e}_j^T = \mathbf{e}_i \left(\mathbf{e}_j \cdot \mathbf{e}_i\right) \mathbf{e}_j^T = \mathbf{0}$          | $\mathbf{e}_i \mathbf{e}_j^T \mathbf{e}_i \mathbf{e}_j^T = \mathbf{e}_i \left(\mathbf{e}_j \cdot \mathbf{e}_i\right) \mathbf{e}_j^T = \mathbf{0}$ , since $i \neq j$ . Since |
| johnston-linear-matrix-atlas-2-F04355 | 452  | 1.000 | $u_1^2 = u_3^2 = 3 / 7$                                                                                                                                 | $u_1^2 = u_3^2 = 3 / 7$                                                                                                                                    | each other first, we find that $u_1^2 = u_3^2 = 3 / 7$ ,                                                                                                                     |
| johnston-linear-matrix-atlas-2-F04356 | 452  | 1.000 | $u_2^2 = 1 / 7$                                                                                                                                         | $u_2^2 = 1 / 7$                                                                                                                                            | $u_2^2 = 1 / 7$ , and $\cos(\theta) = 1 / 8$ . Then focusing                                                                                                                 |
| johnston-linear-matrix-atlas-2-F04357 | 452  | 1.000 | $\cos(\theta) = 1 / 8$                                                                                                                                  | $\cos(\theta) = 1 / 8$                                                                                                                                     | $u_2^2 = 1 / 7$ , and $\cos(\theta) = 1 / 8$ . Then focusing                                                                                                                 |
| johnston-linear-matrix-atlas-2-F04358 | 452  | 1.000 | $\left(\sqrt{3}, 1, \sqrt{3}\right) / \sqrt{7}$                                                                                                         | $\left(\sqrt{3}, 1, \sqrt{3}\right) / \sqrt{7}$                                                                                                            | $\left(\sqrt{3}, 1, \sqrt{3}\right) / \sqrt{7}$ and $\theta = -\arccos(1 / 8) \approx$                                                                                       |
| johnston-linear-matrix-atlas-2-F04359 | 452  | 1.000 | $\theta = -\arccos(1 / 8) \approx$                                                                                                                      | $\theta = -\arccos(1 / 8) \approx$                                                                                                                         | $\left(\sqrt{3}, 1, \sqrt{3}\right) / \sqrt{7}$ and $\theta = -\arccos(1 / 8) \approx$                                                                                       |
| johnston-linear-matrix-atlas-2-F04360 | 452  | 1.000 | $\theta = \arccos(1 / 8) \approx 1.4455$                                                                                                                | $\theta = \arccos(1 / 8) \approx 1.4455$                                                                                                                   | you choose $\theta = \arccos(1 / 8) \approx 1.4455$ then                                                                                                                     |
| johnston-linear-matrix-atlas-2-F04361 | 452  | 1.000 | $\sin(\theta)$                                                                                                                                          | $\sin(\theta)$                                                                                                                                             | the $\sin(\theta)$ terms will have the wrong sign,                                                                                                                           |
| johnston-linear-matrix-atlas-2-F04362 | 453  | 1.000 | $(A B)^3 = (A B)(A B)(A B)$                                                                                                                             | $(A B)^3 = (A B)(A B)(A B)$                                                                                                                                | (c) False. $(A B)^3 = (A B)(A B)(A B)$ and in general                                                                                                                        |
| johnston-linear-matrix-atlas-2-F04363 | 453  | 1.000 | $A^3 B^3 = A A A B B B$                                                                                                                                 | $A^3 B^3 = A A A B B B$                                                                                                                                    | necessarily equal $A^3 B^3 = A A B B B$ .                                                                                                                                    |

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|--------------------------------------|------|-------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-attrib F04364 | 453  | 1.000 | $R(S(T(\mathbf{v})))$                                                                             | $R(S(T(\mathbf{v})))$                                                                             | in $R(S(T(\mathbf{v})))$ .                                                                                                                               |
| johnston-linear-matrix-attrib F04365 | 453  | 0.774 | $\mathbf{u}=(1,0)$                                                                                | $\mathbf{u}=(1,0)$                                                                                | $\mathbf{u}=(1,0)$ .                                                                                                                                     |
| johnston-linear-matrix-attrib F04366 | 453  | 1.000 | $n\theta$                                                                                         | $n\theta$                                                                                         | of $R^\theta$ together results in a rotation by an angle of $n\theta$ ,                                                                                  |
| johnston-linear-matrix-attrib F04367 | 453  | 1.000 | $\ (2,1,0)\times(-2,3,0)\ =\ (0,0,8)\ =8$                                                         | $\ (2,1,0)\times(-2,3,0)\ =\ (0,0,8)\ =8$                                                         | <b>1.A.2</b> (a) $\ (2,1,0)\times(-2,3,0)\ =\ (0,0,8)\ =8$                                                                                               |
| johnston-linear-matrix-attrib F04368 | 453  | 1.000 | $\ (1,2,3)\times(3,-1,2)\ =\ (7,7,-7)\ =7\sqrt{3}$                                                | $\ (1,2,3)\times(3,-1,2)\ =\ (7,7,-7)\ =7\sqrt{3}$                                                | (c) $\ (1,2,3)\times(3,-1,2)\ =\ (7,7,-7)\ =7\sqrt{3}$                                                                                                   |
| johnston-linear-matrix-attrib F04369 | 453  | 1.000 | $\frac{1}{2}\ (0,4,0)\times(1,1,0)\ =\frac{1}{2}\ (0,0,-4)\ =2$                                   | $\frac{1}{2}\ (0,4,0)\times(1,1,0)\ =\frac{1}{2}\ (0,0,-4)\ =2$                                   | (a) $\frac{1}{2}\ (0,4,0)\times(1,1,0)\ =\frac{1}{2}\ (0,0,-4)\ =2$                                                                                      |
| johnston-linear-matrix-attrib F04370 | 453  | 1.000 | $\frac{1}{2}\ (-1,1,-1)\times(3,2,1)\ =\frac{1}{2}\ (3,-2,-5)\ $                                  | $\frac{1}{2}\ (-1,1,-1)\times(3,2,1)\ =\frac{1}{2}\ (3,-2,-5)\ $                                  | (c) $\frac{1}{2}\ (-1,1,-1)\times(3,2,1)\ =\frac{1}{2}\ (3,-2,-5)\ $                                                                                     |
| johnston-linear-matrix-attrib F04371 | 453  | 1.000 | $\frac{\sqrt{38}}{2}$                                                                             | $\frac{\sqrt{38}}{2}$                                                                             | $\frac{\sqrt{38}}{2}$                                                                                                                                    |
| johnston-linear-matrix-attrib F04372 | 453  | 0.826 | $1\cdot 2\cdot 3=6$                                                                               | $1\cdot 2\cdot 3=6$                                                                               | lengths 1, 2, and 3, so its volume is $1\cdot 2\cdot 3=6$ .                                                                                              |
| johnston-linear-matrix-attrib F04373 | 453  | 1.000 | $ \mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) = (1,1,1)\cdot(3,4,-2) =\mid 3+4-$                  | $ \mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) = (1,1,1)\cdot(3,4,-2) = 3+4-$                      | (c) $ \mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) = (1,1,1)\cdot(3,4,-2) = 3+4-$                                                                         |
| johnston-linear-matrix-attrib F04374 | 453  | 0.958 | $2\mid=5$                                                                                         | $2\mid=5$                                                                                         | $2\mid=5$                                                                                                                                                |
| johnston-linear-matrix-attrib F04375 | 453  | 1.000 | $\ (0,2,0)\times(-2,3,0)\ =\ (0,0,4)\ =4$                                                         | $\ (0,2,0)\times(-2,3,0)\ =\ (0,0,4)\ =4$                                                         | (c) $\ (0,2,0)\times(-2,3,0)\ =\ (0,0,4)\ =4$ .                                                                                                          |
| johnston-linear-matrix-attrib F04376 | 453  | 0.996 | $\mid(1,0,0)$                                                                                     | $\mid(1,0,0)$                                                                                     | (e) Now we use Theorem 1.A.3: $\mid(1,0,0)\cdot$                                                                                                         |
| johnston-linear-matrix-attrib F04377 | 453  | 0.998 | $((-3,2,0)\times(2,-7,3))\mid=6$                                                                  | $((-3,2,0)\times(2,-7,3))\mid=6$                                                                  | $((-3,2,0)\times(2,-7,3))\mid=6$ .                                                                                                                       |
| johnston-linear-matrix-attrib F04378 | 453  | 0.999 | $\ \mathbf{v}\times\mathbf{w}\ =\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$                         | $\ \mathbf{v}\times\mathbf{w}\ =\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$                         | <b>1.A.8</b> Recall that $\ \mathbf{v}\times\mathbf{w}\ =\ \mathbf{v}\ \ \mathbf{w}\ \sin(\theta)$ . We are told                                         |
| johnston-linear-matrix-attrib F04379 | 453  | 1.000 | $\ \mathbf{v}\ =\ \mathbf{w}\ =1$                                                                 | $\ \mathbf{v}\ =\ \mathbf{w}\ =1$                                                                 | so $\ \mathbf{v}\ =\ \mathbf{w}\ =1$ . Similarly, we can compute                                                                                         |
| johnston-linear-matrix-attrib F04380 | 453  | 1.000 | $\ \mathbf{v}\times\mathbf{w}\ =\sqrt{(1/3)^2+(2/3)^2+(2/3)^2}=1$                                 | $\ \mathbf{v}\times\mathbf{w}\ =\sqrt{(1/3)^2+(2/3)^2+(2/3)^2}=1$                                 | $\ \mathbf{v}\times\mathbf{w}\ =\sqrt{(1/3)^2+(2/3)^2+(2/3)^2}=1$ . It fol-                                                                              |
| johnston-linear-matrix-attrib F04381 | 453  | 1.000 | $1=\sin(\theta)$                                                                                  | $1=\sin(\theta)$                                                                                  | lows that $1=\sin(\theta)$ , so $\theta=\pi/2$ .                                                                                                         |
| johnston-linear-matrix-attrib F04382 | 454  | 1.000 | $\mathbf{v}=\mathbf{e}_1, \mathbf{w}=\mathbf{e}_2$                                                | $\mathbf{v}=\mathbf{e}_1, \mathbf{w}=\mathbf{e}_2$                                                | $\mathbf{v}=\mathbf{e}_1, \mathbf{w}=\mathbf{e}_2$ , and $\mathbf{x}=\mathbf{e}_2$ . Then $(\mathbf{v}\times\mathbf{w})\times\mathbf{x} =$               |
| johnston-linear-matrix-attrib F04383 | 454  | 1.000 | $\mathbf{x}=\mathbf{e}_2$                                                                         | $\mathbf{x}=\mathbf{e}_2$                                                                         | $\mathbf{v}=\mathbf{e}_1, \mathbf{w}=\mathbf{e}_2$ , and $\mathbf{x}=\mathbf{e}_2$ . Then $(\mathbf{v}\times\mathbf{w})\times\mathbf{x} =$               |
| johnston-linear-matrix-attrib F04384 | 454  | 1.000 | $(\mathbf{v}\times\mathbf{w})\times\mathbf{x} =$                                                  | $(\mathbf{v}\times\mathbf{w})\times\mathbf{x} =$                                                  | $\mathbf{v}=\mathbf{e}_1, \mathbf{w}=\mathbf{e}_2$ , and $\mathbf{x}=\mathbf{e}_2$ . Then $(\mathbf{v}\times\mathbf{w})\times\mathbf{x} =$               |
| johnston-linear-matrix-attrib F04385 | 454  | 1.000 | $(\mathbf{e}_1\times\mathbf{e}_2)\times\mathbf{e}_2=\mathbf{e}_3\times\mathbf{e}_2=-\mathbf{e}_1$ | $(\mathbf{e}_1\times\mathbf{e}_2)\times\mathbf{e}_2=\mathbf{e}_3\times\mathbf{e}_2=-\mathbf{e}_1$ | $(\mathbf{e}_1\times\mathbf{e}_2)\times\mathbf{e}_2=\mathbf{e}_3\times\mathbf{e}_2=-\mathbf{e}_1$ , but $\mathbf{v}\times(\mathbf{w}\times\mathbf{x}) =$ |
| johnston-linear-matrix-attrib F04386 | 454  | 1.000 | $\mathbf{v}\times(\mathbf{w}\times\mathbf{x}) =$                                                  | $\mathbf{v}\times(\mathbf{w}\times\mathbf{x}) =$                                                  | $(\mathbf{e}_1\times\mathbf{e}_2)\times\mathbf{e}_2=\mathbf{e}_3\times\mathbf{e}_2=-\mathbf{e}_1$ , but $\mathbf{v}\times(\mathbf{w}\times\mathbf{x}) =$ |
| johnston-linear-matrix-attrib F04387 | 454  | 1.000 | $\mathbf{e}_1\times(\mathbf{e}_2\times\mathbf{e}_2)=\mathbf{e}_1\times\mathbf{0}=\mathbf{0}$      | $\mathbf{e}_1\times(\mathbf{e}_2\times\mathbf{e}_2)=\mathbf{e}_1\times\mathbf{0}=\mathbf{0}$      | $\mathbf{e}_1\times(\mathbf{e}_2\times\mathbf{e}_2)=\mathbf{e}_1\times\mathbf{0}=\mathbf{0}$ .                                                           |
| johnston-linear-matrix-attrib F04388 | 454  | 1.000 | $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x})=\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$         | $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x})=\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$         | <b>1.A.16</b> (a) To show that $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x})=\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$ we explic-                      |
| johnston-linear-matrix-attrib F04389 | 454  | 1.000 | $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) =$                                                   | $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) =$                                                   | to see that they are equal, so $\mathbf{v}\cdot(\mathbf{w}\times\mathbf{x}) =$                                                                           |
| johnston-linear-matrix-attrib F04390 | 454  | 1.000 | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$                                                     | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$                                                     | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$ . The fact that $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})=\mathbf{x}\cdot(\mathbf{v}\times\mathbf{w})$  |
| johnston-linear-matrix-attrib F04391 | 454  | 1.000 | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})=\mathbf{x}\cdot(\mathbf{v}\times\mathbf{w})$         | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})=\mathbf{x}\cdot(\mathbf{v}\times\mathbf{w})$         | $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})$ . The fact that $\mathbf{w}\cdot(\mathbf{x}\times\mathbf{v})=\mathbf{x}\cdot(\mathbf{v}\times\mathbf{w})$  |

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| johnston-linear-matrix-attached-2<br>F04392 | 455  | 1.000 | $\left[A^2\right]_{1,2}=1$                                                                                                                                           | $[A^2]_{1,2} = 1$                                                                                                                                   | There is thus $[A^2]_{1,2} = 1$ path of length 2 from                                                                                               |
| johnston-linear-matrix-attached-2<br>F04393 | 455  | 1.000 | $\left[A^4\right]_{1,4}=9$                                                                                                                                           | $[A^4]_{1,4} = 9$                                                                                                                                   | There are thus $[A^4]_{1,4} = 9$ paths of length 4                                                                                                  |
| johnston-linear-matrix-attached-2<br>F04394 | 455  | 1.000 | $0 \leq k \leq n-1$                                                                                                                                                  | $0 \leq k \leq n-1$                                                                                                                                 | length $n$ : for any $0 \leq k \leq n-1$ , the self-loop                                                                                            |
| johnston-linear-matrix-attached-2<br>F04395 | 455  | 1.000 | $n-k-1$                                                                                                                                                              | $n-k-1$                                                                                                                                             | another $n-k-1$ times.                                                                                                                              |
| johnston-linear-matrix-attached-2<br>F04396 | 455  | 1.000 | $k \geq 1, \left[A^k\right]_{i,j}$                                                                                                                                   | $k \geq 1, [A^k]_{i,j}$                                                                                                                             | teger $k \geq 1$ , $[A^k]_{i,j}$ counts the number of paths of                                                                                      |
| johnston-linear-matrix-attached-2<br>F04397 | 455  | 0.983 | $\left[A^k\right]_{i,1} a_{1,j}$                                                                                                                                     | $[A^k]_{i,1} a_{1,j}$                                                                                                                               | The first term in the sum above ( $[A^k]_{i,1} a_{1,j}$ ) equals                                                                                    |
| johnston-linear-matrix-attached-2<br>F04398 | 455  | 1.000 | $\left[A^k\right]_{i,1}$                                                                                                                                             | $[A^k]_{i,1}$                                                                                                                                       | $[A^k]_{i,1}$ exactly if there is a path of length $k$ edge be-                                                                                     |
| johnston-linear-matrix-attached-2<br>F04399 | 455  | 1.000 | $k+1$                                                                                                                                                                | $k+1$                                                                                                                                               | counts the number of paths of length $k+1$ between                                                                                                  |
| johnston-linear-matrix-attached-2<br>F04400 | 455  | 1.000 | $\left[A^k\right]_{i,2} a_{2,j}$                                                                                                                                     | $[A^k]_{i,2} a_{2,j}$                                                                                                                               | Similarly, $[A^k]_{i,2} a_{2,j}$ counts the number of paths of                                                                                      |
| johnston-linear-matrix-attached-2<br>F04401 | 455  | 1.000 | $\left[A^{k+1}\right]_{i,j}$                                                                                                                                         | $[A^{k+1}]_{i,j}$                                                                                                                                   | adding these terms up, we see that $[A^{k+1}]_{i,j}$ counts                                                                                         |
| johnston-linear-matrix-attached-2<br>F04402 | 455  | 1.000 | $\left[\begin{array}{ll} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{array}\right]$                                                                                                 | $\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$                                                                                             | (c) $\begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$                                                                                         |
| johnston-linear-matrix-attached-2<br>F04403 | 455  | 1.000 | $\left[\begin{array}{lll} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right]$                                                                                    | $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                                                                 | (e) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                                                                             |
| johnston-linear-matrix-attached-2<br>F04404 | 455  | 1.000 | $\left[\begin{array}{lll} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right]$                                                                       | $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$                                                                    | (g) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$                                                                |
| johnston-linear-matrix-attached-2<br>F04405 | 455  | 1.000 | $\left[\begin{array}{cccccc} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{array}\right]$ | $\begin{bmatrix} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{bmatrix}$ | $\begin{bmatrix} 1 & 0 & 0 & 0 & -4/3 & 2/5 \\ 0 & 1 & 0 & 0 & -4/3 & -4/5 \\ 0 & 0 & 1 & 0 & 7/3 & 9/20 \\ 0 & 0 & 0 & 1 & 2 & 1/10 \end{bmatrix}$ |
| johnston-linear-matrix-attached-2<br>F04406 | 455  | 1.000 | $y: x$                                                                                                                                                               | $y : x$                                                                                                                                             | ables $x$ and $y$ : $x$ is free and $y = 2$ ).                                                                                                      |
| johnston-linear-matrix-attached-2<br>F04407 | 455  | 1.000 | $(x, y)=(1,1)$                                                                                                                                                       | $(x,y) = (1,1)$                                                                                                                                     | 2.1.5 (a) $(x,y) = (1,1)$ .                                                                                                                         |
| johnston-linear-matrix-attached-2<br>F04408 | 456  | 1.000 | $v, w$                                                                                                                                                               | $v, w$                                                                                                                                              | (g) Infinitely many solutions: The variables $v, w$ ,                                                                                               |
| johnston-linear-matrix-attached-2<br>F04409 | 456  | 0.998 | $v, w, x$                                                                                                                                                            | $v, w, x$                                                                                                                                           | 2.1.6 (a) Infinitely many solutions: $v, w, x$ , and $y$                                                                                            |
| johnston-linear-matrix-attached-2<br>F04410 | 456  | 1.000 | $(v, w, x, y, z)=$                                                                                                                                                   | $(v, w, x, y, z) =$                                                                                                                                 | solutions are of the form $(v, w, x, y, z) =$                                                                                                       |
| johnston-linear-matrix-attached-2<br>F04411 | 456  | 0.999 | $(-30, 3, 60, 7, 0)+z(-51, 3, 106, 5, 6) / 6$                                                                                                                        | $(-30, 3, 60, 7, 0) + z(-51, 3, 106, 5, 6)/6$                                                                                                       | $(-30, 3, 60, 7, 0) + z(-51, 3, 106, 5, 6)/6$ .                                                                                                     |
| johnston-linear-matrix-attached-2<br>F04412 | 456  | 0.978 | $(x, y$                                                                                                                                                              | $(x, y$                                                                                                                                             | 2.1.7 (a) True. Each of the variables $(x, y$ , and $z)$ are just                                                                                   |
| johnston-linear-matrix-attached-2<br>F04413 | 456  | 0.976 | $(\sin(1), \sqrt[3]{5})$                                                                                                                                             | $(\sin(1), \sqrt[3]{5})$                                                                                                                            | fact that the scalars $(\sin(1), \sqrt[3]{5})$ are ugly does                                                                                        |
| johnston-linear-matrix-attached-2<br>F04414 | 456  | 0.848 | $(0, 1, 1)$                                                                                                                                                          | $(0, 1, 1)$                                                                                                                                         | through the points $(0, 0, 1)$ and $(0, 1, 1)$ (i.e.,                                                                                               |

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|------------------------------------------|------|-------|---------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------------|
| johnston-linear-matrix-algebra<br>F04415 | 456  | 1.000 | $x=1, y=0$                                                                | $x = 1, y = 0$                                              | is $x = 1, y = 0$ .                                                                |
| johnston-linear-matrix-algebra<br>F04416 | 456  | 1.000 | $t \neq 1 / 2$                                                            | $t \neq 1/2$                                                | if and only if $t \neq 1/2$ . This linear system thus has                          |
| johnston-linear-matrix-algebra<br>F04417 | 456  | 1.000 | $t=1 / 2$                                                                 | $t = 1/2$                                                   | $t = 1/2$ , and (iii) never has infinitely many solutions.                         |
| johnston-linear-matrix-algebra<br>F04418 | 456  | 1.000 | $(w, x, y, z)=(3,-24,30,0)$                                               | $(w, x, y, z) = (3, -24, 30, 0)$                            | and $h = 1.05$ are $(w, x, y, z) = (3, -24, 30, 0)$ ,                              |
| johnston-linear-matrix-algebra<br>F04419 | 456  | 0.999 | $(w, x, y, z)=(-4,60,-180,140)$                                           | $(w, x, y, z) = (-4, 60, -180, 140)$                        | $(w, x, y, z) = (-4, 60, -180, 140)$ , and $(w, x, y, z) =$                        |
| johnston-linear-matrix-algebra<br>F04420 | 456  | 1.000 | $v_{\{2\}}=-2$                                                            | $v_2 = -2$                                                  | $v_3 = 1$ then gives us $v_2 = -2$ and $v_1 = 1$ , so                              |
| johnston-linear-matrix-algebra<br>F04421 | 456  | 1.000 | $v_{\{1\}}=1$                                                             | $v_1 = 1$                                                   | $v_3 = 1$ then gives us $v_2 = -2$ and $v_1 = 1$ , so                              |
| johnston-linear-matrix-algebra<br>F04422 | 456  | 1.000 | $\mathbf{v}=(1,-2,1)$                                                     | $\mathbf{v} = (1, -2, 1)$                                   | the vector $\mathbf{v} = (1, -2, 1)$ is orthogonal to both                         |
| johnston-linear-matrix-algebra<br>F04423 | 456  | 1.000 | $v_{\{4\}}=9$                                                             | $v_4 = 9$                                                   | $v_4 = 9$ then gives us $v_3 = -5, v_2 = -8$ , and                                 |
| johnston-linear-matrix-algebra<br>F04424 | 456  | 1.000 | $v_{\{3\}}=-5, v_{\{2\}}=-8$                                              | $v_3 = -5, v_2 = -8$                                        | $v_4 = 9$ then gives us $v_3 = -5, v_2 = -8$ , and                                 |
| johnston-linear-matrix-algebra<br>F04425 | 456  | 1.000 | $v_{\{1\}}=7$                                                             | $v_1 = 7$                                                   | $v_1 = 7$ , so the vector $\mathbf{v} = (7, -8, -5, 9)$ is or-                     |
| johnston-linear-matrix-algebra<br>F04426 | 456  | 1.000 | $\mathbf{v}=(7,-8,-5,9)$                                                  | $\mathbf{v} = (7, -8, -5, 9)$                               | $v_1 = 7$ , so the vector $\mathbf{v} = (7, -8, -5, 9)$ is or-                     |
| johnston-linear-matrix-algebra<br>F04427 | 456  | 1.000 | $(x, y)=(1,2)$                                                            | $(x, y) = (1, 2)$                                           | so the unique solution is $(x, y) = (1, 2)$ .                                      |
| johnston-linear-matrix-algebra<br>F04428 | 456  | 1.000 | $\mathbf{b}=(1 / 2) \mathbf{v}_{\{1\}}$                                   | $\mathbf{b} = (1/2)\mathbf{v}_1$                            | <b>2.1.17 (a)</b> Yes, $\mathbf{b} = (1/2)\mathbf{v}_1$ .                          |
| johnston-linear-matrix-algebra<br>F04429 | 456  | 1.000 | $\mathbf{b}=\mathbf{v}_{\{1\}}+5 \mathbf{v}_{\{2\}}-6 \mathbf{v}_{\{3\}}$ | $\mathbf{b} = \mathbf{v}_1 + 5\mathbf{v}_2 - 6\mathbf{v}_3$ | <b>(c)</b> Yes, $\mathbf{b} = \mathbf{v}_1 + 5\mathbf{v}_2 - 6\mathbf{v}_3$ .      |
| johnston-linear-matrix-algebra<br>F04430 | 456  | 1.000 | $\mathbf{b}=\mathbf{v}_{\{1\}}+3 \mathbf{v}_{\{2\}}$                      | $\mathbf{b} = \mathbf{v}_1 + 3\mathbf{v}_2$                 | <b>(e)</b> Yes, $\mathbf{b} = \mathbf{v}_1 + 3\mathbf{v}_2$ .                      |
| johnston-linear-matrix-algebra<br>F04431 | 457  | 1.000 | $\mathbf{b}=-\mathbf{v}_{\{1\}}+\mathbf{v}_{\{2\}}-2 \mathbf{v}_{\{3\}}$  | $\mathbf{b} = -\mathbf{v}_1 + \mathbf{v}_2 - 2\mathbf{v}_3$ | <b>(g)</b> Yes, $\mathbf{b} = -\mathbf{v}_1 + \mathbf{v}_2 - 2\mathbf{v}_3$ .      |
| johnston-linear-matrix-algebra<br>F04432 | 457  | 1.000 | $R_{\{1\}}$                                                               | $R_1$                                                       | By replacing $R_1$ with $w_1 R_1$ and then replacing $R_2$                         |
| johnston-linear-matrix-algebra<br>F04433 | 457  | 1.000 | $w_{\{1\}} R_{\{1\}}$                                                     | $w_1 R_1$                                                   | By replacing $R_1$ with $w_1 R_1$ and then replacing $R_2$                         |
| johnston-linear-matrix-algebra<br>F04434 | 457  | 1.000 | $R_{\{2\}}$                                                               | $R_2$                                                       | By replacing $R_1$ with $w_1 R_1$ and then replacing $R_2$                         |
| johnston-linear-matrix-algebra<br>F04435 | 457  | 1.000 | $R_{\{1\}}-v_{\{1\}} R_{\{2\}}$                                           | $R_1 - v_1 R_2$                                             | with $R_1 - v_1 R_2$ , the system becomes                                          |
| johnston-linear-matrix-algebra<br>F04436 | 457  | 1.000 | $w_{\{1\}} \neq 0$                                                        | $w_1 \neq 0$                                                | can only replace $R_1$ by $w_1 R_1$ if $w_1 \neq 0$ . However,                     |
| johnston-linear-matrix-algebra<br>F04437 | 457  | 1.000 | $\mathbf{w} \neq \mathbf{0}$                                              | $\mathbf{w} \neq \mathbf{0}$                                | since $\mathbf{w} \neq \mathbf{0}$ , we can always find <i>some</i> non-zero entry |
| johnston-linear-matrix-algebra<br>F04438 | 457  | 0.999 | $x_{\{1\}}, x_{\{2\}}, x_{\{3\}}$                                         | $x_1, x_2, x_3$                                             | where $x_1, x_2, x_3$ are the variables we are trying to                           |
| johnston-linear-matrix-algebra<br>F04439 | 457  | 1.000 | $x_{\{3\}}=c\left(v_{\{1\}} w_{\{2\}}-\right.$                            | $x_3 = c\left(v_1 w_2 -\right.$                             | If $c \in \mathbb{R}$ is any scalar, and we choose $x_3 = c\left(v_1 w_2 -\right.$ |
| johnston-linear-matrix-algebra<br>F04440 | 457  | 0.865 | $\left.v_{\{2\}} w_{\{1\}}\right)$                                        | $v_2 w_1$                                                   | $v_2 w_1$ ) then we find that $x_1 = c\left(v_2 w_3 - v_3 w_2\right)$ and          |
| johnston-linear-matrix-algebra<br>F04441 | 457  | 0.865 | $x_{\{1\}}=c\left(v_{\{2\}} w_{\{3\}}-v_{\{3\}} w_{\{2\}}\right)$         | $x_1 = c\left(v_2 w_3 - v_3 w_2\right)$                     | $v_2 w_1$ ) then we find that $x_1 = c\left(v_2 w_3 - v_3 w_2\right)$ and          |
| johnston-linear-matrix-algebra<br>F04442 | 457  | 1.000 | $x_{\{2\}}=c\left(v_{\{3\}} w_{\{1\}}-v_{\{1\}} w_{\{3\}}\right)$         | $x_2 = c\left(v_3 w_1 - v_1 w_3\right)$                     | $x_2 = c\left(v_3 w_1 - v_1 w_3\right)$ , so                                       |
| johnston-linear-matrix-algebra<br>F04443 | 457  | 0.950 | $x=45$                                                                    | $x = 45$                                                    | $x = 45$ kilograms of the 13% salt water and $y = 75$                              |

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| johnston-linear-matrix-algebra-F04444 | 457  | ■ 0.950 | y=75                                                            | $y = 75$                                                | $x = 45$ kilograms of the 13% salt water and $y = 75$                                                   |
| johnston-linear-matrix-algebra-F04445 | 457  | ■ 1.000 | w=z                                                             | $w = z$                                                 | Solving this linear system shows that $z$ is free, $w = z$ ,                                            |
| johnston-linear-matrix-algebra-F04446 | 457  | ■ 1.000 | x=3 z / 2                                                       | $x = 3z/2$                                              | $x = 3z/2$ , and $y = z$ . Choosing $z = 2$ so that all of                                              |
| johnston-linear-matrix-algebra-F04447 | 457  | ■ 1.000 | y=z                                                             | $y = z$                                                 | $x = 3z/2$ , and $y = z$ . Choosing $z = 2$ so that all of                                              |
| johnston-linear-matrix-algebra-F04448 | 457  | ■ 1.000 | v_{1}=18-2 v_{2}-2 v_{3}                                        | $v_1 = 18 - 2v_2 - 2v_3$                                | leading variable with $v_1 = 18 - 2v_2 - 2v_3$ . The vec-                                               |
| johnston-linear-matrix-algebra-F04449 | 457  | ■ 1.000 | \mathbf{v}=\left(18-2 v_{2}-2 v_{3}, v_{2}, v_{3}\right)        | $\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$             | form $\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$ , where $v_2, v_3 \in \mathbb{R}$                      |
| johnston-linear-matrix-algebra-F04450 | 457  | ■ 1.000 | v_{2}, v_{3} \in \mathbb{R}                                     | $v_2, v_3 \in \mathbb{R}$                               | form $\mathbf{v} = (18 - 2v_2 - 2v_3, v_2, v_3)$ , where $v_2, v_3 \in \mathbb{R}$                      |
| johnston-linear-matrix-algebra-F04451 | 457  | ■ 0.999 | 0=-1 / 3                                                        | $0 = -1/3$                                              | to the equation $0 = -1/3$ , we see that there                                                          |
| johnston-linear-matrix-algebra-F04452 | 457  | ■ 1.000 | T(1,0)                                                          | $T(1, 0)$                                               | <b>2.1.26</b> (a) If we can find $T(1, 0)$ and $T(0, 1)$ then we can                                    |
| johnston-linear-matrix-algebra-F04453 | 457  | ■ 1.000 | T(0,1)                                                          | $T(0, 1)$                                               | <b>2.1.26</b> (a) If we can find $T(1, 0)$ and $T(0, 1)$ then we can                                    |
| johnston-linear-matrix-algebra-F04454 | 457  | ■ 1.000 | c_{1}=c_{2}=c_{3}=1 / 2                                         | $c_1 = c_2 = c_3 = 1/2$                                 | easily solved to get $c_1 = c_2 = c_3 = 1/2$ and                                                        |
| johnston-linear-matrix-algebra-F04455 | 457  | ■ 1.000 | c_{4}=-1 / 2                                                    | $c_4 = -1/2$                                            | $c_4 = -1/2$ . It follows that                                                                          |
| johnston-linear-matrix-algebra-F04456 | 457  | ■ 1.000 | T(0,1)=                                                         | $T(0, 1) =$                                             | A similar calculation shows that $T(0, 1) =$                                                            |
| johnston-linear-matrix-algebra-F04457 | 458  | ■ 1.000 | u=1 / x, v=1 / y                                                | $u = 1/x, v = 1/y$                                      | (c) Let $u = 1/x$ , $v = 1/y$ , and $w = 1/z$ . Then the                                                |
| johnston-linear-matrix-algebra-F04458 | 458  | ■ 1.000 | w=1 / z                                                         | $w = 1/z$                                               | (c) Let $u = 1/x$ , $v = 1/y$ , and $w = 1/z$ . Then the                                                |
| johnston-linear-matrix-algebra-F04459 | 458  | ■ 1.000 | (u, v, w)=(2, -1,3)                                             | $(u, v, w) = (2, -1, 3)$                                | us the unique solution $(u, v, w) = (2, -1, 3)$ .                                                       |
| johnston-linear-matrix-algebra-F04460 | 458  | ■ 1.000 | (x, y, z)=(1 / 2,-1,1 / 3)                                      | $(x, y, z) = (1/2, -1, 1/3)$                            | $w = 1/z$ then gives us $(x, y, z) = (1/2, -1, 1/3)$                                                    |
| johnston-linear-matrix-algebra-F04461 | 458  | ■ 0.995 | \left[\begin{array}{ll} a & b \\ c & d \end{array}\right]       | $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$          | <b>2.1.28</b> (a) We need to find all matrices $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ such that |
| johnston-linear-matrix-algebra-F04462 | 458  | ■ 1.000 | c=b, d=a                                                        | $c = b, d = a$                                          | $c = b, d = a$ . The matrices that commute with                                                         |
| johnston-linear-matrix-algebra-F04463 | 458  | ■ 0.963 | \left[\begin{array}{cc} a & b \\ b & a \end{array}\right]       | $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$          | $A$ are thus those of the form $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$ .                         |
| johnston-linear-matrix-algebra-F04464 | 458  | ■ 1.000 | c=b, d=a-b                                                      | $c = b, d = a - b$                                      | $c = b, d = a - b$ . The matrices that commute                                                          |
| johnston-linear-matrix-algebra-F04465 | 458  | ■ 0.998 | \left[\begin{array}{cc} a & b \\ b & a-b \end{array}\right]     | $\begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$      | with $C$ are thus those of the form $\begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$ .                |
| johnston-linear-matrix-algebra-F04466 | 458  | ■ 1.000 | B C=C B                                                         | $BC = CB$                                               | in particular $AB = BA$ and $BC = CB$ , where                                                           |
| johnston-linear-matrix-algebra-F04467 | 458  | ■ 1.000 | B=\left[\begin{array}{cc} a & b \\ b & a-b \end{array}\right]   | $B = \begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$  | and (b) of this question, so $B = \begin{bmatrix} a & b \\ b & a - b \end{bmatrix}$                     |
| johnston-linear-matrix-algebra-F04468 | 458  | ■ 1.000 | a-b=a                                                           | $a - b = a$                                             | (by part (b)), and $a - b = a$ (by part (a)), so                                                        |
| johnston-linear-matrix-algebra-F04469 | 458  | ■ 0.689 | B=\left[\begin{array}{ll} a & 0 \\ 0 & a \end{array}\right]=a I | $B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$ | $b = 0$ , so $B = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} = aI$ . On the other hand,               |

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| johnston-linear-matrix-attached-<br>F04470 | 458  | ■ 0.941 | $a \ I$                                                                                                   | $aI$                                                                              | it is clear that the matrix $aI$ commutes with                                                     |
| johnston-linear-matrix-attached-<br>F04471 | 458  | ■ 0.901 | $(a \ I) \ A=a \ A=A(a \ I)$                                                                              | $(aI)A = aA = A(aI)$                                                              | everything in $\mathcal{M}_2$ since $(aI)A = aA = A(aI)$ .                                         |
| johnston-linear-matrix-attached-<br>F04472 | 458  | ■ 1.000 | $(b, \ m)=(-1,3)$                                                                                         | $(b, m) = (-1, 3)$                                                                | has unique solution $(b, m) = (-1, 3)$ , so the                                                    |
| johnston-linear-matrix-attached-<br>F04473 | 458  | ■ 1.000 | $y=3 \ x-1$                                                                                               | $y = 3x - 1$                                                                      | unique line that works is $y = 3x - 1$ .                                                           |
| johnston-linear-matrix-attached-<br>F04474 | 458  | ■ 1.000 | $(a, \ b, \ c)=(2, -3,4)$                                                                                 | $(a, b, c) = (2, -3, 4)$                                                          | has unique solution $(a, b, c) = (2, -3, 4)$ , so the                                              |
| johnston-linear-matrix-attached-<br>F04475 | 458  | ■ 1.000 | $y=2 \ x^{\{2\}}-3 \ x+4$                                                                                 | $y = 2x^2 - 3x + 4$                                                               | unique parabola that works is $y = 2x^2 - 3x + 4$ .                                                |
| johnston-linear-matrix-attached-<br>F04476 | 458  | ■ 0.999 | $d=4$                                                                                                     | $d = 4$                                                                           | choose $d = 4$ , we get exactly the parabola that                                                  |
| johnston-linear-matrix-attached-<br>F04477 | 459  | ■ 0.999 | $\left[\begin{array}{lll}0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0\end{array}\right]$                           | $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$               | (a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$                            |
| johnston-linear-matrix-attached-<br>F04478 | 459  | ■ 1.000 | $\left[\begin{array}{lll}1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]$                           | $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$               | (c) $\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                            |
| johnston-linear-matrix-attached-<br>F04479 | 459  | ■ 1.000 | $\left[\begin{array}{lll}3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1\end{array}\right]$                           | $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$               | (e) $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$                            |
| johnston-linear-matrix-attached-<br>F04480 | 459  | ■ 1.000 | $\left[\begin{array}{cc}-5 / 3 & 2 / 3 \\ 4 / 3 & -1 / 3\end{array}\right]$                               | $\begin{bmatrix} -5/3 & 2/3 \\ 4/3 & -1/3 \end{bmatrix}$                          | (c) $\begin{bmatrix} -5/3 & 2/3 \\ 4/3 & -1/3 \end{bmatrix}$                                       |
| johnston-linear-matrix-attached-<br>F04481 | 459  | ■ 1.000 | $\left[\begin{array}{ccc}0 & -1 / 3 & 2 / 3 \\ 0 & 1 / 3 & 1 / 3 \\ 1 & 1 / 3 & -5 / 3\end{array}\right]$ | $\begin{bmatrix} 0 & -1/3 & 2/3 \\ 0 & 1/3 & 1/3 \\ 1 & 1/3 & -5/3 \end{bmatrix}$ | (e) $\begin{bmatrix} 0 & -1/3 & 2/3 \\ 0 & 1/3 & 1/3 \\ 1 & 1/3 & -5/3 \end{bmatrix}$ (not unique) |
| johnston-linear-matrix-attached-<br>F04482 | 459  | ■ 1.000 | $\left[\begin{array}{ccc}0 & 0 & 1 \\ 0 & -1 / 2 & 3 / 2 \\ 1 & 1 / 2 & -3 / 2\end{array}\right]$         | $\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1/2 & 3/2 \\ 1 & 1/2 & -3/2 \end{bmatrix}$     | (g) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1/2 & 3/2 \\ 1 & 1/2 & -3/2 \end{bmatrix}$ (not unique)     |
| johnston-linear-matrix-attached-<br>F04483 | 459  | ■ 1.000 | $B=-I$                                                                                                    | $B = -I$                                                                          | (g) False. For example, if $A = I$ and $B = -I$ then                                               |
| johnston-linear-matrix-attached-<br>F04484 | 459  | ■ 1.000 | $A+B=0$                                                                                                   | $A + B = O$                                                                       | $A + B = O$ , which is not invertible.                                                             |
| johnston-linear-matrix-attached-<br>F04485 | 459  | ■ 1.000 | $A^{\{-1\} \ 7}$                                                                                          | $A^{-1}7$                                                                         | tiply the equation $A^7 = O$ on the left by $A^{-1} \ 7$                                           |
| johnston-linear-matrix-attached-<br>F04486 | 459  | ■ 1.000 | $I=0$                                                                                                     | $I = O$                                                                           | which simplifies to $I = O$ , which is false. It                                                   |
| johnston-linear-matrix-attached-<br>F04487 | 459  | ■ 1.000 | $X=B \ A^{\{-1\}}$                                                                                        | $X = BA^{-1}$                                                                     | $A^{-1}$ gives $X = BA^{-1}$ , which in general is not                                             |
| johnston-linear-matrix-attached-<br>F04488 | 459  | ■ 1.000 | $A^{\{-1\} \ B}$                                                                                          | $A^{-1}B$                                                                         | the same as $A^{-1}B$ since matrix multiplication is                                               |
| johnston-linear-matrix-attached-<br>F04489 | 459  | ■ 1.000 | $A=(a-b) \ I_{\{n\}}+b \ J_{\{n\}}$                                                                       | $A = (a - b)I_n + bJ_n$                                                           | ery entry equal to 1, then $A = (a - b)I_n + bJ_n$ .                                               |
| johnston-linear-matrix-attached-<br>F04490 | 459  | ■ 1.000 | $B=c \ I_{\{n\}}+d \ J_{\{n\}}$                                                                           | $B = cI_n + dJ_n$                                                                 | trix of the form $B = cI_n + dJ_n$ for some $c, d \in \mathbb{R}$                                  |
| johnston-linear-matrix-attached-<br>F04491 | 460  | ■ 1.000 | $c=1 \ /(a-b)$                                                                                            | $c = 1/(a - b)$                                                                   | choose $c = 1/(a - b)$ and then solve $bc +$                                                       |
| johnston-linear-matrix-attached-<br>F04492 | 460  | ■ 1.000 | $b \ c+$                                                                                                  | $bc+$                                                                             | choose $c = 1/(a - b)$ and then solve $bc +$                                                       |
| johnston-linear-matrix-attached-<br>F04493 | 460  | ■ 0.995 | $d(a-b)+n \ b \ d=0$                                                                                      | $d(a - b) + nbd = 0$                                                              | $d(a - b) + nbd = 0$ to get $d = -b(nd + 1/(a -$                                                   |

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| johnston-linear-matrix-atg6b_u<br>F04494 | 460  | ■ 0.995 | d=-b(n d+1 /(a-                                                                                | $d = -b(nd + 1/(a -$                                                                                                | $d(a - b) + nbd = 0$ to get $d = -b(nd + 1/(a -$                                                                        |
| johnston-linear-matrix-atg6b_u<br>F04495 | 460  | ■ 0.992 | b)) /(a-b)                                                                                     | $b))/ (a - b)$                                                                                                      | $b))/ (a - b)$ . It follows that $A$ is invertible, and                                                                 |
| johnston-linear-matrix-atg6b_u<br>F04496 | 460  | ■ 0.985 | A^{2}=\left[\begin{array}{cc}7 & 4 \\ 12 & 7\end{array}\right]                                 | $A^2 = \left[ \begin{array}{cc} 7 & 4 \\ 12 & 7 \end{array} \right]$                                                | <b>2.2.9</b> (a) $A^2 = \left[ \begin{array}{cc} 7 & 4 \\ 12 & 7 \end{array} \right]$                                   |
| johnston-linear-matrix-atg6b_u<br>F04497 | 460  | ■ 1.000 | A^{-2}=\left[\begin{array}{cc}7 & -4 \\ -12 & 7\end{array}\right]                              | $A^{-2} = \left[ \begin{array}{cc} 7 & -4 \\ -12 & 7 \end{array} \right]$                                           | (c) $A^{-2} = \left[ \begin{array}{cc} 7 & -4 \\ -12 & 7 \end{array} \right]$                                           |
| johnston-linear-matrix-atg6b_u<br>F04498 | 460  | ■ 1.000 | m \times n, A^T A                                                                              | $m \times n, A^T A$                                                                                                 | <b>2.2.11</b> (a) Since $A$ is $m \times n$ , $A^T A$ is $n \times n$ , so $(A^T A)^{-1}$ is                            |
| johnston-linear-matrix-atg6b_u<br>F04499 | 460  | ■ 1.000 | \left(A^T A\right)^{-1}                                                                        | $(A^T A)^{-1}$                                                                                                      | <b>2.2.11</b> (a) Since $A$ is $m \times n$ , $A^T A$ is $n \times n$ , so $(A^T A)^{-1}$ is                            |
| johnston-linear-matrix-atg6b_u<br>F04500 | 460  | ■ 1.000 | P^T=\left(A\left(A^T A\right)^{-1} A^T\right)^T=A^T\left(\left(A^T A\right)^{-1}\right)^T A^T= | $P^T = \left( A \left( A^T A \right)^{-1} A^T \right)^T = (A^T)^T \left( \left( A^T A \right)^{-1} \right)^T A^T =$ | (b) $P^T = \left( A \left( A^T A \right)^{-1} A^T \right)^T = (A^T)^T \left( \left( A^T A \right)^{-1} \right)^T A^T =$ |
| johnston-linear-matrix-atg6b_u<br>F04501 | 460  | ■ 0.968 | A\left(\left(A^T A\right)^T\right)^{-1} A^T=A\left(A^T A\right)^{-1} A^T=P                     | $A \left( \left( A^T A \right)^T \right)^{-1} A^T = A \left( A^T A \right)^{-1} A^T = P$                            | $A \left( \left( A^T A \right)^T \right)^{-1} A^T = A \left( A^T A \right)^{-1} A^T = P$ .                              |
| johnston-linear-matrix-atg6b_u<br>F04502 | 460  | ■ 1.000 | E_{\mathrm{k}} \cdots E_2 E_1 P=I                                                              | $E_k \cdots E_2 E_1 P = I$                                                                                          | then $E_k \cdots E_2 E_1 P = I$ , so $E_k \cdots E_2 E_1 = P^{-1}$ , so the                                             |
| johnston-linear-matrix-atg6b_u<br>F04503 | 460  | ■ 1.000 | E_{\mathrm{k}} \cdots E_2 E_1 =P^{-1}                                                          | $E_k \cdots E_2 E_1 = P^{-1}$                                                                                       | then $E_k \cdots E_2 E_1 P = I$ , so $E_k \cdots E_2 E_1 = P^{-1}$ , so the                                             |
| johnston-linear-matrix-atg6b_u<br>F04504 | 460  | ■ 1.000 | P^{-1} Q                                                                                       | $P^{-1} Q$                                                                                                          | matrix on the right is $P^{-1} Q$ , as claimed.                                                                         |
| johnston-linear-matrix-atg6b_u<br>F04505 | 460  | ■ 1.000 | X=                                                                                             | $X =$                                                                                                               | <b>2.2.13</b> (a) Multiplying on the left by $A^{-1}$ gives $X =$                                                       |
| johnston-linear-matrix-atg6b_u<br>F04506 | 460  | ■ 1.000 | X=A^{-1} I B^{-1}=A^{-1} B^{-1}                                                                | $X = A^{-1} I B^{-1} = A^{-1} B^{-1}$                                                                               | by $B^{-1}$ gives $X = A^{-1} I B^{-1} = A^{-1} B^{-1}$ .                                                               |
| johnston-linear-matrix-atg6b_u<br>F04507 | 460  | ■ 1.000 | a_{\mathrm{n}, \mathrm{n}} x_{\mathrm{n}}=0                                                    | $a_{n,n} x_n = 0$                                                                                                   | substitution. The final equation says $a_{n,n} x_n = 0$ ,                                                               |
| johnston-linear-matrix-atg6b_u<br>F04508 | 460  | ■ 1.000 | x_{\mathrm{n}}=0                                                                               | $x_n = 0$                                                                                                           | which implies $x_n = 0$ as long as $a_{n,n} \neq 0$ . Plug-                                                             |
| johnston-linear-matrix-atg6b_u<br>F04509 | 460  | ■ 1.000 | a_{\mathrm{n}, \mathrm{n}} \neq 0                                                              | $a_{n,n} \neq 0$                                                                                                    | which implies $x_n = 0$ as long as $a_{n,n} \neq 0$ . Plug-                                                             |
| johnston-linear-matrix-atg6b_u<br>F04510 | 460  | ■ 0.997 | a_{\mathrm{n}-1, \mathrm{n}-1} x_{\mathrm{n}-1}=0                                              | $a_{n-1,n-1} x_{n-1} = 0$                                                                                           | $a_{n-1,n-1} x_{n-1} = 0$ , which implies $x_{n-1} = 0$ as                                                              |
| johnston-linear-matrix-atg6b_u<br>F04511 | 460  | ■ 0.997 | x_{\mathrm{n}-1}=0                                                                             | $x_{n-1} = 0$                                                                                                       | $a_{n-1,n-1} x_{n-1} = 0$ , which implies $x_{n-1} = 0$ as                                                              |
| johnston-linear-matrix-atg6b_u<br>F04512 | 460  | ■ 1.000 | a_{\mathrm{n}-1, \mathrm{n}-1} \neq 0                                                          | $a_{n-1,n-1} \neq 0$                                                                                                | long as $a_{n-1,n-1} \neq 0$ . Continuing in this way                                                                   |
| johnston-linear-matrix-atg6b_u<br>F04513 | 460  | ■ 1.000 | a_{\mathrm{j}, \mathrm{j}}=0                                                                   | $a_{j,j} = 0$                                                                                                       | entries equals 0 (let's say $a_{j,j} = 0$ ) then the                                                                    |
| johnston-linear-matrix-atg6b_u<br>F04514 | 460  | ■ 1.000 | a_{\mathrm{j}, \mathrm{j}} x_{\mathrm{j}}=0                                                    | $a_{j,j} x_j = 0$                                                                                                   | equation $a_{j,j} x_j = 0$ gives no restriction on $x_j$ ,                                                              |
| johnston-linear-matrix-atg6b_u<br>F04515 | 460  | ■ 1.000 | x_{\mathrm{j}}                                                                                 | $x_j$                                                                                                               | equation $a_{j,j} x_j = 0$ gives no restriction on $x_j$ ,                                                              |
| johnston-linear-matrix-atg6b_u<br>F04516 | 460  | ■ 1.000 | A \mathbf{x}_{\mathrm{j}}=\mathbf{e}_{\mathrm{j}}                                              | $A \mathbf{x}_j = \mathbf{e}_j$                                                                                     | so $A \mathbf{x}_j = \mathbf{e}_j$ for all $1 \leq j \leq n$ . Since the last                                           |
| johnston-linear-matrix-atg6b_u<br>F04517 | 460  | ■ 0.994 | n-j                                                                                            | $n - j$                                                                                                             | $n - j$ entries of $\mathbf{e}_j$ all equal 0, the last $n - j$                                                         |
| johnston-linear-matrix-atg6b_u<br>F04518 | 460  | ■ 1.000 | \mathbf{x}_{\mathrm{j}}                                                                        | $\mathbf{x}_j$                                                                                                      | entries of $\mathbf{x}_j$ must equal 0 as well, so $A^{-1}$ is                                                          |
| johnston-linear-matrix-atg6b_u<br>F04519 | 460  | ■ 0.999 | b_{\mathrm{1}}, b_{\mathrm{2}}, \ldots, b_{\mathrm{n}}                                         | $b_1, b_2, \dots, b_n$                                                                                              | entries of $A^{-1}$ by $b_1, b_2, \dots, b_n$ , and use aster-                                                          |
| johnston-linear-matrix-atg6b_u<br>F04520 | 460  | ■ 1.000 | b_{\mathrm{1}}=1 / a_{\mathrm{1}}, b_{\mathrm{2}}=1 / a_{\mathrm{2}}                           | $b_1 = 1/a_1, b_2 = 1/a_2$                                                                                          | $b_1 = 1/a_1, b_2 = 1/a_2$ , and so on.                                                                                 |

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| johnston-linear-matrix-algebra-F04521 | 460  | ■ 0.999 | $\left(A^{-1}\right)^{-1}$                                                                                                                 | $\left(A^{-1}\right)^{-1}$                                          | <b>2.2.17</b> (a) An inverse of $A^{-1}$ is a matrix $\left(A^{-1}\right)^{-1}$ such |
| johnston-linear-matrix-algebra-F04522 | 460  | ■ 1.000 | $A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1} A^{-1}=I$<br>$A^{-1}=I$                                                           | $A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1} A^{-1}=I$  | that $A^{-1}\left(A^{-1}\right)^{-1}=\left(A^{-1}\right)^{-1} A^{-1}=I$ . Replac-    |
| johnston-linear-matrix-algebra-F04523 | 460  | ■ 1.000 | $\left(A^{-1}\right)^T A^T=A^T\left(A^{-1}\right)^T=I$<br>$A^T=A^T\left(A^{-1}\right)^T=I$                                                 | $\left(A^{-1}\right)^T A^T=A^T\left(A^{-1}\right)^T=I$              | show that $\left(A^{-1}\right)^T A^T=A^T\left(A^{-1}\right)^T=I$ . Well,             |
| johnston-linear-matrix-algebra-F04524 | 460  | ■ 0.997 | $I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$<br>$A^T=A^T$                                                                     | $I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$           | $I=I^T=\left(A A^{-1}\right)^T=\left(A^{-1}\right)^T A^T$ , and similarly            |
| johnston-linear-matrix-algebra-F04525 | 460  | ■ 1.000 | $I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$<br>$A^T=A^T$                                                                      | $I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$            | $I=I^T=\left(A^{-1} A\right)^T=A^T\left(A^{-1}\right)^T$ , as desired.               |
| johnston-linear-matrix-algebra-F04526 | 460  | ■ 1.000 | $A^{-1} k$                                                                                                                                 | $A^{-1} k$                                                          | (c) Notice that if we multiply $A^k$ by $A^{-1}$ $k$ times                           |
| johnston-linear-matrix-algebra-F04527 | 460  | ■ 1.000 | $A A^{-1}$                                                                                                                                 | $A A^{-1}$                                                          | The central $A A^{-1}$ cancels to give an identity                                   |
| johnston-linear-matrix-algebra-F04528 | 460  | ■ 1.000 | $A^k\left(A^{-1}\right)^k=I$                                                                                                               | $A^k\left(A^{-1}\right)^k=I$                                        | matrix left. It follows that $A^k\left(A^{-1}\right)^k=I$ (and                       |
| johnston-linear-matrix-algebra-F04529 | 460  | ■ 1.000 | $\left(A^{-1}\right)^k A^k=I$                                                                                                              | $\left(A^{-1}\right)^k A^k=I$                                       | the equation $\left(A^{-1}\right)^k A^k=I$ is proved similarly),                     |
| johnston-linear-matrix-algebra-F04530 | 461  | ■ 1.000 | $k, \ell \geq 0$                                                                                                                           | $k, \ell \geq 0$                                                    | <b>2.2.18</b> (a) We already proved this claim when $k, \ell \geq 0$ in              |
| johnston-linear-matrix-algebra-F04531 | 461  | ■ 1.000 | $k, \ell < 0$                                                                                                                              | $k, \ell < 0$                                                       | Exercise 1.3.22, and it follows when $k, \ell < 0$                                   |
| johnston-linear-matrix-algebra-F04532 | 461  | ■ 1.000 | $\ell < 0$                                                                                                                                 | $\ell < 0$                                                          | the case when $k \geq 0$ and $\ell < 0$ . In this case, we                           |
| johnston-linear-matrix-algebra-F04533 | 461  | ■ 1.000 | $A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$<br>$A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$ | $A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$ | notice that $A^k A^\ell=(A A \cdots A)\left(A^{-1} A^{-1} \cdots A^{-1}\right)$ ,    |
| johnston-linear-matrix-algebra-F04534 | 461  | ■ 1.000 | $ \ell =-\ell$                                                                                                                             | $ \ell =-\ell$                                                      | the second has $ \ell =-\ell$ copies of $A^{-1}$ . The                               |
| johnston-linear-matrix-algebra-F04535 | 461  | ■ 1.000 | $k- \ell =k+\ell$                                                                                                                          | $k- \ell =k+\ell$                                                   | ing just $k- \ell =k+\ell$ copies of $A$ (if $k \geq  \ell $ )                       |
| johnston-linear-matrix-algebra-F04536 | 461  | ■ 1.000 | $k \geq \ell $                                                                                                                             | $k \geq \ell $                                                      | ing just $k- \ell =k+\ell$ copies of $A$ (if $k \geq  \ell $ )                       |
| johnston-linear-matrix-algebra-F04537 | 461  | ■ 1.000 | $ \ell -k=-(k+\ell)$                                                                                                                       | $ \ell -k=-(k+\ell)$                                                | or $ \ell -k=-(k+\ell)$ copies of $A^{-1}$ (if $ \ell  \geq k$ ).                    |
| johnston-linear-matrix-algebra-F04538 | 461  | ■ 1.000 | $ \ell >k$                                                                                                                                 | $ \ell >k$                                                          | or $ \ell -k=-(k+\ell)$ copies of $A^{-1}$ (if $ \ell  \geq k$ ).                    |
| johnston-linear-matrix-algebra-F04539 | 461  | ■ 1.000 | $k \geq 0, \ell < 0$                                                                                                                       | $k \geq 0, \ell < 0$                                                | (b) Again, we just explicitly prove the $k \geq 0, \ell < 0$                         |
| johnston-linear-matrix-algebra-F04540 | 461  | ■ 1.000 | $\left(A^k\right)^\ell=$                                                                                                                   | $\left(A^k\right)^\ell=$                                            | similarly. In this case, notice that $\left(A^k\right)^\ell=$                        |
| johnston-linear-matrix-algebra-F04541 | 461  | ■ 0.999 | $(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$<br>$(A A \cdots A)^{-1}$                                                | $(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$  | $(A A \cdots A)^{-1}(A A \cdots A)^{-1} \cdots(A A \cdots A)^{-1}$ ,                 |
| johnston-linear-matrix-algebra-F04542 | 461  | ■ 1.000 | $(A A \cdots A)^{-1}=$                                                                                                                     | $(A A \cdots A)^{-1}=$                                              | are $ \ell =-\ell$ brackets. Since $(A A \cdots A)^{-1}=$                            |
| johnston-linear-matrix-algebra-F04543 | 461  | ■ 1.000 | $A^{-1} A^{-1} \cdots A^{-1}$                                                                                                              | $A^{-1} A^{-1} \cdots A^{-1}$                                       | $A^{-1} A^{-1} \cdots A^{-1}$ , it follows that there are a total                    |
| johnston-linear-matrix-algebra-F04544 | 461  | ■ 1.000 | $k \ell =-k \ell$                                                                                                                          | $k \ell =-k \ell$                                                   | of $k \ell =-k \ell$ copies of $A^{-1}$ , so this product                            |
| johnston-linear-matrix-algebra-F04545 | 461  | ■ 1.000 | $A^{- k \ell }=A^{k \ell}$                                                                                                                 | $A^{- k \ell }=A^{k \ell}$                                          | equals $A^{- k \ell }=A^{k \ell}$ .                                                  |
| johnston-linear-matrix-algebra-F04546 | 461  | ■ 1.000 | $B^T A^T=(A B)^T=I^T=I$                                                                                                                    | $B^T A^T=(A B)^T=I^T=I$                                             | gives $B^T A^T=(A B)^T=I^T=I$ . It follows from the                                  |
| johnston-linear-matrix-algebra-F04547 | 461  | ■ 1.000 | $\left(A^T\right)^{-1}=B^T$                                                                                                                | $\left(A^T\right)^{-1}=B^T$                                         | $A^T$ is invertible and $\left(A^T\right)^{-1}=B^T$ . Using the fact                 |
| johnston-linear-matrix-algebra-F04548 | 461  | ■ 0.992 | $A_{11} B_{1,1}, A_{22} B_{2,2}, \ldots, A_{nn} B_{n,n}$                                                                                   | $A_1 B_{1,1}, A_2 B_{2,2}, \ldots, A_n B_{n,n}$                     | then the products $A_1 B_{1,1}, A_2 B_{2,2}, \ldots, A_n B_{n,n}$                    |

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| johnston-linear-matrix-atlas-<br>F04549 | 461  | 1.000 | $\left[\begin{array}{cc} A & I \\ 0 & I \end{array}\right] = \left[\begin{array}{cc} I & A-A \\ 0 & I \end{array}\right]$ | $\begin{bmatrix} A & I \\ I & O \end{bmatrix} \begin{bmatrix} O & I \\ I & -A \end{bmatrix} = \begin{bmatrix} I & A-A \\ O & I \end{bmatrix} =$ | (a) $\begin{bmatrix} A & I \\ I & O \end{bmatrix} \begin{bmatrix} O & I \\ I & -A \end{bmatrix} = \begin{bmatrix} I & A-A \\ O & I \end{bmatrix} =$ |
| johnston-linear-matrix-atlas-<br>F04550 | 462  | 0.977 | $(\pi, 0)$                                                                                                                | $(\pi, 0)$                                                                                                                                      | (c) Not a subspace, since (for example) $(\pi, 0)$ is in                                                                                            |
| johnston-linear-matrix-atlas-<br>F04551 | 462  | 1.000 | $\frac{1}{2}(\pi, 0) = (\pi/2, 0)$                                                                                        | $\frac{1}{2}(\pi, 0) = (\pi/2, 0)$                                                                                                              | the set but $\frac{1}{2}(\pi, 0) = (\pi/2, 0)$ is not in the set.                                                                                   |
| johnston-linear-matrix-atlas-<br>F04552 | 462  | 1.000 | $\operatorname{span}\{(-2, 1)\}$                                                                                          | $\operatorname{span}\{(-2, 1)\}$                                                                                                                | is $\operatorname{span}\{(-2, 1)\}$ , and spans are always sub-                                                                                     |
| johnston-linear-matrix-atlas-<br>F04553 | 462  | 0.955 | $(0, -1)$                                                                                                                 | $(0, -1)$                                                                                                                                       | $(0, -1)$ are in the set, but $(1, 0) + (0, -1) =$                                                                                                  |
| johnston-linear-matrix-atlas-<br>F04554 | 462  | 0.955 | $(1, 0) + (0, -1) =$                                                                                                      | $(1, 0) + (0, -1) =$                                                                                                                            | $(0, -1)$ are in the set, but $(1, 0) + (0, -1) =$                                                                                                  |
| johnston-linear-matrix-atlas-<br>F04555 | 462  | 1.000 | $x - y = 0$                                                                                                               | $x - y = 0$                                                                                                                                     | line has equation $y = x$ , or $x - y = 0$ .                                                                                                        |
| johnston-linear-matrix-atlas-<br>F04556 | 462  | 1.000 | $-3x + y + z = 0$                                                                                                         | $-3x + y + z = 0$                                                                                                                               | $-3x + y + z = 0$ , which is the equation of a                                                                                                      |
| johnston-linear-matrix-atlas-<br>F04557 | 462  | 1.000 | $y = -x$                                                                                                                  | $y = -x$                                                                                                                                        | $(1, -1)$ (i.e., the line with equation $y = -x$ ).                                                                                                 |
| johnston-linear-matrix-atlas-<br>F04558 | 462  | 0.523 | $z)$                                                                                                                      | $z)$                                                                                                                                            | $z)$ and one leading variable $x$ with $x + y + 2z =$                                                                                               |
| johnston-linear-matrix-atlas-<br>F04559 | 462  | 0.523 | $x + y + 2z =$                                                                                                            | $x + y + 2z =$                                                                                                                                  | $z)$ and one leading variable $x$ with $x + y + 2z =$                                                                                               |
| johnston-linear-matrix-atlas-<br>F04560 | 462  | 1.000 | $(1, 0), \mathbf{v}_2 = (0, 1)$                                                                                           | $(1, 0), \mathbf{v}_2 = (0, 1)$                                                                                                                 | $(1, 0), \mathbf{v}_2 = (0, 1)$ , and $\mathbf{v}_3 = (1, 1)$ .                                                                                     |
| johnston-linear-matrix-atlas-<br>F04561 | 462  | 1.000 | $\mathbf{v}_3 = (1, 1)$                                                                                                   | $\mathbf{v}_3 = (1, 1)$                                                                                                                         | $(1, 0), \mathbf{v}_2 = (0, 1)$ , and $\mathbf{v}_3 = (1, 1)$ .                                                                                     |
| johnston-linear-matrix-atlas-<br>F04562 | 462  | 1.000 | $\mathbf{v} = \mathbf{w} = \mathbf{0}$                                                                                    | $\mathbf{v} = \mathbf{w} = \mathbf{0}$                                                                                                          | $\mathbf{v} = \mathbf{w} = \mathbf{0}$ .                                                                                                            |
| johnston-linear-matrix-atlas-<br>F04563 | 463  | 1.000 | $3 - 3k \neq 0$                                                                                                           | $3 - 3k \neq 0$                                                                                                                                 | If $3 - 3k \neq 0$ (i.e., $k \neq 1$ ), then this system of                                                                                         |
| johnston-linear-matrix-atlas-<br>F04564 | 463  | 1.000 | $k \neq 1$                                                                                                                | $k \neq 1$                                                                                                                                      | If $3 - 3k \neq 0$ (i.e., $k \neq 1$ ), then this system of                                                                                         |
| johnston-linear-matrix-atlas-<br>F04565 | 463  | 1.000 | $y + z - 3x = 0$                                                                                                          | $y + z - 3x = 0$                                                                                                                                | $y + z - 3x = 0$ . This is the equation of a plane,                                                                                                 |
| johnston-linear-matrix-atlas-<br>F04566 | 463  | 1.000 | $(c_0, c_1) = (-3, 2)$                                                                                                    | $(c_0, c_1) = (-3, 2)$                                                                                                                          | $(c_0, c_1) = (-3, 2)$ , so the interpolating polyno-                                                                                               |
| johnston-linear-matrix-atlas-<br>F04567 | 463  | 1.000 | $p(x) = 2x - 3$                                                                                                           | $p(x) = 2x - 3$                                                                                                                                 | mial is $p(x) = 2x - 3$ .                                                                                                                           |
| johnston-linear-matrix-atlas-<br>F04568 | 463  | 1.000 | $(c_0, c_1, c_2) = (2, -2, 1)$                                                                                            | $(c_0, c_1, c_2) = (2, -2, 1)$                                                                                                                  | The unique solution is $(c_0, c_1, c_2) = (2, -2, 1)$ ,                                                                                             |
| johnston-linear-matrix-atlas-<br>F04569 | 463  | 1.000 | $x^2 - 2x + 2$                                                                                                            | $x^2 - 2x + 2$                                                                                                                                  | $x^2 - 2x + 2$ .                                                                                                                                    |
| johnston-linear-matrix-atlas-<br>F04570 | 463  | 1.000 | $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$                                                                                | $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$                                                                                                      | <b>2.3.13</b> Since $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$ , the solution set does not contain                                                  |
| johnston-linear-matrix-atlas-<br>F04571 | 463  | 1.000 | $A_1 = 1$                                                                                                                 | $A_1 = 1$                                                                                                                                       | <b>2.3.14</b> It is straightforward to check that $A_1 = 1$ so $A_1^{-1} = 1$ ,                                                                     |
| johnston-linear-matrix-atlas-<br>F04572 | 463  | 1.000 | $A_1^{-1} = 1$                                                                                                            | $A_1^{-1} = 1$                                                                                                                                  | <b>2.3.14</b> It is straightforward to check that $A_1 = 1$ so $A_1^{-1} = 1$ ,                                                                     |
| johnston-linear-matrix-atlas-<br>F04573 | 463  | 0.996 | $(1, 2, \ldots, n)$                                                                                                       | $(1, 2, \ldots, n)$                                                                                                                             | tice that the first row of $A_n$ is the vector $(1, 2, \ldots, n)$                                                                                  |
| johnston-linear-matrix-atlas-<br>F04574 | 463  | 0.998 | $(n+1, n+2, \ldots, 2n)$                                                                                                  | $(n+1, n+2, \ldots, 2n)$                                                                                                                        | and its second row is the vector $(n+1, n+2, \ldots, 2n)$ .                                                                                         |
| johnston-linear-matrix-atlas-<br>F04575 | 463  | 1.000 | $\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$                                                                             | $\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$                                                                                                   | $\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$ whenever $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$ . Similarly, if we                              |

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| johnston-linear-matrix-attrib F04576 | 463  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$                                                                                                                                   | $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$                                                                                                                                   | $\mathbf{v}_1 + \mathbf{v}_2 \in \mathcal{S}$ whenever $\mathbf{v}_1, \mathbf{v}_2 \in \mathcal{S}$ . Similarly, if we                                                           |
| johnston-linear-matrix-attrib F04577 | 463  | 1.000 | $c_1 \mathbf{v}_1 \in \mathcal{S}$                                                                                                                                             | $c_1 \mathbf{v}_1 \in \mathcal{S}$                                                                                                                                             | choose $k = 1$ then we see that $c_1 \mathbf{v}_1 \in \mathcal{S}$ whenever                                                                                                      |
| johnston-linear-matrix-attrib F04578 | 463  | 1.000 | $\mathbf{v}_1 \in \mathcal{S}$                                                                                                                                                 | $\mathbf{v}_1 \in \mathcal{S}$                                                                                                                                                 | $\mathbf{v}_1 \in \mathcal{S}$ , so both of the defining properties of sub-                                                                                                      |
| johnston-linear-matrix-attrib F04579 | 463  | 0.998 | $c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$                                                                                                    | $c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$                                                                                                    | $c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$ and thus $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$ ,                                     |
| johnston-linear-matrix-attrib F04580 | 463  | 0.998 | $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$                                                                                                                          | $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$                                                                                                                          | $c_2 \mathbf{v}_2 \in \mathcal{S}, \dots, c_k \mathbf{v}_k \in \mathcal{S}$ and thus $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 \in \mathcal{S}$ ,                                     |
| johnston-linear-matrix-attrib F04581 | 463  | 1.000 | $(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$                                                                                                     | $(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$                                                                                                     | so $(c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2) + c_3 \mathbf{v}_3 \in \mathcal{S}$ , and so on until we get                                                                           |
| johnston-linear-matrix-attrib F04582 | 463  | 0.999 | $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$                                                                                                                  | $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$                                                                                                                  | $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \mathcal{S}$ .                                                                                                                  |
| johnston-linear-matrix-attrib F04583 | 463  | 1.000 | $(A - I) \mathbf{v} = \mathbf{0}$                                                                                                                                              | $(A - I) \mathbf{v} = \mathbf{0}$                                                                                                                                              | $(A - I) \mathbf{v} = \mathbf{0}$ . In other words, the fixed-point sub-                                                                                                         |
| johnston-linear-matrix-attrib F04584 | 463  | 0.508 | $\operatorname{null}(A - I)$                                                                                                                                                   | $\operatorname{null}(A - I)$                                                                                                                                                   | space of $A$ is equal to $\operatorname{null}(A - I)$ , and we already                                                                                                           |
| johnston-linear-matrix-attrib F04585 | 463  | 1.000 | $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$                                                                                                                                 | $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$                                                                                                                                 | such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$ . If $c_1 = 0$ then $c_2 \neq 0$                                                                                        |
| johnston-linear-matrix-attrib F04586 | 463  | 1.000 | $c_1 = 0$                                                                                                                                                                      | $c_1 = 0$                                                                                                                                                                      | such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$ . If $c_1 = 0$ then $c_2 \neq 0$                                                                                        |
| johnston-linear-matrix-attrib F04587 | 463  | 1.000 | $c_2 \neq 0$                                                                                                                                                                   | $c_2 \neq 0$                                                                                                                                                                   | such that $c_1 \mathbf{v} + c_2 \mathbf{w} = \mathbf{0}$ . If $c_1 = 0$ then $c_2 \neq 0$                                                                                        |
| johnston-linear-matrix-attrib F04588 | 463  | 1.000 | $c_1 \neq 0$                                                                                                                                                                   | $c_1 \neq 0$                                                                                                                                                                   | then $\mathbf{w} = \mathbf{0}$ . On the other hand, if $c_1 \neq 0$ then                                                                                                         |
| johnston-linear-matrix-attrib F04589 | 463  | 1.000 | $\mathbf{v} = (-c_2 / c_1) \mathbf{w}$                                                                                                                                         | $\mathbf{v} = (-c_2 / c_1) \mathbf{w}$                                                                                                                                         | $\mathbf{v} = (-c_2 / c_1) \mathbf{w}$ .                                                                                                                                         |
| johnston-linear-matrix-attrib F04590 | 463  | 1.000 | $0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$                                                                                                                                     | $0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$                                                                                                                                     | In the other direction, if $\mathbf{w} = \mathbf{0}$ then $0 \mathbf{v} + 1 \mathbf{w} = \mathbf{0}$ ,                                                                           |
| johnston-linear-matrix-attrib F04591 | 463  | 1.000 | $1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$                                                                                                                                  | $1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$                                                                                                                                  | $1 \mathbf{v} + (-c) \mathbf{w} = \mathbf{0}$ , so $\{\mathbf{v}, \mathbf{w}\}$ is linearly dependent.                                                                           |
| johnston-linear-matrix-attrib F04592 | 463  | 1.000 | $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                                                                                                                      | $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                                                                                                                      | <b>2.3.21</b> Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ be the set of vectors. If $S$ is                                                                     |
| johnston-linear-matrix-attrib F04593 | 463  | 1.000 | $c_i \neq 0$                                                                                                                                                                   | $c_i \neq 0$                                                                                                                                                                   | Suppose that $c_i \neq 0$ (which we can do since not all                                                                                                                         |
| johnston-linear-matrix-attrib F04594 | 463  | 0.990 | $\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$ | $\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$ | $\mathbf{v}_i = \frac{-c_1}{c_i} \mathbf{v}_1 + \dots + \frac{-c_{i-1}}{c_i} \mathbf{v}_{i-1} + \frac{-c_{i+1}}{c_i} \mathbf{v}_{i+1} + \dots + \frac{-c_k}{c_i} \mathbf{v}_k$ . |
| johnston-linear-matrix-attrib F04595 | 463  | 1.000 | $B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                                                                                                          | $B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$                                                                                                          | <b>2.3.23</b> Suppose $B = \{\mathbf{0}, \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ . Then                                                                               |
| johnston-linear-matrix-attrib F04596 | 463  | 1.000 | $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$                                                                     | $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$                                                                     | If $C = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k, \mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_m\}$ then we                                                            |
| johnston-linear-matrix-attrib F04597 | 463  | 1.000 | $\mathbf{x} \in \operatorname{range}(A B)$                                                                                                                                     | $\mathbf{x} \in \operatorname{range}(AB)$                                                                                                                                      | <b>2.3.27</b> (a) If $\mathbf{x} \in \operatorname{range}(AB)$ then there exists $\mathbf{y} \in \mathbb{R}^p$                                                                   |
| johnston-linear-matrix-attrib F04598 | 463  | 1.000 | $\mathbf{y} \in \mathbb{R}^p$                                                                                                                                                  | $\mathbf{y} \in \mathbb{R}^p$                                                                                                                                                  | <b>2.3.27</b> (a) If $\mathbf{x} \in \operatorname{range}(AB)$ then there exists $\mathbf{y} \in \mathbb{R}^p$                                                                   |
| johnston-linear-matrix-attrib F04599 | 463  | 1.000 | $\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$                                                                                                                               | $\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$                                                                                                                               | such that $\mathbf{x} = (AB) \mathbf{y} = A(B \mathbf{y})$ . It follows that                                                                                                     |
| johnston-linear-matrix-attrib F04600 | 463  | 0.999 | $\mathbf{x} \in \operatorname{range}(A)$                                                                                                                                       | $\mathbf{x} \in \operatorname{range}(A)$                                                                                                                                       | $\mathbf{x} \in \operatorname{range}(A)$ as well (since it can be written as                                                                                                     |
| johnston-linear-matrix-attrib F04601 | 463  | 1.000 | $\mathbf{x} \in \operatorname{null}(B)$                                                                                                                                        | $\mathbf{x} \in \operatorname{null}(B)$                                                                                                                                        | (b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$ , so $AB \mathbf{x} = A \mathbf{0} =$                                                            |
| johnston-linear-matrix-attrib F04602 | 463  | 1.000 | $B \mathbf{x} = \mathbf{0}$                                                                                                                                                    | $B \mathbf{x} = \mathbf{0}$                                                                                                                                                    | (b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$ , so $AB \mathbf{x} = A \mathbf{0} =$                                                            |
| johnston-linear-matrix-attrib F04603 | 463  | 1.000 | $AB \mathbf{x} = A \mathbf{0} =$                                                                                                                                               | $AB \mathbf{x} = A \mathbf{0} =$                                                                                                                                               | (b) If $\mathbf{x} \in \operatorname{null}(B)$ then $B \mathbf{x} = \mathbf{0}$ , so $AB \mathbf{x} = A \mathbf{0} =$                                                            |

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| johnston-linear-matrix-atq604-F04604 | 463  | ■ 0.998 | $\mathbf{x} \in \operatorname{null}(A \ B)$                                   | $\mathbf{x} \in \operatorname{null}(AB)$                                      | $\mathbf{0}$ , so $\mathbf{x} \in \operatorname{null}(AB)$ . It follows that $\operatorname{null}(B) \subseteq$ |
| johnston-linear-matrix-atq605-F04605 | 463  | ■ 0.998 | $\operatorname{null}(B) \subseteq$                                            | $\operatorname{null}(B) \subseteq$                                            | $\mathbf{0}$ , so $\mathbf{x} \in \operatorname{null}(AB)$ . It follows that $\operatorname{null}(B) \subseteq$ |
| johnston-linear-matrix-atq606-F04606 | 464  | ■ 0.997 | $\operatorname{range}(A \ B)=\operatorname{span}((1,0))$                      | $\operatorname{range}(AB) = \operatorname{span}((1,0))$                       | which has $\operatorname{range}(AB) = \operatorname{span}((1,0))$ , which is                                    |
| johnston-linear-matrix-atq607-F04607 | 464  | ■ 0.999 | $\operatorname{range}(B)=\operatorname{span}((0,1))$                          | $\operatorname{range}(B) = \operatorname{span}((0,1))$                        | not contained within $\operatorname{range}(B) = \operatorname{span}((0,1))$ .                                   |
| johnston-linear-matrix-atq608-F04608 | 464  | ■ 1.000 | $\operatorname{null}(A)=\operatorname{span}((1,0))$                           | $\operatorname{null}(A) = \operatorname{span}((1,0))$                         | Similarly, $\operatorname{null}(A) = \operatorname{span}((1,0))$ , which is not                                 |
| johnston-linear-matrix-atq609-F04609 | 464  | ■ 0.992 | $\operatorname{null}(A \ B)=\operatorname{span}((0,1))$                       | $\operatorname{null}(AB) = \operatorname{span}((0,1))$                        | contained within $\operatorname{null}(AB) = \operatorname{span}((0,1))$ .                                       |
| johnston-linear-matrix-atq610-F04610 | 464  | ■ 1.000 | $(x, \ y)=(1,3)$                                                              | $(x,y) = (1,3)$                                                               | any non-zero vector on the line. $(x,y) = (1,3)$                                                                |
| johnston-linear-matrix-atq611-F04611 | 464  | ■ 1.000 | $B=\{(1,3)\}$                                                                 | $B = \{(1,3)\}$                                                               | works, so $B = \{(1,3)\}$ is a basis of this subspace.                                                          |
| johnston-linear-matrix-atq612-F04612 | 464  | ■ 1.000 | $B=\{(1,1,-3),(1,0,-2)\}$                                                     | $B = \{(1,1,-3), (1,0,-2)\}$                                                  | $B = \{(1,1,-3), (1,0,-2)\}$ is a basis of this                                                                 |
| johnston-linear-matrix-atq613-F04613 | 464  | ■ 1.000 | $z=3$                                                                         | $z = 3$                                                                       | choosing $z = 3$ and solving by back substitution                                                               |
| johnston-linear-matrix-atq614-F04614 | 464  | ■ 1.000 | $y=4$                                                                         | $y = 4$                                                                       | gives $y = 4$ and $x = -1$ . The line points thus in                                                            |
| johnston-linear-matrix-atq615-F04615 | 464  | ■ 1.000 | $(x, \ y, \ z)=(-1,4,3)$                                                      | $(x,y,z) = (-1,4,3)$                                                          | the direction of the vector $(x,y,z) = (-1,4,3)$ ,                                                              |
| johnston-linear-matrix-atq616-F04616 | 464  | ■ 1.000 | $B=\{(-1,4,3)\}$                                                              | $B = \{(-1,4,3)\}$                                                            | so $B = \{(-1,4,3)\}$ is a basis of this line.                                                                  |
| johnston-linear-matrix-atq617-F04617 | 464  | ■ 0.999 | $\{(1,2,3,4),(3,5,7,9)\}$                                                     | $\{(1,2,3,4), (3,5,7,9)\}$                                                    | <b>2.4.3 (a)</b> $\{(1,2,3,4), (3,5,7,9)\}$                                                                     |
| johnston-linear-matrix-atq618-F04618 | 464  | ■ 1.000 | $\{(1,1,4,5,1),(2,-7,4,2,2),(5,2,4,5,3)\}$                                    | $\{(1,1,4,5,1), (2,-7,4,2,2), (5,2,4,5,3)\}$                                  | <b>(c)</b> $\{(1,1,4,5,1), (2,-7,4,2,2), (5,2,4,5,3)\}$                                                         |
| johnston-linear-matrix-atq619-F04619 | 464  | ■ 0.997 | $\{(2,3,1,1),(3,3,2,3),(2,1,2,3),(1,0,0,0)\}$                                 | $\{(2,3,1,1), (3,3,2,3), (2,1,2,3), (1,0,0,0)\}$                              | <b>(a)</b> $\{(2,3,1,1), (3,3,2,3), (2,1,2,3), (1,0,0,0)\}$                                                     |
| johnston-linear-matrix-atq620-F04620 | 464  | ■ 1.000 | $d_1, d_2, \ldots, d_n \in \mathbb{R}$                                        | $d_1, d_2, \ldots, d_n \in \mathbb{R}$                                        | <b>2.3.28</b> Suppose $d_1, d_2, \ldots, d_n \in \mathbb{R}$ are such that                                      |
| johnston-linear-matrix-atq621-F04621 | 464  | ■ 1.000 | $d_1 c_1 = \cdots = d_n c_n = 0$                                              | $d_1 c_1 = \cdots = d_n c_n = 0$                                              | equation implies $d_1 c_1 = \cdots = d_n c_n = 0$ . Since $c_1$ ,                                               |
| johnston-linear-matrix-atq622-F04622 | 464  | ■ 1.000 | $\ldots, c_n$                                                                 | $\ldots, c_n$                                                                 | $\ldots, c_n$ are non-zero, this implies $d_1 = \cdots = d_n = 0$ ,                                             |
| johnston-linear-matrix-atq623-F04623 | 464  | ■ 1.000 | $d_1 = \cdots = d_n = 0$                                                      | $d_1 = \cdots = d_n = 0$                                                      | $\ldots, c_n$ are non-zero, this implies $d_1 = \cdots = d_n = 0$ ,                                             |
| johnston-linear-matrix-atq624-F04624 | 464  | ■ 1.000 | $\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$                                    | $\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$                                    | $\mathbf{v}_j = (1/c_j)(c_j \mathbf{v}_j)$ for each $1 \leq j \leq n$ and then using                            |
| johnston-linear-matrix-atq625-F04625 | 464  | ■ 0.999 | $\{(2,4,3,4),(3,2,-1,-1),(1,0,0,0),(0,1,0,0)\}$                               | $\{(2,4,3,4), (3,2,-1,-1), (1,0,0,0), (0,1,0,0)\}$                            | <b>(c)</b> $\{(2,4,3,4), (3,2,-1,-1), (1,0,0,0), (0,1,0,0)\}$                                                   |
| johnston-linear-matrix-atq626-F04626 | 464  | ■ 0.988 | $(A): \{(1,0),(0,1)\}$                                                        | $(A): \{(1,0), (0,1)\}$                                                       | Basis of $\operatorname{range}(A)$ : $\{(1,0), (0,1)\}$                                                         |
| johnston-linear-matrix-atq627-F04627 | 464  | ■ 1.000 | $\operatorname{null}(A): \{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$     | $\operatorname{null}(A): \{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$     | Basis of $\operatorname{null}(A)$ : $\{ \} (\operatorname{null}(A) = \{\mathbf{0}\})$                           |
| johnston-linear-matrix-atq628-F04628 | 464  | ■ 0.894 | $\operatorname{range}(A^T): \{(1,0),(0,1)\}$                                  | $\operatorname{range}(A^T): \{(1,0), (0,1)\}$                                 | Basis of $\operatorname{range}(A^T)$ : $\{(1,0), (0,1)\}$                                                       |
| johnston-linear-matrix-atq629-F04629 | 464  | ■ 0.994 | $\operatorname{null}(A^T): \{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$ | $\operatorname{null}(A^T): \{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$ | Basis of $\operatorname{null}(A^T)$ : $\{ \} (\operatorname{null}(A^T) = \{\mathbf{0}\})$                       |
| johnston-linear-matrix-atq630-F04630 | 464  | ■ 0.813 | $\operatorname{range}(A): \{(0,1,0)\}$                                        | $\operatorname{range}(A): \{(0,1,0)\}$                                        | Basis of $\operatorname{range}(A)$ : $\{(0,1,0)\}$                                                              |
| johnston-linear-matrix-atq631-F04631 | 464  | ■ 0.999 | $\operatorname{null}(A): \{(1,0,0),(0,-2,1)\}$                                | $\operatorname{null}(A): \{(1,0,0), (0,-2,1)\}$                               | Basis of $\operatorname{null}(A)$ : $\{(1,0,0), (0,-2,1)\}$                                                     |
| johnston-linear-matrix-atq632-F04632 | 464  | ■ 0.966 | $(A^T): \{(0,1,2)\}$                                                          | $(A^T): \{(0,1,2)\}$                                                          | Basis of $\operatorname{range}(A^T)$ : $\{(0,1,2)\}$                                                            |

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| johnston-linear-matrix-attached-<br>F04633 | 464  | ■ 1.000 | $\operatorname{null}(A^T): \{(1,0,0), (0,0,1)\}$               | $\operatorname{null}(A^T) : \{(1,0,0), (0,0,1)\}$              | Basis of $\operatorname{null}(A^T)$ : $\{(1,0,0), (0,0,1)\}$                          |
| johnston-linear-matrix-attached-<br>F04634 | 464  | ■ 0.608 | $\operatorname{range}(A): \{(1,0,0), (0,2,1)\}$                | $\operatorname{range}(A) : \{(1,0,0), (0,2,1)\}$               | (e) Basis of $\operatorname{range}(A)$ : $\{(1,0,0), (0,2,1)\}$                       |
| johnston-linear-matrix-attached-<br>F04635 | 464  | ■ 1.000 | $\operatorname{null}(A^T): \{(0,-1,2)\}$                       | $\operatorname{null}(A^T) : \{(0,-1,2)\}$                      | Basis of $\operatorname{null}(A^T)$ : $\{(0,-1,2)\}$                                  |
| johnston-linear-matrix-attached-<br>F04636 | 464  | ■ 0.928 | $(A): \{(1,0,0), (0,1,0), (0,0,1)\}$                           | $(A) : \{(1,0,0), (0,1,0), (0,0,1)\}$                          | Basis of $\operatorname{range}(A)$ : $\{(1,0,0), (0,1,0), (0,0,1)\}$                  |
| johnston-linear-matrix-attached-<br>F04637 | 464  | ■ 0.699 | $\operatorname{null}(A): \{(-2,1,0,0,0), (-3,0,1,1,0)\}$       | $\operatorname{null}(A) : \{(-2,1,0,0,0), (-3,0,1,1,0)\}$      | Basis of $\operatorname{null}(A)$ : $\{(-2,1,0,0,0), (-3,0,1,1,0)\}$                  |
| johnston-linear-matrix-attached-<br>F04638 | 464  | ■ 0.865 | $\operatorname{null}(A^T): \{(1,2,0,3,0), (0,0,1,-1,0)\}$      | $(A^T) : \{(1,2,0,3,0), (0,0,1,-1,0)\}$                        | Basis of $\operatorname{range}(A^T)$ : $\{(1,2,0,3,0), (0,0,1,-1,0),$                 |
| johnston-linear-matrix-attached-<br>F04639 | 464  | ■ 0.706 | $\operatorname{range}(A): \{(0,-1,-2), (-4,2,0)\}$             | $\operatorname{range}(A) : \{(0,-1,-2), (-4,2,0)\}$            | Basis of $\operatorname{range}(A)$ : $\{(0,-1,-2), (-4,2,0)\}$                        |
| johnston-linear-matrix-attached-<br>F04640 | 464  | ■ 0.725 | $\operatorname{null}(A): \{(1,0,1,0,0), (3,1/2,0,1,0)\}$       | $\operatorname{null}(A) : \{(1,0,1,0,0), (3,1/2,0,1,0)\}$      | Basis of $\operatorname{null}(A)$ : $\{(1,0,1,0,0), (3,1/2,0,1,0),$                   |
| johnston-linear-matrix-attached-<br>F04641 | 464  | ■ 0.546 | $\operatorname{null}(A^T): \{(1,0,-1,-3,-3/2)\}$               | $(A^T) : \{(1,0,-1,-3,-3/2)\}$                                 | Basis of $\operatorname{range}(A^T)$ : $\{(1,0,-1,-3,-3/2),$                          |
| johnston-linear-matrix-attached-<br>F04642 | 464  | ■ 1.000 | $\operatorname{null}(A^T): \{(-1,-2,1)\}$                      | $\operatorname{null}(A^T) : \{(-1,-2,1)\}$                     | Basis of $\operatorname{null}(A^T)$ : $\{(-1,-2,1)\}$                                 |
| johnston-linear-matrix-attached-<br>F04643 | 465  | ■ 0.999 | $\operatorname{rank}(A)=4$                                     | $\operatorname{rank}(A) = 4$                                   | <b>2.4.7</b> (a) If we call this matrix $A$ then $\operatorname{rank}(A) = 4$ ,       |
| johnston-linear-matrix-attached-<br>F04644 | 465  | ■ 1.000 | $\operatorname{nullity}(A)=4$                                  | $\operatorname{nullity}(A) = 4$                                | $\operatorname{nullity}(A) = 4$ , and we have:                                        |
| johnston-linear-matrix-attached-<br>F04645 | 465  | ■ 0.998 | $(A): \{(0,-6,4,-1), (2,-1,-2,0)\}$                            | $(A) : \{(0,-6,4,-1), (2,-1,-2,0)\}$                           | Basis of $\operatorname{range}(A)$ : $\{(0,-6,4,-1), (2,-1,-2,0),$                    |
| johnston-linear-matrix-attached-<br>F04646 | 465  | ■ 1.000 | $\operatorname{null}(A): \{(-1,1,1,0), (-1,6,-2,1)\}$          | $\operatorname{null}(A) : \{(-1,1,1,0), (-1,6,-2,1)\}$         | $\{(-1,1,1,0), (-1,6,-2,1)\}$ (or the standard                                        |
| johnston-linear-matrix-attached-<br>F04647 | 465  | ■ 0.904 | $\operatorname{null}(A): \{(2,-1,0,0,0,0,0,0)\}$               | $\operatorname{null}(A) : \{(2,-1,0,0,0,0,0,0)\}$              | Basis of $\operatorname{null}(A)$ : $\{(2,-1,0,0,0,0,0,0),$                           |
| johnston-linear-matrix-attached-<br>F04648 | 465  | ■ 0.826 | $\operatorname{range}(A^T): \{(1,2,0,1,0,-1,1,0)\}$            | $\operatorname{range}(A^T) : \{(1,2,0,1,0,-1,1,0)\}$           | Basis of $\operatorname{range}(A^T)$ : $\{(1,2,0,1,0,-1,1,0),$                        |
| johnston-linear-matrix-attached-<br>F04649 | 465  | ■ 1.000 | $\operatorname{null}(A): \{(1,1), (2,2)\}$                     | $\operatorname{null}(A) : \{(1,1), (2,2)\}$                    | set $\{(1,1), (2,2)\}$ , which is not even linearly                                   |
| johnston-linear-matrix-attached-<br>F04650 | 465  | ■ 0.978 | $\operatorname{rank}(A+B) \leq$                                | $\operatorname{rank}(A+B) \leq$                                | (i) False. All that we can say is that $\operatorname{rank}(A+B) \leq$                |
| johnston-linear-matrix-attached-<br>F04651 | 465  | ■ 0.855 | $A=B=I_n$                                                      | $A = B = I_n$                                                  | $A = B = I_n$ then $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$ ,       |
| johnston-linear-matrix-attached-<br>F04652 | 465  | ■ 0.855 | $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$     | $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$     | $A = B = I_n$ then $\operatorname{rank}(A+B) = \operatorname{rank}(2I_n) = n$ ,       |
| johnston-linear-matrix-attached-<br>F04653 | 465  | ■ 1.000 | $\operatorname{rank}(A) + \operatorname{rank}(B) = n + n = 2n$ | $\operatorname{rank}(A) + \operatorname{rank}(B) = n + n = 2n$ | but $\operatorname{rank}(A) + \operatorname{rank}(B) = n + n = 2n$ .                  |
| johnston-linear-matrix-attached-<br>F04654 | 465  | ■ 0.992 | $\operatorname{rank}(AB) \leq$                                 | $\operatorname{rank}(AB) \leq$                                 | (k) False. All that we can say is that $\operatorname{rank}(AB) \leq$                 |
| johnston-linear-matrix-attached-<br>F04655 | 465  | ■ 1.000 | $\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$       | $\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$       | $\min\{\operatorname{rank}(A), \operatorname{rank}(B)\}$ , thanks to Theo-            |
| johnston-linear-matrix-attached-<br>F04656 | 465  | ■ 0.983 | $\operatorname{rank}(A_2)=2$                                   | $\operatorname{rank}(A_2) = 2$                                 | <b>2.4.12</b> It is straightforward to check that $\operatorname{rank}(A_2) = 2$ , so |
| johnston-linear-matrix-attached-<br>F04657 | 465  | ■ 1.000 | $k \geq 3$                                                     | $k \geq 3$                                                     | nation of those two rows. In particular, if $k \geq 3$ then                           |
| johnston-linear-matrix-attached-<br>F04658 | 465  | ■ 1.000 | $R_2 - 2R_1$                                                   | $R_2 - 2R_1$                                                   | <b>2.4.13</b> (a) By performing the row operation $R_2 - 2R_1$ , we                   |
| johnston-linear-matrix-attached-<br>F04659 | 465  | ■ 1.000 | $x \neq 2$                                                     | $x \neq 2$                                                     | (i.e., when $x \neq 2$ ).                                                             |
| johnston-linear-matrix-attached-<br>F04660 | 465  | ■ 0.586 | $x \neq \pm 1$                                                 | $x \neq \pm 1$                                                 | the rank is 2. If $x \neq \pm 1$ then the rank is 3.                                  |
| johnston-linear-matrix-attached-<br>F04661 | 465  | ■ 1.000 | $A^{-1}(AB)=B$                                                 | $A^{-1}(AB) = B$                                               | that $A^{-1}(AB) = B$ , so using Theorem 2.4.11                                       |

| Identifier                              | Page | Conf.   | LaTeX source                                                                                                                                                                                                        | Rendered                                                                                                                                                                            | Image                                                                                                                                                                                 |
|-----------------------------------------|------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-atlas-<br>F04662 | 465  | ■ 1.000 | $\operatorname{rank}(B)=\operatorname{rank}\left(A^{-1}(A B)\right) \leq$                                                                                                                                           | $\operatorname{rank}(B)=\operatorname{rank}\left(A^{-1}(A B)\right) \leq$                                                                                                           | again tells us that $\operatorname{rank}(B)=\operatorname{rank}\left(A^{-1}(A B)\right) \leq$                                                                                         |
| johnston-linear-matrix-atlas-<br>F04663 | 466  | ■ 0.523 | $\operatorname{span}\left(\mathbf{v}_{\{1\}}, \ldots, \mathbf{v}_{\{k\}}\right)=\operatorname{span}\left(\mathbf{v}_{\{1\}}, \ldots, \mathbf{v}_{\{i-1\}}, \mathbf{v}_{\{i+1\}}, \ldots, \mathbf{v}_{\{k\}}\right)$ | $\operatorname{span}\left(\mathbf{v}_1, \ldots, \mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k\right)$ | $\operatorname{span}\left(\mathbf{v}_1, \ldots, \mathbf{v}_k\right)=\operatorname{span}\left(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k\right)$ . |
| johnston-linear-matrix-atlas-<br>F04664 | 466  | ■ 1.000 | $\left(\mathbf{v}_{\{1\}}, \ldots, \mathbf{v}_{\{i-1\}}, \mathbf{v}_{\{i+1\}}, \ldots, \mathbf{v}_{\{k\}}\right)$                                                                                                   | $\left\{\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k\right\}$                                                                                     | We then repeat: if $\left\{\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_k\right\}$ is lin-                                                            |
| johnston-linear-matrix-atlas-<br>F04665 | 466  | ■ 1.000 | $\{(1,-1,2),(2,-2,4)\}$                                                                                                                                                                                             | $\{(1,-1,2),(2,-2,4)\}$                                                                                                                                                             | with the set $\{(1,-1,2),(2,-2,4)\}$ , and the                                                                                                                                        |
| johnston-linear-matrix-atlas-<br>F04666 | 466  | ■ 1.000 | $\mathbf{v}=A \mathbf{x}$                                                                                                                                                                                           | $\mathbf{v}=A \mathbf{x}$                                                                                                                                                           | <b>2.4.26 (a)</b> If $\mathbf{v} \in \operatorname{range}(A)$ then we can write it as $\mathbf{v}=A \mathbf{x}$                                                                       |
| johnston-linear-matrix-atlas-<br>F04667 | 466  | ■ 1.000 | $\mathbf{w}=\mathbf{v}$                                                                                                                                                                                             | $\mathbf{w}=\mathbf{v}$                                                                                                                                                             | <b>(b)</b> If $\mathbf{w} \in \operatorname{range}\left(A^T\right)$ then we can write it as $\mathbf{w}=\mathbf{v}$                                                                   |
| johnston-linear-matrix-atlas-<br>F04668 | 466  | ■ 1.000 | $A^T \mathbf{x}$                                                                                                                                                                                                    | $A^T \mathbf{x}$                                                                                                                                                                    | $A^T \mathbf{x}$ for some $\mathbf{x} \in \mathbb{R}^m$ . Then                                                                                                                        |
| johnston-linear-matrix-atlas-<br>F04669 | 466  | ■ 0.998 | $[B \mid \mathbf{0}]$                                                                                                                                                                                               | $[B \mid \mathbf{0}]$                                                                                                                                                               | $[B \mid \mathbf{0}]$ have the same RREF $[R \mid \mathbf{0}]$ , so the lin-                                                                                                          |
| johnston-linear-matrix-atlas-<br>F04670 | 466  | ■ 1.000 | $S \mathbf{x} \neq \mathbf{0}$                                                                                                                                                                                      | $S \mathbf{x} \neq \mathbf{0}$                                                                                                                                                      | that $R \mathbf{x}=\mathbf{0}$ and $S \mathbf{x} \neq \mathbf{0}$ , so $\operatorname{null}(A) \neq \operatorname{null}(B)$ .                                                         |
| johnston-linear-matrix-atlas-<br>F04671 | 466  | ■ 1.000 | $\operatorname{null}(A) \neq \operatorname{null}(B)$                                                                                                                                                                | $\operatorname{null}(A) \neq \operatorname{null}(B)$                                                                                                                                | that $R \mathbf{x}=\mathbf{0}$ and $S \mathbf{x} \neq \mathbf{0}$ , so $\operatorname{null}(A) \neq \operatorname{null}(B)$ .                                                         |
| johnston-linear-matrix-atlas-<br>F04672 | 466  | ■ 1.000 | $\operatorname{range}\left(A^T\right) \neq \operatorname{range}\left(B^T\right)$                                                                                                                                    | $\operatorname{range}\left(A^T\right) \neq \operatorname{range}\left(B^T\right)$                                                                                                    | of $R$ ). It follows that $\operatorname{range}\left(A^T\right) \neq \operatorname{range}\left(B^T\right)$ .                                                                          |
| johnston-linear-matrix-atlas-<br>F04673 | 466  | ■ 1.000 | $\operatorname{rank}(A)=\operatorname{rank}\left(A^T\right)$                                                                                                                                                        | $\operatorname{rank}(A)=\operatorname{rank}\left(A^T\right)$                                                                                                                        | $\operatorname{rank}(A)=\operatorname{rank}\left(A^T\right)$ .                                                                                                                        |
| johnston-linear-matrix-atlas-<br>F04674 | 466  | ■ 1.000 | $\operatorname{rank}(\mathbf{v})=\operatorname{rank}\left(\mathbf{w}^T\right)=1$                                                                                                                                    | $\operatorname{rank}(\mathbf{v})=\operatorname{rank}\left(\mathbf{w}^T\right)=1$                                                                                                    | $\operatorname{rank}(\mathbf{v})=\operatorname{rank}\left(\mathbf{w}^T\right)=1$ . Then Theorem 2.4.11(b)                                                                             |
| johnston-linear-matrix-atlas-<br>F04675 | 466  | ■ 0.997 | $\operatorname{rank}(A)=\operatorname{rank}\left(\mathbf{v w}^T\right) \leq 1$                                                                                                                                      | $\operatorname{rank}(A)=\operatorname{rank}\left(\mathbf{v w}^T\right) \leq 1$                                                                                                      | tells us that $\operatorname{rank}(A)=\operatorname{rank}\left(\mathbf{v w}^T\right) \leq 1$ . Since $\mathbf{v}$ and                                                                 |
| johnston-linear-matrix-atlas-<br>F04676 | 466  | ■ 1.000 | $\operatorname{rank}(A) \neq 0$                                                                                                                                                                                     | $\operatorname{rank}(A) \neq 0$                                                                                                                                                     | $\mathbf{w}$ are non-zero, $A \neq O$ so $\operatorname{rank}(A) \neq 0$ . It follows that                                                                                            |
| johnston-linear-matrix-atlas-<br>F04677 | 466  | ■ 0.748 | $w_2, w_3, \ldots, w_n$                                                                                                                                                                                             | $w_2, w_3, \ldots, w_n$                                                                                                                                                             | is $\mathbf{v}$ then there are constants $w_2, w_3, \ldots, w_n$ such that                                                                                                            |
| johnston-linear-matrix-atlas-<br>F04678 | 466  | ■ 1.000 | $w_2 \mathbf{v}$                                                                                                                                                                                                    | $w_2 \mathbf{v}$                                                                                                                                                                    | the 2nd column is $w_2 \mathbf{v}$ , the 3rd column is $w_3 \mathbf{v}$ , and                                                                                                         |
| johnston-linear-matrix-atlas-<br>F04679 | 466  | ■ 1.000 | $w_3 \mathbf{v}$                                                                                                                                                                                                    | $w_3 \mathbf{v}$                                                                                                                                                                    | the 2nd column is $w_2 \mathbf{v}$ , the 3rd column is $w_3 \mathbf{v}$ , and                                                                                                         |
| johnston-linear-matrix-atlas-<br>F04680 | 466  | ■ 1.000 | $\mathbf{w}^T=\left(1, w_2, w_3, \ldots, w_n\right)$                                                                                                                                                                | $\mathbf{w}^T=\left(1, w_2, w_3, \ldots, w_n\right)$                                                                                                                                | If $\mathbf{w}^T=\left(1, w_2, w_3, \ldots, w_n\right)$ then the above expres-                                                                                                        |
| johnston-linear-matrix-atlas-<br>F04681 | 467  | ■ 0.999 | $[R \mid \mathbf{c}]$                                                                                                                                                                                               | $[R \mid \mathbf{c}]$                                                                                                                                                               | of the form $[R \mid \mathbf{c}]$ , where $\mathbf{c}$ in general may be                                                                                                              |
| johnston-linear-matrix-atlas-<br>F04682 | 467  | ■ 0.974 | $R \mathbf{x}=\mathbf{c}$                                                                                                                                                                                           | $R \mathbf{x}=\mathbf{c}$                                                                                                                                                           | same solution set as $R \mathbf{x}=\mathbf{c}$ for <i>some</i> vector $\mathbf{c}$ ,                                                                                                  |
| johnston-linear-matrix-atlas-<br>F04683 | 467  | ■ 1.000 | $\mathbf{c} \neq \mathbf{b}$                                                                                                                                                                                        | $\mathbf{c} \neq \mathbf{b}$                                                                                                                                                        | but $\mathbf{c} \neq \mathbf{b}$ in general.                                                                                                                                          |
| johnston-linear-matrix-atlas-<br>F04684 | 467  | ■ 0.637 | $\operatorname{range}(A)=\{\mathbf{0}\}$                                                                                                                                                                            | $\operatorname{range}(A)=\{\mathbf{0}\}$                                                                                                                                            | <b>(g)</b> False. If $\operatorname{range}(A)=\{\mathbf{0}\}$ then $A=O$ , which is                                                                                                   |
| johnston-linear-matrix-atlas-<br>F04685 | 467  | ■ 1.000 | $R=E_k \cdots E_2 E_1 A$                                                                                                                                                                                            | $R=E_k \cdots E_2 E_1 A$                                                                                                                                                            | In particular, $R=E_k \cdots E_2 E_1 A$ , so $A=$                                                                                                                                     |
| johnston-linear-matrix-atlas-<br>F04686 | 467  | ■ 1.000 | $E_1^{-1} E_2^{-1} \cdots E_k^{-1} R$                                                                                                                                                                               | $E_1^{-1} E_2^{-1} \cdots E_k^{-1} R$                                                                                                                                               | $E_1^{-1} E_2^{-1} \cdots E_k^{-1} R$ (since elementary matrices are                                                                                                                  |
| johnston-linear-matrix-atlas-<br>F04687 | 467  | ■ 1.000 | $P=E_1^{-1} E_2^{-1} \cdots E_k^{-1}$                                                                                                                                                                               | $P=E_1^{-1} E_2^{-1} \cdots E_k^{-1}$                                                                                                                                               | invertible). We can thus choose $P=E_1^{-1} E_2^{-1} \cdots E_k^{-1}$ ,                                                                                                               |
| johnston-linear-matrix-atlas-<br>F04688 | 467  | ■ 1.000 | $P_1, P_2 \in \mathcal{M}_m$                                                                                                                                                                                        | $P_1, P_2 \in \mathcal{M}_m$                                                                                                                                                        | vertible matrices $P_1, P_2 \in \mathcal{M}_m$ such that $A=P_1 R$                                                                                                                    |
| johnston-linear-matrix-atlas-<br>F04689 | 467  | ■ 1.000 | $A=P_1 R$                                                                                                                                                                                                           | $A=P_1 R$                                                                                                                                                                           | vertible matrices $P_1, P_2 \in \mathcal{M}_m$ such that $A=P_1 R$                                                                                                                    |

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| johnston-linear-matrix-atlas<br>F04690 | 467  | ■ 1.000 | $B=P_{\{2\}} R$                                                               | $B = P_2 R$                              | and $B = P_2 R$ . Then multiplying on the left by $P_1 P_2^{-1}$                                     |
| johnston-linear-matrix-atlas<br>F04691 | 467  | ■ 1.000 | $P_{\{1\}} P_{\{2\}}^{-1}$                                                    | $P_1 P_2^{-1}$                           | and $B = P_2 R$ . Then multiplying on the left by $P_1 P_2^{-1}$                                     |
| johnston-linear-matrix-atlas<br>F04692 | 467  | ■ 1.000 | $P=P_{\{1\}} P_{\{2\}}^{-1}$                                                  | $P = P_1 P_2^{-1}$                       | so we can choose $P = P_1 P_2^{-1}$ .                                                                |
| johnston-linear-matrix-atlas<br>F04693 | 467  | ■ 1.000 | $P=E_{\{1\}} E_{\{2\}} \cdots E_{\{k\}}$                                      | $P = E_1 E_2 \cdots E_k$                 | elementary matrices: $P = E_1 E_2 \cdots E_k$ . It follows that                                      |
| johnston-linear-matrix-atlas<br>F04694 | 467  | ■ 1.000 | $A=E_{\{1\}} E_{\{2\}} \cdots E_{\{k\}} B$                                    | $A = E_1 E_2 \cdots E_k B$               | $A = E_1 E_2 \cdots E_k B$ , so $A$ can be reached from $B$ via                                      |
| johnston-linear-matrix-atlas<br>F04695 | 467  | ■ 1.000 | $(x, y, z)=(0,1,0)$                                                           | $(x,y,z) = (0,1,0)$                      | <b>2.A.1</b> (a) $(x,y,z) = (0,1,0)$ .                                                               |
| johnston-linear-matrix-atlas<br>F04696 | 467  | ■ 0.999 | $(w, x, y, z)=(0,1,0,0)$                                                      | $(w,x,y,z) = (0,1,0,0)$                  | (c) $(w,x,y,z) = (0,1,0,0)$ and $(w,x,y,z) =$                                                        |
| johnston-linear-matrix-atlas<br>F04697 | 467  | ■ 0.998 | $(v, w, x, y, z)=(1,1,0,0,0)$                                                 | $(v,w,x,y,z) = (1,1,0,0,0)$              | (e) $(v,w,x,y,z) = (1,1,0,0,0)$ .                                                                    |
| johnston-linear-matrix-atlas<br>F04698 | 467  | ■ 1.000 | $(x, y, z)=(1,0,1)$                                                           | $(x,y,z) = (1,0,1)$                      | <b>2.A.2</b> (a) $(x,y,z) = (1,0,1)$ .                                                               |
| johnston-linear-matrix-atlas<br>F04699 | 468  | ■ 1.000 | $(v, w, x, y, z)=(0,1,1,0,0)$                                                 | $(v,w,x,y,z) = (0,1,1,0,0)$              | $(v,w,x,y,z) = (0,1,1,0,0)$ , though the original                                                    |
| johnston-linear-matrix-atlas<br>F04700 | 468  | ■ 0.982 | $(v, w, x, y, z)=(4,-1,3,2,8)$                                                | $(v,w,x,y,z) = (4,-1,3,2,8)$             | $(v,w,x,y,z) = (4,-1,3,2,8)$ is another solution).                                                   |
| johnston-linear-matrix-atlas<br>F04701 | 468  | ■ 1.000 | $\mathrm{\mathbf{v}}_{\mathrm{\{v\}}}-\mathrm{\mathbf{v}}_{\mathrm{\{s\}}} =$ | $\mathbf{v_e} - \mathbf{v_s} =$          | need to solve is $A\mathbf{x} = \mathbf{v_e} - \mathbf{v_s}$ , where $\mathbf{v_e} - \mathbf{v_s} =$ |
| johnston-linear-matrix-atlas<br>F04702 | 468  | ■ 1.000 | $\mathrm{\mathbf{x}}=(1,0,0,1,0,1,1,0,1)$                                     | $\mathbf{x} = (1,0,0,1,0,1,1,0,1)$       | part (a) is $\mathbf{x} = (1,0,0,1,0,1,1,0,1)$ , which                                               |
| johnston-linear-matrix-atlas<br>F04703 | 468  | ■ 1.000 | $m_{\{j\}}$                                                                   | $m_j$                                    | the other hand, if the $j$ -th row has $m_j$ ones in                                                 |
| johnston-linear-matrix-atlas<br>F04704 | 468  | ■ 1.000 | $B-I$                                                                         | $B - I$                                  | (b) Just apply part (a) to the matrix $B - I$ .                                                      |
| johnston-linear-matrix-atlas<br>F04705 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(3,0)$                                     | $(x_1,x_2) = (3,0)$                      | $(x_1,x_2) = (3,0)$ .                                                                                |
| johnston-linear-matrix-atlas<br>F04706 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(21,1) / 8$                                | $(x_1,x_2) = (21,1)/8$                   | $(x_1,x_2) = (21,1)/8$ .                                                                             |
| johnston-linear-matrix-atlas<br>F04707 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(3,11,23) / 18$                 | $(x_1,x_2,x_3) = (3,11,23)/18$           | at $(x_1,x_2,x_3) = (3,11,23)/18$ .                                                                  |
| johnston-linear-matrix-atlas<br>F04708 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}\right)=(6,2,0,23) / 30$     | $(x_1,x_2,x_3,x_4) = (6,2,0,23)/30$      | at $(x_1,x_2,x_3,x_4) = (6,2,0,23)/30$ .                                                             |
| johnston-linear-matrix-atlas<br>F04709 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(3,1) / 2$                                 | $(x_1,x_2) = (3,1)/2$                    | $(x_1,x_2) = (3,1)/2$ .                                                                              |
| johnston-linear-matrix-atlas<br>F04710 | 468  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}\right)=(1,0,1)$                        | $(x_1,x_2,x_3) = (1,0,1)$                | $(x_1,x_2,x_3) = (1,0,1)$ .                                                                          |
| johnston-linear-matrix-atlas<br>F04711 | 468  | ■ 0.995 | $7 / 4$                                                                       | $7/4$                                    | (e) The optimal value is $7/4$ , which is attained at                                                |
| johnston-linear-matrix-atlas<br>F04712 | 468  | ■ 0.882 | $\left(x_{\{1\}}, x_{\{2\}}, x_{\{3\}}, x_{\{4\}}\right)=(8,2,0,1) / 8$       | $(x_1,x_2,x_3,x_4) = (8,2,0,1)/8$        | $(x_1,x_2,x_3,x_4) = (8,2,0,1)/8$ .                                                                  |
| johnston-linear-matrix-atlas<br>F04713 | 468  | ■ 1.000 | $\mathrm{\mathbf{x}}=(1,0), \mathrm{\mathbf{b}}=(2,2)$                        | $\mathbf{x} = (1,0), \mathbf{b} = (2,2)$ | <b>2.B.5</b> (a) False. For example, if $\mathbf{x} = (1,0)$ , $\mathbf{b} = (2,2)$ ,                |
| johnston-linear-matrix-atlas<br>F04714 | 469  | ■ 1.000 | $x_{\{j\}} \geq y_{\{j\}}$                                                    | $x_j \geq y_j$                           | (c) True. If $\mathbf{x} \geq \mathbf{y}$ then $x_j \geq y_j$ for each $1 \leq j \leq n$ ,           |
| johnston-linear-matrix-atlas<br>F04715 | 469  | ■ 1.000 | $c_{\{j\}} x_{\{j\}} \geq c_{\{j\}} y_{\{j\}}$                                | $c_j x_j \geq c_j y_j$                   | so $c_j x_j \geq c_j y_j$ for each $j$ , so                                                          |
| johnston-linear-matrix-atlas<br>F04716 | 469  | ■ 1.000 | $\left(x_{\{1\}}, x_{\{2\}}\right)=(0.6,0.8)$                                 | $(x_1,x_2) = (0.6,0.8)$                  | which is attained at $(x_1,x_2) = (0.6,0.8)$ . To see                                                |
| johnston-linear-matrix-atlas<br>F04717 | 469  | ■ 1.000 | $x_{\{2\}} \geq 1 / 2$                                                        | $x_2 \geq 1/2$                           | $x_2 \geq 1/2$ , whereas adding the first constraint to $-2$                                         |
| johnston-linear-matrix-atlas<br>F04718 | 469  | ■ 0.999 | $x_{\{2\}} \leq 4 / 5$                                                        | $x_2 \leq 4/5$                           | times the third constraint shows that $x_2 \leq 4/5$ . In                                            |

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| johnston-linear-matrix-attempt-F04719 | 469  | ■ 1.000 | $2x_2 \leq c$                                                                                                                                                           | $2x_2 \leq c$                                                                                                                                                           | <b>2.B.8</b> (a) Adding the two constraints shows that $2x_2 \leq c$ ,                                                                                                      |
| johnston-linear-matrix-attempt-F04720 | 469  | ■ 1.000 | $x_2 \leq c/2$                                                                                                                                                          | $x_2 \leq c/2$                                                                                                                                                          | so $x_2 \leq c/2$ , so certainly the optimal value of                                                                                                                       |
| johnston-linear-matrix-attempt-F04721 | 469  | ■ 1.000 | $c/2$                                                                                                                                                                   | $c/2$                                                                                                                                                                   | the linear program cannot exceed $c/2$ . On the                                                                                                                             |
| johnston-linear-matrix-attempt-F04722 | 469  | ■ 0.381 | $1/2, c/2$                                                                                                                                                              | $1/2, c/2$                                                                                                                                                              | other hand, $(1/2, c/2)$ is a feasible point of                                                                                                                             |
| johnston-linear-matrix-attempt-F04723 | 469  | ■ 1.000 | $x_1 \leq 1 - x_2/c$                                                                                                                                                    | $x_1 \leq 1 - x_2/c$                                                                                                                                                    | (b) The first constraint tells us that $x_1 \leq 1 - x_2/c$                                                                                                                 |
| johnston-linear-matrix-attempt-F04724 | 469  | ■ 1.000 | $x_1 \geq$                                                                                                                                                              | $x_1 \geq$                                                                                                                                                              | and the second constraint tells us that $x_1 \geq$                                                                                                                          |
| johnston-linear-matrix-attempt-F04725 | 469  | ■ 1.000 | $x_2/c$                                                                                                                                                                 | $x_2/c$                                                                                                                                                                 | $x_2/c$ . Since $x_2 \geq 0$ , it follows that $0 \leq x_1 \leq x_2$ ,                                                                                                      |
| johnston-linear-matrix-attempt-F04726 | 469  | ■ 1.000 | $0 \leq x_1 \leq x_2$                                                                                                                                                   | $0 \leq x_1 \leq x_2$                                                                                                                                                   | $x_2/c$ . Since $x_2 \geq 0$ , it follows that $0 \leq x_1 \leq x_2$ ,                                                                                                      |
| johnston-linear-matrix-attempt-F04727 | 469  | ■ 1.000 | $A^T \mathbf{1} = \mathbf{1}$                                                                                                                                           | $A^T \mathbf{1} = \mathbf{1}$                                                                                                                                           | Notice that the rows of $A^T$ sum to 1 so $A^T \mathbf{1} = \mathbf{1}$                                                                                                     |
| johnston-linear-matrix-attempt-F04728 | 469  | ■ 1.000 | $\mathbf{1} = (1, 1, \dots, 1)$                                                                                                                                         | $\mathbf{1} = (1, 1, \dots, 1)$                                                                                                                                         | (where $\mathbf{1} = (1, 1, \dots, 1)$ ). It follows that the latter                                                                                                        |
| johnston-linear-matrix-attempt-F04729 | 469  | ■ 1.000 | $\mathbf{y} = \mathbf{1}$                                                                                                                                               | $\mathbf{y} = \mathbf{1}$                                                                                                                                               | problem is feasible, since we can choose $\mathbf{y} = \mathbf{1}$ in it.                                                                                                   |
| johnston-linear-matrix-attempt-F04730 | 469  | ■ 1.000 | $\left[ \begin{array}{c} 1 \\ -1 \end{array} \right] \left[ \begin{array}{cc} 1 & -1 \end{array} \right]$                                                               | $\left[ \begin{array}{c} 1 \\ -1 \end{array} \right] \left[ \begin{array}{cc} 1 & -1 \end{array} \right]$                                                               | (a) $\left[ \begin{array}{c} 1 \\ -1 \end{array} \right] \left[ \begin{array}{cc} 1 & -1 \end{array} \right]$                                                               |
| johnston-linear-matrix-attempt-F04731 | 469  | ■ 0.924 | $\left[ \begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[ \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$ | $\left[ \begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[ \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$ | (c) $\left[ \begin{array}{ccc} 2 & 4 & 0 \\ 1 & -2 & 0 \\ 2 & 0 & -1 \end{array} \right] \left[ \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$ |
| johnston-linear-matrix-attempt-F04732 | 469  | ■ 1.000 | $\left[ \begin{array}{c} 3 \\ 1 \end{array} \right] \left[ \begin{array}{cc} 2 & 1 \end{array} \right]$                                                                 | $\left[ \begin{array}{c} 3 \\ 1 \end{array} \right] \left[ \begin{array}{cc} 2 & 1 \end{array} \right]$                                                                 | (e) $\left[ \begin{array}{c} 3 \\ 1 \end{array} \right] \left[ \begin{array}{cc} 2 & 1 \end{array} \right]$                                                                 |
| johnston-linear-matrix-attempt-F04733 | 469  | ■ 1.000 | $\left[ \begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[ \begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$         | $\left[ \begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[ \begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$         | (g) $\left[ \begin{array}{cc} 2 & 6 \\ 5 & 4 \\ 3 & -2 \end{array} \right] \left[ \begin{array}{cccc} 1 & 0 & 2 & 5/11 \\ 0 & 1 & -1/2 & -7/22 \end{array} \right]$         |
| johnston-linear-matrix-attempt-F04734 | 470  | ■ 0.994 | $\tilde{C} \in \mathcal{M}_{m,r}$                                                                                                                                       | $\tilde{C} \in \mathcal{M}_{m,r}$                                                                                                                                       | <b>2.C.6</b> It is clear that $\tilde{C} \in \mathcal{M}_{m,r}$ and $\tilde{R} \in \mathcal{M}_{r,n}$ , so we just                                                          |
| johnston-linear-matrix-attempt-F04735 | 470  | ■ 0.994 | $\tilde{R} \in \mathcal{M}_{r,n}$                                                                                                                                       | $\tilde{R} \in \mathcal{M}_{r,n}$                                                                                                                                       | <b>2.C.6</b> It is clear that $\tilde{C} \in \mathcal{M}_{m,r}$ and $\tilde{R} \in \mathcal{M}_{r,n}$ , so we just                                                          |
| johnston-linear-matrix-attempt-F04736 | 470  | ■ 1.000 | $\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$                                                                                                                              | $\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$                                                                                                                              | pute $\tilde{C} \tilde{R} = CP^{-1}PR = CR = A$ .                                                                                                                           |
| johnston-linear-matrix-attempt-F04737 | 470  | ■ 1.000 | $PR$                                                                                                                                                                    | $PR$                                                                                                                                                                    | $PR$ , where $P \in \mathcal{M}_m$ is invertible and $R \in \mathcal{M}_{m,n}$                                                                                              |
| johnston-linear-matrix-attempt-F04738 | 470  | ■ 1.000 | $P_1, P_2, Q_1$                                                                                                                                                         | $P_1, P_2, Q_1$                                                                                                                                                         | find invertible matrices $P_1, P_2, Q_1$ , and $Q_2$ such                                                                                                                   |
| johnston-linear-matrix-attempt-F04739 | 470  | ■ 1.000 | $Q_2$                                                                                                                                                                   | $Q_2$                                                                                                                                                                   | find invertible matrices $P_1, P_2, Q_1$ , and $Q_2$ such                                                                                                                   |
| johnston-linear-matrix-attempt-F04740 | 470  | ■ 1.000 | $P_2^{-1}$                                                                                                                                                              | $P_2^{-1}$                                                                                                                                                              | Multiplying on the left by $P_2^{-1}$ and on the right                                                                                                                      |
| johnston-linear-matrix-attempt-F04741 | 470  | ■ 1.000 | $Q_2^{-1}$                                                                                                                                                              | $Q_2^{-1}$                                                                                                                                                              | by $Q_2^{-1}$ shows that                                                                                                                                                    |
| johnston-linear-matrix-attempt-F04742 | 470  | ■ 1.000 | $Q = Q_2^{-1}Q_1$                                                                                                                                                       | $Q = Q_2^{-1}Q_1$                                                                                                                                                       | so we can choose $P = P_1P_2^{-1}$ and $Q = Q_2^{-1}Q_1$                                                                                                                    |

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| johnston-linear-matrix-atgevrn<br>F04743 | 470  | ■ 0.951 | $L=\left[\begin{array}{ll}1 & 0 \\ 2 & 1\end{array}\right]$ , $U=\left[\begin{array}{ll}1 & 2 \\ 0 & 1\end{array}\right]$                                                               | $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                   | <b>2.D.1 (a)</b> $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ .                                                   |
| johnston-linear-matrix-atgevrn<br>F04744 | 470  | ■ 0.994 | $L=\left[\begin{array}{cc}1 & 0 \\ -1 & 1\end{array}\right]$ , $U=\left[\begin{array}{ccc}3 & 1 & 2 \\ 0 & -2 & 1\end{array}\right]$                                                    | $L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$                                         | <b>(c)</b> $L = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -2 & 1 \end{bmatrix}$ .                                               |
| johnston-linear-matrix-atgevrn<br>F04745 | 470  | ■ 1.000 | $L=\left[\begin{array}{lll}1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1\end{array}\right]$ , $U=\left[\begin{array}{cc}1 & 2 \\ 0 & 0\end{array}\right]$                                         | $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$                    | <b>(e)</b> $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 3 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}$ .                          |
| johnston-linear-matrix-atgevrn<br>F04746 | 470  | ■ 1.000 | $L=\left[\begin{array}{ccc}1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1\end{array}\right]$ , $U=\left[\begin{array}{ccc}1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1\end{array}\right]$               | $L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1 \end{bmatrix}$     | <b>(g)</b> $L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ -2 & -1 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & -4 & 5 \\ 0 & 3 & -7 \\ 0 & 0 & 1 \end{bmatrix}$ .           |
| johnston-linear-matrix-atgevrn<br>F04747 | 470  | ■ 0.995 | $P=\left[\begin{array}{ll}0 & 1 \\ 1 & 0\end{array}\right]$ , $L=\left[\begin{array}{ll}1 & 0 \\ 0 & 1\end{array}\right]$ , $U=\left[\begin{array}{ll}1 & 3 \\ 0 & 2\end{array}\right]$ | $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$ | <b>2.D.2 (a)</b> $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix}$ . |
| johnston-linear-matrix-atgevrn<br>F04748 | 470  | ■ 0.999 | $P=\left[\begin{array}{lll}1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0\end{array}\right]$ , $L=\left[\begin{array}{lll}1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1\end{array}\right]$ , $U=$            | $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, U =$    | <b>(c)</b> $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, U =$            |
| johnston-linear-matrix-atgevrn<br>F04749 | 470  | ■ 0.760 | $\left[\begin{array}{lll}1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1\end{array}\right]$                                                                                                         | $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$                                                                                    | $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$ .                                                                                                     |
| johnston-linear-matrix-atgevrn<br>F04750 | 470  | ■ 1.000 | $(x, y)=(5, -3)$                                                                                                                                                                        | $(x, y) = (5, -3)$                                                                                                                                     | <b>2.D.3 (a)</b> $(x, y) = (5, -3)$ .                                                                                                                                     |
| johnston-linear-matrix-atgevrn<br>F04751 | 470  | ■ 1.000 | $(x, y, z)=(-3, 6, -2)$                                                                                                                                                                 | $(x, y, z) = (-3, 6, -2)$                                                                                                                              | <b>(c)</b> $(x, y, z) = (-3, 6, -2)$ .                                                                                                                                    |
| johnston-linear-matrix-atgevrn<br>F04752 | 470  | ■ 1.000 | $A=[B \mid C]$                                                                                                                                                                          | $A = [B \mid C]$                                                                                                                                       | wider than it is tall). If we write $A = [B \mid C]$ and let                                                                                                              |
| johnston-linear-matrix-atgevrn<br>F04753 | 470  | ■ 1.000 | $\widetilde{U}=\left[U \mid L^{-1} C\right]$                                                                                                                                            | $\widetilde{U} = [U \mid L^{-1}C]$                                                                                                                     | $\widetilde{U} = [U \mid L^{-1}C]$ (note that $L$ is invertible since its                                                                                                 |
| johnston-linear-matrix-atgevrn<br>F04754 | 471  | ■ 1.000 | $A=L_{\{1\}} U_{\{1\}}=L_{\{2\}} U_{\{2\}}$                                                                                                                                             | $A = L_1 U_1 = L_2 U_2$                                                                                                                                | sitions $A = L_1 U_1 = L_2 U_2$ . Since $A$ is invert-                                                                                                                    |
| johnston-linear-matrix-atgevrn<br>F04755 | 471  | ■ 1.000 | $L_{\{1\}}, L_{\{2\}}, U_{\{1\}}$                                                                                                                                                       | $L_1, L_2, U_1$                                                                                                                                        | of $L_1, L_2, U_1$ , and $U_2$ must be invertible as                                                                                                                      |
| johnston-linear-matrix-atgevrn<br>F04756 | 471  | ■ 1.000 | $U_{\{2\}}$                                                                                                                                                                             | $U_2$                                                                                                                                                  | of $L_1, L_2, U_1$ , and $U_2$ must be invertible as                                                                                                                      |
| johnston-linear-matrix-atgevrn<br>F04757 | 471  | ■ 1.000 | $L_{\{1\}}^{-1} L_{\{2\}}=U_{\{1\}} U_{\{2\}}^{-1}$                                                                                                                                     | $L_1^{-1} L_2 = U_1 U_2^{-1}$                                                                                                                          | $L_1^{-1} L_2 = U_1 U_2^{-1}$ .                                                                                                                                           |
| johnston-linear-matrix-atgevrn<br>F04758 | 471  | ■ 1.000 | $L_{\{1\}}^{-1} L_{\{2\}}$                                                                                                                                                              | $L_1^{-1} L_2$                                                                                                                                         | lar, it follows that $L_1^{-1} L_2$ and $U_1 U_2^{-1}$ must in                                                                                                            |
| johnston-linear-matrix-atgevrn<br>F04759 | 471  | ■ 1.000 | $U_{\{1\}} U_{\{2\}}^{-1}$                                                                                                                                                              | $U_1 U_2^{-1}$                                                                                                                                         | lar, it follows that $L_1^{-1} L_2$ and $U_1 U_2^{-1}$ must in                                                                                                            |
| johnston-linear-matrix-atgevrn<br>F04760 | 471  | ■ 1.000 | $L_{\{1\}}^{-1}$                                                                                                                                                                        | $L_1^{-1}$                                                                                                                                             | entries of $L_1^{-1}$ and $L_2$ all equal 1, it follows                                                                                                                   |
| johnston-linear-matrix-atgevrn<br>F04761 | 471  | ■ 1.000 | $L_{\{2\}}$                                                                                                                                                                             | $L_2$                                                                                                                                                  | entries of $L_1^{-1}$ and $L_2$ all equal 1, it follows                                                                                                                   |

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| johnston-linear-matrix-algebra-F04762 | 471  | 1.000 | $L_{\{1\}^{\wedge\{-1\}}L_{\{2\}}=I$                                                                                                                                                                                                                         | $L_1^{-1}L_2 = I$                                                                                                                                                                                                                              | that $L_1^{-1}L_2 = I$ , so $L_1 = L_2$ . Multiplying the                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F04763 | 471  | 1.000 | $L_{\{1\}}=L_{\{2\}}$                                                                                                                                                                                                                                        | $L_1 = L_2$                                                                                                                                                                                                                                    | that $L_1^{-1}L_2 = I$ , so $L_1 = L_2$ . Multiplying the                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F04764 | 471  | 1.000 | $L_{\{1\}}U_{\{1\}}=L_{\{2\}}U_{\{2\}}$                                                                                                                                                                                                                      | $L_1U_1 = L_2U_2$                                                                                                                                                                                                                              | equation $L_1U_1 = L_2U_2$ on the left by $L_1^{-1}$ then                                                                                                                                                                                          |
| johnston-linear-matrix-algebra-F04765 | 471  | 1.000 | $U_{\{1\}}=U_{\{2\}}$                                                                                                                                                                                                                                        | $U_1 = U_2$                                                                                                                                                                                                                                    | shows that $U_1 = U_2$ as well, so these two unit                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F04766 | 471  | 1.000 | $\underset{\sim}{\sim}\{k\} \times k$                                                                                                                                                                                                                        | $\tilde{k} \times k$                                                                                                                                                                                                                           | matrix multiplication shows that the top-left $\tilde{k} \times k$                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F04767 | 471  | 1.000 | $\widetilde{L} \widetilde{U}$                                                                                                                                                                                                                                | $\tilde{L}\tilde{U}$                                                                                                                                                                                                                           | submatrix of $\hat{A}$ has the form $\tilde{L}\tilde{U}$ , where $\hat{L}$ and $\tilde{U}$ are                                                                                                                                                     |
| johnston-linear-matrix-algebra-F04768 | 471  | 1.000 | $\widetilde{L}$                                                                                                                                                                                                                                              | $\tilde{L}$                                                                                                                                                                                                                                    | submatrix of $\hat{A}$ has the form $\tilde{L}\tilde{U}$ , where $\hat{L}$ and $\tilde{U}$ are                                                                                                                                                     |
| johnston-linear-matrix-algebra-F04769 | 471  | 1.000 | $\widetilde{U}$                                                                                                                                                                                                                                              | $\tilde{U}$                                                                                                                                                                                                                                    | submatrix of $\hat{A}$ has the form $\tilde{L}\tilde{U}$ , where $\hat{L}$ and $\tilde{U}$ are                                                                                                                                                     |
| johnston-linear-matrix-algebra-F04770 | 471  | 1.000 | $\frac{1}{275}\left[\begin{array}{cccccc}171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 1 & 2 & 21 & 82 & 32 & 16 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185\end{array}\right]$ | $\frac{1}{275} \begin{bmatrix} 171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 1 & 2 & 21 & 82 & 32 & 16 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185 \end{bmatrix}$ | $(b) \frac{1}{275} \begin{bmatrix} 171 & 67 & 16 & -3 & -28 & -14 \\ 67 & 134 & 32 & -6 & -56 & -28 \\ 17 & 34 & 82 & 19 & -6 & -3 \\ 1 & 2 & 21 & 82 & 32 & 16 \\ -13 & -26 & 2 & 34 & 134 & 67 \\ -40 & -80 & -15 & 20 & 95 & 185 \end{bmatrix}$ |
| johnston-linear-matrix-algebra-F04771 | 471  | 1.000 | $\widetilde{U}=D U$                                                                                                                                                                                                                                          | $\tilde{U} = DU$                                                                                                                                                                                                                               | <b>2.D.10</b> If $A = LDU$ then setting $\tilde{U} = DU$ gives $A = L\tilde{U}$ as a                                                                                                                                                               |
| johnston-linear-matrix-algebra-F04772 | 471  | 1.000 | $A=L \widetilde{U}$                                                                                                                                                                                                                                          | $A = L\tilde{U}$                                                                                                                                                                                                                               | <b>2.D.10</b> If $A = LDU$ then setting $\tilde{U} = DU$ gives $A = L\tilde{U}$ as a                                                                                                                                                               |
| johnston-linear-matrix-algebra-F04773 | 471  | 1.000 | $\widetilde{U}=D^{\wedge\{-1\}} U$                                                                                                                                                                                                                           | $\tilde{U} = D^{-1}U$                                                                                                                                                                                                                          | $\tilde{U} = D^{-1}U$ to get $A = \tilde{L}\tilde{U}$ as an LDU decomposi-                                                                                                                                                                         |
| johnston-linear-matrix-algebra-F04774 | 471  | 1.000 | $A=L D \widetilde{U}$                                                                                                                                                                                                                                        | $A = L\tilde{D}\tilde{U}$                                                                                                                                                                                                                      | $\tilde{U} = D^{-1}U$ to get $A = \tilde{L}\tilde{D}\tilde{U}$ as an LDU decomposi-                                                                                                                                                                |
| johnston-linear-matrix-algebra-F04775 | 471  | 1.000 | $L D \widetilde{U}=L D D^{\wedge\{-1\}} U=L U=A$                                                                                                                                                                                                             | $L\tilde{D}\tilde{U} = L\tilde{D}\tilde{D}^{-1}U = LU = A$                                                                                                                                                                                     | tion of $A$ (since $L\tilde{D}\tilde{U} = L\tilde{D}\tilde{D}^{-1}U = LU = A$ ). This                                                                                                                                                              |
| johnston-linear-matrix-algebra-F04776 | 471  | 1.000 | $L=\left[\begin{array}{ll}1 & 0 \\ 2 & 1\end{array}\right], D=\left[\begin{array}{ll}1 & 0 \\ 0 & 1\end{array}\right], U=\left[\begin{array}{ll}1 & 2 \\ 0 & 1\end{array}\right]$                                                                            | $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$                                                                                         | <b>2.D.11</b> (a) $L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, U = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$ .                                                                         |
| johnston-linear-matrix-algebra-F04777 | 471  | 1.000 | $R_{\{1\}}+c R_{\{2\}}$                                                                                                                                                                                                                                      | $R_1 + cR_2$                                                                                                                                                                                                                                   | top-left corner are $R_1 \leftrightarrow R_2$ or $R_1 + cR_2$ , so                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F04778 | 472  | 1.000 | $u_{\{1,1\}}=0$                                                                                                                                                                                                                                              | $u_{1,1} = 0$                                                                                                                                                                                                                                  | shows that $u_{1,1} = 0$ . However, this then im-                                                                                                                                                                                                  |
| johnston-linear-matrix-algebra-F04779 | 472  | 1.000 | $[\mathbf{v}]_B=(2,1 / 2)$                                                                                                                                                                                                                                   | $[\mathbf{v}]_B = (2, 1/2)$                                                                                                                                                                                                                    | <b>3.1.1</b> (a) $[\mathbf{v}]_B = (2, 1/2)$ , since $2(3, 0) + \frac{1}{2}(0, 2) = (6, 1)$ .                                                                                                                                                      |
| johnston-linear-matrix-algebra-F04780 | 472  | 1.000 | $2(3,0)+\frac{1}{2}(0,2)=(6,1)$                                                                                                                                                                                                                              | $2(3, 0) + \frac{1}{2}(0, 2) = (6, 1)$                                                                                                                                                                                                         | <b>3.1.1</b> (a) $[\mathbf{v}]_B = (2, 1/2)$ , since $2(3, 0) + \frac{1}{2}(0, 2) = (6, 1)$ .                                                                                                                                                      |
| johnston-linear-matrix-algebra-F04781 | 472  | 1.000 | $(6,1)=c_{\{1\}}(3,0)+c_{\{2\}}(0,2)$                                                                                                                                                                                                                        | $(6, 1) = c_1(3, 0) + c_2(0, 2)$                                                                                                                                                                                                               | $(6, 1) = c_1(3, 0) + c_2(0, 2)$ .                                                                                                                                                                                                                 |
| johnston-linear-matrix-algebra-F04782 | 472  | 1.000 | $[\mathbf{v}]_B=(1 / 2,3 / 2)$                                                                                                                                                                                                                               | $[\mathbf{v}]_B = (1/2, 3/2)$                                                                                                                                                                                                                  | (c) $[\mathbf{v}]_B = (1/2, 3/2)$ since $(2, 3, 1) =$                                                                                                                                                                                              |
| johnston-linear-matrix-algebra-F04783 | 472  | 1.000 | $(2,3,1)=$                                                                                                                                                                                                                                                   | $(2, 3, 1) =$                                                                                                                                                                                                                                  | (c) $[\mathbf{v}]_B = (1/2, 3/2)$ since $(2, 3, 1) =$                                                                                                                                                                                              |
| johnston-linear-matrix-algebra-F04784 | 472  | 1.000 | $\frac{1}{2}(1,0,-1)+\frac{3}{2}(1,2,1)$                                                                                                                                                                                                                     | $\frac{1}{2}(1, 0, -1) + \frac{3}{2}(1, 2, 1)$                                                                                                                                                                                                 | $\frac{1}{2}(1, 0, -1) + \frac{3}{2}(1, 2, 1)$ .                                                                                                                                                                                                   |

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| johnston-linear-matrix-atgevr2-F04785 | 472  | 1.000 | $B=\{(2,3),(2,-1)\}$                                                                   | $B = \{(2,3), (2,-1)\}$                                                                | <b>3.1.2 (a)</b> $B = \{(2,3), (2,-1)\}$ works.                                                |
| johnston-linear-matrix-atgevr2-F04786 | 472  | 1.000 | $\left[\begin{array}{ll}1 & 3 \\ 2 & 4\end{array}\right]$                              | $\left[\begin{array}{cc}1 & 3 \\ 2 & 4\end{array}\right]$                              | <b>(a)</b> $\left[\begin{array}{cc}1 & 3 \\ 2 & 4\end{array}\right]$                           |
| johnston-linear-matrix-atgevr2-F04787 | 472  | 0.999 | $\left[\begin{array}{cc}11/10 & 5/2 \\ 3/10 & 1/2\end{array}\right]$                   | $\left[\begin{array}{cc}11/10 & 5/2 \\ 3/10 & 1/2\end{array}\right]$                   | <b>(c)</b> $\left[\begin{array}{cc}11/10 & 5/2 \\ 3/10 & 1/2\end{array}\right]$                |
| johnston-linear-matrix-atgevr2-F04788 | 472  | 1.000 | $\left[\begin{array}{cc}2 & 0 \\ -5 & 2\end{array}\right]$                             | $\left[\begin{array}{cc}2 & 0 \\ -5 & 2\end{array}\right]$                             | <b>(a)</b> $\left[\begin{array}{cc}2 & 0 \\ -5 & 2\end{array}\right]$                          |
| johnston-linear-matrix-atgevr2-F04789 | 472  | 1.000 | $\left[\begin{array}{cc}2 & 0 \\ 0 & -3\end{array}\right]$                             | $\left[\begin{array}{cc}2 & 0 \\ 0 & -3\end{array}\right]$                             | <b>(c)</b> $\left[\begin{array}{cc}2 & 0 \\ 0 & -3\end{array}\right]$                          |
| johnston-linear-matrix-atgevr2-F04790 | 472  | 1.000 | $\left[\begin{array}{ccc}10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1\end{array}\right]$ | $\left[\begin{array}{ccc}10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1\end{array}\right]$ | $\left[\begin{array}{ccc}10 & -5 & 0 \\ 5 & -6 & -4 \\ -11 & 5 & -1\end{array}\right]$         |
| johnston-linear-matrix-atgevr2-F04791 | 472  | 1.000 | $\operatorname{tr}(A)=\operatorname{tr}(B)=2$                                          | $\operatorname{tr}(A) = \operatorname{tr}(B) = 2$                                      | $\operatorname{tr}(A) = \operatorname{tr}(B) = 2$ .                                            |
| johnston-linear-matrix-atgevr2-F04792 | 472  | 1.000 | $\operatorname{tr}(A)=6$                                                               | $\operatorname{tr}(A) = 6$                                                             | <b>(c)</b> $\operatorname{tr}(A) = 6$ and $\operatorname{tr}(B) = 3$ , so they are not similar |
| johnston-linear-matrix-atgevr2-F04793 | 472  | 1.000 | $\operatorname{tr}(B)=3$                                                               | $\operatorname{tr}(B) = 3$                                                             | <b>(c)</b> $\operatorname{tr}(A) = 6$ and $\operatorname{tr}(B) = 3$ , so they are not similar |
| johnston-linear-matrix-atgevr2-F04794 | 472  | 0.995 | $n!$                                                                                   | $n!$                                                                                   | unit PLU decomposition for each of the $n!$ dif-                                               |
| johnston-linear-matrix-atgevr2-F04795 | 472  | 0.999 | $P^T P$                                                                                | $P^T P$                                                                                | $(i,j)$ -entry of $P^T P$ is                                                                   |
| johnston-linear-matrix-atgevr2-F04796 | 472  | 0.549 | $\mathbf{p}_i \cdot \mathbf{p}_j = 1$                                                  | $\mathbf{p}_i \cdot \mathbf{p}_j = 1$                                                  | other entries equal to 0, we conclude that $\mathbf{p}_i \cdot \mathbf{p}_j = 1$               |
| johnston-linear-matrix-atgevr2-F04797 | 472  | 1.000 | $\mathbf{p}_i$                                                                         | $\mathbf{p}_i$                                                                         | the other hand, if $i \neq j$ then the columns $\mathbf{p}_i$ and $\mathbf{p}_j$               |
| johnston-linear-matrix-atgevr2-F04798 | 472  | 1.000 | $\mathbf{p}_i \cdot \mathbf{p}_j = 0$                                                  | $\mathbf{p}_i \cdot \mathbf{p}_j = 0$                                                  | would be a row of $P$ with two 1s in it), so $\mathbf{p}_i \cdot \mathbf{p}_j = 0$             |
| johnston-linear-matrix-atgevr2-F04799 | 472  | 0.953 | $P^T P = I$                                                                            | $P^T P = I$                                                                            | $P^T P$ equal 0. It follows that $P^T P = I$ , so $P^T = P^{-1}$ .                             |
| johnston-linear-matrix-atgevr2-F04800 | 472  | 0.953 | $P^T = P^{-1}$                                                                         | $P^T = P^{-1}$                                                                         | $P^T P$ equal 0. It follows that $P^T P = I$ , so $P^T = P^{-1}$ .                             |
| johnston-linear-matrix-atgevr2-F04801 | 472  | 1.000 | $p_{1,2}$                                                                              | $p_{1,2}$                                                                              | its solutions have $p_{1,2}$ and $p_{2,2}$ free, with                                          |
| johnston-linear-matrix-atgevr2-F04802 | 472  | 1.000 | $p_{2,2}$                                                                              | $p_{2,2}$                                                                              | its solutions have $p_{1,2}$ and $p_{2,2}$ free, with                                          |
| johnston-linear-matrix-atgevr2-F04803 | 472  | 1.000 | $p_{1,1} = -p_{1,2} / 2$                                                               | $p_{1,1} = -p_{1,2}/2$                                                                 | $p_{1,1} = -p_{1,2}/2$ and $p_{2,1} = -p_{2,2}$ . We just need                                 |
| johnston-linear-matrix-atgevr2-F04804 | 472  | 1.000 | $p_{2,1} = -p_{2,2}$                                                                   | $p_{2,1} = -p_{2,2}$                                                                   | $p_{1,1} = -p_{1,2}/2$ and $p_{2,1} = -p_{2,2}$ . We just need                                 |
| johnston-linear-matrix-atgevr2-F04805 | 472  | 1.000 | $P_{B \leftarrow B}$                                                                   | $P_{B \leftarrow B}$                                                                   | of $P_{B \leftarrow B}$ is thus $\mathbf{e}_j$ , so $P_{B \leftarrow B} = I$ .                 |
| johnston-linear-matrix-atgevr2-F04806 | 472  | 1.000 | $P^{-1} P = I$                                                                         | $P^{-1} P = I$                                                                         | <b>(c)</b> True. This follows from the fact that $P^{-1} P = I$                                |
| johnston-linear-matrix-atgevr2-F04807 | 472  | 1.000 | $P P^{-1} = I$                                                                         | $P P^{-1} = I$                                                                         | $P P^{-1} = I$ for all invertible matrices $P$ .                                               |
| johnston-linear-matrix-atgevr2-F04808 | 472  | 1.000 | $P A P^{-1} = A$                                                                       | $P A P^{-1} = A$                                                                       | $P A P^{-1} = A$ for all invertible $P$ (i.e., $A$ is only                                     |
| johnston-linear-matrix-atgevr2-F04809 | 472  | 1.000 | $A=B=I \in \mathcal{M}_2$                                                              | $A = B = I \in \mathcal{M}_2$                                                          | <b>(g)</b> False. For example, if $A = B = I \in \mathcal{M}_2$ then                           |
| johnston-linear-matrix-atgevr2-F04810 | 472  | 1.000 | $\operatorname{tr}(A B)=2$                                                             | $\operatorname{tr}(A B) = 2$                                                           | $\operatorname{tr}(A B) = 2$ but $\operatorname{tr}(A) \operatorname{tr}(B) = 4$ .             |

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| johnston-linear-matrix-atgevr2<br>F04811 | 472  | ■ 1.000 | $\operatorname{tr}(A) \operatorname{tr}(B)=4$                                                           | $\operatorname{tr}(A) \operatorname{tr}(B) = 4$              | $\operatorname{tr}(AB) = 2$ but $\operatorname{tr}(A)\operatorname{tr}(B) = 4$ .                              |
| johnston-linear-matrix-atgevr2<br>F04812 | 472  | ■ 1.000 | $\operatorname{tr}(A)=\operatorname{tr}(B)=$                                                            | $\operatorname{tr}(A) = \operatorname{tr}(B) =$              | (a) $\operatorname{rank}(A) = \operatorname{rank}(B) = 2$ and $\operatorname{tr}(A) = \operatorname{tr}(B) =$ |
| johnston-linear-matrix-atgevr2<br>F04813 | 472  | ■ 1.000 | $P \in \mathcal{M}_2$                                                                                   | $P \in \mathcal{M}_2$                                        | matrix $P \in \mathcal{M}_2$ such that                                                                        |
| johnston-linear-matrix-atgevr2<br>F04814 | 472  | ■ 1.000 | $d_{\{j\}}$                                                                                             | $d_j$                                                        | Since not all of the $d_j$ 's are zero, this is a different                                                   |
| johnston-linear-matrix-atgevr2<br>F04815 | 473  | ■ 1.000 | $[\mathbf{w}]_B : [\mathbf{v}]_B =$                                                                     | $[\mathbf{w}]_B : [\mathbf{v}]_B =$                          | to the entries of $[\mathbf{v}]_B$ and $[\mathbf{w}]_B$ : $[\mathbf{v}]_B =$                                  |
| johnston-linear-matrix-atgevr2<br>F04816 | 473  | ■ 0.999 | $\left(c_{\{1\}}, c_{\{2\}}, \ldots, c_{\{k\}}\right)$                                                  | $(c_1, c_2, \ldots, c_k)$                                    | $(c_1, c_2, \ldots, c_k)$ and $[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$ .                                    |
| johnston-linear-matrix-atgevr2<br>F04817 | 473  | ■ 0.999 | $[\mathbf{w}]_B=\left(d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{k\}}\right)$                                   | $[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$                   | $(c_1, c_2, \ldots, c_k)$ and $[\mathbf{w}]_B = (d_1, d_2, \ldots, d_k)$ .                                    |
| johnston-linear-matrix-atgevr2<br>F04818 | 473  | ■ 1.000 | $[\mathbf{v}+\mathbf{w}]_B=\left(c_{\{1\}}+d_{\{1\}}, c_{\{2\}}+\right.$<br>$\left.\right.$             | $[\mathbf{v}+\mathbf{w}]_B = (c_1 + d_1, c_2 +$              | which means that $[\mathbf{v}+\mathbf{w}]_B = (c_1 + d_1, c_2 +$                                              |
| johnston-linear-matrix-atgevr2<br>F04819 | 473  | ■ 0.566 | $\left.d_{\{2\}}, \ldots, c_{\{k\}}+d_{\{k\}}\right)$                                                   | $d_2, \ldots, c_k + d_k)$                                    | $d_2, \ldots, c_k + d_k)$ , which is the same as $[\mathbf{v}]_B +$                                           |
| johnston-linear-matrix-atgevr2<br>F04820 | 473  | ■ 0.566 | $[\mathbf{v}]_B+$                                                                                       | $[\mathbf{v}]_B +$                                           | $d_2, \ldots, c_k + d_k)$ , which is the same as $[\mathbf{v}]_B +$                                           |
| johnston-linear-matrix-atgevr2<br>F04821 | 473  | ■ 1.000 | $[\mathbf{v}]_B=\left(d_{\{1\}}, d_{\{2\}}, \ldots, d_{\{k\}}\right)$                                   | $[\mathbf{v}]_B = (d_1, d_2, \ldots, d_k)$                   | $[\mathbf{v}]_B = (d_1, d_2, \ldots, d_k)$ . This means that                                                  |
| johnston-linear-matrix-atgevr2<br>F04822 | 473  | ■ 0.956 | $[c \mathbf{v}]_B=\left(c d_{\{1\}}, c d_{\{2\}}, \ldots, c\right.$<br>$\left.d_{\{k\}}\right)$         | $[c\mathbf{v}]_B = (cd_1, cd_2, \ldots, cd_k)$               | which means that $[c\mathbf{v}]_B = (cd_1, cd_2, \ldots, cd_k)$ ,                                             |
| johnston-linear-matrix-atgevr2<br>F04823 | 473  | ■ 1.000 | $c[\mathbf{v}]_B$                                                                                       | $c[\mathbf{v}]_B$                                            | which is the same as $c[\mathbf{v}]_B$ .                                                                      |
| johnston-linear-matrix-atgevr2<br>F04824 | 473  | ■ 1.000 | $[\mathbf{v}]_B=\mathbf{0}$                                                                             | $[\mathbf{v}]_B = \mathbf{0}$                                | We just notice that $[\mathbf{v}]_B = \mathbf{0}$ if and only if $\mathbf{v} = \mathbf{0}$ ,                  |
| johnston-linear-matrix-atgevr2<br>F04825 | 473  | ■ 1.000 | $c_{\{1\}}=\cdots=c_{\{m\}}=0$                                                                          | $c_1 = \cdots = c_m = 0$                                     | In particular, $c_1 = \cdots = c_m = 0$ is the only so-                                                       |
| johnston-linear-matrix-atgevr2<br>F04826 | 473  | ■ 0.997 | $[\mathbf{v}]_B \in \mathbb{R}^k$                                                                       | $[\mathbf{v}]_B \in \mathbb{R}^k$                            | nate vector $[\mathbf{v}]_B \in \mathbb{R}^k$ , but conversely every vector                                   |
| johnston-linear-matrix-atgevr2<br>F04827 | 473  | ■ 1.000 | $\mathcal{S}: \mathbf{x}=[\mathbf{v}]_B$                                                                | $\mathcal{S}: \mathbf{x} = [\mathbf{v}]_B$                   | some vector from $\mathcal{S}$ : $\mathbf{x} = [\mathbf{v}]_B$ for some $\mathbf{v} \in \mathcal{S}$ .        |
| johnston-linear-matrix-atgevr2<br>F04828 | 473  | ■ 0.998 | $C=\left\{\mathbf{v}_{\{1\}}, \mathbf{v}_{\{2\}}, \ldots,\right.$<br>$\left.\mathbf{v}_{\{k\}}\right\}$ | $C = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$   | that $C = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\}$ is a basis of $\mathcal{S}$ , and (by         |
| johnston-linear-matrix-atgevr2<br>F04829 | 473  | ■ 1.000 | $P_{C \leftarrow E} P_{E \leftarrow B}=P_{C \leftarrow B}$                                              | $P_{C \leftarrow E} P_{E \leftarrow B} = P_{C \leftarrow B}$ | $P_{C \leftarrow E} P_{E \leftarrow B} = P_{C \leftarrow B}$ .                                                |
| johnston-linear-matrix-atgevr2<br>F04830 | 473  | ■ 1.000 | $[T]_B[\mathbf{v}]_B=[T(\mathbf{v})]_B$                                                                 | $[T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B$                    | To see that $[T]_B[\mathbf{v}]_B = [T(\mathbf{v})]_B$ , first write $[\mathbf{v}]_B =$                        |
| johnston-linear-matrix-atgevr2<br>F04831 | 473  | ■ 1.000 | $\left(c_{\{1\}}, \ldots, c_{\{n\}}\right)$                                                             | $(c_1, \ldots, c_n)$                                         | $(c_1, \ldots, c_n)$ and then use block matrix multiplication:                                                |
| johnston-linear-matrix-atgevr2<br>F04832 | 473  | ■ 1.000 | $[T(\mathbf{v})]_B=A[\mathbf{v}]_B$                                                                     | $[T(\mathbf{v})]_B = A[\mathbf{v}]_B$                        | any matrix such that $[T(\mathbf{v})]_B = A[\mathbf{v}]_B$ for all $\mathbf{v} \in \mathbb{R}^n$ .            |
| johnston-linear-matrix-atgevr2<br>F04833 | 473  | ■ 1.000 | $[T]_B[\mathbf{v}]_B=A[\mathbf{v}]_B$                                                                   | $[T]_B[\mathbf{v}]_B = A[\mathbf{v}]_B$                      | Then $[T]_B[\mathbf{v}]_B = A[\mathbf{v}]_B$ for all $\mathbf{v} \in \mathbb{R}^n$ . It follows from          |
| johnston-linear-matrix-atgevr2<br>F04834 | 473  | ■ 1.000 | $A=[T] \ B$                                                                                             | $A = [T]B$                                                   | Exercise 1.3.18(d) that $A = [T]B$ , which completes                                                          |
| johnston-linear-matrix-atgevr2<br>F04835 | 473  | ■ 1.000 | $\operatorname{nullity}(A)=n-\operatorname{rank}(A)$                                                    | $\operatorname{nullity}(A) = n - \operatorname{rank}(A)$     | have $\operatorname{nullity}(A) = n - \operatorname{rank}(A)$ . Since we already                              |
| johnston-linear-matrix-atgevr2<br>F04836 | 474  | ■ 1.000 | $\operatorname{tr}(A+B)$                                                                                | $\operatorname{tr}(A+B)$                                     | 3.1.19 (a) We compute $\operatorname{tr}(A+B)$ directly from the defini-                                      |
| johnston-linear-matrix-atgevr2<br>F04837 | 474  | ■ 1.000 | $\operatorname{tr}(c A)$                                                                                | $\operatorname{tr}(cA)$                                      | (b) Similarly, we now compute $\operatorname{tr}(cA)$ directly                                                |
| johnston-linear-matrix-atgevr2<br>F04838 | 474  | ■ 1.000 | $\mathbf{e}_{\{i\}} \mathbf{e}_{\{j\}}^T$                                                               | $\mathbf{e}_i \mathbf{e}_j^T$                                | $\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some $i, j, k$ with $i \neq k$ :      |
| johnston-linear-matrix-atgevr2<br>F04839 | 474  | ■ 1.000 | $C=\mathbf{e}_{\{j\}} \mathbf{e}_{\{k\}}^T$                                                             | $C = \mathbf{e}_j \mathbf{e}_k^T$                            | $\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some $i, j, k$ with $i \neq k$ :      |

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| johnston-linear-matrix-atgevrnF04840 | 474  | 1.000 | $i, j, k$                                                                                                                        | $i, j, k$                                                                                          | $\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some $i, j, k$ with $i \neq k$ :      |
| johnston-linear-matrix-atgevrnF04841 | 474  | 1.000 | $i \neq k$                                                                                                                       | $i \neq k$                                                                                         | $\mathbf{e}_i \mathbf{e}_j^T$ and $C = \mathbf{e}_j \mathbf{e}_k^T$ for some $i, j, k$ with $i \neq k$ :      |
| johnston-linear-matrix-atgevrnF04842 | 474  | 1.000 | $f(A \ C \ B)=f(A \ B \ C)$                                                                                                      | $f(ACB) = f(ABC)$                                                                                  | Since $f(ACB) = f(ABC)$ , it follows that $f(A\mathbf{e}_i \mathbf{e}_k^T) =$                                 |
| johnston-linear-matrix-atgevrnF04843 | 474  | 1.000 | $f\left(A \ \mathbf{e}_i \ \mathbf{e}_k^T\right)=$                                                                               | $f\left(A \mathbf{e}_i \mathbf{e}_k^T\right) =$                                                    | Since $f(ACB) = f(ABC)$ , it follows that $f(A\mathbf{e}_i \mathbf{e}_k^T) =$                                 |
| johnston-linear-matrix-atgevrnF04844 | 474  | 0.999 | $A=\mathbf{e}_j \ \mathbf{e}_i^T$                                                                                                | $A = \mathbf{e}_j \mathbf{e}_i^T$                                                                  | 0. By now choosing $A = \mathbf{e}_j \mathbf{e}_i^T$ , we see that                                            |
| johnston-linear-matrix-atgevrnF04845 | 474  | 1.000 | $j, k$                                                                                                                           | $j, k$                                                                                             | for all $j, k$ . Since $f$ is linear and every matrix $\underline{A} \in \mathcal{M}_n$                       |
| johnston-linear-matrix-atgevrnF04846 | 474  | 1.000 | $A=\sum_{j, k=1}^n a_{j, k} \ \mathbf{e}_j \ \mathbf{e}_k^T$                                                                     | $A = \sum_{j, k=1}^n a_{j, k} \mathbf{e}_j \mathbf{e}_k^T$                                         | can be written in the form $A = \sum_{j, k=1}^n a_{j, k} \mathbf{e}_j \mathbf{e}_k^T$ , it fol-               |
| johnston-linear-matrix-atgevrnF04847 | 474  | 0.999 | $B=\left\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\right\},\left[\mathbf{v}\right]_B=\left(c_1, c_2, \ldots, c_k\right)$ | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}, [\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$ | $B = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}, [\mathbf{v}]_B = (c_1, c_2, \dots, c_k)$ and        |
| johnston-linear-matrix-atgevrnF04848 | 474  | 1.000 | $\mathbf{v}_i \cdot \mathbf{v}_j=1$                                                                                              | $\mathbf{v}_i \cdot \mathbf{v}_j = 1$                                                              | that $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ if $i \neq j$ |
| johnston-linear-matrix-atgevrnF04849 | 474  | 1.000 | $\mathbf{v}_i \cdot \mathbf{v}_j=0$                                                                                              | $\mathbf{v}_i \cdot \mathbf{v}_j = 0$                                                              | that $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ if $i \neq j$ |
| johnston-linear-matrix-atgevrnF04850 | 474  | 1.000 | $B=\{(1,1),(0,1)\}$                                                                                                              | $B = \{(1, 1), (0, 1)\}$                                                                           | For example, if $B = \{(1, 1), (0, 1)\}$ and $\mathbf{v} =$                                                   |
| johnston-linear-matrix-atgevrnF04851 | 474  | 1.000 | $(1,0), \mathbf{w}=(0,1)$                                                                                                        | $(1, 0), \mathbf{w} = (0, 1)$                                                                      | $(1, 0), \mathbf{w} = (0, 1)$ then $[\mathbf{v}]_B = (1, -1)$ and                                             |
| johnston-linear-matrix-atgevrnF04852 | 474  | 1.000 | $[\mathbf{v}]_B=(1,-1)$                                                                                                          | $[\mathbf{v}]_B = (1, -1)$                                                                         | $(1, 0), \mathbf{w} = (0, 1)$ then $[\mathbf{v}]_B = (1, -1)$ and                                             |
| johnston-linear-matrix-atgevrnF04853 | 474  | 1.000 | $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in B$                                                                         | $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in B$                                            | <b>3.1.22</b> We start by supposing that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in B$ and          |
| johnston-linear-matrix-atgevrnF04854 | 474  | 1.000 | $\mathbf{v}_1 \cdot \mathbf{0}$                                                                                                  | $\mathbf{v}_1 \cdot \mathbf{0}$                                                                    | this end, we start by computing $\mathbf{v}_1 \cdot \mathbf{0}$ in two different                              |
| johnston-linear-matrix-atgevrnF04855 | 474  | 1.000 | $\left\ \mathbf{v}_1\right\  \neq 0$                                                                                             | $\ \mathbf{v}_1\  \neq 0$                                                                          | that $\ \mathbf{v}_1\  \neq 0$ , so this implies $c_1 = 0$ .                                                  |
| johnston-linear-matrix-atgevrnF04856 | 474  | 1.000 | $\mathbf{v}_2 \cdot \mathbf{0}$                                                                                                  | $\mathbf{v}_2 \cdot \mathbf{0}$                                                                    | A similar computation involving $\mathbf{v}_2 \cdot \mathbf{0}$ shows that                                    |
| johnston-linear-matrix-atgevrnF04857 | 474  | 1.000 | $c_2=0$                                                                                                                          | $c_2 = 0$                                                                                          | $c_2 = 0$ , and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$ ,                          |
| johnston-linear-matrix-atgevrnF04858 | 474  | 1.000 | $\mathbf{v}_k \cdot \mathbf{0}$                                                                                                  | $\mathbf{v}_k \cdot \mathbf{0}$                                                                    | $c_2 = 0$ , and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$ ,                          |
| johnston-linear-matrix-atgevrnF04859 | 474  | 1.000 | $c_k=0$                                                                                                                          | $c_k = 0$                                                                                          | $c_2 = 0$ , and so on up to $\mathbf{v}_k \cdot \mathbf{0}$ showing that $c_k = 0$ ,                          |
| johnston-linear-matrix-atgevrnF04860 | 474  | 0.976 | $\sqrt{\sin ^2(\theta)+\cos ^2(\theta)}=\sqrt{1}=$                                                                               | $\sqrt{\sin^2(\theta) + \cos^2(\theta)} = \sqrt{1} =$                                              | which has length $\sqrt{\sin^2(\theta) + \cos^2(\theta)} = \sqrt{1} =$                                        |
| johnston-linear-matrix-atgevrnF04861 | 474  | 1.000 | $\left\{\mathbf{u}_1, \mathbf{u}_2\right\}$                                                                                      | $\{\mathbf{u}_1, \mathbf{u}_2\}$                                                                   | <b>(b)</b> Suppose that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthonormal ba-                                |
| johnston-linear-matrix-atgevrnF04862 | 474  | 1.000 | $\mathbf{u}_1=(a, b)$                                                                                                            | $\mathbf{u}_1 = (a, b)$                                                                            | sis of $\mathbb{R}^2$ . Since $\mathbf{u}_1 = (a, b)$ is a unit vector,                                       |
| johnston-linear-matrix-atgevrnF04863 | 474  | 1.000 | $ a  \leq 1$                                                                                                                     | $ a  \leq 1$                                                                                       | we know that $ a  \leq 1$ , so we can find $\theta$ such                                                      |
| johnston-linear-matrix-atgevrnF04864 | 474  | 0.999 | $\cos (\theta)=a$                                                                                                                | $\cos(\theta) = a$                                                                                 | that $\cos(\theta) = a$ . Then $b = \pm \sqrt{1 - \cos^2(\theta)} =$                                          |
| johnston-linear-matrix-atgevrnF04865 | 474  | 0.999 | $b=\pm \sqrt{1-\cos ^2(\theta)}=$                                                                                                | $b = \pm \sqrt{1 - \cos^2(\theta)} =$                                                              | that $\cos(\theta) = a$ . Then $b = \pm \sqrt{1 - \cos^2(\theta)} =$                                          |
| johnston-linear-matrix-atgevrnF04866 | 474  | 1.000 | $\pm \sin (\theta) $                                                                                                             | $\pm \sin(\theta) $                                                                                | $\pm \sin(\theta) $ . If $b$ is positive, we have writ-                                                       |
| johnston-linear-matrix-atgevrnF04867 | 474  | 1.000 | $\mathbf{u}_1=(\cos (\theta), \sin (\theta))$                                                                                    | $\mathbf{u}_1 = (\cos(\theta), \sin(\theta))$                                                      | ten $\mathbf{u}_1 = (\cos(\theta), \sin(\theta))$ . If $b$ is negative,                                       |

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|------------------------------------------|------|-------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| johnston-linear-matrix-atgevrq<br>F04868 | 474  | 1.000 | $-\theta$                                         | $-\theta$                                         | then replace $\theta$ by $-\theta$ so that we have $\mathbf{u}_1 =$                                    |
| johnston-linear-matrix-atgevrq<br>F04869 | 474  | 1.000 | $\mathbf{u}_1 =$                                  | $\mathbf{u}_1 =$                                  | then replace $\theta$ by $-\theta$ so that we have $\mathbf{u}_1 =$                                    |
| johnston-linear-matrix-atgevrq<br>F04870 | 475  | 1.000 | $\mathbf{u}_2$                                    | $\mathbf{u}_2$                                    | Next, since $\mathbf{u}_2$ is orthogonal to $\mathbf{u}_1$ , we know                                   |
| johnston-linear-matrix-atgevrq<br>F04871 | 475  | 1.000 | $\mathbf{u}_1$                                    | $\mathbf{u}_1$                                    | Next, since $\mathbf{u}_2$ is orthogonal to $\mathbf{u}_1$ , we know                                   |
| johnston-linear-matrix-atgevrq<br>F04872 | 475  | 0.953 | $(-\sin(\theta), \cos(\theta))$                   | $(-\sin(\theta), \cos(\theta))$                   | $\mathbf{u}_2$ is a multiple of $(-\sin(\theta), \cos(\theta))$ . Since                                |
| johnston-linear-matrix-atgevrq<br>F04873 | 475  | 0.817 | $\mathbf{u}_2 =$                                  | $\mathbf{u}_2 =$                                  | are 1 and $-1$ , so we know that either $\mathbf{u}_2 =$                                               |
| johnston-linear-matrix-atgevrq<br>F04874 | 475  | 1.000 | $2^7 3^3 = 3456$                                  | $2^7 3^3 = 3456$                                  | <a href="#">(g)</a> $2^7 3^3 = 3456$                                                                   |
| johnston-linear-matrix-atgevrq<br>F04875 | 475  | 1.000 | $\det(-A) = (-1)^n \det(A)$                       | $\det(-A) = (-1)^n \det(A)$                       | <a href="#">(a)</a> False. $\det(-A) = (-1)^n \det(A)$ , so the given                                  |
| johnston-linear-matrix-atgevrq<br>F04876 | 475  | 1.000 | $A = I_2$                                         | $A = I_2$                                         | <a href="#">(c)</a> False. For example, if $A = I_2$ and $B = -I_2$                                    |
| johnston-linear-matrix-atgevrq<br>F04877 | 475  | 1.000 | $B = -I_2$                                        | $B = -I_2$                                        | <a href="#">(c)</a> False. For example, if $A = I_2$ and $B = -I_2$                                    |
| johnston-linear-matrix-atgevrq<br>F04878 | 475  | 1.000 | $\det(A+B) = \det(O) = 0$                         | $\det(A+B) = \det(O) = 0$                         | then $\det(A+B) = \det(O) = 0$ , but $\det(A) +$                                                       |
| johnston-linear-matrix-atgevrq<br>F04879 | 475  | 1.000 | $\det(A) +$                                       | $\det(A) +$                                       | then $\det(A+B) = \det(O) = 0$ , but $\det(A) +$                                                       |
| johnston-linear-matrix-atgevrq<br>F04880 | 475  | 1.000 | $\det(B) = 1 + 1 = 2$                             | $\det(B) = 1 + 1 = 2$                             | $\det(B) = 1 + 1 = 2$ .                                                                                |
| johnston-linear-matrix-atgevrq<br>F04881 | 475  | 0.980 | $\det([P_{\mathbf{u}}]) = 0$                      | $\det([P_{\mathbf{u}}]) = 0$                      | $\det([P_{\mathbf{u}}]) = 0$ .                                                                         |
| johnston-linear-matrix-atgevrq<br>F04882 | 475  | 1.000 | $\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$    | $\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$    | $(-\sin(\theta), \cos(\theta))$ or $\mathbf{u}_2 = (\sin(\theta), -\cos(\theta))$ .                    |
| johnston-linear-matrix-atgevrq<br>F04883 | 475  | 1.000 | $\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$           | $\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$           | In the former case, we are done: $\mathbf{u}_1 = R^\theta(\mathbf{e}_1)$                               |
| johnston-linear-matrix-atgevrq<br>F04884 | 475  | 1.000 | $\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$           | $\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$           | and $\mathbf{u}_2 = R^\theta(\mathbf{e}_2)$ . In the latter case, we use                               |
| johnston-linear-matrix-atgevrq<br>F04885 | 475  | 1.000 | $\sin(\theta) = \cos(\theta - \pi/2)$             | $\sin(\theta) = \cos(\theta - \pi/2)$             | the facts that $\sin(\theta) = \cos(\theta - \pi/2)$ and                                               |
| johnston-linear-matrix-atgevrq<br>F04886 | 475  | 0.650 | $-\cos(\theta) = \sin(\theta - \pi/2)$            | $-\cos(\theta) = \sin(\theta - \pi/2)$            | $-\cos(\theta) = \sin(\theta - \pi/2)$ to see that $\mathbf{u}_1 =$                                    |
| johnston-linear-matrix-atgevrq<br>F04887 | 475  | 1.000 | $R_{\theta - \pi/2}(\mathbf{e}_2)$                | $R_{\theta - \pi/2}(\mathbf{e}_2)$                | $R_{\theta - \pi/2}(\mathbf{e}_2)$ and $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$ , so again we |
| johnston-linear-matrix-atgevrq<br>F04888 | 475  | 1.000 | $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$ | $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$ | $R_{\theta - \pi/2}(\mathbf{e}_2)$ and $\mathbf{u}_2 = R^{\theta - \pi/2}(\mathbf{e}_1)$ , so again we |
| johnston-linear-matrix-atgevrq<br>F04889 | 475  | 1.000 | $\det(A^T) = 0$                                   | $\det(A^T) = 0$                                   | must also not be invertible, so $\det(A^T) = 0$ as                                                     |
| johnston-linear-matrix-atgevrq<br>F04890 | 475  | 1.000 | $E_1 E_2 \cdots E_k$                              | $E_1 E_2 \cdots E_k$                              | $E_1 E_2 \cdots E_k$ . Then $\det(A) = \det(E_1 E_2 \cdots E_k) =$                                     |
| johnston-linear-matrix-atgevrq<br>F04891 | 475  | 1.000 | $\det(A) = \det(E_1 E_2 \cdots E_k) =$            | $\det(A) = \det(E_1 E_2 \cdots E_k) =$            | $E_1 E_2 \cdots E_k$ . Then $\det(A) = \det(E_1 E_2 \cdots E_k) =$                                     |
| johnston-linear-matrix-atgevrq<br>F04892 | 475  | 1.000 | $\det(E_1) \cdots \det(E_k)$                      | $\det(E_1) \cdots \det(E_k)$                      | $\det(E_1) \cdots \det(E_k)$ . Also, $A^T = E_k^T \cdots E_2^T E_1^T$ ,                                |
| johnston-linear-matrix-atgevrq<br>F04893 | 475  | 1.000 | $A^T = E_k^T \cdots E_2^T E_1^T$                  | $A^T = E_k^T \cdots E_2^T E_1^T$                  | $\det(E_1) \cdots \det(E_k)$ . Also, $A^T = E_k^T \cdots E_2^T E_1^T$ ,                                |
| johnston-linear-matrix-atgevrq<br>F04894 | 475  | 1.000 | $\det(A^T) = \det(E_k^T \cdots E_2^T E_1^T) =$    | $\det(A^T) = \det(E_k^T \cdots E_2^T E_1^T) =$    | so $\det(A^T) = \det(E_k^T \cdots E_2^T E_1^T) =$                                                      |
| johnston-linear-matrix-atgevrq<br>F04895 | 475  | 1.000 | $\det(E_k^T) \cdots \det(E_2^T) \det(E_1^T)$      | $\det(E_k^T) \cdots \det(E_2^T) \det(E_1^T)$      | $\det(E_k^T) \cdots \det(E_2^T) \det(E_1^T)$ , so it suffices to                                       |

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| johnston-linear-matrix-atgevr_a<br>F04896 | 475  | 1.000 | $\operatorname{det}(E)=\operatorname{det}(\left(E^{\mathrm{T}}\right.\left.\right)$                          | $\det(E)=\det(E^T)$                                          | show that $\det(E)=\det(E^T)$ whenever $E$ is an                                 |
| johnston-linear-matrix-atgevr_a<br>F04897 | 475  | 1.000 | $E^{\mathrm{T}}=E$                                                                                           | $E^T=E$                                                      | operation $cR_i$ then $E^T=E$ , so it is trivially                               |
| johnston-linear-matrix-atgevr_a<br>F04898 | 475  | 1.000 | $\operatorname{det}(\left(E^{\mathrm{T}}\right)\operatorname{det}(E)$                                        | $\det(E^T)=\det(E)$                                          | the case that $\det(E^T)=\det(E)$ . For the one                                  |
| johnston-linear-matrix-atgevr_a<br>F04899 | 475  | 1.000 | $\operatorname{det}(C)$                                                                                      | $\det(C)$                                                    | on the right have determinants equal to $\det(C)$                                |
| johnston-linear-matrix-atgevr_a<br>F04900 | 476  | 1.000 | $\widetilde{B}$                                                                                              | $\widetilde{B}$                                              | where $\widetilde{B}$ is the matrix obtained from $B$ by re-                     |
| johnston-linear-matrix-atgevr_a<br>F04901 | 476  | 1.000 | $\operatorname{det}(\left(I_m+A\right.\left.\right)$<br>$\operatorname{det}(\left(I_n+B\right.\left.\right)$ | $\det(I_m+AB)=\det(I_n+BA)$                                  | We thus conclude that $\det(I_m+AB)=\det(I_n+BA)$ ,                              |
| johnston-linear-matrix-atgevr_a<br>F04902 | 476  | 1.000 | $\operatorname{det}(\left(I_n+\mathbf{vw}^{\mathrm{T}}\right)=1+\mathbf{w}^{\mathrm{T}}\mathbf{v}$           | $\det(I_n+\mathbf{vw}^T)=1+\mathbf{w}^T\mathbf{v}$           | so $\det(I_n+\mathbf{vw}^T)=1+\mathbf{w}^T\mathbf{v}$ .                          |
| johnston-linear-matrix-atgevr_a<br>F04903 | 477  | 1.000 | $\mathbf{v}=(5,-4)$                                                                                          | $\mathbf{v}=(5,-4)$                                          | <b>3.3.2 (a)</b> A corresponding eigenvector is $\mathbf{v}=(5,-4)$ .            |
| johnston-linear-matrix-atgevr_a<br>F04904 | 477  | 1.000 | $\mathbf{v}=(3,2,0)$                                                                                         | $\mathbf{v}=(3,2,0)$                                         | <b>(c)</b> $\mathbf{v}=(3,2,0)$                                                  |
| johnston-linear-matrix-atgevr_a<br>F04905 | 477  | 1.000 | $\mathbf{v}=(-1,1,0,1)$                                                                                      | $\mathbf{v}=(-1,1,0,1)$                                      | <b>(e)</b> $\mathbf{v}=(-1,1,0,1)$                                               |
| johnston-linear-matrix-atgevr_a<br>F04906 | 477  | 1.000 | $(A-\mathbf{0}\ I)\mathbf{v}=\mathbf{0}$                                                                     | $(A-0I)\mathbf{v}=\mathbf{0}$                                | $(A-0I)\mathbf{v}=\mathbf{0}$ :                                                  |
| johnston-linear-matrix-atgevr_a<br>F04907 | 477  | 1.000 | $v_1=-2\ v_2$                                                                                                | $v_1=-2v_2$                                                  | $v_1=-2v_2$ , so all eigenvectors are of the form                                |
| johnston-linear-matrix-atgevr_a<br>F04908 | 477  | 1.000 | $\mathbf{v}=v_2(-2,1)$                                                                                       | $\mathbf{v}=v_2(-2,1)$                                       | $\mathbf{v}=v_2(-2,1)$ . It follows that $\{(-2,1)\}$ is a ba-                   |
| johnston-linear-matrix-atgevr_a<br>F04909 | 477  | 1.000 | $\{(-2,1)\}$                                                                                                 | $\{(-2,1)\}$                                                 | $\mathbf{v}=v_2(-2,1)$ . It follows that $\{(-2,1)\}$ is a ba-                   |
| johnston-linear-matrix-atgevr_a<br>F04910 | 477  | 1.000 | $\mathbf{v}=v_2(-1,1)$                                                                                       | $\mathbf{v}=v_2(-1,1)$                                       | $\mathbf{v}=v_2(-1,1)$ . It follows that $\{(-1,1)\}$ is a                       |
| johnston-linear-matrix-atgevr_a<br>F04911 | 477  | 1.000 | $\{(-1,1)\}$                                                                                                 | $\{(-1,1)\}$                                                 | $\mathbf{v}=v_2(-1,1)$ . It follows that $\{(-1,1)\}$ is a                       |
| johnston-linear-matrix-atgevr_a<br>F04912 | 477  | 1.000 | $\{(1,0)\}$                                                                                                  | $\{(1,0)\}$                                                  | eigenspace is $\{(1,0)\}$ , so this eigenvalue has                               |
| johnston-linear-matrix-atgevr_a<br>F04913 | 477  | 1.000 | $\left\{\mathbf{e}_1\right\},\left\{\mathbf{e}_2\right\}$                                                    | $\{\mathbf{e}_1\},\{\mathbf{e}_2\}$                          | are $\{\mathbf{e}_1\},\{\mathbf{e}_2\}$ , and $\{\mathbf{e}_3\}$ , respectively. |
| johnston-linear-matrix-atgevr_a<br>F04914 | 477  | 1.000 | $\{(1,0,0,0)\},\{(1,-5,0,0)\}$                                                                               | $\{(1,0,0,0)\},\{(1,-5,0,0)\}$                               | eigenspaces are $\{(1,0,0,0)\},\{(1,-5,0,0)\}$ ,                                 |
| johnston-linear-matrix-atgevr_a<br>F04915 | 477  | 0.998 | $\{(1,-1,-2,0)\}$                                                                                            | $\{(1,-1,-2,0)\}$                                            | and $\{(1,-1,-2,0)\}$ , respectively, so all of the                              |
| johnston-linear-matrix-atgevr_a<br>F04916 | 477  | 1.000 | $\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$                                                                            | $\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$                            | are $\{(1,1,-1,-1)\},\{(2,-1,2,-1)\}$ , and                                      |
| johnston-linear-matrix-atgevr_a<br>F04917 | 477  | 1.000 | $\{(1,1,0,-1),(0,1,-2,0)\}$                                                                                  | $\{(1,1,0,-1),(0,1,-2,0)\}$                                  | $\{(1,1,0,-1),(0,1,-2,0)\}$ , respectively.                                      |
| johnston-linear-matrix-atgevr_a<br>F04918 | 477  | 1.000 | $\{(0,0,1,0,-1),(1,0,0,2,0)\}$                                                                               | $\{(0,0,1,0,-1),(1,0,0,2,0)\}$                               | eigenspaces are $\{(0,0,1,0,-1),(1,0,0,2,0)\}$ ,                                 |
| johnston-linear-matrix-atgevr_a<br>F04919 | 477  | 0.998 | $\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$                                                                             | $\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$                             | $\{(1,-1,-1,0,1),(2,-2,0,1,0)\}$ , and                                           |
| johnston-linear-matrix-atgevr_a<br>F04920 | 477  | 0.999 | $\{(0,2,2,-1,0)\}$                                                                                           | $\{(0,2,2,-1,0)\}$                                           | $\{(0,2,2,-1,0)\}$ , respectively.                                               |
| johnston-linear-matrix-atgevr_a<br>F04921 | 477  | 1.000 | $\operatorname{det}(A-\lambda I)=(-1)^n\operatorname{det}(\lambda I-A)$                                      | $\det(A-\lambda I)=(-1)^n\det(\lambda I-A)$                  | made is $\det(A-\lambda I)=(-1)^n\det(\lambda I-A)$ .                            |
| johnston-linear-matrix-atgevr_a<br>F04922 | 477  | 1.000 | $\lambda=\frac{1}{2}(1+k)\pm\sqrt{(1+k)^2-4k-4}$                                                             | $\lambda=\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{(1+k)^2-4k-4}=$ | $\lambda$ , we get $\lambda=\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{(1+k)^2-4k-4}=$  |
| johnston-linear-matrix-atgevr_a<br>F04923 | 477  | 1.000 | $\frac{1}{2}(1+k)\pm\sqrt{k^2-2k-3}$                                                                         | $\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{k^2-2k-3}$              | $\frac{1}{2}(1+k)\pm\frac{1}{2}\sqrt{k^2-2k-3}$ . We get two distinct            |
| johnston-linear-matrix-atgevr_a<br>F04924 | 477  | 1.000 | $k^2-2k-3>0$                                                                                                 | $k^2-2k-3>0$                                                 | $k^2-2k-3>0$ . Similarly, we get only one real                                   |

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| johnston-linear-matrix-algebra-F04925 | 477  | 1.000 | $k^2 - 2k - 3$                                                                                  | $k^2 - 2k - 3$                                                                                  | The discriminant $k^2 - 2k - 3$ factors as $(k - 3)(k + 1)$ ,                                                                           |
| johnston-linear-matrix-algebra-F04926 | 477  | 1.000 | $(k - 3)(k + 1)$                                                                                | $(k - 3)(k + 1)$                                                                                | The discriminant $k^2 - 2k - 3$ factors as $(k - 3)(k + 1)$ ,                                                                           |
| johnston-linear-matrix-algebra-F04927 | 477  | 1.000 | $k > 3$                                                                                         | $k > 3$                                                                                         | when $k > 3$ or $k < -1$ , (ii) we get exactly one distinct                                                                             |
| johnston-linear-matrix-algebra-F04928 | 477  | 1.000 | $k < -1$                                                                                        | $k < -1$                                                                                        | when $k > 3$ or $k < -1$ , (ii) we get exactly one distinct                                                                             |
| johnston-linear-matrix-algebra-F04929 | 477  | 1.000 | $k = 3$                                                                                         | $k = 3$                                                                                         | eigenvalue when $k = 3$ or $k = -1$ , and (iii) we get                                                                                  |
| johnston-linear-matrix-algebra-F04930 | 477  | 1.000 | $k = -1$                                                                                        | $k = -1$                                                                                        | eigenvalue when $k = 3$ or $k = -1$ , and (iii) we get                                                                                  |
| johnston-linear-matrix-algebra-F04931 | 477  | 1.000 | $-1 < k < 3$                                                                                    | $-1 < k < 3$                                                                                    | no real eigenvalues when $-1 < k < 3$ .                                                                                                 |
| johnston-linear-matrix-algebra-F04932 | 477  | 1.000 | $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \operatorname{det}(A)$                | $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \operatorname{det}(A)$                | matrix $A$ is $p_A(\lambda) = \lambda^2 - \operatorname{tr}(A)\lambda + \operatorname{det}(A)$ . Applying                               |
| johnston-linear-matrix-algebra-F04933 | 478  | 1.000 | $\lambda = \pm 1$                                                                               | $\lambda = \pm 1$                                                                               | <b>3.3.12</b> The only possible eigenvalues are $\lambda = \pm 1$ and                                                                   |
| johnston-linear-matrix-algebra-F04934 | 478  | 1.000 | $\lambda = \pm i$                                                                               | $\lambda = \pm i$                                                                               | $\lambda = \pm i$ . The fact that these eigenvalues are all pos-                                                                        |
| johnston-linear-matrix-algebra-F04935 | 478  | 1.000 | $A^4 \mathbf{v} = \lambda^4 \mathbf{v}$                                                         | $A^4 \mathbf{v} = \lambda^4 \mathbf{v}$                                                         | shows that $A^4 \mathbf{v} = \lambda^4 \mathbf{v}$ . On the other hand, $A^4 = I$ , so                                                  |
| johnston-linear-matrix-algebra-F04936 | 478  | 0.999 | $A^4 \mathbf{v} = \mathbf{v}$                                                                   | $A^4 \mathbf{v} = \mathbf{v}$                                                                   | $A^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 = 1$ .                                           |
| johnston-linear-matrix-algebra-F04937 | 478  | 0.999 | $\lambda^4 \mathbf{v} = \mathbf{v}$                                                             | $\lambda^4 \mathbf{v} = \mathbf{v}$                                                             | $A^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 = 1$ .                                           |
| johnston-linear-matrix-algebra-F04938 | 478  | 0.999 | $\lambda^4 = 1$                                                                                 | $\lambda^4 = 1$                                                                                 | $A^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 \mathbf{v} = \mathbf{v}$ , so $\lambda^4 = 1$ .                                           |
| johnston-linear-matrix-algebra-F04939 | 478  | 1.000 | $\det([R^\theta] - \lambda I) = 0$                                                              | $\det([R^\theta] - \lambda I) = 0$                                                              | <b>3.3.13</b> (a) Setting $\det([R^\theta] - \lambda I) = 0$ gives                                                                      |
| johnston-linear-matrix-algebra-F04940 | 478  | 1.000 | $\lambda = \cos(\theta) \pm \sqrt{\cos^2(\theta) - 1}$                                          | $\lambda = \cos(\theta) \pm \sqrt{\cos^2(\theta) - 1}$                                          | mula) gives $\lambda = \cos(\theta) \pm \sqrt{\cos^2(\theta) - 1}$ .                                                                    |
| johnston-linear-matrix-algebra-F04941 | 478  | 1.000 | $\lambda = \cos(\theta) \pm$                                                                    | $\lambda = \cos(\theta) \pm$                                                                    | (b) We found the eigenvalues to be $\lambda = \cos(\theta) \pm$                                                                         |
| johnston-linear-matrix-algebra-F04942 | 478  | 0.978 | $\sqrt{\cos^2(\theta) - 1}$                                                                     | $\sqrt{\cos^2(\theta) - 1}$                                                                     | $\sqrt{\cos^2(\theta) - 1}$ , which are real if and only if                                                                             |
| johnston-linear-matrix-algebra-F04943 | 478  | 0.999 | $\cos^2(\theta) - 1 \geq 0$                                                                     | $\cos^2(\theta) - 1 \geq 0$                                                                     | $\cos^2(\theta) - 1 \geq 0$ , i.e., $\cos(\theta) = \pm 1$ . In other                                                                   |
| johnston-linear-matrix-algebra-F04944 | 478  | 0.999 | $\cos(\theta) = \pm 1$                                                                          | $\cos(\theta) = \pm 1$                                                                          | $\cos^2(\theta) - 1 \geq 0$ , i.e., $\cos(\theta) = \pm 1$ . In other                                                                   |
| johnston-linear-matrix-algebra-F04945 | 478  | 1.000 | $\theta = k\pi$                                                                                 | $\theta = k\pi$                                                                                 | words, $\theta = k\pi$ for some integer $k$ .                                                                                           |
| johnston-linear-matrix-algebra-F04946 | 478  | 1.000 | $\theta = 2k\pi$                                                                                | $\theta = 2k\pi$                                                                                | $\theta = 2k\pi$ then all vectors are being rotated back                                                                                |
| johnston-linear-matrix-algebra-F04947 | 478  | 0.999 | $\theta = (2k + 1)\pi$                                                                          | $\theta = (2k + 1)\pi$                                                                          | value 1). If $\theta = (2k + 1)\pi$ then every vector is                                                                                |
| johnston-linear-matrix-algebra-F04948 | 478  | 1.000 | $\theta \neq k\pi$                                                                              | $\theta \neq k\pi$                                                                              | if $\theta \neq k\pi$ , then every vector is rotated so that                                                                            |
| johnston-linear-matrix-algebra-F04949 | 478  | 1.000 | $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^* \mathbf{w}$                                         | $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^* \mathbf{w}$                                         | tors then $\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^* \mathbf{w}$ . It follows that $\mathbf{x} \cdot (A\mathbf{y}) =$                  |
| johnston-linear-matrix-algebra-F04950 | 478  | 1.000 | $\mathbf{x}^* A \mathbf{y}$                                                                     | $\mathbf{x}^* A \mathbf{y}$                                                                     | $\mathbf{x}^* A \mathbf{y}$ and $(A^* \mathbf{x}) \cdot \mathbf{y} = (A^* \mathbf{x})^* \mathbf{y} = \mathbf{x}^* A \mathbf{y}$ . Since |
| johnston-linear-matrix-algebra-F04951 | 478  | 1.000 | $(A^* \mathbf{x}) \cdot \mathbf{y} = (A^* \mathbf{x})^* \mathbf{y} = \mathbf{x}^* A \mathbf{y}$ | $(A^* \mathbf{x}) \cdot \mathbf{y} = (A^* \mathbf{x})^* \mathbf{y} = \mathbf{x}^* A \mathbf{y}$ | $\mathbf{x}^* A \mathbf{y}$ and $(A^* \mathbf{x}) \cdot \mathbf{y} = (A^* \mathbf{x})^* \mathbf{y} = \mathbf{x}^* A \mathbf{y}$ . Since |
| johnston-linear-matrix-algebra-F04952 | 478  | 1.000 | $\mathbf{x} \cdot (A \mathbf{y}) = (A^* \mathbf{x}) \cdot \mathbf{y}$                           | $\mathbf{x} \cdot (A \mathbf{y}) = (A^* \mathbf{x}) \cdot \mathbf{y}$                           | $\mathbf{x} \cdot (A \mathbf{y}) = (A^* \mathbf{x}) \cdot \mathbf{y}$ .                                                                 |

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| johnston-linear-matrix-atgevr0<br>F04953 | 478  | ■ 1.000 | $\mathbf{x} \cdot (A \mathbf{y}) = \mathbf{x}^* A \mathbf{y}$             | $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^* A\mathbf{y}$               | (b) Just like in part (a), $\mathbf{x} \cdot (A\mathbf{y}) = \mathbf{x}^* A\mathbf{y}$ and                          |
| johnston-linear-matrix-atgevr0<br>F04954 | 478  | ■ 1.000 | $(B \mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$           | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$            | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$ . We thus know that $\mathbf{x}^* A\mathbf{y} =$     |
| johnston-linear-matrix-atgevr0<br>F04955 | 478  | ■ 1.000 | $\mathbf{x}^* A \mathbf{y} =$                                             | $\mathbf{x}^* A\mathbf{y} =$                                              | $(B\mathbf{x}) \cdot \mathbf{y} = \mathbf{x}^* B^* \mathbf{y}$ . We thus know that $\mathbf{x}^* A\mathbf{y} =$     |
| johnston-linear-matrix-atgevr0<br>F04956 | 478  | ■ 0.999 | $\mathbf{x}^* B^* \mathbf{y}$                                             | $\mathbf{x}^* B^* \mathbf{y}$                                             | $\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$ . However, if |
| johnston-linear-matrix-atgevr0<br>F04957 | 478  | ■ 0.999 | $\mathbf{x} \in \mathbb{C}^m$                                             | $\mathbf{x} \in \mathbb{C}^m$                                             | $\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$ . However, if |
| johnston-linear-matrix-atgevr0<br>F04958 | 478  | ■ 0.999 | $\mathbf{y} \in \mathbb{C}^n$                                             | $\mathbf{y} \in \mathbb{C}^n$                                             | $\mathbf{x}^* B^* \mathbf{y}$ for all $\mathbf{x} \in \mathbb{C}^m$ and $\mathbf{y} \in \mathbb{C}^n$ . However, if |
| johnston-linear-matrix-atgevr0<br>F04959 | 478  | ■ 1.000 | $\mathbf{x}^* A \mathbf{y} = a_{i,j}$                                     | $\mathbf{x}^* A\mathbf{y} = a_{i,j}$                                      | we choose $\mathbf{x} = \mathbf{e}_i$ and $\mathbf{y} = \mathbf{e}_j$ then $\mathbf{x}^* A\mathbf{y} = a_{i,j}$     |
| johnston-linear-matrix-atgevr0<br>F04960 | 478  | ■ 1.000 | $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$                                   | $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$                                   | and $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$ , so we see that $a_{i,j} = \overline{b_{j,i}}$ for                     |
| johnston-linear-matrix-atgevr0<br>F04961 | 478  | ■ 1.000 | $a_{i,j} = \overline{b_{j,i}}$                                            | $a_{i,j} = \overline{b_{j,i}}$                                            | and $\mathbf{x}^* B^* \mathbf{y} = b_{j,i}$ , so we see that $a_{i,j} = \overline{b_{j,i}}$ for                     |
| johnston-linear-matrix-atgevr0<br>F04962 | 478  | ■ 0.975 | $A \mathbf{v} = 0 \mathbf{v} = \mathbf{0}$                                | $A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$                                  | exists a non-zero vector $\mathbf{v}$ such that $A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$ ,                          |
| johnston-linear-matrix-atgevr0<br>F04963 | 478  | ■ 1.000 | $(A^*)^* = A$                                                             | $(A^*)^* = A$                                                             | lows from the fact that $(A^*)^* = A$ so if $\bar{\lambda}$ is an                                                   |
| johnston-linear-matrix-atgevr0<br>F04964 | 478  | ■ 1.000 | $\overline{\bar{\lambda}} = \lambda$                                      | $\bar{\bar{\lambda}} = \lambda$                                           | eigenvalue of $A^*$ then $\bar{\bar{\lambda}} = \lambda$ is an eigenvalue                                           |
| johnston-linear-matrix-atgevr0<br>F04965 | 478  | ■ 1.000 | $\mathbf{w} \cdot (A \mathbf{v})$                                         | $\mathbf{w} \cdot (A\mathbf{v})$                                          | (b) We compute $\mathbf{w} \cdot (A\mathbf{v})$ in two different ways:                                              |
| johnston-linear-matrix-atgevr0<br>F04966 | 478  | ■ 1.000 | $\mathbf{w} \cdot \mathbf{v} = 0$                                         | $\mathbf{w} \cdot \mathbf{v} = 0$                                         | is if $\mathbf{w} \cdot \mathbf{v} = 0$ (or equivalently, $\mathbf{v} \cdot \mathbf{w} = 0$ ).                      |
| johnston-linear-matrix-atgevr0<br>F04967 | 478  | ■ 1.000 | $\lambda_2, \lambda_3, \dots, \lambda_n$                                  | $\lambda_2, \lambda_3, \dots, \lambda_n$                                  | its other eigenvalues are $\lambda_2, \lambda_3, \dots, \lambda_n$ . To this                                        |
| johnston-linear-matrix-atgevr0<br>F04968 | 478  | ■ 1.000 | $\overline{\lambda_j}$                                                    | $\overline{\lambda_j}$                                                    | ing to the eigenvalue $\overline{\lambda_j}$ for $2 \leq j \leq n$ . Then                                           |
| johnston-linear-matrix-atgevr0<br>F04969 | 478  | ■ 1.000 | $\lambda_1 \neq$                                                          | $\lambda_1 \neq$                                                          | Exercise 3.3.21 and the assumption that $\lambda_1 \neq$                                                            |
| johnston-linear-matrix-atgevr0<br>F04970 | 478  | ■ 1.000 | $\lambda_1 = \lambda_j$                                                   | $\lambda_1 = \lambda_j$                                                   | $\lambda_j$ . The statement is still true even if $\lambda_1 = \lambda_j$ ,                                         |
| johnston-linear-matrix-atgevr0<br>F04971 | 478  | ■ 1.000 | $B^*$                                                                     | $B^*$                                                                     | It follows that $B^*$ has $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ as eigen-       |
| johnston-linear-matrix-atgevr0<br>F04972 | 478  | ■ 1.000 | $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ | $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ | It follows that $B^*$ has $\overline{\lambda_2}, \overline{\lambda_3}, \dots, \overline{\lambda_n}$ as eigen-       |
| johnston-linear-matrix-atgevr0<br>F04973 | 478  | ■ 1.000 | $\lambda_1 = 1$                                                           | $\lambda_1 = 1$                                                           | which has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ with                                                      |
| johnston-linear-matrix-atgevr0<br>F04974 | 478  | ■ 1.000 | $\lambda_2 = 2$                                                           | $\lambda_2 = 2$                                                           | which has eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$ with                                                      |
| johnston-linear-matrix-atgevr0<br>F04975 | 478  | ■ 1.000 | $\mathbf{v}_1 = (1, 0)$                                                   | $\mathbf{v}_1 = (1, 0)$                                                   | corresponding unit eigenvectors $\mathbf{v}_1 = (1, 0)$                                                             |
| johnston-linear-matrix-atgevr0<br>F04976 | 478  | ■ 1.000 | $\mathbf{v}_2 = (1, 1) / \sqrt{2}$                                        | $\mathbf{v}_2 = (1, 1) / \sqrt{2}$                                        | and $\mathbf{v}_2 = (1, 1) / \sqrt{2}$ , respectively. Then                                                         |
| johnston-linear-matrix-atgevr0<br>F04977 | 478  | ■ 1.000 | $\mathbf{v}_2 = (1, 2)$                                                   | $\mathbf{v}_2 = (1, 2)$                                                   | multiples of $\mathbf{v}_2 = (1, 2)$ .                                                                              |
| johnston-linear-matrix-atgevr0<br>F04978 | 479  | ■ 1.000 | $A \mathbf{1} = \mathbf{1}$                                               | $A\mathbf{1} = \mathbf{1}$                                                | $A\mathbf{1} = \mathbf{1}$ , since $A\mathbf{1}$ is simply the vector whose                                         |
| johnston-linear-matrix-atgevr0<br>F04979 | 479  | ■ 1.000 | $A \mathbf{1}$                                                            | $A\mathbf{1}$                                                             | $A\mathbf{1} = \mathbf{1}$ , since $A\mathbf{1}$ is simply the vector whose                                         |
| johnston-linear-matrix-atgevr0<br>F04980 | 479  | ■ 0.973 | $ v_j  \leq 1$                                                            | $ v_j  \leq 1$                                                            | fact that we scaled $\mathbf{v}$ so that $ v_j  \leq 1$ for all $j$ .                                               |
| johnston-linear-matrix-atgevr0<br>F04981 | 479  | ■ 1.000 | $\lambda = i$                                                             | $\lambda = i$                                                             | (g) Not diagonalizable, since its eigenvalue $\lambda = i$                                                          |

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| johnston-linear-matrix-attrib F04982 | 480  | ■ 1.000 | 2+i, 2+i, 2-i, 2-i                                                                                            | $2 + i, 2 + i, 2 - i, 2 - i$                                                                               | with diagonal entries $2 + i, 2 + i, 2 - i, 2 - i$ ,                                                      |
| johnston-linear-matrix-attrib F04983 | 480  | ■ 0.999 | $\left[\begin{array}{ll}\sin (2) / 2 & \sin (2) / 2 \\ \sin (2) / 2 & \sin (2) / 2\end{array}\right]$         | $\left[\begin{array}{cc}\sin (2) / 2 & \sin (2) / 2 \\ \sin (2) / 2 & \sin (2) / 2\end{array}\right]$      | (b) $\left[\begin{array}{cc}\sin (2) / 2 & \sin (2) / 2 \\ \sin (2) / 2 & \sin (2) / 2\end{array}\right]$ |
| johnston-linear-matrix-attrib F04984 | 480  | ■ 1.000 | P=D=0                                                                                                         | $P=D=O$                                                                                                    | choose $P=D=O$ . To make this statement                                                                   |
| johnston-linear-matrix-attrib F04985 | 480  | ■ 1.000 | A=P D P^{-1}=P(\lambda I) P^{-1}=\lambda P P^{-1}=\lambda I                                                   | $A=PDP^{-1}=P(\lambda I)P^{-1}=\lambda PP^{-1}=\lambda I$                                                  | $A=PDP^{-1}=P(\lambda I)P^{-1}=\lambda PP^{-1}=\lambda I$ .                                               |
| johnston-linear-matrix-attrib F04986 | 480  | ■ 1.000 | L_{0}=2                                                                                                       | $L_0=2$                                                                                                    | except for the initial condition $L_0=2$ instead of                                                       |
| johnston-linear-matrix-attrib F04987 | 480  | ■ 0.999 | 2 \phi -1=\sqrt{5}                                                                                            | $2\phi -1=\sqrt{5}$                                                                                        | where we used the fact that $2\phi -1=\sqrt{5}$ in the fourth                                             |
| johnston-linear-matrix-attrib F04988 | 480  | ■ 1.000 | $\operatorname{rank}(A)=\operatorname{rank}\left(P D P^{-1}\right)=\operatorname{rank}\left(D P^{-1}\right)=$ | $\operatorname{rank}(A)=\operatorname{rank}\left(PDP^{-1}\right)=\operatorname{rank}\left(DP^{-1}\right)=$ | $\operatorname{rank}(A)=\operatorname{rank}(PDP^{-1})=\operatorname{rank}(DP^{-1})=$                      |
| johnston-linear-matrix-attrib F04989 | 480  | ■ 0.996 | $\operatorname{rank}(D)$                                                                                      | $\operatorname{rank}(D)$                                                                                   | $\operatorname{rank}(D)$ . Since $D$ is diagonal, its rank is the                                         |
| johnston-linear-matrix-attrib F04990 | 480  | ■ 1.000 | B=P D P^{-1}                                                                                                  | $B=PDP^{-1}$                                                                                               | is, $B=PDP^{-1}$ too, so $\tilde{A}=B$ .                                                                  |
| johnston-linear-matrix-attrib F04991 | 481  | ■ 0.913 | $(\{(1,0)\})$                                                                                                 | $(\{(1,0)\})$                                                                                              | tors $(\{(1,0)\})$ is a basis of the corresponding                                                        |
| johnston-linear-matrix-attrib F04992 | 481  | ■ 1.000 | P \widetilde{D} P^{-1}                                                                                        | $\tilde{P}\tilde{D}P^{-1}$                                                                                 | $k^n$ distinct $k$ -th roots of the form $\tilde{P}\tilde{D}P^{-1}$ , where                               |
| johnston-linear-matrix-attrib F04993 | 481  | ■ 1.000 | $k^n$ $k$                                                                                                     | $k^n k$                                                                                                    | there are <i>exactly</i> $k^n$ $k$ -th roots), notice that if                                             |
| johnston-linear-matrix-attrib F04994 | 481  | ■ 1.000 | P=                                                                                                            | $P=$                                                                                                       | we will call $\lambda$ , and we partition $P$ as $P=$                                                     |
| johnston-linear-matrix-attrib F04995 | 481  | ■ 1.000 | $\left \mathbf{v}_1\right  \left \mathbf{v}_2\right  P_2$                                                     | $\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid P_2\right]$                                                     | $\left[\mathbf{v}_1 \mid \mathbf{v}_2 \mid P_2\right]$ . If we let                                        |
| johnston-linear-matrix-attrib F04996 | 481  | ■ 1.000 | $\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}$                         | $\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}$                      | be such that $\left\{c_1 \mathbf{v}_1+c_2 \mathbf{v}_2, d_1 \mathbf{v}_1+d_2 \mathbf{v}_2\right\}$ is     |
| johnston-linear-matrix-attrib F04997 | 481  | ■ 1.000 | A=Q D Q^{-1}                                                                                                  | $A=QDQ^{-1}$                                                                                               | $A=QDQ^{-1}$ as well. If we let $\tilde{D}$ be a $k$ -th root                                             |
| johnston-linear-matrix-attrib F04998 | 481  | ■ 1.000 | $\widetilde{D}$                                                                                               | $\tilde{D}$                                                                                                | $A=QDQ^{-1}$ as well. If we let $\tilde{D}$ be a $k$ -th root                                             |
| johnston-linear-matrix-attrib F04999 | 481  | ■ 1.000 | Q \widetilde{D} Q^{-1}                                                                                        | $Q\tilde{D}Q^{-1}$                                                                                         | matrices of the form $Q\tilde{D}Q^{-1}$ (by varying the                                                   |
| johnston-linear-matrix-attrib F05000 | 481  | ■ 1.000 | $c_1, c_2, d_1, d_2$                                                                                          | $c_1, c_2, d_1, d_2$                                                                                       | coefficients $c_1, c_2, d_1, d_2$ in the definition of $Q$ ),                                             |
| johnston-linear-matrix-attrib F05001 | 481  | ■ 1.000 | $A^2 \mathbf{v}=\lambda A \mathbf{v}=\lambda^2 \mathbf{v}$                                                    | $A^2 \mathbf{v}=\lambda A \mathbf{v}=\lambda^2 \mathbf{v}$                                                 | that $A^2 \mathbf{v}=\lambda A \mathbf{v}=\lambda^2 \mathbf{v}$ .                                         |
| johnston-linear-matrix-attrib F05002 | 481  | ■ 1.000 | $\lambda^0=1$                                                                                                 | $\lambda^0=1$                                                                                              | $A^0=I$ with eigenvalue $\lambda^0=1$ .                                                                   |
| johnston-linear-matrix-attrib F05003 | 481  | ■ 1.000 | $p(x)=c_k x^k+\cdots+c_1 x+c_0$                                                                               | $p(x)=c_k x^k+\cdots+c_1 x+c_0$                                                                            | $p(x)=c_k x^k+\cdots+c_1 x+c_0$ then                                                                      |
| johnston-linear-matrix-attrib F05004 | 481  | ■ 1.000 | $x^r x^s=x^{r+s}$                                                                                             | $x^r x^s=x^{r+s}$                                                                                          | works entrywise and $x^r x^s=x^{r+s}$ for all scalars                                                     |
| johnston-linear-matrix-attrib F05005 | 481  | ■ 1.000 | $\left(x^r\right)^s=x^{r s}$                                                                                  | $\left(x^r\right)^s=x^{r s}$                                                                               | works entrywise and $\left(x^r\right)^s=x^{r s}$ for all scalars                                          |
| johnston-linear-matrix-attrib F05006 | 481  | ■ 1.000 | $\operatorname{det}\left(e^{A+B}\right)=e^{\operatorname{tr}(A+B)}$                                           | $\operatorname{det}\left(e^{A+B}\right)=e^{\operatorname{tr}(A+B)}$                                        | similarly $\operatorname{det}\left(e^{A+B}\right)=e^{\operatorname{tr}(A+B)}$ . We thus conclude          |
| johnston-linear-matrix-attrib F05007 | 481  | ■ 1.000 | $-a_0-a_1 \lambda_j-a_2 \lambda_j^2-$                                                                         | $-a_0-a_1 \lambda_j-a_2 \lambda_j^2-$                                                                      | where the final entry $-a_0-a_1 \lambda_j-a_2 \lambda_j^2-$                                               |
| johnston-linear-matrix-attrib F05008 | 481  | ■ 1.000 | $\cdots-a_{n-1} \lambda_j^{n-1}$                                                                              | $\cdots-a_{n-1} \lambda_j^{n-1}$                                                                           | $\cdots-a_{n-1} \lambda_j^{n-1}$ equals $\lambda_j^n$ , since $\lambda_j$ is a root                       |
| johnston-linear-matrix-attrib F05009 | 481  | ■ 1.000 | $\lambda_j^n$                                                                                                 | $\lambda_j^n$                                                                                              | $\cdots-a_{n-1} \lambda_j^{n-1}$ equals $\lambda_j^n$ , since $\lambda_j$ is a root                       |

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| johnston-linear-matrix-atlas-482-F05010 | 482  | 1.000 | $C \, \mathbf{v} = \lambda \mathbf{v}$                                                                                       | $C\mathbf{v} = \lambda\mathbf{v}$                                                                                            | value $\lambda$ . Then $C\mathbf{v} = \lambda\mathbf{v}$ implies                                                                    |
| johnston-linear-matrix-atlas-482-F05011 | 482  | 1.000 | $\operatorname{tr}(B) =$                                                                                                     | $\operatorname{tr}(B) =$                                                                                                     | (c) Not similar. For example, $\operatorname{tr}(A) = 6$ but $\operatorname{tr}(B) =$                                               |
| johnston-linear-matrix-atlas-482-F05012 | 482  | 1.000 | $(\lambda - 2)(\lambda - 3)$                                                                                                 | $(\lambda - 2)(\lambda - 3)$                                                                                                 | istic polynomials $(\lambda - 2)(\lambda - 3)$ . Explicitly,                                                                        |
| johnston-linear-matrix-atlas-482-F05013 | 482  | 1.000 | $(\lambda - 1)(\lambda - 3)^2$                                                                                               | $(\lambda - 1)(\lambda - 3)^2$                                                                                               | istic polynomials $(\lambda - 1)(\lambda - 3)^2$ . Explicitly,                                                                      |
| johnston-linear-matrix-atlas-482-F05014 | 482  | 1.000 | $A^2 \, \mathbf{v} = 3 \, A \, \mathbf{v} = 9 \, \mathbf{v}$                                                                 | $A^2\mathbf{v} = 3A\mathbf{v} = 9\mathbf{v}$                                                                                 | by $A$ gives $A^2\mathbf{v} = 3A\mathbf{v} = 9\mathbf{v}$ .                                                                         |
| johnston-linear-matrix-atlas-482-F05015 | 482  | 1.000 | $p_A(\lambda) = p_B(\lambda) = \lambda^4$                                                                                    | $p_A(\lambda) = p_B(\lambda) = \lambda^4$                                                                                    | is $p_A(\lambda) = p_B(\lambda) = \lambda^4$ .                                                                                      |
| johnston-linear-matrix-atlas-482-F05016 | 482  | 0.728 | $\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$                                                                          | $\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$                                                                          | $\operatorname{span}\{(1, 0, 0, 0), (0, 1, 0, 0)\}$ . In particular, for                                                            |
| johnston-linear-matrix-atlas-482-F05017 | 482  | 1.000 | $v_2 = \lambda v_1$                                                                                                          | $v_2 = \lambda v_1$                                                                                                          | $v_2 = \lambda v_1$ , the next entry tells us that $v_3 = \lambda v_2$ ,                                                            |
| johnston-linear-matrix-atlas-482-F05018 | 482  | 1.000 | $v_3 = \lambda v_2$                                                                                                          | $v_3 = \lambda v_2$                                                                                                          | $v_2 = \lambda v_1$ , the next entry tells us that $v_3 = \lambda v_2$ ,                                                            |
| johnston-linear-matrix-atlas-482-F05019 | 482  | 1.000 | $v_n = \lambda v_{n-1}$                                                                                                      | $v_n = \lambda v_{n-1}$                                                                                                      | and so on up to $v_n = \lambda v_{n-1}$ . It follows that                                                                           |
| johnston-linear-matrix-atlas-482-F05020 | 482  | 1.000 | $v_j = \lambda^{j-1} v_1$                                                                                                    | $v_j = \lambda^{j-1} v_1$                                                                                                    | $v_j = \lambda^{j-1} v_1$ for $1 \leq j \leq n - 1$ . Then the bot-                                                                 |
| johnston-linear-matrix-atlas-482-F05021 | 482  | 1.000 | $1 \leq j \leq n - 1$                                                                                                        | $1 \leq j \leq n - 1$                                                                                                        | $v_j = \lambda^{j-1} v_1$ for $1 \leq j \leq n - 1$ . Then the bot-                                                                 |
| johnston-linear-matrix-atlas-482-F05022 | 482  | 1.000 | $v_n = \lambda^{n-1} v_1$                                                                                                    | $v_n = \lambda^{n-1} v_1$                                                                                                    | so $v_n = \lambda^{n-1} v_1$ . We have thus shown                                                                                   |
| johnston-linear-matrix-atlas-482-F05023 | 482  | 1.000 | $v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$                                                                           | $v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$                                                                           | $v_1(1, \lambda, \lambda^2, \dots, \lambda^{n-1})$ . This set of vectors is                                                         |
| johnston-linear-matrix-atlas-482-F05024 | 482  | 1.000 | $P \in \mathcal{M}_4$                                                                                                        | $P \in \mathcal{M}_4$                                                                                                        | (d) Following the hint, let $P \in \mathcal{M}_4$ and write the                                                                     |
| johnston-linear-matrix-atlas-482-F05025 | 482  | 1.000 | $a + d = 0$                                                                                                                  | $a + d = 0$                                                                                                                  | that either $c = 0$ or $a + d = 0$ . However, if $a + d = 0$                                                                        |
| johnston-linear-matrix-atlas-482-F05026 | 482  | 1.000 | $b(0 + 0) = 1$                                                                                                               | $b(0 + 0) = 1$                                                                                                               | gives us a contradiction since it says $b(0 + 0) = 1$ . It                                                                          |
| johnston-linear-matrix-atlas-483-F05027 | 483  | 0.979 | $\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$                                     | $\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$                                     | (a) $\operatorname{det}(A) = -3, \operatorname{det}(A_1) = -3, \operatorname{det}(A_2) = -3$ ,                                      |
| johnston-linear-matrix-atlas-483-F05028 | 483  | 0.971 | $\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$                                        | $\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$                                        | (c) $\operatorname{det}(A) = 4, \operatorname{det}(A_1) = 2, \operatorname{det}(A_2) = 8$ ,                                         |
| johnston-linear-matrix-atlas-483-F05029 | 483  | 1.000 | $\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$                                                                     | $\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$                                                                     | $\operatorname{det}(A_3) = 6, (x, y, z) = (1/2, 2, 3/2)$ .                                                                          |
| johnston-linear-matrix-atlas-483-F05030 | 483  | 0.998 | $\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$                                      | $\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$                                      | (e) $\operatorname{det}(A) = -2, \operatorname{det}(A_1) = 1, \operatorname{det}(A_2) = -2$ ,                                       |
| johnston-linear-matrix-atlas-483-F05031 | 483  | 1.000 | $\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$                                                                   | $\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$                                                                   | $\operatorname{det}(A_3) = -1, (x, y, z) = (-1/2, 1, 1/2)$ .                                                                        |
| johnston-linear-matrix-atlas-483-F05032 | 483  | 0.999 | $\operatorname{rank}(\operatorname{cof}(A)) = n$                                                                             | $\operatorname{rank}(\operatorname{cof}(A)) = n$                                                                             | then $\operatorname{rank}(\operatorname{cof}(A)) = n$ by part (a) of this exer-                                                     |
| johnston-linear-matrix-atlas-483-F05033 | 483  | 0.741 | $\boldsymbol{l} \circ \boldsymbol{l} = \boldsymbol{l}$                                                                       | (not rendered)                                                                                                               | (g) True. This follows from the fact that $\boldsymbol{l} \circ \boldsymbol{l} = \boldsymbol{l}$ .                                  |
| johnston-linear-matrix-atlas-483-F05034 | 483  | 1.000 | $n = 5$                                                                                                                      | $n = 5$                                                                                                                      | $n = 5$ , there are $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ per-                                                             |
| johnston-linear-matrix-atlas-483-F05035 | 483  | 1.000 | $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$                                                                               | $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$                                                                               | $n = 5$ , there are $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$ per-                                                             |
| johnston-linear-matrix-atlas-483-F05036 | 483  | 1.000 | $\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$ | $\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$ | that $\operatorname{cof}(A^2) = \operatorname{cof}(AA) = \operatorname{cof}(A) \operatorname{cof}(A) = (\operatorname{cof}(A))^2$ . |
| johnston-linear-matrix-atlas-483-F05037 | 483  | 1.000 | $\operatorname{cof}(A^{-1}) =$                                                                                               | $\operatorname{cof}(A^{-1}) =$                                                                                               | If $k = -1$ then we notice that $\operatorname{cof}(A^{-1}) =$                                                                      |

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| johnston-linear-matrix-atlas-04-F05038 | 483  | 1.000 | $A^T / \operatorname{det}(A) = (\operatorname{cof}(A))^{-1}$        | $A^T / \det(A) = (\operatorname{cof}(A))^{-1}$                      | $A^T / \det(A) = (\operatorname{cof}(A))^{-1}$ . For $k < -1$ we just re-                                 |
| johnston-linear-matrix-atlas-04-F05039 | 483  | 1.000 | $\operatorname{cof}(A) =$                                           | $\operatorname{cof}(A) =$                                           | <b>3.A.10</b> We use Theorem 3.A.1 to see that $\operatorname{cof}(A) =$                                  |
| johnston-linear-matrix-atlas-04-F05040 | 483  | 1.000 | $\operatorname{det}(A) \left( A^T \right)^{-1}$                     | $\det(A) \left( A^T \right)^{-1}$                                   | $\det(A) \left( A^T \right)^{-1}$ . Taking the determinant of both sides                                  |
| johnston-linear-matrix-atlas-04-F05041 | 483  | 1.000 | $\tau = (21345 \cdots n)$                                           | $\tau = (21345 \cdots n)$                                           | <b>(b)</b> Let $\tau = (2 \ 1 \ 3 \ 4 \ 5 \ \cdots \ n)$ be the permutation                               |
| johnston-linear-matrix-atlas-04-F05042 | 483  | 1.000 | $\operatorname{sgn}(\tau) = -1$                                     | $\operatorname{sgn}(\tau) = -1$                                     | alone. Then $\operatorname{sgn}(\tau) = -1$ and as $\sigma$ ranges over                                   |
| johnston-linear-matrix-atlas-04-F05043 | 483  | 0.993 | $\operatorname{rank}(A) \leq n < m$                                 | $\operatorname{rank}(A) \leq n < m$                                 | <b>3.A.14</b> If $m > n$ then $\operatorname{rank}(A) \leq n < m$ so $\operatorname{rank}(AB) \leq n < m$ |
| johnston-linear-matrix-atlas-04-F05044 | 483  | 0.993 | $\operatorname{rank}(A \ B) \leq n < m$                             | $\operatorname{rank}(AB) \leq n < m$                                | <b>3.A.14</b> If $m > n$ then $\operatorname{rank}(A) \leq n < m$ so $\operatorname{rank}(AB) \leq n < m$ |
| johnston-linear-matrix-atlas-04-F05045 | 483  | 1.000 | $\operatorname{det}(A \ B) = 0$                                     | $\det(AB) = 0$                                                      | so it is thus not invertible and $\det(AB) = 0$ .                                                         |
| johnston-linear-matrix-atlas-04-F05046 | 483  | 0.999 | $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$             | $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$             | then $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$ , but $\operatorname{per}(AB) = 48$ .        |
| johnston-linear-matrix-atlas-04-F05047 | 483  | 0.999 | $\operatorname{per}(A \ B) = 48$                                    | $\operatorname{per}(AB) = 48$                                       | then $\operatorname{per}(A) = 2, \operatorname{per}(B) = 10$ , but $\operatorname{per}(AB) = 48$ .        |
| johnston-linear-matrix-atlas-04-F05048 | 484  | 0.881 | $1 \quad$                                                           | 1                                                                   | <b>3.B.1</b> <b>(a)</b> $\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$                           |
| johnston-linear-matrix-atlas-04-F05049 | 484  | 0.881 | $\quad \lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$       | $\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$             | <b>3.B.1</b> <b>(a)</b> $\lambda \approx 5.37, \mathbf{v} \approx (0.42, 0.91)$                           |
| johnston-linear-matrix-atlas-04-F05050 | 484  | 1.000 | $\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$    | $\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$    | <b>(c)</b> $\lambda \approx -3.33, \mathbf{v} \approx (0.73, -0.42, -0.55)$                               |
| johnston-linear-matrix-atlas-04-F05051 | 484  | 1.000 | $\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$ | $\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$ | <b>(e)</b> $\lambda \approx 7.92, \mathbf{v} \approx (0.73, 0.29, 0.57, 0.24)$                            |
| johnston-linear-matrix-atlas-04-F05052 | 484  | 1.000 | $2 \quad$                                                           | 2                                                                   | <b>3.B.2</b> <b>(a)</b> $\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$                       |
| johnston-linear-matrix-atlas-04-F05053 | 484  | 1.000 | $\quad \lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$   | $\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$         | <b>3.B.2</b> <b>(a)</b> $\lambda_1 \approx 5.37, \mathbf{v}_1 \approx (0.42, 0.91)$                       |
| johnston-linear-matrix-atlas-04-F05054 | 484  | 1.000 | $\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$       | $\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$       | $\lambda_2 \approx -0.37, \mathbf{v}_2 \approx (0.82, -0.57)$                                             |
| johnston-linear-matrix-atlas-04-F05055 | 484  | 1.000 | $\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$  | $\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$  | <b>(c)</b> $\lambda_1 \approx 16.12, \mathbf{v}_1 \approx (0.23, 0.53, 0.82)$                             |
| johnston-linear-matrix-atlas-04-F05056 | 484  | 1.000 | $\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$ | $\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$ | $\lambda_2 \approx -1.12, \mathbf{v}_2 \approx (0.79, 0.09, -0.61)$                                       |
| johnston-linear-matrix-atlas-04-F05057 | 484  | 1.000 | $\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$  | $\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$  | $\lambda_3 \approx 0.00, \mathbf{v}_2 \approx (0.41, -0.82, 0.41)$                                        |
| johnston-linear-matrix-atlas-04-F05058 | 484  | 0.520 | $3 \quad$                                                           | 3                                                                   | <b>3.B.3</b> <b>(a)</b> $\lambda \approx 11.58,$                                                          |
| johnston-linear-matrix-atlas-04-F05059 | 484  | 0.520 | $\quad \lambda \approx 11.58$                                       | $\lambda \approx 11.58$                                             | <b>3.B.3</b> <b>(a)</b> $\lambda \approx 11.58,$                                                          |
| johnston-linear-matrix-atlas-04-F05060 | 484  | 1.000 | $\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$    | $\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$    | $\mathbf{v} \approx (0.15, 0.25, 0.01, 0.54, -0.27, 0.46, 0.58)$                                          |
| johnston-linear-matrix-atlas-04-F05061 | 484  | 1.000 | $A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$                     | $A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$                     | ing to the dominant eigenvalue of $A$ , so $A \mathbf{v}_k \approx \lambda_1 \mathbf{v}_k$                |
| johnston-linear-matrix-atlas-04-F05062 | 484  | 1.000 | $\ A \mathbf{v}_k\  \approx \ \lambda_1 \mathbf{v}_k\  = \lambda_1$ | $\ A \mathbf{v}_k\  \approx \ \lambda_1 \mathbf{v}_k\  = \lambda_1$ | when $k$ is large, so $\ A \mathbf{v}_k\  \approx \ \lambda_1 \mathbf{v}_k\  = \lambda_1$ when $k$ is     |
| johnston-linear-matrix-atlas-04-F05063 | 484  | 0.994 | $k - 2 \cdot 4^2 -$                                                 | $k - 2 \cdot 4^2 -$                                                 | $k - 2 \cdot 4^2 -$                                                                                       |
| johnston-linear-matrix-atlas-04-F05064 | 484  | 1.000 | $k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$         | $k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$         | <b>(c)</b> $k = 4: A^4 = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$                                    |

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| johnston-linear-matrix-atlas-04-F05065 | 484  | 0.993 | $k=4: A^{\{4\}}=\left[\begin{array}{lll}5 & 4 & 1 \\ 1 & 3 & 2\end{array}\right]$                                                              | $k = 4: A^4 = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 3 \\ 1 & 3 & 2 \end{bmatrix}$                                                       | (e) $k = 4: A^4 = \begin{bmatrix} 5 & 4 & 1 \\ 4 & 6 & 3 \\ 1 & 3 & 2 \end{bmatrix}$                                                               |
| johnston-linear-matrix-atlas-04-F05066 | 484  | 0.445 | $\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$                                                                                        | $\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$                                                                                | 3.B.10 (a) $\lambda \approx 1.62, \mathbf{v} \approx (0.53, 0.85)$                                                                                 |
| johnston-linear-matrix-atlas-04-F05067 | 484  | 1.000 | $\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$                                                                                       | $\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$                                                                               | (c) $\lambda \approx 2.00i, \mathbf{v} \approx (0.71, 0.71)$                                                                                       |
| johnston-linear-matrix-atlas-04-F05068 | 484  | 1.000 | $\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$                                                                                  | $\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$                                                                          | (e) $\lambda \approx 1.80, \mathbf{v} \approx (0.59, 0.74, 0.33)$                                                                                  |
| johnston-linear-matrix-atlas-04-F05069 | 484  | 1.000 | $\left[R^{\pi/2}\right]$                                                                                                                       | $\left[R^{\pi/2}\right]$                                                                                                               | 3.C.1 (a) We recognize this as the rotation matrix $\left[R^{\pi/2}\right]$ ,                                                                      |
| johnston-linear-matrix-atlas-04-F05070 | 484  | 1.000 | $Q=I, r=1$                                                                                                                                     | $Q = I, r = 1$                                                                                                                         | just from choosing $Q = I, r = 1$ , and $\theta = \pi/2$ .                                                                                         |
| johnston-linear-matrix-atlas-04-F05071 | 484  | 1.000 | $1 \pm i = \sqrt{2} e^{i \pi/4}$                                                                                                               | $1 \pm i = \sqrt{2} e^{\pm i \pi/4}$                                                                                                   | (c) This matrix has eigenvalues $1 \pm i = \sqrt{2} e^{\pm i \pi/4}$                                                                               |
| johnston-linear-matrix-atlas-04-F05072 | 484  | 0.999 | $(1, -1 \pm i)$                                                                                                                                | $(1, -1 \pm i)$                                                                                                                        | with corresponding eigenvectors $(1, -1 \pm i)$ , so                                                                                               |
| johnston-linear-matrix-atlas-04-F05073 | 484  | 1.000 | $r = \sqrt{2}, \theta = \pi/4$                                                                                                                 | $r = \sqrt{2}, \theta = \pi/4$                                                                                                         | we can choose $r = \sqrt{2}, \theta = \pi/4$ , and                                                                                                 |
| johnston-linear-matrix-atlas-04-F05074 | 484  | 0.922 | $B=\operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q=\left[\begin{array}{ccc}1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1\end{array}\right]$ | $B = \operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ | 3.C.2 (a) $B = \operatorname{diag}\left(1,\left[R^{\pi/2}\right]\right), Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ . |
| johnston-linear-matrix-atlas-04-F05075 | 484  | 1.000 | $B=\operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$                                                                         | $B = \operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$                                                               | (c) $B = \operatorname{diag}\left(1,2,\sqrt{2}\left[R^{\pi/4}\right]\right)$ ,                                                                     |
| johnston-linear-matrix-atlas-04-F05076 | 484  | 0.998 | $B=\operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$                                          | $B = \operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$                                | 3.C.3 (a) $B = \operatorname{diag}\left(2\sqrt{2}\left[R^{\pi/4}\right],2\sqrt{2}\left[R^{\pi/4}\right],4\right)$ ,                                |
| johnston-linear-matrix-atlas-04-F05077 | 485  | 0.391 | $2^k(\cos(k\pi/6) - \sin(k\pi/6))$                                                                                                             | $2^k(\cos(k\pi/6) - \sin(k\pi/6))$                                                                                                     | $2^k(\cos(k\pi/6) - \sin(k\pi/6))$ , top-right                                                                                                     |
| johnston-linear-matrix-atlas-04-F05078 | 485  | 1.000 | $2^k \sin(k\pi/6)$                                                                                                                             | $2^k \sin(k\pi/6)$                                                                                                                     | entry $2^k \sin(k\pi/6)$ , bottom-left entry                                                                                                       |
| johnston-linear-matrix-atlas-04-F05079 | 485  | 1.000 | $-2^{k+1} \sin(k\pi/6)$                                                                                                                        | $-2^{k+1} \sin(k\pi/6)$                                                                                                                | $-2^{k+1} \sin(k\pi/6)$ , and bottom-right entry                                                                                                   |
| johnston-linear-matrix-atlas-04-F05080 | 485  | 1.000 | $2^k(\cos(k\pi/6) + \sin(k\pi/6))$                                                                                                             | $2^k(\cos(k\pi/6) + \sin(k\pi/6))$                                                                                                     | $2^k(\cos(k\pi/6) + \sin(k\pi/6))$ .                                                                                                               |
| johnston-linear-matrix-atlas-04-F05081 | 485  | 1.000 | $\lambda^2 - \lambda - 6$                                                                                                                      | $\lambda^2 - \lambda - 6$                                                                                                              | currence relation is $\lambda^2 - \lambda - 6$ , which has roots                                                                                   |
| johnston-linear-matrix-atlas-04-F05082 | 485  | 0.998 | $\lambda_0=3$                                                                                                                                  | $\lambda_0 = 3$                                                                                                                        | $\lambda_0 = 3$ and $\lambda_1 = -2$ , each with multiplicity 1.                                                                                   |
| johnston-linear-matrix-atlas-04-F05083 | 485  | 0.998 | $\lambda_1=-2$                                                                                                                                 | $\lambda_1 = -2$                                                                                                                       | $\lambda_0 = 3$ and $\lambda_1 = -2$ , each with multiplicity 1.                                                                                   |
| johnston-linear-matrix-atlas-04-F05084 | 485  | 1.000 | $3 \pm \sqrt{5}$                                                                                                                               | $3 \pm \sqrt{5}$                                                                                                                       | (c) $3 \pm \sqrt{5}$ , each with multiplicity 1.                                                                                                   |
| johnston-linear-matrix-atlas-04-F05085 | 485  | 0.999 | $1 \pm \sqrt{3}$                                                                                                                               | $1 \pm \sqrt{3}$                                                                                                                       | (i) $3$ and $1 \pm \sqrt{3}$ , each with multiplicity 1.                                                                                           |
| johnston-linear-matrix-atlas-04-F05086 | 485  | 1.000 | $\left[\begin{array}{ll}0 & 1 \\ 6 & 1\end{array}\right]$                                                                                      | $\begin{bmatrix} 0 & 1 \\ 6 & 1 \end{bmatrix}$                                                                                         | (a) $\begin{bmatrix} 0 & 1 \\ 6 & 1 \end{bmatrix}$                                                                                                 |
| johnston-linear-matrix-atlas-04-F05087 | 485  | 1.000 | $\left[\begin{array}{cc}0 & 1 \\ -4 & 6\end{array}\right]$                                                                                     | $\begin{bmatrix} 0 & 1 \\ -4 & 6 \end{bmatrix}$                                                                                        | (c) $\begin{bmatrix} 0 & 1 \\ -4 & 6 \end{bmatrix}$                                                                                                |
| johnston-linear-matrix-atlas-04-F05088 | 485  | 1.000 | $\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6\end{array}\right]$                                                              | $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$                                                                  | (e) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$                                                                          |

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|-----------------------------------------|------|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| johnston-linear-matrix-atlas-<br>F05089 | 485  | 1.000 | $\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6\end{array}\right]$ | $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6 \end{bmatrix}$ | (g) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 8 & -12 & 6 \end{bmatrix}$       |
| johnston-linear-matrix-atlas-<br>F05090 | 485  | 1.000 | $\left[\begin{array}{ccc}0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5\end{array}\right]$ | $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5 \end{bmatrix}$ | (i) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -4 & 5 \end{bmatrix}$       |
| johnston-linear-matrix-atlas-<br>F05091 | 485  | 1.000 | $\lambda^2 z - \lambda - 6$                                                       | $\lambda^2 z - \lambda - 6$                                           | currence relation is $\lambda^2 z - \lambda - 6$ , which has roots              |
| johnston-linear-matrix-atlas-<br>F05092 | 485  | 1.000 | $c_{\{0\}}$                                                                       | $c_0$                                                                 | scalars $c_0$ and $c_1$ such that $x_n = c_0 3^n + c_1 (-2)^n$                  |
| johnston-linear-matrix-atlas-<br>F05093 | 485  | 1.000 | $x_{\{n\}} = c_{\{0\}} 3^n + c_{\{1\}} (-2)^n$                                    | $x_n = c_0 3^n + c_1 (-2)^n$                                          | scalars $c_0$ and $c_1$ such that $x_n = c_0 3^n + c_1 (-2)^n$                  |
| johnston-linear-matrix-atlas-<br>F05094 | 485  | 1.000 | $x_{\{0\}} = 4$                                                                   | $x_0 = 4$                                                             | for all $n$ . Using the fact that $x_0 = 4$ and $x_1 = -3$                      |
| johnston-linear-matrix-atlas-<br>F05095 | 485  | 1.000 | $x_{\{1\}} = -3$                                                                  | $x_1 = -3$                                                            | for all $n$ . Using the fact that $x_0 = 4$ and $x_1 = -3$                      |
| johnston-linear-matrix-atlas-<br>F05096 | 485  | 1.000 | $x_{\{n\}} = 3^n +$                                                               | $x_n = 3^n +$                                                         | shows that $c_0 = 1$ and $c_1 = 3$ , so $x_n = 3^n +$                           |
| johnston-linear-matrix-atlas-<br>F05097 | 485  | 1.000 | $3(-2)^n$                                                                         | $3(-2)^n$                                                             | $3(-2)^n$ .                                                                     |
| johnston-linear-matrix-atlas-<br>F05098 | 485  | 1.000 | $x_{\{n\}} = i^n + (-i)^n$                                                        | $x_n = i^n + (-i)^n$                                                  | (c) $x_n = i^n + (-i)^n$ . Alternatively, a formula that                        |
| johnston-linear-matrix-atlas-<br>F05099 | 485  | 1.000 | $x_{\{n\}} =$                                                                     | $x_n =$                                                               | does not involve complex numbers is $x_n =$                                     |
| johnston-linear-matrix-atlas-<br>F05100 | 485  | 1.000 | $2 \cos(\pi n / 2)$                                                               | $2 \cos(\pi n / 2)$                                                   | $2 \cos(\pi n / 2)$ .                                                           |
| johnston-linear-matrix-atlas-<br>F05101 | 485  | 1.000 | $x_{\{n\}} = 3^n - 1$                                                             | $x_n = 3^n - 1$                                                       | (e) $x_n = 3^n - 1$ .                                                           |
| johnston-linear-matrix-atlas-<br>F05102 | 485  | 1.000 | $x_{\{n\}} = (2n - 3) + 2^n$                                                      | $x_n = (2n - 3) + 2^n$                                                | (g) $x_n = (2n - 3) + 2^n$ .                                                    |
| johnston-linear-matrix-atlas-<br>F05103 | 485  | 1.000 | $\lambda^2 + 1$                                                                   | $\lambda^2 + 1$                                                       | mial $\lambda^2 + 1$ , which has roots $\pm i$ .                                |
| johnston-linear-matrix-atlas-<br>F05104 | 485  | 1.000 | $\pm i$                                                                           | $\pm i$                                                               | mial $\lambda^2 + 1$ , which has roots $\pm i$ .                                |
| johnston-linear-matrix-atlas-<br>F05105 | 485  | 1.000 | $n^2 - n^3$                                                                       | $n^2 - n^3$                                                           | $n^2 - n^3$ has degree $4 - 1 = 3$ ). One such lin-                             |
| johnston-linear-matrix-atlas-<br>F05106 | 485  | 1.000 | $4 - 1 = 3$                                                                       | $4 - 1 = 3$                                                           | $n^2 - n^3$ has degree $4 - 1 = 3$ ). One such lin-                             |
| johnston-linear-matrix-atlas-<br>F05107 | 485  | 1.000 | $x_{\{n\}} = 4x_{\{n-1\}} - 6x_{\{n-2\}} +$                                       | $x_n = 4x_{n-1} - 6x_{n-2} +$                                         | ear recurrence relation is $x_n = 4x_{n-1} - 6x_{n-2} +$                        |
| johnston-linear-matrix-atlas-<br>F05108 | 485  | 1.000 | $4x_{\{n-3\}} - x_{\{n-4\}}$                                                      | $4x_{n-3} - x_{n-4}$                                                  | $4x_{n-3} - x_{n-4}$ , which has characteristic polyno-                         |
| johnston-linear-matrix-atlas-<br>F05109 | 485  | 1.000 | $(\lambda - 1)^4$                                                                 | $(\lambda - 1)^4$                                                     | mial $(\lambda - 1)^4$ .                                                        |
| johnston-linear-matrix-atlas-<br>F05110 | 485  | 0.998 | $p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$              | $p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$  | it is $p(\lambda) = (\lambda - 2)(\lambda - 3) = \lambda^2 - 5\lambda + 6$ , so |
| johnston-linear-matrix-atlas-<br>F05111 | 485  | 1.000 | $x_{\{n\}} = 9x_{\{n-1\}} - 26x_{\{n-2\}} + 24x_{\{n-3\}}$                        | $x_n = 9x_{n-1} - 26x_{n-2} + 24x_{n-3}$                              | (c) $x_n = 9x_{n-1} - 26x_{n-2} + 24x_{n-3}$                                    |
| johnston-linear-matrix-atlas-<br>F05112 | 485  | 1.000 | $x_{\{n\}} = 9x_{\{n-1\}} - 30x_{\{n-2\}} + 44x_{\{n-3\}} - 24x_{\{n-4\}}$        | $x_n = 9x_{n-1} - 30x_{n-2} + 44x_{n-3} - 24x_{n-4}$                  | (e) $x_n = 9x_{n-1} - 30x_{n-2} + 44x_{n-3} - 24x_{n-4}$                        |
| johnston-linear-matrix-atlas-<br>F05113 | 485  | 1.000 | $x_{\{n\}} = 3x_{\{n-1\}} - 3x_{\{n-2\}} + x_{\{n-3\}}$                           | $x_n = 3x_{n-1} - 3x_{n-2} + x_{n-3}$                                 | (g) $x_n = 3x_{n-1} - 3x_{n-2} + x_{n-3}$                                       |
| johnston-linear-matrix-atlas-<br>F05114 | 485  | 1.000 | $x_{\{n\}} = 8x_{\{n-1\}} - 26x_{\{n-2\}} + 48x_{\{n-3\}} - 45x_{\{n-4\}}$        | $x_n = 8x_{n-1} - 26x_{n-2} + 48x_{n-3} - 45x_{n-4}$                  | (i) $x_n = 8x_{n-1} - 26x_{n-2} + 48x_{n-3} - 45x_{n-4}$                        |
| johnston-linear-matrix-atlas-<br>F05115 | 485  | 1.000 | $\pm i =$                                                                         | $\pm i =$                                                             | rence relation is $\lambda^2 + 1$ , which has roots $\pm i =$                   |

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| johnston-linear-matrix-atlas-2019-F05116 | 485  | 1.000 | $e^{\pm i \pi / 2}$                                | $e^{\pm i \pi / 2}$                                | $e^{\pm i \pi / 2}$ . After solving for the coefficients in the                    |
| johnston-linear-matrix-atlas-2019-F05117 | 485  | 1.000 | $(\pm i)^n = \left( e^{\pm i \pi / 2} \right)^n =$ | $(\pm i)^n = \left( e^{\pm i \pi / 2} \right)^n =$ | tuting in the polar forms $(\pm i)^n = (e^{\pm i \pi / 2})^n =$                    |
| johnston-linear-matrix-atlas-2019-F05118 | 485  | 1.000 | $e^{\pm i n \pi / 2}$                              | $e^{\pm i n \pi / 2}$                              | $e^{\pm i n \pi / 2}$ , we see that                                                |
| johnston-linear-matrix-atlas-2019-F05119 | 485  | 1.000 | $x_n = 1 + 2^n \cos(n \pi / 2)$                    | $x_n = 1 + 2^{n+1} \cos(n \pi / 2)$                | (c) $x_n = 1 + 2^{n+1} \cos(n \pi / 2)$ .                                          |
| johnston-linear-matrix-atlas-2019-F05120 | 485  | 1.000 | $x_0 = x_1 = 0$                                    | $x_0 = x_1 = 0$                                    | trivial since we were told that $x_0 = x_1 = 0$ . For the in-                      |
| johnston-linear-matrix-atlas-2019-F05121 | 485  | 1.000 | $x_{n-1} = x_{n-2}$                                | $x_{n-1} = x_{n-2}$                                | ductive step, suppose $x_{n-1} = x_{n-2}$ . Then using the re-                     |
| johnston-linear-matrix-atlas-2019-F05122 | 485  | 1.000 | $x_{n+1} = x_n - 4 x_{n-1} + 4 x_{n-2} =$          | $x_{n+1} = x_n - 4 x_{n-1} + 4 x_{n-2} =$          | currence relation gives $x_{n+1} = x_n - 4 x_{n-1} + 4 x_{n-2} =$                  |
| johnston-linear-matrix-atlas-2019-F05123 | 485  | 1.000 | $x_n - 4 x_{n-1} + 4 x_{n-1} = x_n$                | $x_n - 4 x_{n-1} + 4 x_{n-1} = x_n$                | $x_n - 4 x_{n-1} + 4 x_{n-1} = x_n$ , which completes the induc-                   |
| johnston-linear-matrix-atlas-2019-F05124 | 492  | 1.000 | $O_{m,n}$ $m \times n$                             | $O_{m,n}$ $m \times n$                             | $O_{m,n}$ $m \times n$ zero matrix                                                 |
| johnston-linear-matrix-atlas-2019-F05125 | 492  | 1.000 | $O_n$ $n \times n$                                 | $O_n$ $n \times n$                                 | $O_n$ $n \times n$ zero matrix                                                     |
| johnston-linear-matrix-atlas-2019-F05126 | 492  | 1.000 | $[A]_{i,j}$ $(i,j)$                                | $[A]_{i,j}$ $(i,j)$                                | $[A]_{i,j}$ $(i,j)$ -entry of the matrix $A$                                       |
| johnston-linear-matrix-atlas-2019-F05127 | 492  | 0.788 | $[T]$                                              | $[T]$                                              | $[T]$ standard matrix of $T$ (with respect to the standard basis)                  |
| johnston-linear-matrix-atlas-2019-F05128 | 492  | 1.000 | $[T]_B$                                            | $[T]_B$                                            | $[T]_B$ standard matrix of $T$ with respect to basis $B$                           |
| johnston-linear-matrix-atlas-2019-F05129 | 492  | 0.623 | $\ \cdot\ $                                        | $\ \cdot\ $                                        | $\ \cdot\ $ the length of a vector                                                 |
| johnston-linear-matrix-atlas-2019-F05130 | 492  | 1.000 | $\overline{A}$                                     | $\overline{A}$                                     | $\overline{A}$ entrywise complex conjugate of the matrix or number $A$             |
| johnston-linear-matrix-atlas-2019-F05131 | 492  | 0.957 | $\det(A)$ $\det(A)$                                | $\det(A)$ $\det(A)$                                | $\det(A)$ determinant of the matrix $A$                                            |
| johnston-linear-matrix-atlas-2019-F05132 | 492  | 0.629 | $\mathbf{e}_j$ $j$                                 | $\mathbf{e}_j$ $j$                                 | $\mathbf{e}_j$ $j$ -th standard basis vector (1 in its $j$ -th entry, 0 elsewhere) |
| johnston-linear-matrix-atlas-2019-F05133 | 492  | 1.000 | $I_n$ $n \times n$                                 | $I_n$ $n \times n$                                 | $I_n$ $n \times n$ identity matrix                                                 |
| johnston-linear-matrix-atlas-2019-F05134 | 492  | 1.000 | $\mathcal{M}_{m,n}$ $m \times n$                   | $\mathcal{M}_{m,n}$ $m \times n$                   | $\mathcal{M}_{m,n}$ $m \times n$ matrices                                          |
| johnston-linear-matrix-atlas-2019-F05135 | 492  | 1.000 | $\mathcal{M}_n$ $n \times n$                       | $\mathcal{M}_n$ $n \times n$                       | $\mathcal{M}_n$ $n \times n$ matrices                                              |
| johnston-linear-matrix-atlas-2019-F05136 | 492  | 1.000 | $\mathcal{M}_n(\mathbb{C})$ $n \times n$           | $\mathcal{M}_n(\mathbb{C})$ $n \times n$           | $\mathcal{M}_n(\mathbb{C})$ $n \times n$ matrices with complex entries             |
| johnston-linear-matrix-atlas-2019-F05137 | 492  | 1.000 | $\mathcal{M}_n(\mathbb{R})$ $n \times n$           | $\mathcal{M}_n(\mathbb{R})$ $n \times n$           | $\mathcal{M}_n(\mathbb{R})$ $n \times n$ matrices with real entries                |
| johnston-linear-matrix-atlas-2019-F05138 | 492  | 1.000 | $\mathbb{R}^n$                                     | $\mathbb{R}^n$                                     | $\mathbb{R}^n$ vectors/tuples with $n$ real entries                                |
| johnston-linear-matrix-atlas-2019-F05139 | 492  | 1.000 | $\mathrm{tr}(A)$                                   | $\mathrm{tr}(A)$                                   | $\mathrm{tr}(A)$ trace of the matrix $A$                                           |
| johnston-linear-matrix-atlas-2019-F05140 | 492  | 0.644 | $\mathbb{Z}_p$                                     | $\mathbb{Z}_p$                                     | $\mathbb{Z}_p$ field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod $p$   |
| johnston-linear-matrix-atlas-2019-F05141 | 492  | 0.644 | $\{0, 1, \dots, p-1\}$                             | $\{0, 1, \dots, p-1\}$                             | $\mathbb{Z}_p$ field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod $p$   |
| johnston-linear-matrix-atlas-2019-F05142 | 492  | 0.644 | $\bmod p$                                          | $\bmod p$                                          | $\mathbb{Z}_p$ field of integers $\{0, 1, \dots, p-1\}$ under arithmetic mod $p$   |
| johnston-linear-matrix-atlas-2019-F05143 | 492  | 1.000 | $\mathbb{Z}_p^n$                                   | $\mathbb{Z}_p^n$                                   | $\mathbb{Z}_p^n$ vectors/tuples with $n$ entries from $\mathbb{Z}_p$               |